

Economic study on the dynamic effects of mergers

Volume 1 – Main report

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¹ European Commission (2025), '[Study on dynamic merger effects](#)', accessed 21 July 2026.

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Executive summary

Competition is a dynamic process of rivalry, where firms compete across a number of dimensions—including price, quality, innovation, investment, and product variety—to make attractive offers to consumers, win sales from their rivals, and increase their profits. A merger may hinder this competitive process by eliminating rivalry between firms that were previously competing, at the expense of consumers. It may also provide merging firms with the ability and incentives to foreclose rivals, thus reducing the competitive constraint exercised by these rivals. However, a merger may also promote rivalry by making the merged entity a more effective competitor, to the benefit of consumers. These effects may apply to current competition on existing products and to future competition on new products and technologies.

Dynamic elements of competition—including rivalry on innovation and investment, and entry by firms challenging the position of established incumbents—are recognised as central to technological progress, long-run growth, and global competitiveness. Dynamic considerations are increasingly relevant in a modern economy that is characterised by disruptive innovation and rapid technological cycles.

This European Commission study, prepared by Oxera, is an input into the Commission's ongoing review of its 2004 and 2008 merger guidelines. This review follows the recommendations of the 2024 Draghi Report and its call for greater weight to be given to innovation, investment, and future competition in merger control. Although competition authorities, including the Commission, have increasingly reflected dynamic considerations in their merger control decisions over the past decade, the framework for analysing dynamic effects is less settled than the one for the effects of mergers on static competition.

Building on a comprehensive synopsis of the relevant academic literature, this Study puts forward a rigorous analytical foundation for the economic assessment of dynamic merger effects. It focuses on mergers between competitors ('horizontal mergers') and also covers mergers between producers of complementary products ('non-horizontal mergers').

Main findings

- **Dynamic effects should be defined broadly, to include the overall effect of mergers on future product market competition**—capturing their impact on investment, innovation, entry, and the prices and availability of new or improved products.
- **Dynamic effects can amplify, offset, or countervail static effects**, increasing the scope for both anticompetitive and procompetitive effects. **This raises the stakes in merger control decisions** in cases with a significant dynamic dimension.
- **Our conceptual framework evaluates mergers giving rise to dynamic effects under an Expected Consumer Welfare (ECW) standard** that accounts for the presence of uncertainty in some of the underlying drivers of future competitive outcomes (notably the success of R&D and innovation efforts, the introduction of new products, entry by potential competitors, and the realisation of dynamic synergies). This standard takes into account both the probability and the magnitude of outcomes from a merger.
- **Horizontal mergers lead to a loss of competition between the merging firms in future product markets. Absent spillovers and synergies, this is likely to reduce ECW**, as a result of the internalisation of negative investment and price externalities. This is the case even in those situations where mergers increase investment incentives by softening future price competition. Greater

market power in future product markets is therefore not a suitable basis for an 'innovation defence'.

- **Horizontal mergers may lead to procompetitive dynamic effects** by leading to dynamic synergies or internalising spillovers. Dynamic synergies may arise through a number of channels, including greater technology diffusion; reductions in incremental investment costs or increases in the effectiveness of investments, through economies of scale and complementarities; or easing of financial constraints. A merger may also internalise spillovers between the merging parties, potentially enhancing innovation incentives. However, the internalisation of spillovers to non-merging firms may create opposing effects, reducing innovation incentives.
- **Non-horizontal mergers may enable the merged entity to engage in dynamic foreclosure conduct**, reducing investments or deterring entry by its rivals, to the detriment of consumers. **Non-horizontal mergers may also give rise to procompetitive dynamic effects.** These can result from the internalisation of price or investment externalities, dynamic synergies, or the mitigation of contractual frictions. Since non-horizontal mergers tend to generate efficiencies from the integration of complementary activities, potentially harmful and beneficial mechanisms are often concurrent and intertwined.
- **In addition to leading to potentially harmful dynamic horizontal and dynamic foreclosure effects, mergers may also entrench the position of an incumbent with significant market power**, by increasing barriers to entry and expansion by rivals, to the ultimate detriment of consumers. The recent academic literature identifies a number of specific dynamic mechanisms that may give rise to entrenchment. When a specific entrenchment mechanism is identified, balancing against any procompetitive effects of a merger may still be required, especially in the case of acquisitions of complementary assets.
- **The three-limb approach to efficiencies** (*verifiability, merger specificity, and consumer benefit*) **provides a suitable substantive framework to assess procompetitive dynamic effects, under an ECW standard. This standard allows for symmetric balancing of adverse and beneficial dynamic effects, where both are present. A sliding scale approach is appropriate for balancing**, with evidence of stronger and more likely beneficial effects required in those cases where harm can be expected to be more significant (e.g. because the merger brings together two close or significant competitors) or immediate.
- **Our conceptual framework can be mapped to five categories of canonical merger cases with potential dynamic effects:** (i) mergers giving rise to overlaps in product or project pipelines (including R&D pipeline-to-product overlaps, R&D pipeline-to-pipeline overlaps, and overlapping physical investment projects); (ii) mergers giving rise to capability overlaps; (iii) acquisitions of nascent competitors; (iv) mergers raising dynamic foreclosure concerns; and (v) mergers raising entrenchment concerns.

Literature synopsis

Our synopsis of the relevant literature integrates theoretical and empirical insights from over 450 publications in industrial economics, financial economics, macroeconomics, and the business literature. We also review relevant competition policy publications and relevant guidance from competition authorities, the International Competition Network, and the OECD.

We first consider the well-established literature on **competition and innovation**. While this literature does not focus on mergers (which are a specific form of change in competition), it nonetheless provides useful insights for merger policy. Models of patent races show that rivalry at the innovation stage tends to stimulate innovation, as firms compete to be the first to introduce a new product. By contrast, the effect of product market competition on innovation incentives is ambiguous. It depends on its impact on the *difference* between pre- and post-innovation profits. Weaker product market competition *before* innovation takes place may dampen innovation incentives, by increasing the cannibalisation of existing profits and lowering the gains from ‘escaping’ competition (an *Arrovian* effect). By contrast, weaker product market competition *after* successful innovation may strengthen incentives, through greater appropriability (a *Schumpeterian* effect). The presence of these contrasting effects underlines the need to assess innovation competition and product market competition jointly, as a merger affects both dimensions of competition. Empirically, competition tends to raise productivity through both a ‘treatment’ effect (i.e. existing firms becoming more productive) and a ‘selection’ effect (i.e. more productive firms entering and less productive firms leaving the market), but its effect on innovation is mixed, in line with the theoretical literature.

Recent policy debates (notably in Europe) have prompted a substantial **theoretical literature on horizontal mergers between established competitors** and their effects on investment and consumer welfare. While stylised, models in this literature provide useful insights for merger policy. In settings where incumbents compete to improve products, reduce costs, or innovate to enter new markets, this literature shows that mergers may either reduce or increase investment, absent knowledge spillovers and synergies. While the literature indicates that both effects can arise under plausible conditions, it also suggests that mergers reduce investments more often than they increase them. Despite the mixed results on investment, the literature also indicates that mergers are likely to reduce consumer welfare, absent spillovers and synergies. The literature also considers additional settings where mergers combine weaker or lagging firms or lead to a reallocation of investment across projects. In these settings, a merger may benefit consumers. A merger may also internalise spillovers between the merging parties, potentially enhancing innovation incentives. However, the internalisation of spillovers to non-merging firms may create opposing effects, reducing innovation incentives. Finally, the literature shows that merger-induced dynamic synergies can mitigate or offset dynamic harm from a merger.

The **theoretical literature on the acquisition of potential competitors** indicates that mergers can affect innovation both *ex post*, following the transaction, and *ex ante*, through incentives created by the prospect of acquisition by an incumbent (‘innovation-for-buyout’). Absent synergies and spillovers, *ex post* effects tend to be anticompetitive where the acquisition suppresses the target’s innovation, eliminates future price competition, or entrenches incumbency (e.g. by allowing the acquirer to serve a larger base of captive consumers). These concerns are strongest where the incumbent has significant market power, and the target is a credible competitor (either on a standalone basis or through acquisitions by alternative buyers raising fewer competition concerns than the incumbent). However, acquisitions may also enable synergies, by allowing smaller innovative companies to scale, increasing technology diffusion, or combining complementary assets. *Ex ante* effects can stimulate entry by new innovative companies by providing a more profitable exit route, but have mixed effects on the direction of their innovation (i.e. what types of technologies or research pathways are pursued).

The literature also includes several **empirical studies** examining the dynamic impact of horizontal mergers, across a range of sectors. While some of this literature is affected by methodological limitations, its results are nonetheless indicative of the empirical effects of mergers. Overall, the literature indicates mixed empirical effects of mergers on investment or innovation inputs (e.g. R&D expenditure) and outputs (e.g. patents), with limited evidence on the effects of mergers on consumer welfare. The empirical corporate finance

literature shows that mergers often increase combined-firm value—although these benefits usually accrue to targets rather than acquirers, and may reflect market power effects.

The theoretical literature on **non-horizontal mergers (vertical or conglomerate)** shows that these can promote investment and benefit consumers by eliminating double markups, internalising positive investment externalities, combining complementary capabilities, resolving contracting frictions, or mitigating hold-up concerns. However, they may also harm competition by providing or increasing the incentives and ability to engage in conduct leading to the dynamic foreclosure of rivals, including conduct aimed at raising the cost of rivals, or at weakening their incentives or ability to invest, expand, or enter. While harms and benefits from non-horizontal mergers are often concurrent (leading to an ambiguous net effect on consumers), the literature has identified some anticompetitive mechanisms that are not associated with a countervailing efficiency. Empirical evidence confirms that non-horizontal mergers may enhance investment and innovation (e.g. through improved coordination of R&D decisions or knowledge transfers across complementary businesses), but may also reduce entry and innovation by third parties.

The **literature on financial market frictions** shows that mergers may affect dynamic competition by relaxing financing constraints on investment or innovation, including through ‘liquidity mergers’ that preserve distressed but productive assets. However, these benefits are not always merger-specific, as financing constraints may be addressed through equity issuance, private equity or venture capital funding, or bank lending. Mergers that allow firms to diversify their activities may reduce distress risk and financing costs through coinsurance and internal capital markets, but can also create inefficient capital allocation, cross-subsidisation, agency problems, opacity, or risk contamination.

Conceptual framework

The conceptual framework for the evaluation of dynamic merger effects is the core of this Study. Its aim is to provide a comprehensive characterisation of the potential anticompetitive and procompetitive dynamic effects of mergers. We also discuss how case examples and practical evidence fit within the conceptual framework.

Definitions and substantive standard

Dynamic merger effects are defined to reflect the impact of mergers on competition in future product markets, resulting from changes in product features (e.g. price, quality, and availability of new or improved products), costs, or the number of firms. The effects of a merger on future product markets may differ from those on current product markets due to investments carried out by firms to improve their future competitive position. We adopt a wide notion of investment, covering investments in R&D or infrastructure by incumbents, and efforts by nascent competitors to enter a market. Our definition of dynamic merger effects is also broad as it captures the impact of mergers both on incentives to invest and on future product market competition (e.g. price competition on new or improved products), for a given level of investment. A broad definition is required to ensure consistency of treatment across different potential determinants of competition in future product markets, and to capture their interaction. Our definition of dynamic effects only includes post-merger effects, in line with merger enforcement policy. Ex ante effects from the prospect of a future acquisition are therefore outside the scope of our conceptual framework, although the policy stance adopted by competition authorities can affect innovation incentives through this channel as well.

We evaluate dynamic merger effects under a **consumer welfare** standard, in line with prevailing legal and decisional practice. Consumer welfare provides a helpful yardstick to evaluate the overall impact of a merger when multiple effects need to be netted off against each other (e.g. higher prices, but also higher quality or product variety). Adopting a consumer welfare standard implies that an ‘innovation defence’ under merger control applies only when consumers benefit sufficiently from higher innovation, offsetting any

concurrent adverse dynamic effects of a merger (e.g. higher prices for innovative products).

To account for the presence of uncertainty in investment (notably in R&D activities), we adopt an **Expected Consumer Welfare (ECW) standard**. Under this standard, a merger leads to adverse effects on competition if it results in a reduction in ECW relative to the situation that would arise absent the merger (i.e. $\Delta ECW < 0$). In practice, the ECW standard operates largely as a conceptual tool that structures the analysis rather than as a measure that is explicitly evaluated or quantified. Assessment under an ECW standard can reflect uncertain individual effects of a merger that may have a large impact on consumer welfare. This is particularly relevant in the case of dynamic merger effects, for both anticompetitive effects (e.g. a merger eliminating potential but uncertain competition by a nascent competitor) and procompetitive effects (e.g. a large but uncertain benefit from a merger). Accounting for dynamic effects broadens the range of possible outcomes from a merger, potentially magnifying both harms and benefits relative to a static analysis, and raising the stakes in merger control decisions.

Dynamic effects from horizontal mergers

We group dynamic effects of horizontal mergers into two categories: 'Box 1' covering *anticompetitive* effects, and 'Box 2' covering *procompetitive* effects, including internalisation of spillovers and synergies. This is a substantive categorisation of economic effects, which is not intended to prejudge procedural questions on burden of proof in merger review processes or legal questions on what constitutes a Significant Impediment to Effective Competition under the EU Merger Regulation (both of which are outside the scope of this Study).

Box 1 captures the potential **anticompetitive dynamic effects of horizontal mergers**, arising from the elimination of rivalry between the merging firms, in relation to both investment and future product market competition (for a given level of investment). In our baseline exposition, we focus on the canonical case of product innovation followed by competition in prices, but results on the overall impact of a merger under Box 1 apply more broadly to other forms of investment (e.g. process innovation, investment in infrastructure) and product market competition (e.g. quantity competition). The internalisation of negative price and investment externalities resulting from a merger softens the merged entity's overall incentive to compete. Box 1 captures this effect through three distinct channels: (i) *direct effects from the loss of price competition*, resulting in higher prices for future products; (ii) *direct investment effects* from the internalisation of negative investment externalities (which tend to reduce investment incentives, through a 'cannibalisation' effect); and (iii) *indirect investment effects* from the reduction in price competition (which tends to raise investment incentives, by increasing 'appropriability'). As shown by the economic literature, the combined impact of the three channels may increase or reduce investment incentives. While the literature indicates that both effects can arise under plausible conditions, it also suggests that mergers reduce investments more often than they increase them. This does not support a general presumption regarding the impact of a merger on investment under Box 1, but indicates that the combined impact of Box 1 effects on investment gives rise to a substantial concern that a merger may lead to a reduction in investments (absent countervailing effects under Box 2). The joint impact of Box 1 effects on ECW is more clear-cut: the combination of price and investment effects is likely to harm consumers, given that direct price effects harm consumers and direct investment effects tend to reduce consumer welfare, while indirect investment effects rest on higher prices. As a result, *no balancing* between higher prices and any resulting increase in investment is generally required under Box 1, absent spillovers and synergies. There are, however, specific circumstances where a finding of consumer harm under Box 1 may not apply (for example, if a merger brings together two weaker or lagging firms), even absent countervailing effects under Box 2.

Box 2 captures the potential **procompetitive dynamic effects of horizontal mergers**, including mechanisms through which a merger may enhance the merging firms' incentives and ability to invest, to the benefit of consumers. Box 2 distinguishes between the internalisation of spillovers (Box 2A) and dynamic synergies (Box 2B).

- **Box 2A** reflects situations where the fruits of innovation are partially non-excludable and spill over to competitors. In this case, a merger may internalise positive externalities between the merging firms, boosting innovation incentives and benefiting consumers. However, spillovers may also benefit non-merging firms. Where this is the case, the merging firms may become more concerned about spillovers to their rivals after the merger, thus weakening innovation incentives.
- **Box 2B** concerns dynamic synergies, reflecting an improvement in the capabilities of the merged entity. These may arise from (i) technology transfers between merging parties that would not otherwise have occurred, (ii) lower R&D costs or greater R&D effectiveness, (iii) lower investment costs or greater investment effectiveness (including through shared infrastructure), (iv) production cost efficiencies, and (v) easing of financial constraints that would otherwise hinder an acquired firm from bringing innovation to market. Box 2B effects, if present, boost investment and can also increase the diffusion of innovation, to the benefit of consumers.

The standard **three-limb test** for efficiencies provides a suitable *substantive* framework for assessing Box 2 effects: to counter adverse effects under Box 1, dynamic benefits should be *merger-specific*, *verifiable*, and *passed on to consumers*. In our conceptual framework, the assessment is carried out under an ECW standard, to ensure consistency with the evaluation of Box 1 effects.

Balancing between Box 1 and Box 2 effects is required where both are established to the requisite standard, using a consistent approach to evidence, timeframe, and the treatment of uncertainty. The ECW standard provides an appropriate yardstick for balancing: benefits from Box 2 need to offset any adverse impact on ECW from Box 1 effects, including both future price and investment effects. This implies that a net increase in investment incentives alone is insufficient to conclude that a merger does not reduce ECW, as it may not entirely offset the harm to consumers from higher prices for new or improved products. Depending on the case, both quantitative tools (e.g. merger simulations) and qualitative methods can be used to balance dynamic harms and benefits. Balancing between static and dynamic effects may also be required where these point in opposite directions. The ECW framework can accommodate this through appropriate probability weighting and discounting of static and dynamic effects. A **sliding scale approach** is appropriate for balancing, with evidence of stronger and more certain beneficial effects under Box 2 required in those cases where Box 1 effects can be expected to be more significant or immediate (e.g. a horizontal merger involving a dominant incumbent, or two significant or close competitors in a concentrated market).

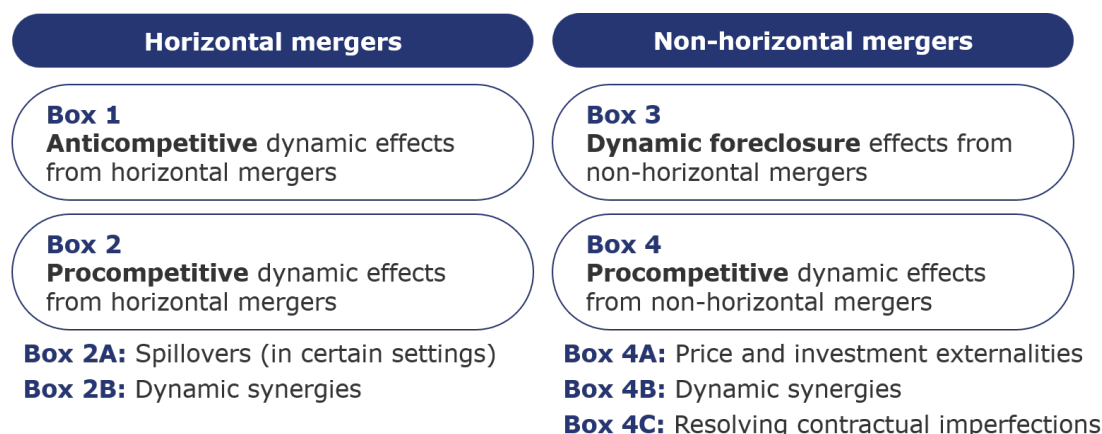
Non-horizontal effects

As with horizontal mergers, we group dynamic effects of non-horizontal mergers into two categories: 'Box 3' covering *anticompetitive dynamic foreclosure effects*, and 'Box 4' covering *procompetitive non-horizontal dynamic effects*.

Dynamic foreclosure effects under **Box 3** capture situations where the merger gives rise to or strengthens the ability and incentives of the merged entity to engage in conduct that reduces investments by its rivals, to the detriment of consumers. Potential theories of harm under Box 3 include hold-up of rival investment, raising the cost of rivals through a refusal to deal, deterring multi-level or multi-product entry, dynamic leveraging to defend the rents of the acquirer in its core market, and depriving rivals of a valuable input or complement to prevent them from becoming stronger competitors. Competitive harm under Box 3 is less direct than under Box 1, as non-horizontal mergers do not lead to a loss of competition between competitors in the same market.

Procompetitive dynamic effects from non-horizontal mergers under **Box 4** include the internalisation of positive price and investment externalities (Box 4A), dynamic synergies (Box 4B), and the resolution of contractual imperfections (Box 4C). Since non-horizontal mergers tend to generate efficiencies from the integration of complementary activities, potentially harmful and beneficial mechanisms are often concurrent and intertwined.

The figure below summarises the core horizontal and non-horizontal dynamic effects of a merger captured in our framework.



Entrenchment

Our conceptual framework also accounts for *entrenchment* as a potential adverse dynamic effect that is not captured under Box 1 or Box 3. This reflects situations where an acquisition involving a firm with significant or substantial market power raises barriers to entry or expansion by rivals in a way that weakens the future competitive constraint that they would otherwise exercise, to the ultimate detriment of consumers. We identify reasons why, from a policy and economic perspective, it is appropriate to limit entrenchment concerns to firms with a high degree of market power. We identify potential entrenchment mechanisms based on recent findings of the academic literature. These may operate for example by reducing innovation-for-buyout incentives for future entrants, or by leading to so-called ‘kill zones’. We discuss the case where the strengthening of the position of a dominant acquirer and the concurrent reduction in rivals’ sales caused by procompetitive effects of the merger in the short run may nonetheless lead to net competitive harm over time. The economic literature and economic principles suggest that circumstances in which this may occur are rare.

Canonical cases

We map effects under our conceptual framework to identify categories of ‘canonical’ merger cases. For each category, we explain which dynamic effects matter, provide examples and case studies, and set out practical evidence to assess effects. The categories—summarised in the table below—are distinguished by (i) the nature of the overlap between merging parties (e.g. specific product or project pipelines vs broader capabilities), (ii) the nature of the target, (iii) the presence of an acquirer with a high degree of market power, and (iv) the type of investment involved (R&D vs physical investment). Categories A to C capture Boxes 1 and 2, Category D captures Boxes 3 and 4, and Category E refers to entrenchment. A specific merger may fall into more than one of these categories.

Category	Key features and case examples
A1 Pipeline overlaps: R&D pipeline-to-product	One party’s R&D pipeline project competes with the other’s existing product. <i>Example: Novartis/GlaxoSmithKline Oncology Business (2015, EU)</i>

A2	Pipeline overlaps: R&D pipeline-to-pipeline	Both parties work on parallel and competing R&D pipeline projects. <i>Example: Adobe/Figma (2023; EU, UK)</i>
A3	Pipeline overlaps: physical investment project	One or both parties work on specific physical investment projects, overlapping with an existing product or investment project of the other firm (including geographic presence). <i>Example: ENI/EDP/GDP (2004, EU)</i>
B1	Capability overlaps: innovation capabilities	Two out of a limited number of firms have significant overlapping R&D capabilities. <i>Example: Dow/DuPont (2017, EU)</i>
B2	Capability overlaps: investment capabilities	Two out of a limited number of firms have significant overlapping investment capabilities required to compete for future customer needs or enter markets where the other firm is present. <i>Examples: Ryanair/Aer Lingus I and III (2007, 2013; EU)</i>
C	Acquisitions of nascent competitors	Incumbent acquires a nascent competitor with emerging pipeline projects or capabilities. <i>Example: Facebook/Instagram/WhatsApp (2020, US)</i>
D	Dynamic foreclosure	Merger may provide or increase the ability and incentives to foreclose investment by rivals. <i>Example: Illumina/GRAIL (2022, EU; 2023, US)</i>
E	Entrenchment	Merger may strengthen an acquirer with significant market power and harm consumers by reducing contestability in the longer term. <i>Example: Booking/eTraveli (2023, EU)</i>

Sectoral analysis

The Study concludes by considering the relevance of merger control and our proposed framework in three strategic sectors, examining the main bottlenecks to investment and innovation in the EU and the role of different policy tools, including merger control.

- In **biopharmaceuticals**, the EU is experiencing a widening productivity and R&D investment gap relative to the US and China. Innovation in the sector is also shifting towards emerging biopharmaceutical companies. This makes the treatment of acquisitions of small innovative companies an important policy consideration. While acquisitions of small innovative companies may accelerate pipeline development, they may also suppress R&D diversity, inventor productivity, and nascent competition. Merger control needs to account for these opposing potential dynamic effects.
- In **quantum computing**, the EU has strong research capabilities but faces challenges in translating scientific excellence into commercial scale, as entrants remain younger, smaller, and less well-funded than their US counterparts. Although merger control does not yet play a major role in this sector, it may become more relevant as consolidation increases.
- Finally, in **clean technologies**, where R&D, learning-curve effects, and scale are central to competition, Europe faces strong pressure from Chinese manufacturers and supply-chain dependencies. Merger control can play a role by enabling firms to capture economies of scale, while guarding against 'killer acquisitions' and foreclosure concerns in relation to promising new technologies, and by recognising that the ability to decarbonise energy production effectively is an important dimension of current and future cost competitiveness.

Résumé exécutif

La concurrence est un processus dynamique de rivalité, dans lequel les entreprises s'affrontent sur plusieurs dimensions — notamment le prix, la qualité, l'innovation, l'investissement et la diversité des produits — afin de proposer des offres attractives aux consommateurs, de gagner des ventes aux dépens de leurs concurrents et d'accroître leurs profits. Une concentration peut entraver ce processus concurrentiel en éliminant la rivalité entre des entreprises qui se faisaient auparavant concurrence, au détriment des consommateurs. Elle peut également conférer aux parties à la concentration la capacité et les incitations pour verrouiller le marché aux entreprises rivales, réduisant ainsi la contrainte concurrentielle exercée par ces dernières. Toutefois, une concentration peut aussi favoriser la rivalité en faisant de l'entité issue de l'opération un concurrent plus efficace, au bénéfice des consommateurs. Ces effets peuvent s'appliquer aussi bien à la concurrence actuelle sur les produits existants qu'à la concurrence future sur de nouveaux produits et de nouvelles technologies.

Les éléments dynamiques de la concurrence — notamment la rivalité en matière d'innovation et d'investissement ainsi que l'entrée d'entreprises venant contester la position des entreprises en place — sont reconnus comme essentiels au progrès technologique, à la croissance à long terme et à la compétitivité mondiale. Les considérations dynamiques revêtent une importance croissante dans une économie moderne caractérisée par l'innovation de rupture et des cycles technologiques rapides.

Cette étude de la Commission européenne, préparée par Oxera, constitue une contribution au réexamen en cours par la Commission européenne de ses lignes directrices sur les concentrations de 2004 et 2008. Cette révision fait suite aux recommandations du rapport Draghi de 2024 et à son appel à accorder davantage de poids à l'innovation, à l'investissement et à la concurrence future dans le contrôle des concentrations. Bien que les autorités de concurrence, y compris la Commission européenne, aient progressivement intégré les considérations dynamiques dans leurs décisions en matière de contrôle des concentrations au cours de la dernière décennie, le cadre d'analyse des effets dynamiques est moins établi que celui relatif aux effets des concentrations sur la concurrence statique.

S'appuyant sur une synthèse exhaustive de la littérature académique pertinente, la présente étude propose un fondement analytique rigoureux pour l'appréciation économique des effets dynamiques des concentrations. Elle se concentre sur les concentrations entre concurrents (« concentrations horizontales »), tout en couvrant également les concentrations entre producteurs de produits complémentaires (« concentrations non horizontales »).

Principales conclusions

- **Les effets dynamiques devraient être définis au sens large, afin d'inclure l'effet global des concentrations sur la concurrence future sur les marchés de produits** — en prenant en compte leur impact sur l'investissement, l'innovation, l'entrée, ainsi que sur les prix et la disponibilité de produits nouveaux ou améliorés.
- **Les effets dynamiques peuvent amplifier, compenser ou contrebalancer les effets statiques**, élargissant le champ des effets tant anticoncurrentiels que pro-concurrentiels. **Les enjeux des décisions de contrôle des concentrations s'en trouvent accrus** dans les cas présentant une dimension dynamique significative.
- **Notre cadre conceptuel apprécie les concentrations produisant des effets dynamiques au regard d'un critère de bien-être attendu des consommateurs (« Expected Consumer Welfare », ou ECW), qui tient**

compte de l'incertitude entourant certains des déterminants sous-jacents des résultats concurrentiels futurs (notamment le succès des efforts de R&D et d'innovation, l'introduction de nouveaux produits, l'entrée de concurrents potentiels et la réalisation de synergies dynamiques). Ce critère prend en compte à la fois la probabilité et l'ampleur des conséquences d'une concentration.

- **Les concentrations horizontales entraînent une perte de concurrence entre les parties à la concentration sur les futurs marchés de produits. En l'absence de retombées (« spillovers ») et de synergies, il est probable que cette perte réduise l'ECW**, en raison de l'internalisation des externalités négatives d'investissement et de prix. C'est le cas même dans les situations où les concentrations augmentent les incitations à investir en atténuant la concurrence future par les prix. Un pouvoir de marché accru sur les futurs marchés de produits ne constitue donc pas un fondement approprié pour une « défense fondée sur l'innovation ».
- **Les concentrations horizontales peuvent produire des effets dynamiques pro-concurrentiels** en générant des synergies dynamiques ou en internalisant des retombées. Les synergies dynamiques peuvent émerger par le biais de plusieurs canaux, notamment une diffusion technologique accrue, des réductions des coûts d'investissement incrémentaux ou des augmentations de l'efficacité des investissements grâce aux économies d'échelle et aux complémentarités, ou encore un assouplissement des contraintes financières. Une concentration peut également internaliser les retombées entre les parties à la concentration, renforçant potentiellement les incitations à innover. Cependant, l'internalisation des retombées vers les entreprises non parties à la concentration peut créer des effets opposés, réduisant les incitations à innover.
- **Les concentrations non horizontales peuvent permettre à l'entité issue de l'opération d'adopter un comportement de verrouillage dynamique**, réduisant les investissements ou dissuadant l'entrée de ses concurrents, au détriment des consommateurs. **Les concentrations non horizontales peuvent également donner lieu à des effets dynamiques pro-concurrentiels**. Ceux-ci peuvent résulter de l'internalisation d'externalités de prix ou d'investissement, de synergies dynamiques ou de l'atténuation de frictions contractuelles. Puisque les concentrations non horizontales tendent à générer des gains d'efficacité par l'intégration d'activités complémentaires, les mécanismes potentiellement préjudiciables et bénéfiques sont souvent concomitants et entremêlés.
- **Outre le fait qu'elles puissent entraîner des effets dynamiques horizontaux et des effets de verrouillage dynamique potentiellement préjudiciables, les concentrations peuvent également renforcer la position d'une entreprise en place dotée d'un pouvoir de marché significatif**, en renforçant les barrières à l'entrée et à l'expansion des concurrents, au détriment final des consommateurs. La littérature académique récente identifie plusieurs mécanismes dynamiques spécifiques susceptibles de donner lieu à un renforcement de position. Lorsqu'un mécanisme spécifique de renforcement de position est identifié, une mise en balance avec les éventuels effets pro-concurrentiels d'une concentration peut tout de même être requise, notamment dans le cas d'acquisitions d'actifs complémentaires.
- **L'approche en trois volets applicable aux gains d'efficacité (vérifiabilité, spécificité à la concentration et bénéfice pour les consommateurs) constitue un cadre d'analyse approprié pour apprécier les effets dynamiques pro-concurrentiels, dans le cadre d'un critère ECW. Ce critère permet une mise en balance symétrique des effets dynamiques préjudiciables et bénéfiques, lorsque les deux sont présents. Une approche graduée est**

appropriée pour cette mise en balance, une preuve d'effets bénéfiques plus solides et plus probables étant requise dans les cas où le préjudice est susceptible d'être plus significatif (par exemple, car la concentration rapproche deux concurrents proches ou significatifs) ou immédiat.

- **Notre cadre conceptuel peut être articulé autour de cinq catégories de cas canoniques de concentrations présentant des effets dynamiques potentiels** : (i) les concentrations donnant lieu à des chevauchements entre projets ou produits en développement (y compris les chevauchements de R&D entre projet en développement et produit existant, les chevauchements de R&D entre plusieurs projets en développement, et les chevauchements entre projets d'investissement physique) ; (ii) les concentrations donnant lieu à des chevauchements de capacité ; (iii) les acquisitions de concurrents émergents ; (iv) les concentrations soulevant des préoccupations de verrouillage dynamique ; et (v) les concentrations soulevant des préoccupations de renforcement de position.

Synthèse de la littérature

Notre synthèse de la littérature pertinente intègre les enseignements théoriques et empiriques de plus de 450 publications académiques relevant de l'économie industrielle, de l'économie financière, de la macroéconomie et des sciences de gestion. Nous examinons également les publications pertinentes en matière de politique de concurrence ainsi que les orientations pertinentes émanant des autorités de concurrence, du Réseau International de la Concurrence et de l'OCDE.

Nous examinons d'abord la littérature bien établie sur **la concurrence et l'innovation**. Bien que cette littérature ne porte pas spécifiquement sur les concentrations (qui constituent une forme particulière de modification de la concurrence), elle fournit néanmoins des enseignements utiles pour la politique de contrôle des concentrations. Les modèles de courses aux brevets montrent que la rivalité au stade de l'innovation tend à stimuler celle-ci, les entreprises se faisant concurrence pour être les premières à introduire un nouveau produit. En revanche, l'effet de la concurrence sur les marchés de produits sur les incitations à innover est ambigu. Cela dépend de son impact sur la *différence* entre les profits avant et après l'innovation. Une concurrence plus faible sur les marchés de produits *avant* que l'innovation ne se matérialise peut affaiblir les incitations à innover, en accroissant la cannibalisation des profits existants et en réduisant les gains liés au fait d'« échapper » à la concurrence (effet d'Arrow). À l'inverse, une concurrence plus faible sur les marchés de produits *après* une innovation réussie peut renforcer les incitations grâce à une meilleure appropriabilité (effet *schumpétérien*). La coexistence de ces effets opposés souligne la nécessité d'apprécier conjointement la concurrence en matière d'innovation et la concurrence sur les marchés de produits, puisqu'une concentration affecte ces deux dimensions. Sur le plan empirique, la concurrence tend à accroître la productivité à la fois à travers un effet de « traitement » (les entreprises existantes deviennent plus productives) et un effet de « sélection » (les entreprises plus productives entrent sur le marché et les moins productives le quittent), mais son effet sur l'innovation est contrasté, conformément à la littérature théorique.

Les récents débats politiques (notamment en Europe) ont suscité une abondante **littérature théorique sur les concentrations horizontales entre concurrents établis** et leurs effets sur l'investissement et le bien-être des consommateurs. Bien que stylisés, les modèles provenant de cette littérature apportent des enseignements utiles pour la politique de contrôle des concentrations. Dans les situations où les entreprises en place se font concurrence pour améliorer leurs produits, réduire les coûts ou innover afin d'entrer sur de nouveaux marchés, cette littérature montre que les concentrations peuvent aussi bien réduire qu'accroître l'investissement, en l'absence de retombées de connaissances (« knowledge spillovers ») et de synergies. Si la littérature indique que les

deux effets peuvent survenir dans des conditions plausibles, elle suggère également que les concentrations réduisent les investissements plus souvent qu'elles ne les augmentent. Malgré ces résultats contrastés concernant l'investissement, la littérature indique également qu'il est probable que les concentrations réduisent le bien-être des consommateurs, en l'absence de retombées et de synergies. La littérature examine aussi des configurations supplémentaires dans lesquelles les concentrations réunissent des entreprises plus faibles ou moins avancées, ou conduisent à la réallocation des investissements entre différents projets. Dans ces configurations, une concentration peut être bénéfique pour les consommateurs. Une concentration peut également internaliser les retombées entre les parties à la concentration, pouvant potentiellement renforcer les incitations à innover. Toutefois, l'internalisation des retombées vers des entreprises non-parties à la concentration peut produire des effets opposés, réduisant les incitations à l'innovation. Enfin, la littérature montre que les synergies dynamiques induites par une concentration peuvent atténuer ou compenser le préjudice dynamique résultant de celle-ci.

La **littérature théorique relative à l'acquisition de concurrents potentiels** indique que les concentrations peuvent à la fois affecter l'innovation *ex post*, suite à l'opération, et *ex ante*, par les incitations créées par la perspective d'une acquisition par un acteur en place (« innovation-for-buyout », c'est-à-dire l'innovation motivée par la perspective d'un rachat). En l'absence de synergies et de retombées, les effets *ex post* tendent à être anticoncurrentiels lorsque l'acquisition supprime l'innovation de la cible, élimine la concurrence future par les prix ou renforce la position de l'entreprise en place (par exemple, en permettant à l'acquéreur de servir une base élargie de consommateurs captifs). Ces préoccupations sont les plus fortes lorsque l'opérateur en place dispose d'un pouvoir de marché significatif et que la cible constitue un concurrent crédible (soit de manière individuelle, soit à la suite d'une acquisition par d'autres acheteurs soulevant moins de préoccupations de concurrence que l'entreprise en place). Toutefois, les acquisitions peuvent également permettre des synergies, en donnant aux petites entreprises innovantes les moyens de changer d'échelle, en accroissant la diffusion technologique ou en combinant des actifs complémentaires. Les effets *ex ante* peuvent stimuler l'entrée de nouvelles entreprises innovantes en offrant une voie de sortie plus rentable, mais ont des effets contrastés sur la direction de leur innovation (c'est-à-dire sur les types de technologies ou de trajectoires de recherche poursuivies).

La littérature comprend également plusieurs **études empiriques** examinant l'impact dynamique des concentrations horizontales dans divers secteurs. Bien qu'une partie de cette littérature présente des limites méthodologiques, ses résultats donnent néanmoins des indications sur les effets empiriques des concentrations. Dans l'ensemble, la littérature fait état d'effets empiriques contrastés des concentrations sur les efforts d'investissement ou d'innovation (par exemple, les dépenses de R&D) et sur leurs résultats (par exemple, les brevets), tandis que les preuves empiriques relatives aux effets des concentrations sur le bien-être des consommateurs demeurent limitées. La littérature empirique en finance d'entreprise montre que les concentrations augmentent souvent la valeur pour l'entité combinée – même si ces bénéfices reviennent généralement aux entreprises cibles plutôt qu'aux acquéreurs, et peuvent également refléter des effets de pouvoir de marché.

La littérature théorique sur les **concentrations non horizontales (verticales ou conglomerates)** montre qu'elles peuvent favoriser l'investissement et bénéficier aux consommateurs en éliminant les doubles marges, en internalisant les externalités positives d'investissement, en combinant des capacités complémentaires, en résolvant des frictions contractuelles ou en atténuant les problèmes de hold-up (c'est-à-dire les risques d'opportunisme post-contractuel). Elles peuvent toutefois aussi nuire à la concurrence en créant ou en renforçant les incitations et la capacité à adopter des comportements conduisant au verrouillage dynamique du marché pour les concurrents, y compris des comportements visant à augmenter les coûts de ces derniers ou à affaiblir leurs incitations ou leur capacité à investir, à se développer ou à entrer sur le marché. Si les préjudices et bénéfices des concentrations non horizontales sont souvent concomitants (rendant l'effet

net sur les consommateurs ambigus), la littérature a néanmoins identifié certains mécanismes anticoncurrentiels qui ne sont pas associés à un gain d'efficacité compensatoire. Les données empiriques confirment que les concentrations non horizontales peuvent accroître l'investissement et l'innovation (par exemple, par une meilleure coordination des décisions de R&D ou des transferts de connaissances entre activités complémentaires), mais qu'elles peuvent aussi réduire l'entrée et l'innovation d'entreprises tierces.

La **littérature relative aux frictions sur les marchés financiers** montre que les concentrations peuvent influencer la concurrence dynamique en assouplissant les contraintes de financement pesant sur l'investissement ou l'innovation, notamment par le biais de « liquidity mergers » (c'est-à-dire des concentrations motivées par des considérations de liquidité) qui préservent des actifs en difficulté mais productifs. Ces bénéfices ne sont toutefois pas toujours spécifiques à la concentration, les contraintes de financement pouvant être levées par une émission d'actions, un financement par capital-investissement ou par capital-risque, ou un financement bancaire. Les concentrations qui permettent aux entreprises de diversifier leurs activités peuvent réduire le risque de défaillance et les coûts de financement grâce à la coassurance et aux marchés internes de capitaux, mais elles peuvent aussi engendrer une allocation inefficace du capital, des subventions croisées, des problèmes d'agence, un manque de transparence ou une contamination des risques.

Cadre conceptuel

Le cadre conceptuel d'appréciation des effets dynamiques des concentrations constitue le cœur de cette étude. Il vise à fournir une caractérisation complète des effets dynamiques anticoncurrentiels et pro-concurrentiels potentiels des concentrations. Nous examinons également la manière dont les exemples de cas et les éléments de preuve concrets s'inscrivent dans ce cadre conceptuel.

Définitions et critère substantiel

Les effets dynamiques des concentrations sont définis de manière à refléter l'impact des concentrations sur la concurrence dans les futurs marchés de produits, résultant de modifications des caractéristiques des produits (par exemple, le prix, la qualité et la disponibilité de produits nouveaux ou améliorés), des coûts ou du nombre d'entreprises. Les effets d'une concentration sur les futurs marchés de produits peuvent différer de ceux sur les marchés actuels en raison des investissements réalisés par les entreprises pour améliorer leur future position concurrentielle. Nous retenons une notion large de l'investissement, couvrant les investissements en R&D ou en infrastructures réalisés par les opérateurs en place, ainsi que les efforts déployés par les concurrents émergents pour entrer sur un marché. Notre définition des effets dynamiques des concentrations est également large, en ce qu'elle englobe l'impact des concentrations à la fois sur les incitations à investir et sur la concurrence future sur les marchés de produits (par exemple, la concurrence par les prix sur des produits nouveaux ou améliorés) pour un niveau d'investissement donné. Une définition large est nécessaire afin d'assurer la cohérence du traitement des différents déterminants potentiels de la concurrence sur les futurs marchés de produits, et de saisir leurs interactions. Conformément à la politique de mise en œuvre du contrôle des concentrations, notre définition des effets dynamiques ne prend en compte que les effets postérieurs à la concentration. Les effets *ex ante* liés à la perspective d'une acquisition future se situent donc hors du champ de notre cadre conceptuel, même si l'orientation adoptée par les autorités de concurrence peut également affecter les incitations à innover par ce canal.

Nous évaluons les effets dynamiques des concentrations au regard d'un critère de **bien-être des consommateurs**, conformément à la pratique juridique et décisionnelle dominante. Le bien-être des consommateurs fournit un critère de référence utile pour

évaluer l'impact global d'une concentration lorsque des effets multiples doivent être mis en balance (par exemple, des prix plus élevés mais aussi une qualité ou une variété de produits accrues). L'adoption d'un critère de bien-être des consommateurs implique qu'une « défense fondée sur l'innovation » dans le cadre du contrôle des concentrations ne s'applique que lorsque les consommateurs bénéficient suffisamment d'une innovation accrue, de manière à compenser les éventuels effets dynamiques préjudiciables d'une concentration (par exemple, des prix plus élevés pour les produits innovants).

Afin de tenir compte de l'incertitude liée à l'investissement (notamment dans les activités de R&D), nous adoptons un **critère de bien-être attendu des consommateurs (« Expected Consumer Welfare », ou ECW)**. Selon ce critère, une concentration produit des effets préjudiciables sur la concurrence si elle entraîne une diminution de l'ECW par rapport à la situation qui prévaudrait en l'absence de la concentration (c'est-à-dire $\Delta ECW < 0$). En pratique, le critère ECW fonctionne principalement comme un outil conceptuel structurant l'analyse plutôt que comme une mesure explicitement calculée ou quantifiée. L'appréciation selon un critère ECW peut refléter les effets individuels incertains d'une concentration susceptibles d'avoir un impact important sur le bien-être des consommateurs. Cela est particulièrement pertinent pour les effets dynamiques des concentrations, qu'ils soient anticoncurrentiels (par exemple, une concentration éliminant une concurrence potentielle mais incertaine provenant d'un concurrent émergent) ou pro-concurrentiels (par exemple, un bénéfice important mais incertain résultant d'une concentration). La prise en compte des effets dynamiques élargit l'éventail des résultats possibles d'une concentration, amplifiant potentiellement tant les préjudices que les bénéfices par rapport à une analyse statique, et accroissant les enjeux des décisions en matière de contrôle des concentrations.

Effets dynamiques des concentrations horizontales

Nous regroupons les effets dynamiques des concentrations horizontales en deux catégories : l'« Encadré 1 », qui couvre les effets *anticoncurrentiels*, et l'« Encadré 2 », qui couvre les effets *pro-concurrentiels*, y compris l'internalisation des retombées et les synergies. Il s'agit d'une catégorisation substantielle des effets économiques, qui n'entend pas préjuger des questions procédurales relatives à la charge de la preuve dans les procédures de contrôle des concentrations ou des questions juridiques concernant ce qui constitue une entrave significative à une concurrence effective au sens du règlement de l'UE sur les concentrations (ces deux aspects se situant en dehors du champ de la présente étude).

L'Encadré 1 couvre les **effets dynamiques anticoncurrentiels potentiels des concentrations horizontales**, résultant de l'élimination de la rivalité entre les parties à la concentration, s'agissant tant de l'investissement que de la concurrence future sur les marchés de produits (pour un niveau d'investissement donné). Dans notre présentation de référence, nous nous concentrons sur le cas canonique de l'innovation de produit suivie d'une concurrence par les prix, mais les résultats relatifs à l'impact global d'une concentration de l'Encadré 1 s'appliquent plus largement à d'autres formes d'investissement (par exemple, l'innovation de procédé ou l'investissement dans les infrastructures), ainsi qu'à d'autres formes de concurrence sur les marchés de produits (par exemple, la concurrence par les quantités). L'internalisation des externalités négatives de prix et d'investissement résultant d'une concentration réduit l'incitation globale de l'entité issue de l'opération à concourir. L'Encadré 1 saisit cet effet à travers trois canaux distincts : (i) *les effets directs de la perte de concurrence par les prix*, qui entraînent des prix plus élevés pour les futurs produits ; (ii) *les effets directs sur l'investissement*, liés à l'internalisation des externalités négatives sur l'investissement (qui tendent à réduire les incitations à investir, par le biais d'un effet de « cannibalisation ») ; et (iii) *les effets indirects sur l'investissement*, liés à la réduction de la concurrence par les prix (qui tendent à accroître les incitations à investir, en augmentant l'« appropriabilité »). Comme le montre la littérature économique, l'effet combiné de ces trois canaux peut augmenter ou réduire les incitations à investir. Bien que la littérature indique que ces deux effets peuvent

survenir dans des conditions plausibles, elle suggère également que les concentrations réduisent les investissements plus fréquemment qu'elles ne les augmentent. Cela ne justifie pas une présomption générale quant à l'impact d'une concentration sur l'investissement au titre de l'Encadré 1, mais indique que l'impact combiné des effets au titre de l'Encadré 1 sur l'investissement soulève une préoccupation sérieuse selon laquelle une concentration pourrait conduire à une réduction des investissements (en l'absence d'effets compensatoires au titre de l'Encadré 2). L'effet conjoint de l'Encadré 1 sur l'ECW est plus tranché : il est probable que la combinaison des effets de prix et d'investissement nuise aux consommateurs, car les effets directs de prix leur sont préjudiciables et que les effets directs d'investissement tendent à réduire leur bien-être, tandis que les effets indirects d'investissement reposent sur des prix plus élevés. Par conséquent, il n'est généralement *pas nécessaire de mettre en balance* des prix plus élevés et toute augmentation de l'investissement qui en résulterait, au titre de l'Encadré 1, en l'absence de retombées et de synergies. Il existe toutefois des circonstances particulières dans lesquelles une conclusion de préjudice pour le consommateur au titre de l'Encadré 1 peut ne pas s'appliquer (par exemple, si une concentration réunit deux entreprises plus faibles ou moins avancées), même en l'absence d'effets compensatoires au titre de l'Encadré 2.

L'Encadré 2 couvre les effets dynamiques pro-concurrentiels potentiels des concentrations horizontales, notamment les mécanismes par lesquels une concentration peut renforcer la capacité et les incitations des parties à la concentration à investir, au bénéfice des consommateurs. L'Encadré 2 distingue l'internalisation des retombées (Encadré 2A) et les synergies dynamiques (Encadré 2B).

- **L'Encadré 2A** couvre les situations dans lesquelles les fruits de l'innovation sont partiellement non exclusifs et bénéficient également aux concurrents sous forme de retombées. Dans ce cas, une concentration peut internaliser les externalités positives entre les parties à la concentration, renforçant les incitations à innover et bénéficiant aux consommateurs. Toutefois, les retombées peuvent également profiter aux entreprises non parties à la concentration. Dans ce cas, les parties à la concentration peuvent devenir plus soucieuses des retombées au profit de leurs concurrents après la concentration, ce qui affaiblirait les incitations à innover.
- **L'Encadré 2B** traite des synergies dynamiques, reflétant une amélioration des ressources et compétences de l'entité issue de la concentration. Elles peuvent découler (i) de transferts de technologies entre les parties à la concentration qui n'auraient pas eu lieu autrement, (ii) d'une réduction des coûts de R&D ou d'une plus grande efficacité de la R&D, (iii) d'une réduction des coûts d'investissement ou d'une plus grande efficacité des investissements (notamment grâce au partage d'infrastructures), (iv) d'économies sur les coûts de production et (v) d'un assouplissement des contraintes financières qui empêcheraient autrement une entreprise ayant été acquise de commercialiser une innovation. Lorsqu'ils existent, les effets relevant de l'Encadré 2B stimulent l'investissement et peuvent aussi accroître la diffusion de l'innovation, au bénéfice des consommateurs.

Le **test classique en trois volets** applicable aux gains d'efficacité fournit un cadre d'analyse approprié pour apprécier les effets de l'Encadré 2 : pour compenser les effets préjudiciables de l'Encadré 1, les bénéfices dynamiques doivent être *spécifiques à la concentration, vérifiables et répercutés sur les consommateurs*. Dans notre cadre conceptuel, l'appréciation est menée au regard du critère de l'ECW afin d'assurer la cohérence avec l'appréciation des effets relevant de l'Encadré 1.

Une mise en balance des effets relevant des Encadrés 1 et 2 s'impose lorsque les deux sont établis au niveau de preuve requis, en adoptant une approche cohérente quant aux éléments de preuve, à l'horizon temporel et au traitement de l'incertitude. Le critère de l'ECW fournit une référence appropriée pour cette mise en balance : les bénéfices relevant de l'Encadré 2 doivent compenser tout impact préjudiciable sur l'ECW résultant des effets de l'Encadré 1, y compris les futurs effets de prix et d'investissement. Une

augmentation nette des incitations à investir ne suffit donc pas, à elle seule, à conclure qu'une concentration ne réduit pas l'ECW, dans la mesure où il est possible qu'elle ne vienne pas compenser intégralement le préjudice pour les consommateurs résultant de prix plus élevés pour des produits nouveaux ou améliorés. Selon les cas, des outils quantitatifs (par exemple, des simulations de concentrations) et des méthodes qualitatives peuvent être utilisés pour mettre en balance les préjudices et les bénéfices dynamiques. Une mise en balance des effets statiques et dynamiques peut également s'avérer nécessaire lorsque ceux-ci vont dans des directions opposées. Le cadre d'analyse ECW permet de le faire au moyen d'une pondération par les probabilités et d'une actualisation adéquates des effets statiques et dynamiques. Une **approche graduée** est adaptée à cette mise en balance, avec des éléments établissant des effets bénéfiques plus marqués et plus certains au titre de l'Encadré 2 étant requis dans les cas où les effets de l'Encadré 1 sont susceptibles d'être plus importants ou plus immédiats (par exemple, une concentration horizontale impliquant une entreprise dominante en place ou deux concurrents importants ou proches sur un marché concentré).

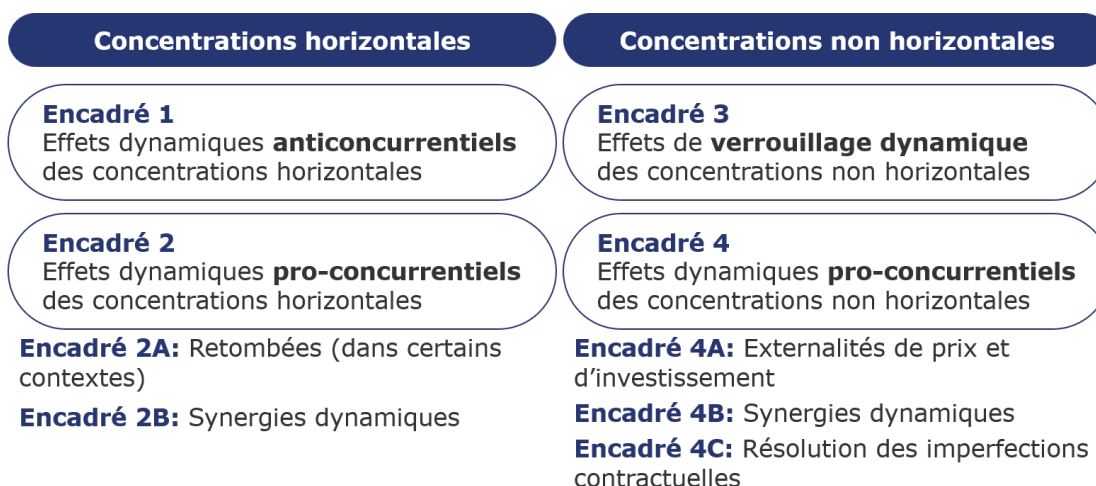
Effets non horizontaux

Comme pour les concentrations horizontales, nous regroupons les effets dynamiques des concentrations non horizontales en deux catégories : l'« Encadré 3 », qui couvre les *effets dynamiques anticoncurrentiels de verrouillage de marché*, et l'« Encadré 4 », qui couvre les *effets dynamiques pro-concurrentiels non horizontaux*.

Les **effets de verrouillage dynamique** relevant de l'**Encadré 3** couvrent les situations dans lesquelles la concentration confère ou renforce la capacité et les incitations de l'entité issue de l'opération à adopter un comportement qui réduit les investissements de ses concurrents, au détriment des consommateurs. Les théories du préjudice envisageables au titre de l'Encadré 3 comprennent le hold-up des investissements des concurrents, l'augmentation du coût des concurrents par un refus de contracter, la dissuasion à l'entrée à plusieurs niveaux ou sur plusieurs produits, l'exploitation dynamique d'un levier afin de protéger les rentes de l'acquéreur sur son marché principal, ainsi que la privation d'un intrant ou d'un complément important pour faire obstacle à la montée en puissance de concurrents. Le préjudice concurrentiel au titre de l'Encadré 3 est moins direct que celui au titre de l'Encadré 1, les concentrations non horizontales n'entraînant pas de perte de concurrence entre concurrents présents sur un même marché.

Les **effets dynamiques pro-concurrentiels des concentrations non horizontales** relevant de l'**Encadré 4** comprennent l'internalisation des externalités positives de prix et d'investissement (Encadré 4A), les synergies dynamiques (Encadré 4B) et la résolution des imperfections contractuelles (Encadré 4C). Dans la mesure où les concentrations non horizontales tendent à générer des gains d'efficacité issus de l'intégration d'activités complémentaires, les mécanismes potentiellement préjudiciables et bénéfiques sont souvent concomitants et entremêlés.

La figure ci-dessous résume les principaux effets dynamiques horizontaux et non horizontaux d'une concentration couverts dans notre cadre d'analyse.



Renforcement de position

Notre cadre conceptuel tient également compte du *renforcement de position* comme un potentiel effet dynamique préjudiciable, qui n'est pas couvert par l'Encadré 1 ou l'Encadré 3. Celui-ci reflète les situations dans lesquelles une acquisition impliquant une entreprise disposant d'un pouvoir de marché significatif ou substantiel augmente les barrières à l'entrée ou à l'expansion des concurrents d'une manière qui affaiblit la contrainte concurrentielle future que ceux-ci auraient autrement exercée, au détriment final des consommateurs. Nous identifions les raisons pour lesquelles il convient, d'un point de vue politique et économique, de limiter les préoccupations liées au renforcement de position aux entreprises disposant d'un haut degré de pouvoir de marché. Nous identifions les mécanismes potentiels de renforcement de position à partir des travaux récents de la littérature académique. Ces mécanismes peuvent opérer, par exemple, en réduisant les incitations à innover en vue d'un rachat (ou « innovation-for-buyout ») pour de futurs entrants, ou en conduisant à des « zones d'évitement » (ou « kill zones »). Nous discutons le cas dans lequel le renforcement de la position d'un acquéreur dominant et la réduction concomitante des ventes de ses concurrents, causés à court terme par les effets pro-concurrentiels de la concentration, peuvent néanmoins entraîner un préjudice concurrentiel net à plus long terme. La littérature et les principes économiques suggèrent que les circonstances dans lesquelles cela peut se produire sont rares.

Cas canoniques

Nous établissons une correspondance entre les effets identifiés dans notre cadre conceptuel et des catégories de cas de concentrations « canoniques ». Pour chaque catégorie, nous expliquons quels effets dynamiques sont pertinents, fournissons des exemples et des études de cas et présentons les éléments de preuve concrets permettant d'apprécier ces effets. Les catégories, résumées dans le tableau ci-dessous, se distinguent par (i) la nature du chevauchement entre les parties à la concentration (par exemple, des projets ou des produits spécifiques en développement *versus* des compétences plus générales), (ii) la nature de la cible, (iii) la présence d'un acquéreur bénéficiant d'un fort pouvoir de marché et (iv) le type d'investissement en cause (R&D *versus* investissement physique). Les catégories A à C correspondent aux Encadrés 1 et 2, la catégorie D correspond aux Encadrés 3 et 4 et la catégorie E correspond au renforcement de position. Une concentration donnée peut relever de plusieurs de ces catégories.

Catégorie	Principales caractéristiques et exemples de cas
A1 Chevauchements de projets ou produits en développement : R&D	Le projet de R&D en développement d'une partie à la concentration concurrence le produit existant de l'autre. <i>Exemple : Novartis/GlaxoSmithKline Oncology Business (2015, EU)</i>

	projet en développement sur produit existant	
A2	Chevauchements de projets ou produits en développement : R&D projet en développement sur projet en développement	Les deux parties travaillent sur des projets de R&D en développement, parallèles et concurrents. <i>Exemple : Adobe/Figma (2023 ; EU, UK)</i>
A3	Chevauchements de projets ou produits en développement : projet d'investissement physique	L'une des parties à la concentration ou les deux travaillent sur des projets d'investissement physique spécifiques qui se recoupent avec un produit existant ou un projet d'investissement de l'autre entreprise (y compris en termes de présence géographique). <i>Exemple : ENI/EDP/GDP (2004, EU)</i>
B1	Chevauchements de ressources et compétences : capacités d'innovation	Deux entreprises parmi un nombre limité disposent d'importantes capacités de R&D qui se recoupent. <i>Exemple : Dow/DuPont (2017, EU)</i>
B2	Chevauchements de ressources et compétences : capacités d'investissement	Deux entreprises parmi un nombre limité disposent d'importantes capacités d'investissement qui se recoupent et sont nécessaires pour concourir et répondre aux besoins futurs des clients ou entrer sur des marchés où l'autre entreprise est présente. <i>Exemples : Ryanair/Aer Lingus I et III (2007, 2013 ; EU)</i>
C	Acquisitions de concurrents émergents	Une entreprise en place acquiert un concurrent émergent disposant de projets ou de capacités en développement. <i>Exemple : Facebook/Instagram/WhatsApp (2020, US)</i>
D	Verrouillage dynamique	La concentration peut créer ou renforcer la capacité et les incitations à verrouiller les investissements des concurrents. <i>Exemple : Illumina/GRAIL (2022, EU ; 2023, US)</i>
E	Renforcement de position	La concentration peut renforcer la position d'un acquéreur disposant d'un pouvoir de marché significatif et nuire aux consommateurs en réduisant la contestabilité à long terme. <i>Exemple : Booking/eTraveli (2023, EU)</i>

Analyse sectorielle

L'étude se conclut par un examen de la pertinence du contrôle des concentrations et du cadre d'analyse que nous proposons dans trois secteurs stratégiques, en analysant les principaux obstacles à l'investissement et à l'innovation dans l'UE, ainsi que le rôle des différents instruments de politique publique, dont le contrôle des concentrations.

- Dans le **secteur biopharmaceutique**, l'UE fait face à un écart croissant en termes de productivité et d'investissement en R&D par rapport aux États-Unis et à la Chine. L'innovation dans le secteur se déplace également vers des entreprises biopharmaceutiques émergentes. Cela fait du traitement des acquisitions de petites entreprises innovantes une considération politique importante. Si les acquisitions de petites entreprises innovantes peuvent accélérer les projets de développement en cours, elles peuvent également freiner la diversité de la R&D, la productivité des inventeurs et la concurrence émergente. Le contrôle des concentrations doit tenir compte de ces effets dynamiques potentiels opposés.
- Dans **l'informatique quantique**, l'UE dispose de solides capacités de recherche, mais peine à traduire son excellence scientifique à l'échelle commerciale, les nouveaux entrants étant plus récents, plus petits et moins bien financés que leurs homologues américains. Bien que le contrôle des concentrations ne joue pas encore un rôle majeur dans ce secteur, son importance pourrait croître à mesure que la consolidation de ce secteur s'accroît.

- Enfin, dans **les technologies propres**, où la R&D, les effets de courbe d'apprentissage et l'échelle sont essentiels à la concurrence, l'Europe fait face à une forte pression des fabricants chinois et à des dépendances au niveau des chaînes d'approvisionnement. Le contrôle des concentrations peut jouer un rôle en permettant aux entreprises de réaliser des économies d'échelle, tout en se prémunissant contre les acquisitions prédatrices (ou « killer acquisitions ») et les préoccupations de verrouillage du marché liées aux nouvelles technologies prometteuses, et en reconnaissant que la capacité à décarboniser efficacement la production d'énergie constitue une dimension importante de la compétitivité-coût actuelle et future.

Kurzzusammenfassung

Wettbewerb ist ein dynamischer Rivalitätsprozess, in dem Unternehmen über verschiedene Parameter — darunter Preis, Qualität, Innovation, Investitionen und Produktvielfalt — konkurrieren, um Verbrauchern attraktive Angebote zu unterbreiten, Absatz von ihren Wettbewerbern zu gewinnen und ihre Gewinne zu steigern. Ein Zusammenschluss kann diesen Wettbewerbsprozess beeinträchtigen, indem er die Rivalität zwischen zuvor miteinander konkurrierenden Unternehmen zum Nachteil der Verbraucher beseitigt. Er kann den beteiligten Unternehmen außerdem die Fähigkeit und die Anreize geben, Wettbewerber abzuschotten und damit den von ihnen ausgehenden Wettbewerbsdruck zu verringern. Ein Zusammenschluss kann die Rivalität jedoch auch fördern, wenn das fusionierte Unternehmen zu einem wirksameren Wettbewerber wird, was den Verbrauchern zugutekommt. Diese Effekte können den gegenwärtigen Wettbewerb bei bestehenden Produkten und den künftigen Wettbewerb bei neuen Produkten und Technologien betreffen.

Dynamische Elemente des Wettbewerbs — darunter der Innovations- und Investitionswettbewerb sowie der Markteintritt von Unternehmen, die die Stellung etablierter Marktteilnehmer angreifen — gelten als zentral für technologischen Fortschritt, langfristiges Wachstum und globale Wettbewerbsfähigkeit. Dynamische Erwägungen gewinnen in einer modernen Wirtschaft an Bedeutung, die durch disruptive Innovationen und schnelle Technologiezyklen gekennzeichnet ist.

Diese von Oxera erstellte Studie der Europäischen Kommission ist ein Beitrag zur fortlaufenden Überarbeitung der Fusionskontrollleitlinien der Kommission aus den Jahren 2004 und 2008. Diese Überarbeitung folgt den Empfehlungen des Draghi-Berichts 2024 und dessen Forderung, Innovation, Investitionen und dem künftigen Wettbewerb in der Fusionskontrolle größeres Gewicht beizumessen. Zwar haben Wettbewerbsbehörden, einschließlich der Kommission, dynamische Erwägungen im vergangenen Jahrzehnt zunehmend in ihren fusionskontrollrechtlichen Entscheidungen berücksichtigt; der Rahmen zur Analyse dynamischer Effekte ist jedoch weniger gefestigt als der Rahmen für die Effekte von Zusammenschlüssen auf den statischen Wettbewerb.

Aufbauend auf einer umfassenden Synthese der einschlägigen wissenschaftlichen Literatur entwickelt diese Studie eine belastbare analytische Grundlage für die ökonomische Beurteilung dynamischer Zusammenschlusseffekte. Der Schwerpunkt liegt auf Zusammenschlüssen zwischen Wettbewerbern („horizontale Zusammenschlüsse“), und behandelt werden auch Zusammenschlüsse zwischen Herstellern komplementärer Produkte („nicht-horizontale Zusammenschlüsse“).

Zentrale Ergebnisse

- **Dynamische Effekte sollten breit definiert werden und die Gesamtwirkung von Zusammenschlüssen auf den künftigen Produktmarktwettbewerb umfassen** — einschließlich ihrer Auswirkungen auf Investitionen, Innovationen, Markteintritt sowie Preise und Verfügbarkeit neuer oder verbesserter Produkte.
- **Dynamische Effekte können statische Effekte verstärken, ausgleichen oder ihnen entgegenwirken** und damit das Ausmaß sowohl von wettbewerbswidrigen als auch wettbewerbsfördernden Effekten erweitern. **Dies erhöht die Tragweite fusionskontrollrechtlicher Entscheidungen** in Fällen mit einer erheblichen dynamischen Komponente.
- **Unser konzeptioneller Rahmen beurteilt Zusammenschlüsse mit dynamischen Effekten anhand eines Standards der erwarteten Konsumentenwohlfahrt (EKW)**, der Unsicherheit bei einigen Treibern künftiger Wettbewerbsergebnisse berücksichtigt (insbesondere in Bezug auf den Erfolg von

Forschungs- und Entwicklungsanstrengungen (F&E) und Innovationsanstrengungen, die Einführung neuer Produkte, den Eintritt potenzieller Wettbewerber und die Realisierung dynamischer Synergien). Dieser Standard berücksichtigt sowohl die Wahrscheinlichkeit als auch das Ausmaß der Auswirkungen einer Fusion.

- **Horizontale Zusammenschlüsse beseitigen Wettbewerb zwischen den beteiligten Unternehmen auf künftigen Produktmärkten. Ohne Spillover-Effekte und Synergien verringert dies die EKW wahrscheinlich,** weil negative Investitions- und Preisexternalitäten internalisiert werden. Dies gilt auch, wenn der Zusammenschluss Investitionsanreize durch eine Abschwächung des künftigen Preiswettbewerbs erhöht. Größere Marktmacht auf künftigen Produktmärkten ist daher keine geeignete Grundlage für eine „Innovation Defence“.
- **Horizontale Zusammenschlüsse können wettbewerbsfördernd wirken,** indem sie zu dynamischen Synergien oder zur Internalisierung von Spillover-Effekten führen. Dynamische Synergien können durch mehrere Kanäle entstehen, darunter eine stärkere Technologiediffusion, niedrigere inkrementelle Investitionskosten oder eine höhere Investitionswirksamkeit durch Skaleneffekte und Komplementaritäten sowie eine Lockerung finanzieller Einschränkungen. Eine Fusion kann auch Spillover-Effekte zwischen den fusionierenden Parteien internalisieren und dadurch potenziell die Innovationsanreize stärken. Die Internalisierung von Spillover-Effekten gegenüber nicht an der Fusion beteiligten Unternehmen kann jedoch gegenteilige Wirkungen erzeugen und die Innovationsanreize verringern.
- **Nicht-horizontale Zusammenschlüsse können dem fusionierten Unternehmen dynamische Abschottungsstrategien ermöglichen,** die Investitionen von Wettbewerbern reduzieren oder deren Markteintritt abschrecken und damit Verbrauchern schaden. **Nicht-horizontale Zusammenschlüsse können auch wettbewerbsfördernde dynamische Effekte haben.** Diese können aus der Internalisierung von Preis- oder Investitionsexternalitäten, aus dynamischen Synergien oder aus der Verringerung vertraglicher Friktionen entstehen. Da nicht-horizontale Zusammenschlüsse tendenziell Effizienzgewinne aus der Integration komplementärer Tätigkeiten hervorbringen, treten potenziell nachteilige und vorteilhafte Mechanismen häufig gleichzeitig auf und sind miteinander verflochten.
- **Neben potenziell schädlichen dynamischen horizontalen Effekten und dynamischen Abschottungseffekten können Fusionen auch die Stellung eines etablierten Marktteilnehmers mit erheblicher Marktmacht festigen,** indem sie Eintritts- und Expansionsbarrieren für Wettbewerber erhöhen und damit letztlich Verbrauchern schaden. Die jüngere wissenschaftliche Literatur benennt mehrere spezifische dynamische Mechanismen, die Verfestigung hervorrufen. Wenn ein spezifischer Verfestigungsmechanismus identifiziert wird, kann dennoch eine Abwägung gegenüber etwaigen wettbewerbsfördernden Auswirkungen einer Fusion erforderlich sein, insbesondere im Fall des Erwerbs komplementärer Ressourcen.
- **Der dreigliedrige Ansatz zur Bewertung von Effizienzgewinnen** (Nachprüfbarkeit, Fusionsspezifität und Verbrauchervorteil) **bietet einen geeigneten materiellen Rahmen zur Bewertung wettbewerbsfördernder dynamischer Effekte im Rahmen eines EKW-Standards. Dieser Standard ermöglicht eine symmetrische Abwägung nachteiliger und vorteilhafter dynamischer Effekte, sofern beide vorliegen. Ein gradueller Ansatz ist für die Abwägung geeignet,** wobei in den Fällen, in denen erheblichere (z. B. weil

die Fusion zwei enge oder bedeutende Wettbewerber zusammenführt) oder unmittelbare Schäden zu erwarten sind, der Nachweis stärkerer und wahrscheinlicher positiver Effekte erforderlich ist.

- **Unser konzeptioneller Rahmen wird auf fünf Kategorien typischer Zusammenschlussfälle mit dynamischen Effekten übertragen:** (i) Zusammenschlüsse mit Produkt- oder Projekt-Pipeline-Überschneidungen (einschließlich F&E Pipeline-zu-Produkt-Überschneidungen, F&E Pipeline-zu-Pipeline-Überschneidungen und Überschneidungen physischer Investitionsprojekte); (ii) Zusammenschlüsse mit Kompetenzüberschneidungen; (iii) Übernahmen junger Wettbewerber; (iv) Zusammenschlüsse mit Bedenken hinsichtlich dynamischer Abschottung; und (v) Zusammenschlüsse mit Bedenken hinsichtlich Verfestigungseffekten.

Literaturüberblick

Unser Überblick über die einschlägige Literatur bündelt theoretische und empirische Erkenntnisse aus mehr als 450 Publikationen der Industrieökonomik, Finanzökonomik, Makroökonomik und Betriebswirtschaftslehre. Zudem berücksichtigen wir einschlägige wettbewerbspolitische Veröffentlichungen und relevante Leitlinien von Wettbewerbsbehörden, dem International Competition Network und der OECD.

Zunächst betrachten wir die etablierte Literatur zu **Wettbewerb und Innovation**. Der Schwerpunkt dieser Literatur liegt zwar nicht auf Zusammenschlüssen (die eine besondere Form der Veränderung von Wettbewerbsverhältnissen darstellen), sie liefert jedoch wichtige Erkenntnisse für die Fusionskontrollpolitik. Modelle zu Patentrennen zeigen, dass Rivalität auf der Innovationsstufe Innovationen tendenziell fördert, da Unternehmen darum konkurrieren, als Erste ein neues Produkt einzuführen. Demgegenüber ist die Wirkung des Produktmarktwettbewerbs auf Innovationsanreize nicht eindeutig. Sie hängt davon ab, wie sich der Wettbewerb auf die Differenz zwischen den Gewinnen vor und nach der Innovation auswirkt. Schwächerer Produktmarktwettbewerb vor einer Innovation kann Innovationsanreize dämpfen, weil er die Kannibalisierung bestehender Gewinne verstärkt und die Erträge eines „Entkommens“ aus dem Wettbewerb mindert (Arrowscher Effekt). Schwächerer Produktmarktwettbewerb nach einer erfolgreichen Innovation kann die Anreize hingegen durch eine bessere Aneignung der Erträge stärken (Schumpeterscher Effekt). Diese gegenläufigen Effekte verdeutlichen, dass Innovationswettbewerb und Produktmarktwettbewerb gemeinsam beurteilt werden müssen, da ein Zusammenschluss beide Wettbewerbsdimensionen beeinflusst. Empirisch erhöht Wettbewerb die Produktivität tendenziell durch den sogenannten „Behandlungseffekt“ (d.h. bestehende Unternehmen werden produktiver) und den „Selektionseffekt“ (d.h. produktivere Unternehmen treten in den Markt ein und weniger produktive Unternehmen verlassen ihn); seine Wirkung auf Innovation ist jedoch nicht eindeutig, im Einklang mit der theoretischen Literatur.

Jüngere politische Debatten, insbesondere in Europa, haben eine umfangreiche **theoretische Literatur zu horizontalen Zusammenschlüssen zwischen etablierten Wettbewerbern** und deren Auswirkungen auf Investitionen und Konsumentenwohlfahrt hervorgebracht. Die Modelle dieser Literatur sind zwar stilisiert, liefern aber nützliche Erkenntnisse für die Fusionskontrollpolitik. Unter Bedingungen, in denen etablierte Unternehmen um Produktverbesserungen, Kostensenkungen oder Innovationen für den Eintritt in neue Märkte konkurrieren, zeigt die Literatur, dass Zusammenschlüsse Investitionen ohne Wissens-Spillover-Effekte und Synergien sowohl verringern als auch erhöhen können. Während die Literatur darauf hindeutet, dass beide Effekte unter plausiblen Bedingungen auftreten können, legt sie auch nahe, dass Fusionen Investitionen häufiger verringern als erhöhen. Trotz der gemischten Ergebnisse zu Investitionen weist die Literatur außerdem darauf hin, dass Zusammenschlüsse ohne Spillover-Effekte und

Synergien die Konsumentenwohlfahrt wahrscheinlich mindern. Untersucht werden auch Bedingungen, in denen Zusammenschlüsse schwächere oder nachzügeln Unternehmen („laggards“) verbinden oder zu einer Neuverteilung von Investitionen zwischen Projekten führen. In solchen Fällen kann ein Zusammenschluss den Verbrauchern auch ohne Spillover-Effekte und Synergien zugutekommen. Ein Zusammenschluss kann auch Spillover-Effekte zwischen den beteiligten Unternehmen internalisieren und dadurch Innovationsanreize stärken. Die Internalisierung von Spillover-Effekten zugunsten nicht beteiligter Unternehmen kann jedoch gegenläufige Effekte kreieren und Innovationsanreize reduzieren. Schließlich zeigt die Literatur, dass zusammenschlussbedingte dynamische Synergien dynamische Schäden eines Zusammenschlusses mindern oder ausgleichen können.

Die **theoretische Literatur zur Übernahme potenzieller Wettbewerber** zeigt, dass Zusammenschlüsse Innovationen sowohl ex post, infolge der Transaktion, als auch ex ante, durch Anreize, die aus der Aussicht auf eine Übernahme durch ein etabliertes Unternehmen entstehen („Innovation-for-Buyout“), beeinflussen können. Ohne Synergien und Spillover-Effekte sind die Ex-post-Effekte tendenziell wettbewerbswidrig, wenn die Übernahme die Innovationstätigkeit des Zielunternehmens unterdrückt, künftigen Preiswettbewerb beseitigt, oder die Stellung des etablierten Unternehmens verfestigt (z.B. indem es dem Erwerber ermöglicht wird, eine größere Basis gebundener Verbraucher zu bedienen). Diese Bedenken sind am stärksten, wenn das etablierte Unternehmen über erhebliche Marktmacht verfügt und das Zielunternehmen ein glaubwürdiger Wettbewerber ist – entweder eigenständig oder im Rahmen einer Übernahme durch einen alternativen Käufer, der geringere Wettbewerbsbedenken aufwirft. Übernahmen können jedoch auch Synergien ermöglichen, indem sie kleineren innovativen Unternehmen die Skalierung erlauben, die Technologiediffusion erhöhen oder komplementäre Ressourcen verbinden. Ex-ante-Effekte können den Markteintritt neuer innovativer Unternehmen fördern, indem sie einen profitableren Ausstieg ermöglichen; die Auswirkungen auf die Ausrichtung ihrer Innovationen, also auf die verfolgten Technologien oder Forschungspfade, sind jedoch nicht eindeutig.

Die Literatur umfasst außerdem mehrere **empirische Studien** zu den dynamischen Auswirkungen horizontaler Zusammenschlüsse in verschiedenen Sektoren. Ein Teil dieser Arbeiten weist zwar methodische Einschränkungen auf, dennoch geben die Ergebnisse Hinweise auf die empirischen Wirkungen von Zusammenschlüssen. Insgesamt zeigt die Literatur gemischte empirische Effekte auf Investitions- oder Innovationsinputs (z.B. F&E-Ausgaben) und auf Outputs (z.B. Patente); zu den Auswirkungen auf die Konsumentenwohlfahrt liegen nur begrenzte Erkenntnisse vor. Die empirische Corporate-Finance-Literatur zeigt, dass Zusammenschlüsse häufig Wert für das fusionierte Unternehmen steigern, wobei diese Vorteile in der Regel den Anteilseignern der Zielunternehmen und nicht denen der Erwerber zufließen und zugleich Marktmachteeffekte reflektieren können.

Die theoretische Literatur zu **nicht-horizontalen (vertikalen oder konglomeraten) Zusammenschlüssen** zeigt, dass diese Investitionen fördern und Verbrauchern zugutekommen können, indem sie doppelte Preisaufschläge beseitigen, positive Investitionsexternalitäten internalisieren, komplementäre Fähigkeiten verbinden, Vertragsfraktionen lösen oder Hold-up-Probleme mindern. Sie können den Wettbewerb jedoch auch beeinträchtigen, indem sie die Fähigkeit oder die Anreize schaffen beziehungsweise verstärken, Verhaltensweisen einzusetzen, die zu einer dynamischen Abschottung von Wettbewerbern führen. Dazu gehören Maßnahmen, die die Kosten der Wettbewerber erhöhen oder deren Fähigkeit oder Anreize schwächen, zu investieren, zu expandieren oder in den Markt einzutreten. Da Schäden und Vorteile nicht-horizontaler Zusammenschlüsse häufig gleichzeitig auftreten, ist ihre Gesamtwirkung auf Verbraucher oft unklar. Die Literatur hat allerdings einige wettbewerbswidrige Mechanismen identifiziert, denen keine gegenläufige Effizienz gegenübersteht. Empirische Erkenntnisse bestätigen, dass nicht-horizontale Zusammenschlüsse Investitionen und Innovationen erhöhen können (z.B. durch eine bessere Koordinierung von F&E-Entscheidungen oder

Wissenstransfers zwischen komplementären Geschäftsbereichen), zugleich aber den Markteintritt und die Innovationstätigkeit Dritter verringern können.

Die **Literatur zu Friktionen auf Finanzmärkten** zeigt, dass Zusammenschlüsse den dynamischen Wettbewerb beeinflussen können, indem sie Finanzierungsbeschränkungen für Investitionen oder Innovationen lockern. Dazu gehören „Liquiditätszusammenschlüsse“, die notleidende, aber produktive Vermögenswerte erhalten. Diese Vorteile sind jedoch nicht immer zusammenschlusspezifisch, da Finanzierungsengpässe auch durch Aktienemissionen, Private-Equity- oder Venture-Capital-Finanzierungen oder Bankkredite behoben werden können. Zusammenschlüsse, die Unternehmen eine Diversifizierung ihrer Tätigkeiten ermöglichen, können durch wechselnde Risikoabsicherung („Coinsurance“) und interne Kapitalmärkte das Krisenrisiko und die Finanzierungskosten senken; sie können aber auch zu ineffizienter Kapitalallokation, Quersubventionierung, Agency-Problemen, Intransparenz oder Risikoübertragung führen.

Konzeptioneller Rahmen

Der konzeptionelle Rahmen für die Bewertung dynamischer Zusammenschlusseffekte bildet den Kern dieser Studie. Ziel ist eine umfassende Charakterisierung der potenziell wettbewerbswidrigen und wettbewerbsfördernden dynamischen Effekte von Zusammenschlüssen. Zudem erörtern wir, wie Fallbeispiele und Evidenz aus der Praxis in den konzeptionellen Rahmen einzuordnen sind.

Definitionen und materieller Prüfungsmaßstab

Dynamische Zusammenschlusseffekte werden so definiert, dass sie die Auswirkungen von Zusammenschlüssen auf den Wettbewerb in künftigen Produktmärkten erfassen, die aus Änderungen von Produkteigenschaften (z.B. Preis, Qualität und Verfügbarkeit neuer oder verbesserter Produkte), Kosten oder der Zahl der Unternehmen resultieren. Die Effekte eines Zusammenschlusses auf künftige Produktmärkte können von jenen auf gegenwärtige Produktmärkte abweichen, weil Unternehmen investieren, um ihre künftige Wettbewerbsposition zu verbessern. Wir legen einen breiten Investitionsbegriff zugrunde, der F&E- und Infrastrukturinvestitionen etablierter Unternehmen sowie die Bemühungen junger Wettbewerber um einen Markteintritt umfasst. Unsere Definition dynamischer Zusammenschlusseffekte ist auch deshalb breit, weil sie sowohl die Auswirkungen auf Investitionsanreize als auch auf den künftigen Produktmarktwettbewerb (z.B. Preiswettbewerb bei neuen oder verbesserten Produkten) bei gegebenem Investitionsniveau erfasst. Eine breite Definition ist erforderlich, um unterschiedliche potenzielle Bestimmungsfaktoren des Wettbewerbs auf künftigen Produktmärkten einheitlich zu behandeln und ihr Zusammenspiel abzubilden. Entsprechend der Fusionskontrollpraxis umfasst unsere Definition dynamischer Effekte nur Post-Fusions-Effekte. Ex-ante-Effekte aus der Aussicht auf eine künftige Übernahme liegen daher außerhalb des konzeptionellen Rahmens, auch wenn die Haltung der Wettbewerbsbehörden Innovationsanreize über diesen Kanal ebenfalls beeinflussen kann.

Wir beurteilen dynamische Zusammenschlusseffekte anhand eines **Konsumentenwohlfahrtsstandards**, im Einklang mit der geltenden Rechts- und Entscheidungspraxis. Die Konsumentenwohlfahrt dient als hilfreicher Maßstab für die Gesamtwirkung eines Zusammenschlusses, wenn mehrere Wirkungen gegeneinander aufgerechnet werden müssen (z.B. höhere Preise einerseits und höhere Qualität oder größere Produktvielfalt andererseits). Die Anwendung eines Konsumentenwohlfahrtsstandards bedeutet, dass eine „Innovation Defence“ in der Fusionskontrolle nur dann greift, wenn die Verbraucher hinreichend von einer höheren Innovationsleistung profitieren und damit gleichzeitig eintretende nachteilige dynamische Effekte des Zusammenschlusses (z.B. höhere Preise für innovative Produkte), ausgeglichen werden.

Um der Unsicherheit bei Investitionen, insbesondere bei F&E-Aktivitäten, Rechnung zu tragen, verwenden wir einen **Standard der erwarteten Konsumentenwohlfaht (EKW)**. Nach diesem Standard führt ein Zusammenschluss zu nachteiligen Auswirkungen auf den Wettbewerb, wenn er zu einer Verringerung der EKW im Vergleich zur Situation ohne den Zusammenschluss führt (d. h. $\Delta EKW < 0$). Eine formale Bewertung von ΔEKW erfordert, die Auswirkungen des Zusammenschlusses in den relevanten tatsächlichen und kontrafaktischen Szenarien nach Eintrittswahrscheinlichkeit zu gewichten. In der Praxis dient der EKW-Standard überwiegend als konzeptionelles Instrument zur Strukturierung der Analyse und weniger als ausdrücklich berechnete oder quantifizierte Größe. Dadurch kann die Prüfung nach EKW-Standard unsichere Einzeleffekte eines Zusammenschlusses widerspiegeln, die erhebliche Auswirkungen auf das Verbraucherwohl haben können. Dies ist bei dynamischen Zusammenschlusswirkungen besonders relevant, sowohl bei wettbewerbswidrigen Effekten (z.B. der Beseitigung potenziellen, aber ungewissen Wettbewerbs durch einen aufstrebenden Wettbewerber) als auch bei wettbewerbsfördernden Wirkungen (z.B. einem großen, aber ungewissen Vorteil des Zusammenschlusses). Die Berücksichtigung dynamischer Effekte erweitert die Bandbreite möglicher Ergebnisse eines Zusammenschlusses, kann Schäden und Vorteile gegenüber einer statischen Analyse vergrößern und erhöht damit die Bedeutung fusionskontrollrechtlicher Entscheidungen.

Dynamische Wirkungen horizontaler Zusammenschlüsse

Wir ordnen die dynamischen Effekte horizontaler Zusammenschlüsse zwei Kategorien zu: „Box 1“ erfasst wettbewerbswidrige Effekte und „Box 2“ wettbewerbsfördernde Effekte einschließlich der Internalisierung von Spillover-Effekten und Synergien. Diese inhaltliche Kategorisierung ökonomischer Effekte soll weder verfahrensrechtliche Fragen zur Beweislast in der Fusionskontrolle noch die Rechtsfrage vorwegnehmen, wann eine erhebliche Behinderung wirksamen Wettbewerbs im Sinne der EU-Fusionskontrollverordnung vorliegt; beides liegt außerhalb des Untersuchungsgegenstands dieser Studie.

Box 1 erfasst die potenziell **wettbewerbswidrigen dynamischen Wirkungen horizontaler Zusammenschlüsse**, die aus der Beseitigung der Rivalität zwischen den beteiligten Unternehmen sowohl bei Investitionen als auch beim künftigen Produktmarktwettbewerb bei gegebenem Investitionsniveau entstehen. In unserer Ausgangsdarstellung konzentrieren wir uns auf den Standardfall einer Produktinnovation mit anschließendem Preiswettbewerb. Die Ergebnisse zur Gesamtwirkung eines Zusammenschlusses unter Box 1 gelten jedoch breiter auch für andere Investitionsformen (z.B. Prozessinnovationen oder Infrastrukturinvestitionen) und andere Formen des Produktmarktwettbewerbs (z.B. Mengenwettbewerb). Die Internalisierung negativer Preis- und Investitionsexternalitäten, die aus einem Zusammenschluss resultieren, schwächt den Gesamtanreiz des zusammengeschlossenen Unternehmens zum Wettbewerb. Box 1 erfasst diesen Effekt über drei unterschiedliche Kanäle: (i) *direkte Effekte* aus dem Verlust des Preiswettbewerbs, die zu höheren Preisen für künftige Produkte führen; (ii) *direkte Investitionseffekte* aus der Internalisierung negativer Investitionsexternalitäten (die dazu neigen, Investitionsanreize durch einen „Kannibalisierung“-Effekt zu verringern); und (iii) *indirekte Investitionseffekte* aus der Verringerung des Preiswettbewerbs (die dazu neigen, Investitionsanreize durch eine erhöhte „Appropriierbarkeit“ zu steigern). Wie die wirtschaftswissenschaftliche Literatur zeigt, kann die kombinierte Wirkung der drei Kanäle die Investitionsanreize erhöhen oder verringern. Während die Literatur darauf hindeutet, dass beide Effekte unter plausiblen Bedingungen auftreten können, legt sie auch nahe, dass Fusionen Investitionen häufiger verringern als erhöhen. Dies stützt keine allgemeine Vermutung hinsichtlich der Auswirkung eines Zusammenschlusses auf Investitionen im Rahmen von Box 1, deutet jedoch darauf hin, dass die kombinierte Wirkung der Box-1-Effekte auf Investitionen Anlass zu erheblicher Besorgnis gibt, dass ein Zusammenschluss zu einer Verringerung der Investitionen führen kann (sofern keine gegenläufigen Effekte im Rahmen von Box 2 vorliegen). Die gemeinsame Wirkung der Box-1-Effekte auf die EKW

ist klarer: Die Kombination aus Preis- und Investitionseffekten dürfte den Verbrauchern schaden, da direkte Preiseffekte den Verbrauchern schaden und direkte Investitionseffekte dazu neigen, die Konsumentenwohlfahrt zu verringern, während indirekte Investitionseffekte auf höheren Preisen beruhen. Daher ist im Rahmen von Box 1 grundsätzlich *keine Abwägung* zwischen höheren Preisen und einem daraus resultierenden Anstieg der Investitionen erforderlich, sofern keine Spillover-Effekte und Synergien vorliegen. Es gibt jedoch spezifische Umstände, unter denen eine Feststellung eines Verbraucherschadens im Rahmen von Box 1 möglicherweise nicht zutrifft (beispielsweise wenn ein Zusammenschluss zwei schwächere oder nachzügeln Unternehmen zusammenführt), selbst wenn keine gegenläufigen Effekte im Rahmen von Box 2 vorliegen.

Box 2 erfasst die **potenziell wettbewerbsfördernden dynamischen Wirkungen horizontaler Zusammenschlüsse**, einschließlich der Mechanismen, durch die ein Zusammenschluss die Fähigkeit und die Anreize der beteiligten Unternehmen zu Investitionen zum Vorteil der Verbraucher stärken kann. Box 2 unterscheidet zwischen der Internalisierung von Spillover-Effekten (Box 2A) und dynamischen Synergien (Box 2B).

- **Box 2A** betrifft Situationen, in denen die Innovationserträge teilweise nicht ausschließbar sind und auf Wettbewerber übergreifen. In diesem Fall kann ein Zusammenschluss positive Externalitäten zwischen den beteiligten Unternehmen internalisieren, Innovationsanreize erhöhen und Verbrauchern zugutekommen. Spillover-Effekte können jedoch auch nicht an der Fusion beteiligten Unternehmen zugutekommen. Ist dies der Fall, könnten die fusionierenden Unternehmen nach dem Zusammenschluss stärker besorgt über Spillover-Effekte zugunsten ihrer Wettbewerber sein, was die Innovationsanreize schwächt.
- **Box 2B** betrifft dynamische Synergien, die eine Verbesserung der Fähigkeiten des fusionierten Unternehmens widerspiegeln. Diese können entstehen durch (i) Technologietransfers zwischen den beteiligten Unternehmen, die andernfalls nicht stattgefunden hätten, (ii) niedrigere F&E-Kosten oder höhere F&E-Wirksamkeit, (iii) niedrigere Investitionskosten oder höhere Investitionswirksamkeit, einschließlich gemeinsam genutzter Infrastruktur, (iv) Produktionskosteneffizienzen und (v) eine Lockerung finanzieller Beschränkungen, die ein erworbenes Unternehmen sonst daran hindern würden, eine Innovation auf den Markt zu bringen. Soweit vorhanden, erhöhen Box-2B-Wirkungen die Investitionen und können zudem die Verbreitung von Innovationen zum Vorteil der Verbraucher fördern.

Der übliche **dreigliedrige Effizienztest** bietet einen geeigneten materiellen Rahmen für die Prüfung von Box-2-Wirkungen: Um nachteilige Wirkungen aus Box 1 auszugleichen, müssen dynamische Vorteile zusammenschlusspezifisch sein, nachprüfbar sein und an die Verbraucher weitergegeben werden. In unserem konzeptionellen Rahmen erfolgt die Prüfung anhand des EKW-Standards, um Konsistenz mit der Bewertung der Box-1-Wirkungen sicherzustellen.

Eine **Abwägung der Wirkungen aus Box 1 und Box 2** ist erforderlich, wenn beide nach dem jeweils erforderlichen Maßstab nachgewiesen sind; dabei sind Evidenz, Zeithorizont und Unsicherheit einheitlich zu behandeln. Der EKW-Standard ist ein geeigneter Abwägungsmaßstab: Vorteile aus Box 2 müssen sämtliche nachteiligen Auswirkungen der Box-1-Effekte auf die EKW ausgleichen, einschließlich künftiger Preis- und Investitionseffekte. Ein Nettoanstieg der Investitionsanreize allein reicht daher nicht für die Schlussfolgerung aus, dass ein Zusammenschluss die EKW nicht mindert, da er den Schaden, der sich für Konsumenten aus höheren Preisen für neue oder verbesserte Produkte ergibt, möglicherweise nicht vollständig kompensiert. Je nach Einzelfall können quantitative Instrumente (z.B. Fusionssimulationen) und qualitative Methoden eingesetzt werden, um dynamische Schäden und Vorteile abzuwägen. Eine Abwägung statischer und dynamischer Effekte kann ebenfalls erforderlich sein, wenn sie in entgegengesetzte Richtungen weisen. Der EKW-Rahmen ermöglicht dies durch eine angemessene Wahrscheinlichkeitsgewichtung und Abzinsung statischer und dynamischer Effekte. Für die Abwägung ist ein **gradueller Ansatz** sachgerecht: Nachweise stärkerer und gewisserer

vorteilhafter Wirkungen aus Box 2 sind erforderlich, wenn die Wirkungen aus Box 1 voraussichtlich größer oder unmittelbarer sind (z.B. bei einem horizontalen Zusammenschluss unter Beteiligung eines marktbeherrschenden etablierten Unternehmens oder zweier bedeutender beziehungsweise enger Wettbewerber in einem konzentrierten Markt).

Nicht-horizontale Wirkungen

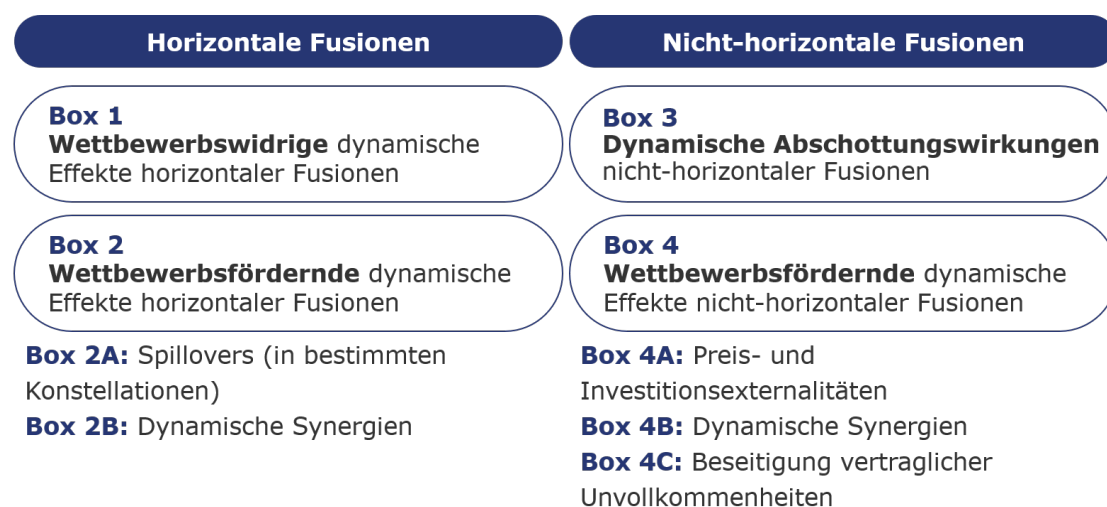
Wie bei horizontalen Zusammenschlüssen ordnen wir die dynamischen Wirkungen nicht-horizontaler Zusammenschlüsse zwei Kategorien zu: „Box 3“ erfasst *wettbewerbswidrige dynamische Abschottungswirkungen* und „Box 4“ *wettbewerbsfördernde nicht-horizontale dynamische Wirkungen*.

Dynamische Abschottungswirkungen unter **Box 3** erfassen Situationen, in denen der Zusammenschluss dem fusionierten Unternehmen die Fähigkeit und die Anreize verschafft oder verstärkt, Verhaltensweisen einzusetzen, die die Investitionen seiner Wettbewerber zum Nachteil der Verbraucher verringern. Mögliche Schadenstheorien unter Box 3 umfassen Hold-up bei Investitionen von Wettbewerbern, die Erhöhung ihrer Kosten durch eine Lieferverweigerung, die Abschreckung eines Markt- oder Produkteintritts auf mehreren Ebenen, eine dynamische Hebelwirkung zur Absicherung der ökonomischen Renten des Erwerbers in seinem Kernmarkt sowie die Vorenthaltung eines wertvollen Inputs oder Komplements, um Wettbewerber daran zu hindern, stärker zu werden. Der Wettbewerbsschaden unter Box 3 ist weniger unmittelbar als unter Box 1, da nicht-horizontale Zusammenschlüsse keinen Wettbewerb zwischen Unternehmen desselben Marktes beseitigen.

Zu den **wettbewerbsfördernden dynamischen Effekten nicht-horizontaler Zusammenschlüsse** unter **Box 4** gehören die Internalisierung positiver Preis- und Investitionsexternalitäten (Box 4A), dynamische Synergien (Box 4B) und die Beseitigung vertraglicher Unvollkommenheiten (Box 4C).

Weil nicht-horizontale Zusammenschlüsse durch die Integration komplementärer Tätigkeiten typischerweise Effizienzen erzeugen, treten potenziell schädliche und vorteilhafte Mechanismen häufig gleichzeitig auf und sind miteinander verflochten.

Die nachstehende Abbildung fasst die wesentlichen horizontalen und nicht-horizontalen dynamischen Effekte eines Zusammenschlusses zusammen, die in unserem Rahmen erfasst werden.



Verfestigung („Entrenchment“)

Unser konzeptioneller Rahmen berücksichtigt die Verfestigung außerdem als möglichen gegenläufigen dynamischen Effekt, der nicht unter Box 1 oder Box 3 erfasst wird. Erfasst werden Situationen, in denen der Zusammenschluss unter Beteiligung eines Unternehmens mit erheblicher oder substanzieller Marktmacht Eintritts- oder Expansionsbarrieren für Wettbewerber so erhöht, dass der künftige Wettbewerbsdruck, den diese andernfalls ausüben würden, zum letztlichen Nachteil der Verbraucher geschwächt wird. Wir zeigen, warum es aus wettbewerbspolitischer und ökonomischer Sicht sachgerecht ist, Verfestigungsbedenken auf Unternehmen mit hoher Marktmacht zu beschränken. Auf Grundlage neuerer wissenschaftlicher Erkenntnisse identifizieren wir mögliche Verfestigungsmechanismen. Beispielsweise können Innovationsanreize künftiger Marktteilnehmer im Hinblick auf einen späteren Verkauf verringert oder sogenannte „Kill-Zones“ entstehen. Wir erörtern den Fall, in dem die durch kurzfristig wettbewerbsfördernde Effekte eines Zusammenschlusses bewirkte Stärkung eines marktbeherrschenden Erwerbers und der gleichzeitige Umsatzrückgang seiner Wettbewerber langfristig dennoch zu einem Nettowettbewerbsschaden führen können. Die ökonomische Literatur und ökonomische Prinzipien deuten darauf hin, dass solche Konstellationen selten sind.

Typische Fallkonstellationen

Wir ordnen die Wirkungen unseres konzeptionellen Rahmens Kategorien typischer Zusammenschlussfälle zu. Für jede Kategorie erläutern wir die relevanten dynamischen Effekte, geben Beispiele und Fallstudien und benennen praktische Beweismittel für die Prüfung. Die in der nachstehenden Tabelle zusammengefassten Kategorien unterscheiden sich nach (i) der Art der Überschneidung zwischen den beteiligten Unternehmen (z.B. bestimmte Produkt- oder Projektpipelines oder umfassendere Fähigkeiten), (ii) der Art des Zielunternehmens, (iii) dem Vorliegen eines Erwerbers mit hoher Marktmacht und (iv) der Art der betroffenen Investition (F&E vs. physische Investitionen). Die Kategorien A bis C erfassen Box 1 und Box 2, Kategorie D Box 3 und Box 4 und Kategorie E die Verfestigung. Ein bestimmter Zusammenschluss kann mehreren Kategorien zugeordnet werden.

Kategorie	Zentrale Merkmale und Fallbeispiele
A1 Pipeline-Überschneidungen: F&E Pipeline-zu-Produkt	Das F&E-Pipeline-Projekt eines Unternehmens konkurriert mit dem bestehenden Produkt des anderen. Beispiel: <i>Novartis/GlaxoSmithKline Oncology Business (2015, EU)</i>
A2 Pipeline-Überschneidungen: F&E Pipeline-zu-Pipeline	Beide Unternehmen arbeiten an parallelen und konkurrierenden F&E-Pipeline-Projekten. Beispiel: <i>Adobe/Figma (2023; EU, UK)</i>
A3 Pipeline-Überschneidungen: physisches Investitionsprojekt	Eines oder beide Unternehmen arbeiten an bestimmten physischen Investitionsprojekten, die sich mit einem bestehenden Produkt oder Investitionsprojekt des anderen Unternehmens überschneiden, einschließlich geografischer Präsenz. Beispiel: <i>ENI/EDP/GDP (2004, EU)</i>
B1 Überschneidungen von Fähigkeiten: Innovationsfähigkeiten	Zwei von nur wenigen Unternehmen verfügen über erhebliche, sich überschneidende F&E-Fähigkeiten. Beispiel: <i>Dow/DuPont (2017, EU)</i>
B2 Überschneidung von Fähigkeiten: Investitionsfähigkeiten	Zwei von nur wenigen Unternehmen verfügen über erhebliche, sich überschneidende Investitionsfähigkeiten, die für die Deckung künftiger Kundenbedürfnisse oder für den Eintritt in Märkte erforderlich sind, in denen das andere Unternehmen tätig ist. Beispiele: <i>Ryanair/Aer Lingus I und III (2007 und 2013)</i>
C Erwerb aufstrebender Wettbewerber	Ein etabliertes Unternehmen erwirbt einen aufstrebenden Wettbewerber mit entstehenden Pipeline-Projekten oder Fähigkeiten. Beispiel: <i>Facebook/Instagram/WhatsApp (2020, USA)</i>

D	Dynamische Abschottung	Der Zusammenschluss kann die Fähigkeit und die Anreize schaffen oder verstärken, Investitionen von Wettbewerbern abzuschotten. Beispiel: <i>Illumina/GRAIL (2022, EU; 2023, USA)</i>
E	Verfestigung	Der Zusammenschluss kann einen Erwerber mit erheblicher Marktmacht stärken und Verbrauchern schaden, indem er die Bestreitbarkeit langfristig verringert. Beispiel: <i>Booking/eTraveli (2023, EU)</i>

Sektoranalyse

Abschließend untersucht die Studie die Bedeutung der Fusionskontrolle und des vorgeschlagenen Rahmens in drei strategischen Sektoren. Dabei werden die wichtigsten Investitions- und Innovationshemmnisse in der EU sowie die Rolle verschiedener politischer Instrumente einschließlich der Fusionskontrolle betrachtet.

- Im Bereich der **Biopharmazeutika** verzeichnet die EU im Vergleich zu den USA und China eine zunehmende Lücke bei der Produktivität sowie bei den Investitionen in F&E. Gleichzeitig verlagert sich die Innovation in diesem Sektor zunehmend hin zu aufstrebenden Biopharmaunternehmen. Dies macht den Umgang mit Übernahmen kleiner innovativer Unternehmen zu einer wichtigen politischen Erwägung. Solche Übernahmen können die Pipeline-Entwicklung beschleunigen, zugleich aber die Vielfalt der F&E, die Produktivität von Erfindern und den entstehenden Wettbewerb beeinträchtigen. Die Fusionskontrolle muss diese gegenläufigen potenziellen dynamischen Wirkungen berücksichtigen.
- Im **Quantencomputing** verfügt die EU über starke Forschungskapazitäten, hat jedoch Schwierigkeiten, wissenschaftliche Exzellenz in kommerzielle Größenordnungen zu überführen, da Marktteilnehmer jünger, kleiner und schwächer finanziert sind als ihre US-amerikanischen Pendanten. Obwohl die Fusionskontrolle in diesem Sektor bislang keine große Rolle spielt, könnte ihre Bedeutung mit zunehmender Konsolidierung steigen.
- Bei **sauberen Technologien**, bei denen F&E, Lernkurveneffekte und Skalierung für den Wettbewerb entscheidend sind, steht Europa unter starkem Druck chinesischer Hersteller und unter Lieferkettenabhängigkeiten. Die Fusionskontrolle kann es Unternehmen ermöglichen, Skaleneffekte zu erzeugen, und zugleich vor „Killer Acquisitions“ und Abschottungsrisiken hinsichtlich vielversprechender neuer Technologien schützen sowie anerkennen, dass die Fähigkeit zur wirksamen Dekarbonisierung der Energieerzeugung einen wichtigen Faktor der gegenwärtigen und künftigen Kostenwettbewerbsfähigkeit darstellt.

PART A INTRODUCTION AND BACKGROUND

1 Introduction and overview

- 1.1 In August 2025, the European Commission (Directorate-General for Competition) contracted Oxera to undertake an Economic Study to provide an analytical foundation for the assessment of dynamic merger effects in merger control (the Study).² The Study is carried out by Oxera in collaboration with Professor Otto Toivanen as Lead Academic Researcher, and Professors Yassine Lefouili and Leonardo Madio as Economic Advisers.

1.1 The context

- 1.2 Merger control is one of the principal tools by which competition authorities maintain and promote competitive markets. Merger review decisions often focus on the static effects of mergers—their impact on prices and output in existing product markets. However, mergers may also affect *dynamic* competition between firms. The dynamic effects of mergers can encompass a broad range of competitive parameters, including the effects of mergers on the ability and incentives of merging firms to innovate, invest, and enter new markets, on the ability and incentives to foreclose investment by rivals, on entrenchment of a dominant position, and on price effects in relation to future, innovative products. Dynamic merger effects may include both anticompetitive effects (e.g. a reduction of innovation, or an increase in the prices of innovative products, to the detriment of consumers), and procompetitive effects (e.g. greater ability and incentives by merging firms to bring an innovative product to market, to the benefit of consumers).
- 1.3 Dynamic merger effects can amplify or counteract static effects. A merger may result in reduced price competition for existing products while at the same time increasing innovation competition by leading to dynamic synergies (e.g. lower R&D costs or greater diffusion of innovation). Conversely, a merger may lead to both static harm in relation to existing products, and dynamic harm in the form of lower incentives to innovate, and higher prices for future innovative products. The dynamic effects of mergers imply that the stakes from merger control decisions are larger, with greater potential for competitive harm from a merger, but also for more significant procompetitive effects.
- 1.4 Dynamic effects are increasingly viewed as a central concern for merger control. The Commission has recognised the importance of dynamic effects in its merger control practice for a number of years, in several prominent mergers including merger prohibitions upheld by the European Courts.³ The same holds for significant cases in the US over the past three decades or so.⁴
- 1.5 There is currently renewed impetus to review the treatment of dynamic merger effects, to ensure that these are properly recognised in merger control practice and guidelines. This renewed focus on dynamic merger effects has been prompted by recent policy recommendations, most notably those contained in the 2024 Draghi Report on Europe’s competitiveness (Draghi, 2024). The Draghi Report urges that more weight be given to innovation and future competition in updated merger control guidelines and in decision practice. This recommendation was followed in the Mission Letter from Commission President Von der Leyen to Executive Vice

² The Tender Specifications for the Study are set out in the call for tenders, European Commission (2025), ‘[Study on dynamic merger effects](#)’, accessed 21 July 2026.

³ See discussion in Chapter 2.

⁴ See discussion in Chapter 2.

President Ribera,⁵ which calls for a modernisation of competition policy to support European firms to innovate and compete globally, and for a review of the merger control guidelines to give adequate importance to innovation and efficiency. These initiatives reflect the growing importance of innovation and non-price parameters of competition in the modern economy (e.g. as captured by the growth of high-tech sectors), and Europe's lack of competitiveness in highly innovative sectors.

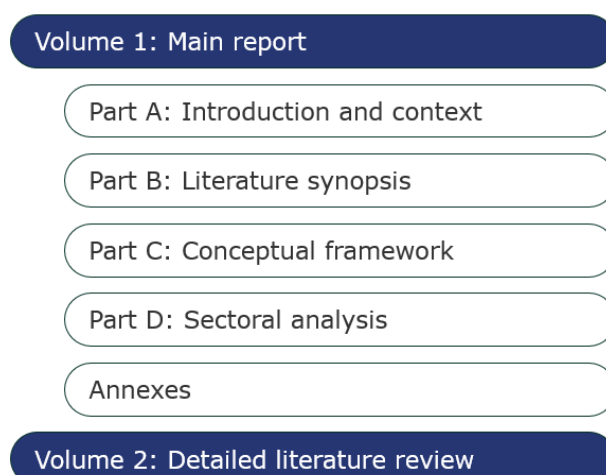
- 1.6 The objective of the Study is to provide the Commission with a solid analytical foundation to assess the dynamic effects of mergers, and any potential trade-offs with the static effects. This includes cases where a dynamic analysis amplifies static effects from a merger (including both negative and positive effects), cases where there is a trade-off between positive dynamic effects and adverse static effects, and cases with a trade-off between adverse dynamic effects and beneficial static effects. The Study covers the dynamic effects of both horizontal and non-horizontal mergers (i.e. mergers between producers of substitutes, and mergers between producers of complements, respectively), with a primary focus on horizontal effects, in line with the Tender Specifications.
- 1.7 The Study rests on a comprehensive review of the relevant theoretical and empirical literature. There is a substantial body of academic work which focuses on the impact of competition on innovation. Recent academic work also looks specifically at the effects of mergers on investment and on potential competition, often spurred by debates on competition policy and merger control. This literature spans industrial economics, financial economics, macroeconomics, and business. Taken in its entirety, the academic literature provides important insights into the effect of mergers, even though not all of its findings can be directly transposed to merger control. The Study relies on the literature to provide a rigorous and comprehensive conceptual framework for the evaluation of dynamic merger effects.
- 1.8 The Study intends to serve as input into the ongoing revision of the EU Horizontal Merger Guidelines and Non-Horizontal Merger Guidelines, which was launched by the Commission in May 2025. A draft of the new merger guidelines was published by the Commission on 30 April 2026, with final guidelines expected by the end of 2026.

1.2 Contents and structure of the Study

- 1.9 The Study is structured in two volumes. Volume 1 constitutes the main report of the Study, while Volume 2 provides a detailed paper-by-paper review of the relevant academic literature. Figure 1.1 provides an overview of the two volumes. Volume 1 contains four parts: Parts A, B, C, and D. Parts B, C, and D are each introduced in more detail in the main body of Volume 1 (in Chapters 3, 12, and 21 respectively).

⁵ European Commission (2024), 'Mission Letter to Teresa Ribera Rodríguez, Executive Vice-President for a Clean, Just and Competitive Transition', September.

Figure 1.1 Outline of the Study



Source: Oxera.

- 1.10 **Part A introduces the Study**, setting out its structure and scope, as well as providing an overview of the relevant policy and conceptual background.
- 1.11 **Part B contains the synopsis of the relevant academic literature.** It covers the literature on competition and innovation; horizontal mergers between established competitors (from a theoretical perspective); acquisitions of potential competitors (from a theoretical perspective); the empirical evidence on the impact of horizontal mergers on investment, innovation, and consumer welfare; non-horizontal mergers; financial market frictions and liquidity mergers; and competition policy. It also contains a brief review of relevant guidelines and recommended practices by major competition authorities and international organisations. Volume 2 contains a detailed review of the academic papers and other publications covered in the synopsis.
- 1.12 **Part C contains our conceptual framework for the evaluation of dynamic merger effects.** This covers both horizontal and non-horizontal effects (with a focus on the former). It also includes a review of canonical merger cases, with practical examples, building on the review of case studies contained in Annex A2. In line with the Tender Specifications, it represents the core part of the Study.
- 1.13 **Part D presents an evaluation of three strategic sectors: biopharmaceuticals, quantum computing, and clean technologies.** Its purpose is to provide a sector-specific assessment of the current challenges and bottlenecks to investment and innovation in the EU, and of how different policy tools including merger control could help address such challenges.
- 1.14 **Annexes provide supporting material for the analysis in the main report.** Annex A1 summarises the European Commission's four-layer framework for assessing innovation competition, Annex A2 presents detailed case studies of selected merger reviews involving dynamic effects, Annex A3 provides an overview of Commission merger cases in which dynamic effects were relevant, and Annexes A4 and A5 list, respectively, the cases and the merger guidelines and recommended practices cited throughout the Study.

2 Conceptual and policy background

- 2.1 This Chapter provides the policy and conceptual background to the Study. It covers several interrelated strands, which we review in turn.
- 2.2 The first is the long-standing debate in economics on the relationship between competition and innovation, which dates back to the contributions of Schumpeter (1942) and Arrow (1962), and which has generated a wide-ranging subsequent literature in industrial organisation, macroeconomics, and growth theory. This debate (concerning whether competitive pressure stimulates or dampens innovation incentives, and under what conditions) is prominent in policy discussion on merger control, and it remains relevant for the analysis of dynamic merger effects developed in the Study.
- 2.3 The second is the growing importance of innovation and technological change in the modern economy. In fast-moving, high-technology sectors, dynamic competition—rivalry over future products, services, and technologies—is a significant dimension of competitive interaction. This shift also has direct implications for the competitiveness of European firms and for the ability of European markets to generate and sustain innovation at the technological frontier.
- 2.4 The third is the recent evolution of merger control enforcement. Dynamic effects—including the impact of mergers on investment incentives, innovation competition, and competition by nascent firms—have become increasingly prominent in decisional practice, both in the European Union and in other major jurisdictions. They have also received growing attention in merger guidelines by competition authorities, reflecting a broader recognition that a purely static framework is insufficient to capture the competitive significance of mergers in innovation-intensive industries. While these developments are not yet reflected in the current European Commission merger guidelines, the Commission has signalled its intention to expand the treatment of dynamic effects in its revised guidelines, and the draft merger guidelines published by the Commission on 30 April 2026 contain an extensive discussion of dynamic effects.⁶
- 2.5 The fourth is the set of recent policy recommendations calling for greater prominence to be given to dynamic merger effects in EU merger control. Most notably, the Draghi Report identifies the assessment of innovation and future competition in merger review as a priority area for reform, calling for updated guidelines that explain how the Commission assesses the impact of competition on innovation incentives and that provide a clear framework for an ‘innovation defence’ (Draghi, 2024).

2.1 Relationship between competition and innovation: ‘Arrow vs Schumpeter’?

- 2.6 The debate on the relationship between competition, competition policy, and innovation is often characterised as a contrast between the perspectives associated with the contributions of Kenneth Arrow and Joseph Schumpeter.⁷ However, as discussed below, these perspectives are not necessarily inconsistent.

⁶ European Commission (2026), ‘Draft Communication: Guidelines on the assessment of mergers under Council Regulation (EC) No 139/2004 on the control of concentrations between undertakings’, 30 April.

⁷ For a comprehensive summary of this debate, see for example Gilbert (2020).

2.1.1 The Arrovian view that competition promotes innovation

- 2.7 Arrow's finding in his seminal article of 1962⁸ is that greater competition in the product market spurs innovation by limiting pre-innovation profits. This enhances the incentive to innovate in order to 'escape competition' and earn incremental profits. Arrow's result captures the notion that innovation 'cannibalises' (or replaces) existing profits, and that these profits therefore act as a disincentive to innovate for incumbents who enjoy a degree of market power. This finding also translates, at least in part, to the context of merger control, because when two firms merge they increase the level of overall existing profits at risk of cannibalisation if one of the two merging firms innovates.
- 2.8 Arrow's framework does not fully capture how a merger may affect competition between rival innovators. As discussed in more depth in Chapter 4 of this volume, Arrow's setting assumes that there is only one potential innovator, and hence there is no innovation competition between rivals. Arrow's model also considers the case of process innovation, while the impact of product market competition on innovation incentives differs in the case of product innovation.
- 2.9 Notwithstanding these caveats, Arrow's work is broadly associated with the notion that greater rivalry between firms tends to stimulate innovation, and that competition is beneficial not only because it leads to lower prices (promoting 'allocative' efficiency), but also because it incentivises firms to improve production processes and develop new and innovative products and solutions (hence enhancing 'productive' and 'dynamic' efficiency as well).
- 2.10 This view is generally echoed by competition authorities when setting out the benefits of competition policy. For example, in its Communication accompanying the revised framework for competition policy which came into force in May 2004 (including the revised, and still applicable, Merger Regulation), the Commission stated that 'vigorous competition in a supportive business environment is a key driver of productivity growth and competitiveness' and that 'competition also puts pressure on firms to innovate and to reorganise their business activities in order to continuously improve their cost structure and reap productivity gains'.⁹
- 2.11 Along similar lines, the revised 2023 US Merger Guidelines note that '[c]ompetition is a process of rivalry that incentivizes businesses to offer lower prices, improve wages and working conditions, enhance quality and resiliency, innovate, and expand choice, among many other benefits. Mergers that substantially lessen competition or tend to create a monopoly increase, extend, or entrench market power and deprive the public of these benefits'.¹⁰

2.1.2 The Schumpeterian view that competition may discourage innovation

- 2.12 The second school of thought on the relationship between competition and innovation is generally associated with the work of Schumpeter.
- 2.13 In *Capitalism, Socialism and Democracy* (1942), Schumpeter argued that successful innovation should be rewarded with a degree of temporary market power, and that a market where prices are driven down to cost would not generate the returns necessary to incentivise investment in innovation, and hence would lead to

⁸ Arrow (1962).

⁹ See European Commission (2004), 'A pro-active Competition Policy for a Competitive Europe', COM(2004) 293, April, p. 3.

¹⁰ See US DOJ & FTC (2023), 'Merger Guidelines', 18 December, p. 1. See Chapter 11, and Volume 2, Chapter 9 for a review of these guidelines.

inefficient outcomes over time.¹¹ This proposition provides a foundation for the system of intellectual property (IP) rights, which rewards successful innovators with at least a degree of temporary market power through patents. A related strand of Schumpeter's work suggests that larger firms may be better placed to innovate than smaller ones, owing to their greater access to resources (including financial resources).^{12,13}

- 2.14 Schumpeter's work is also associated with the concept of 'creative destruction', whereby new entrants innovate and render the products of existing incumbents obsolete in a process of continuous renewal. In this dynamic process, rivalry operates not only through price competition in existing markets but also through the displacement of incumbents by superior innovations. Schumpeter argued that this process of creative destruction (e.g. competition *for* the market) is a fundamental determinant of economic performance, and more important than static competition (e.g. competition *in* the market).¹⁴

2.1.3 Reconciling Arrow and Schumpeter

- 2.15 While Arrow's and Schumpeter's perspectives have often been characterised as opposed to each other, there are elements of consistency between the two.
- 2.16 As Shapiro (2012) argued, the propositions that (pre-invention) product market competition spurs innovation (Arrow), and that the prospect of market power and large scale also enhances innovation (Schumpeter) are compatible with each other.¹⁵ Shapiro reconciles them by identifying a 'contestability' principle (defined by Shapiro as 'the prospect of gaining or protecting profitable sales by providing greater value to customers spurs innovation') and an 'appropriability' principle (defined by Shapiro as 'increased appropriability spurs innovation'). The contestability principle is consistent with both Arrow's and Schumpeter's positions, while the appropriability principle is in line with Schumpeter's main thesis.
- 2.17 There are, however, different implications from the application of Arrow's and Schumpeter's perspectives, including in relation to the dynamic impact of mergers.

¹¹ 'A system—any system, economic or other—that at every given point of time fully utilizes its possibilities to the best advantage may yet in the long-run be inferior to a system that does so at no given point of time, because the latter's failure to do so may be a condition for the level or speed of long-run performance'; and 'In this respect, perfect competition is not only impossible but inferior, and has no title to being set up as a model of ideal efficiency' (Schumpeter, 1942, p. 83 and p. 92).

¹² 'As soon as we go into details and inquire into the individual items in which progress was most conspicuous, the trail leads not to the doors of those firms that work under conditions of comparatively free competition but precisely to the doors of the large concerns . . . and a shocking suspicion dawns upon us that big business may have had more to do with creating that standard of life than with keeping it down' (Schumpeter, 1942, p. 82).

¹³ Tirole (1988) sets out both of these elements of Schumpeter's work (p. 390), noting that the first rests on the notion that 'if one wants to induce firms to undertake R&D, one must accept the creation of monopolies as a necessary evil'. By contrast, Tirole argues that the second argument (that 'monopolies are natural breeding grounds for R&D') is controversial and not central to Schumpeter's thesis.

¹⁴ 'Economists are at long last emerging from the stage in which price competition was all they saw. [...] But in capitalist reality as distinguished from its textbook picture, it is not that kind of competition which counts but the competition from the new commodity, the new technology, the new source of supply, the new type of organization [...]—competition which commands a decisive cost or quality advantage and which strikes not at the margins of the profits and the outputs of the existing firms but at their foundations and their very lives. This kind of competition is as much more effective than the other as a bombardment is in comparison with forcing a door.' (Schumpeter, 1942, p. 74).

¹⁵ For a discussion of these policy contributions, see Chapter 10 and Volume 2, Chapter 8.

- 2.18 The impact of mergers on innovation incentives is determined by its effect on the pre- and post-innovation profits of the merging parties. This implies that the Arrowian and Schumpeterian effects are typically at work at the same time, given that Arrow focuses on pre-innovation competition, while Schumpeter focuses on post-innovation competition.
- 2.19 In particular, a merger may increase the appropriability of the benefits of innovation, by reducing product market competition on innovative products or by reducing the risk of imitation by rivals. This effect can *boost* innovation incentives. At the same time, successful innovation by each of the merging firms cannibalises the profits of the other merging partner (a mechanism that is related to the replacement of pre-innovation profits emphasised by Arrow). This channel *softens* incentives to innovate. These contrasting forces introduce a trade-off between the effects of a merger on innovation incentives. Whether the appropriability effect is sufficient to outweigh the cannibalisation effects, and whether the net effect on innovation incentives can in turn benefit consumers, is one of the central questions for the assessment of dynamic merger effects. We analyse this interaction in depth in Part B and Part C of the main report.
- 2.20 Schumpeter also emphasised that the *ability* to innovate may be fostered by firms with greater resources. It is therefore possible that mergers may promote innovation, by enhancing firms' capability to invest, despite leading to stronger cannibalisation effects.
- 2.21 Subsequent academic work has sought to combine the perspectives of Arrow and Schumpeter. Aghion and Howitt, in a seminal paper published in 1992 and in subsequent work by them and their co-authors, develop a model of endogenous growth in which creative destruction from innovative entrants is an important driver of innovation and growth.¹⁶ At the same time, more intense product market competition may dampen those incentives. Aghion and Howitt distinguish between two cases to study the impact of product market competition. In the first, firms innovate in order to pull ahead of their rivals, and thereby escape product market competition. In this setting greater competitive intensity acts as a spur to innovate (as in Arrow's model). In the second setting, firms innovate principally to catch up with more advanced rivals, and in this case greater product market competition between firms of similar technological standing acts as a disincentive to invest, since the returns to catching up are eroded by the intensity of rivalry (this can be seen as a Schumpeterian effect). The distinction between these two settings generates the influential inverted-U relationship between product market competition and innovation (Aghion, Bloom, Blundell, Griffith, and Howitt, 2005), whereby competitive pressure stimulates innovation at low levels of competition but dampens it in a more competitive market.
- 2.22 As we discuss in Chapter 4, one needs to be cautious when applying the Aghion–Howitt framework to merger assessment, as this approach does not consider the impact of mergers explicitly. A merger is a specific form of change in competitive conditions, which entails two competing firms coming together to coordinate their investment and pricing decisions. Its impact is not well proxied by a market-wide variation in the intensity of product market competition, or even by an exogenous change in the number of competing products and firms.

¹⁶ See Aghion and Howitt (1992) and subsequent work building on this framework (e.g. Aghion, Harris, Howitt, and Vickers, 2001; Aghion, Bloom, Blundell, Griffith, and Howitt, 2005). Aghion and Howitt were awarded the Nobel Prize in Economics in 2025 for their contribution. For a detailed review, see Royal Swedish Academy of Sciences (2025) and Aghion (2026).

- 2.23 The academic literature has built further on the findings of Schumpeter, Arrow, Aghion, and Howitt, providing an extensive characterisation of the impact of mergers on investment and product market competition. We take stock of the insights of this literature in Part B of the main report.

2.2 Growing importance of innovation competition in the modern economy

- 2.24 Innovation and technological change are the primary drivers of sustained long-run economic growth. This was established in the seminal work of Solow, which demonstrated that economic growth is not primarily driven by the accumulation of physical or human capital, but by technological progress.¹⁷ Endogenous growth theories (e.g. as developed in the work of Romer, Aghion, and Howitt) have emphasised the role of competitive dynamics and market power in explaining incentives to innovate.¹⁸
- 2.25 The practical significance of these insights has grown considerably in recent decades, as private R&D has become an increasingly central feature of modern economies. In the US, private sector R&D accounts for around 70% of total R&D expenditure, and R&D intensity is particularly high in sectors such as pharmaceuticals and computing.¹⁹ The rise of large technology platforms reflects the growing economic weight of sectors characterised by fast innovation cycles, disruptive business models (e.g. in advertising, content distribution, and publishing), strong network effects, and high barriers to entry. Moreover, acquisitions have increasingly displaced initial public offerings (IPOs) as the dominant exit route for venture capital, and the principal channel through which early-stage innovation is commercialised²⁰—a shift reflected in the volume of acquisitions by the major technology firms.²¹
- 2.26 Europe’s position in this evolving landscape has been a source of concern. Business expenditure on R&D in the EU stands at approximately 1.2% of GDP—roughly half the US figure of 2.3% of GDP.²² European business R&D is concentrated in mid-technology sectors, such as the automotive industry, which compete by applying existing technological advances to production but do not require the same R&D intensity or offer the same growth potential as high-technology industries that produce the newest technologies. By contrast, the EU industry invests less than its peers in R&D in software, hardware, and artificial intelligence (AI).²³
- 2.27 The consequences are visible in aggregate productivity data. As the Draghi Report documents, the gap in GDP between the EU and the US has widened over the past 20 years, driven mainly by a more pronounced slowdown in productivity growth in Europe. Europe’s failure to capitalise on the digital revolution led by the internet—both in generating new technology companies and in diffusing digital technology across the economy—accounts for the bulk of the productivity gap. The Draghi Report also finds that the EU is lagging in the breakthrough digital technologies that are likely to drive future growth, with the US accounting for the bulk of foundational AI models and the global cloud market.

¹⁷ Solow (1956).

¹⁸ Romer (1990) and Aghion and Howitt (1992).

¹⁹ Gilbert (2020), Chapter 2.

²⁰ See for example Ederer and Pellegrino (2023).

²¹ For a summary of recent trends, see Fumagalli, Jullien, Lefouili, and Polo (2026).

²² Fuest, Gros, Mengel, Presidente, and Tirole (2024).

²³ Fuest et al. (2024).

- 2.28 These trends have implications for the current debate on merger control. In high-technology sectors characterised by rapid innovation cycles, strong network effects, and acquisition of early-stage innovators, the dynamic effects of mergers—including their impact on investment incentives, innovation competition, and the prospects of nascent competitors—are often more significant for consumers and for long-run welfare than static price effects.
- 2.29 Moreover, in a context where innovation is an increasingly important parameter of competition and productivity in the EU is lagging behind compared to the US, ensuring that Commission merger control fosters innovation in Europe becomes even more critical for the competitiveness of the European economy.

2.3 Trends in the treatment of dynamic effects in merger control

- 2.30 In part in response to the industry trends reviewed in the previous Section, over the past two to three decades competition authorities—including the European Commission, the UK Competition and Markets Authority (CMA), and the US Department of Justice (DOJ) and Federal Trade Commission (FTC)—have given increasing prominence to dynamic merger effects in their reviews of mergers.
- 2.31 This shows that competition authorities have generally been responsive to calls for greater attention to the dynamic effects of mergers. At the same time, the majority of merger control decisions still focus primarily on static effects on price. Moreover, where dynamic effects have been extensively discussed by the Commission, this has generally been in support of theories of harm, rather than theories of benefits.

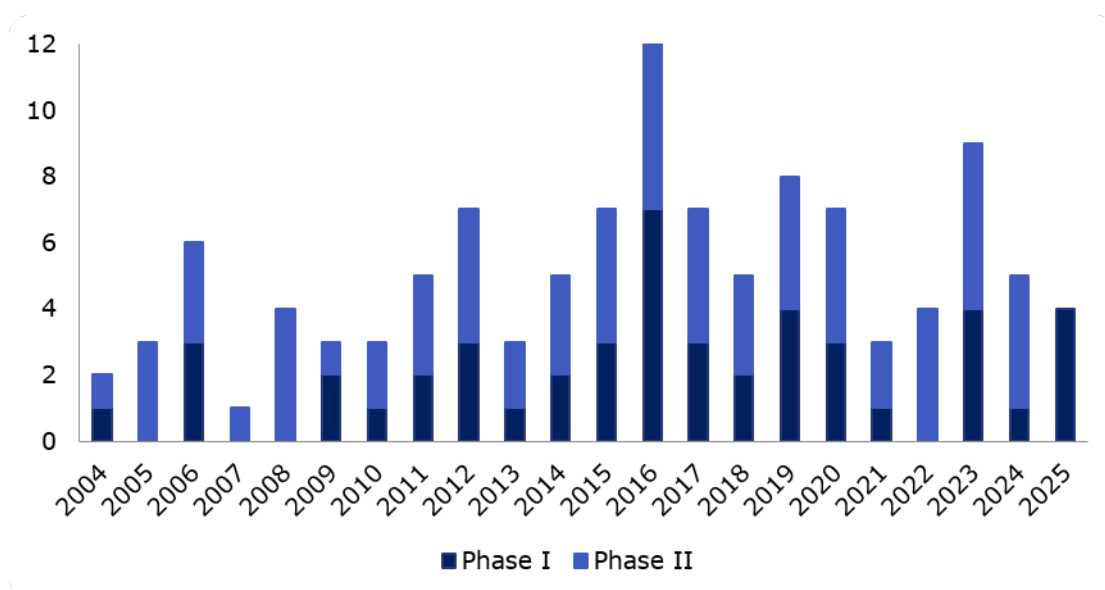
2.3.1 Merger control trends at the European Commission

- 2.32 While the Commission's current Horizontal Merger Guidelines devote relatively limited attention to dynamic merger effects (see discussion in Chapter 11), its recent decisional practice has increasingly focused on the impact of mergers on investment, innovation, and potential competition.²⁴
- 2.33 In their review of theories of harm in Commission merger decisions between 1990 and 2023, Duso, Bernhardt, and Piechucka (2024) find that theories of harm relating to innovation concerns have become more frequent over the period, both in non-simplified Phase I decisions and in Phase II decisions. Before 2004, innovation issues were discussed in around 10–15% of decisions and in 70–80% of Phase II decisions, while after 2004 this reached around 40% of all decisions. By the end of the 2000s, innovation concerns were mentioned in virtually all Phase II decisions. The authors also show that innovation concerns have become more structured over time, with a focus on the concept of innovation competition and pipeline products.

²⁴ Moreover, the revised Notice on Market Definition (European Commission (2024), 'Commission Notice on the relevant market for the purposes of Union competition law', C/2024/1645, 22 February) includes an extensive discussion of dynamic elements of market definition, including a discussion of 'innovation spaces' and R&D (see also Annex A1 on the Commission's 'four-layer' approach). The Horizontal Cooperation Agreement Guidelines (European Commission (2023), 'Guidelines on the applicability of Article 101 of the Treaty on the Functioning of the European Union to horizontal co-operation agreements', C/2023/4572, July) also contain a detailed discussion of potential innovation effects (see also Chapter 11). This was the case already in the 2011 version of these guidelines. The Commission has also published a number of Competition Policy Briefs and Merger Briefs on dynamic effects in merger control (e.g. European Commission (2016), 'EU Merger control and innovation', Competition Policy Brief, 2016-01, October; and European Commission (2024), 'Non-Price Competition: EU Merger Control Framework and Case Practice', Competition Policy Brief, 2024-01, April).

- 2.34 Using a large language model (LLM), our own analysis of the Commission’s decisions since 2004 (i.e. under the current European Union Merger Regulation (EUMR)) and until 2025 broadly shows a similar upward trend.²⁵ Considering both harm and efficiencies, we identified 47 Phase I decisions, and 66 Phase II decisions (including withdrawn cases) in which dynamic effects were an important element of the Commission’s assessment. This represents around 6.1% of all non-simplified procedures since 2004 and around 48% of all Phase II cases. A full list of the identified cases is provided in Annex A3.
- 2.35 Figure 2.1 below illustrates the evolution of cases in which the Commission assessed dynamic effects in its review over time, and shows that there was a broad upward trend in the number of relevant cases between 2004 and 2020, with fewer cases since 2021, except for a spike in 2023.

Figure 2.1 Number of EU merger cases where dynamic effects were relevant



Note: This figure shows the evolution in the number of European Commission merger cases in which dynamic effects were an important element from 2004 to 2025, differentiating between Phase I and Phase II cases. Phase II cases include cases withdrawn during the course of Phase II.

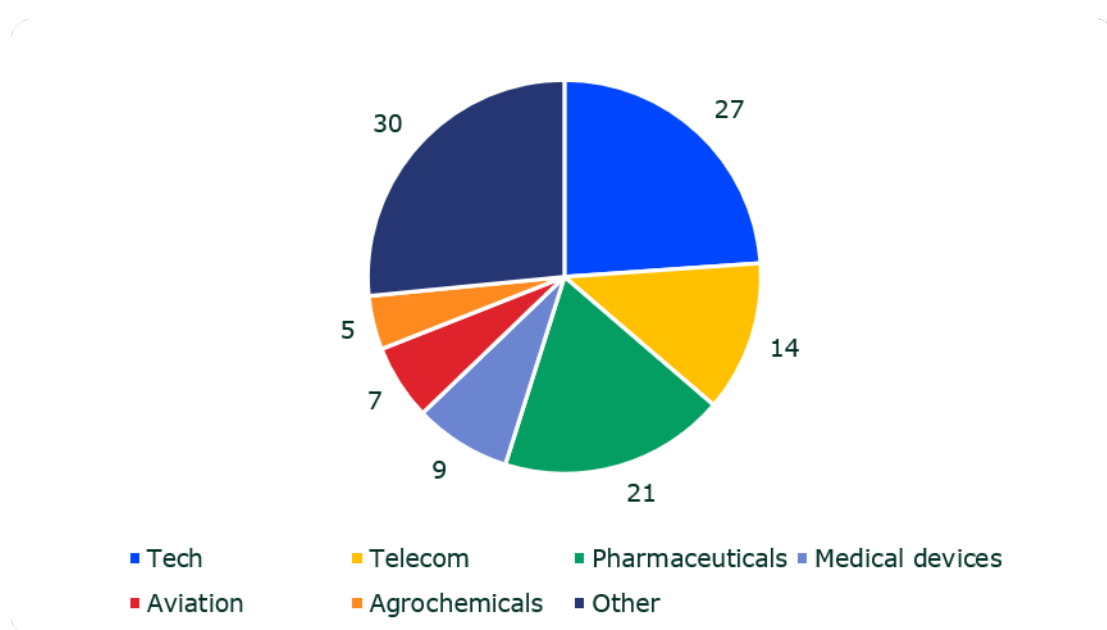
Source: Oxera.

- 2.36 Figure 2.2 below gives a breakdown of dynamic effects cases by sector. As can be seen, a significant proportion of relevant cases is concentrated in pharmaceuticals, medical devices, high-tech markets (computing and digital markets), agrochemicals, aviation, and telecommunications.²⁶ Other relevant sectors aggregated under 'Other' include a wide range of other industries (see the list provided in Annex A3).

²⁵ Relevant decisions were identified using an LLM-based classification process, complemented by manual review. A detailed description of the methodology is provided in Annex A3.

²⁶ In this classification pharmaceuticals cases include pharmaceutical products and preparations. Tech covers activities such as programming, computer manufacturing, and software publishing. Agrochemicals, aviation, medical devices, and telecommunications follow the corresponding European Commission Statistical Classification of Economic Activities in the European Community (EC NACE) categories. 'Others' cover all the remainder.

Figure 2.2 Dynamic effects cases by sector



Note: This figure illustrates the distribution of sectors across 113 innovation-related cases from 2004 to 2025, differentiating between tech, telecommunications, pharmaceuticals, medical devices, aviation, agrochemicals, and others.

Source: Oxera.

- 2.37 Notable, high-profile cases where the Commission has looked at innovation and investment effects in depth cover a broad range of industries including for example:
- Ryanair/Aer Lingus I and III (2007, 2013)—airlines;
 - Novartis/GlaxoSmithKline Oncology Business (2015)—pharmaceuticals;
 - General Electric/Alstom (Thermal Power – Renewable Power & Grid Business) (2015)—gas turbines;
 - Dow/DuPont (2017) and Bayer/Monsanto (2018)—agrochemicals; and
 - several cases in national mobile telecommunication markets, mostly between 2015 and 2024.²⁷
- 2.38 A prominent example of a merger decision with a significant focus on innovation concerns is the Dow/DuPont decision of 2017, where the Commission developed a structured ‘four-layer’ approach to the assessment of innovation competition (see Annex A1). Dynamic concerns were also part of merger prohibitions upheld by the European Courts.²⁸
- 2.39 In recent years, the Commission has also increased its attention to acquisitions by large digital platforms. In 2019, it commissioned a study on digital markets (Crémer, de Montjoye, and Schweitzer, 2019) which advocated for stricter merger control in digital markets, in particular for the acquisition of nascent rivals.²⁹ Recent cases involving digital platforms or large technology firms include for example

²⁷ Including notably mergers in the Irish, German, Danish, UK, Italian, Spanish, and Dutch national markets. An overview of all cases is provided in Volume 2, Table 5.1.

²⁸ Notably ENI/EDP/GDP (2004); Judgment of the Court of First Instance (2005), EDP v Commission, T-87/05, September; and Deutsche Börse/NYSE Euronext (2012); Judgment of the General Court (2015), Deutsche Börse v Commission, T-175/12, March. Several of these cases mentioned above are reviewed as case examples or case studies in Part C of the Study.

²⁹ See discussion in Chapter 10 and Volume 2, Chapter 8.

Google/Fitbit (2020), Meta/Kustomer (2022), Booking Holdings/eTraveli Group (2023), Microsoft/Activision Blizzard (2023), and Amazon/iRobot (2024).

2.3.2 Merger control trends in the US and in the UK

- 2.40 Dynamic merger effects are discussed and analysed in significant decisions and in merger guidelines also in the US and in the UK.
- 2.41 Since the mid-1990s, there has been a long-standing debate in the US on how best to reflect innovation concerns in competition policy (including merger control). In 1995 the DOJ and FTC published the IP Licensing Guidelines which provided a structured approach to innovation concerns, based on the concept of 'innovation markets'.³⁰ As Gilbert and Tom (2001) and Gilbert and Greene (2015) document, merger challenges involving innovation concerns have increased since the mid-1990s (with only 3% of merger challenges alleging innovation effects during the 1990–'94 period, 17.5% of cases during 1995–'99, and 34% during the 2004–'14 period). Prominent cases where innovation concerns were raised by US authorities include General Motors/ZF (1993), Genzyme/Novazyme (2004), Thoratec/Heartware (2009), Applied Materials/Tokyo Electron (2015), Halliburton/Baker Hughes (2016), and Bayer/Monsanto (2018).³¹
- 2.42 The 2010 US Guidelines on Horizontal Mergers also included a relatively detailed section on the impact of mergers on innovation and product variety, which was retained in the 2023 revision, and complemented by specific treatment of potential competition concerns and entrenchment.³² A discussion of this is provided in Chapter 11.
- 2.43 The UK CMA has also increasingly focused on dynamic effects in its recent practice, both in terms of their guidelines (revised in 2021—see Chapter 11) and merger control decisions.³³ This includes specific attention to acquisitions by digital platforms, following the publication of a report for the UK government³⁴ and an economic study for the CMA on the impact of past digital mergers.³⁵

2.4 Recent policy initiatives and recommendations

- 2.44 Partially in response to concerns about Europe's lack of competitiveness in the global race for technological leadership, a number of policy initiatives have re-considered the role of merger control as one of the policy tools that can foster innovation and growth in Europe. These initiatives have provided the political impetus for the European Commission's ongoing review of its Horizontal and Non-Horizontal Merger Guidelines.

³⁰ At approximately the same time, in 1996, the FTC published an extensive report on competition policy in high technology markets (including the role of merger control), reflecting a series of public hearings (FTC (1996), 'Anticipating the 21st Century: Competition Policy in the New High-Tech, Global Marketplace', May).

³¹ For a review of some of these cases see Gilbert (2020) and Federico, Scott Morton, and Shapiro (2020). Some of the cases are discussed in Chapter 20 and Annex A2.

³² US DOJ & FTC (2010), 'Horizontal Merger Guidelines', 19 August; US DOJ & FTC (2023), 'Merger Guidelines', 18 December.

³³ Prominent UK merger control decisions on dynamic effects include Facebook/GIPHY (2021) and Adobe/Figma (2023) (see discussion in Chapter 20 and Annex A2).

³⁴ Coyle, Fletcher, Furman, Marsden, and McAuley (2019). See discussion in Volume 2, Chapter 8.

³⁵ Lear (2019). See discussion in Volume 2, Chapter 8.

2.4.1 The Draghi Report

- 2.45 The 2024 Draghi Report provides a comprehensive assessment of Europe's competitiveness challenges (Draghi, 2024). As discussed in Section 2.2, one of its main findings is that the EU faces a deepening productivity gap relative to the US, driven by structural weaknesses including high energy costs, fragmented capital markets, underinvestment in research and innovation, and a general inability by firms across industries to scale up.
- 2.46 The Draghi Report contains a set of 'horizontal' policy recommendations intended to address these structural challenges. It devotes a chapter to 'revamping competition' as one of these recommendations.³⁶ This chapter notes that stronger competition lowers prices and tends to foster productivity, investment, and innovation.³⁷ The chapter also notes that there are cases where competition can be harmful to innovation by eroding the profits from innovation (echoing Schumpeter's arguments).³⁸ Moreover, given the changing business landscape (with a shift to high-tech industries where innovation and scale matter more than price), competition policy needs to be more agile and forward-looking.
- 2.47 On merger control, the Draghi Report notes that merger control decisions in the last decade have started to consider more than just price impacts on consumers and to evaluate other dimensions, such as quality and innovation.³⁹ However, the Report recommends that the European Commission should place greater weight on innovation and future competition in its decisions, moving beyond the backward-looking assessment of existing markets which sometimes characterises merger reviews.
- 2.48 The Draghi Report also calls for updated guidelines clarifying how the Commission evaluates the impact of competition on the incentive to innovate.⁴⁰ Additionally, these guidelines should set out the conditions for an 'innovation defence', allowing merging parties to demonstrate that a merger enhances innovation—for instance, by enabling the pooling of resources to cover large fixed costs and compete globally. To prevent abuse, parties invoking this defence should commit to investment levels that can be monitored, with disincentives for non-compliance, and the defence should not be available to already-dominant firms or where concentration risks entrenching dominance.
- 2.49 Finally, the Draghi Report notes that while benefits from greater scale may boost short-term innovation, they must be weighed against longer-term reductions in competitive incentives.⁴¹ Since foreign competition constrains behaviour in tradable sectors, competition enforcement needs to be especially vigilant against the risks of concentration in non-tradable sectors.

³⁶ Draghi (2024), Part B, Section 2, Chapter 4.

³⁷ The Draghi Report notes that 'A summary of the current empirical evidence overwhelmingly shows that stronger competition generally not only delivers lower prices, but also tends to stimulate greater productivity, investment, and innovation', by reference to an extensive report by the European Commission titled 'Protecting competition in a changing world' (European Commission, 2024). See Draghi (2024), Part B, p. 298. This is further discussed in Volume 2, Chapter 8.

³⁸ The Draghi Report puts it as follows: 'Although stronger competition will in theory generally both lower prices and foster innovation, there are cases where it can be harmful to innovation. Schumpeter worried that tough competition would erode the profit rents from innovation and thus disincentivise R&D.', Draghi (2024), Part B, p. 298.

³⁹ Draghi (2024), Part B, Section 2, Chapter 4.

⁴⁰ Draghi (2024), Part B, Section 2, Chapter 4.

⁴¹ Draghi (2024), Part B, Section 2, Chapter 4.

- 2.50 The Draghi Report specifically discusses the role of competition policy and regulation in the telecommunication sector.⁴² It notes that EU telecommunication operators face fragmented national markets, excessive ex ante regulation, and spectrum licensing rules that constrain scale. It also contends that the current regulatory and competition framework has deterred the cross-border consolidation needed to finance next-generation network investment, including 5G and 6G infrastructures. In the context of merger review, the Draghi Report recommends defining telecommunication markets at EU rather than Member State level to facilitate cross-border integration, and giving greater weight to innovation and investment commitments in clearance decisions (rather than focusing remedies on partial de-consolidation or asset divestitures).

2.4.2 The Mission Letter to Executive Vice-President Ribera

- 2.51 The current College of Commissioners put forward by President von der Leyen in September 2024 took office in December 2024. As per standard practice, in September 2024 President von der Leyen published her Mission Letter to Teresa Ribera, the Executive Vice-President for a Clean, Just and Competitive Transition.⁴³ The Mission Letter tasks Ribera with modernising EU competition policy to better support companies in innovating, scaling up, and competing globally, while serving broader objectives including decarbonisation, a just transition, and resilience.
- 2.52 On merger control specifically, the Letter calls for a review of the Horizontal Merger Control Guidelines. This review is to give ‘adequate weight’ to the EU economy’s needs in respect of resilience, efficiency, and innovation, the time horizons and investment intensity of competition in certain strategic sectors, and the changed defence and security environment. The Letter also highlights the risk of ‘killer acquisitions’ by foreign companies targeting European small and medium-sized enterprises (SMEs) and small midcaps in the EU as potential future competitors.
- 2.53 The Mission Letter therefore directly addresses the recommendation in the Draghi Report that more attention be given to innovation and dynamic effects more broadly in merger control, reinforcing the trend seen in recent years.

2.4.3 The Competitiveness Compass

- 2.54 The Competitiveness Compass, a Communication published by the Commission in January 2025 as its direct strategic response to the Draghi Report, identifies closing the innovation gap as one of Europe’s three transformational imperatives.⁴⁴ It frames Europe’s core problem as a failure to translate research into marketable technologies and to allow innovative firms to scale—a situation partly driven by market fragmentation, capital constraints, and insufficient innovation support.
- 2.55 The Compass notes that competition policy is also an important lever to strengthen Europe’s competitiveness, and that rigorous and effective antitrust and merger enforcement protects fair competition and incentivises companies to innovate and become more efficient.⁴⁵ At the same time, the Compass follows the recommendation of the Draghi Report and the direction set out in the Mission Letter, indicating that competition policy must be ‘more supportive of companies scaling up in global markets’ and ‘better geared to common goals, including

⁴² Draghi (2024), Part A, p. 35 and Part B, Section 2, Chapter 3.1.

⁴³ European Commission (2024) (full reference in footnote 5).

⁴⁴ European Commission (2025), ‘A competitiveness compass for the EU’, COM(2025) 30 final, 29 January.

⁴⁵ European Commission (2025), p. 7 (full reference in footnote 44).

decarbonisation'. Finally, the Compass calls for revised merger guidelines to ensure that innovation, resilience, and the investment intensity of competition in certain strategic sectors are given adequate weight in light of the European economy's needs.⁴⁶

2.4.4 Review of European Commission merger guidelines

- 2.56 In May 2025 the Commission launched a review of its Horizontal Merger Guidelines and the Non-Horizontal Merger Guidelines. The review aims at modernising the framework within which the Commission assesses mergers, to reflect the changed geopolitical and trade context, and the fact that industrial scale, global competitiveness, innovation, and investment have become increasingly important, and sustainability and resilience have become relevant parameters of competition.
- 2.57 In launching the review, the Commission published a series of background papers, including one on innovation and dynamic effects.⁴⁷ In this paper, it explains that firms compete not only through short-term pricing decisions but also through longer-term investments in innovation, productive capacity, and cost-reducing technologies, and through their impact on potential competition. The paper also recognises that mergers may enhance the merged entity's capacity to innovate, but equally may harm innovation competition and thereby reduce firms' incentives to invest in R&D. The paper also notes that the effects of mergers on innovation are often more difficult to predict than those on prices, and the challenge for the revised Guidelines is to develop an administrable framework for assessing dynamic merger effects.
- 2.58 The call for tender for this Study was published by the Commission in March 2025, shortly before the launch of the review of the guidelines. As set out by the Commission, the Study aims to provide analytical foundations to assess whether a merger has a positive or negative impact on dynamic effects (including on firms' ability and incentive to invest and innovate, as well as to enter or exit a market in the mid-to-long term), and how dynamic effects trade off against static factors—such as changes in prices or output.
- 2.59 As set out in the Tender Specifications,⁴⁸ the Study is requested to cover four elements:
- an inter-disciplinary synopsis of the relevant academic literature;
 - a conceptual framework for the identification of the circumstances when mergers may be likely to generate dynamic benefits or dynamic harm. This is the core part of the Study;
 - an applied part that illustrates, with case examples, how the conceptual framework can be practically applied in the assessment of dynamic merger effects; and
 - a sectoral evaluation part, examining how dynamic merger effects may apply in specific strategic sectors, within a broader context of the determinants of investment and innovation in these sectors.
- 2.60 The Commission published a draft of its revised merger guidelines on 30 April 2026.⁴⁹ This draft devotes substantial attention to the assessment of dynamic

⁴⁶ European Commission (2025), p. 7 (full reference in footnote 44).

⁴⁷ European Commission (2025), '[Topic C: Innovation and other dynamic elements in merger control](#)', accessed 21 July 2026.

⁴⁸ European Commission (2025), '[Study on dynamic merger effects](#)', accessed 21 July 2026.

⁴⁹ European Commission (2026) (full reference in footnote 6).

effects in the context of both horizontal and non-horizontal mergers, and for both theories of harm and theories of benefit. At the time of publication of this Study, the final guidelines are expected to be published by the end of 2026.

PART B SYNOPSIS OF THE ACADEMIC LITERATURE

3 Approach and structure of the academic literature synopsis

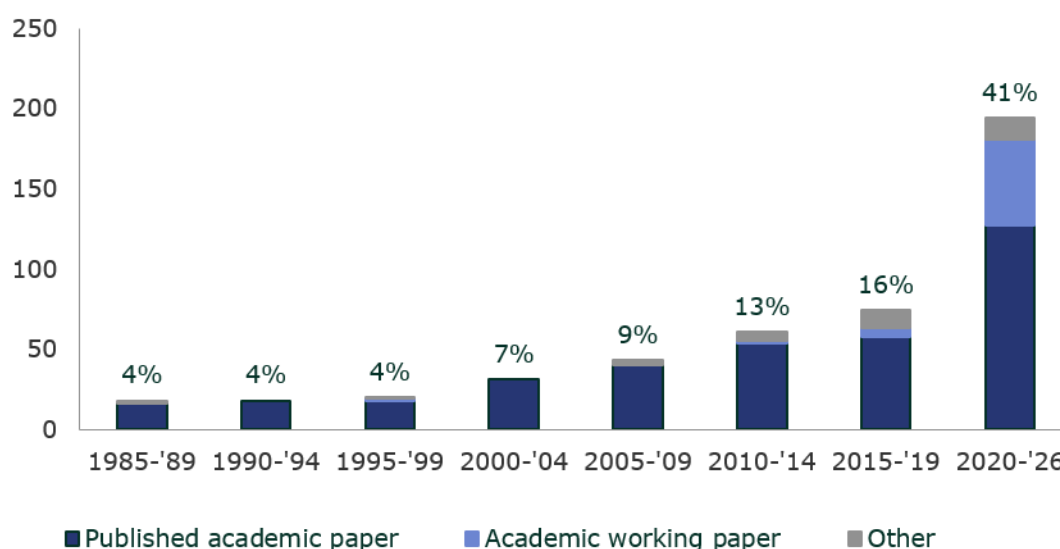
- 3.1 Part B of the main report provides a synopsis of the academic literature on the dynamic effects of mergers. The purpose of this literature synopsis is to take stock of the insights from the literature and provide a rigorous and comprehensive analytical foundation for the conceptual framework developed in Part C of the main report. We review both the older and well-established literature on competition and innovation, and the more recent work looking specifically at mergers—including work prompted by competition policy debates and by past merger decisions.
- 3.2 In this introductory Chapter, we set out the general approach and structure of the synopsis. Each chapter of the literature synopsis below provides our summary insights into the literature. A more detailed paper-by-paper review is provided in Volume 2. Volume 2, Chapter 1 sets out the methodology and the scope of the papers reviewed.

3.1 General approach and scope

- 3.3 The literature synopsis aims to be both interdisciplinary and comprehensive, including results and insights from the broader economics literature, as well as business and management studies. To make the most of the interdisciplinary nature of our review, the synopsis is generally organised by theme rather than academic field. In terms of academic fields covered, this includes theoretical and empirical industrial organisation and microeconomics, applied econometrics, macroeconomics, and growth, corporate finance, business and management studies, and competition policy.
- 3.4 For business and management studies, we focus on papers directly related to merger control. This includes in particular (empirical) papers from top business and management journals. For competition policy papers, we focus on those directly linked to the economic literature (e.g. we generally do not include legal contributions).
- 3.5 Our review focuses on peer-reviewed academic publications, as well as relevant working papers reporting ongoing and recent academic research. Additionally, we include several policy reports, book chapters, and other non-academic publications that relate directly to the academic papers reviewed.
- 3.6 The final set of papers included in the synopsis consists of 475 publications and working papers: 375 (79%) published, peer-reviewed academic papers; 61 (13%) academic working papers; and the remaining 39 (8%) policy reports, books, book chapters, and other non-academic publications.⁵⁰ Figure 3.1 provides the distribution of the literature by date of publication, showing that around 41% of all publications and working papers reviewed date from 2020 or later. This shows that a large part of the literature that we consider relevant consists of fairly recent publications and working papers.

⁵⁰ Although we do not make a distinction here between different types of academic working papers, we note that some working paper series included—e.g. NBER and CEPR—are generally regarded to be of higher quality than average.

Figure 3.1 Distribution of the literature over time by publication type



Note: This figure shows the distribution of literature since 1985 by publication type. 'Other' includes in particular books (including textbooks) and policy papers and reports.
Source: Oxera.

- 3.7 Volume 2, Annex A2 provides an overview of relevant disclosures of the papers reviewed. These disclosures generally include whether the article or study was (i) commissioned or funded by an interested party (including private or public organisations) and/or (ii) authored by someone directly affiliated with an interested or involved party at the time of publication.
- 3.8 Our literature synopsis builds on several existing surveys of the economic literature, including Gilbert (2020), Haucap and Stiebale (2026), and Lefouili and Madio (2026).

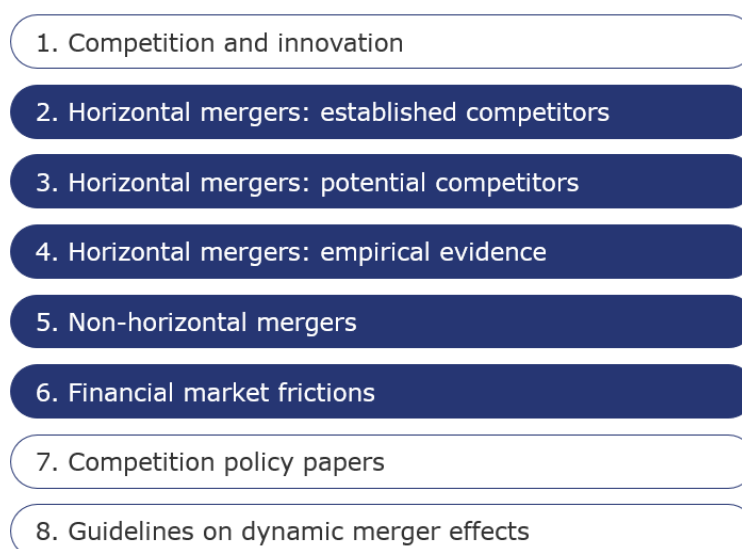
3.2 Focus of our synopsis

- 3.9 We focus our synopsis of the literature on those findings that are most relevant to the assessment of the dynamic effects of mergers. We adopt a broad definition of dynamic effects, which includes both effects on *investment* and on *prices of future products*. This is in line with the definition of dynamic effects that we introduce and explain in our conceptual framework, in Part C of the main report. We also adopt a broad definition of the notion of investment, including investment in R&D and innovation (both product and process innovation), in physical infrastructure, and investments to enter new markets.
- 3.10 We also pay specific attention to the implications of the literature for the impact of mergers on consumer welfare (where this is analysed). Consumer welfare is a well-established criterion for evaluating an impediment to effective competition, and is the substantive standard that we adopt in our conceptual framework (see discussion in Part C).
- 3.11 Where possible, we focus explicitly on the dynamic impact of mergers as opposed to the impact of other forms of change in competition (e.g. variations in the level of product differentiation, or changes in regulation). Depending on the paper, this is not always possible, especially for the literature reviewed in Chapter 4 that focuses on the broader relationship between competition and innovation.

3.3 Structure of the synopsis

- 3.12 The literature synopsis is divided into eight Chapters, each covering a different general theme (see Figure 3.2). Chapter 4 considers the literature on the general relationship between competition and innovation, providing relevant context for the rest of the synopsis. Chapters 5 to 9 cover the literature specifically on mergers and dynamic effects—looking in turn at the theoretical literature on horizontal mergers between incumbents and acquisitions of potential competitors, the empirical literature on horizontal mergers and acquisitions (M&As), non-horizontal mergers, and financial market frictions. These Chapters comprise the core material of the synopsis. Chapter 10 covers additional competition policy papers and Chapter 11 relevant insights from existing guidelines and recommendations.

Figure 3.2 Outline and structure of the literature synopsis



Note: The literature synopsis consists of five core chapters, on horizontal mergers, non-horizontal mergers, and financial market frictions. The other chapters provide further relevant context and considerations.

Source: Oxera.

- 3.13 Below, we provide a more detailed summary of what is covered in each Chapter in the literature synopsis.
- 3.14 **Chapter 4, competition and innovation**, covers the theoretical and empirical literature on the relationship between competition and innovation—drawing in particular from the literature on industrial organisation, macroeconomics, and growth. It is important to note at the outset that the literature on competition and innovation cannot be directly applied to the context of mergers, principally because it does not focus on the impact of a merger on innovation and on consumers. However, some of the underlying mechanisms studied in this literature apply to mergers. Moreover, findings from this literature have been influential in the policy debates on the relationship between competition policy and innovation, and it is therefore relevant to review these papers and frame them correctly for the purposes of the evaluation of dynamic merger effects.
- 3.15 **Chapter 5, horizontal mergers—mergers between established competitors**, covers the theoretical literature on how horizontal mergers between established competitors (i.e. existing firms offering substitute products) affect investment and innovation. This is a relatively recent literature, spurred and motivated by debates

in merger control, including high-profile merger control decisions by the European Commission over the past decade or so.

- 3.16 **Chapter 6, horizontal mergers—acquisitions of potential competitors,** covers the theoretical literature on acquisitions by incumbents of potential competitors. In contrast to Chapter 5, this Chapter focuses on acquisitions by (large) incumbents of significantly smaller firms. This part of the literature also considers issues around (i) entry- or innovation-for-buyout (the possibility that new firms enter or innovate because of the prospect of being acquired by an incumbent, potentially to the benefit of consumers) and (ii) potential adverse dynamic effects on third parties through ‘entrenchment’ of incumbents with significant market power (which we define as a mechanism raising barriers to entry and expansion by competitors to the merging firms, to the ultimate detriment of consumers). Much of this literature is also relatively recent, and has been motivated in part by policy debates on acquisitions by large digital platforms of small innovative companies (often referred to in the literature as ‘startups’).
- 3.17 **Chapter 7, horizontal mergers—empirical evidence,** reviews the available empirical evidence on the relationship between horizontal M&As and innovation. Particular attention is paid to sector-specific studies—with most of the papers reviewed relating to pharmaceuticals, mobile telecommunications, and digital markets. However, we also consider cross-sectoral studies from business studies, industrial organisation, and corporate finance.
- 3.18 **Chapter 8, non-horizontal mergers,** covers the literature on non-horizontal mergers and innovation or investment. It recalls some of the seminal economics literature on the boundaries of the firm, including motivations for non-horizontal integration as a means to overcome contracting frictions and informational asymmetries. We then cover the literature that explicitly considers how vertical and conglomerate mergers can affect investment, innovation, entry, or expansion by competitors—sometimes to the detriment of consumers despite procompetitive effects on the conduct of the merging parties. We also review the general literature on non-horizontal foreclosure by firms that are already integrated, and consider the relevant empirical literature.
- 3.19 **Chapter 9, financial market frictions,** covers the corporate finance literature related to financial market frictions and how M&As may alleviate or amplify such frictions. We outline which types of financial market friction may exist, as a result of informational asymmetries. We then discuss how both non-diversifying and diversifying mergers (defined as M&As occurring or not occurring within the same industry, respectively) may affect these frictions, in some cases leading to increased financing capabilities of one or both of the merging parties relative to a scenario without the merger.
- 3.20 **Chapter 10, competition policy papers,** covers recent competition policy contributions on dynamic effects in mergers. These papers typically do not provide a formal analysis of the relevant economic mechanisms. Nevertheless, they contain relevant economic considerations and policy recommendations, and represent a useful cross-check on how best to summarise the insights from the academic literature.
- 3.21 **Chapter 11, guidelines on dynamic merger effects,** provides a descriptive overview of existing guidelines by competition agencies on dynamic merger effects (also covering, where relevant, topics that are indirectly linked to dynamic merger effects). This includes relevant guidelines from the Commission, guidelines from the UK and the US, and recommendations from the International Competition Network (ICN) and Organisation for Economic Co-operation and Development (OECD).

4 Competition and innovation

Key points

- The literature on market structure, competition, and innovation does not directly model mergers. It relies on proxies for the intensity of competition that do not necessarily capture merger effects. However, insights from this literature remain instructive for merger policy.
- One main insight is that innovation incentives depend on the prospective change in profits resulting from innovation. All else being equal, weaker competition pre-innovation reduces the innovation incentive (due to Arrow's *replacement* or *cannibalisation* effect) but weaker competition post-innovation increases it (a Schumpeterian effect).
- The literature on patent races emphasises that the change in profits for the winner following successful innovation differs significantly from the change in profits for the loser, generating business-stealing effects between competitors. Greater rivalry at the innovation stage tends to spur innovation by increasing the threat posed by rivals.
- Where incumbents compete for a patent with new entrants, they invest more than entrants for non-drastic innovations (to pre-empt future product market competition). This can help explain defensive acquisitions of potential competitors. However, for drastic innovations, entrants may invest more (due to Arrow's replacement effect).
- The endogenous growth literature pioneered by Aghion and Howitt shows that greater product market competition has ambiguous effects on incentives to innovate—either increasing incentives (via lower pre-innovation profits) or reducing them (via lower post-innovation profits). At the same time, greater rivalry at the innovation stage (e.g. more rival innovators, lower entry costs) broadly boosts innovation, as in the literature on patent races. This shows that it is important to evaluate simultaneously the impact of both innovation competition and product market competition. This joint assessment is directly relevant to horizontal mergers between rival innovators, since mergers affect competition in both areas.
- The empirical evidence indicates that competition raises productivity, through both a 'treatment' effect (i.e. existing firms becoming more productive) and a 'selection' effect (i.e. more productive firms entering and less productive firms leaving the market). However, the evidence on the effects of competition on innovation is mixed and depends on firm characteristics, market structure, and measurement. The empirical evidence also indicates that while larger firms spend more on innovation, they also tend to have lower innovation productivity.

- 4.1 This Chapter includes a synopsis of the key theoretical and empirical papers from the academic literature on competition, innovation, and spillovers. This excludes papers focusing specifically on mergers, which are discussed in Chapters 5 and 6. In addition to papers from the industrial organisation literature, this Chapter pays attention to contributions from the macroeconomics and growth literature.
- 4.2 While this literature does not model mergers explicitly, it studies underlying economic mechanisms that, as we will see in later chapters, are relevant for the impact of mergers. This literature captures, for example, the impact on innovation

incentives of changes in market structure at the upstream innovation stage (e.g. an increase in the number of rival innovators), or of changes in the intensity of product market competition (e.g. through lower product differentiation). Both effects are generally present also when examining the dynamic impact of mergers. It is helpful to review the results from this literature as they feature prominently in the policy debates on the impact of competition policy on dynamic effects (including the Schumpeterian notion that excessive product market competition may reduce innovation incentives).⁵¹

- 4.3 However, it is important to bear in mind that the results from this literature cannot be *directly* applied to the analysis of mergers, since mergers are a *specific form* of change in competitive conditions. In particular, the theoretical literature on competition and innovation typically considers market-wide changes in competitive parameters, such as the degree of product differentiation, the level of demand elasticity, the degree of coordination in the product market, or exogenous changes in the number of firms in a market. None of these measures directly captures the impact of a merger (and resulting coordination of behaviour and assets) between two out of a given number of firms. Even a reduction in the number of firms in an industry does not capture the impact of a merger, as it typically assumes that a firm simply ‘disappears’ from the market (e.g. its assets and products are no longer available). By contrast a merger allows two previously competing firms to *combine* their assets and *coordinate* their conduct on the relevant competitive variables (prices, investment, etc.). Therefore, while the mechanisms highlighted in this area of the literature are broadly relevant to the analysis of mergers, their specific application to mergers will vary. We return to this point in Chapter 5 in relation to the discussion of specific models of mergers and investment.
- 4.4 Similarly, the empirical literature often measures competition by looking at changes in profit margins or changes in market structure due to deregulation, trade liberalisation, and other variations of competitive conditions. These measures are, however, imperfect proxies for the impact of a merger.
- 4.5 Following Tirole (1988, Chapter 10), we start our summary of the literature with a discussion of incentives to innovate as a function of market structure, and consider the comparison between monopoly and competition for the cases of process innovation (originally considered by Arrow) and product innovation. We also cover here the case of an ‘insecure’ monopolist threatened by entry (as in Gilbert and Newbery, 1982). We then turn to the literature on patent races, which explicitly models the impact of changes in market structure at the (upstream) innovation stage. We then look at the literature on the impact of changes in both innovation competition and product market competition, starting from the endogenous growth literature associated with Aghion and Howitt (1992), and subsequent papers tackling related issues.
- 4.6 The final section provides a summary of the relevant empirical literature. We consider both the empirical evidence on how competition affects innovation and the evidence on the relationship between competition and productivity. In doing so, we consider both innovation inputs (for example, as measured by R&D expenditure or intensity) and innovation outputs (for example, as measured by patent counts or number of citations). We conclude with a review of the literature that considers the relationship between firm size and innovation.

⁵¹ The European Commission reviewed this literature, and its applicability to horizontal mergers in its decision on Dow/DuPont (2017). See Annex 4 in this decision.

- 4.7 In the remainder of this Chapter, we summarise the key insights and lessons from this area of the literature. A more detailed paper-by-paper review is contained in Volume 2, Chapter 2.

4.1 Innovation incentives as a function of market structure

- 4.8 The theoretical literature on the impact of market structure on innovation incentives shows that market structure and market power affect firms' willingness to acquire and deploy new technologies, but the direction of the effect depends on the type of innovation and on the competitive context.
- 4.9 The literature shows that innovation incentives depend on the prospective *change* in profit from innovation. This implies that pre-invention profits act as a disincentive to innovate, due to Arrow's *replacement* effect (also referred to as a *cannibalisation* or *diversion* effect). This effect captures the fact that an incumbent with market power internalises the loss of pre-innovation rents when adopting a new technology, which weakens its net gain from innovating relative to firms starting from competitive conditions.⁵²
- 4.10 Arrow's seminal 1962 paper models the case of process innovation with homogeneous goods, i.e. a reduction in the (constant) marginal cost of production. Because monopoly output is lower than competitive output, the incremental profit from a cost reduction is smaller for a monopolist, even when it can capture greater post-innovation rents through higher prices (e.g. as in the case of non-drastic innovation).⁵³ In this setting, as noted by Arrow, the 'only ground for arguing that monopoly may create superior incentives to invest is that appropriability may be higher under monopoly' (p. 622). We return to the notion of appropriability later in this Chapter, and in Part C of the main report.
- 4.11 Arrow's replacement effect is present in multiple economic settings (including several models of product innovation, with differentiated products), and is an important determinant of the impact of market power on innovation.
- 4.12 However, the replacement effect is not the only mechanism at work in determining innovation incentives. Competition also affects post-innovation profits, and hence influences the incremental profit from innovation. In settings with differentiated products and product innovation, a firm with market power is able to jointly set the prices of substitute products under its ownership. This allows it to optimally set the prices of the products in its portfolio, capturing greater profits from the introduction of a new product (compared to a single-product firm).⁵⁴
- 4.13 The effect from less intense post-innovation competition may outweigh the replacement effect in generating incentives to invest, depending on the specific setting. For example, in the Hotelling model of horizontal differentiation considered by Chen and Schwartz (2013), a monopolist's ability to coordinate prices across its product portfolio outweighs the cannibalisation effect (Arrow) if the new product is of higher quality than the existing product, implying that innovation incentives by a monopolist exceed those under duopoly.⁵⁵ In Chen and Schwartz's baseline model, consumer welfare remains lower under monopoly than duopoly, given that post-

⁵² Arrow (1962), Tirole (1988). More generally, the Arrow replacement effect captures the point that by increasing *current* profit flows, a merger might blunt innovation incentives so as to offset or at least diminish the spur to innovation from greater *future* profit.

⁵³ Arrow (1962).

⁵⁴ Greenstein and Ramey (1998), Chen and Schwartz (2013).

⁵⁵ However, this is not the case in the model of vertical product differentiation considered by Greenstein and Ramey (1998).

innovation prices are higher with monopoly. The higher innovation under monopoly therefore does not translate into a better outcome for consumers.

- 4.14 Although this literature does not directly study the effects of mergers, it is of direct relevance in some settings. For example, it is informative for the evaluation of a merger between an incumbent with an existing product and a potential competitor that could enter with an innovative product (in this scenario the incumbent does not innovate, and hence there is no competition at the innovation stage). A merger of this type would have an ambiguous effect on innovation incentives (depending on the underlying model of product differentiation). This literature also suggests that such a merger is likely to harm consumers (absent spillovers and synergies), even in cases where it results in more innovation.
- 4.15 Innovation incentives are also affected by the threat of entry.⁵⁶ A threatened monopolist may face stronger innovation incentives than a new entrant or a secure monopolist (even though the resulting patent may not be deployed, resulting in a 'sleeping patent'). This points to another important competitive channel that can stimulate innovation: rivalry at the innovation stage. In some circumstances it can also explain the persistence of market power, through defensive acquisitions of potential competitors by incumbents.

4.2 Patent races

- 4.16 The literature on patent races looks at firms that compete with each other to be the first to discover a given innovation. This can be particularly informative for settings where competition has 'winner-take-all' features (e.g. a market with first mover advantages or strong network effects). This literature also does not model mergers directly, so its results are only indirectly applicable to dynamic merger effects.
- 4.17 This literature shows that greater rivalry at the innovation stage of the market tends to increase the speed of innovation.⁵⁷ This is because firms engaged in a patent race do not take into account the adverse impact of their innovation efforts on rivals (as this results in rivals facing a lower probability of winning the race). This is a *negative externality* (or *business-stealing effect*) between rival innovators. More rivalry between firms (e.g. as proxied by a higher number of independent and competing innovators) strengthens these externalities, and by doing so it accelerates the arrival of innovation—to the benefit of consumers.
- 4.18 Depending on the model, innovation efforts may be strategic complements (i.e. greater efforts by one firm may push others to invest more in R&D)⁵⁸ or strategic substitutes (greater efforts by one firm may lead others to invest less).⁵⁹ An increase in innovation efforts by one firm (say, following a reduction in its R&D costs) therefore may or may not lead to an increase in innovation by rivals.
- 4.19 The literature also shows that in a patent race setting, entrants may invest more than incumbents in the case of drastic innovation (due to Arrow's replacement effect), while in the case of non-drastic innovation, pre-emption effects may prevail, leading to the persistence of incumbency.⁶⁰ However, these conclusions are not universal, as the persistence of monopoly under uncertain innovation depends on how competition for innovation is modelled.⁶¹ Imperfect appropriability (e.g.

⁵⁶ Gilbert and Newbery (1982).

⁵⁷ Loury (1979), Lee and Wilde (1980), Reinganum (1983).

⁵⁸ Lee and Wilde (1980).

⁵⁹ Loury (1979).

⁶⁰ Reinganum (1983).

⁶¹ Yi (1995).

through spillovers) tends to reduce innovation incentives (compared to the socially optimal level).⁶²

4.3 R&D competition, product market competition, and innovation

- 4.20 The economics literature has considered the impact of R&D competition and product market competition in more complex dynamic oligopoly models, showing that changes in the degree of innovation competition or product market competition can have theoretically ambiguous and context-dependent effects on innovation incentives. Nonetheless, some important principles emerge.
- 4.21 It is useful to distinguish between changes in (upstream) *innovation competition* (e.g. as proxied by the number of rival innovators or lower entry costs in R&D) and changes in (downstream) *product market competition* (e.g. as proxied by factors affecting profitability in the product market). These two dimensions operate through distinct mechanisms and can generate opposing effects on innovation.
- 4.22 As seen in the patent race literature, greater rivalry at the innovation stage tends to boost the overall rate of innovation by enhancing business-stealing effects, though this may be offset by knowledge spillovers.⁶³ By contrast, greater competition at the product market stage has an ambiguous effect on innovation incentives.
- 4.23 These two mechanisms are both at work in the seminal work of Aghion and Howitt (1992). In their model, more *competition in innovation* results in greater 'creative destruction' with new products replacing existing products and boosting growth.⁶⁴ At the same time, greater *product market competition* reduces innovation by limiting post-innovation rents. The result on the impact of product market competition is instead ambiguous in richer models where both leaders and laggards can innovate (while in the original Aghion and Howitt (1992) model only laggards innovate, to leapfrog the incumbent).⁶⁵ The impact of product market competition generally depends on a number of factors, including the identity of the innovator,⁶⁶ the nature of product market competition,⁶⁷ and the degree of asymmetries between firms.⁶⁸ In some cases, more intense product market competition boosts innovation incentives through its negative effect on pre-innovation profits (and the resulting increase in the incremental profitability of innovating).⁶⁹ In other cases, competition lowers innovation incentives because of its detrimental effect on post-innovation profits.⁷⁰ Which of these two effects prevails depends on the nature of the specific economic setting under consideration.
- 4.24 For example, in influential work by Aghion, Harris, Howitt, and Vickers (2001) and Aghion, Bloom, Blundell, Griffith, and Howitt (2005), both effects are present, as firms innovate either to *escape competition* or to *catch up* with rivals.⁷¹ Greater product market competition increases the incentives to *escape competition* for neck-

⁶² For a discussion, see Tirole (1988).

⁶³ Aghion and Howitt (1992), Aghion and Howitt (1998), Vives (2008), Marshall and Parra (2019).

⁶⁴ Aghion and Howitt (1998).

⁶⁵ See Harris and Vickers (1985), Harris and Vickers (1987), Budd, Harris, and Vickers (1993), Aghion, Harris, and Vickers (1997), and discussion in Aghion (2026).

⁶⁶ Boone (2001), Aghion, Bloom, Blundell, Griffith, and Howitt (2005), Gilbert, Riis, and Riis (2018).

⁶⁷ Aghion, Harris, Howitt, and Vickers (2001), Vives (2008), Schmutzler (2013), Gilbert, Riis, and Riis (2018), Marshall and Parra (2019).

⁶⁸ Boone (2000), Boone (2001), Aghion et al. (2005), Schmutzler (2013).

⁶⁹ Aghion et al. (2001), Aghion et al. (2005).

⁷⁰ Aghion and Howitt (1992), Boone (2000), Aghion et al. (2005), Hashmi (2013), Schmutzler (2013).

⁷¹ Aghion et al. (2001), Aghion et al. (2005).

and-neck firms, as competition reduces their *pre-innovation* profits. Conversely, stronger competition reduces the incentives by lagging firms to *catch up* with rivals as it lowers *post-innovation* profits (this has been described as a *Schumpeterian* effect). This interaction can generate an inverted-U relationship between competition and innovation, where competition at low levels stimulates innovation, but at higher levels discourages it (e.g. Aghion et al., 2005).

- 4.25 Which effect dominates depends on the distribution of technological gaps across firms (i.e. whether firms are in *neck-and-neck* competition or whether there is typically a technology leader followed by a laggard), the size of the gap,⁷² the level of technological progress (i.e. the amount of investment required to achieve a marginal cost reduction or quality improvement),⁷³ and the prevailing market structure. In step-by-step innovation, state-dependent IP rights may be optimal, with stronger protection for frontier firms to incentivise innovation.⁷⁴
- 4.26 A key insight from this literature is that it is important to consider *jointly* the impact of competition at the innovation stage and at the product market stage. This is directly relevant for the assessment of horizontal mergers between rival innovators, as a merger typically has an impact on the intensity and nature of competition at both stages of competition.

4.4 Empirical evidence

- 4.27 The empirical papers on the effects of competition have generally considered two main output variables: innovation (typically measured in terms of input such as R&D expenditure or intensity, or output such as patent counts or citations) or productivity (typically defined as the amount of inputs required to produce a certain level of output).
- 4.28 Considering first innovation, some of the empirical literature indicates the presence of an inverted-U relationship, with greater product market competition increasing innovation when competition is weak,⁷⁵ but reducing it when competition is strong.⁷⁶ This effect has been explored with reference to the theoretical models of product market competition we review in Section 4.3 and specifically the trade-off between the *escape competition effect* and the *catching-up effect*.
- 4.29 The evidence on the existence of an inverted-U relationship is mixed, however. While some studies find evidence consistent with such a relationship,⁷⁷ others find a positive relationship between competition and innovation,⁷⁸ or a weak negative relationship.⁷⁹ There is also evidence that the curve becomes flatter when the R&D cost relative to firm size increases, and disappears for small firms.⁸⁰ Papers looking at competition from foreign imports generally also find ambiguous effects on firm innovation: it can either stimulate technological upgrading and reallocation towards

⁷² Hashmi (2013).

⁷³ Jeanjean (2021).

⁷⁴ Acemoglu and Akcigit (2012).

⁷⁵ Blundell, Griffith, and Van Reenen (1999), Aghion et al. (2005), Bloom, Draca, and Van Reenen (2016), Aghion, Akcigit, Hyytinen, and Toivanen (2018).

⁷⁶ Scherer (1967), Aghion et al. (2005), Hashmi (2013), Hounghon and Jeanjean (2016), Ciriani and Jeanjean (2020).

⁷⁷ Aghion et al. (2005), Bos, Kolari, and van Lamoen (2013), Hounghon and Jeanjean (2016), Garfinkel and Hammoudeh (2025).

⁷⁸ Correa (2012), Correa and Ornaghi (2014), Beneito, Rochina-Barrachina, and Sanchis (2017), Gutiérrez and Philippon (2017).

⁷⁹ Hashmi (2013).

⁸⁰ Askenazy, Cahn, and Irac (2013).

more productive firms⁸¹ or hinder innovation by reducing firm resources,⁸² with outcomes depending on factors such as firms' technological position, competitive exposure, and measurement of rivalry.

- 4.30 Most of the papers that have studied this relationship rely on reduced-form estimations, which do not allow for inferences on the effects of changes in competition on consumer surplus overall (including both price and innovation effects). One notable exception is a study that uses a structural model for the computer microprocessor industry (looking at competition between Intel and AMD between 1993 and 2004).⁸³ This finds that an increase in innovation due to weaker product market competition is associated with a reduction in consumer welfare (because of a concurrent increase in prices). Another paper using structural modelling in the hard-disk drive industry between 1981 and 1997 finds that the Arrow replacement effect dominates pre-emption and innovation cost advantages, with a reduction in competition (measured as a decrease in the number of active firms in the market) leading to less innovation.⁸⁴ A third paper looking at the global automobile industry between 1982 and 2006 finds that entry generally increases aggregate innovation even when it reduces incumbent innovation.⁸⁵
- 4.31 Papers considering the effect of competition-enhancing remedies following antitrust intervention find that structural breakups and compulsory licensing regimes generally increase innovation by intensifying rivalry and enabling knowledge spillovers and follow-on research.⁸⁶ However, effects typically depend on complementary conditions such as access to downstream markets, and may be uneven across technologies, sectors, and types of firms.
- 4.32 Looking instead at the more general relationship between competition and productivity, the empirical literature generally provides robust evidence that greater product market competition increases productivity—with some evidence of both a *treatment effect*, in which competition pushes firms to increase their productivity,⁸⁷ and a *selection effect*, in which low-productivity firms exit or shrink, and demand shifts from low- to high-productivity firms.⁸⁸
- 4.33 The empirical literature on firm size and innovation finds that while innovation input (e.g. R&D spending) often increases in proportion to firm size, innovation productivity (e.g. the number of patents per unit of R&D spending) typically rises less than proportionately with size.⁸⁹ Moreover, the share of innovation activity devoted to process (rather than product) innovation increases with firm size.⁹⁰ One explanation is that larger firms are better able to exploit incremental or marginal innovation opportunities, which are more often process rather than product innovations, because they can capture the benefits of innovation over a larger output base than smaller firms can. Additionally, papers based on detailed within-firm data suggest that innovation productivity is driven mainly by economies of

⁸¹ Bloom, Draca, and Van Reenen (2016), Xia, Cassiman, and Wehrheim (2025).

⁸² Autor, Dorn, Hanson, Pisano, and Shu (2020), Liu, Lu, Lu, and Luong (2021).

⁸³ Goettler and Gordon (2011).

⁸⁴ Igami (2017).

⁸⁵ Hashmi and van Biesebroeck (2016).

⁸⁶ Watzinger, Fackler, Nagler, and Schnitzer (2020), Watzinger and Schnitzer (2022), Poege (2022), Mamrak (2026).

⁸⁷ Nickell (1996), Fabrizio, Rose, and Wolfram (2007), Holmes and Schmitz (2010), Van Reenen (2011), Backus (2020), Sadun, Schuh, Hartley, Van Reenen, and Bloom (2025).

⁸⁸ Olley and Pakes (1996), Syverson (2004).

⁸⁹ Scherer (1965), Bound, Cummins, Griliches, Hall, and Jaffe (1984), Cohen, Levin, and Mowery (1987), Cohen and Klepper (1996a), Fritsch and Meschede (2001), Cohen (2010).

⁹⁰ Cohen and Klepper (1996b), Fritsch and Meschede (2001).

scope, including knowledge complementarities across research activities and internal and external spillovers, rather than by firm size or economies of scale per se.⁹¹

⁹¹ Henderson and Cockburn (1996), Cockburn and Henderson (2001).

5 Horizontal mergers: established competitors (theory)

Key points

- Recent policy debates on merger control have spurred a developing theoretical economics literature on the impact of horizontal mergers on innovation, investment, and consumer welfare. Many of these models assume away potentially important aspects, such as the presence of non-merging rivals, asymmetries between firms, multiproduct firms, spillovers, and synergies. Despite these limitations, this literature offers useful insights for merger control.
- In settings where incumbents compete to improve their products or reduce their costs (including both models of stochastic R&D and deterministic investment), mergers internalise negative investment and price externalities.
- The literature establishes that mergers in these settings may either reduce or increase investment, even in the absence of spillovers or synergies. While the literature indicates that both effects can arise under plausible conditions, it also suggests that mergers reduce investments more often than they increase them.
- This literature indicates that mergers are likely to harm consumers, *absent spillovers and synergies*. In these settings, consumers are harmed through reduced product market competition (e.g. higher prices for innovative products) and, where applicable, lower investments. This result holds also in cases where mergers spur investments by reducing product market competition.
- These insights extend to settings involving mergers between competing firms that innovate to enter a new market.
- The literature has also considered cases with asymmetric firms, and diversity of innovation paths or projects. This shows that mergers may benefit consumers when bringing together two laggards or when inducing the merging firms to reallocate their total investment across R&D projects (even absent synergies).
- A merger can also spur investments and benefit consumers if it leads to R&D synergies, including technology or knowledge sharing.
- Additionally, a merger can increase investment and benefit consumers if it leads to the internalisation of spillovers. However, in settings where non-merging firms also benefit from spillovers, opposing effects may arise, making the overall effect on consumers potentially negative.
- Theoretical work on overlapping ownership between firms broadly aligns with the findings of the literature on mergers and investment.

- 5.1 This Chapter presents an overview of the relevant theoretical literature on the effects of horizontal mergers, when innovation and investment are important parameters of competition. Much of this literature is relatively new, and has been motivated by recent policy debates on the impact of mergers on innovation and investment, including notable decisions by the European Commission (see discussion in Chapter 2).
- 5.2 We structure the review of this part of the literature by first considering a baseline case of mergers between firms that are investing to improve their products or reduce their costs in markets where they are already present, with existing

products. It is helpful to start with this setting to isolate some of the main effects of a horizontal merger. Much of this literature focuses on how the elimination of competition between merging firms may affect the level of innovation or consumer welfare, abstracting from cost or R&D synergies, and from the internalisation of spillovers.⁹² While this is an instructive exercise, it typically abstracts from possible procompetitive effects from mergers.

- 5.3 We then consider a number of additional settings, including: investments made by existing firms to enter new markets; mergers between asymmetric firms; mergers in settings with sequential innovation; the impact of horizontal mergers on diversity and direction of innovation; mergers leading to dynamic benefits, including voluntary technology or knowledge transfers and the internalisation of involuntary spillovers; and mergers in procurement markets.⁹³ In some of these additional settings, a merger can lead to an increase in consumer welfare, through a variety of mechanisms that we review below.
- 5.4 We note at the outset that several papers in the literature focus on a duopoly scenario (where the only two, often symmetric, firms in a market merge), for analytical tractability and to illustrate effects starkly.⁹⁴ In practice, mergers reviewed by competition authorities are rarely mergers-to-monopoly, or mergers between symmetric firms. The literature has often used numerical simulations to consider mergers in markets with more than two firms. We review and comment on these simulations in our review below, seeking to obtain robust insights for merger control policy. In particular, where simulations report the results for a given range of parameters, we seek to infer from this (where possible) how likely it is that a merger may result in a specific outcome (say, an increase or decrease in expected consumer welfare).
- 5.5 We also summarise in this Chapter some of the literature on horizontal cooperation (including overlapping ownership) and the related literature on research cooperation, as some of the underlying mechanisms are similar to those at work in a horizontal merger. It is therefore instructive to consider this literature as well.
- 5.6 This Chapter does not consider papers that study acquisitions of potential or nascent competitors, which are covered in the next Chapter. Conceptually, the papers discussed in this Chapter overlap to some extent with the literature on potential competitors, as some of the mechanisms at play are common to both (e.g. the internalisation of negative externalities brought about by a merger). The main conceptual difference is that horizontal mergers between incumbents generally involve cases where the two merging firms are not radically asymmetric (e.g. they have a broadly similar position in the market, in terms of degree of incumbency and access to existing products), whereas acquisitions of potential competitors generally involve highly asymmetric cases (e.g. a large incumbent firm acquiring a small innovative company).
- 5.7 In the remainder of this Chapter, we summarise the key insights and lessons from this literature. A more detailed paper-by-paper review is contained in Volume 2, Chapter 3.

⁹² In doing so, the literature mostly looks at how mergers between rivals internalise negative externalities.

⁹³ We also consider calibrated computational models of horizontal mergers.

⁹⁴ In addition, many of the models in the literature are mostly 'static', in the sense that they focus on stylised two-stage models of investment and pricing competition.

5.1 Investment in existing markets

- 5.8 This literature considers competition between incumbent firms that sell existing products in a market, and can invest to either improve the quality of their products or reduce their costs, assuming that innovations are non-drastic. It looks at stochastic R&D (e.g. investment to increase the probability of discovery of a new product), at deterministic R&D (e.g. investments to reduce the costs of production or increase quality, with no uncertainty on the likelihood of success), and at investments to increase capacity (e.g. investments to reduce congestion on a network, and improve user experience). Papers in this area have considered both simpler duopoly settings (where the merger leads to a monopoly) and more complex oligopoly settings (where two out of a given number of firms merge). The papers also consider different models of consumer demand (where consumer utility from a product varies with the underlying parameters, including price, quality, and variety). These papers consider a setting with parallel innovation efforts, where innovation by one firm does not negatively affect the probability that other firms will successfully innovate.⁹⁵
- 5.9 Absent the merger, competition between firms leads to negative externalities (or business-stealing effects), in relation to both price and investment. That is, lower prices or greater investment efforts by each firm will tend to capture sales from its competitors, reducing their profits. A merger internalises both price and investment externalities. The combined impact of this internalisation (including interaction effects across the two externalities) determines the impact of the merger on overall investment incentives, and on expected consumer welfare.
- 5.10 The literature establishes that, absent synergies and spillovers, mergers may reduce or increase investment. While the literature indicates that both effects can arise under plausible conditions, it also suggests that mergers reduce investments more often than they increase them.
- 5.11 In several of the theoretical settings studied in the literature (including both product and process innovation), the combined internalisation of price and investment externalities brought about by a horizontal merger raises prices, and *reduces* overall investment.⁹⁶ In these settings, consumers are harmed by both higher prices and lower innovation, with the two effects going in the same direction in terms of their impact on consumer welfare.
- 5.12 The literature also shows that there are settings where a merger between incumbents instead results in *higher* innovation incentives (even absent synergies and spillovers).⁹⁷ For example, for the case of product innovation, this may follow from the fact that the merger increases post-innovation prices and margins, thus increasing the incentive to expand demand through innovation. In some settings, the increase in post-innovation prices from the merger dominates the internalisation of negative innovation externalities, resulting in a *net increase* in innovation incentives.⁹⁸ In relation to the impact on consumer welfare, the merger therefore gives rise to a trade-off between higher innovation incentives and higher future prices. Depending on which effect prevails, the overall impact on consumer welfare

⁹⁵ This is distinct from a patent race environment, where success for one firm precludes success by others for the same innovation.

⁹⁶ Federico, Langus, and Valletti (2018), Motta and Tarantino (2021), Bourreau, Jullien, and Lefouili (2026), Moraga-González and Motchenkova (2026), and López (2025).

⁹⁷ Bourreau, Jullien, and Lefouili (2026) and Moraga-González and Motchenkova (2026).

⁹⁸ This can be the case also in sequential models of process innovation (Moraga-González and Motchenkova, 2026). However, it is not the case in simultaneous models of process innovation (Motta and Tarantino, 2021; Bourreau, Jullien, and Lefouili, 2026).

is a priori ambiguous. Nevertheless, the papers examining these settings find, through numerical simulations, that consumers are generally harmed by the merger. A key mechanism behind this finding is that the net increase in investment incentives rests on higher future prices.⁹⁹

- 5.13 The papers in this area of the literature therefore suggest that, while the impact of a horizontal merger on innovation may be positive or negative, its effect on consumer welfare is generally negative (again, absent spillovers and synergies). This insight applies to both models of product innovation and process innovation. It also applies to models of capacity expansion which are more appropriate for network industries (e.g. telecommunications).¹⁰⁰
- 5.14 This area of the literature has also considered innovation efficiencies, in the form of lower R&D costs or more effective R&D. Such efficiencies, if sufficiently strong, can overturn the adverse effects of a merger on innovation (for cases where the merger reduces innovation), and on consumer welfare.¹⁰¹ Similarly, if greater innovation yields benefits in other markets where only one of the merging parties is active (i.e. out-of-market effects), then consumers may benefit (in aggregate) from an increase in innovation due to a merger, even in the absence of in-market efficiencies.¹⁰² Finally, in some settings, the existence of spillovers can overturn the adverse effect of a merger on innovation when such an adverse effect exists (see further discussion below).

5.2 Additional settings

- 5.15 The previous section considered a baseline case of mergers between firms that are investing to improve their products or reduce their costs in markets where they are already present, with existing products. In this section, we consider additional settings.

5.2.1 Investments to enter new markets

- 5.16 A number of papers have looked at the case where two or more firms compete to introduce a product in a new market.¹⁰³ We look at this case as a merger between incumbents (i.e. there is no strong asymmetry between firms, unlike in the case of an incumbent acquiring a small innovative company).
- 5.17 Papers in this area of the literature broadly confirm the findings of the models where incumbent firms invest to improve their existing products. In a number of settings, a merger reduces innovation and increases the prices of future products.¹⁰⁴ However, there are several settings where the internalisation of price externalities from a merger is sufficiently strong as to induce an increase in innovation incentives.¹⁰⁵ Simulations in the corresponding studies find that, despite the increase in innovation incentives, consumers are likely to be harmed due to the

⁹⁹ Bourreau, Jullien, and Lefouili (2026), and Moraga-González and Motchenkova (2026). Moraga-González and Motchenkova (2026) find that in some of the models considered in their paper consumer surplus may increase, but this is the case in a very restricted range of parameters.

¹⁰⁰ López (2025).

¹⁰¹ Federico, Langus, and Valletti (2018), López (2025), Bourreau, Jullien, and Lefouili (2026), Moraga-González and Motchenkova (2026).

¹⁰² Moraga-González and Motchenkova (2026).

¹⁰³ These models can also be interpreted as models of drastic innovation in existing markets.

¹⁰⁴ Federico, Langus, and Valletti (2017), Gilbert (2019), D'Annunzio, Jullien, Lefouili, and Madio (2026a).

¹⁰⁵ Bourreau and Jullien (2018), D'Annunzio et al. (2026a), Moraga-González and Motchenkova (2026).

impact of higher prices. The literature has also identified situations with asymmetric outcomes following a merger-to-monopoly (where the R&D efforts of one of the merging parties are discontinued post-merger, and all R&D is carried out by the other merging firm), which can result in greater innovation post-merger, by avoiding duplication of R&D investments addressing a similar market need.¹⁰⁶

5.2.2 Mergers in markets with asymmetric firms

- 5.18 While most of the papers summarised above assume that firms are symmetric pre-merger (even though the merger introduces an asymmetry between competitors, except for the merger-to-monopoly case), some of the papers explicitly consider the case where firms are asymmetric pre-merger. One study shows that if asymmetries are significant (e.g. the merger involves two smaller high-cost firms competing against a larger low-cost incumbent), then a merger can increase investment and, if cost asymmetries are sufficiently strong, also consumer surplus.¹⁰⁷ Another study finds that this may also be the case if the merger brings together two laggards that are yet to enter a market (or market segment), and where the merger can help them resolve a coordination failure.¹⁰⁸

5.2.3 Spillovers and technology transfers

- 5.19 Technology (or knowledge) transfers refer to the situation in which innovation by one of the merging firms is voluntarily shared with another firm, e.g. in the context of licensing or following a merger. Spillovers refer to the situation where firms may be able to imitate or otherwise use (at least partially) innovation carried out by other (potentially rival) firms without compensating them. In this case there is involuntary technology, information, or knowledge sharing between competing firms. A merger would internalise these spillovers to the extent that these take place between the merging parties.
- 5.20 As discussed in Box 5.1, empirical evidence shows that knowledge diffusion between firms through spillovers can be substantial, implying that private returns to innovation typically fall short of the social returns. Market-based transactions such as licensing can instead support technology or knowledge transfers and diffusion between firms.



Box 5.1 Empirical evidence on spillovers and technology transfers

Empirical research on spillovers finds that, in general, these are large, leading to social returns to R&D that are higher than private returns. This also implies underinvestment in innovation from a social welfare perspective, relative to a situation with no spillovers.¹⁰⁹ Evidence also shows that geographic barriers to knowledge diffusion (i.e. spillovers, measured by patent citations) have declined over time, suggesting that firms increasingly benefit from cross-border spillovers.¹¹⁰ While this creates social benefits in terms of wider diffusion of knowledge and technology, it also creates further underinvestment in this knowledge from a social welfare perspective if these spillovers are not internalised.

¹⁰⁶ Denicolò and Polo (2018). In this model, a monopoly outcome always prevails in the product market (even absent the merger). Therefore, the effect of the merger on consumer surplus has the same sign as the effect on innovation.

¹⁰⁷ Vuong and Baranes (2021).

¹⁰⁸ Padilla, Piccolo, and Reynolds (2024).

¹⁰⁹ Bloom, Schankerman, and Van Reenen (2013), Adhami (2026).

¹¹⁰ Griffith, Lee, and Van Reenen (2011).

There is also evidence suggesting that the increasing gap between frontier and laggard firms in the US and reduced business dynamism and productivity growth can in large part be explained by declining spillovers and technology diffusion between firms.¹¹¹

Knowledge diffusion does not occur only through spillovers, however. Empirical evidence shows that a significant portion of knowledge diffusion occurs through market-based transactions such as licensing.¹¹² This helps narrow the gap between private and social returns and may offer alternatives to mergers for knowledge sharing.

Source: Oxera.

- 5.21 The theoretical economics literature has considered the impact of mergers resulting from technology transfers and spillovers. Both mechanisms relate to the public good properties of knowledge and technology, namely that it is non-rivalrous (i.e. access to knowledge by one party does not reduce its availability or usefulness to others), and possibly non-excludable (i.e. knowledge acquired by one firm may spill over to others, depending in particular on the nature and strength of any associated IP rights).
- 5.22 In models of process innovation with increasing marginal costs and homogeneous products, technology transfers between merging parties can counteract the standard output contraction effect of mergers (by allowing the merged entity to apply a given cost-reduction technology to a larger output base), leading to greater innovation and output.¹¹³ The same can hold in a setting where products are differentiated, depending on the intensity of product market competition, and the extent of asymmetries between firms.¹¹⁴ In general terms, technology transfers between merging firms can have both a positive *incentive* effect (by increasing the rewards obtained from future innovations) and a positive *diffusion* effect (even with no change in incentives), by broadening the reach of a given innovation. The feasibility of licensing technology between competitors (absent a merger) is an important determinant of the impact of technology transfer resulting from a merger.¹¹⁵
- 5.23 A merger may also internalise spillovers between competing firms.¹¹⁶ This is a positive externality between competitors that is at work absent the merger. Before a merger, spillovers reduce a firm's incentives to invest because they imply that a given investment results in a smaller advantage over the innovator's rivals (compared to a benchmark without spillovers). After the merger, each innovator recaptures the spillovers to the other merging firm (which boosts innovation incentives). However, the merged entity still suffers from spillovers to the remaining rivals, and may actually internalise this effect to an even greater extent (compared to the pre-merger situation). This follows from the fact that each merging firm now takes into account the fact that spillovers to non-merging firms are also detrimental to its merging partner. As a result, spillovers from the merging firms to non-merging rivals create a channel through which the merger may dampen the merging firms' innovation incentives. Therefore, the presence of industry-wide spillovers may make

¹¹¹ Akcigit and Ates (2023).

¹¹² Arqué-Castells and Spulber (2022).

¹¹³ Denicolò and Polo (2021).

¹¹⁴ Haucap, Rasch, and Stiebale (2019), Mukherjee (2022).

¹¹⁵ Régibeau and Rockett (2019).

¹¹⁶ Atallah (2016), Régibeau and Rockett (2019), Salinger (2019), Pietola, Tarantino, and Zenger (2026), Hwangbo and Lee (2026), D'Annunzio, Jullien, Lefouili, and Madio (2026b).

a merger either more or less likely to increase innovation incentives. The literature suggests that the greater the pre-merger spillovers between the merging firms, and the lower the spillovers to non-merging firms, the more likely it is that the merger will increase innovation incentives.¹¹⁷

5.2.4 Sequential innovation

- 5.24 Recent work on mergers considers sequential settings, where firms compete to introduce multiple and sequential innovations, and can learn from the success of early innovators (i.e. this is another form of information spillover across firms).¹¹⁸ In this case a merger has ambiguous effects on innovation incentives, due to the conflicting effect of *cannibalisation* (which reduces incentives to invest in later innovation, so as not to divert sales from early innovation), and *appropriability* (which captures the increase in incentives for early innovation, by protecting it from competing follow-on innovation). In this setting the impact of a merger on consumers is ambiguous, and depends in part on the strength of the information spillovers internalised by the merger.

5.2.5 Diversity of innovation paths

- 5.25 The recent literature has also analysed the impact of mergers not just on the level of innovation, but also on its direction. One study shows that competition can result in greater R&D diversity, as firms invest in riskier projects to insulate themselves from competitive pressure.¹¹⁹ A merger (proxied by an exogenous reduction in the number of firms) can therefore reduce the diversity of R&D portfolios. Conversely, if firms are subject to an increasing marginal cost of *aggregate* R&D efforts, R&D competition between them leads to both negative and positive externalities, depending on their allocation of R&D spending across alternative projects.¹²⁰ In this case, a merger can lead to a superior distribution of R&D efforts across projects, and increase consumer surplus (even when total innovation falls). The presence of adverse price effects from a merger reduces the scope for beneficial consumer effects (especially if the merger leads to future price effects on innovative products, as in the case of non-drastic innovation).

5.2.6 Procurement settings

- 5.26 Horizontal mergers can also affect innovation incentives in procurement auctions. In efficient procurement mechanisms with unit demand, absent efficiencies and spillovers, mergers do not affect innovation incentives, but raise prices (in tenders where the merging parties are ranked first and second) and hence reduce expected consumer surplus.¹²¹ In this setting, a merger-to-monopoly is anticompetitive even in the presence of efficiencies, as these are not passed through to consumers. More generally, higher concentration makes it less likely that efficiencies will offset harmful effects. With symmetric industry-wide spillovers, the partial internalisation of spillovers brought about by a merger is not sufficient to benefit consumers if the number of merger outsiders is sufficiently large (since external spillovers from merging firms to outsiders are more significant in this case), and can actually harm consumers.

¹¹⁷ Atallah (2016), Pietola, Tarantino, and Zenger (2026), D’Annunzio et al. (2026b).

¹¹⁸ Das, Mayskaya, and Nikandrova (2026).

¹¹⁹ Letina (2016).

¹²⁰ Moraga-González, Motchenkova, and Nevrekar (2022).

¹²¹ Pietola, Tarantino, and Zenger (2026). In first-price auctions, mergers reduce investment and consumer surplus (absent efficiencies and spillovers).

5.2.7 Calibrated dynamic models

- 5.27 We have identified four articles in the academic literature that use calibrated dynamic models of competition to consider the impact of mergers on investment by both incumbents and new entrants. Three of these studies find that a merger control policy based on a static consumer welfare standard also delivers significant dynamic benefits, in terms of higher investment and productivity.¹²² In the fourth study, however, the prospect of a merger (innovation-for-buyout) leads to greater innovation and can outweigh the short-term harm to consumers from a merger (e.g. this is the case if entry costs are low, entrants and incumbents can both innovate rapidly, and the degree of horizontal product differentiation is low).¹²³

5.3 R&D cooperation and overlapping ownership

- 5.28 The literature on R&D cooperation precedes the recent literature on horizontal mergers, but provides useful insights into dynamic merger effects as some of the underlying mechanisms are similar to those at work in a horizontal merger. The same is the case for the more recent literature on overlapping ownership—which includes both minority shareholding in competing firms (i.e. corporate equity stakes between firms) and common ownership (i.e. common shareholders across firms).
- 5.29 The R&D cooperation literature has studied the impact of various scenarios (R&D cooperation, output and R&D cooperation, and no-cooperation) in a duopoly setting.¹²⁴ This is directly informative for the case of merger-to-monopoly (but not beyond it). This work suggests that competition (i.e. no-cooperation in either output or R&D) delivers superior welfare outcomes to those in the cooperative benchmarks when spillovers are low. R&D cooperation is instead preferable to no-cooperation when spillovers are higher, and dominates full cooperation (which can be seen as equivalent to a merger). This has implications for the evaluation of mergers, compared with less anticompetitive alternatives such as R&D joint ventures. A caveat to keep in mind is that these theoretical models typically assume away product-market coordination that may be induced by R&D joint ventures, whereas the empirical evidence suggests that such an effect may exist.¹²⁵
- 5.30 The recent theoretical literature on overlapping ownership (mostly motivated by recent policy debates on common ownership) reaches similar conclusions. It shows that in a wide range of oligopoly settings, overlapping ownership can lead to greater R&D and higher consumer welfare if (and only if) there are sufficiently high spillovers (with the level of required spillover for an increase in consumer welfare lying above the corresponding level required for an increase in R&D).¹²⁶ In these models, overlapping ownership *de facto* captures partial coordination across *all firms* in a market, and as such it reflects the internalisation of price and innovation externalities brought about by a merger to monopoly. However, it does not capture the fundamentally asymmetric nature of mergers (where only two out of a number of rivals coordinate their conduct). Additional theoretical research on common ownership indicates that it may improve the direction of R&D,¹²⁷ it may benefit consumers for the case of asymmetric partial cross-ownership (which, however,

¹²² Mermelstein, Nocke, Satterthwaite, and Whinston (2020), Cavenaile, Celik, and Tian (2021), Guthmann and Rahman (2026). The second and third papers calibrate their simulations with aggregate data from the US market and are also considered in Chapter 7 on empirical evidence.

¹²³ Hollenbeck (2020).

¹²⁴ D'Aspremont and Jacquemin (1988), Leahy and Neary (1997).

¹²⁵ Sovinsky (2022).

¹²⁶ López and Vives (2019), Antón, Ederer, Giné, and Schmalz (2025).

¹²⁷ Nevrekar (2022).

only has limited implications for mergers),¹²⁸ but it may slow down the adoption of a new technology.¹²⁹

¹²⁸ Shelegia and Spiegel (2024).

¹²⁹ Hurkens and López (2026).

6 Horizontal mergers: acquisitions of potential competitors (theory)

Key points

- The acquisition of potential competitors can affect innovation both *ex post* (i.e. after the transaction) and *ex ante* (i.e. in anticipation of a transaction).
- Ex post effects may be negative or positive. Acquisitions of innovative targets may reduce or suppress both innovation (potentially resulting in a ‘killer acquisition’) and future price competition. At the same time, such acquisitions may also enable small innovative companies to scale effectively (by exploiting the greater access to market of incumbent acquirers), lead to greater diffusion of the technology of the target, or combine complementary assets with those of the acquirer, resulting in procompetitive effects.
- Consumer harm is most likely to occur when the incumbent has strong market power, the target is a credible standalone competitor or there are alternative buyers that raise fewer competitive concerns, and/or there are limited synergies.
- Acquisitions of potential innovative competitors may also entrench incumbency by raising barriers to entry for future rivals—e.g. via larger captive user bases, defensive strategies which discourage entry by creating ‘kill zones’, or acquihires.
- Ex ante effects may also be positive or negative. Acquisitions may provide new firms with an important exit route, stimulating entry and investment through an *investment-for-buyout* effect. This potentially leads to a trade-off between ex post and ex ante effects on innovation (if the target can capture some of the incremental benefits from an acquisition by an incumbent).
- The prospect of acquisition may also affect the direction of innovation. It may steer incentives towards incremental, ‘me-too’, innovation paths rather than more disruptive ones, reducing competitive pressure on incumbents. However, this is not always the case and ex ante effects may also direct innovation portfolios towards markets with higher consumer benefits.

- 6.1 While the previous Chapter focused on literature on mergers between established incumbents, this Chapter looks at the theoretical literature on the acquisition of potential, or nascent, competitors. Some of this literature is relatively recent, and has largely been motivated by policy debates on the acquisition of small innovative companies (often referred to as ‘startups’) by large digital platforms or pharmaceutical companies over the past two decades.¹³⁰
- 6.2 Potential competitors are firms that are currently not active in the relevant product market of the potential acquirer, but may be so in the future. Unlike the case of horizontal mergers involving incumbents, the literature has modelled acquisitions involving potential competitors by considering settings with significant asymmetry between the acquirer and the target. As discussed in this Chapter, these acquisitions can have a variety of effects on the incentive to innovate for the acquirer, the target,

¹³⁰ Some of these policy debates have centred on where competition authorities have or should have jurisdiction to consider these acquisitions. We do not cover this issue in the Study.

and (potential) third parties. These effects may depend on whether one considers only ex post effects (effects on investment and innovation and on future prices *after* the acquisition by an incumbent) or ex ante effects (effects on investment by a potential competitor or acquirer in anticipation of a possible merger). As we note below, there may be legal limits on whether or how ex ante effects are considered under merger control.

- 6.3 In the remainder of this Chapter, we summarise the key insights and lessons from this literature. A detailed paper-by-paper review is contained in Volume 2, Chapter 4. Empirical papers on acquisitions of potential competitors are covered in Chapter 7.

6.1 Killer acquisitions and the loss of future product market competition

- 6.4 Several papers in the literature consider the simple case where an incumbent can acquire a target with a promising investment project (e.g. an R&D project to introduce a new, differentiated, product). This type of merger can harm consumers by: (i) leading to the discontinuation or delay of the target's R&D project (a 'killer acquisition'); (ii) leading to the discontinuation or delay of the acquirer's own R&D project (a so-called 'reverse killer acquisition'); and/or (iii) suppressing future (price) competition between the existing product of the incumbent and the new product of the target.¹³¹
- 6.5 The risk of discontinuation of the target's investment project after a merger is a manifestation of Arrow's replacement effect: the new product, if successfully introduced, diverts profits from the incumbent's existing product, reducing its incentive to develop it relative to an entrant.¹³² Even if the target's product is not discontinued (say, because the remaining development costs at the time of the merger are limited, or because the products are sufficiently differentiated), a loss of competition (absent efficiencies) is still expected due to the incumbent and the target not competing on price in the future product market. Competitive concerns about the loss of expected future competition (e.g. in the form of a 'killer acquisition'¹³³) are likely to be more significant the higher the pre-existing market power of the acquirer.
- 6.6 The papers in this literature also show that mergers may enable various synergies or efficiencies. These include development or commercialisation synergies that increase the success probability for the target's innovative project¹³⁴ or resource complementarities whereby an incumbent with sufficient resources to develop a project (including financial capital but also data, marketing, and distribution capabilities, and human capital) acquires a startup with weaker access to resources.¹³⁵ If sufficiently strong, such synergies or efficiencies can outweigh the loss of competition from acquisitions of potential competitors, and increase consumer welfare.
- 6.7 The acquisition price offered to the target may reveal important information on the viability of the target on a standalone basis, and on the competitive effects of the

¹³¹ Cunningham, Ederer, and Ma (2021), Motta and Peitz (2021).

¹³² As we saw in Chapter 4, Arrow's replacement effect is not always present, depending on the model of product differentiation. However, even when it is not present, a merger between an incumbent and a potential competitor with an innovative product may result in harm to consumers because of the reduction in future price competition—see for example Chen and Schwartz (2013), as discussed in Chapter 4.

¹³³ Cunningham, Ederer, and Ma (2021).

¹³⁴ Cunningham, Ederer, and Ma (2021).

¹³⁵ Motta and Peitz (2021), Fumagalli, Motta, and Tarantino (2022).

transaction. For example, a high acquisition price may (among other factors) indicate that the target is viable and a source of future competitive pressure (on a standalone basis or through acquisition by an outside investor).¹³⁶ However, it may also indicate that the smaller firm has a promising new technology that complements the assets of the acquirer.

6.2 Entry- or investment-for-buyout

- 6.8 The prospect of being bought out by an incumbent may increase the investment incentives of small innovative companies, and may be a natural exit route. This generates an *ex ante*, or pre-acquisition, effect on innovation (investment-for-buyout effect).¹³⁷
- 6.9 At the same time, the investment-for-buyout effect may result from the sharing of the market power rents created by the merger, as the merger avoids a dissipation of industry profits by removing future competition between the incumbent and the entrant. This mechanism may therefore rest on a reduction in the level of *ex post* (or *post-acquisition*) competition. There can therefore be a trade-off between a beneficial *ex ante* effect of more innovation, and an adverse *ex post* effect of less competition in the product market (in the form of discontinuation of R&D efforts or higher future prices).¹³⁸ The *ex ante* investment-for-buyout effect is stronger the greater the bargaining power of the potential competitor when negotiating the acquisition price with the incumbent.¹³⁹
- 6.10 Investment-for-buyout effects can also result from merger-specific diffusion synergies, for example the possibility that an incumbent can apply the product innovation of the target to the (larger) sales of its existing product.¹⁴⁰ This mechanism mitigates the potential *ex post* loss of competition from the merger.
- 6.11 Effects similar to innovation-for-buyout are at work if struggling competitors are allowed to exit a market by merging with incumbents (which has implications in particular for the standard of assessment under the 'failing-firm defence'). In this case, the prospect of being able to exit can stimulate *ex ante* investment by encouraging potential competitors to enter despite demand uncertainty.¹⁴¹ In this case too, there is a trade-off between *ex ante* and *ex post* effects. The bargaining power of the potential competitor when negotiating over the acquisition price also plays a role in this setting (with higher bargaining power associated with a more lenient optimal merger control policy). The analysis is more complex under conditions of asymmetric information on the level of demand for the target's products, as the entrant can then still extract rents due to private information about demand. In this case, a more lenient exit policy may stimulate investment and benefit consumers (relative to the counterfactual of a stricter exit policy, which deters entry) also if the potential competitor has no or limited bargaining power.¹⁴² Separately, in dynamic environments where market participants learn about the

¹³⁶ Fumagalli, Motta, and Tarantino (2022).

¹³⁷ Rasmusen (1988), Mason and Weeds (2013), Phillips and Zhdanov (2013).

¹³⁸ Letina, Schmutzler, and Seibel (2024), Motta and Peitz (2021), Fumagalli, Motta, and Tarantino (2022). Teh and Wang (2026) present a related stylised discussion.

¹³⁹ Mason and Weeds (2013), Phillips and Zhdanov (2013), Letina, Schmutzler, and Seibel (2024), Teh and Wang (2026).

¹⁴⁰ Mason and Weeds (2013), Phillips and Zhdanov (2013).

¹⁴¹ Mason and Weeds (2013).

¹⁴² Bisceglia, Padilla, Perkins, and Piccolo (2024).

success probability of a target's innovation over time, informational frictions can distort both the timing of acquisitions and innovation incentives.¹⁴³

- 6.12 The literature on investment-for-buyout effects assumes that the competition authority can pre-commit to either a stricter or more permissive merger review stance (e.g. via legislation or policy guidelines) in order to stimulate ex ante investment. However, if the legal standard requires the impact of each merger to be examined in isolation (as is typically the case), it may not be feasible in practice to apply a stance that takes into account ex ante investment-for-buyout effects that have already materialised by the time an acquisition is reviewed by an authority. However, the literature can shed light on the potential costs of under- and over-enforcement in relation to acquisitions of nascent competitors.

6.3 Strengthening of dominance, entrenchment, and kill zone concerns

- 6.13 Acquisitions of potential competitors can strengthen the position of incumbent firms, by increasing barriers to entry or expansion of *future* potential competitors. This can over time dilute or even reverse beneficial ex ante investment-for-buyout effects, and support a stricter merger control stance. The literature has identified several mechanisms whereby the acquisition of a potential competitor may lead to the *strengthening* or *entrenchment* of market power by discouraging entry, expansion, or investment by *future* potential competitors, to the detriment of consumers. Some of these mechanisms rest on the impact of mergers on investment-for-buyout effects in relation to *future* potential competitors.
- 6.14 One mechanism is based on the expansion of an incumbent's captive customer base enabled by a merger, which may in turn worsen the outside option of future entrants when seeking to be acquired by the incumbent and bargaining over the acquisition price.¹⁴⁴ This can lead to lower investment by future potential competitors and greater market power by incumbents over time, by reducing investment-for-buyout effects.
- 6.15 Incumbents may be able to suppress entrants' profits through a range of actions (e.g. input foreclosure, degradation of interoperability, tying), reducing entrants' standalone viability and their outside option, thereby weakening pre-merger innovation incentives.¹⁴⁵ An incumbent may also undertake investment to imitate a rival, which can worsen an entrant's outside option and discourage investment (a so-called 'incumbency-for-buyout' effect).¹⁴⁶
- 6.16 A merger with a potential competitor that has access to an investment project may also enable an incumbent to engage in a defensive strategy that deters other entrants from innovating in the incumbent's core market. This can lead to a so-called 'kill zone', where an incumbent firm's threat of counter-development discourages other entrants from developing products that would compete with the incumbent's core market, thereby diverting their R&D to other projects. This effect may co-exist with a killer acquisition by the incumbent (where the acquired investment project is discontinued). However, under some conditions, acquisition by the incumbent and subsequent shelving of the target's R&D project reduces the incumbent's competitive threat and creates a 'safe zone' inducing other firms to enter the core market of the incumbent.¹⁴⁷

¹⁴³ Guéron and Lee (2025).

¹⁴⁴ Denicolò and Polo (2026).

¹⁴⁵ Motta and Peitz (2021).

¹⁴⁶ Katz (2021).

¹⁴⁷ Teh, Banerjee, and Wang (2026).

- 6.17 Additionally, the presence of network effects can lead to entrenchment via mergers.¹⁴⁸ The anticipation of an acquisition by an incumbent platform can deter the adoption by users and developers of a platform competing with the incumbent, lowering its standalone value and expected acquisition price, and hence discouraging ex ante investment. In this case, stricter merger control raises the outside option of the potential competitor but delays any synergies from platform integration.
- 6.18 Finally, firms may engage in pre-emptive ‘acquihires’, with the purpose and effect of blocking rivals’ access to talent rather than to realise efficiencies.¹⁴⁹ Whether this talent hoarding is harmful depends on the acquiring firm’s match quality (how well the target’s employees fit into and raise productivity at the acquirer) and market structure (e.g. number of rivals, presence of a dominant firm, and sequential vs. simultaneous bidding), and it can raise hiring and layoff volatility for the acquired employees.

6.4 Direction of innovation

- 6.19 The prospect of acquisition may affect not only the level of innovation by potential competitors, but also its direction. This is another ex ante mechanism through which the prospect of a merger (and merger control policy) may affect consumers.
- 6.20 The literature identifies several mechanisms in this area. In some theoretical settings, the prospect of a merger with an incumbent leads potential competitors to innovate in a more incremental and less disruptive (or novel) direction, in order to maximise the rents from being acquired by the incumbent.¹⁵⁰ The innovation-for-buyout effect may also steer innovation towards solutions that benefit market leaders rather than weaker competitors, hence potentially leading to entrenchment.¹⁵¹
- 6.21 However, this mechanism is not always present, and depends on market settings. For example, a stricter (e.g. balance of harm) stance on the direction of innovation may actually encourage innovation aimed at challenging the position of the incumbent, while still benefiting consumers compared to a more lenient (e.g. balance of probabilities) approach.¹⁵² There are also settings where a permissive merger policy encourages more radical innovation by a potential competitor that competes with multiple incumbents engaged in R&D, as it increases the expected acquisition payoff for a successful disruptive technology relative to more incremental innovations.¹⁵³ Under a dynamic merger policy that accounts for firms’ pre-merger innovation choices, it may be optimal to approve mergers with positive probability in order to encourage innovation in a direction that intensifies product competition when a merger ends up being prohibited.¹⁵⁴
- 6.22 Relatedly, when multiple firms bid for a small innovative company, innovation can be strategically directed to shape bidding competition: prospective targets may tilt their innovation towards a lower-quality competitor to intensify rivalry and increase the expected acquisition price. When the weaker firm has greater absorption capability than the leader, this can enable the weaker firm to acquire the asset,

¹⁴⁸ Kamepalli, Rajan, and Zingales (2022).

¹⁴⁹ Benkert, Letina, and Liu (2025).

¹⁵⁰ Cabral (2018), Callander and Matouschek (2022).

¹⁵¹ Bryan and Hovenkamp (2020).

¹⁵² Cabral (2025).

¹⁵³ Bourreau and Gautier (2026).

¹⁵⁴ Gilbert and Katz (2022).

thereby leapfrogging the dominant firm. In other cases it can instead induce pre-emptive acquisitions by the leader.¹⁵⁵

- 6.23 Acquisitions can also affect the allocation of R&D across different markets (e.g. in a stylised setting where firms have fixed R&D budgets). By changing relative returns, acquisitions may redirect innovation towards markets with higher consumer surplus (potentially leading to beneficial out-of-market effects), or may instead affect the direction of innovation to the detriment of consumers.¹⁵⁶ This R&D portfolio effect can worsen or mitigate the ex post harm from higher prices, depending on the setting. In a setting where both an incumbent and an entrant choose the direction of their innovation, there is an additional strategic (incumbency-for-buyout) effect, which may outweigh the innovation-for-buyout effect (e.g. if the bargaining power of the entrant is small), increasing investment by the incumbent towards a contestable market, and tilting investment by the entrant towards a non-contestable market.¹⁵⁷
- 6.24 As with innovation-for-buyout, incorporating potential ex ante effects on the direction of innovation in the assessment of a given merger may not be feasible on legal grounds.

¹⁵⁵ Bedre Defolie, Biglaiser, and Jullien (2026).

¹⁵⁶ Dijk, Moraga-González, and Motchenkova (2024, 2025).

¹⁵⁷ Dijk, Moraga-González, and Motchenkova (2025).

7 Horizontal mergers: empirical evidence

Key points

- There is a large empirical literature on the dynamic effects of mergers. This includes both sector-specific and cross-sectoral studies. Methods vary from simpler correlational analyses to more robust causal inference and structural models (allowing for counterfactual analysis).
- Overall, the empirical papers reviewed find mixed effects from mergers on various input and output metrics for investment, with some papers finding positive effects and others negative effects. On consumer welfare effects, there is very limited evidence.
- Many of the papers reviewed have methodological limitations. Additionally, the estimated effects are typically averaged across mergers, making it difficult to draw inferences about any specific transaction. Moreover, empirical papers come with standard external validity concerns.
- Looking at pharmaceutical markets specifically, there is evidence that M&As may reduce early-stage research and R&D productivity, but can enhance later-stage development or innovation under some conditions (e.g. technological complementarities, presence of scale economies).
- In mobile telecommunications, some of the evidence shows mixed effects of mergers on investment (e.g. measured via capital expenditure), but causal inference is often limited by the implementation of merger remedies and broader market-wide changes. One structural econometric study finds that, assuming a fixed infrastructure network, short-run efficiency-driven quality gains (following economies of pooling and density) are insufficient to offset harm to consumers from increased prices.
- In digital hardware markets (e.g. semiconductors, hard-disk drives), empirical findings show that mergers and joint ventures may reduce patenting (including duplicative or defensive patenting that adds little innovation value) or inventor productivity, but may also generate synergies and increase R&D investment. In digital software markets, acquisitions may reduce innovation output by targets, although overall innovation (as captured by overall patenting activity) may still increase. Acquisitions by incumbents may harm consumers by deterring future entry from other small innovative companies, offsetting ex ante entry-for-buyout incentives. In the app market, there is also evidence that M&As may steer innovation away from adding new products and towards improving existing products.
- Big Tech acquisitions typically involve young and adjacent firms and may involve integration or discontinuation of the target. The empirical evidence on digital 'killer acquisitions' and innovation effects is mixed: some studies indicate reduced entry and venture capital funding (kill zones), although causality cannot always be established and some evidence points in the opposite direction.
- Merger effects in other sectors (e.g. general manufacturing, aviation, medical devices, banking, and display technologies) vary across contexts. The outcome depends on the specific industry, country, degree of technological overlap, and pre-merger competition between parties: mergers can lead to reduced competition, higher prices, or weaker innovation incentives, but in some cases this may be offset by efficiencies arising from increased innovation performance,

operational investments, learning-by-doing, coordination, and quality improvements.

- Cross-sectoral business studies consider the importance of organisational context, knowledge complementarity, and governance, although often with weaker causal inference. Industrial organisation studies focus more on causal inference, finding that mergers may reduce R&D intensity and increase market power but also may drive restructuring and efficiencies.
- Cross-sector corporate finance studies find that mergers usually benefit shareholders of the target and often create (modest) combined-firm value. Acquirer performance is however mixed and frequently weak—particularly when deals are unrelated, stock-financed, highly contested, or driven by poorly performing or overconfident managers. These studies shed light on potential synergies, even though the financial benefits identified in this literature may also reflect market power effects.
- Overlapping ownership may increase innovation when involuntary spillovers or technology transfers between firms owned by the same common shareholder are important, but reduce it when the internalisation of business-stealing effects dominates.
- Cross-sector studies looking at the acquisition of potential competitors find that these may reduce post-acquisition innovation and competition and entrench incumbent firms, but also spur pre-acquisition entry and innovation, with effects depending on the extent of overlap.

- 7.1 This Chapter presents an overview of the relevant empirical literature on the effects of horizontal mergers on investments and innovation—considering both mergers between incumbents and acquisitions of potential competitors.¹⁵⁸ This review focuses on academic publications and working papers, although non-academic publications are also reviewed where relevant. The papers discussed include both cross-sector and sector-specific studies. Particular attention is paid to sector-specific studies and the lessons drawn from them, as cross-sector studies necessarily face challenges in accommodating (sector-specific) heterogeneity in dynamic effects induced by mergers.
- 7.2 Given their diversity in issues related to dynamic effects and the number of relevant empirical papers, we look in particular at three sectors: **pharmaceuticals**, where dynamic considerations typically relate to innovation input and output in terms of R&D expenditure and intensity, patenting behaviour, and progression in clinical trials; **mobile telecommunications**, where dynamic considerations typically relate to network investments; and **digital markets**, where dynamic considerations typically relate to software development, platform integration, talent acquisitions, and venture capital (VC) financing (for this literature, we split the review between hardware markets, software markets, and acquisitions by Big Tech firms). Additionally, we examine a range of sector-specific studies that cover **other sectors**, including general manufacturing, aviation, medical devices, banking, and display technologies (LCD). Following the sector-specific studies, we review **cross-sector studies** from business and management studies, industrial organisation, and finance. We conclude with a review of empirical papers specifically on **potential**

¹⁵⁸ Some of the papers reviewed in this Chapter also consider non-horizontal mergers. This is pointed out where relevant.

competition and startup acquisitions (where the literature generally uses the term 'startup' to refer to small innovative companies).

- 7.3 Out of the 156 papers reviewed in this Chapter, 85 rely on descriptive statistics, qualitative empirical analysis (e.g. based on structured interviews), or cross-sectional or panel data regressions, often with various control variables to reduce concerns around omitted variable bias.¹⁵⁹ Separately, 57 papers use difference-in-difference (DiD) designs, where post-merger changes in the 'treatment group' are compared to a 'control group'—often simply taking non-merging firms as a control group. To reduce concerns around selection bias in treatment and control, 29 of these DiD papers construct a control group using matching techniques, such as propensity score matching, that go beyond simply comparing merging and non-merging firms,¹⁶⁰ of which two papers explicitly exclude direct competitors from the control group, to prevent equilibrium effects from having an impact on the estimated treatment effect of mergers on the outcome variables through the effect of mergers on the control group.¹⁶¹ Finally, 17 papers (also) use structural empirical methods, where a theoretical model is formulated and estimated or calibrated using actual data and counterfactual simulations of the effects of mergers on (equilibrium) outcomes.^{162,163} The distribution of empirical methods across the reviewed studies is summarised in Figure 7.1.

¹⁵⁹ This also includes event study papers from corporate finance that estimate the relationship between various variables and the merging firms' cumulative abnormal stock returns around merger announcement date.

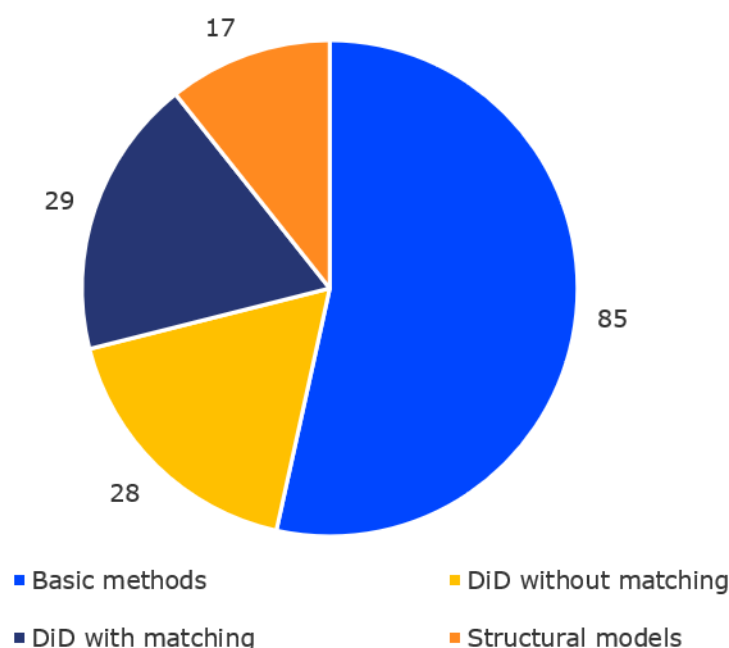
¹⁶⁰ Danzon, Epstein, and Nicholson (2007), Bertrand (2009), Ornaghi (2009), Zhao (2009), Desyllas and Hughes (2010), Aggarwal and Hsu (2014), Davis et al. (2014), Banerjee and Nayak (2015), Stiebale (2016), Entezarkheir and Moshiri (2018), Lee, Mauer, and Xu (2018), Haucap, Rasch, and Stiebale (2019), Informa Pharma Consulting (2020), Stiebale and Szücs (2022), Gibbon and Schain (2023), Schutz (2023), Alezra and Berquier (2024), Bonaimé and Wang (2024), Wang (2024), Berger, Calligaris, Greppi, and Kirpichev (2025), Gugler, Szücs, and Wohak (2025), Jin, Leccese, and Wagman (2025), He and Xue (2025), Doan and Mariuzzo (2026), Kummer, Mariuzzo, and Zhang (2026), Feyler, Heim, Spiegel, and Szücs (2026), Newham, Jaggi, and Posth (2026).

¹⁶¹ Haucap, Rasch, and Stiebale (2019) and Grajek, Gugler, Kretschmer, and Mişcişin (2019). Additionally, several references looking at mobile telecommunication mergers explicitly exclude competitors by looking at effects at a country-level. See for example Aimene, Jeanjean, and Liang (2021), as well as some of the policy reports discussed in Volume 2.

¹⁶² Papers undertaking structural estimation (i.e. econometric estimation of model parameters) include Li, Taylor, and Wang (2018), An and Zhao (2019), Doi and Ohashi (2019), Hounghonon and Jeanjean (2019), Igami and Uetake (2020), Wang and Wu (2020), David (2021), Igami, Kusaka, Qiu, and Tran (2025), Einfeld (2025), Elliott, Hounghonon, Ivaldi, and Scott (2025), Wabiszewski (2025), Dubois and Majewska (2026), and Lin, Tang, and Xiao (2026); and structural calibration (i.e. calibrating a model to data moments) include Cavenaile, Celik, and Tian (2021), Fons-Rosen, Roldan-Blanco, and Schmitz (2024), and Guthmann and Rahman (2026).

¹⁶³ The sum of papers across the different categories (159) exceeds the total number of papers (156). This is because three papers—Doi and Ohashi (2019), Einfeld (2025), and Dubois and Majewska (2026)—belong to two categories: they use both a difference-in-differences approach and structural estimation.

Figure 7.1 Distribution of empirical methods used in reviewed papers



Note: This figure illustrates the distribution of empirical design approaches used across the papers analysed in this chapter. The basic methods category covers correlational studies and other descriptive approaches. The DiD with matching category includes DiD designs that use matching techniques such as propensity score matching. The DiD without matching category covers all remaining DiD papers. Structural models refers to papers that formulate and estimate or calibrate a theoretical model.

Source: Oxera analysis.

- 7.4 Even when relying on quasi-experimental designs or structural estimation, empirical studies may suffer from methodological limitations which mean their results should be interpreted with caution. Box 7.1 sets out the key methodological considerations when reviewing and interpreting the results of these papers. A more detailed discussion of the key methodological considerations is provided in Volume 2, Annex A1.
- 7.5 Additionally, and more specifically in the context of innovation, many studies use patent-based measures for inventive output, which are not without problems. In particular, patents capture only a subset of inventive activity and vary in their technical and economic significance. These challenges have been known for a long time.¹⁶⁴ Schumpeter's distinction between inventions (new ideas) and innovations (new products and services) is useful in this context. Work continues on understanding the link between patents and innovations.¹⁶⁵



Box 7.1 Methodological considerations in empirical papers

Empirical studies provide an important source of evidence on the dynamic effects of mergers, including their impact on innovation, investment, and potentially consumer welfare. However, the reliability, interpretation, and policy relevance of such studies depend critically on their methodological design, identification

¹⁶⁴ Schmookler (1954), Griliches (1998).

¹⁶⁵ See e.g. Beneish, Harvey, Tseng, and Vorst (2022) and Gilbert and Villas-Boas (2025).

strategy, and data quality. This box summarises the key considerations relevant when assessing the strength of empirical evidence in the context of this Study.

A first fundamental distinction concerns internal and external validity.

- **Internal validity** refers to the question of whether the study at hand has been executed such that its results are correct for the studied setting and can be given the intended interpretation. In the case of merger studies, this means that the results are reliable for the set of firms, markets, and countries studied, for the time period that the data covers. In evaluating internal validity both the quantity and quality of the data must be considered, as well as the econometric methods that have been applied.
- **External validity** concerns the extent to which the findings can be generalised beyond the studied context, including to different industries, time periods, or institutional settings. Even well-identified results may have limited applicability if market conditions differ materially.

A large share of empirical merger studies relies on **difference-in-differences (DiD) methods**, which compare outcomes for merging firms (the treatment group) and an assumed to be unaffected control group before and after the merger. The credibility of such approaches depends on several key assumptions:

- **parallel trends:** absent the merger, the outcomes for the treatment and control firms would have evolved similarly;
- **appropriate control group selection:** while observation units in the control group should be as similar as possible to those in the treatment group, at the same time they should not be affected by the merger as this would endanger causal interpretation;
- **treatment timing:** mergers often occur at different points in time (staggered treatment), which—based on recent methodological insights—can lead to biased estimates unless an appropriate estimator developed for such a setting is applied;
- **treatment heterogeneity:** mergers differ in size, scope, and market context, yet empirical studies often estimate average treatment effects (i.e. without conditioning the average treatment effect on any observables), which may mask important variation across transactions.

Another key methodological concern is **endogeneity**, which arises when merger decisions are correlated with unobserved factors that also affect the outcomes such as innovation or investment. In a DiD setting, this problem may materialise as a violation of the common trends assumption unless the control group is carefully chosen. In other settings **instrumental variable (IV) methods** can address this problem, provided that the instrument satisfies two conditions:

- **instrument relevance:** the instrument is strongly correlated with merger occurrence;
- **instrument exogeneity:** the instrument affects the outcome variable of interest (e.g. patenting behaviour) only through its effect on the merger, and not directly.

Where valid instruments are available, the IV approach can strengthen causal inference. However, the resulting estimates may capture only the 'local treatment

effect', which is the average causal effect for those firms whose treatment status (i.e. whether the firm merged, in our case) was affected by the instrument, rather than the average effect for all firms. This may limit their general applicability and their policy relevance.

Some studies instead rely on a **structural model**. These combine economic theory with observed data to estimate behavioural and market parameters and simulate counterfactual merger scenarios. Such models allow researchers to explicitly model firm behaviour and competitive interactions, analyse mechanisms and welfare effects, and simulate alternative policy scenarios. However, their conclusions depend on modelling assumptions, functional-form choices, and identification strategies. Similarly, **calibrated general equilibrium models** allow analysis of broader, industry- or economy-wide effects, but typically rely more heavily on a restricted set of internally calibrated parameters and may capture institutional detail less precisely.

Finally, several additional methodological considerations are relevant when interpreting empirical findings.

- **Interpretation of mechanisms:** empirical studies typically identify reduced-form relationships and (in contrast to theoretical papers in industrial organisation) may not isolate specific economic mechanisms.
- **Average versus case-specific effects:** estimated average effects do not preclude substantial variation across individual mergers, to the extent that in an individual merger, the impact may be in the opposite direction of the estimated average treatment effect.
- **Measurement choices:** bias may be introduced through various common transformations of outcome variables (e.g. taking the log of a non-negative variable, to look at the percentage rather than absolute change, and adding one to the variable to avoid taking the log of zero, which is undefined) or use of proxy measures.
- **Statistical precision:** when interpreting the results of individual empirical studies, confidence intervals (while generally abstracted from in our review) can provide important information on the range of plausible effects, including when estimates are not statistically significant.

Source: Oxera.

- 7.6 In the remainder of this Chapter, we summarise the key insights and lessons from this area of the literature. A more detailed paper-by-paper review is contained in Volume 2, Chapter 5.

7.1 Pharmaceuticals

- 7.7 The pharmaceutical sector is one of the most studied sectors for the purposes of assessing the effect of M&As on innovation. This sector is characterised by continuous R&D efforts, innovations, new product launches, and by a high degree of M&A activity. Papers reviewed in this section consider both mergers between established incumbents and acquisitions of smaller, potentially nascent targets. They also study the impact of mergers on both innovation input and output at the

- firm level.¹⁶⁶ When looking at firm-specific indicators of innovation, the literature examines input metrics like R&D expenditure and/or output metrics like new drug approvals or patent counts. Looking at project-specific effects, several papers consider also clinical trial progressions or discontinuations.¹⁶⁷
- 7.8 There is a much wider empirical literature that also looks at the effect of M&As on (future) prices in pharmaceutical markets. We focus here on innovation effects, and only consider (future) price effects if they are also explicitly covered in the papers that we review.
- 7.9 The literature considers several potential channels or features through which pharmaceutical M&As can affect innovation, including reductions in research diversity following the elimination of parallel development paths,¹⁶⁸ organisational and human-capital disruptions,¹⁶⁹ anticompetitive motives (e.g. killer acquisitions),¹⁷⁰ proximity in the product market and technology space (including the mechanism of action (MoA)),¹⁷¹ and economies of scale.¹⁷²
- 7.10 Most of the empirical papers rely on a within-firm pre-/post-merger comparison, often using a control group to estimate the average treatment effect. However, most of the papers that use a control group do not explicitly exclude competitors from these controls, which means that the actual effect of a merger may differ from the estimated effect in either direction, depending on the effect of the merger on competitors.¹⁷³ Additionally, while all papers with a DiD approach use data that would necessitate the use of a 'staggered' design (as the mergers happen at different points in time), they may not use state-of-the-art methodology that addresses potential biases in such an approach. This methodological issue is discussed in more detail in Volume 2, Annex A1.
- 7.11 Nevertheless, there are several studies that rely on DiD designs with a matched control group that is plausibly unaffected by the impact of the merger on the treated group.¹⁷⁴ This is because the matched non-merging control group explicitly excludes competitors¹⁷⁵ or because the objects of interest are acquisitions of small firms.¹⁷⁶ The review also includes one study with a structural analysis.¹⁷⁷
- 7.12 Evidence from the studies reviewed indicates that pharmaceutical M&A, on average, can lead to both positive and negative effects on innovation. Some of the research shows that, on average across the sample, M&A activity leads to reduced

¹⁶⁶ E.g. Ornaghi (2009), Haucap, Rasch, and Stiebale (2019), Schutz (2023), Bonaimé and Wang (2024).

¹⁶⁷ E.g. Cunningham, Ederer, and Ma (2021), Malek, Seldeslachts, and Veugelers (2025), Wabiszewski (2025).

¹⁶⁸ Comanor and Scherer (2013).

¹⁶⁹ Paruchuri, Nerkar, and Hambrick (2006), Aggarwal and Hsu (2014), Verginer and Riccaboni (2025), Ornaghi and Cassi (2026).

¹⁷⁰ Cunningham, Ederer, and Ma (2021), Lear (2024).

¹⁷¹ Malek, Seldeslachts, and Veugelers (2025). Mechanism of action is the biological interaction involved in the drug achieving its desired end.

¹⁷² Wabiszewski (2025).

¹⁷³ Some papers that suffer from this limitation but are otherwise robust include Ornaghi (2009), Schutz (2023), Bonaimé and Wang (2024), Malek, Seldeslachts, and Veugelers (2025).

¹⁷⁴ This is also known as the stable unit treatment value assumption, or SUTVA.

¹⁷⁵ Haucap, Rasch, and Stiebale (2019); but also papers looking at mobile telecommunications, where performance by firms in other countries is used as comparators.

¹⁷⁶ See e.g. Cunningham, Ederer, and Ma (2021).

¹⁷⁷ Wabiszewski (2025).

patenting,¹⁷⁸ higher project termination,¹⁷⁹ and lower R&D productivity.¹⁸⁰ Positive effects of pharmaceutical M&A have been found in specific therapeutic areas (e.g. oncology) and late-stage development contexts (for example driven by economies of scale);¹⁸¹ or when technologically similar incumbents acquire related projects, creating technological synergies (e.g. combining similar patent-based knowledge and R&D expertise to improve development and generate more follow-on patents).¹⁸² M&As have also been found to lead to an increased focus on secondary development (i.e. incremental innovations focused on changing an existing product's label or manufacturing rather than discovering new ones).¹⁸³ Research has also emphasised that whether acquisitions spur or stifle innovation depends on the characteristics of the acquirer and the target not only in the product market but also in technology markets.¹⁸⁴ There is also evidence that common ownership (i.e. shared shareholders across firms) is negatively associated with market entry in the pharmaceutical industry.¹⁸⁵

- 7.13 Finally, there is empirical evidence indicating that not only merging entities, but also their rivals may reduce their innovation activities following a merger due to a reduction in competitive pressure.¹⁸⁶ Other evidence suggests that rivals to merging firms increase innovation to exploit the change in behaviour of merging firms engaged in post-acquisition integration.¹⁸⁷

7.2 Mobile telecommunications

- 7.14 Another sector that has been extensively examined with respect to the effect of mergers is the mobile telecommunication sector, particularly in Europe. One of the reasons is that European countries have experienced a number of high-profile mergers in national telecommunication markets, many of which reduced the number of mobile network operators (MNOs) from four to three.¹⁸⁸ Several of these mergers have been subject to merger control decisions by the European Commission (often resulting in remedies).
- 7.15 Studies that focus on investment (rather than only price) in mobile telecommunication mergers have focused on investment input measured by capital expenditure (CAPEX) and quality outcomes in terms of download speed or penetration. Much of this work relates concentration to investment input by exploiting variation across countries and over time in market structure (e.g. changes in the number of MNOs) and examining how these shifts correlate with operator- or country-level CAPEX.¹⁸⁹ By contrast, only a subset of papers examines investment outputs directly (for example, realised network performance or

¹⁷⁸ Haucap, Rasch, and Stiebale (2019), Schutz (2023).

¹⁷⁹ Cunningham, Ederer, and Ma (2021), Malek, Seldeslachts, and Veugelers (2025).

¹⁸⁰ Ornaghi (2009).

¹⁸¹ Wabiszewski (2025).

¹⁸² Malek, Seldeslachts, and Veugelers (2025).

¹⁸³ Schutz (2023), Bonaimé and Wang (2024).

¹⁸⁴ Malek, Seldeslachts, and Veugelers (2025).

¹⁸⁵ Newham, Seldeslachts, and Banal-Estañol (2025).

¹⁸⁶ Haucap, Rasch, and Stiebale (2019).

¹⁸⁷ Valentini (2016).

¹⁸⁸ An overview of recent European Commission mobile telecommunication mergers is provided in Volume 2, Table 5.1.

¹⁸⁹ Jeanjean and Hounghonon (2017), Genakos, Valletti, and Verboven (2018), Grajek, Gugler, Kretschmer, and Mişcişin (2019), Hounghonon and Jeanjean (2019), Bahia and Castells (2023).

coverage expansion), rather than treating CAPEX as a sufficient proxy for investment.¹⁹⁰

- 7.16 The European mobile telecommunication sector is unusual in the breadth of empirical work assessing the effects of mergers on both prices and investment. This includes peer-reviewed academic studies, some of which are written by authors affiliated with industry participants. Additionally, there is a body of reports produced by, or commissioned for, public institutions (competition authorities and sector regulators) and industry participants (telecommunication operators and industry associations such as the GSMA).¹⁹¹ We present, separately, peer-reviewed academic publications and reports from public institutions and industry participants.
- 7.17 As with pharmaceutical mergers, the wider literature on telecommunication mergers focuses primarily on price effects. Here we examine the effect of mergers on investment and quality outcomes, and consider price effects only if they are also explicitly covered in the papers reviewed.
- 7.18 We also note that a sector-specific identification challenge is that many of the mergers studied were implemented subject to remedies imposed by competition authorities (e.g. divestitures of spectrum and sites, or mandated access to mobile virtual network operators (MVNOs)). Estimated effects therefore typically capture the impact of the transaction subject to remedies, and cannot be interpreted as the effect of the merger absent those remedies. Since remedies may attenuate the effects on market power and/or constrain potential efficiency gains, this complicates comparisons across studies and the interpretation of average effects.
- 7.19 Starting with structural empirical papers, a recent empirical model using French data from October 2015 explicitly models demand and supply to assess the effect of market consolidation on investment and consumer welfare.¹⁹² The paper distinguishes between a short-run analysis, in which infrastructure is fixed, and a long-run analysis, in which infrastructure investment is endogenous. In the long-run analysis, the authors consider a setting with symmetric firms, in which total industry bandwidth is allocated equally across operators, and analyse how the number of firms affects prices, infrastructure investment, and download speeds. The reduction in the number of (symmetric) firms has a negative effect on consumer surplus when the market has fewer than five firms and a small effect when five or more firms are present. Consumer surplus is maximised when the market consists of eight firms.¹⁹³ By contrast, total surplus is maximised at four firms. In the short-run analysis (in which infrastructure is assumed to be fixed), the authors find that all hypothetical 4-to-3 mergers in their context reduce consumer surplus. While mergers generate efficiencies through economies of pooling¹⁹⁴ and, to a much lesser extent, economies of density,¹⁹⁵ leading in particular to improvements in network

¹⁹⁰ Wellmann (2019), Woroch (2020), Elliott et al. (2025), Lin, Tang, and Xiao (2026).

¹⁹¹ For all relevant disclosures, see Volume 2, Annex A2.

¹⁹² Elliott et al. (2025).

¹⁹³ As the authors note, these long-run counterfactuals should not be interpreted as merger analyses because they assume firms have symmetric bandwidth allocations and plan offerings. A merger between two firms would instead generally involve asymmetries. This is the same point as discussed in the context of the literature on competition and innovation in general—see in particular para. 4.3.

¹⁹⁴ Economies of *pooling* refer to the gains from combining networks and user bases so that stochastic demand fluctuations are smoothed, reducing congestion and allowing the same fixed network investment to serve more users or deliver higher quality (or to deliver the same service at lower cost).

¹⁹⁵ Economies of *density* refer to the improvements from serving a denser customer base—reducing average transmission distances and unit costs for a given level of network investment, which raises performance (e.g. download speed) or lowers incremental costs.

performance reflected in higher download speeds, these direct efficiencies are outweighed by the increase in market power from a consumer welfare perspective.¹⁹⁶

- 7.20 Separately, another recent structural empirical paper looking at 4G-LTE deployment in the US finds that a hypothetical T-Mobile/Sprint merger in 2016 would lead to asset complementarities and roll-out synergies from combining the parties' existing 3G and neighbouring 4G-LTE coverage.¹⁹⁷ However, the subsequent reduced competitive pressure would on balance reduce 4G-LTE deployment, despite the identified asset complementarities.¹⁹⁸
- 7.21 Reduced-form empirical academic papers looking at concentration (e.g. Herfindahl-Hirschman Index (HHI)) or consolidation (e.g. number of operators) and investment levels find that operator-level investment tends to rise with concentration or the loss of an independent operator.¹⁹⁹ The evidence on overall investment at the industry level, however, is less conclusive: one paper finds that industry-wide investment increases in the long run,²⁰⁰ whereas others find no conclusive evidence on its impact.²⁰¹ Finally, one study finds that in five specific ex post merger reviews, the merging and rival firms' prices generally moved in the same direction as investment—i.e. investment by all operators tends to increase only if their prices increase, whereas investment by all operators tends to decrease only if prices decrease.²⁰²
- 7.22 These studies primarily measure investment inputs (generally CAPEX), rather than output (e.g. coverage, speed). As a result, they provide limited direct evidence on whether and in what way consolidation affects consumer-relevant performance in terms of coverage, reliability, or download speeds (let alone balance these potential consumer benefits against potential consumer harms). The link between market-level CAPEX and consumer quality can be affected by different factors. For example, there may be significant economies of scale (e.g. economies of density or pooling) that can lead to increases in performance even if industry-wide investment does not increase. Additionally, CAPEX includes investment not related to the roll-out of mobile telecommunication infrastructure, and may follow investment cycles that further confound causal interpretations.
- 7.23 Studies that examine quality outcomes more directly find that increases in concentration (HHI) on average increase consumer quality in terms of wider coverage,²⁰³ improved network quality,²⁰⁴ or increased data efficiency and subsequent lower price per data.²⁰⁵ That said, this literature—as with the previous papers looking at consolidation and investment—often includes mergers that were

¹⁹⁶ Because infrastructure is treated as exogenous, the analysis does not capture the effects of mergers on investment incentives, whether positive (e.g. increased incentives to invest resulting from economies of pooling and density) or negative (e.g. the internalisation of negative investment externalities).

¹⁹⁷ Lin, Tang, and Xiao (2026).

¹⁹⁸ Lin, Tang, and Xiao (2026).

¹⁹⁹ Jeanjean and Hounghonon (2017), Genakos, Valletti, and Verboven (2018), Bahia and Castells (2023); see also GSMA (2020) for a similar finding by the mobile telecommunication industry association.

²⁰⁰ Jeanjean and Hounghonon (2017).

²⁰¹ Genakos, Valletti, and Verboven (2018), Bahia and Castells (2023); see also Frontier Economics (2015, 2021) and GSMA (2020) for a similar statistically inconclusive finding, and Lear et al. (2024) for a non-academic study that finds that industry-wide investment levels decrease.

²⁰² Grajek et al. (2019).

²⁰³ Woroch (2020).

²⁰⁴ Wellmann (2019).

²⁰⁵ Aimene, Jeanjean, and Liang (2021).

subject to remedies by competition authorities.²⁰⁶ As noted above, this affects the interpretation of any inferred causal effect. Additionally, some papers consider only changes in market concentration which, while directly affected by mergers, are also affected by other changes.²⁰⁷

- 7.24 Finally, even if quality indicators such as coverage or speed increase, this still leaves open the question of how such quality indicators are valued by consumers relative to any associated price effects.²⁰⁸ In general, the reduced-form designs discussed, while informative of potentially relevant effects, cannot recover the net consumer-welfare effects when quality improvements coincide with price increases.

7.3 Digital markets (excluding Big Tech acquisitions)

- 7.25 Another sector that has been studied significantly, particularly in recent years, is the digital sector. We first look at empirical papers studying the relationship between mergers and acquisitions on the one hand, and innovation on the other hand, in so far as they do not explicitly relate to acquisitions by Big Tech. In Section 7.4, we look at papers that analyse Big Tech acquisitions.
- 7.26 The papers considered span both digital hardware (electronic equipment, semiconductors, telecommunication equipment, and hard-disk drives) and digital software (e.g. cloud computing, enterprise software, or VC-backed technology startups that are predominantly software-focused). Methodologically, the studies range from early correlational panel regressions to more recent causal designs—including DiD methods.²⁰⁹ Innovation itself is measured in different ways, including R&D intensity (R&D expenditure over revenues), patent counts, patent citations, inventor productivity, new product introductions, cost reductions, entry responses, and VC funding. This variety in innovation metrics also indicates the measurement challenges and heterogeneity that exist when considering different digital technologies, making it hard to directly compare results across papers.
- 7.27 Looking at digital hardware, early studies on the semiconductor sector find that: higher R&D intensity is associated with fewer acquisitions (consistent with a make-or-buy trade-off);²¹⁰ both mergers and research joint ventures (RJVs) can generate synergies (reflected in increases in market share), with RJVs often achieving similar benefits with fewer competition concerns;²¹¹ and acquisitions are associated with reduced individual inventor output—especially when organisational mismatch or size disparities are large.²¹²
- 7.28 In the context of the telecommunication equipment industry, a reduced-form study finds that mergers improve firms' innovation performance (measured through patent-based indicators and R&D intensity), particularly for firms with strong absorptive capacity or weak prior R&D productivity.²¹³ This is consistent with acquisitions serving as a mechanism to access complementary technologies and overcome internal innovation constraints.

²⁰⁶ Jeanjean and Hounghonon (2017), Genakos, Valletti, and Verboven (2018), Wellmann (2019), Aimene, Jeanjean, and Liang (2021), Bahia and Castells (2023).

²⁰⁷ Woroch (2020), Bahia and Castells (2023).

²⁰⁸ For studies looking at the valuation of internet speed and quality or the drivers of broadband traffic, see Kenny, Kenny, and Gehan (2023) and Rabbani, Bogulski, Eswaran, and Hayes (2024).

²⁰⁹ Including matching, matrix completion, instrumental variables, and structural dynamic models.

²¹⁰ Blonigen and Taylor (2000).

²¹¹ Gugler and Siebert (2007).

²¹² Kapoor and Lim (2007).

²¹³ Gantumur and Stephan (2012).

- 7.29 Structural and quasi-experimental analyses of the hard-disk drive industry suggest that consolidation raises both market power and efficiency.²¹⁴ One structural study finds negative effects on R&D investment if the number of firms falls below four, with the effects plateauing if the number stays above four. Considering the effects on consumer welfare from the merger control stance, this study finds negative effects from adopting a prohibition threshold that allows for consolidation below three firms (with the effects plateauing with stricter merger prohibition thresholds).²¹⁵ Another reduced-form study of mergers in the hard-disk drive industry examines both innovation inputs and outputs—including R&D intensity, patents, new models, and quality-adjusted prices—and finds that R&D intensity increases post-merger, whereas patent intensity decreases on average, potentially reflecting reduced duplication and defensive patenting.²¹⁶ Citation intensity of the patents, however, increases post-merger. These heterogeneous effects may also be driven by the presence of merger remedies and the extent to which mergers allowed firms to realise technological synergies and eliminate duplicative R&D.
- 7.30 In digital software markets, the literature shows that M&As can affect innovation, entry, and investment through multiple channels and that results are heterogeneous across settings. One study finds that acquisitions are associated with increases in combined-patent counts, with stronger effects when the acquirer is a leading firm and a multi-sided platform—suggesting positive cross-group network effects.²¹⁷
- 7.31 Other studies adopting quasi-experimental methods find that acquisitions of innovative targets reduce target-level patenting, particularly for highly innovative firms.²¹⁸ However, this may also reflect IP centralisation, changes in appropriation strategies, or the discontinuation of overlapping projects (in ways that do not harm overall innovation outcomes) rather than a suppression of innovation, which are not separately identified.
- 7.32 Beyond patents, a recent study looking at enterprise software highlights dynamic entry effects: acquisitions by strategic incumbents can deter future startup entry (kill zone), but the prospect of being acquired encourages entry ex ante (entry-for-buyout).²¹⁹ The net effect (on entry) is found to be negative in case of a strategic acquirer (i.e. companies that are active producers of enterprise software such as the Big Tech firms Alphabet (Google), Apple, Meta (Facebook), Amazon, and Microsoft (often collectively referred to as 'GAFAM')), but not in the case of financial buyers or industry outsiders. This is irrespective of whether the acquirer and target are active within the same space. The same study estimates a structural model to identify counterfactual merger policies: blocking all acquisitions would reduce startup entry (by about 16%), but blocking only high-value transactions by strategic incumbents would increase entry slightly (by roughly 1–1.5%). At the same time,

²¹⁴ Igami and Uetake (2020), Bennato, Davies, Mariuzzo, and Ormosi (2021).

²¹⁵ Igami and Uetake (2020). Note that the adoption of a merger prohibition threshold at a particular level is not necessarily the same as an industry actually consolidating to that level, as the threshold constrains future mergers rather than directly determining market structure, which may endogenously settle at a higher or lower number of firms.

²¹⁶ Bennato et al. (2021). Defensive patenting refers to the practice of obtaining patents primarily to protect incumbents' market position by blocking, deterring, or complicating rivals' innovation—for example to deny competitors freedom-to-operate, discourage challenges, or secure bargaining power—rather than to advance new commercially used technologies.

²¹⁷ Doan and Mariuzzo (2026).

²¹⁸ Alezra and Berquier (2024), Bello, Rabellotti, Ravanos, and Smallenbroek (2025).

²¹⁹ Eisfeld (2025).

acquisitions can also generate strong signalling effects that boost venture capital investment and entrepreneurial activity in related niches.²²⁰

- 7.33 Finally, another recent study finds that M&As involving developers active in the US Google Play Store mostly affect non-price channels, rather than prices.²²¹ It finds that such M&As shift developer efforts away from expanding app portfolios and towards improving existing apps—thereby reducing new launches and acquisitions while slightly increasing updates.

7.4 Big Tech acquisitions

- 7.34 In this section we consider specifically papers that look at acquisitions by Alphabet (Google), Apple, Meta (Facebook), Amazon, and Microsoft, which we collectively refer to as GAFAM or Big Tech. Generally, these papers consider one or both of the two themes: potential concerns around killer acquisitions by Big Tech firms, where acquisitions lead to the discontinuation of the target's potential as a direct competitor to the acquirer; and effects on ex post entry activity and VC funding.
- 7.35 A descriptive study indicates that acquisitions are a common feature of the tech industry as a whole, and not only of GAFAM.²²² Several studies show that GAFAM firms tend to acquire targets that are young, active in adjacent or complementary technological or functional spaces (rather than direct horizontal overlaps), and which are frequently integrated or discontinued post-merger—with discontinuation more likely for younger, less-funded firms and for deals closer to the acquirer's core markets.²²³ A study found that around 26 of the 470 GAFAM acquisitions studied involved nascent competitors and hence potential killer acquisitions, with the authors concluding that the risk of killer acquisitions in digital markets appears relatively infrequent although it is non-negligible.²²⁴ However, these descriptive patterns do not, on their own, allow anticompetitive suppression of innovation to be distinguished from efficient integration or branding strategies, nor do they establish causal effects on innovation.
- 7.36 Papers that look specifically at innovation outcomes from Big Tech acquisitions, while generally aiming to isolate causal effects, do not provide a consistent picture. One patent-based analysis shows only temporary boosts in acquirer citations (relative to a matched control group of comparable non-acquired patents), followed by declines in domains where incumbents are already strong, consistent with possible defensive motives and anticompetitive effects.²²⁵ Another paper shows that acquisitions in technology markets have positive effects on patenting, and finds that the positive effect on innovation persists in technological domains where acquisitions are clustered. Moreover, the study finds evidence of positive effects on citations, which gets stronger with follow-on acquisitions.²²⁶ Finally, some papers find a non-trivial minority of plausibly harmful deals.²²⁷
- 7.37 Separately, two of the reviewed papers look at the effect on potential entrants, documenting that—relative to different control groups—acquisitions can deter subsequent entry and early-stage VC funding in technologically proximate

²²⁰ Jin, Leccese, and Wagman (2025).

²²¹ Kummer, Mariuzzo, and Zhang (2026).

²²² Jin, Leccese, and Wagman (2023).

²²³ Affeldt and Kesler (2021), Argentesi et al. (2021), Cunningham, Ederer, and Ma (2021), Gautier and Lamesch (2021), Jin, Leccese, and Wagman (2023), Gautier and Maitry (2024).

²²⁴ de Melo e Lemos and Mendes Resende (2025).

²²⁵ de Barys and Gautier (2024).

²²⁶ Ederer, Seibel, and Simcoe (2025).

²²⁷ Gugler, Szücs, and Wohak (2025), de Melo e Lemos and Mendes Resende (2025).

markets.²²⁸ This leads to concerns about potential 'kill zones', where market dynamism is discouraged because of strong incumbency. Some evidence points to persistent long-run declines in entry or financing (at the technology group level), despite possible short-run efficiencies.²²⁹ However, this may also be driven by market saturation effects.²³⁰ Moreover, other empirical evidence finds that GAFAM entry into a technology category through acquisitions is not associated with a subsequent slowdown in entry by other firms through M&A in the same category, providing no evidence of a 'kill-zone' effect on entry through acquisitions.²³¹

7.5 Other specific sectors

- 7.38 In addition to studies on pharmaceuticals, mobile telecommunications, and digital markets, there are also empirical studies on mergers and acquisitions and dynamic effects in other sectors.
- 7.39 The literature includes studies looking at the chemical industry, broad manufacturing, aviation (both aircraft manufacturing and airline networks), the medical sector (e.g. medical devices and procurement markets), banking, and display technologies (LCD panels). Across these papers, innovation is less uniformly centred on formal R&D pipelines, CAPEX, or software development, and more often linked to process innovation, learning-by-doing, organisational integration, and plant- or firm-level productivity.
- 7.40 Across these other sectors, the effects of mergers on innovation and performance are found to be highly context-dependent. For example, one early descriptive analysis of the chemical sector finds that innovation outcomes tend to be more positive when acquired knowledge bases are complementary and related to each other to some extent, while very large or highly overlapping technological portfolios are associated with weaker post-merger innovation, consistent with integration frictions and organisational disruption (though the study does not link this to cannibalisation effects, for example).²³²
- 7.41 Further evidence from manufacturing likewise points to mixed effects. A specific study on Swedish manufacturing data finds that productivity and cost efficiencies sometimes arise in association with a merger, particularly at the plant level, but often alongside downsizing and reallocation, suggesting selection and rationalisation rather than strong increases in innovative effort.²³³ For Dutch manufacturing firms, M&A involvement is associated with greater innovation performance, particularly for SMEs.²³⁴ By contrast, evidence from Spanish manufacturing firms suggests that M&As are associated with reductions in both internal and external R&D investment.²³⁵ At a more aggregated level, recent evidence from Chinese high-tech manufacturing suggests that M&A activity can raise average industry productivity through technology diffusion, although gains appear stronger for acquiring firms than for non-acquirers and may simultaneously widen productivity gaps.²³⁶
- 7.42 Studies of cross-border acquisitions, in which acquiring and target firms are located in different countries, add a further dimension, suggesting that acquisitions of

²²⁸ Kamepalli, Rajan, and Zingales (2022), Gugler, Szücs, and Wohak (2025).

²²⁹ Gugler, Szücs, and Wohak (2025), Koski, Kässi, and Braesemann (2026).

²³⁰ Koski, Kässi, and Braesemann (2026).

²³¹ Jin, Leccese, and Wagman (2023).

²³² Ahuja and Katila (2001).

²³³ Siegel and Simons (2010).

²³⁴ Cefis and Marsili (2015).

²³⁵ Cefis and Triguero (2016).

²³⁶ Wei, Wang, and Ma (2025).

foreign companies can increase R&D spending, innovation,²³⁷ productivity, and the adoption of new technologies and organisational practices in target firms,²³⁸ although the effects vary across industries and measures. Related evidence on horizontal common ownership points in a similar direction. Using data on manufacturing firms from 35 European countries, one study investigates common ownership (i.e. the same investors holding shares in multiple competing firms) between competing firms. It finds that common ownership is associated with higher mark-ups and, in industries that have high technological spillovers, with higher innovation output.²³⁹

- 7.43 Findings from other industrial settings are broadly consistent with evidence from manufacturing. In electricity generation, a recent structural study of acquisitions of US power plants finds efficiency gains, driven primarily by operational improvements, especially management changes, rather than capital investment, with more productive firms acquiring and improving underperforming assets.²⁴⁰ This suggests that, outside classic R&D-intensive settings, the dynamic effects of mergers may operate through operational improvement and asset reallocation, rather than through increased innovation incentives.
- 7.44 Similarly, structural studies in the aviation industry suggest that mergers can generate dynamic efficiencies through learning-by-doing, coordination, and quality improvements, such as changes in network structure (connecting versus nonstop flights). Specifically, one study shows that the merger generates dynamic efficiencies through learning-by-doing, leading to an overall increase in consumer surplus.²⁴¹ Another study shows that merger-induced changes in routing quality increase consumer surplus in non-overlapping markets, whereas deteriorations in routing quality reduce consumer surplus in overlapping markets.²⁴² A related study finds that a 2002 Japanese airline merger led to increases in flight frequency (a proxy for quality) of both the merging parties and non-merging parties and lower prices, leading to an increase in both consumer and social welfare in the simulated mergers (with statistically significant consumer welfare increases observed only on routes unaffected by the merger—i.e. non-overlapping routes), except on merger-to-monopoly routes, in which consumers faced increased prices.²⁴³
- 7.45 In the global LCD panel industry, a recent structural study covering the period between 2001 and 2011 finds that two thirds of all simulated mergers (including 7-to-6, 6-to-5, etc.) reduce innovation incentives, while one third of all simulated mergers increase individual firms' investment incentives.²⁴⁴
- 7.46 In the medical devices sector, one study finds that acquisitions in the US medical sector are associated with substantially greater post-acquisition reconfiguration, including product-line addition, retention, and deletion.²⁴⁵ Other work on medical devices and related equipment suggests that M&A is associated with higher patent counts but lower patent quality.²⁴⁶ Similarly, an ex post evaluation using

²³⁷ Bertrand and Zuniga (2006), Bertrand (2009).

²³⁸ Guadalupe, Kuzmina, and Thomas (2012).

²³⁹ Gibbon and Schain (2023).

²⁴⁰ Demirer and Karaduman (2024).

²⁴¹ An and Zhao (2019).

²⁴² Chen and Gayle (2019).

²⁴³ Doi and Ohashi (2019).

²⁴⁴ Igami et al. (2025).

²⁴⁵ Karim and Mitchell (2000).

²⁴⁶ Valentini (2012).

procurement data for medical equipment finds higher prices after merger, but no corresponding improvement in clinical quality.²⁴⁷

- 7.47 Finally, in the banking sector, several studies find that consolidation can lead to measurable cost reductions, such as eliminating overlapping operations and consolidating backroom operations,²⁴⁸ and that small, within-market mergers with modestly sized banks (e.g. <10% market share) can lower interest rates.²⁴⁹ However, an earlier study found that mergers involving banks with significant local market share increase interest rates due to market power.²⁵⁰ That said, the literature on M&As in banking has generally placed limited emphasis on potential dynamic effects.

7.6 Cross-sectoral studies

- 7.48 Several papers have investigated how M&As lead to dynamic effects, pooling data across sectors. In this section, we discuss these papers, segmented by field of study. First, we consider business and management studies, which typically focus on underlying organisational processes and concepts like ability and knowledge relatedness, but less on robust causal inference. Second, we discuss empirical papers from industrial organisation, which focus more on causal inference (e.g. by relying on matching and DiD approaches), though often with control groups that include competitors, or with a focus on clearly measurable proxies for innovation that may fail to capture relevant complexities. Finally, we discuss papers from the corporate finance literature, which focus on overall corporate performance and how this is linked to M&As. Unlike the industrial organisation literature, however, the papers in corporate finance generally do not distinguish between increased performance from efficiencies to the benefit of consumers and from effects from greater market power (e.g. higher prices)—looking instead at overall ‘synergies’ when considering stock market valuation, cash flows, or value-based productivity measures.

7.6.1 Business and management studies

- 7.49 The business and management literature discusses how M&As reshape firms’ internal organisational context in ways that are closely linked to subsequent innovation outcomes, with a central role for knowledge relatedness, governance structures, and absorptive capacity.
- 7.50 Some of the earlier literature finds that acquisitions are associated with declines in internally generated R&D intensity and patenting,²⁵¹ consistent with a shift of managerial attention and resources away from internal ‘championing’ of new ideas towards acquisition activity and with governance changes that emphasise short-term financial controls (e.g. quarterly earnings) over longer-term strategic controls (e.g. competitive positioning and strategic initiatives).²⁵²
- 7.51 At the same time, some studies discuss how the innovation effects of M&A are shown to be contingent on the nature of the knowledge bases being combined. Mergers between technologically or scientifically complementary firms, or those where the acquirer has a sufficiently large and diverse prior knowledge base, are often

²⁴⁷ Atella, Ceschin, and Decarolis (2025).

²⁴⁸ Houston, James, and Ryngaert (2001), Coccoresse and Ferri (2020).

²⁴⁹ Sapienza (2002).

²⁵⁰ Sapienza (2002).

²⁵¹ Hitt, Hoskisson, Ireland, and Harrison (1991), Hitt, Hoskisson, Johnson, and Moesel (1996), Cassiman, Colombo, Garrone, and Veugelers (2005).

²⁵² Hitt et al. (1991), Hitt et al. (1996).

associated with gains in R&D productivity, higher-quality or more novel inventions, and scope-based synergies, whereas mergers between highly substitutive or weakly related knowledge bases are more often associated with R&D cutbacks, rationalisation of overlapping projects, loss of key personnel, shorter innovation horizons, and integration frictions.²⁵³ One study also suggests nonlinear relationships, where moderate knowledge overlap is most highly associated with higher post-merger patent counts.²⁵⁴

- 7.52 Finally, financial structure and governance (e.g. leverage or takeover protections) can shape whether post-merger incentives support longer-term innovative investment or induce short-termism.²⁵⁵
- 7.53 Most of these studies rely on small and non-random samples, case studies, surveys, or within-firm pre-/post-comparisons without well-defined counterfactuals, all of which limits causal interpretation and generalisation.

7.6.2 Industrial organisation

- 7.54 The industrial organisation studies contain a set of papers that analyse mergers primarily as shocks to market structure and competitive incentives. They use large panel data sets and quasi-experimental designs to estimate causal effects on innovation proxies such as patents and R&D expenditure.
- 7.55 Generally, industrial organisation studies look at specific sectors, to reduce concerns around average results being driven by sector-specific omitted variables. Although these sector-specific studies were discussed earlier, there are also some cross-sectoral studies.
- 7.56 One study finds that acquisitions are not associated with a deterioration in internal innovation, unlike a matched control group of unsuccessful deals, suggesting the ability to offset internal innovative shortfalls.²⁵⁶ Additionally, mergers have been found to drive within-firm restructuring (in terms of narrower domestic product scope, reallocation towards core lines, and export expansion), consistent with efficiency motives.²⁵⁷ However, post-merger adjustments are also found to be asymmetric, with one study finding reductions relative to the control group in the R&D growth rate for the targets but not the acquirers, while the R&D intensity declines for both targets and acquirers, again relative to the control group.²⁵⁸ Additionally, an increase of the US merger notification threshold in late 2000 has been linked to drops in patent quantity and quality in horizontal deals that would previously have been scrutinised, suggesting that enforcement constrains transactions which would otherwise harm innovation.²⁵⁹ Another study finds that acquisitions redirect acquirers' innovation towards their targets' technologies, with 'dirty' acquirers of clean-technology targets (i.e. firms with technologies concentrated in high emission technologies) shifting most strongly.²⁶⁰ There is also evidence that technological acquisitions raise price markups mainly through the

²⁵³ Cassiman et al. (2005), Cloudt, Hagedoorn, and Kranenburg (2006), Desyllas and Hughes (2010), Makri, Hitt, and Lane (2010), Sears and Hitt (2023).

²⁵⁴ Cloudt, Hagedoorn, and Kranenburg (2006).

²⁵⁵ Hitt et al. (1996), Desyllas and Hughes (2010), Humphery-Jenner (2014).

²⁵⁶ Zhao (2009).

²⁵⁷ Chan, Irlacher, and Koch (2025).

²⁵⁸ Szücs (2014).

²⁵⁹ Morzenti (2023).

²⁶⁰ Newham, Jaggi, and Posth (2026).

consolidation of patent stocks rather than new inventive output, pointing to gains in market power via IP control.²⁶¹

- 7.57 Finally, two studies use general equilibrium models calibrated to real-world data to assess the long-run effects of mergers and merger policy on innovation and welfare.²⁶² One general equilibrium model calibrated on US merger policy finds that allowing all mergers reduces welfare by around 0.5% and lowers long-run annual growth by around 0.02 percentage points (from around 2.20%), as weaker enforcement reduces innovation by large, market-leading firms (referred to as superstars) and shifts M&A activity towards less synergistic mergers.²⁶³ Stricter merger control (defined as a 10% reduction in HHI investigation thresholds) increases welfare by around 2% and raises long-run annual growth by around 0.08 percentage points. Antitrust enforcement through merger control has only a modest effect on merger activity, with the annual merger rate increasing by around 0.02 percentage points (from around 3.83%) when all mergers are allowed.
- 7.58 Another paper similarly calibrates its general equilibrium model with US data but finds substantially larger effects of merger policy.²⁶⁴ Relative to the status quo, allowing all mergers reduces growth by around 0.25 percentage points (from around 2.34%) and increases the merger rate by around 7.15 percentage points (from around 4.45% to around 11.60%). The model predicts that both absolute innovation investment per firm and innovation intensity relative to firm value follow an inverted-U relationship with the number of firms, with absolute innovation investment highest in duopoly markets, while innovation intensity over value peaks around five firms. Among simple market-structure rules, welfare is maximised by a relatively strict rule that blocks mergers leaving fewer than five firms in a market, although welfare differences relative to thresholds of six or seven firms yield almost identical welfare levels. In addition, welfare differences relative to thresholds of three or four firms are also modest (especially compared to a policy of allowing all mergers).²⁶⁵ The main mechanism is that stricter merger control preserves innovation incentives, productivity growth from knowledge spillovers, and selection through entry and exit, whereas unrestricted mergers lead to excessive concentration and slower growth. The study also shows that simple price policies, by reflecting information on synergies and demand, outperform non-price policies based on concentration or number of firms.
- 7.59 Finally, one paper develops and estimates a structural general equilibrium model of M&As and analyses aggregate implications of M&A for the economy, showing that both complementarities between merging firms and productivity improvements are needed to explain observed merger patterns, and that M&A enhances aggregate productivity and output through resource reallocation and firm selection.²⁶⁶
- 7.60 Going beyond the dynamic effects of M&As on merging firms, one study focuses instead on the role of their non-merging rivals and finds them to reduce investment and innovation following a merger.²⁶⁷ Other studies focus on the role of RJVs as another form of inter-firm R&D collaboration and show that such arrangements can

²⁶¹ Martinez Cillero, Napolitano, Rentocchini, Seri, and Zaurino (2025).

²⁶² Cavenaile, Celik, and Tian (2021), Guthmann and Rahman (2026). Both papers and their theoretical models are also discussed in Chapter 5.

²⁶³ Cavenaile, Celik, and Tian (2021).

²⁶⁴ Guthmann and Rahman (2026).

²⁶⁵ Similar to e.g. Igami and Uetake (2020), this rule does not imply that every market consolidates to this threshold number of firms, with market structure jointly determined by entry, exit, and mergers.

²⁶⁶ David (2021).

²⁶⁷ Stiebale and Szücs (2022).

enable firms to internalise spillovers and improve performance through better absorption of external R&D.²⁶⁸

- 7.61 Finally, there is cross-sectoral evidence on cross-border M&A that such transactions can increase innovation in the acquiring²⁶⁹ and combined firm, particularly where the acquirer and target have strong pre-merger patent stocks,²⁷⁰ but that this innovation is reallocated geographically, with patenting rising in the acquirer's country and falling in the target's country.²⁷¹

7.6.3 Corporate finance

- 7.62 Finally, looking at corporate finance, the literature treats mergers as capital allocation and corporate control events, evaluating value creation mainly through stock returns, changes in cash flow, and productivity estimates.²⁷² This literature also sheds light on whether mergers succeed, from a shareholder perspective.
- 7.63 At the same time, shareholder benefits do not necessarily imply consumer benefits when market power effects cannot be excluded. Additionally, as with most other empirical papers, causal inference is limited by endogeneity and selection issues and average estimated merger effects may not generalise to individual transactions or typical competition authority cases.
- 7.64 Below, we summarise the key insights from the corporate finance literature on M&As and corporate performance, followed by insights specifically related to private equity acquisitions, common ownership, and labour market considerations.

Do mergers succeed? M&As and corporate performance

- 7.65 Whether M&As increase shareholder value depends on whether you consider the perspective of the target, acquirer, or both. Early literature shows that shareholders of targets consistently receive large positive stock price returns upon announcement, and capture most of the gains from M&As,²⁷³ with stronger bidding competition increasing premiums and shifting more value towards the target.²⁷⁴ Innovative targets and firms with valuable technological capabilities also experience higher stock price premiums.²⁷⁵ However, gains by targets do not necessarily represent equivalent aggregate value creation, as some may reflect transfers from acquirers caused by overpayment or managerial hubris.²⁷⁶
- 7.66 Considering the combined-firm level, announcement returns are generally positive, indicating expected aggregate value creation on average.²⁷⁷ Gains tend to be larger when 'strong' acquirers purchase 'weaker' targets (in terms of profitability relative

²⁶⁸ Röller, Tombak, and Siebert (1997), Banal-Estañol, Duso, Seldeslachts, and Szücs (2022).

²⁶⁹ Stiebale (2013).

²⁷⁰ Stiebale (2016).

²⁷¹ Stiebale and Reize (2011), Stiebale (2025).

²⁷² The corporate finance literature discussed here relates to mergers and corporate performance in general. A separate stream within the corporate finance literature considers how mergers affect firms' financing constraints, which is instead covered in Chapter 9.

²⁷³ Jensen and Ruback (1983), Bradley, Desai, and Kim (1988), Andrade, Mitchell, and Stafford (2001), Mulherin, Netter, and Poulsen (2017), Renneboog and Vansteenkiste (2019), Cumming, Jindal, Kumar, and Pandey (2023).

²⁷⁴ See in particular Bradley, Desai, and Kim (1988).

²⁷⁵ Wu and Chung (2019).

²⁷⁶ Roll (1986), Ismail and Mavis (2022).

²⁷⁷ Bradley, Desai, and Kim (1988), Andrade, Mitchell, and Stafford (2001), Mulherin, Netter, and Poulsen, (2017), Cumming et al. (2023).

to asset replacement value)²⁷⁸ and when operations, assets, or technologies are related or complementary.²⁷⁹ Nevertheless, the underlying source of these gains is difficult to identify, as they may reflect market power,²⁸⁰ as well as operating efficiencies or tax benefits.²⁸¹

- 7.67 Considering gains by shareholders of acquirers on announcement, these are usually close to zero on average, but vary substantially across deals.²⁸² Returns are weaker for unrelated diversification,²⁸³ acquisitions by poorly performing bidders,²⁸⁴ highly contested deals,²⁸⁵ and transactions associated with overconfidence or private benefits of control.²⁸⁶ They are stronger for experienced acquirers,²⁸⁷ cash-financed deals (which increase monitoring and control relative to stock-financed deals),²⁸⁸ and transactions involving specific technological gains.²⁸⁹ Earlier results on acquirer returns over the long run are mixed: some studies find underperformance, especially after stock-financed mergers,²⁹⁰ while others show that this result weakens with appropriate risk adjustment.²⁹¹
- 7.68 Considering instead corporate performance beyond stock price effects, results remain mixed. Some studies find improvements in cash flows, asset turnover, and operating margins, particularly in related mergers, but these effects are often weaker after controlling for firms' superior pre-merger performance.²⁹² Plant-level studies provide stronger evidence of efficiency gains, showing that assets move towards more productive owners and that retained plants improve productivity and margins.²⁹³ These gains often result from lower capital expenditure, employment, wages, and working capital rather than higher output.²⁹⁴ M&As can also improve innovation when firms combine complementary technologies,²⁹⁵ although some of the measured performance gains may instead reflect higher mark-ups or reduced competition.²⁹⁶ One study shows that a higher ex ante implied return-on-equity improvement (IRI) of a deal (which measures the minimum increase in target return on equity required for the acquirer to break even on the purchase price) predicts lower post-acquisition acquirer performance, with IRI also associated with CEO overconfidence, incentive misalignment, limited due diligence, and difficulty estimating synergies.²⁹⁷

²⁷⁸ Servaes (1991).

²⁷⁹ Healy, Palepu, and Ruback (1992), Devos, Kadapakkam, and Krishnamurthy (2009), Hoberg and Phillips (2010), Bena and Li (2014), Alhenawi and Krishnaswami (2015).

²⁸⁰ Devos, Kadapakkam, and Krishnamurthy (2009), Mulherin, Netter, and Poulsen (2017).

²⁸¹ Devos, Kadapakkam, and Krishnamurthy (2009).

²⁸² Jensen and Ruback (1983), Bradley, Desai, and Kim (1988), Andrade, Mitchell, and Stafford (2001), Mulherin, Netter, and Poulsen (2017), Renneboog and Vansteenkiste (2019), Cumming et al. (2023).

²⁸³ Morck, Shleifer, and Vishny (1990).

²⁸⁴ Morck, Shleifer, and Vishny (1990).

²⁸⁵ Bradley, Desai, and Kim (1988).

²⁸⁶ Roll (1986), Wang and Wu (2020), Ismail and Mavis (2022).

²⁸⁷ Hu, Li, Li, and Wang (2020).

²⁸⁸ Servaes (1991), Ghosh (2001).

²⁸⁹ Li (2013), Wu and Chung (2019).

²⁹⁰ Agrawal, Jaffe, and Mandelker (1992), Loughran and Vijh (1997).

²⁹¹ Franks, Harris, and Titman (1991).

²⁹² Healy, Palepu, and Ruback (1992), Andrade, Mitchell, and Stafford (2001), Ghosh (2001).

²⁹³ Maksimovic and Phillips (2001), Maksimovic, Phillips, and Prabhala (2011).

²⁹⁴ Devos, Kadapakkam, and Krishnamurthy (2009), Li (2013).

²⁹⁵ Hoberg and Phillips (2010), Bena and Li (2014), Alhenawi and Krishnaswami (2015).

²⁹⁶ Kepler, McClure, and Stewart (2026), see also Mulherin, Netter, and Poulsen (2017).

²⁹⁷ Ellahie, Hsieh, and Zhang (2025).

- 7.69 Several studies also show that pre-merger similarities between the acquirer and target in products and innovation activities influence the likelihood of a merger or acquisition taking place.²⁹⁸ This demonstrates the importance of using a robust control group when looking at post-merger behaviour. One of these studies also finds that mergers between technologically related firms are followed by higher post-merger patenting relative to comparable withdrawn bids.²⁹⁹
- 7.70 Considering whether patented or unpatented innovation drives merger synergies, one study finds that unpatented innovation (measured as the residual book value of acquired innovation minus patented innovation) explains a larger share of synergies (defined as abnormal stock price returns following merger announcement) than patented innovation.³⁰⁰ This highlights the limits of using patent-based measures of innovation.
- 7.71 Another study shows that the exclusion of internally generated intangible assets from US accounting rules enables some M&A deals to bypass US merger notification, and these non-reportable transactions are linked to higher deal premia, substantial post-merger markup increases (around 26 percentage points over three years), and greater discontinuation of overlapping innovation projects—particularly in intangibles-intensive sectors—indicating a softening of competition and increased market power.³⁰¹

Private equity acquisitions

- 7.72 Corporate finance studies of private-equity acquisitions provide related evidence on merger efficiencies in settings with little or no horizontal overlap, helping to distinguish operational improvements from market-power effects. One study finds that private-equity buyouts create value not only through leverage but also through higher operating performance, with stronger sales growth, higher margins, higher valuation multiples, and better outcomes where the expertise of the deal-partner matches the deal type (ex-consultants or ex-industry managers in the case of deals that focus on internal value creation; ex-bankers or ex-accountants in the case of deals that focus on external M&As).³⁰² A related study finds that private-equity buyouts mainly increase job and asset reallocation through closures, divestitures, and expansion in more productive units, with limited net employment losses overall and productivity gains in manufacturing.³⁰³

Common ownership

- 7.73 Other studies investigate overlapping ownership—in particular common ownership, whereby a common shareholder owns equity in competing firms—as another governance channel through which firms' innovation incentives may be affected.³⁰⁴ This literature finds that common ownership can facilitate technology transfers and the internalisation of spillovers across firms (improving information diffusion and increasing innovation), but can dampen innovation incentives by reducing the gains from business-stealing (especially where firms compete closely in product markets). The empirical evidence indicates that some firms experience increased innovation

²⁹⁸ Hoberg and Phillips (2010), Bena and Li (2014), Wu and Chung (2019).

²⁹⁹ Bena and Li (2014).

³⁰⁰ Beneish et al. (2022).

³⁰¹ Kepler, McClure, and Stewart (2026).

³⁰² Acharya, Gottschalg, Hahn, and Kehoe (2013).

³⁰³ Davis, Haltiwanger, Handley, Jarmin, Lerner, and Miranda (2014).

³⁰⁴ Kostovetsky and Manconi (2020), Kini, Lee, and Shen (2024), Antón, Ederer, Giné, and Schmalz (2025). Sector-specific studies that look at common ownership include Gibbon and Schain (2023) (European manufacturing) and Newham, Seldeslachts, and Banal-Estañol (2025) (pharmaceuticals).

and product dynamism with greater common ownership, while others reduce innovative effort. Relatedly, one study investigates the effect of horizontal minority stake acquisitions and finds that they lead to a persistent decline in innovation for both the acquiring and the target firm.³⁰⁵

Human capital, acquihires, and labour market power in M&As

- 7.74 There is also evidence from the corporate finance literature that mergers can serve as mechanisms for reallocating skilled labour and inventor talent through acquihires³⁰⁶ with stronger merger gains and post-merger productivity where firms' human capital is more closely related.³⁰⁷ Another study finds that increased labour market power after a merger can also reduce inventor mobility, earnings, and innovative output, because the merger weakens competition for talent, lowers inventors' outside options, and thereby reduces their incentives to innovate.³⁰⁸

7.7 Potential competition and acquisitions of small innovative companies

- 7.75 The sector-specific synopsis provided above includes papers involving potential competition and startup acquisitions—where the literature typically uses the term 'startup' to refer to small innovative companies. There are also additional cross-sectional studies examining how the prospect of entry or acquisition affects competitive behaviour, startup innovation incentives, post-acquisition innovation outcomes, overall growth, and consumer welfare. Overall, the evidence suggests that acquisitions of potential competitors can generate both procompetitive and anticompetitive effects: while the prospect of acquisition may encourage entry and innovation-for-buyout and ultimately increase consumer welfare,³⁰⁹ acquisitions may also reduce startup innovation, limit the implementation of acquired ideas, and entrench incumbent firms.³¹⁰ The balance of these effects is context-specific, and depends in particular on the degree of overlap between the merging firms and on whether ex ante innovation incentives outweigh the loss of future competitive pressure.³¹¹

³⁰⁵ Feyler, Heim, Spiegel, and Szücs (2026).

³⁰⁶ Ouimet and Zarutskie (2020).

³⁰⁷ Lee, Mauer, and Xu (2018).

³⁰⁸ He and Xue (2025).

³⁰⁹ Phillips and Zhdanov (2013), Fons-Rosen, Roldan-Blanco, and Schmitz (2024).

³¹⁰ Fons-Rosen, Roldan-Blanco, and Schmitz (2024), Berger et al. (2025).

³¹¹ See also Goolsbee and Syverson (2008) on the impact of future market entry on competitive behaviour by incumbents.

8 Non-horizontal mergers

Key points

- Non-horizontal mergers (vertical or conglomerate) differ from horizontal mergers in that they do not directly reduce competition between competing firms, including innovation rivalry.
- They can generate dynamic efficiencies to the extent that they eliminate double markups, resolve contracting frictions, mitigate hold-up concerns, and internalise positive investment externalities. These mechanisms can all enhance the investment incentives of the merged entity, to the benefit of consumers.
- Additionally, conglomerate acquisitions may help firms enter new markets or acquire capabilities (including data) that are complementary across products.
- However, non-horizontal mergers may reduce competition if integration raises the costs of rivals of the merged entity or weakens their incentives to invest, innovate, enter, or expand. This can happen either as a by-product of mechanisms that are in the first instance procompetitive, or through channels not associated with procompetitive effects (e.g. increasing rivals' risk of hold-up by independent suppliers). The theoretical literature shows that the balance of the pro- and anticompetitive effects is often ambiguous, and case-specific.
- Empirical evidence indicates that non-horizontal mergers can enhance investment and innovation—e.g. through improved coordination of interdependent R&D efforts, knowledge transfers across complementary businesses, or commercialisation of innovation—but that there are also circumstances where they may reduce entry and innovation by third parties.

- 8.1 This Chapter presents an overview of the literature on the dynamic effects of non-horizontal mergers—defined here as mergers involving producers of complementary products. This includes vertical mergers (between input providers and downstream customers) and conglomerate mergers (defined here as mergers of firms offering complementary goods to final consumers³¹²).³¹³
- 8.2 Non-horizontal mergers do not lead to a direct loss of price or investment competition between the merging parties. This means that some of the dynamic concerns discussed in the context of horizontal mergers do not apply. However, merging parties that offer complementary products at the time of the merger review may have become future direct competitors over time (in part because of their access to complementary assets). In that case, the merger can (also) have dynamic horizontal aspects, and the literature and insights reviewed in Chapters 5 and 6 of the main report apply.
- 8.3 Non-horizontal mergers may lead to potential benefits associated with the removal of contracting frictions between providers of complementary goods, including the

³¹² Although conglomerate mergers also include mergers involving unrelated or weakly substitutable products, the literature on conglomerate mergers generally considers the case of complementary products.

³¹³ A given merger may result in both horizontal and non-horizontal effects, e.g. it may involve a potential competitor that is currently active in a different part of the value chain, and hence may also lead to non-horizontal effects (see discussion in Part C).

mitigation of ‘hold-up’ problems, the internalisation of positive R&D externalities, and the elimination of double markups in the case of linear price contracting. Although horizontal mergers may also bring together complementary assets, this is more commonly the case in non-horizontal mergers.

- 8.4 That said, non-horizontal mergers may lead to adverse dynamic effects through the foreclosure of competitors of the merging firms—including both actual and potential competitors. This generally occurs when the merger reduces the incentives or ability of rivals to the merged entity to invest, innovate, enter a market, or expand. While there is a broad literature on the incentives for dominant integrated companies to foreclose competitors (which we briefly cover at the end of this Chapter), a key question in the context of merger control is the *overall* dynamic effect of a non-horizontal merger, taking into account both the potential efficiencies captured by the merged entity and any adverse foreclosure effects on non-merging parties.
- 8.5 Non-horizontal integration can suppress rival innovation or entry through different mechanisms, potentially leading to consumer harm. The literature identifies mechanisms that are harmful to consumers, and not associated with any countervailing benefits. For example, vertical integration involving one of only a few suppliers can expose downstream rivals to upstream holdup by independent suppliers, benefiting the merging parties independently of any merger efficiencies. However, in several settings, the harmful effects are closely associated with mechanisms that are in the first instance procompetitive. For example, vertical integration can eliminate double margins or increase supply assurance, which benefit the merging parties and may increase their ability or incentives to invest and/or innovate, but at the same time also suppress or reduce rivals’ ability or incentive to do so. A balancing of pro- and anticompetitive effects is therefore required in such cases.
- 8.6 In the remainder of this Chapter, we summarise the key insights and lessons from this literature. A more detailed paper-by-paper review is contained in Volume 2, Chapter 6.

8.1 Firm boundaries and resolving external contracting frictions

- 8.7 Much of the early economic literature on non-horizontal integration relates to the fundamental question of what determines the boundaries of the firm—i.e. which complementary activities are brought under the internal control of a single entity and which are instead coordinated externally via market transactions.
- 8.8 Building on the organisational economics literature, there are two related general perspectives: the **transaction cost economics (TCE)** literature, which explains the firm’s boundaries with reference to the reduction of transaction costs (e.g. contracting and governance costs) under conditions of incomplete contracts, uncertainty, and asset specificity;³¹⁴ and the **property rights theory (PRT)** literature, which explains the firm’s boundaries with reference to the allocation of residual control rights given their effect on ex ante investment incentives.³¹⁵ These two perspectives are closely related, as they both rest on the existence of incomplete contracts. The implication for merger control from this literature is that it provides a potential procompetitive motivation for non-horizontal integration, independent of any adverse strategic impact on competitors.

³¹⁴ Coase (1937), Williamson (1975, 1985), Klein, Crawford, and Alchian (1978).

³¹⁵ Grossman and Hart (1986), Hart and Moore (1990), Aghion and Tirole (1994).

8.2 Non-horizontal integration and static foreclosure

- 8.9 The industrial organisation literature considers the potential effects of non-horizontal integration on foreclosure of non-integrated rivals.³¹⁶ This literature shows that vertical integration can harm downstream competitors through an increase in their costs, potentially at the expense of consumers. This can be the case if the merger creates incentives to foreclose and if the merged entity can credibly commit not to supply rivals.³¹⁷ An alternative mechanism for anticompetitive foreclosure of rivals rests on the presence of opportunism by an unintegrated supplier absent a merger, which limits its ability to exploit its market power. In this setting, vertical integration is a mechanism to *restore* market power, to the detriment of consumers.³¹⁸ This may have adverse dynamic effects, notably if it involves a market for an important R&D input. At the same time, in the presence of uncertainty about retailers' costs and two-part tariffs, vertical integration can be procompetitive if it involves integration with a less efficient retailer and subsequent output shifting to the more efficient retailer through lower unit pricing.³¹⁹
- 8.10 Most of the early literature considers 'static' foreclosure in isolation—in the sense that the integration leads to an increase in rivals' costs and higher profits for the merged entity, independent of any exit or changes to investment by rivals. However, this type of foreclosure may also be dynamic in that a fall in the profit of rivals (e.g. through input or customer foreclosure) may also cause a competitor to reduce its investments or exit a market, as well as reduce incentives to enter a market.

8.3 Non-horizontal integration and incentives to invest

- 8.11 Some of the theoretical literature specifically considers the impact of non-horizontal mergers on incentives to invest and shows that such mergers are typically associated with trade-offs in terms of their positive dynamic effect on the merging parties and potential negative dynamic effects on non-merging parties ('outsiders'). The overall impact of these two effects on consumers is typically mixed, even though there are some settings where the literature shows that vertical integration is more likely to harm consumers.³²⁰
- 8.12 The literature on vertical mergers shows that these mergers may increase supply assurance, eliminate double markups, and mitigate hold-up problems,³²¹ thereby increasing appropriability of returns and encouraging investment by the merging

³¹⁶ Early papers looking at strategic motivations for vertical integration include Salinger (1988), Ordover, Saloner, and Salop (1990), and Hart and Tirole (1990). An overview is provided by Rey and Tirole (2007).

³¹⁷ Ordover, Saloner, and Salop (1990).

³¹⁸ Hart and Tirole (1990), Rey and Tirole (2007). Extending the framework of Hart and Tirole (1990), Baake, Kamecke, and Normann (2004) show that prohibiting vertical integration may reduce total welfare, despite preventing monopolisation, by disincentivising the upstream monopolist from properly investing in marginal production cost reductions. However, output still decreases and prices increase, which implies that consumer surplus decreases.

³¹⁹ Reisinger and Tarantino (2015).

³²⁰ See e.g. Bolton and Whinston (1993), Innes (2006), Buehler and Schmutzler (2008), Allain, Chambolle, and Rey (2016), Loertscher and Riordan (2019), D'Annunzio, Russo, and Shekhar (2024)—results discussed in more detail in the remainder of this section.

³²¹ Bolton and Whinston (1993), Buehler and Schmutzler (2008), Chen and Sappington (2010), Liu (2016), Chambolle and Guignard (2026), Israel and O'Brien (2025).

parties. Vertical integration may also be an efficient mechanism to increase appropriability and investment incentives in the presence of licensing frictions.³²²

- 8.13 At the same time, vertical mergers can reduce investments by non-integrated rivals. This may result from 'intimidation' or discouragement effects, where they face a more efficient integrated competitor which lowers expected returns from innovation,³²³ or from an increased exposure to a less competitive upstream market or opportunistic behaviour by other suppliers.³²⁴
- 8.14 In much of the literature on vertical mergers and investment, adverse effects on competitors derive from the fact that vertical integration allows the merged entity to capture efficiencies. As a result, the net effect of the merger on overall investment and consumer welfare is mixed. There are, however, settings where consumers suffer from a vertical merger, depending on market structure, contractual frictions, the type of conduct enabled by the merger (e.g. tying), and the stage at which innovation occurs (i.e. upstream or downstream).³²⁵
- 8.15 While much of the literature has focused on vertical mergers, comparable mechanisms and conclusions apply to conglomerate mergers. Integration between producers of complementary goods or platform participants can increase the merged entity's investment incentives, but may also reduce incentives for entrants or independent complementors to innovate.³²⁶ This can occur, for example, when integration reduces the profitability of parallel entry or raises rivals' costs in complementary products.³²⁷ In digital and platform settings, integration may allow firms to internalise network effects and strategically steer demand through their control over 'access points' (where consumers start their online journey and may be influenced by a platform to consume other complementary products). This can suppress external innovation and lead to underinvestment by third parties.³²⁸
- 8.16 Papers examining data-related conglomerate mergers and acquisitions find that their competitive effects are context-dependent. Acquiring or combining data can intensify competition, lower prices, and improve product quality by increasing incentives to expand data collection, attract users, and invest in data-enabled personalisation or quality improvements.³²⁹ However, it may also facilitate surplus extraction, defensive data acquisitions, tipping, and longer-run market entrenchment by enabling targeted pricing or advertising, allowing incumbents to deny valuable datasets to rivals, and reinforcing data-driven learning and feedback advantages.³³⁰ A merger is less likely to benefit consumers if data trade is possible in the absence of a merger.³³¹ Literature that looks more broadly at 'platform envelopment' (a strategy whereby a platform enters another platform market by bundling its existing services with those of the target market to leverage shared users, data, or technological components) shows how firms can leverage their assets to enter and dominate adjacent platform markets. They can do this through

³²² Brocas (2003).

³²³ Bolton and Whinston (1993), Innes (2006), Buehler and Schmutzler (2008), Allain, Chambolle, and Rey (2016), Loertscher and Riordan (2019), D'Annunzio, Russo, and Shekhar (2024).

³²⁴ Allain, Chambolle, and Rey (2016).

³²⁵ Buehler and Schmutzler (2008), Schmidt (2014), Liu (2016), Bryan and Hovenkamp (2020), D'Annunzio, Russo, and Shekhar (2024), Chambolle and Guignard (2026), Faulí-Oller and Sandonís (2025).

³²⁶ Choi (2008), Etro (2019), Heidhues, Köster, and Kőszegi (2024).

³²⁷ Choi (2008), Etro (2019).

³²⁸ Heidhues, Köster, and Kőszegi (2024).

³²⁹ Chen, Choe, Cong, and Matsushima (2022), de Cornière and Taylor (2024, 2025).

³³⁰ Chen et al. (2022), Hagiu and Wright (2023), de Cornière and Taylor (2024).

³³¹ De Cornière and Taylor (2024).

bundling or data integration strategies that may simultaneously create efficiencies and deter entry into the primary (data-rich) market (potentially to the detriment of competition and consumers over time).³³²

- 8.17 Separately, a broader capabilities-based perspective finds that mergers are driven not only by product complementarities, but by the accumulation of transferable, scarce capabilities, such as data, technology, and user relationships, which can enable entry into related markets.³³³ While combining distinct capabilities may enhance innovation and consumer benefits, mergers involving overlapping and scarce capabilities (including both horizontal and non-horizontal mergers) are more likely to weaken competition and limit future market entry and innovation.
- 8.18 In many of the papers reviewed, concerns around foreclosure and entrenchment through non-horizontal M&As are a by-product of a concurrent procompetitive effect, as noted above. However, the literature has also identified anticompetitive foreclosure mechanisms that are not associated with a concurrent procompetitive effect. For example, a vertical merger may result in degradation of the quality of access and support offered by the merged entity to downstream rivals.³³⁴ This makes those rivals more exposed to rival upstream firms, thereby generating hold-up of investment by the downstream rivals. Likewise, a market leader may acquire a complementary input provider, even though in the counterfactual the laggard may have instead acquired the input provider, leading to a more competitive outcome (in terms both of optimal diffusion of technology and incentives for the input provider to innovate in a direction that benefits the laggard rather than the leader).³³⁵

8.4 Foreclosure by integrated firms

- 8.19 In addition to the literature that specifically considers the impact of non-horizontal mergers, there is a broader literature that examines dynamic foreclosure by dominant and integrated firms.³³⁶ Taking vertical or conglomerate integration as *given*, this literature investigates the conditions under which foreclosure conduct is profitable for the integrated firm, and anticompetitive. For example, the literature shows how bundling and tying of complementary goods can deter entry by ensuring that the success of an entrant's innovation in one market becomes dependent on the success of innovation in the complementary market (by either the same or an independent entrant),³³⁷ or how foreclosure of competitors in an adjacent or vertically related secondary market can help prevent subsequent entry in the integrated firm's core market.³³⁸
- 8.20 This broader literature on pre-existing integration and dynamic anticompetitive foreclosure is instructive for the purposes of merger control as it analyses some of the foreclosure strategies that an integrated firm may implement in a post-merger situation. However, it does not provide for a full characterisation of the impact of a

³³² Eisenmann, Parker, and Van Alstyne (2011), Condorelli and Padilla (2020, 2024).

³³³ Chen, Elliott, and Koh (2023), Boa, Elliott, and Foster (2023). Boa, Elliott, and Foster (2023) present a capability-based approach that they explain is particularly valuable in non-horizontal mergers and innovation-driven settings, where conventional product-market analysis may provide limited guidance. However, the paper is primarily framed around horizontal mergers, and therefore is also relevant to those.

³³⁴ Allain, Chambolle, and Rey (2016).

³³⁵ Bryan and Hovenkamp (2020).

³³⁶ For literature reviews, see in particular Rey and Tirole (2007) and Fumagalli, Motta, and Calcagno (2018).

³³⁷ Choi and Stefanadis (2001), Choi (2004).

³³⁸ Carlton and Waldman (2002), Fumagalli and Motta (2020).

non-horizontal merger, as it typically starts from a counterfactual where the merger has already taken place. This literature also explores whether firms may rely on contractual solutions other than a merger to foreclose rivals (e.g. exclusive dealing).³³⁹

8.5 Empirical evidence

- 8.21 Empirical evidence shows that non-horizontal integration can affect investment incentives, although any effects are context-dependent. For example, one early study shows how corporate diversification can shift managerial attention towards newly acquired segments at the expense of existing operations.³⁴⁰ Another study finds that vertically integrated US hospitals and physicians invest more in environments with stronger contracting frictions (e.g. due to asset specificity and ex post bargaining), consistent with improved coordination and reduced hold-up.³⁴¹ One empirical test of the property rights theory confirms that integration strengthens investment incentives for the owning party while weakening those of the non-integrated partner.³⁴² Consistent with this mechanism, integration patterns vary with the relative technological importance of upstream versus downstream activities: taking a situation where downstream firms acquire upstream firms (i.e. backward integration), integration is more likely when innovation on the downstream side is more important and less likely when innovation on the upstream side is more important. Additional findings indicate that integration often facilitates the transfer of intangible capabilities rather than physical inputs.³⁴³
- 8.22 At the same time, some empirical evidence indicates a non-linear relationship: moderate integration enhances innovation through better appropriability and knowledge transfer, while high levels of integration can reduce it due to organisational rigidity and weaker absorption of external knowledge.³⁴⁴ Evidence on acquisitions and market dynamics further suggests that integration can support the commercialisation of existing innovation.³⁴⁵ Yet demand-side dynamics may alter the balance between efficiency gains and foreclosure effects, particularly in dynamic settings with strong competitive interactions,³⁴⁶ and acquisitions by strategic buyers may weaken ex post incentives for innovative entry.³⁴⁷
- 8.23 Papers examining the choice between acquisitions versus internal development in entering new markets find that firms may use acquisitions to fill capability gaps or extend into areas more closely related to their primary business domain, while they rely more on internal development for building new capabilities and pursuing more radical or distant diversification.³⁴⁸

³³⁹ Fumagalli and Motta (2020).

³⁴⁰ Schoar (2002).

³⁴¹ Ciliberto (2006).

³⁴² Acemoglu, Griffith, Aghion, and Zilibotti (2010)

³⁴³ Atalay, Hortaçsu, and Syverson (2014).

³⁴⁴ Li and Tang (2010).

³⁴⁵ Frésard, Hoberg, and Phillips (2020).

³⁴⁶ Sandiumenge-i-Boy (2025).

³⁴⁷ Eisfeld (2025).

³⁴⁸ Lee and Lieberman (2010, 2024).

9 Financial market frictions

Key points

- Mergers may improve firms' financing conditions by alleviating financial market frictions. This can improve the ability to invest, innovate, and enter or expand. However, these benefits are not always unique to mergers, and mergers may also lead to increased financing inefficiencies.
- Mergers may improve access to internal or external financing via the acquirer. For example, 'liquidity mergers'—defined as acquisitions in which a financially distressed firm is acquired by a liquid industry rival—may relax financing constraints and preserve productive assets that might otherwise be inefficiently liquidated.
- However, financial relief may not always require merging with an industry peer. Other forms of corporate financing that may resolve financing concerns also exist, without necessarily raising competition concerns (e.g. new equity issuance, private equity or VC funding, or bank lending).
- Diversifying mergers may create coinsurance by pooling imperfectly correlated cash flows, as well as internal capital market benefits by reallocating funds within the firm, potentially reducing distress risk and financing costs.
- Diversification can also generate costs through inefficient internal capital allocation, cross-subsidisation, agency problems, opacity, and, in some settings, risk contamination, contributing instead to a 'diversification discount'.

- 9.1 This Chapter presents an overview of the relevant literature on financial market frictions and how horizontal or non-horizontal mergers may potentially affect such frictions and the subsequent effects on corporate performance, including investment and innovation.
- 9.2 Financial market frictions arise because asymmetric information prevents capital from flowing to the most productive uses. The literature shows that asymmetric information with **adverse selection** (inability of the lender to observe, at least to the same extent as the borrower, the quality of the borrower's project) and with **moral hazard** (inability of the lender to observe post-contract actions of the borrower that affect the profitability of the project) may generate inefficient outcomes in the form of financial constraints and an increase in financing costs.³⁴⁹ Additionally, concerns around **debt overhang** may further discourage investment decisions, by discouraging firms with high existing debt burdens from undertaking projects with a positive net present value (NPV), since a share of the resulting gains accrues to existing creditors rather than shareholders.³⁵⁰
- 9.3 Depending on the extent of financial market frictions, M&As that increase a firm's scale or scope may resolve contracting problems and relax **external and internal financing constraints** by reducing informational frictions, making contracting solutions to these frictions easier to implement, or resolving debt overhang.

³⁴⁹ Seminal papers on adverse selection and moral hazard include for example Akerlof (1970), Jensen and Meckling (1976), Stiglitz and Weiss (1981), and Myers and Majluf (1984).

³⁵⁰ Myers (1977).

- 9.4 The majority of the papers discussed in this Chapter are from the corporate finance literature, which is particularly informative on the effects of internal capital financing, including those arising from mergers. This literature comes with several methodological caveats, similar to those discussed in, for example, Chapter 7 when considering the empirical evidence on horizontal M&A. First, whether an acquisition is ‘non-diversifying’ (i.e. horizontal) or ‘diversifying’ (i.e. non-horizontal) is generally determined based on whether firms are active in the same or different industries (e.g. based on two- or three-digit SIC codes). These classifications are based on supply-side similarities, with generally only a single classification per firm, rather than actual substitution patterns. Second, several empirical papers look at abnormal stock market returns as indicators of expected merger ‘synergies’, which may also—depending on the case—reflect potential anticompetitive effects. Third, many of the studies simply compare merging firms to non-merging firms, which can lead to concerns around self-selection and omitted variable bias. Some studies use matched controls to reduce these concerns, though they generally do not ensure that direct competitors are excluded from the control group (and hence any measured effect may reflect within-market equilibrium effects).
- 9.5 In the remainder of this Chapter, we summarise the key insights and lessons from this literature. A more detailed paper-by-paper review is contained in Volume 2, Chapter 7.

9.1 Non-diversifying mergers and acquisitions

- 9.6 The literature on non-diversifying mergers (defined in the reviewed papers as M&As taking place within the same two- or three-digit SIC code³⁵¹) generally focuses on **liquidity mergers**, defined as within-industry acquisitions in which a financially distressed firm is acquired by a liquid industry rival, with the effect of reallocating liquidity and avoiding inefficient liquidation.³⁵²
- 9.7 Although liquidity may also be provided by non-competitors, one paper finds that liquidity mergers are more common in industries where productive assets are highly specific to the industry, yet readily re-deployable among firms within that industry.³⁵³ This may, depending on the case, require an industry participant to acquire a financially constrained target. However, within-industry acquirers are not always necessary to realise these benefits, since improved access to external financing through the parent company and risk-pooling benefits may also arise in diversifying mergers.³⁵⁴
- 9.8 In principle, the concept of a liquidity merger resembles that of a failing firm within EU merger control, where assets would be inefficiently liquidated absent the transaction. However, one paper extends the definition of liquidity mergers to also include acquisitions that simply allow the acquirer to pursue otherwise forgone positive-NPV projects by the target.³⁵⁵ This relates also to the literature on startup financing through acquisitions—discussed in Section 9.4.
- 9.9 A related concern is that of fire sales, in which a shock forces distressed firms to liquidate productive assets, potentially to less-specialised, non-industry buyers at

³⁵¹ As noted above, this suffers from the limitation that actual rivals are not necessarily in the same industry code, and vice versa.

³⁵² Almeida, Campello, and Hackbarth (2011), Erel, Jang, and Weisbach (2015).

³⁵³ Almeida, Campello, and Hackbarth (2011).

³⁵⁴ Erel, Jang, and Weisbach (2015).

³⁵⁵ Duan and Jin (2019).

discounted prices.³⁵⁶ This is likely to happen when an entire industry is distressed and cash constraints prevent buyers from within the industry from bidding for the assets. The fire sale price involves a deadweight cost when buyers from outside the industry are lower productivity users than the industry sellers. A timely acquisition by industry participants may prevent such an outcome and keep assets with owners that are relatively high productivity users.

9.2 Diversifying mergers and acquisitions

- 9.10 Internal capital markets (ICM) refer to the process by which a firm allocates financial resources within its own organisational boundaries, rather than relying solely on external financing (e.g. banks or public markets). The corporate finance literature has long recognised that ICM activity affects how resources are allocated within diversified firms under financial frictions.³⁵⁷ Diversified firms can use control rights to reallocate internal funds more efficiently than external markets. This may take the form of ‘winner-picking’, where internal resources are channelled towards higher-return projects.³⁵⁸ It can also take the form of ‘efficient cross-subsidisation’ from more profitable business units to less profitable units, whereby abundant internal resources are reallocated in favour of the latter.³⁵⁹ However, informational asymmetries and agency problems exist within firms, implying that ICM can also magnify distortions through inefficient cross-subsidisation, rent-seeking and distorted investment driven by internal bargaining or managerial ‘empire-building’ incentives.³⁶⁰
- 9.11 Diversifying mergers create or expand the scope for ICM activity and give rise to a fundamental trade-off. On the one hand, combining firms with imperfectly correlated cash flows can generate **coinsurance gains** by reducing default risk, increasing debt capacity, creating additional tax shields,³⁶¹ and lowering the cost of capital.³⁶² These benefits are shown to be strongest when financially unconstrained firms acquire constrained targets,³⁶³ when acquisitions expand access to credit markets,³⁶⁴ or when firms face severe negative market shocks.³⁶⁵
- 9.12 On the other hand, diversification can potentially introduce significant inefficiencies within internal capital markets, including cross-subsidisation, rent-seeking, and distorted investment driven by internal bargaining or managerial incentives. These are mechanisms commonly associated with the so-called **diversification discount**, where conglomerate firms would trade at a discount on the stock market relative to what they would trade at if they were separate.³⁶⁶ Additional frictions arise from risk contamination, where joint financing can increase expected distress costs.³⁶⁷

³⁵⁶ Shleifer and Vishny (1992), Pulvino (1998, 1999), Meier and Servaes (2019), Franks, Seth, Sussman, and Vig (2022).

³⁵⁷ Stein (1997, 2003).

³⁵⁸ Stein (1997).

³⁵⁹ Cestone and Fumagalli (2005)

³⁶⁰ Stein (2003).

³⁶¹ Tax shield refers to the reduction in taxable income and tax liability resulting from deductible expenses, such as interest payments or depreciation.

³⁶² Lewellen (1971), Hubbard and Palia (1999), Fluck and Lynch (1999), Hann, Ogneva, and Ozbas (2013), Bielstein, Fischer, and Kaserer (2018).

³⁶³ Hubbard and Palia (1999), Williamson and Yang (2021).

³⁶⁴ Cornaggia and Li (2019).

³⁶⁵ Kuppuswamy and Villalonga (2016).

³⁶⁶ Rajan, Servaes, and Zingales (2000), Scharfstein and Stein (2000).

³⁶⁷ Banal-Estañol, Ottaviani, and Winton (2013).

- 9.13 Empirical evidence suggests that the net effect of diversification is heterogeneous: some papers find that diversified firms achieve more efficient capital reallocation and lower financing costs,³⁶⁸ others find that the benefits only arise during severe negative market shocks (e.g. during the 2008 financial crisis),³⁶⁹ and yet others that firms experience value destruction due to agency problems and organisational complexity.³⁷⁰ This last effect explains common cases of shareholder activism pushing for the unbundling of conglomerates.
- 9.14 The effects of ICM activity can extend to product market outcomes. There is evidence that entry into markets where incumbents are affiliated with cash-richer groups is harder, while entry by subsidiaries backed by cash richer groups is more likely to be successful. This provides support for the hypothesis that a group's deep pockets mitigate a firm's financial constraints and can enhance its competitive strength.³⁷¹

9.3 Mergers and acquisitions in general

- 9.15 The literature that looks at M&As in general (as opposed to specifically non-diversifying or diversifying deals) shows that such deals may alleviate targets' financial constraints through both informational and financing channels. Targets that face greater informational frictions and limited access to external finance tend to benefit more from being acquired, as integration with a larger or more established firm improves credibility and access to capital.³⁷² One study finds that corporate minority block acquisitions (defined as acquiring between a 5% and 50% equity stake) can ease financing frictions and increase investment, by providing certification and facilitating access to external funding, particularly when the acquirer has relevant industry expertise.³⁷³ Based on evidence from European firms, acquisitions have been found to enhance financial flexibility not through higher post-takeover earnings, but by easing internal financing constraints (through greater earnings retention and interest-free intra-group borrowing) as well as external financing constraints (strengthened via lower borrowing costs, higher debt issuance, more supplier credit, and faster receivables collection).³⁷⁴

9.4 Venture capital financing and exit

- 9.16 Increasingly, small innovative companies (often referred to as 'startups' in the literature) rely on acquisitions, as opposed to IPOs, to provide additional liquidity and facilitate venture capital exits, with evidence suggesting that IPOs occur less frequently and at a later stage of startup development.³⁷⁵ Evidence further shows that startups in more stable technological areas are more likely to be acquired, consistent with greater integration synergies arising from technological complementarity and incremental innovation.³⁷⁶ Moreover, startup acquisitions are shown to play an important corrective role in the presence of financing frictions, providing liquidity for startups that cannot secure follow-on funding and preventing

³⁶⁸ Doukas and Kan (2008), Hann, Ogneva, and Ozbas (2013), Bielstein, Fischer, and Kaserer (2018).

³⁶⁹ Kuppuswamy and Villalonga (2016).

³⁷⁰ Seru (2014).

³⁷¹ Boutin, Cestone, Fumagalli, Pica, and Serrano-Velarde (2013).

³⁷² Mantecon (2008).

³⁷³ Liao (2014).

³⁷⁴ El Ghoul, Gong, and Guedhami (2025).

³⁷⁵ Lerner and Nanda (2020), Ewens and Farre-Mensa (2020), Ederer and Pellegrino (2023).

³⁷⁶ Bowen, Frésard, and Hoberg (2023).

a premature shutdown of otherwise viable projects, especially in sectors or regions where financing constraints are more severe.³⁷⁷

³⁷⁷ Parry (2025).

10 Competition policy papers

Key points

- There are several papers that translate insights from the academic economics literature into competition policy recommendations for the assessment of dynamic merger effects.
- Papers in this policy literature generally build on the Arrowian and Schumpeterian perspectives on innovation and competition, and seek to establish guiding principles for the assessment of dynamic merger effects (e.g. focusing on the effects of mergers on *contestability*, *appropriability*, and *synergies*).
- The policy literature describes how mergers can harm innovation through different channels—including through internalised R&D externalities and reduced rivalry in R&D races. At the same time, the literature discusses how mergers can stimulate innovation through the combination of complementary assets, scale, technology transfers, greater appropriability, and R&D coordination.
- Some of these policy contributions call for a neutral, case-specific approach to innovation effects in merger control. Others emphasise that absent sufficient synergies or merger-specific internalisation of spillovers, the combination of competing innovators is likely to harm consumers, when considering the overall impact of both innovation and future price effects.
- The competition policy papers we review support an effects-based, evidence-led assessment of mergers, emphasising the need to focus on market conditions that can help determine whether a merger is likely to harm innovation or not, and the underlying mechanisms for dynamic harm and benefits.

- 10.1 The preceding Chapters considered the academic literature on competition, mergers, and dynamic effects, including theoretical and empirical literature from industrial organisation, business and management studies, corporate finance, and macroeconomics and growth theory. In this Chapter, we review selected competition policy contributions. We focus on competition policy contributions that are directly linked to the economic literature on dynamic merger effects. This means that we do not include here contributions which focus primarily on legal aspects of competition policy.
- 10.2 We make a distinction between earlier contributions, starting with papers by US scholars during the 2000s and early 2010s, and later literature looking specifically at the ‘innovation theory of harm’ motivated by recent merger decisions by the European Commission (notably Dow/DuPont (2017)). We also discuss subsequent competition policy contributions that have considered a broader set of dynamic concerns, including the acquisition of small innovative companies.
- 10.3 In the remainder of this Chapter, we summarise the key insights and lessons from this literature. A more detailed paper-by-paper review is contained in Volume 2, Chapter 8.

10.1 Academic work by US scholars (2000–‘15)

- 10.4 Competition policy work by US scholars in the period 2000–‘15 generally builds on the contributions of Arrow (whereby greater competitive pressure strengthens incentives to innovate) and Schumpeter (whereby appropriability and the prospect

of market power stimulate innovation efforts)—see discussion in Chapter 2.³⁷⁸ In an influential paper, Shapiro (2012) reconciles the two positions by emphasising that innovation outcomes depend on the interaction between three broad principles (*contestability*, *appropriability*, and *synergies*). More generally, this literature suggests that competition tends to stimulate innovation where firms face credible threats of displacement, where sales are contestable, or where firms are engaged in R&D races, while intense ex post competition and rapid imitation (e.g. due to weak intellectual property protection) may reduce expected returns and dampen investment incentives.

- 10.5 The literature distinguishes between product and process innovation, noting that effects can differ across settings:³⁷⁹ process innovations often generate gains in proportion to the innovator's existing output and may therefore be greater for larger incumbents, whereas product innovations more often depend on demand substitution, entry, and rivalry for new sales, so competitive pressure may play a stronger role in stimulating them.
- 10.6 Related specifically to merger control, some papers discuss a number of factors which can determine the impact of a merger on innovation.³⁸⁰ This includes in particular the following (not mutually exclusive) factors: (i) loss of rivalry between close innovators, (ii) internalisation of R&D and pricing externalities, (iii) combination of complementary assets and capabilities, (iv) changes in appropriability, (v) the scope for efficiencies such as improved scale or funding capacity, (vi) the availability of less anticompetitive alternatives, and (vii) the likely reactions of non-merging firms. These papers also emphasise the importance of evidence on pipeline overlap, innovation capabilities, entry conditions, commercialisation assets, and whether the merging firms are among a small number of important direct R&D rivals.
- 10.7 These early contributions differ in terms of their overall policy stance: whereas Katz and Shelanski (2007a) for example favour a broadly neutral presumption on innovation effects in mergers (except for mergers to monopoly), both Baker (2007) and Shapiro (2012) articulate a rebuttable presumption that mergers between a small number of important and direct R&D rivals are likely to harm innovation. All reviewed papers, however, support the proposition that innovation effects are economically important and should be assessed explicitly in merger review.

10.2 Policy papers relating to the 'innovation theory of harm' in Dow/DuPont

- 10.8 Since the European Commission's 2017 decision in Dow/DuPont, further competition policy papers have emerged—particularly from Europe—that looked specifically at a so-called 'innovation theory of harm' in merger control.³⁸¹ These papers generally focus on horizontal mergers between incumbents.
- 10.9 In general, these contributions set out how horizontal mergers between rival innovators may affect innovation and consumer welfare through several channels in both beneficial and harmful ways. On the harm side, a merger may reduce innovation rivalry and soften incentives to invest by internalising business-stealing effects between the merging firms, where innovation would otherwise divert (or

³⁷⁸ Gilbert (2006), Baker (2007), Katz and Shelanski (2007a), Shapiro (2012), Gilbert and Greene (2015). For an earlier contribution see Gilbert and Sunshine (1995).

³⁷⁹ For example Gilbert (2006).

³⁸⁰ Baker (2007), Katz and Shelanski (2007a), Shapiro (2012), Gilbert and Greene (2015).

³⁸¹ Federico (2017), Jullien and Lefouili (2018), Régibeau and Rockett (2019, 2022), Denicolò and Polo (2019), Federico, Scott Morton, and Shapiro (2020), Valletti (2025).

cannibalise) competing sales from a rival.³⁸² Mergers may also raise prices or reduce output, which can in turn lower incentives for process innovation (as the benefits of process innovation are typically proportional to output).³⁸³

- 10.10 This literature also identifies different settings in which harm to innovation may arise, including mergers involving overlapping R&D pipelines or future products, mergers between firms with similar innovation capabilities even absent current product overlaps, acquisitions of nascent or disruptive challengers by incumbents, and cases where firms compete to discover commercially valuable knowledge before products exist (as in a patent race). Concerns are generally strongest where there are only a small number of credible innovators, high barriers to entry, or durable first-mover advantages.
- 10.11 However, potential offsetting benefits from mergers are also identified. For example, this literature recognises that mergers may increase innovation by combining complementary R&D assets or generating economies of scale or scope in R&D, internalising involuntary technological spillovers, or facilitating voluntary technology or knowledge transfers between firms. Additional pro-innovation mechanisms discussed include: the elimination of duplicative research, which may allow firms to coordinate R&D efforts more efficiently across projects;³⁸⁴ and an increase in the appropriability of investment in product innovation, through higher margins in the product market.³⁸⁵
- 10.12 Some contributions argue that the potential for offsetting benefits negates the basis for adopting a presumption of harm to innovation from mergers,³⁸⁶ or may favour a more lenient treatment of some types of innovation-intensive mergers given the greater scope for efficiencies.³⁸⁷
- 10.13 Other papers emphasise that mergers which eliminate important innovation rivalry or future competitive threats are likely to result in consumer harm (taking into account detrimental effects on both innovation and future prices), in the absence of merger-specific innovation synergies, or the internalisation of spillovers.³⁸⁸

10.3 Further policy contributions

- 10.14 Later competition policy publications similarly highlight that innovation harms may arise where mergers remove actual or potential innovation rivalry, especially in concentrated R&D markets or where a dominant firm acquires next-generation capabilities.³⁸⁹ As one recent contribution summarises, mergers can harm innovation through three principal channels: the elimination of independent research paths, the restriction of rivals' access to key inputs, and the weakening of incentives to adopt and diffuse new technologies.³⁹⁰ Another recent paper, however, stresses that many mergers generate efficiencies and innovation effects are often ambiguous, suggesting that enforcement should not adopt a presumption that mergers harm innovation.³⁹¹ Instead, merger assessment should focus on case-

³⁸² See e.g. Federico (2017), Jullien and Lefouili (2018).

³⁸³ See e.g. Jullien and Lefouili (2018).

³⁸⁴ Denicolò and Polo (2019).

³⁸⁵ Jullien and Lefouili (2018).

³⁸⁶ Jullien and Lefouili (2018), Denicolò and Polo (2019), Régibeau and Rockett (2019, 2022).

³⁸⁷ Régibeau and Rockett (2019).

³⁸⁸ Federico, Scott Morton, and Shapiro (2020), Valletti (2025).

³⁸⁹ Baker (2019), Gilbert (2020), Padilla (2026), Peitz (2026). See also BRG (2026) for a recent policy study on the topic.

³⁹⁰ Peitz (2026).

³⁹¹ Padilla (2026).

specific conditions that can help distinguish innovation-reducing from innovation-enhancing transactions (including whether the merger creates complementarities, redirects R&D efficiently, strengthens lagging competitors, reallocates capital productively, or improves resilience).

- 10.15 Other recent contributions focus specifically on concerns relating to non-horizontal acquisitions and the acquisition of small innovative companies—referred to as ‘startups’.³⁹² One of these contributions argues that cross-market acquisitions may bring substantial efficiencies such as economies of scale and scope, faster entry, capability diffusion, complementarities, and stronger challenges to incumbents.³⁹³ In relation to innovation effects in startup acquisitions, a recent paper argues in favour of a clear distinction between pre- and post-acquisition innovation effects.³⁹⁴ It emphasises that killer acquisitions (a post-acquisition effect) are possible but should not be presumed, and that merger policy also shapes startups’ ex ante incentives to enter, invest, and innovate given the importance of acquisition as an exit route.
- 10.16 Specifically in relation to digital markets, a recent paper argues that merger assessment in this sector should place greater emphasis on dynamic competition, platform economics, data, and ecosystem effects, while adopting a case-specific approach that recognises both the potential anticompetitive risks and the potential efficiency gains of digital acquisitions.³⁹⁵

³⁹² Shapiro (2025), Teh and Wang (2026), Peitz (2026).

³⁹³ Shapiro (2025).

³⁹⁴ Teh and Wang (2026).

³⁹⁵ Fumagalli, Jullien, Lefouili, and Polo (2026).

11 Guidelines on dynamic merger effects and related principles

Key points

- Innovation is consistently recognised in guidelines by competition authorities as playing a central role in the process of competition between firms, to the benefit of consumers. In innovation-intensive markets, the assessment of the competitive impact of a merger needs to be forward-looking and consider dynamic effects to capture the full competitive significance of the merger.
- Market definition, and related concentration measures, need to be adapted when the theory of harm concerns innovation. Product markets may not always be defined around specific products, whether existing or new, but instead may need to be delineated by reference to the nature, scope, or objectives of, or assets involved in, innovation efforts.
- All major merger guidelines identify dynamic anticompetitive effects that may arise from horizontal and non-horizontal mergers.
- Guidelines recognise that anticompetitive horizontal effects may arise from the loss of competition in innovation between incumbents or from the loss of competition from a potential entrant, including both adverse effects on competition in innovation and on future product market competition.
- Anticompetitive non-horizontal effects may result from dynamic foreclosure and the associated increase in barriers to entry and expansion for potential rivals, but also from the merged entity's access to commercially sensitive information of its competitors (through its role as their supplier or customer).
- Currently only the US merger guidelines identify entrenchment of a dominant position as a standalone theory of harm which may lead to anticompetitive effects through mechanisms such as the increase in barriers to entry or expansion, and the elimination of a nascent competitive threat.
- Regarding procompetitive dynamic effects, existing merger guidelines typically recognise that mergers can generate dynamic efficiencies such as increased innovation from combining complementary R&D assets, aligning investment incentives, or reducing incremental innovation costs. Dynamic efficiencies are assessed within the well-established framework for procompetitive effects from mergers (i.e. they must be merger-specific, timely, and verifiable).

- 11.1 In this Chapter we provide an overview of the frameworks relevant for the assessment of dynamic merger effects in horizontal and non-horizontal merger guidelines. The focus is on the frameworks set out in the current horizontal and non-horizontal merger guidelines of the EU (2004 and 2008, respectively), the UK (2021), and the US (2023), as well as in the EC Market Definition Notice (2024), alongside the recommendations for the assessment of dynamic effects provided by the ICN (2025) and the OECD (2025). We also draw some insights related to dynamic competitive harm and benefits from the EC Horizontal Cooperation Agreement Guidelines (2023), specifically in relation to R&D agreements, and the US DOJ/FTC guidelines on IP licensing (2017). We do not discuss the draft merger guidelines published by the European Commission on 30 April 2026, since this Study

is part of the same review process, and will be an input in the finalisation of the guidelines, expected by the end of 2026.

- 11.2 In what follows, we first set out considerations around the role of innovation as a key dimension of competition and the implications for market definition and concentration measures in innovation-intensive markets. We then review the frameworks to assess dynamic competitive harm in horizontal and non-horizontal mergers respectively, concluding with a discussion of dynamic efficiencies. A more detailed review of each set of guidelines and recommendations is contained in Volume 2, Chapter 9.

11.1 Innovation as a parameter of competition and the competitive process

- 11.3 Competition authorities across jurisdictions have consistently recognised that competition is not limited to price rivalry but encompasses a broader process of rivalry in which innovation plays a central and increasingly prominent role. The European Commission's 2004 Horizontal Merger Guidelines (EC Horizontal Merger Guidelines) identify innovation among the benefits that effective competition delivers to consumers, and discuss the risk that mergers significantly increasing market power may diminish innovation.³⁹⁶ The 2008 Non-Horizontal Merger Guidelines (EC Non-Horizontal Merger Guidelines) confirm this position, treating a reduction in innovation as a potential adverse effect in both horizontal and non-horizontal contexts.³⁹⁷
- 11.4 The UK CMA's 2021 Merger Assessment Guidelines (CMA Merger Guidelines) recognise that mergers can harm the 'process of rivalry' between firms, undermining their incentives not only to cut prices but also to improve non-price dimensions of their offerings—including the introduction of new and better products—with innovation considered to be central to some merger cases.³⁹⁸ The 2023 US DOJ/FTC Merger Guidelines (US Merger Guidelines) adopt an analogous framing, noting that competition is a process of rivalry that incentivises firms to innovate, and that mergers may inhibit this process to the detriment of consumers.³⁹⁹ The 2025 ICN Recommended Practices for Merger Analysis (ICN Recommended Practices) place a significant emphasis on innovation as a parameter of competition, recognising that mergers can negatively affect the level of innovation through both horizontal and non-horizontal effects.⁴⁰⁰
- 11.5 Though not directly merger-specific instruments, both the 2023 EC Horizontal Cooperation Agreement Guidelines and the 2017 US DOJ/FTC IP Licensing Guidelines (US IP Guidelines) offer relevant conceptual building blocks. The EC Horizontal Cooperation Agreement Guidelines, in assessing horizontal R&D cooperation under Article 101, explicitly identify innovation competition as a distinct dimension of competition alongside product market competition, and distinguish between R&D directed at improving existing products and R&D aimed at developing new products or technologies—an analytical distinction that is directly applicable to

³⁹⁶ European Commission (2004), 'Guidelines on the assessment of horizontal mergers under the Council Regulation on the control of concentrations between undertakings', 5 February, 2004/C 31/03, para. 8.

³⁹⁷ European Commission (2008), 'Guidelines on the assessment of non-horizontal mergers under the Council Regulation on the control of concentrations between undertakings', 2008/C 265/07, 18 October, para. 10.

³⁹⁸ CMA (2021), 'Merger Assessment Guidelines', 18 March, paras 2.2 and 2.5.

³⁹⁹ US DOJ & FTC (2023), 'Merger Guidelines', 18 December, p. 1.

⁴⁰⁰ ICN (2025), 'ICN Recommended Practices for Merger Analysis', June. See also OECD (2025), 'Recommendation of the Council on Merger Review', June.

merger assessments.⁴⁰¹ The US IP Guidelines similarly emphasise that IP arrangements must be assessed not only for their static effects on price and output, but also for their potential to change innovation incentives and R&D competition.

- 11.6 Together, these instruments reflect a shared understanding that the competitive process is forward-looking and dynamic, and that static price-based analysis may be insufficient to capture the full competitive significance of a merger in innovation-intensive markets.

11.2 Considerations on market definition and market concentration

- 11.7 Defining the relevant market in dynamic contexts presents specific challenges, particularly where the competitive interaction between firms concerns pipeline products or early-stage R&D efforts rather than products already on the market.
- 11.8 The 2024 EC Market Definition Notice (EC Market Definition Notice) includes a dedicated framework for market definition in R&D-intensive cases. It distinguishes between: (i) pipeline products at an advanced stage, which may fit within existing product markets or constitute entirely new ones; and (ii) early-stage innovation efforts or research, where market boundaries are delineated by reference to the nature and scope of innovation efforts, the objectives of different lines of research, the specialisation of teams involved, and past innovation outcomes.⁴⁰² The latter concept draws on the notion of 'innovation spaces', developed most notably in the Dow/DuPont (2017) case. The EC Market Definition Notice also acknowledges that relevant geographic markets for pipeline products or early innovation efforts are often broader than those for already-commercialised products and may be global in scope.⁴⁰³
- 11.9 The US IP Guidelines offer a complementary framework from the IP perspective. Where IP rights are marketed separately from the products in which they are used, the US IP Guidelines recognise that a distinct technology market may be defined, consisting of the licensed technology and its close substitutes.⁴⁰⁴ Additionally, where R&D efforts concern the identification of a commercialisable product or the development of new or improved goods or processes, the US IP Guidelines acknowledge the appropriateness of defining a separate R&D market, comprising relevant R&D assets and their substitutes. Such a market should include firms with the capability and incentive to pursue closely substitutable R&D.⁴⁰⁵
- 11.10 With respect to market shares, the EC Market Definition Notice explains that in R&D-intensive contexts these may be measured by indicators such as R&D spending, patent counts, or patent citations, in addition to, or in lieu of, traditional output-based metrics.⁴⁰⁶ The US IP Guidelines similarly note that market share and concentration thresholds in innovation contexts may be assessed with reference to R&D capabilities.⁴⁰⁷ The US Merger Guidelines similarly acknowledge that markets may be defined around products that could result from successful innovation, even

⁴⁰¹ European Commission (2023), 'Guidelines on the applicability of Art. 101 of the Treaty on the Functioning of the European Union to horizontal co-operation agreements', 2023/C 259/01, 22 July, paras 136–138.

⁴⁰² European Commission (2024), 'Commission Notice on the definition of the relevant market for the purposes of Union competition law', C/2024/1645, 22 February, para. 92.

⁴⁰³ EC Market Definition Notice, paras 91–92.

⁴⁰⁴ US DOJ & FTC (2017), 'Antitrust Guidelines for the Licensing of Intellectual Property', 13 January, Subsection 3.2.2.

⁴⁰⁵ US IP Guidelines, Subsection 3.2.3.

⁴⁰⁶ EC Market Definition Notice, para. 109.

⁴⁰⁷ US IP Guidelines, Section 4.3.

if those products do not yet exist, consistent with the concept of pipeline products in the EC Market Definition Notice and the CMA Merger Guidelines.⁴⁰⁸ The US Merger Guidelines further note that separate relevant markets may be defined when assessing the innovation dimension of competition as opposed to other dimensions.

- 11.11 These considerations reflect a broader methodological shift in existing guidelines: traditional market definition tools (as applicable to existing products) need to be adapted when the theory of harm concerns innovation rather than static product price and quality effects.

11.3 Anticompetitive horizontal effects

- 11.12 This section considers horizontal effects arising from the loss of innovation competition and the loss of potential competition as identified in the guidelines (noting that not all merger guidelines set out these concerns separately).

11.3.1 Loss of innovation competition

- 11.13 Existing guidelines recognise that a merger between firms competing on innovation can eliminate or significantly reduce the intensity of innovation rivalry, even where the products in question are not yet commercially available. The US Merger Guidelines articulate this mechanism by explaining that a merged firm may have reduced incentives to continue or initiate development of new products that would have competed with those of the other merging party, since post-merger such products would ‘cannibalise’ the sales of the combined entity. This concern is expected to be greater when the merger involves two of a small number of firms with innovation capabilities in a specific area.⁴⁰⁹
- 11.14 A loss of innovation competition may relate to products already on the market, pipeline products at an advanced or early stage of development, or early R&D efforts relating to technologies still years away from commercialisation. The EC Horizontal Merger Guidelines similarly acknowledge that competition can be significantly reduced by a merger between two important innovators, particularly where those firms have competing pipeline products.⁴¹⁰ The EC Horizontal Cooperation Agreement Guidelines—though focused on cooperation agreements rather than mergers—articulate analogous dynamic competition concerns that are informative in the merger context. In particular, R&D cooperation (or, by analogy, a merger) may: (i) reduce or slow innovation;⁴¹¹ (ii) eliminate or reduce alternative independent R&D poles, removing alternative technological paths;⁴¹² and (iii) entrench market power by amplifying effects from the innovation stage into product markets.⁴¹³ The US IP Guidelines echo these concerns, noting that horizontal arrangements may result in harm to competition if they pose a significant risk of restricting the development of new or improved goods or processes, and giving as a specific example of harm an arrangement that coordinates the activities of two actual or potential competitors in R&D in a relevant field.⁴¹⁴
- 11.15 The CMA Merger Guidelines focus on the related concept of a loss of dynamic competition. This concerns the weakening of the ongoing incentives of the merging firms to pursue entry, expansion, or innovation efforts, including where the specific

⁴⁰⁸ US Merger Guidelines, p. 48.

⁴⁰⁹ US Merger Guidelines, pp 6–8 and p. 39.

⁴¹⁰ EC Horizontal Merger Guidelines, para. 38.

⁴¹¹ EC Horizontal Cooperation Agreement Guidelines, para. 54.

⁴¹² EC Horizontal Cooperation Agreement Guidelines, para. 168.

⁴¹³ EC Horizontal Cooperation Agreement Guidelines, paras 145–149.

⁴¹⁴ US IP Guidelines, Sections 3.1 and 3.3.

future product overlaps are not yet identifiable.⁴¹⁵ The UK CMA emphasises that even where entry or expansion is not itself ‘more likely than not’, the elimination of dynamic competition can give rise to a substantial lessening of competition (SLC), because the process of dynamic rivalry itself has present economic value—as it increases the likelihood of new innovations or products being made available.⁴¹⁶

11.3.2 Loss of potential competition

- 11.16 The EC Horizontal Merger Guidelines envisage two ways in which a merger with a potential competitor may harm competition: first, the potential competitor may already significantly constrain incumbents via a credible threat of entry (where entry does not entail significant sunk costs); and second, the potential competitor may constrain incumbents following actual entry (where a significant likelihood of entry exists). On this basis, the EC Horizontal Merger Guidelines identify two cumulative conditions for such a theory of harm to arise: (i) the potential competitor must already exert a significant constraining influence or there must be a significant likelihood that it would grow into an effective competitive force; and (ii) there must not be a sufficient number of other potential competitors capable of maintaining sufficient competitive pressure after the merger.⁴¹⁷ The EC Horizontal Merger Guidelines also recognise that competition can be harmed by a merger with a recent entrant that is likely to become a significant competitive force in the future.⁴¹⁸
- 11.17 The CMA Merger Guidelines distinguish between the loss of ongoing competition between an incumbent and a potential competitor before entry, which, as noted above, is referred to as dynamic competition, and the loss of future competition.⁴¹⁹ The latter arises where a merger eliminates competition between the merging parties at a future point in time following expected entry or expansion. The CMA Merger Guidelines assess the loss of future competition on a ‘more likely than not’ standard, examining whether either merging party would have entered or expanded absent the merger and whether the loss of that future competition would lead to an SLC.⁴²⁰ A sliding scale principle applies: the fewer the existing competitive constraints, the more significant the competitive impact of a potential entrant, such that the acquisition of a potential entrant by a firm with a strong market position may be concerning even if that potential entrant is expected to be small.⁴²¹
- 11.18 The US Merger Guidelines adopt a structurally similar framework under Guidelines 4 and 6. Guideline 4 distinguishes between the elimination of ‘actual’ potential competition (where one or both merging firms had a reasonable probability of entering or expanding into a concentrated market, including in markets where commercial products do not yet exist, thereby preventing future procompetitive effects from arising)⁴²² and the elimination of ‘perceived’ potential competition, where a firm perceived as a potential entrant exerts current competitive pressure on incumbents, for instance by prompting them to invest or improve their products.⁴²³ For both theories, a sliding scale applies: the higher the market concentration, the lower the required probability of entry to give rise to concern, and even an ‘attenuated’ source of competition may substantially lessen competition

⁴¹⁵ CMA Merger Guidelines, paras 5.17–5.23.

⁴¹⁶ CMA Merger Guidelines, para. 5.20.

⁴¹⁷ EC Horizontal Merger Guidelines, paras 58–60.

⁴¹⁸ EC Horizontal Merger Guidelines, para. 37.

⁴¹⁹ CMA Merger Guidelines, paras 5.2–5.3.

⁴²⁰ CMA Merger Guidelines, paras 5.7–5.16.

⁴²¹ CMA Merger Guidelines, para. 5.15.

⁴²² US Merger Guidelines, pp 11–12.

⁴²³ US Merger Guidelines, pp 12–13.

in highly concentrated markets.⁴²⁴ Guideline 6 addresses the elimination of nascent competitive threats by dominant firms, including situations where nascent competitors serve only narrow customer segments or offer partially overlapping services, and where the acquisition may be designed to prevent the rival from gaining sufficient customers and posing a stronger competitive constraint in the future.⁴²⁵

- 11.19 The US IP Guidelines are also instructive in that a firm is treated as a potential competitor if it is reasonably probable that it would have become a competitor in the absence of the IP licensing arrangement (or, by analogy, the merger), having regard to its capabilities and incentives.⁴²⁶

11.4 Anticompetitive non-horizontal effects

- 11.20 This Section considers guidelines on dynamic anticompetitive effects that may arise in non-horizontal mergers, in particular as a result of dynamic foreclosure, access to commercially sensitive information of rivals, or entrenchment.

11.4.1 Dynamic foreclosure

- 11.21 The EC Non-Horizontal Merger Guidelines acknowledge that non-horizontal mergers are less likely to eliminate competition directly between the merging parties, but recognise the risk of foreclosure effects—both static and dynamic. The well-established ‘Ability-Incentive-Effects’ framework is employed to assess whether the merged entity would have the ability and incentive to foreclose rivals, and what the market-wide effects of such foreclosure would be. Dynamic foreclosure concerns are particularly relevant where foreclosure raises entry barriers for potential competitors—for instance by compelling them to enter simultaneously at multiple levels of the supply chain (multi-level entry), or in regulated industries that are in the process of opening up to competition.⁴²⁷ Customer foreclosure may imply an additional dynamic dimension: by reducing the revenue streams of upstream rivals, it may erode their ability and incentive to invest in R&D, cost reduction, and product quality, potentially causing their exit in the long run.⁴²⁸
- 11.22 The EC Horizontal Cooperation Agreement Guidelines identify analogous foreclosure concerns in the R&D cooperation context, e.g. due to exclusive licensing.⁴²⁹ By analogy, a merger giving the combined entity control over key IP or technology assets—or enabling it to restrict access to such assets—may raise equivalent dynamic foreclosure concerns.
- 11.23 The US Merger Guidelines similarly address foreclosure of potential rivals under Guideline 5, noting that a merged entity limiting access to products, services, or routes to market may deter potential rivals from entering—particularly when entry would require parallel investment both in the relevant market and in substitutes for foreclosed inputs (multi-level entry).⁴³⁰ The threat of limited access can deter potential rivals from investing to enter, and Guideline 5 broadens the traditional framework by specifying that such effects may arise regardless of the merged firm’s intent or incentive to foreclose. The US IP Guidelines similarly note that in vertical relationships a licensing restraint may harm competition by anticompetitively

⁴²⁴ US Merger Guidelines, p. 12.

⁴²⁵ US Merger Guidelines, pp 18–21.

⁴²⁶ US IP Guidelines, Section 3.1, footnote 27.

⁴²⁷ EC Non-Horizontal Merger Guidelines, paras 49, 64, 75, 101, and 112.

⁴²⁸ EC Non-Horizontal Merger Guidelines, para. 65.

⁴²⁹ EC Horizontal Cooperation Agreement Guidelines, paras 54, 123(b), and 137.

⁴³⁰ US Merger Guidelines, pp 13–18.

foreclosing access to, or increasing competitors' costs of obtaining, important inputs—with the risk of foreclosure related to the proportion of markets affected, market concentration, difficulty of entry, and the duration of the arrangement.⁴³¹

- 11.24 The CMA Merger Guidelines note that foreclosure strategies (input, customer, and conglomerate) can affect not only current rivals but also potential ones, by raising barriers to entry, with adverse effects on customers potentially taking time to materialise fully.⁴³² Conglomerate effects are considered particularly pronounced in nascent and digital markets, where scaling is critical, rivals may be more easily marginalised, and the long-term competitive significance of market control is especially large.

11.4.2 Access to commercially sensitive information

- 11.25 A further non-horizontal dynamic concern—recognised by the EC Non-Horizontal Merger Guidelines, the CMA Merger Guidelines, and the US Merger Guidelines—is the risk that the merged entity gains access to commercially sensitive information of its upstream or downstream rivals through its role as their customer or supplier. This information may include rivals' innovation plans and R&D strategies, and access to it may allow the merged entity to compete less aggressively and deter rivals from innovating or entering, thus impairing the longer-term competitive process.⁴³³ The US IP Guidelines are also relevant in this context, noting that visibility into rivals' product improvements, innovations, or entry plans—which a vertical arrangement (or merger) may provide—can undermine rivals' ability or incentives to compete aggressively.

11.5 Entrenchment

- 11.26 The US Merger Guidelines identify 'entrenchment' of a dominant firm's market position as a standalone theory of harm. This includes a number of potential mechanisms including: raising barriers to entry and competition (e.g. via increasing switching costs, interfering with the use of competing alternatives and depriving rivals of scale economies or network effects); eliminating a nascent competitive threat; and extending a dominant position into another market (e.g. through a leveraging strategy).⁴³⁴ These mechanisms include both horizontal and non-horizontal effects, partially overlapping with the theories of harm described above.
- 11.27 The US Merger Guidelines note that, if there are entrenchment effects, a merger may lead to long-term harm to competition even if it may provide some short-term benefits, and that the merged entity may be willing to sacrifice short-term profits to increase its profits from dominance in the longer term.
- 11.28 The US Merger Guidelines distinguish anticompetitive entrenchment from the growth or development of a dominant firm as a consequence of increased capabilities or incentives.

11.6 Dynamic efficiencies

- 11.29 Existing merger guidelines typically acknowledge that mergers can generate dynamic efficiencies—procompetitive benefits going beyond static benefits from

⁴³¹ US IP Guidelines, Section 3.4.

⁴³² CMA Merger Guidelines, paras 7.2, 7.20, 7.29, and 7.36.

⁴³³ EC Non-Horizontal Merger Guidelines, para. 78, CMA Merger Guidelines, para. 7.3, US Merger Guidelines, p. 17.

⁴³⁴ US Merger Guidelines, pp 18–21, Subsection 2.6.A.

variable or marginal cost savings—and that these must be assessed alongside potential harms. The EC Horizontal Merger Guidelines highlight that corporate reorganisations in the form of mergers may enhance dynamic competition and increase the competitiveness of an industry.⁴³⁵ Consumers may benefit not only from lower prices but also from new or improved products and services stemming from efficiency gains in the area of R&D and innovation.⁴³⁶ A particular challenge for the assessment is that dynamic efficiencies may take longer to materialise than static efficiencies and may be harder to quantify, making it more difficult to satisfy the standard requirements of timeliness and verifiability.⁴³⁷

- 11.30 The EC Non-Horizontal Merger Guidelines recognise the scope for dynamic efficiencies in non-horizontal mergers, including the internalisation of positive externalities between complementary activities—for example, where integration at one level of the supply chain increases incentives to improve services or step up innovation at another level.⁴³⁸ Non-horizontal integration may also align investment incentives and avoid free-rider concerns in vertical relationships where competing downstream providers would otherwise underinvest.⁴³⁹ The EC Horizontal Cooperation Agreement Guidelines similarly recognise that R&D agreements—and by analogy mergers combining complementary R&D assets—can bring about efficiencies by combining complementary skills and assets, resulting in improved or new products and technologies being developed and marketed more rapidly, and may also lead to a wider dissemination of knowledge that triggers further innovation.⁴⁴⁰ In innovation-related efficiency claims, the EC Horizontal Cooperation Agreement Guidelines indicate that relevant evidence may include R&D spending, patents, estimated project timelines, and market impact.⁴⁴¹
- 11.31 The CMA Merger Guidelines treat dynamic efficiencies within the broader framework of rivalry-enhancing efficiencies, i.e. efficiencies that change the merging firms' incentives so as to make them stronger competitors, potentially offsetting anticompetitive effects and preventing an SLC from arising.^{442,443} The guidelines acknowledge dynamic efficiencies, such as increased innovation from combining complementary R&D assets or reducing incremental innovation costs.⁴⁴⁴ In addition, the CMA Merger Guidelines provide for 'relevant customer benefits'—including out-of-market innovation benefits—which, while not preventing an SLC, may outweigh it and its adverse effects.⁴⁴⁵
- 11.32 The US Merger Guidelines are comparatively succinct in their treatment of dynamic efficiencies, omitting the explicit recognition of R&D-based innovation efficiencies that featured in the previous version of the guidelines (of 2010). Out-of-market innovation benefits are effectively disregarded under the current US Merger Guidelines framework, since benefits outside the relevant market will not be

⁴³⁵ EC Horizontal Merger Guidelines, para. 76.

⁴³⁶ EC Horizontal Merger Guidelines, para. 81.

⁴³⁷ EC Horizontal Merger Guidelines, paras 83 and 86.

⁴³⁸ EC Non-Horizontal Merger Guidelines, para. 13.

⁴³⁹ EC Non-Horizontal Merger Guidelines, paras 14 and 57.

⁴⁴⁰ EC Horizontal Cooperation Agreement Guidelines, para. 141.

⁴⁴¹ EC Horizontal Cooperation Agreement Guidelines, paras 141–144.

⁴⁴² The CMA is currently reviewing its approach to assessing rivalry-enhancing efficiencies in merger cases, with revised draft guidelines published in June 2026, and revised guidelines expected to be adopted later in 2026. The main change expected will be greater emphasis and details on the assessment of dynamic efficiencies. See Volume 2, Chapter 9 for more details.

⁴⁴³ CMA Merger Guidelines, para. 8.4.

⁴⁴⁴ CMA Merger Guidelines, para. 8.11.

⁴⁴⁵ CMA Merger Guidelines, paras 8.4–8.5.

credited if there is a lessening of competition within it.⁴⁴⁶ The US IP Guidelines, by contrast, include a more nuanced approach to dynamic efficiencies. They note that horizontal agreements may yield integrative efficiencies, including economies of scale and the integration of complementary R&D, production, and marketing capabilities.⁴⁴⁷ In the case of non-horizontal arrangements, a combination of the licensor's IP with the licensee's complementary factors of production may align incentives to promote the development and marketing of licensed technology, or enhance competition by substantially reducing transaction costs.⁴⁴⁸ Across both horizontal and non-horizontal contexts, the US IP Guidelines apply a sliding scale principle: while the balancing of harms and efficiencies is necessarily qualitative, greater evidence of efficiencies is required as expected anticompetitive effects increase, and efficiencies cannot be credited if they could be achieved by significantly less restrictive means.⁴⁴⁹

- 11.33 Across all frameworks, dynamic efficiencies need to be assessed within the well-established framework for procompetitive effects from mergers: they must be merger-specific, timely, and verifiable. Given the inherent uncertainty surrounding innovation outcomes and the extended time horizons over which such efficiencies typically materialise, satisfying these requirements in practice may be challenging.

⁴⁴⁶ US Merger Guidelines, p. 32.

⁴⁴⁷ US IP Guidelines, Section 5.1.

⁴⁴⁸ US IP Guidelines, Section 3.4.

⁴⁴⁹ US IP Guidelines, Sections 3.4 and 4.2.

PART C CONCEPTUAL FRAMEWORK

12 Introduction and overview of conceptual framework

Key points

- Part C of the main report provides a conceptual framework for analysing the potential dynamic effects of mergers, grounded in the relevant academic literature.
- The framework adopts an *Expected Consumer Welfare* (ECW) standard, accounting for uncertainty in the evaluation of dynamic merger effects.
- The framework is built around four main categories of dynamic merger effects:
 1. **Box 1:** *Anticompetitive dynamic effects from horizontal mergers.* These effects reflect the fact that a horizontal merger eliminates competition between the merging firms both in investment and in future product markets (for a given level of investment). The resulting internalisation of negative investment and price externalities is likely to result in dynamic harm to consumers, absent spillovers and synergies.
 2. **Box 2:** *Procompetitive dynamic effects from horizontal mergers,* reflecting the fact that a merger may allow firms to internalise spillovers, potentially increasing their innovation incentives (**Box 2A**), and that it may improve the ability of the merged entity to invest (**Box 2B**).
 3. **Box 3:** *Dynamic foreclosure effects from non-horizontal mergers,* whereby a merger of complementary activities gives rise to or strengthens the ability and incentive of the merged entity to engage in conduct that reduces rivals' investment, to the detriment of consumers.
 4. **Box 4:** *Procompetitive dynamic effects from non-horizontal mergers,* including the internalisation of positive price and investment externalities (**Box 4A**), dynamic synergies (**Box 4B**), and investment-enhancing effects from the resolution of contractual imperfections (**Box 4C**).
- The conceptual framework separately considers potential *entrenchment effects*, capturing the fact that a merger may entrench the position of an acquirer with significant or substantial market power through anticompetitive mechanisms beyond those captured under Boxes 1 and 3. These mechanisms may reduce contestability and discourage future innovation and entry by rivals, even absent explicit foreclosure conduct, and may lead to consumer harm.
- The conceptual framework is applied to specific categories of canonical mergers, identifying the most relevant economic mechanisms, qualitative and quantitative indicators, and practical sources of evidence for each type of case.

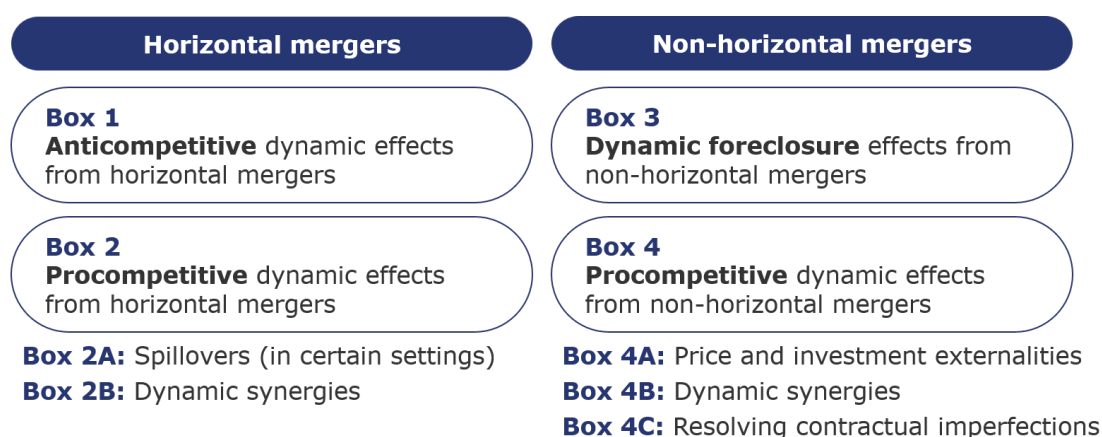
- 12.1 Part C of the main report sets out a conceptual framework for the analysis of dynamic merger effects. The objective of this framework is to provide a comprehensive and coherent characterisation of the possible dynamic effects of mergers, including both potential anticompetitive and procompetitive effects. The framework is the core of this Study, in line with the requirements of the Tender Specifications.
- 12.2 The conceptual framework rests on the academic literature reviewed in Part B of the main report. It builds in particular on the theoretical insights from the literature,

with specific attention to research that has looked explicitly at the dynamic effects of mergers (as opposed to the impact of changes in competitive conditions other than those brought about by mergers).

- 12.3 The conceptual framework is also grounded in the reality of merger control enforcement and builds on case work and guidelines (including guidelines from jurisdictions other than the EU). The objective is to offer a framework that can be applied in case practice, and which can inform and support the ongoing revision of the European Commission's merger guidelines.
- 12.4 In accordance with the Tender Specifications for this Study, the conceptual part of the Study seeks to identify the circumstances when mergers may be likely to generate dynamic benefits or dynamic harm, illustrating the economic mechanisms at work. It also reviews possible qualitative and quantitative criteria to assess these mechanisms in practice.
- 12.5 The framework we develop provides guidance on how to balance the pro- and anticompetitive effects of mergers. This includes balancing pro- and anticompetitive effects that are both dynamic in nature, and includes asymmetric cases where dynamic harm needs to be balanced with static benefits, or where dynamic benefits need to be balanced with static harm. Our conceptual framework focuses on the *substantive* assessment of the dynamic effects of mergers. It does not address procedural and legal issues, for example on how the investigation of the dynamic effects of a merger should be structured, on the burden of proof for establishing specific effects (e.g. procompetitive dynamic effects), and on how to interpret the meaning of a Significant Impediment to Effective Competition under the EU Merger Regulation.⁴⁵⁰ This is outside the scope of the Study. Nonetheless, guidance and clarity on the conceptual framework for dynamic merger effects can provide insights into how to structure the investigation and evaluation of these effects.
- 12.6 Our conceptual framework relies on a distinction between the anticompetitive dynamic effects of horizontal mergers or dynamic foreclosure effects from non-horizontal mergers on the one hand, and procompetitive dynamic effects of either horizontal or non-horizontal mergers on the other. These effects are grouped in 'boxes'. Figure 12.1 provides an overview of these boxes, which are then further discussed in the remainder of this Chapter. A given merger may raise both horizontal and non-horizontal dynamic effects.

⁴⁵⁰ We therefore abstract from issues that may arise from an asymmetry of information or of resources between a competition authority and market participants (i.e. the merging parties and their competitors, suppliers, and customers).

Figure 12.1 Categories of dynamic effects



Note: This provides an overview of the core dynamic effects considered in our conceptual framework. In the discussion of our framework we separately reflect on entrenchment.
Source: Oxera.

- 12.7 The conceptual framework developed in this Part of the main report is structured as follows.
- 12.8 **Chapter 13, introduction and overview of conceptual framework**, presents our definition of dynamic merger effects. It is important to define at the outset the scope of the effects captured by our conceptual framework and to include the main sources of potential dynamic harms and dynamic benefits from a merger.
- 12.9 **Chapter 14, substantive standard**, outlines the substantive standard that we adopt for the evaluation of dynamic merger effects. In line with the prevailing legal and policy framework, we rely on a consumer welfare standard, suitably adapted to capture the role of uncertainty in the evaluation of dynamic effects.
- 12.10 **Chapters 15, 16, and 17, on horizontal mergers**, set out the main dynamic effects captured in our framework as applicable to mergers between producers of substitute products, including potential or nascent competitors. This is the core of Part C of the main report, as the primary focus of the Study is on horizontal mergers.
- 12.11 Chapters 15 and 16 respectively describe the two broad categories ('Boxes') of potential dynamic effects from horizontal mergers captured in our framework:
- **Box 1:** anticompetitive dynamic effects from horizontal mergers, flowing from a loss of competition between the merging firms in relation to both investment and future product market competition (for a given level of investment), and the resulting internalisation of negative externalities brought about by the merger.
 - **Box 2:** procompetitive dynamic effects from horizontal mergers, including potential benefits flowing from the internalisation of spillovers between the merging firms (Box 2A), and benefits from dynamic synergies from a merger (Box 2B).⁴⁵¹
- 12.12 In Chapter 15 we review the anticompetitive dynamic effects that may result from a horizontal merger (under Box 1 of our framework). These reflect the overall

⁴⁵¹ In line with the discussion in Chapter 5, we use 'spillovers' to refer to involuntary knowledge or information spillovers between competing firms. We instead refer to 'technology transfers' (or 'knowledge sharing') to refer to voluntary transfers of knowledge between firms, either through licensing agreements or as a result of a merger.

impact of the loss of competition between the merging parties, with respect to both investment (including investment in R&D, innovation, or physical infrastructure) and future product markets (for a given level of investment). For analytical simplicity, we focus in our framework on the impact of a merger on the prices of future (innovative) products, as our proxy for the effects of the merger on future product market competition, for a given level of investment.⁴⁵²

- 12.13 Effects under Box 1 therefore capture the internalisation of negative investment and future price externalities (or business-stealing effects) resulting from the merger.⁴⁵³ We consider here the impact of the loss of competition between the merging parties on both overall investment, and on expected consumer welfare (in line with our substantive standard).
- 12.14 As we will explain, the elimination of competition between competing firms, and the resulting internalisation of negative externalities, is likely to lead to dynamic harm to consumers, absent spillovers and synergies. This in turn implies that under our framework a balancing of harmful effects with potential countervailing effects is generally not required for effects captured *within* Box 1 (even though, as we discuss in Chapter 15, there are specific circumstances where a finding of consumer harm under Box 1 does not apply, even absent beneficial effects under Box 2).
- 12.15 In Chapter 16 we examine the procompetitive dynamic effects that may result from a horizontal merger. We identify two subcategories of procompetitive dynamic effects:
- **Box 2A:** internalisation of spillovers between the merging parties. This captures the fact that the loss of competition between merging firms may enable them to internalise spillovers, and so increase their incentives to innovate. However, the presence of spillovers to non-merging rivals affects the importance and even the direction of this effect, implying that effects under Box 2A are not always procompetitive.
 - **Box 2B:** dynamic synergies. This captures improvements in the capabilities of the merging parties which can benefit consumers (e.g. reductions in R&D costs, voluntary technology transfers, beneficial diffusion effects, and the impact of complementarities in research, development, and distribution capabilities).⁴⁵⁴
- 12.16 We also discuss in this Chapter how the standard substantive three-limbed test for the assessment of efficiencies under the existing Commission merger guidelines can be applied to procompetitive dynamic effects under our economic framework. As already noted, we focus on the substantive assessment, and do not enter into the legal and procedural question of whether the competition agencies or the merging parties should bear the burden of proof for establishing specific competitive effects.
- 12.17 In Chapter 17 we discuss how to balance anti- and procompetitive effects from horizontal mergers, in cases where a balancing is required to determine the overall

⁴⁵² Firms may also compete in the product markets in quantities, and not prices. Our framework equally applies to this case. We note, however, that most of the guidance from the academic literature applies to the canonical case of price competition between differentiated products.

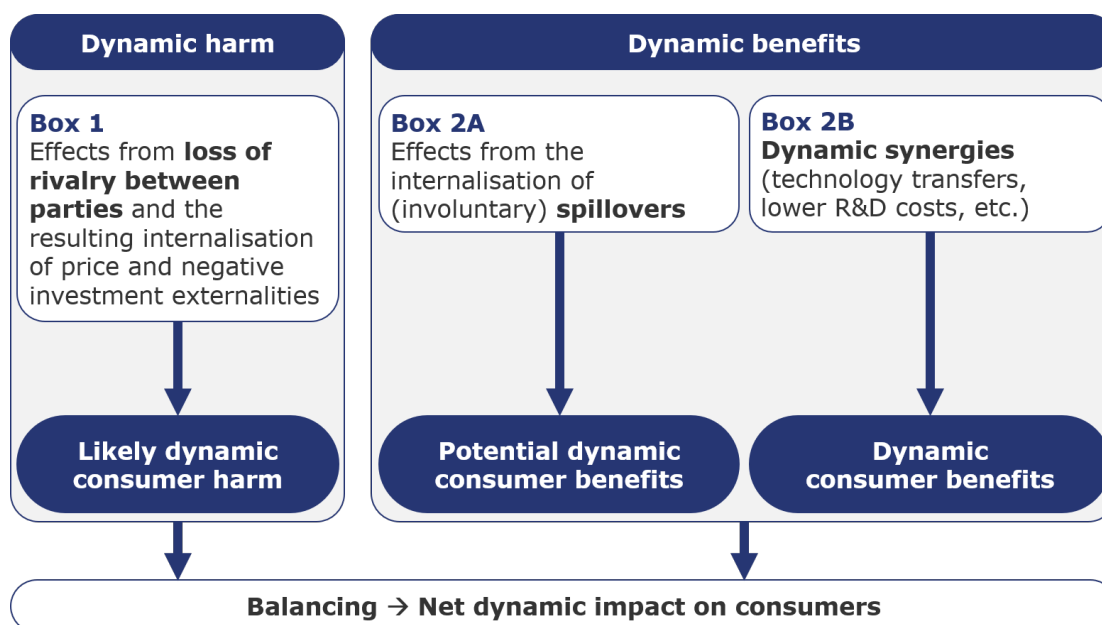
⁴⁵³ Competition between rival firms generates what are commonly known in economics as 'negative externalities'. In a setting where firms compete with each other, a negative externality describes the negative impact of an action by a given firm on the profits of its competitors. These negative externalities between competitors typically benefit consumers, by inducing firms to compete more aggressively.

⁴⁵⁴ Formally, Box 2A captures the fact that two previously separate objective functions are now brought together, while Box 2B reflects the impact of changes in the parameters of the optimisation problem faced by firms (e.g. lower costs, or greater impact from innovation).

impact of a merger (e.g. where the analysis indicates the presence of both anticompetitive and procompetitive effects). We discuss how, applying a consistent Expected Consumer Welfare framework to harms and benefits, larger expected effects on one side require correspondingly larger offsetting effects on the other.

- 12.18 Figure 12.2 provides a schematic overview of our categories of dynamic effects from horizontal mergers.

Figure 12.2 Categories of dynamic effects in horizontal mergers



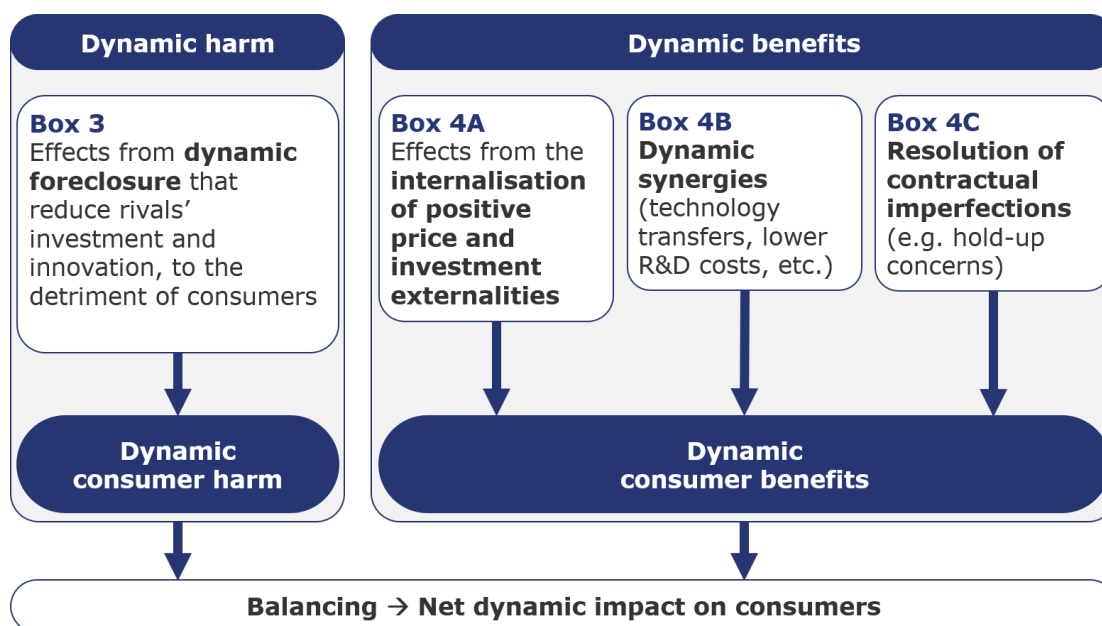
Note: This figure represents the two main categories of dynamic merger effects in horizontal mergers: those arising from the internalisation of negative externalities (Box 1) and those arising from spillovers and dynamic synergies (Box 2).

Source: Oxera.

- 12.19 **Chapter 18, on non-horizontal mergers**, applies and extends the conceptual framework to the case of mergers between firms producing complementary products, distinguishing potential dynamic foreclosure effects (**Box 3**) and potential procompetitive dynamic effects from non-horizontal mergers (**Box 4**).⁴⁵⁵ Dynamic foreclosure effects under Box 3 capture situations where the merged entity has the ability and incentives to engage in conduct that reduces investment by rival firms, to the detriment of consumers. Procompetitive effects under Box 4 capture the internalisation of positive price and investment externalities resulting from a merger of complementary activities (**Box 4A**), dynamic synergies conceptually similar to those associated with horizontal mergers (**Box 4B**), and additional investment-enhancing effects from the resolution of contractual imperfections (**Box 4C**).
- 12.20 Figure 12.3 provides a schematic overview of our categories of dynamic effects from non-horizontal mergers.

⁴⁵⁵ As the main body of Part C discusses, horizontal and non-horizontal effects are not mutually exclusive, and a single merger may give rise to both. This is the case, for example, where the target operates at a different level of the value chain from the acquirer—making the merger non-horizontal in form—but is also a potential competitor in the acquirer's primary market, giving rise to horizontal dynamic effects as well.

Figure 12.3 Categories of dynamic effects in non-horizontal mergers



Note: This figure represents the two main categories of dynamic merger effects in non-horizontal mergers: those arising from dynamic foreclosure of rivals (Box 3) and those arising from internalisation of positive externalities, synergies and resolution of contractual imperfections (Box 4).

Source: Oxera.

- 12.21 **Chapter 19, on entrenchment**, extends the conceptual framework to capture the potential effects of mergers that raise barriers to entry and expansion by competitors to the merging firms, to the ultimate detriment of consumers. This includes additional mechanisms that may discourage future innovation by rivals to an incumbent with significant or substantial market power, to the detriment of consumers. We define entrenchment to reflect anticompetitive effects not already captured under Box 1 (e.g. elimination of the threat from a nascent competitor) and Box 3 (e.g. foreclosure of future competitive threats).
- 12.22 **Chapter 20, application to canonical merger cases**, applies the conceptual framework to canonical cases of mergers giving rise to dynamic effects. This includes mergers involving pipeline overlaps, mergers involving capability overlaps, acquisitions of nascent competitors, cases raising dynamic foreclosure concerns, and cases raising entrenchment concerns. For each canonical case, we review the elements of the conceptual framework that are likely to be most relevant, and consider potential sources of evidence and practical indicators to assess dynamic merger effects. We consider both qualitative and quantitative indicators that a competition agency could use in the evaluation of a merger.
- 12.23 Throughout the description of the conceptual framework we use both hypothetical examples and concrete cases from past merger reviews, to illustrate the relevant mechanisms and effects. This includes the following past merger control decisions for which more detailed case study background is provided in Annex A2 (Facebook/GIPHY (2021, UK), Adobe/Figma (2023; EU, UK), Novartis/GlaxoSmithKline Oncology Business (2015, EU), Elanco Animal Health/Bayer Animal Health Division (2020, EU), Pfizer/Seagen (2023, EU), Bayer/Monsanto (2018, US), Illumina/PacBio (2019, UK; 2020, US), Illumina/GRAIL (2023, US), Orange/MásMóvil (2024, EU), and Vodafone/Three (2024, UK)).

13 Defining dynamic merger effects

Key points

- *Dynamic merger effects* reflect the impact of a merger on *future product markets*, capturing its effect on product features (e.g. prices, quality, availability), costs, and the number of firms. This contrasts with static merger effects, which take into account current market conditions only.
- A dynamic analysis of merger effects is necessary where a static analysis would provide an imperfect indicator for the effect of a merger on future product markets. This is the case if market participants are carrying out investments (in the counterfactual absent the merger and/or with the merger). As a result, the change in expected competition between the merger scenario and the counterfactual varies over time, requiring a dynamic assessment.
- We adopt a broad notion of investment, including investment in R&D and physical infrastructure, which applies equally to incumbent firms and to nascent competitors seeking to enter markets.
- Our definition of dynamic merger effects captures the impact of mergers both on investment incentives and on product market competition on future products (for a given level of investment). It therefore does not consider the impact of a merger on investment in isolation. A broad definition is necessary because individual dynamic effects can interact with one another.
- The static versus dynamic distinction that we adopt is not equivalent to a distinction between short- and long-term effects. Static effects may persist over the medium or long term, in which case current market conditions remain a good proxy for future conditions. Nor is it equivalent to a distinction between price and non-price effects, as dynamic effects may include price effects on future products, while static effects may include non-price effects on existing products.
- Dynamic effects are most relevant where investment is a key competitive parameter, markets are subject to significant technological change or transition, or where current products are becoming obsolete.
- Our conceptual framework concentrates on ex post effects—that is, the outcomes of a specific merger relative to the counterfactual at the moment the merger is reviewed. However, it recognises the fact that the merger policy stance adopted by competition authorities can shape ex ante investment incentives, particularly for small innovative entrants and nascent competitors.

- 13.1 Before presenting our framework for the analysis of the dynamic effects of mergers, we set out in this Chapter a definition of what constitutes a *dynamic merger effect*. A clear definition is necessary to delineate the scope of the conceptual framework, and to ensure a consistent approach in the treatment of the potential multiple dynamic effects of a merger (including interactions between separate individual effects).

13.1 Static vs dynamic merger effects

- 13.2 A merger effect captures the change in expected competition between a situation where the merger takes place (the 'merger scenario') and the situation without the merger (the 'counterfactual').⁴⁵⁶ Merger effects may include both *static* and *dynamic* effects, and both harmful and beneficial effects.
- 13.3 We define the dynamic effects of mergers by contrast to static effects.
- Static merger effects capture the impact of a merger in current product markets, taking into account existing products, cost technologies, and firms.⁴⁵⁷ These effects may persist over time and hence apply to product markets in the future, if market conditions in current product markets are a reasonable proxy for future market conditions.
 - By contrast, *dynamic merger effects* describe the effects of a merger on product markets in the future (insofar as these differ from static effects). These effects capture changes in product features (availability, quality, and prices of future products), cost technologies, and number of firms (through entry or exit). Effects on future product markets differ from effects on current product markets because of the investments carried out by firms. This may be the case in the counterfactual absent the merger and/or with the merger. Dynamic merger effects reflect the elimination of competition (and resulting internalisation of externalities) between the merging firms with respect to both investment and future product market competition (for a given level of investment), and any merger-enabled dynamic synergies. These effects also capture the reaction to the merger by non-merging firms.⁴⁵⁸
- 13.4 In our definition, we focus explicitly on the dynamic effects of a merger between two previously independent firms, and the resulting changes in market outcomes. We consider the impact of the merger on coordination of the conduct of the two merging firms resulting from the *change in ownership* (with respect to prices, investment, and other relevant competitive variables) and of any dynamic synergies that they are able to realise through the merger. In doing this, we build on the literature on the impact of mergers (reviewed in Part B, Chapters 5–9), and only indirectly draw from the literature on the role of competition in general (reviewed in Part B, Chapter 4).⁴⁵⁹
- 13.5 Our definition captures both horizontal and non-horizontal dynamic effects of mergers. However, there are some important differences between the two categories of effects, as we set out in the corresponding chapters (see Chapters 15–18 and in particular Chapter 18). In line with the Tender Specifications for this Study, the main focus of our Study is on dynamic effects from horizontal mergers.

⁴⁵⁶ Both the merger scenario and the counterfactual may include multiple possible outcomes, if the underlying competitive parameters are uncertain when the merger is being evaluated. We return to this issue in the discussion of the substantive standard in Chapter 14.

⁴⁵⁷ These capture the impact of a merger on both allocative and productive efficiency (including static cost synergies caused by the merger). Static cost synergies may, however, also affect investment incentives, and hence have an impact on dynamic merger effects.

⁴⁵⁸ In doing so, we restrict our definition of dynamic effects to non-coordinated (or unilateral) effects—see further discussion in Section 13.2.

⁴⁵⁹ We therefore do not include in our definition the effects of other variations in the conditions of competition on investment and on consumers (e.g. changes in the intensity of product market competition not due to a merger; changes in market structure other than through a merger; changes in market size or in entry conditions). We discuss cooperative arrangements other than a full merger in the assessment of the impact of spillovers and dynamic synergies, in relation to merger-specificity.

We accordingly devote more attention and analysis to this category of dynamic merger effects.

- 13.6 Our definition of dynamic merger effects is broad, as it captures the impact of mergers on *future product market competition*, and not just on investments. As noted in the previous Chapter, in our exposition we focus on the case of price competition in the product market, as a proxy for the impact of mergers on other forms of product market competition between rivals (e.g. competition in quantity).
- 13.7 Our definition of future products in turn includes both *new* products resulting from product innovation efforts, and *improved* products (e.g. capturing technological improvements resulting from product or process innovation). It also includes new or improved products resulting from investments in physical infrastructure or investment to enter new markets.
- 13.8 A focus on the impact of mergers on competition in future product markets is appropriate because consumers benefit from any changes in dynamic parameters of competition (e.g. greater investments) through their impact on future product markets (e.g. in the form of higher quality of products, lower costs resulting in lower prices, or greater or faster availability of improved products).
- 13.9 A holistic definition of dynamic effects is important to ensure that all relevant sources of dynamic harm and benefits are covered by the conceptual framework. This facilitates the consistency of treatment of different types of dynamic effect. A broad definition is necessary also because individual dynamic merger effects can interact with each other, and no single effect should be looked at in isolation.
- 13.10 We employ the term *investment* in a general sense, to include both investments in R&D (or in innovation) and in physical infrastructure. These include investments to improve existing products, introduce new products (in product or geographic markets), reduce costs, or increase capacity. They also include investments by existing firms (incumbents) and investments by new firms to enter a market (potential or nascent competitors). In line with the academic literature, we frequently refer to R&D or innovation as a leading example of the effects of a merger on investment incentives, while recognising that innovation has specific features that do not necessarily apply to other forms of investment (e.g. because of the non-rivalrous and in some cases non-excludable nature of knowledge). Investment has an intrinsic dynamic element, since investing firms incur an upfront cost in order to obtain a future benefit (e.g. in the form of lower costs, better products, or additional capacity). Box 13.1 sets out the categories of investment captured by our definition.



Box 13.1 Categories of investment

Dynamic merger effects include the effects of mergers on investments. Our definition of investments is broad, and captures a number of dimensions. The manifestation of dynamic merger effects may vary depending on the type of investment that is affected by a merger.

We make the following broad categorisation of investments. The three categories are intended to be mutually exclusive even though in some cases the delineation between them may not be straightforward.

- **Investment in *product innovation*:** this captures investment in an R&D process to invent a new product, or improve an existing one, to attract additional demand.
- **Investment in *process innovation*:** this includes investment in an R&D process to reduce the costs of an existing product, and to become more

competitive. It is distinguished from a reduction in costs not based on an investment (e.g. due to economies of scale, or to integration of complementary assets).

- **Investment in physical infrastructure or other assets:** this captures deterministic (i.e. non-stochastic) investment in assets, including production capacity, distribution, or marketing assets, network infrastructure, licences, etc. Such investments can enable a firm to increase the quality of its offering (e.g. higher speed of transmission in telecommunication networks), reduce its effective costs (by expanding capacity) or enter a new market (e.g. by building a production facility in a new geographic market, or obtaining a licence).

All three categories may include investments in underlying capabilities (e.g. human capital; physical facilities) and in specific projects (e.g. an R&D pipeline product; or the roll-out of a new technology in a specific area). However, they differ in a number of dimensions.

One is the **role of uncertainty** in the investment process. The prospects of success of product and process innovation may be subject to significant uncertainty, depending on the phase of the R&D process (with early investment associated with greater uncertainty and lower probabilities of success). By contrast, investment in physical infrastructure is typically deterministic (firms know that by investing a given amount they will be able to improve their product offering—even though they may still face significant commercial uncertainty on the profitability of their investment). This in turn matters for the analysis of potential dynamic effects of a merger.

A second key distinction relates to the **nature of the investment outcome**—in particular whether the outcome is non-rivalrous (meaning that its use by one party does not reduce its availability or usefulness to others) and/or non-excludable (meaning that others cannot be prevented from accessing or using it). Product and process innovation typically centre on the discovery of new solutions and advancements in useful knowledge. This implies that the fruits of successful innovation are typically non-rivalrous, and potentially non-excludable (depending on the effectiveness of the IP regime).⁴⁶⁰ If innovation is non-excludable, then innovation by one firm will benefit other firms (in the same or different product markets). This will also generate a positive externality, via spillovers, that is not taken into account by the innovator. The degree to which innovation is non-excludable depends on the strength of IP rights, and reliance on know-how and trade secrets. By contrast, the output of investment in physical infrastructure typically does not lead to a new, innovative, solution, is excludable and is often rivalrous. This has implications for the nature of potential procompetitive merger effects—see discussion in Chapter 16.

Source: Oxera.

- 13.11 All types of investment described here may relate equally to incumbent firms in an existing market, and to smaller firms (including small innovative entrants and potential competitors) seeking to enter new markets. Our definition of dynamic effects captures the impact of a merger on competition both on incumbent firms

⁴⁶⁰ Non-rivalry refers to the fact that multiple parties can exploit an innovation at the same time, without reducing the availability or usefulness of the innovation for others. Non-excludability refers to the fact it may not be possible to stop others from accessing knowledge or innovation once it is produced.

active in the same market, and on competition from potential rivals (including concerns about the loss of potential competition and 'killer acquisitions').

- 13.12 Our definition of dynamic merger effects applies both to markets where competition in innovation is a key parameter of competition (e.g. certain pharmaceutical or technology markets), and to seemingly more static markets where firms possess important dynamic capabilities that enable them to continuously improve or expand their product offerings (e.g. airline markets where firms compete to expand their network of routes; or retail markets where firms compete on new store locations). We illustrate the importance of this dimension in our review of canonical cases and past examples of mergers with dynamic effects (see Chapter 20).
- 13.13 Finally, dynamic merger effects are more likely to be relevant for the overall assessment of a merger if investment is an important parameter of competition, and the merging parties are engaged in significant investment activities. This may be the case if a market is subject to significant technological change, or is in the midst of a technology transition (e.g. from an old to a new generation of products). It may also be the case if consumers derive significant value from product quality and innovative features, and the industry displays a pattern of significant and continuous innovation. Dynamic effects are also more likely if current products are becoming obsolete (e.g. due to changes in regulation, in consumer preferences, or in the effectiveness of current solutions). In all of these cases, a static analysis of the merger would omit important implications of the merger, and it may lead to an inaccurate prediction of the overall effect of a merger.
- 13.14 We define *adverse* and *beneficial* dynamic merger effects in Chapter 14 of the main report, by reference to our substantive standard for assessment.

13.2 Other dimensions of merger effects

- 13.15 There are additional dimensions of merger effects which relate to dynamic effects and partially overlap with our definition. We discuss these in this Section, as it helps to clarify the scope and implications of our definition.
- 13.16 Our distinction between static and dynamic effects is *not* a distinction between *price* and *non-price* effects. A merger may suppress price competition between future, innovative products. By our definition, this is a dynamic merger effect that is captured in our conceptual framework as it has an impact on dynamic harm to consumers (see Chapter 14 for our discussion of the relevant substantive standard). Conversely, a merger may lead to a change in incentives to compete on non-price dimensions of existing products (e.g. quality), without affecting investment incentives. In our definition this is treated as a static merger effect.
- 13.17 Our distinction is also *not* between the *short-term* and *medium-/long-term* impact of a merger. A static analysis captures both the short- and medium-/long-term effects of a merger in markets that are reasonably stable, where both the merger scenario and the counterfactual are not expected to change significantly over time. As such, a static analysis of current product markets is a good proxy for the impact of the merger on product markets in the future as well.⁴⁶¹ A dynamic analysis would not be required in such a case.
- 13.18 By contrast, an assessment of dynamic merger effects is necessary where a static analysis would provide an imperfect indicator for the effect of a merger on future

⁴⁶¹ Our definition of static effects captures also static cost synergies (e.g. reduction of production costs of existing products, which do not require an investment and which do not affect investment incentives).

product markets. This is the case if the change in competition between the merger scenario and the counterfactual is expected to vary over time, notably as a result of investments carried out by merging parties.⁴⁶² For example, the counterfactual may reflect greater competition in the future if one of the merging firms is expected to introduce a new product or enter a new market. A static comparison between the merger and the counterfactual (based on the *status quo*) would omit this dynamic effect. Similarly, the future merger scenario may reflect synergies leading to greater investment by the merged entity. A dynamic merger effect captures this effect by comparing a forward-looking merger scenario with a counterfactual without the synergies. We further elaborate on this definition of dynamic effects, by reference to our substantive standard, in the next Chapter.

- 13.19 While our definition of dynamic merger effects is relatively general, it focuses on the impact of mergers on traditional dynamic parameters of competition (e.g. investments, innovation, future prices, and entry) and does not directly reflect other dimensions of competition which over time may be affected by a merger (e.g. resilience or sustainability). This is in line with the treatment of dynamic effects in the EU Merger Regulation,⁴⁶³ the EC Horizontal Merger Guidelines,⁴⁶⁴ and the Draghi Report.⁴⁶⁵ It is also consistent with DG Competition's discussion of dynamic effects in one of the background papers for the ongoing revision of the Merger Guidelines.⁴⁶⁶ However, changes in investments caused by the merger may also have an impact on consumers through their effect on broader parameters of competition, including resilience of supply chains. For example, lower investment in capacity following a merger may increase the risk of supply shortages and undermine resilience. Conversely, dynamic synergies leading to higher investments may be beneficial by strengthening resilience. Our substantive standard indirectly captures these effects (see Chapter 14).
- 13.20 Our definition encompasses a wide range of product improvements which firms can achieve by investing. For example, firms may be competing in green technologies in markets where consumers place a strong value on sustainable solutions or where market signals are in place to encourage green investments (e.g. through carbon pricing). In such a scenario, (changes in) competition in green innovation would be captured by our definition of dynamic merger effects.
- 13.21 Finally, the focus of our conceptual framework is on non-coordinated (or unilateral) effects. As a result, we do not cover coordinated effects. That is, we assume that the merger does not affect the nature of competition between firms, enabling them to coordinate their conduct.⁴⁶⁷ This is in line with the assessment of dynamic merger

⁴⁶² Exogenous changes to both the merger scenario and the counterfactual (say, entry by non-merging firms, or cost shocks) are not captured in our definition of dynamic merger effects, even though they are dynamic elements which may influence the assessment of the magnitude of merger effects over time (e.g. by reducing market concentration or affecting margins). For a recent discussion, see Nocke, Peitz, and Schutz (2026a) and Nocke, Peitz, and Schutz (2026b). A 'failing firm' situation (where the future counterfactual differs from the static one due to the exit of one of the merging firms) is not the focus of our analysis of dynamic merger effects. However, several of the considerations developed in our framework on the definition of dynamic effects and the relevant substantive standard equally apply to this situation.

⁴⁶³ European Commission (2004), 'Council Regulation on the control of concentrations between undertakings', No 139/2004, January. See recital 4 (with reference to dynamic competition and competitiveness) and Article 2(1)(b) (with reference to the notion of development of technical and economic progress).

⁴⁶⁴ See discussion in Chapter 10.

⁴⁶⁵ See discussion in Chapter 2.

⁴⁶⁶ See European Commission (2025), '[Topic C: Innovation and other dynamic elements in merger control](#)', accessed 21 July 2026.

⁴⁶⁷ We also do not consider the case where a merger makes pre-existing coordination more effective.

effects in the recent academic literature and case practice. That said, coordinated effects also represent a relevant form of dynamic effect of a merger, which may be important in a specific case.

13.3 Ex ante vs ex post dynamic effects

- 13.22 Our definition of the dynamic effects of mergers captures the outcomes of a *specific* merger (*ex post effects*). These outcomes are considered relative to the counterfactual and relate to the impact of the merger on the conduct of the merging parties, and its indirect effect on their current and future competitors. The focus on ex post effects is in line with the approach in the EU Merger Regulation and in the EC Horizontal Merger Guidelines. Our categorisation of the dynamic effects of mergers (notably in Chapters 14–19) accordingly considers only ex post effects.
- 13.23 In the academic literature on dynamic merger effects, attention has also been devoted to *ex ante effects*.⁴⁶⁸ Ex ante effects capture the impact of the *prospect* of an acquisition on the innovation incentives of firms, including firms that may not be directly involved in (or affected by) a specific merger. This effect may be particularly relevant for nascent competitors, including startups, for which merging with or being acquired by an established firm may represent one of the main means of monetising an initial investment. The academic literature identifies situations where ex ante effects of mergers encourage innovation efforts. The *policy stance* adopted by competition authorities can therefore affect these ex ante incentives (in terms of both the intensity and direction of innovation).⁴⁶⁹ However, in our conceptual framework we follow the approach taken under merger enforcement practice and do not consider these effects, as they take place before a specific merger is reviewed, and hence do not directly affect the impact of that merger on future product market competition. Ex ante effects are therefore outside the scope of our conceptual framework.

⁴⁶⁸ See for example the review of the theoretical literature on investment-for-buyout effects in Section 6.2.

⁴⁶⁹ For example, if the competition agency can pre-commit to a given merger control policy stance, it may be able to encourage ex ante innovation efforts by nascent competitors.

14 Substantive standard

Key points

- Our conceptual framework evaluates dynamic merger effects through a consumer welfare standard, under which a merger is anticompetitive if it reduces consumer welfare (e.g. due to higher prices, lower quality, or reduced product variety) and procompetitive if it increases it. We adopt a ‘trading partner’ definition of consumer welfare, which includes the impact of a merger on business customers.
- Consumer welfare provides a suitable yardstick (or metric) to evaluate the combined impact of a merger across several possible parameters of competition, capturing both anti- and procompetitive effects.
- Under a consumer welfare standard, an increase in investment incentives does not automatically constitute a procompetitive overall dynamic effect—it must be accompanied by a net benefit to consumers. Higher investment incentives resulting from higher future prices for new or improved products may not yield a net consumer welfare gain. The ‘innovation defence’ suggested in the Draghi Report therefore applies only where consumers benefit sufficiently from the fruits of higher investment brought about by a merger (in those cases where a merger increases investments).
- To address the inherently probabilistic nature of dynamic competition—particularly R&D and innovation—the framework adopts an Expected Consumer Welfare (ECW) standard, formally defined as the probability-weighted average of consumer welfare across relevant future scenarios, with and without the merger.
- Precise quantification of probabilities and outcomes is not required to implement an ECW standard. ECW serves as a conceptual benchmark to ensure that individual scenarios with a material probability of occurrence and significant consumer impact are not overlooked. It is often possible to focus directly on the change in ECW (ΔECW), which can simplify the assessment.
- Our framework applies a balance of probabilities standard to ΔECW itself (rather than to individual competitive outcomes). This implies that a merger leads to adverse effects if it is likely to result in a negative ΔECW .
- Accounting for dynamic effects broadens the range of possible outcomes from a merger, potentially magnifying both harms and benefits relative to a static analysis, and underscoring the need for careful, case-by-case assessment of mergers with dynamic effects.
- We identify a taxonomy of four possible cases in relation to the interaction between static and dynamic effects. This includes cases where static and dynamic effects go in the same direction, and where there is a trade-off between the two categories.

14.1 Consumer welfare standard

- 14.1 In our conceptual framework, we evaluate the dynamic effects of mergers under a *consumer welfare* standard, in line with current decisional practice.⁴⁷⁰
- 14.2 Evaluating the dynamic effects of mergers under a consumer welfare standard means that a merger results in anticompetitive or *adverse*⁴⁷¹ static or dynamic effects if it reduces consumer welfare (e.g. through higher prices or lower quality or product variety). This approach views competition as a dynamic *process* of rivalry where firms face incentives to offer *better deals* to consumers (e.g. lower prices, higher quality, greater variety), to win profitable business away from competitors.⁴⁷² Greater rivalry tends to intensify this process, as firms have more opportunities and incentives to take business away from competitors, and a greater threat of losing sales if they are not competitive. A merger may hinder this competitive process, by eliminating rivalry between the merging parties, thus harming consumers.
- 14.3 By contrast, a merger results in procompetitive or *beneficial* effects if it increases consumer welfare, for example if it leads to dynamic synergies or (under some conditions) internalises knowledge spillovers between the merging firms, leading to higher innovation and a resulting benefit to consumers in terms of higher quality or product variety.
- 14.4 While our notion of adverse or beneficial effects applies to both static and dynamic effects, we focus on dynamic merger effects given the scope of the Study. However, the overall impact of a merger on competition is determined by the *combined* impact of static and dynamic effects. These may go in the same direction in terms of consumer welfare (e.g. dynamic effects may amplify static effects) or in opposite directions (see the taxonomy of cases discussed in Section 14.4).
- 14.5 Adverse and beneficial effects may apply at the same time to a given merger (say, because a merger eliminates price competition on existing products between the merging firms, but also increases the incentives to innovate by bringing together complementary assets). In this case a *balancing* exercise is required, where adverse and beneficial effects are netted off against each other.
- 14.6 We refer to a *loss of dynamic competition* from a merger if it leads to an *overall* (or net) adverse dynamic effect (i.e. adverse dynamic effects dominate the beneficial effects); and to an *increase in dynamic competition* if it instead leads to an *overall* (or net) beneficial dynamic effect. We do not discuss when adverse dynamic effects (or, more generally, a reduction in consumer welfare) imply that a merger results in a Significant Impediment to Effective Competition, under the EU Merger Regulation. This is a legal question which is outside the scope of this Study. We do, however, comment on determinants of the magnitude of adverse and beneficial dynamic effects.

⁴⁷⁰ We therefore consider the impact of a merger on consumer surplus, and do not account for any changes in producer surplus (or profits). This means that we examine the impact of the merger on consumer welfare rather than total welfare or total efficiency.

⁴⁷¹ We also refer to these as *harmful* dynamic effects.

⁴⁷² See Vickers (1995), Vickers (2002), CMA Merger Guidelines, para. 2.2, and 2023 US Merger Guidelines, p. 1. As Vickers (2002, p. 4) puts it: 'Competition is good for consumers for the simple reason that it impels producers to offer better deals – lower prices, better quality, new products, and more choice'. The competitive process may also lead to beneficial effects via the *discovery* of new ideas (Hayek, 1945; 1968).

- 14.7 While we refer to consumers for convenience in our discussion of the substantive standard, the consumer welfare standard captures the harm from a merger on *trading parties* located on the other side of the market relative to the merging firms. In the case of a merger between suppliers, the trading parties may be business customers or end consumers, while in the case of a merger between buyers, the trading parties are upstream suppliers.⁴⁷³
- 14.8 We discuss in Section 14.3 how to apply this standard in the presence of uncertainty about future market developments, which is particularly relevant in the case of dynamic merger effects.
- 14.9 There are valid reasons from an economic perspective to adopt a consumer welfare standard in merger control (Vickers, 2025). In particular, a merger control *stance* which seeks to prevent reductions in consumer welfare may also promote total welfare and growth, by inducing positive selection effects (i.e. by encouraging firms to propose mergers with stronger efficiencies).
- 14.10 Adopting a consumer welfare standard does not imply focusing only on the static effects of a merger on prices of current products. While prices are one of the determinants of consumer welfare, they are not the only ones. The consumer welfare standard is sufficiently broad to accommodate other parameters of competition, including quality and variety of current and future products.
- 14.11 It is important to note that adopting a consumer welfare standard does not mean that the specific impact of a merger on consumer welfare needs to be quantified. Rather, the consumer welfare standard is an analytical tool to understand the impact of a merger on competition, and help determine whether a merger promotes or hinders the competitive process. This may be particularly relevant in cases where a merger leads to multiple effects, including both anticompetitive and procompetitive ones (see discussion in the next Section).
- 14.12 The consumer welfare standard is often implemented in practice by formulating and substantiating an economically coherent *theory of harm*, and examining the relevant evidence on the competitive process. This can include an assessment of relatively simple measures such as market shares or R&D overlaps, rather than a specific evaluation of the exact form and level of consumer harm that may result from a merger. We discuss relevant indicators on the possible magnitude of anticompetitive and procompetitive effects in Chapters 15–19.

14.2 Implications of a consumer welfare standard for dynamic effects

- 14.13 Adopting a consumer welfare standard in the context of dynamic merger effects implies that a merger is not evaluated according to whether it enhances a specific dynamic parameter of competition (e.g. innovation), but rather on its overall impact on consumer welfare, taking into account all the relevant determinants of consumer welfare affected by the merger. Thus, if, for example, innovation incentives are boosted by a merger, but prices for innovative products are also increased, then under our definition the merger would result in an increase in dynamic competition (i.e. a net beneficial dynamic effect) only if the higher likelihood of innovation more than compensates consumers for the higher prices of future products (even before the weighing of static effects). This is not just an abstract example, as in several settings identified in the academic literature a merger may lead to *both* higher prices for future innovative products and higher innovation, even in the absence of synergies or other procompetitive mechanisms (we discuss this further in Chapter

⁴⁷³ For a discussion, see Shapiro (2017) and Hemphill and Rose (2018).

15). This implies that the 'innovation defence' recommended in the Draghi Report would apply to merger control only if an increase in innovation due to a merger is associated with a corresponding increase in consumer welfare (i.e. a sufficient share of the benefits from higher innovation is passed on to consumers).

- 14.14 An important benefit of the consumer welfare standard is that it provides a suitable benchmark or *yardstick* for the assessment of potentially conflicting dynamic effects of a merger (including efficiencies). In the example set out in the previous paragraph (where a merger increases the likelihood that new products are introduced, but at the same time increases their prices), using a consumer welfare standard enables a trade-off of the two conflicting effects and an assessment of the overall effect on dynamic competition.
- 14.15 The impact on consumer welfare in any specific case depends on the model of competition and consumer demand that best fits the market features, and this will depend on the characteristics of the market in question. As noted in Section 14.1, a precise or quantitative assessment of the impact on consumer welfare will frequently be difficult to carry out in practice, and will often not be necessary. Nevertheless, the consumer welfare standard provides a suitable conceptual framework that allows for these trade-offs to be evaluated, in an internally consistent fashion.
- 14.16 For similar reasons, a consumer welfare standard allows for an effective balancing of static and dynamic effects, in cases where they go in opposite directions in terms of their impact on consumers (see Section 14.4).

14.3 Adapting the consumer welfare standard to account for uncertainty

- 14.17 The impact of a merger on consumer welfare needs to be evaluated at the time of the merger assessment adopting a *forward-looking* approach. In particular in the context of dynamic effects, it needs to take into account the presence of uncertainty on the evolution of future competitive parameters (e.g. successful introduction of a new product by one of the merging parties, with and without the merger; the likelihood of successful entry by the target of a merger, under the counterfactual; the risk of higher future prices). Uncertainty is likely to be relevant both in the counterfactual absent a merger (e.g. would a potential competitor successfully enter a market and compete with an incumbent?) and in the scenario with the merger (e.g. would a given innovation synergy be realised?).
- 14.18 Uncertainty is particularly relevant in the case of dynamic merger effects since these often relate to competitive processes that are inherently probabilistic or stochastic, including notably investment in R&D and innovation.
- 14.19 Moreover, dynamic merger effects are intended to capture the future impact of a merger in a way that cannot be easily proxied by looking at current, and more readily observable, market conditions. Competition agencies may therefore need to rely on imperfect evidentiary proxies to evaluate the dynamic impact of a merger. This means that they typically face uncertainty about the distribution of possible future consumer outcomes from a merger.
- 14.20 Competition authorities may therefore face two distinct sources of uncertainty on the dynamic effects of a merger. The first captures the *stochastic* nature of some of the relevant competitive processes (e.g. R&D). The second relates to the fact that the competition agency will typically face imperfect information on some of the underlying parameters of competition and may not be able to accurately assess the probability distribution of the impact of a merger on consumers. This adds another

dimension of uncertainty in the assessment of the dynamic effects of the merger.⁴⁷⁴ We discuss the implications of both sources of uncertainty below (in Sections 14.3.1 and 14.3.3 respectively).

14.3.1 The Expected Consumer Welfare standard

- 14.21 To address the first type of uncertainty, in our conceptual framework we adopt a forward-looking ECW standard. We therefore capture the impact of a merger by reference to the change in ECW (i.e. ΔECW), defined as the difference in ECW between the merger scenario and the counterfactual.
- 14.22 The notion of ECW captures the presence of uncertainty about the impact of the merger on consumers, as this depends on the realisation of some of the underlying stochastic parameters of competition (e.g. entry, innovation, efficiencies).
- 14.23 Technically, ECW captures the level of consumer welfare across multiple possible relevant outcomes, weighted by the probability of each outcome.⁴⁷⁵ If, for example, in the counterfactual consumer welfare in a given market can take one of two values (say, €100 million or €120 million) depending on the probability of entry by a potential competitor, ECW is a weighted average of €100 million and €120 million, where the weights reflect the probability of successful entry. A merger between an incumbent and the potential competitor harms consumers in expectation if it eliminates the possibility of independent entry (absent synergies). This would result in a reduction in ECW (i.e. a negative ΔECW). Box 14.1 provides a simple numerical exposition of this example, to illustrate the concept of ECW and of ΔECW .
- 14.24 In practice, it will typically be difficult (and often not necessary) to place specific values on the probabilities and associated outcomes of specific competitive events, and compute the level of ECW with and without the merger. As stated earlier in the discussion of the consumer welfare standard, reliance on ECW should be primarily seen as an *analytical benchmark* to evaluate the impact of a merger in situations where the future evolution of important competitive variables is uncertain. Moreover, in any given case, the competition agency might have good grounds for confidence that a merger would worsen or improve the distribution of consumer welfare outcomes, without knowing what the ex ante distribution actually is. It hence can focus its analysis on the sign of ΔECW , as the primary object of interest to determine whether the merger impedes or enhances competition.

⁴⁷⁴ This can be seen as a distinction between *aleatory uncertainty* (or *risk*) arising from stochastic variation in an outcome, conditional on a known probability distribution (e.g. a stochastic R&D process); and *model/parameter uncertainty*, reflecting imperfect knowledge by the competition authority on the shape of the probability distribution or some of its underlying parameters. The first element reflects underlying uncertainty directly arising from the outcome-generating process, while the second reflects the additional uncertainty faced by the competition authority due to imperfect observability.

⁴⁷⁵ Technically, if there are a number of outcomes indexed with i , (either in the counterfactual or with the merger), each with an associated consumer welfare denoted by c_i and probability p_i , then $ECW = \sum_i c_i \cdot p_i$. If consumer welfare outcomes vary over time, then more distant outcomes need to be discounted relative to more imminent outcomes. This may matter in particular for trading off static and dynamic effects (see discussion in Chapter 17).



Box 14.1 Illustration of Expected Consumer Welfare (ECW)

Assume that in the counterfactual absent the merger, there is a market with an incumbent firm (Firm A) and that there is a potential competitor (Firm B) that may enter the market with probability X . The value of X captures the fact that to enter the market, the potential competitor needs to invent a new product, and the associated R&D process is subject to uncertainty. With probability X the R&D process succeeds, and Firm B enters the market, while with probability $1 - X$ the R&D process fails, and there is no entry. In practice, it may be difficult to quantify the value of the probability X . In this example, we assume this is possible to illustrate the properties of ECW and of ΔECW .

Assume that there is a merger involving Firm A acquiring Firm B, and that Firm A would discontinue the R&D project of Firm B with certainty, to avoid cannibalising its existing product.

Assume also that consumer welfare in the market is €100 million if Firm B does not succeed in entering the market, and €120 million if Firm B enters. That is, entry intensifies competition, and by doing so it naturally benefits consumers (through lower prices and greater product variety).

ECW in the counterfactual absent the merger is the average level of consumer welfare across the two scenarios, weighted by the probability of each scenario. If X is 25%, ECW equals $75\% \times \text{€}100 \text{ million} + 25\% \times \text{€}120 \text{ million} = \text{€}105 \text{ million}$.

ECW in the merger scenario is simply the level of consumer welfare in the no entry case (by assumption). ECW equals **€100 million** in this case.

The merger therefore leads to a reduction of ECW (or a negative ΔECW) compared to the counterfactual. In this example, **$\Delta\text{ECW} = -\text{€}5 \text{ million}$** .

In this stylised example, the presence of merger synergies can change the prediction on ECW. Assume for example that the R&D project of Firm B can be applied to the product of Firm A, improving its quality with probability Y . In this case, if Y is large enough, the merger would not have a negative impact on ECW.

Source: Oxera.

- 14.25 ECW reflects the underlying uncertainty in the competitive process by weighting each possible outcome by both its probability of occurrence and its magnitude. A more likely outcome therefore contributes more to ECW than a less likely one; and, for a given probability, an outcome associated with greater consumer benefit contributes more than one with a smaller effect on consumers.
- 14.26 The ECW standard therefore embodies a 'sliding-scale' approach: a low-probability outcome can weigh heavily in the assessment of a merger if its potential effect on consumers is large. This is illustrated by the case of a merger between an incumbent with significant market power and a potential competitor, where in the counterfactual the potential competitor could have entered the market in competition with the incumbent—an outcome that may be uncertain but, if it materialises, could be of considerable benefit to consumers.⁴⁷⁶ We return to this

⁴⁷⁶ This is in line with the approach on potential competition concerns in the US Merger Guidelines: 'A merger that eliminates a potential entrant into a concentrated market can substantially lessen competition or tend to create a monopoly. The more concentrated the market, the greater the magnitude of harm to competition from any lost potential entry and the greater the tendency to

point in the context of acquisitions of potential or nascent competitors, where it is particularly relevant (see Chapter 20).

14.3.2 Benefits of an Expected Consumer Welfare standard

- 14.27 An ECW approach is appropriate because a merger may be associated with significant dynamic merger effects even if those effects are largely driven by an individual outcome that is uncertain and unlikely to occur, but yet with a reasonable probability of being realised.⁴⁷⁷ To take the example above where an incumbent acquires a nascent competitor, even though consumers would not be harmed by the merger in the most likely outcome from the merger (where entry does *not* take place), they may be harmed in expected terms (on an *ex ante* basis), before the uncertainty on the relevant competitive outcomes is resolved. The change in ECW from the merger captures this source of competitive harm.
- 14.28 The same reasoning applies to possible synergies from a merger. That is, an uncertain procompetitive dynamic merger effect could still result in a positive change in ECW ($\Delta\text{ECW} > 0$) and would hence be taken into account under an ECW standard (see the illustration in Box 14.1). Also in this case, its positive impact on ECW depends both on the likelihood of the procompetitive effect, and its magnitude.
- 14.29 One of the advantages of using an ECW standard is that the mere fact that a competitive process is subject to considerable uncertainty about its ultimate outcome does not preclude a finding that a specific merger may lead to adverse or beneficial merger effects. An effective competitive process where firms try to outcompete each other with innovative efforts is valuable for consumers, as it can be expected to generate more competitive future product markets (i.e. markets offering innovative products at competitive conditions). Moreover, a competitive race for the next innovation can push firms to try harder to be the winner of that race, boosting innovation incentives, and accelerating innovation.⁴⁷⁸ The fact that the race is subject to uncertainty about the ultimate 'winner' or whether the innovation will ultimately reach the product market does not imply that the underlying competitive process is not valuable to consumers, in *expected* terms. An ECW standard is able to capture the value to consumers from a competitive process subject to uncertainty.⁴⁷⁹ It also preserves the benefits of an effective competitive process in handling uncertainty.
- 14.30 ECW also reflects how markets and businesses work in practice. When firms make risky investments, they look at expected returns from those investments, over the relevant scenarios. Thus, even scenarios that are relatively unlikely to occur are taken into account in the overall decision-making process, especially if they are associated with large profits or large losses. The academic literature on the impact of mergers involving stochastic R&D processes also relies on the notion of ECW to

create a monopoly. Accordingly, for mergers involving one or more potential entrants, the higher the market concentration, the lower the probability of entry that gives rise to concern' (Guideline 4, p. 10).

⁴⁷⁷ By analogy, this approach is comparable with the European Commission jurisprudence on 'pay-for-delay' under Article 101 (see Judgement of the European Court of Justice (2020), *Generics (UK) Ltd and Others v Competition and Markets Authority*, C-307/18, January, paras 112–122). It is also consistent with the US jurisprudence on similar conduct (see Judgement of the US Supreme Court (2013), *Federal Trade Commission v. Actavis, Inc.*, No. 12-416, June) finding that reverse payments may seek to 'prevent the risk of competition'. For an academic discussion of pay-for-delay and patent settlements, and the suitability of an ECW standard in these cases, see Shapiro (2003).

⁴⁷⁸ See review of the patent race literature in Chapter 4.

⁴⁷⁹ See discussion of this point in the CMA Merger Guidelines (see Chapter 11).

assess the impact of mergers on consumers.⁴⁸⁰ This is the standard way to account for probabilistic parameters of competition in economic modelling of dynamic merger effects.⁴⁸¹ Additionally, in cost-benefit analyses (CBA) in the context of public policy more generally, the use of expected value reasoning is well established.⁴⁸²

- 14.31 The discussion in the competition policy literature on the use of an ‘error-cost framework’ and decision theory is closely linked to the use of an ECW standard for merger control (see Box 14.2).⁴⁸³ The ECW standard also relates to a ‘balance of harm’ approach, whereby mergers are evaluated considering both the likelihood and magnitude of their impact.⁴⁸⁴



Box 14.2 Error-cost and decision theory in dynamic merger effects

The **error-cost framework** in merger control applies decision theory to the assessment of mergers by competition agencies.⁴⁸⁵ Because competition agencies decide cases under uncertainty, they can make two types of mistakes: **false positives**, or Type I errors, where procompetitive mergers are wrongly prohibited⁴⁸⁶; and **false negatives**, or Type II errors, where anticompetitive mergers are wrongly permitted. The framework asks which decision (or legal standard) minimises the expected social cost of these errors, and hence selects the decision with the lowest expected error cost.

Example

⁴⁸⁰ See for example the work of Federico, Langus, and Valletti (2018), D’Annunzio, Jullien, Lefouili, and Madio (2026a), and Moraga-González and Motchenkova (2026).

⁴⁸¹ From a policy perspective, a number of papers and reports have argued in favour of an ECW standard in merger control, as reviewed in Chapter 10, and Volume 2, Chapter 8. This is related to the expected value approach put forward by Katz and Shelanski (2007a). For a recent application see also the discussion of the Steris/Synergy (2015, US) by Sanford (2026).

⁴⁸² See in particular references in the European Commission’s CBA Guide to the probability distribution of the NPV of projects, including the expected value of NPV, in Section 2.9, and to probabilistic risk analysis in Annex VIII. European Commission (2014), ‘Guide to Cost-Benefit Analysis of Investment Projects: Economic appraisal tool for cohesion policy 2014–2020’, December.

⁴⁸³ This is also related to the US Merger Guidelines, and its treatment of merger analysis as a risk assessment exercise, allowing dynamic theories of harm to be evaluated without precise quantification of effects—see Athey and Nevo (2023) and Athey, Gross, Marinescu, and Shanefelter (2024). As Athey and Nevo (2023) put it: ‘Treating merger analysis as an input to decision under uncertainty also facilitates a more comprehensive approach to theories of harm to competition, particularly theories of harm that are difficult to quantify with any degree of certainty, such as the dynamic or long-term effects of a merger. For example, a merger that creates barriers to entry or dampens incentives for innovation may lead to substantial harm, harm that can be clearly described with frameworks from modern economic theory. However, it may be difficult to specify precisely which firms or products would enter in the future. Acknowledging that considering these harms contributes to the analysis of a merger despite the inherent uncertainty about specific outcomes enables economists and other experts to incorporate modern economic thinking, and consider the impact on future competition, not just today’s competition, even when doing so is inherently uncertain’.

⁴⁸⁴ See the discussion and recommendations in Coyle, Fletcher, Furman, Marsden, and McAuley (2019).

⁴⁸⁵ For a discussion of the application of decision theory to merger control, see Katz and Shelanski (2007b), Baker (2019), and Salop (2026).

⁴⁸⁶ We abstract for simplicity from the possibility of imposing a remedy to prevent harm from a merger.

Take for example the case of a merger associated with a simple binary outcome: it either leads to (net) harm (H) with probability p , or to (net) benefits (B) with probability $1 - p$.

The two costs in this example are:

- **False positive** (prohibition of a merger that was in fact benign). The cost is the benefit foregone, B .
- **False negative** (clearance of a merger that in fact lessens competition). The cost is the harm incurred, H .

The expected cost of a false positive is therefore $(1 - p) \times B$, whilst the expected cost of a false negative is $p \times H$. An error-cost framework implies that a merger should be prohibited if the expected cost of the false negative exceeds that of the false positive ($p \times H > (1 - p) \times B$ which is equivalent to $p > B/(H + B)$), and approved otherwise.

This standard therefore weighs each error by both its probability and its magnitude. In the symmetric case where the magnitude of benefits and harm is the same ($B = H$), the error-cost framework implies that the merger should be blocked if the probability of the scenario where the merger leads to harm is greater than 0.5 (i.e. 50%). In general however, this is not the optimal rule, and the error cost approach implies instead putting more weight on the costlier error. If $H > B$, the blocking threshold $B/(H + B)$ falls below 0.5, so a probability of the scenario where there is harm of below 0.5 can justify prohibition. If $B > H$ the threshold rises above 0.5, so even a more-likely-than-not prospect of the scenario with harm does not warrant blocking the merger.⁴⁸⁷ The weighing of error costs can also take into account the irreversibility of alternative decisions, and the option value attached to preserving flexibility under uncertainty.

The Δ ECW approach that we describe in the main text is equivalent to an error-cost framework. As set out, under a Δ ECW standard a competition agency considers the question of whether the change in expected welfare from the merger is positive or negative (weighting benefit and harm by their respective probabilities), and intervenes to block the merger if Δ ECW < 0 . The error-cost formulation is simply the Δ ECW calculation viewed from the standpoint of the cost of an erroneous decision (relative to the decision that would have been correct ex post). The baseline is therefore different: it is not the pre-merger outcome, but the best decision in each state of the world (allowing the merger when it is in fact beneficial; or blocking it when it is in fact harmful). An error cost approach measures consumer welfare foregone relative to that benchmark.

Relying on an error-cost framework or the Δ ECW approach does not require exact quantification of magnitudes and probabilities. Instead, both are analytical benchmarks to ensure a consistent approach to decision making under uncertainty.

Source: Oxera.

⁴⁸⁷ Salop (2026) discusses the statutory provisions in the US, and related US jurisprudence, and interprets them as suggesting a greater concern with under-deterrence than over-deterrence. Under an error-cost framework, this can justify a 'reasonable likelihood' (or 'appreciable danger') standard for a plaintiff seeking to block a merger (i.e. evidence of a probability of harm of less than 50% would be sufficient to block), and a stricter standard for merging parties to demonstrate efficiencies ('a clear showing' or 'clear and convincing evidence').

14.3.3 Dealing with uncertainty over the distribution of ΔECW

- 14.32 As noted above, the competition agency may have imperfect information on the distribution of some of the relevant competitive parameters. This source of uncertainty is distinct from (and additional to) uncertainty about the outcome of an underlying stochastic process (e.g. a stochastic R&D technology).
- 14.33 For example, the agency may not know the level of probability of entry in the counterfactual (this was simply assumed in the illustrative example in Box 14.1). Similarly, in the context of the same example, it may not observe the distribution of possible cost of production of the entrant or the quality of its new product (both of which affect the consumer benefits from competitive entry). This creates uncertainty around the impact of the merger on the ECW, implying that there is a *distribution* of possible outcomes for ΔECW .⁴⁸⁸
- 14.34 To handle this second source of uncertainty, in our conceptual framework we apply a 'balance of probabilities' approach to the concept of ΔECW itself. That is, we consider that a merger harms consumers if, across the various possible outcomes for ΔECW , it is likely to lead to a reduction in ECW.⁴⁸⁹
- 14.35 This approach preserves the benefits of using an ECW benchmark (as described in the previous Section), but also accounts for the fact that the competition authority may face uncertainty on the impact of a merger on the ECW. This approach has the benefit of avoiding having to apply a balance of probabilities standard to individual probabilistic outcomes (say, uncertain entry by a potential competitor), which would undermine the benefits of using an ECW standard.
- 14.36 Applying the balance of probabilities approach to ECW should again be seen as a conceptual benchmark. In practice, in any particular case, one does not expect a competition agency to have data and information to compute a distribution of possible outcomes for ECW, with and without the merger. In practice, simpler evidentiary proxies (e.g. overlaps, market shares, margins) can therefore be used to inform the assessment.

14.3.4 Applying an Expected Consumer Welfare standard in practice

- 14.37 As set out above, using the change in ECW as a substantive benchmark to assess dynamic merger effects conceptually requires the consideration of multiple scenarios (with and without the merger), reflecting the probability of each scenario occurring. In practice, using an ECW standard does not mean that a competition authority would need to carry out a specific probabilistic assessment of a merger, assigning probabilities to multiple scenarios and undertaking a separate competitive assessment for each. This is not realistic or appropriate in most cases.

⁴⁸⁸ By contrast, in the example in **Box 14.1**, by construction the competition authority would know *with certainty* that the merger leads to a reduction in ECW of €5 million (absent synergies). That is, while the underlying competitive process may be uncertain, the merger review process could establish with certainty that ECW would be reduced by a specific amount. This is of course just an illustrative example where it is simply assumed that the distribution of the relevant competitive parameters can be observed by the competition authority. This is not realistic in practice.

⁴⁸⁹ As explained, this is equivalent to saying that the median value for ΔECW is negative. An alternative standard would be to consider the mean (or expected value) of the distribution of ΔECW outcomes, rather than the median, and hence give more weight to outcomes with a significant impact on ECW but with low probability. While both approaches are internally coherent, we consider the first to be more consistent with the prevailing decisional practice.

- 14.38 The value of relying on an ECW standard is that it shows that careful consideration needs to be given to the *possibility* that a merger may result in alternative outcomes, and that these may be associated with significantly different effects for consumers. This is particularly relevant if the alternative outcomes under consideration have a material probability of taking place (i.e. scenarios with a negligible or merely hypothetical probability of being realised can be discarded, and those with similar competitive implications can be aggregated and looked at together). To make the ECW approach practical and operational, we suggest focusing in the conceptual framework only on scenarios with a reasonable probability of being realised, omitting outcomes that are merely hypothetical or negligible.⁴⁹⁰
- 14.39 Reliance on an ECW standard requires a holistic analysis of the facts and evidence, rather than focusing on a single counterfactual or merger scenario. In some cases, focusing on the most likely merger scenario (compared with the most likely counterfactual) may provide a good proxy for the overall effects of the merger (e.g. because other scenarios are unlikely, or are not associated with very different competition outcomes). However, this needs to be verified and evaluated, on a case-by-case basis.
- 14.40 In assessing the effects of a merger on ECW, relatively simple evidentiary proxies can be used to infer the expected outcome of the merger. For example, the degree of market power of the merging parties, the extent of R&D overlaps between the merging firms, and the competitive significance of entry or innovation will all typically play a significant role in the assessment, as they are important determinants of the impact of the merger on consumers. Similarly, the impact of the merger on the incentives and ability of merging parties to invest is an important overarching consideration. We discuss evidentiary proxies for dynamic merger effects under an ECW standard in Chapters 15, 16, and 17 (as well as Chapter 20).

14.4 Merger effects from a static and dynamic perspective

- 14.41 As discussed, a merger may give rise to static and dynamic effects at the same time. Based on our definition of static and dynamic effects (set out in Chapter 13), and our substantive standard (set out earlier in this Chapter), a static effect captures the change in ECW from the merger on the basis of current market conditions (which may or may not persist over time). By contrast, a dynamic effect includes additional future changes in ECW due to the merger, that are not captured by static effects.
- 14.42 Depending on the nature of static and dynamic effects, four broad categories of cases are possible (see Figure 14.1).⁴⁹¹
- **Case A:** Adverse dynamic merger effects may amplify adverse static effects if the merger leads to a loss of competition in future products (as a result of lower innovation or higher prices for future products), over and above any loss of competition in current products.

⁴⁹⁰ This too is in line with the European Commission jurisprudence on ‘pay-for-delay’, and the requirement for ‘real and concrete’ possibilities of entry, in the context of an assessment of potential competition (see Judgement of the European Court of Justice (2020), Generics (UK) Ltd and Others v Competition and Markets Authority, C-307/18, January). It is also consistent with the notion of a ‘reasonable probability of entry’ in the discussion of potential competition in the US Merger Guidelines (see Guideline 4). See also discussion in Cabral (2025) of a similar approach followed in the 2024 Australian Merger guidelines.

⁴⁹¹ We assume for simplicity that mergers have a non-zero impact on static and dynamic effects.

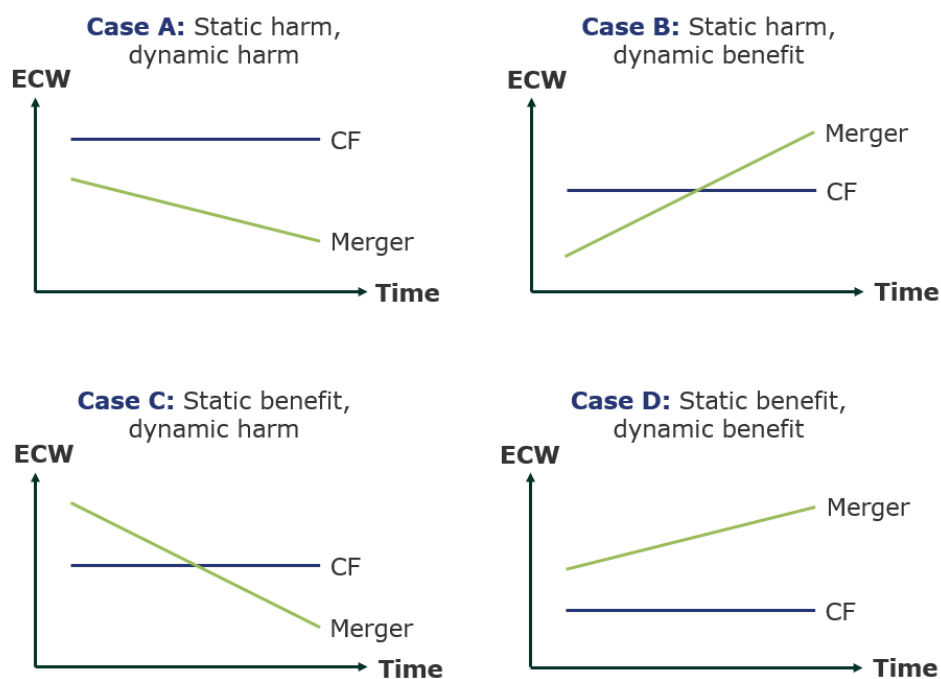
- **Case B:** Dynamic merger effects alleviate static competition concerns from a merger if they lead to dynamic benefits (e.g. net benefits to consumers arising from the introduction of new products and stronger dynamic competition), offsetting part or all of the harm from any loss of competition on current products. This would require a balancing between static harm and dynamic benefits.
- **Case C:** A merger may be beneficial in the short term, but problematic dynamically, e.g. if it entrenches a strong incumbency position, or if merging parties are not current competitors, but are potential competitors in one or several markets. A balancing exercise may be required in such cases.
- **Case D:** A merger may be beneficial in terms of both static and dynamic effects, with dynamic benefits amplifying static benefits.

Figure 14.1 Static and dynamic effects of mergers

	Dynamic harm	Dynamic benefits
Static harm	Case A Adverse dynamic effects amplify static concerns	Case B Balancing required between adverse static effects and beneficial dynamic effects
Static benefits	Case C Balancing required between beneficial static effects and adverse dynamic effects	Case D Static and dynamic effects are both beneficial, with no concerns from the merger

Source: Oxera.

- 14.43 Figure 14.2 below illustrates these four cases by reference to the level of ECW under the merger scenario and counterfactual respectively, and the resulting evolution of Δ ECW over time.
- 14.44 We return to the question of balancing adverse and beneficial effects of mergers, both *within* a dynamic assessment, and *between* a static and a dynamic assessment, in Chapter 17 of the main report.
- 14.45 An important implication of taking into account dynamic merger effects is that they typically broaden the range of the possible outcomes of the merger. They can magnify adverse effects (compared to a static analysis), as consumers benefit from effective dynamic competition in new and innovative products. Consumers therefore have more to lose from a merger that would deprive them not only of static benefits, but also of dynamic benefits. Conversely, accounting for dynamic merger effects can magnify beneficial effects. Dynamic efficiencies typically apply to a broader range of competitive parameters (e.g. not just production costs, but also R&D), and may positively affect the likelihood and effectiveness of innovation. If sufficiently large, these dynamic benefits can outweigh static and dynamic concerns on the loss of current and future price competition. Overall, the *competitive stakes are higher* in the case of mergers with dynamic effects, indicating the need for a careful and detailed assessment of competitive effects.

Figure 14.2 Taxonomy of cases on the impact of a merger on ΔECW 

Note: This illustrates the four cases in which static and dynamic effects can interact and their effect on ΔECW —i.e. the difference in expected consumer welfare (ECW) under the merger scenario (Merger) and counterfactual scenario (CF). Although the slopes are depicted here as either flat (for the counterfactual) or up- or downwards sloping (for the merger), this is just for illustrative purposes: what matters is the difference between the Merger and CF scenarios over time.

Source: Oxera.

15 **Box 1: anticompetitive dynamic effects from horizontal mergers**

Key points

- *Core mechanisms behind Box 1.* A horizontal merger eliminates rivalry between the merging parties, in relation to both investment and future product market competition (for a given level of investment). The resulting internalisation of negative externalities softens the merged entity's incentives to compete. For the canonical case of product innovation followed by price competition in the product market, effects under Box 1 can be decomposed into three effects:
- *Price effects on future products.* The merger reduces future price competition in *new* or *improved* products, independently of its impact on investment. This dynamic price effect is larger where the merging parties are close competitors in future markets, earn high margins on new or improved products, or where one party is a well-placed potential entrant into a concentrated market where the other party is an incumbent.
- *Direct investment effects.* The internalisation of negative investment externalities—whereby one firm's R&D or capacity investment diverts profitable sales in future product markets from the other—tends to reduce each party's incentive to engage in demand-enhancing investments. This effect is stronger where the merging parties are close innovators, in a market with limited alternatives.
- *Indirect investment effects.* The merger-induced increase in prices can raise incentives to engage in demand-enhancing investment by allowing the merged entity to appropriate a greater share of the returns from its investment.
- *Overall impact on investment.* The academic literature shows that, absent synergies and spillovers, a horizontal merger may reduce or increase investments. While the literature indicates that both effects can arise under plausible conditions, it also suggests that mergers reduce investments more often than they increase them. This does not support a general presumption regarding the direction of the investment effect, but it indicates that the combined impact of Box 1 effects gives rise to a substantial concern that a merger may lead to a reduction in investments (absent countervailing effects under Box 2).
- *Overall impact on Expected Consumer Welfare.* The combination of price and investment effects under Box 1 is likely to lead to a reduction in ECW. As a result, no balancing between higher prices and any resulting increase in investment captured within Box 1 is generally required. There are, however, specific circumstances where a finding of harm under Box 1 may not apply (even absent procompetitive effects under Box 2), for example if a merger brings together two weaker or lagging firms.
- Box 1 effects apply to product innovation, process innovation, and physical infrastructure investment, though the precise articulation of direct and indirect effects varies.
- Non-merging rivals often respond to a merger by increasing their own investment, in cases where the merging firms reduce their investment. This reaction is unlikely to be sufficient to prevent an overall reduction in ECW.

- Adverse dynamic effects under Box 1 are larger when the merging parties have significant competitive overlap in future markets, operate in concentrated markets, earn high profit margins, and are close and significant innovators.

- 15.1 Box 1 captures the impact of the elimination of dynamic competition between the merging parties, and the resulting adverse effects on consumers. A merger blunts the incentives of the merging firms to compete through the offer of new or improved products or the adoption of new technologies. It does so by internalising the negative externalities (or business-stealing effects) that the merging firms previously imposed on each other through competition. These externalities may apply both to the price of current and future products, and to investment.⁴⁹² The *combined* effect of the internalisation of future price and investment externalities determines the overall dynamic impact of a merger on ECW, in the absence of countervailing procompetitive effects (which we discuss in Chapter 16). For reasons that we explain in depth in this Chapter, the loss of competition between merging parties and the resulting internalisation of negative externalities is likely to lead to an overall reduction in ECW. This is why we refer to effects under Box 1 as *anticompetitive* dynamic effects of mergers.
- 15.2 The logic of the mechanism underlying effects under Box 1 rests on the fact that more intense competition by Firm A, in the form of lower prices of current or future products or of higher investment (leading to lower costs or better products in the future), increases the attractiveness to consumers of the products of Firm A, at the expense of its rivals. It therefore diverts current or future product market sales from rival firms, leading to a negative effect on their profits, and a positive effect for consumers. From the perspective of Firm A this is an 'external' effect, which it does not take into account when it sets its optimal course of action absent a merger (i.e. the adverse effect on rival firms is not reflected in Firm A's payoffs). While this external effect is negative on rivals, it is good for consumers, as it leads to Firm A providing a more attractive offering (e.g. in terms of lower prices or higher quality).
- 15.3 If two rivals merge (say, Firm A and Firm B), they internalise the negative profit externalities present between them in the absence of the merger. That is, Firm A now recognises that some of the profits diverted from its rivals through more aggressive competition (in price or investments) would come at the expense of the profits of Firm B. In other words, after the merger more intense competition by Firm A would simply *cannibalise* part of the current or future sales of Firm B, which is now part of the same merged entity. This effect reduces the incremental profits captured by the merged entity if Firm A lowers its price or increases its investment (relative to the counterfactual). The merged entity therefore faces lower incentives to lower prices or increase investment than the standalone Firm A. This results in less intense competition by Firm A. The same logic applies to the conduct of Firm B. The change in conduct by Firm A and Firm B has knock-on effects on non-merging parties, which in turn lead to feedback effects in equilibrium. We discuss the dynamic impact of mergers on non-merging parties in Section 15.6.
- 15.4 The loss of competition between Firm A and Firm B is larger the closer the two merging parties are in product or technology space (that is, the greater is the diversion of sales in the product market between the two firms when one of the two firms increases the competitive intensity of its conduct), and the more profitable

⁴⁹² As set out in Chapter 13, we focus on the case of price competition to illustrate the impact of a reduction in product market competition due to a merger, for a given level of investments.

are the sales diverted when the two firms compete (including both current sales and future sales from new or improved products).

- 15.5 A static analysis of unilateral effects in price illustrates the impact of a loss of competition between merging firms (and the resulting internalisation of negative pre-merger externalities).⁴⁹³ The static analysis captures the fact that a merger reduces the incentives to engage in price competition, leading to an upward pressure on pricing compared with the counterfactual. The extent to which the merging parties are close to each other (i.e. there is significant diversion of sales between them in response to changes in relative prices) plays a central role in the assessment of the significance of potential price effects.⁴⁹⁴
- 15.6 The analysis of adverse dynamic merger effects requires extending well-established concepts from the static analysis of price effects to a broader set of competitive variables, including in particular prices of future products (i.e. new or improved products) and investment (e.g. in product innovation, process innovation, or greater physical capacity).
- 15.7 The fact that in a dynamic setting competition leads to negative externalities across multiple competitive variables makes the analysis more complex relative to a static assessment, primarily because of the interactions between these variables, and the existence of indirect effects (e.g. less intense future price competition has an indirect effect on the incentives to engage in investment competition).
- 15.8 A key element in the analysis is evidence on *competitive overlaps* between the merging firms, in particular from a forward-looking perspective. That is, are the firms expected to compete with each other in future product markets absent the merger, and capture profitable future sales from each other, over and above competition on current (or static) product overlaps? Depending on the case, evidence of future competitive overlaps may comprise evidence on overlaps in pipeline products, in projects, and in upstream dynamic capabilities. We expand on these evidentiary elements in our taxonomy of possible cases and in the discussion of relevant evidence in Chapter 20.
- 15.9 Dynamic merger effects under Box 1 relate to the *direct* change in the *incentives* brought about by the merger. That is, for a given set of competitive conditions and parameters prevailing absent the merger, the mere fact that two firms come under joint ownership and coordinate their competitive conduct after a merger affects their incentives and as such results in changes in their conduct.⁴⁹⁵ However, this may not be the only change in incentives of the merging parties. The merger may also allow the merging parties to internalise *positive externalities* from investments (in the form of spillovers), potentially leading to an enhanced incentive to invest, depending on the setting. We discuss this second category of merger-induced changes in incentives in Section 16.2, as they are part of the potential beneficial effects of mergers.
- 15.10 In the rest of this Section, we break down the possible components of the analysis of adverse dynamic effects included in Box 1. While these components all apply

⁴⁹³ This can be evaluated using standard upward pricing pressure analysis or merger simulations.

⁴⁹⁴ This is well recognised in the EC Horizontal Merger Guidelines (see para. 28), and in the US Merger Guidelines (see Guideline 2 and Section 4.2, in particular 4.2.B in relation to price competition).

⁴⁹⁵ Formally, these effects reflect the change in the optimisation problem of the merging firms that results from a merger (i.e. a combination of the firms' objective functions and hence change in the 'first order conditions' of their profit maximisation problem), with no other changes in the underlying parameters of competition.

concurrently for a given merger, it is useful to discuss them individually for analytical convenience. We discuss in turn:

- Section 15.1: the direct effect on prices from the reduction of competition in future product markets (and the resulting internalisation of price externalities);
- Section 15.2: the direct effect on investments from the loss of investment competition (and the resulting internalisation of investment externalities);
- Section 15.3: the indirect effect on investments from the loss of competition between the merging parties in future product markets; and
- Section 15.4: the overall impact on investment incentives and on ECW under Box 1.

- 15.11 Throughout this Chapter, we illustrate dynamic merger effects mostly by reference to the canonical case of non-drastic product innovation which has been studied extensively in the economic literature. We discuss applicability to other types of investment (e.g. process innovation and investment in physical infrastructure) in Section 15.5.⁴⁹⁶
- 15.12 We then discuss other relevant elements of the analysis of adverse dynamic effects, including the reaction of non-merging parties (Section 15.6); and determinants of the magnitude of adverse dynamic merger effects (Section 15.7).
- 15.13 Our discussion of Box 1 effects is based on the existing findings of the academic literature, notably the theoretical literature on mergers between incumbents (reviewed in Chapter 5). We capture in our discussion the various settings that have been modelled in the literature, and hence reflect a variety of possible circumstances. We do so to present a relatively broad account of possible effects under Box 1, while putting more weight on the more common mechanisms identified in the literature, as these are likely to be the most relevant from a policy perspective.

15.1 Direct effect on prices from reduction in future product market competition

- 15.14 This part of the analysis of adverse dynamic effects captures the loss of price competition in future product markets, for a given level of investment by each firm. It therefore reflects the impact of the merger on prices for a given set of future, new or improved, products, a given set of firms, and a given level of cost reduction efforts. It therefore assumes, purely as an analytical step, that innovation and other dynamic parameters of competition (e.g. quality, investment, cost reduction efforts, entry) are not affected by the merger. Our definition of future products in turn includes both *new* products resulting from product innovation efforts, and *improved* products (e.g. capturing technological improvements resulting from product or process innovation).
- 15.15 Dynamic effects on the prices of future products are the result of the internalisation of pricing externalities, i.e. the fact that a reduction in the price of a new innovative product by one of the merging parties diverts profits from the other merging party. This is the analogue of well-established unilateral effects in prices of existing products.

⁴⁹⁶ Non-drastic innovation refers to product improvements which are not so strong as to allow a firm to monopolise a market and thus eliminate competition from rivals. By contrast, drastic innovation implies that the innovator is no longer constrained by competitors (at least for a period, until the next innovation is introduced).

- 15.16 Adverse dynamic price effects are larger if the merger brings together two important innovators that are likely to be close competitors in future product markets or if it removes competition from a well-placed potential entrant in an otherwise highly concentrated market. These price effects are larger also if the merging parties are expected to earn high margins from future innovative products, and hence the value of diverted sales is higher.
- 15.17 Dynamic price effects may need to be considered together with static price effects if a merger brings together existing competitors at the same time as raising dynamic concerns. Price effects may therefore include both a static and a dynamic element. For the purposes of our definition of dynamic merger effects, we only focus on price effects in future product markets, to the extent that these are not captured by a static analysis.
- 15.18 When looked at in isolation, dynamic price effects *reduce* ECW, by depriving consumers of the benefit of price competition in new or improved products (over and above the effect on static competition). This price effect may, however, have a positive effect on innovation incentives (as discussed in Section 15.3).
- 15.19 The following two examples provide a concrete illustration of adverse dynamic price effects. These examples assume that the merger does not affect investment incentives, and that there are no beneficial dynamic effects under Box 2 (such as those described in Chapter 16 of the main report).
- **Example 1:** consider a merger between a dominant incumbent and a potential competitor with a new, innovative product subject to a stochastic R&D process. Assume that the merger does not lead to a discontinuation of the R&D process (say, because R&D costs are low or because the product is highly valuable so discontinuation would be too costly for the merged entity). In this case the only dynamic harm from the merger is the loss of price competition in the future product market between the incumbent and the potential competitor, in the scenario where the potential competitor successfully introduces the new product. This loss is greater if the incumbent possesses significant market power absent the merger or if the new product is a close substitute for the product of the incumbent, as product market competition from the potential competitor is more valuable for consumers in these cases.
 - **Example 2:** consider a merger between two rival innovators that are working on rival R&D projects, but that have no existing products. Assume that R&D efforts remain constant after the merger (see Section 15.2 for a discussion of effects on R&D) and that innovation is not a winner-take-all race (i.e. innovators can offer competing and differentiated innovative products). In this case the dynamic harm from the merger results from the loss of future price competition between the two parties in states of the world where both R&D projects are successful. The merger leads to a reduction in ECW, as long as there is a positive probability that both projects succeed. As in Example 1, the only source of adverse dynamic effects is the loss of future price competition. The magnitude of the expected harm will depend on the probability of successful innovation, and the intensity of price competition between the merging parties in the counterfactual.

15.2 Direct effect on investments from loss of investment competition

- 15.20 We refer to *direct* investment effects as those resulting from loss of investment competition between the merging firms, and the resulting internalisation of negative investment externalities, for a given level of prices. These externalities stem from the fact that investment (e.g. product innovation and the resulting introduction of

a new or improved product; or entry into a market) by one of the merging parties diverts expected profits from the other merging party, and vice versa.

- 15.21 Internalisation of investment externalities tends to reduce the direct incentives to innovate of both of the merging parties, in a manner which is analogous to the internalisation of negative pricing externalities.⁴⁹⁷ The downward impact on investment will be stronger if the merger brings together close innovators in a concentrated market (i.e. if investment by one firm is expected to divert a significant proportion of profits from the other).
- 15.22 This investment diversion effect may relate to profits from *current* products (e.g. if successful innovation by Firm A diverts profitable sales that Firm B would have otherwise made from its existing products) or profits from *future* products (e.g. if successful innovation by Firm A diverts profitable sales that Firm B would have otherwise made from its future innovative products, in scenarios where Firm B also successfully innovates).
- 15.23 As in the case of negative price externalities, competitive overlaps between the merging parties in future product markets are a central element in the assessment of the likelihood and magnitude of any adverse investment effects arising from the internalisation of negative investment externalities. For these effects to be present, the merging parties need to expect to divert profitable sales from each other in the counterfactual.
- 15.24 Negative investment externalities are also at work in patent race settings, where competing firms race to be the first to successfully introduce a given innovation. In such settings, investment efforts by Firm A reduce the probability of competing firms winning the race. If Firm A merges with another competitor in the race, this externality is internalised, and the merged entity can be expected to reduce its investment efforts (see discussion in Chapter 4). In such settings, innovation externalities arise directly at the innovation stage of competition, rather than being mediated through the effects of innovation in the future product market.
- 15.25 The concept of negative investment externalities is extensively analysed in the academic literature on competition, mergers⁴⁹⁸ and innovation, including in policy work. It is closely related to the economic notions of ‘business-stealing’ effects,⁴⁹⁹ ‘cannibalisation’⁵⁰⁰ and ‘innovation diversion’.⁵⁰¹
- 15.26 The notion of a negative investment externality is easiest to see in a sequential setting, where firms first invest to improve their products or invent a new product (period 1), and then compete in price on the basis of their product range (period 2). In this setting, greater investment by Firm A in period 1 makes its product offering more attractive in period 2, diverting expected sales from competing firms.

⁴⁹⁷ This is the case if one rules out a strongly asymmetric impact on the post-merger distribution of R&D efforts (with one of the merging parties entirely stopping their R&D activities), and a possible increase in total innovation through this channel. See discussion of Denicolò and Polo (2018) in Volume 2, Chapter 3.

⁴⁹⁸ See Federico, Langus, and Valletti (2018) (which refers to an ‘innovation externality channel’), Bourreau, Jullien, and Lefouili (2026) (which refers to an ‘innovation diversion effect’), and Moraga-González and Motchenkova (2026) (which also refers to ‘innovation externalities’).

⁴⁹⁹ For example, as identified in the patent race literature from the 1970s and 1980s, and discussed in Aghion and Howitt (1992), Aghion and Howitt (1998), and Tirole (1988, Chapter 10). See discussion in Volume 2, Chapter 2.

⁵⁰⁰ For example, see 2023 US Merger Guidelines (Subsection 4.2.E, in relation to Innovation and Product Variety Competition).

⁵⁰¹ For example, see Farrell and Shapiro (2010), and the related ‘diversion effect’ in Chen and Schwartz (2013).

The same logic moreover applies in simultaneous settings, where firms set investments and prices at the same time (this is equivalent to an assumption that investments by rivals are not observable when firms set prices).

- 15.27 The following example provides an illustration of the impact of the internalisation of negative investment externalities. To look at this effect in isolation, we assume that the merger does not affect prices, and that there are no procompetitive dynamic effects from the merger.

- **Example 3:** suppose that two rivals that are pursuing competing but differentiated innovations merge in an industry with fixed prices. Conditional on the innovation outcomes, consumers are not affected by the merger on the assumption of fixed prices. The effect of the merger then comes solely through the changed R&D investments, which will affect the probability of one or both firms succeeding in innovation. In particular, the merger will tend to reduce the R&D efforts of both merging parties due to the internalisation of innovation externalities, reducing the likelihood of successful innovation. Ex post (once R&D uncertainty is resolved) it may be that R&D efforts by each party are successful (despite the reduction in R&D efforts). For this realisation of R&D outcomes, the merger has no impact on consumers (given the assumption of fixed prices). There will also be situations where neither merging party would have innovated, either with or without the merger. The merger is also neutral for consumers in these cases. However, due to the reduction in R&D efforts, there will also be states of the world where one or both of the merging parties would have successfully innovated without the merger, but would not innovate post-merger. Consumers are harmed in this third set of cases, due to the loss of product variety and/or lower quality of products. In expected terms (i.e. from an ex ante perspective, before the realisation of R&D outcomes), consumer welfare falls compared with the counterfactual.

15.3 Indirect effect on investments from the loss of product market competition between the merging parties

- 15.28 Investment incentives are driven not only by the internalisation of investment externalities, but are also affected by changes in prices due to the merger. Because of the internalisation of negative price externalities, a merger can be expected to increase prices both absent investment and with investment.
- 15.29 In the case of non-drastic product innovation (which is our focus here), the merger affects prices of both the existing and the improved products, since the merging firm internalises negative price externalities both with and without innovation. Price effects from a merger therefore affect the incremental profits that firms can expect to obtain from investment (a 'difference-in-differences'). The loss of price competition between merging parties results in an additional, indirect, effect on investment incentives which is not captured by the internalisation of negative investment externalities.
- 15.30 Moreover, in a sequential setting (where investment outcomes are observable once prices are set) there is also an additional 'strategic effect', which tends to reduce investment incentives pre-merger (as firms react to successful innovation by rivals by reducing their prices to protect their sales). The presence of this strategic effect generates a positive effect of the merger on incentives to engage in product innovation.
- 15.31 One can think about the indirect effect on investment incentives of changes in prices due to a merger as an aspect of Schumpeterian *appropriability*. Notably, a merger, by allowing merging firms to optimise their pricing across the joint product portfolio of the merged entity, may allow them to appropriate a greater share of the value

from their investment, and thus increase investment incentives (all else being equal). This is not the only effect of a merger on appropriability. We return to the impact of a merger on appropriability in the discussion of the procompetitive dynamic effects of mergers, in Chapter 16.

- 15.32 The effect on investment incentives of this aspect of appropriability is in principle ambiguous, since a merger increases prices both pre- and post-investment, and hence may actually reduce the incremental gain from investment (i.e. if the effect on pre-investment profits is larger than the one on post-investment profits). However, based on the existing economic literature on the impact of mergers on investment (for the case of product innovation), in commonly used economic settings in theoretical models, the impact of a merger on prices *increases investment incentives*, by allowing firms to capture a higher share of the benefits resulting from their investment (i.e. the merger increases appropriability). A common economic mechanism behind this effect is the fact that successful product innovation boosts demand (implying that a given price increase results in more profits from innovation), and that joint price setting on multiple products enables the merging firms to optimally divert demand to their high-quality innovative products.⁵⁰²
- 15.33 In the context of merger control, it is important to note that the mechanism behind this indirect positive effect on investment incentives is based on higher prices and greater rent extraction from consumers. This has implications for the overall impact of the merger on ECW, as we discuss in more detail in Section 15.4. Notably, the higher investment incentive induced by this channel (even in those cases where it is sufficient to outweigh the impact of the internalisation of negative externalities and results in higher overall investment) is unlikely to be sufficient to result in an increase in ECW, absent positive effects captured in Box 2.⁵⁰³ This is because the reason for the positive effect on incentives from this mechanism rests directly on higher prices, which harm consumers. Greater appropriability of investment due to higher pricing power following a merger is therefore not a suitable basis for an ‘innovation defence’ under an ECW standard.

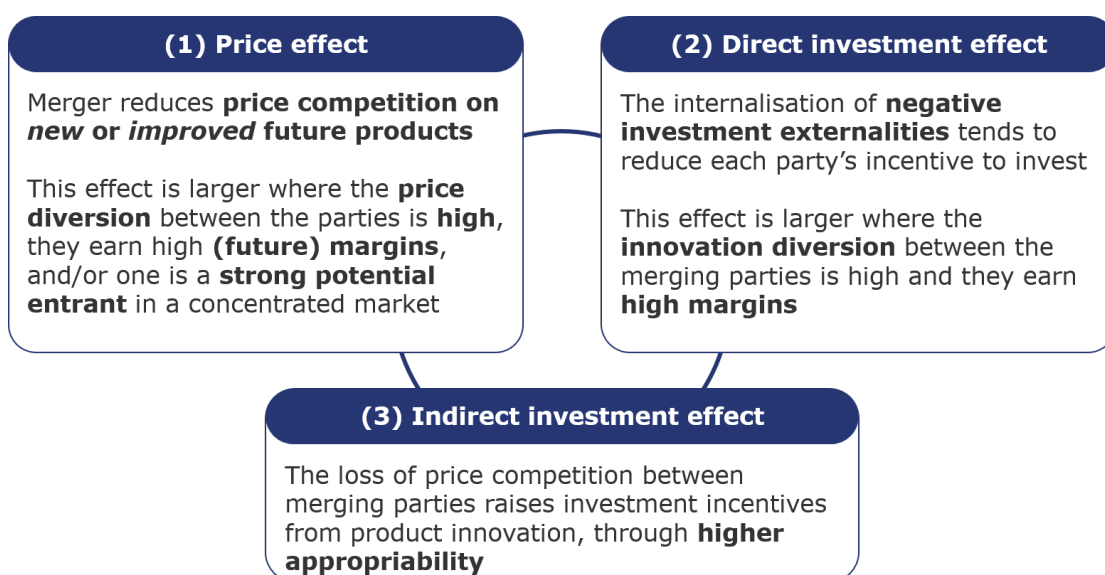
15.4 Overall impact on investment and expected consumer welfare

- 15.34 In this section of the Chapter we discuss the overall impact of the three effects described above on investment and on ECW. Figure 15.1 summarises the nature of the three effects, for the case of product innovation.

⁵⁰² See discussions of the impact of mergers on product innovation in Bourreau, Jullien, and Lefouili (2026), Federico, Langus, and Valletti (2018), and Moraga-González and Motchenkova (2026) (the first paper emphasises the ‘demand expansion’ effect from mergers, whilst the second and third papers stress the effect of ‘price coordination’ between the merging parties).

⁵⁰³ This discussion only relates to the incentives of merging firms. If customers are not final consumers but instead business consumers that also invest, their investment incentives will also be affected by the merger. If the prices of the merging firms increase, this will typically reduce the incentive to invest by business customers, by reducing the margins from output expansion. This is the analogue of the greater investment incentives faced by merging parties due to higher prices.

Figure 15.1 Box 1 effects for product innovation



Note: This figure sets out the three components that go into the overall anticompetitive dynamic effects from horizontal mergers, for the case of product innovation.
 Source: Oxera.

- 15.35 We first discuss the impact on the overall investment incentives of the merging firms, before turning to a discussion of the overall impact of a merger on consumers. This is relevant in light of the attention that this issue has received in the academic and policy literature, and because investment is an important dynamic determinant of overall consumer welfare, and growth. It is also relevant in terms of the evidentiary requirements to support potential dynamic concerns from a merger. However, the substantive standard for the assessment of mergers in our conceptual framework is based on their impact on ECW, rather than investment (or innovation) alone, as set out in Chapter 14.

15.4.1 Overall impact on investment incentives

- 15.36 Dynamic merger effects under Box 1 have two distinct effects on investment incentives: the impact of the internalisation of negative investment externalities (*direct effect*); and the impact of the changes in prices due to the internalisation of negative price externalities (*indirect effect*). For reasons that we have explained in Sections 15.2 and 15.3, for the case of product innovation, the first, direct, effect tends to *reduce* investment incentives, while the indirect effect typically *increases* investment incentives.⁵⁰⁴
- 15.37 In several economic models that have been studied in the theoretical economic literature, the direct effect is stronger than the indirect effect, and the internalisation of negative externalities from a merger leads to a reduction in *net* investment incentives.⁵⁰⁵ In these models, while a merger increases the *overall* profits of the

⁵⁰⁴ Conversely, the indirect effect can lower innovation incentives in the case of process innovation, as we discuss in Section 15.2 below.

⁵⁰⁵ See, for example, Federico, Langus, and Valletti (2017), Federico, Langus, and Valletti (2018), Gilbert (2019), Motta and Tarantino (2021), Cunningham, Ederer, and Ma (2021), Motta and Peitz (2021), López (2025), D'Annunzio, Jullien, Lefouili, and Madio (2026a), Bourreau, Jullien, and Lefouili (2026), and Moraga-González and Motchenkova (2026).

merging parties (due to the reduction in competition), it reduces the profitability of investment efforts, and hence leads to *lower investment* by the merging parties.⁵⁰⁶

- 15.38 However, there are also a number of settings where the indirect effects from higher prices outweigh the direct effect from the internalisation of investment externalities, implying that a merger results in *stronger* incentives to invest in product innovation.⁵⁰⁷ This is more likely to be the case if the merger results in a significant increase in prices, and hence has a particularly pronounced positive effect on appropriability. Based on existing results from the academic literature, this is in turn more likely to be the case if: (i) consumer demand is more responsive to an increase in quality at lower quantities (implying that a merger, by leading to higher prices and lower quantities, increases the returns from investment);⁵⁰⁸ (ii) the price diversion ratio between merging firms is larger than the innovation diversion ratio (e.g. firms are closer along the price dimension than the quality dimension);⁵⁰⁹ or (iii) the outcome of investments is observable by rivals (implying that there are strategic effects at work pre-merger, which reduce the returns from investment).⁵¹⁰
- 15.39 Overall, the academic literature shows that, absent synergies and spillovers, the overall impact of effects under Box 1 may reduce or increase investments. While the literature indicates that both effects can arise under plausible conditions, it also suggests that mergers reduce investments more often than they increase them.⁵¹¹ This does not support a presumption regarding the sign of the investment effect, but it indicates that there is a substantial concern that a merger between rival innovators would reduce their investment incentives (absent spillovers and synergies under Box 2).

15.4.2 Overall impact on expected consumer welfare

- 15.40 The overall impact on expected consumer welfare (ECW) of dynamic merger effects under Box 1 reflects the combined effect of future price effects and investment effects.
- 15.41 In cases where the internalisation of negative externalities leads to an increase in both current and future prices, and to a reduction in investment, the impact on expected consumer welfare is clearly negative, as the merger affects both sets of competitive parameters in a way that is detrimental to consumers.

⁵⁰⁶ In some settings (e.g. procurement markets with efficient mechanisms), the impact of a merger on innovation is instead neutral—see Pietola, Tarantino, and Zenger (2026).

⁵⁰⁷ See, for example, Bourreau and Jullien (2018), Gilbert (2019), Vuong and Baranes (2021), D’Annunzio et al. (2026a), Bourreau, Jullien, and Lefouili (2026), Moraga-González and Motchenkova (2026). Denicolò and Polo (2018) also find that a merger can increase innovation output in a setting with duplicative R&D.

⁵⁰⁸ See Moraga-González and Motchenkova (2026).

⁵⁰⁹ Bourreau, Jullien, and Lefouili (2026) show that the comparison between the size of the innovation and price diversion ratios is informative about the direction of the effect of a merger on investment incentives. This measures the relative strength of price and investment externalities, by considering the closeness between the merging parties in the product space, along the price dimension and the dimension affected by investment (e.g. quality) respectively. A merger is more likely to reduce investment if the innovation diversion ratio exceeds the price diversion ratio.

⁵¹⁰ See Moraga-González and Motchenkova (2026) and Bourreau, Jullien, and Lefouili (2026).

⁵¹¹ As an illustration, a recent structural study in the global LCD panel industry (Igami, Kusaka, Qiu, and Tran, 2025) finds that two thirds of simulated mergers—including all possible combination of firms (7-to-6, 6-to-5, up to 3-to-2)—reduce innovation incentives, and one third increase them. Results on the impact of mergers on innovation incentives are stable across the various types of mergers that do not result in a monopoly.

- 15.42 In settings where the internalisation of negative externalities brought about by a merger instead leads to both higher prices and higher innovation, there are two opposing effects on ECW. However, in the large majority of theoretical settings and examples studied in the academic literature so far, the impact of a merger on expected consumer surplus remains negative also in cases where the merger leads to greater investments (in the absence of countervailing procompetitive effects under Box 2). That is, the detrimental impact of higher current and future prices tends to outweigh the beneficial effects from higher investment. The underlying intuition for this result rests on the fact that the higher investment induced by the merger under this scenario is the result of higher prices induced by the merger, which directly harm consumers. This price effect needs to be particularly strong to outweigh the impact of the internalisation of the negative investment externality and therefore induce higher investment. Since strong price effects harm consumers, it is unlikely that in this scenario consumers benefit from the merger, notwithstanding the increase in investment incentives.
- 15.43 The dynamic harm in both scenarios (those where investment falls post-merger, and those where it does not) is due to less intense competition between merging parties in new or improved products. The evidence that is required to show that the dynamic harm from the merger is significant is also similar in both scenarios: the merger tends to lead to adverse dynamic effects if it brings together two important innovation competitors in a market with a limited number of rivals, or if it eliminates a potential entrant in a highly concentrated market where the other merging firm is an established incumbent. This evidence would suggest that the negative price and innovation externalities internalised by the merger are significant, and hence that the merger would lead to a significant loss of dynamic competition between the parties. We return to the issue of how to examine the magnitude of adverse effects in Section 15.7 below.
- 15.44 Based on the above, evidence of a reduction in R&D efforts, the discontinuation of R&D projects, and a reduction in investments is a *sufficient* condition for a finding of adverse dynamic effects, but it is *not necessary*. This is why it is appropriate to adopt a broad view of the impact of mergers on dynamic competition, and not focus on innovation effects alone. Evidence of a reduction in investment efforts may represent a specific and concrete type of evidence of dynamic competitive harm that a competition agency has access to in a given case, and therefore may have a specific evidentiary role (compared with, for example, evidence of a loss of future price competition over the medium term). However, evidence of this type may not be available or may not be reliable in a given case.
- 15.45 Given that the loss of competition between the merging parties (and resulting internalisation of negative externalities) is likely to lead to an overall adverse dynamic effect and a reduction in ECW, under our conceptual framework we consider that *no balancing is generally required between effects captured within Box 1*. That is, no balancing would be necessary between the future price increase caused by the merger and a (potential) resulting increase in investment. This implies in turn that there is generally no need to specifically evaluate the potentially countervailing effects from higher innovation *within Box 1*, to reach a conclusion on the likely anticompetitive impact of the merger. The main evidentiary issue for effects under Box 1 is to evaluate their magnitude (see Section 15.7). However, any beneficial effects under Box 2 would need to be balanced against potential adverse effects under Box 1 (see Chapters 16 and 17).
- 15.46 The academic literature has identified specific circumstances where a finding of consumer harm under Box 1 does not apply (even absent spillovers and synergies). This includes the case of a merger that brings together two weaker or lagging

firms,⁵¹² or that improves the distribution of innovation efforts across R&D projects.⁵¹³ In some of these settings, the direct effect of the loss of innovation competition between the merging parties results in greater innovation output. The existence of these economic mechanisms could be relevant to the assessment of the overall impact of competitive effects under Box 1, depending on the specific merger at hand.⁵¹⁴

- 15.47 Our approach to dynamic effects under Box 1 is consistent with the current treatment of potential countervailing effects that result from an anticompetitive mechanism. As set out in the current EC Horizontal Merger Guidelines (see para. 80—by analogy to cost reductions resulting from anticompetitive conduct) and US Merger Guidelines (see p. 33, third paragraph⁵¹⁵), countervailing effects based on anticompetitive conduct are not accepted as efficiencies. This means that additional investment incentives resulting purely from higher prices due to market power would not be credited as a potential efficiency or countervailing effect.
- 15.48 Figure 15.2 below summarises the overall impact on investment and on consumer welfare from effects under Box 1, based on the reasoning developed in this Section. Box 15.1 (at the end of this Section) provides a formal example of a merger which may increase or reduce investment, but which results in consumer harm.
- 15.49 The hypothetical examples below illustrate the overall impact of a merger on ECW in two hypothetical cases where the merger leads to a reduction in innovation efforts and an increase in prices through the internalisation of negative externalities (we again rule out here any beneficial effects under Box 2).
- **Example 4:** consider a merger between parties that compete on existing products, and that are also working on rival R&D projects addressing the same market. R&D efforts increase the probability of quality improvement in the existing product. Assume that the direct effect on investment incentives from the internalisation of negative innovation externalities outweighs the indirect effect from higher prices. In this case, the merger reduces innovation efforts and also increases current and future prices. Consumers are harmed both through lower innovation and higher prices. We can distinguish between three scenarios: (i) in states of the world where R&D efforts by both parties are successful, the harm to consumers comes only from higher prices on future innovative products; (ii) in states of the world where R&D efforts by both parties fail, the harm to consumers comes only from higher prices on current products, and is fully captured by a static assessment of competition on existing products; and (iii) in states of the world where one of the merging parties innovates and the other one does not, consumers are harmed by higher prices for existing and future products. Due to the reduction of R&D efforts, the distribution of these states changes after the merger, making it less likely that consumers will have access to innovative products, which is an additional source of consumer harm. If we consider these possible states of the world on an ex ante basis, ECW is reduced by the merger, due to the combination of lower R&D efforts and higher prices.

⁵¹² See Vuong and Baranes (2021) (where the merger brings together two higher cost firms) and Padilla, Piccolo, and Reynolds (2024) (where the merger involves two challengers ('laggards') that have not yet entered the market).

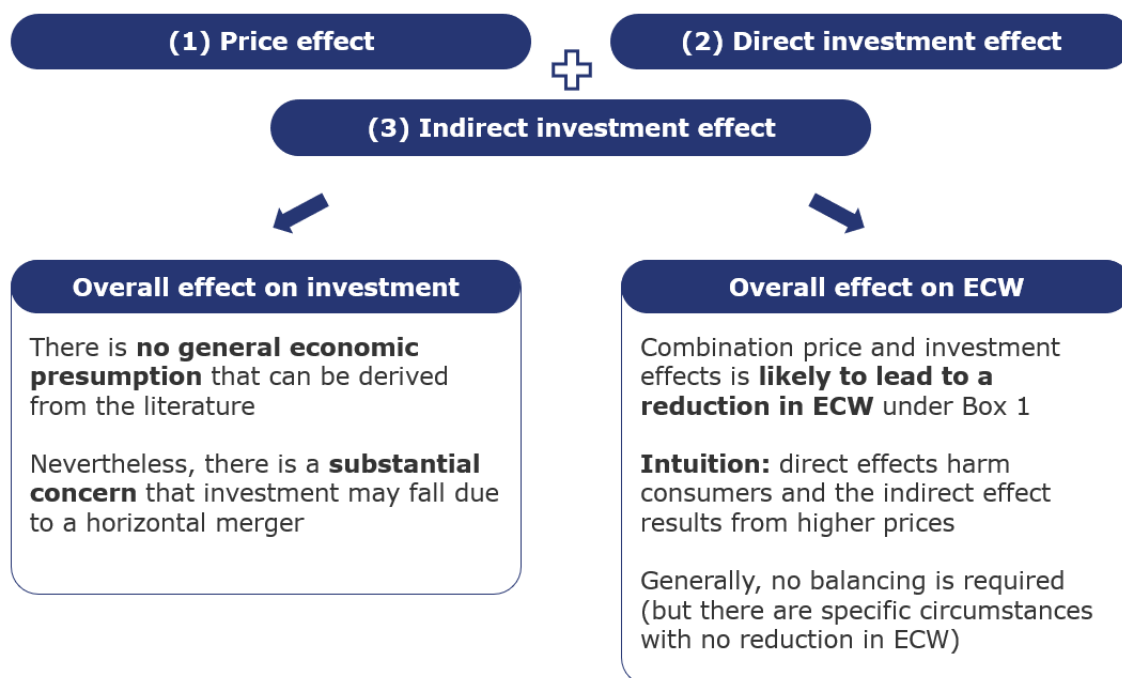
⁵¹³ See Denicolò and Polo (2018) and Moraga-González, Motchenkova, and Nevrekar (2022).

⁵¹⁴ We do not enter into a procedural discussion of whether the competition authority or the merging parties should bear the burden of proof for establishing the applicability of these effects.

⁵¹⁵ US Merger Guidelines: 'Any benefits claimed by the merging parties are cognizable only if they do not result from the anticompetitive worsening of terms for the merged firm's trading partners' (p. 33).

- **Example 5:** consider a merger between an incumbent and a potential competitor with a specific innovative project that has sizeable development costs. Assume that absent the merger, the potential competitor develops the project, as the profits from successful innovation exceed the development costs. In the counterfactual, consumers would therefore benefit from greater competition and more product variety. Post-merger the incumbent instead decides to shelve the project since the diversion of sales from its existing product lowers the gross profit from developing the project, bringing it below the cost of development (this is an example of a 'killer acquisition'). Harm in this case is due to the loss of a potential competitor. Consumers are harmed due to the loss of product variety and the higher prices set by the incumbent (relative to the counterfactual with entry by the potential competitor).

Figure 15.2 Box 1 effects on investment and ECW for product innovation



Note: This figure sets out the overall effect on investment and ECW following the three components that go into the anticompetitive dynamic effects of horizontal mergers (i.e. Box 1), considering specifically the case of product innovation.

Source: Oxera.



Box 15.1 Mergers and product innovation incentives in a sequential model (*technical material*)

This example provides a technical illustration of the dynamic effects of a merger under Box 1 of our conceptual framework. It focuses on the case of product innovation in a sequential setting. We illustrate how a merger may reduce or increase incentives to engage in product innovation, and how it affects consumer welfare.⁵¹⁶

⁵¹⁶ This illustration is based on the two-stage merger model of Federico, Langus, and Valletti (2018) and Moraga-González and Motchenkova (2026), for the duopoly case. Moraga-González and Motchenkova (2026) refer to this as a case with 'endogenous probability of success' (distinguishing it from the case with 'endogenous innovation outcomes').

Set-up

Consider a setting where firms first invest in stochastic R&D, and then compete in prices over horizontally differentiated products. R&D effort raises the probability of enhancing the quality of the product offered to consumers. Pre-merger each firm offers only one product, implying that the merged entity offers two products. As set out in the main body of the main report, in this setting a merger's effect on R&D incentives combines the *direct effect* from the loss of investment competition between the merging parties (and the resulting internalisation of negative investment externalities), and the *indirect effect* from the elimination of price competition between the two products post-merger.⁵¹⁷

The effect of a merger on incentives to invest in product innovation can be signed by considering the *gross profit* obtained by the merged entity from successful innovation on each product, conditional on the outcome of innovation on the other product. The gross profit is given by the difference between the post-merger and pre-merger incremental profit from innovation on each product (a *difference-in-differences*), net of the cannibalisation of profits that successful innovation imposes on the other product of the merged entity.⁵¹⁸ This is a gross profit measure as it does not consider the cost of R&D, and only captures profits earned in the product market stage of the two-stage game.

The gross profit gain on each product can be defined as follows⁵¹⁹, in either the 'catch-up' state (where innovation on the other product of the merged entity *succeeds*) or the 'escape' state (where innovation on the other product of the merged entity *fails*):

$$K_1 = (\hat{\pi}_i^{ss} - \hat{\pi}_i^{fs}) - (\pi_i^{ss} - \pi_i^{fs}) - (\hat{\pi}_j^{sf} - \hat{\pi}_j^{ss}) \quad (\text{catch-up state})$$

$$K_2 = (\hat{\pi}_i^{sf} - \hat{\pi}_i^{ff}) - (\pi_i^{sf} - \pi_i^{ff}) - (\hat{\pi}_j^{ff} - \hat{\pi}_j^{fs}) \quad (\text{escape state})$$

where

- the two products are denoted as i and j ;
- pre- and post-merger product market profits on each product are denoted by π and $\hat{\pi}$ respectively;
- the two entries in the superscript of the profit term denote the innovation outcome, with the first entry referring to the innovation outcome of the product in question, the second entry to the innovation outcome on the other product, s denoting whether innovation is successful, and f denoting that innovation has failed; and
- the first term in brackets in the expressions for each of K_1 and K_2 reflects the post-merger incremental profit on product i from successful innovation on product i ; the second term reflects the pre-merger incremental profit on product i from successful innovation on product i ; and the third term reflects the

⁵¹⁷ These are referred to as the innovation externality channel and the price coordination channel in Federico, Langus, and Valletti (2018). Moraga-González and Motchenkova (2026) use similar terminology.

⁵¹⁸ In Federico, Langus, and Valletti (2018) this measure is referred to as the 'impetus' faced by each firm to change innovation effort following a merger (see equation 7), whilst in Moraga-González and Motchenkova (2026) it is referred to as the difference between post-merger and pre-merger *replacement effects*, net of the externality on the merging partner.

⁵¹⁹ This adopts the notation of Moraga-González and Motchenkova (2026).

externality on profits earned on product j from successful innovation on product i .

If both K_1 and K_2 are negative, then a merger reduces R&D incentives. If both K_1 and K_2 are positive, a merger increases R&D incentives. If the signs of K_1 and K_2 are different, then more information is required to sign the impact of a merger on R&D incentives.

Numerical illustration with logit demand (from Moraga-González and Motchenkova (2026))

We illustrate the values of K_1 and K_2 , and the overall impact of a merger on innovation incentives and consumer surplus, by using the model with logit demand included in Moraga-González and Motchenkova (2026).⁵²⁰ Under this demand model the parameter κ captures the price sensitivity of demand (with a higher value indicating a more elastic demand). We consider the cases $\kappa = 2$ (high price elasticity) and $\kappa = 0.5$ (low price elasticity). The tables below report product market profits for each product i and j , pre-merger (top line in each cell) and post-merger (bottom line in each cell), for the specific parameter values considered in Moraga-González and Motchenkova (2026).⁵²¹ Below each table we also report the decomposition and values for K_1 and K_2 respectively.

Case $\kappa = 2$ (high price elasticity): merger reduces R&D incentives, lowers consumer welfare. Product market profits in each state.

Product i / j	j succeeds	j fails
i succeeds	Pre: 0.300, 0.300 Post: 0.523, 0.523	Pre: 0.417, 0.205 Post: 0.714, 0.263
i fails	Pre: 0.205, 0.417 Post: 0.263, 0.714	Pre: 0.287, 0.287 Post: 0.431, 0.431

which implies

$$K_1 = (0.523 - 0.263) - (0.300 - 0.205) - (0.714 - 0.523) \approx -0.0262$$

$$K_2 = (0.714 - 0.431) - (0.417 - 0.287) - (0.431 - 0.263) \approx -0.0145$$

In this case, both K_1 and K_2 are *negative*, and the merger *reduces* R&D incentives. As a result, the merger also reduces consumer welfare due to the combination of lower R&D and higher prices in the product market.

Case $\kappa = 0.5$ (low price elasticity): merger raises R&D incentives, lowers consumer welfare. Product market profits in each state.

Product i / j	j succeeds	j fails
i succeeds	Pre: 7.055, 7.055 Post: 11.668, 11.668	Pre: 9.022, 4.654 Post: 14.936, 5.495

⁵²⁰ See notably Appendix B.1 of the paper.

⁵²¹ As in Moraga-González and Motchenkova (2026), we assume $c = 1$, $s_f = 5$, and $s_s = 6$, where c reflects production costs, and s quality levels.

i* fails*Pre:** 4.654, 9.022
Post: 5.495, 14.936**Pre:** 5.841, 5.841
Post: 8.089, 8.089

which implies

$$K_1 = (11.668 - 5.495) - (7.055 - 4.654) - (14.936 - 11.668) \approx 0.506$$

$$K_2 = (14.936 - 8.089) - (9.022 - 5.841) - (8.089 - 5.495) \approx 1.073$$

In this case, both K_1 and K_2 are *positive*, and the merger *increases* R&D incentives. However, also in this case, Moraga-González and Motchenkova (2026) report that the merger reduces consumer welfare, as the impact of higher prices in the product market dominates the impact from higher R&D. In the specific numerical example illustrated in the paper,⁵²² expected consumer surplus drops from 2.23 pre-merger to 0.99 post-merger, whilst R&D investment increases from 0.295 to 0.375.

Source: Oxera. This is a simplified example based on the sequential duopoly model of Moraga-González and Motchenkova (2026).

15.5 Relevance of effects under Box 1 to different types of investment

- 15.50 Dynamic merger effects under Box 1 apply to competition in several forms of investment. The discussion so far focused on the canonical case of non-drastic product innovation. We discuss here how dynamic effects under Box 1 apply specifically to some of the other main types of investments, including innovation to enter a new market, patent races, process innovation, and investment in physical infrastructure.

15.5.1 Product innovation

- 15.51 The setting discussed so far is the common case of product innovation, where merging firms are engaging in parallel innovation efforts, to enhance the quality of their existing products.
- 15.52 Similar considerations as those set out in Sections 15.1–15.4 apply to cases where the merger involves firms seeking to enter a *new market* with a product innovation. In this case, a merger leads to no pre-innovation price effects (as the firms are not present absent the innovation), and the indirect effect operates only through the increase in price of the innovative products (which tends to enhance innovation incentives). Overall product innovation incentives may increase or decrease post-merger, due to the countervailing effect of the internalisation of investment externalities.⁵²³ Conversely, consumers are harmed by the increase in future prices resulting from the merger.
- 15.53 Similar effects are also at play in patent race settings, where firms compete to be the first to successfully introduce a new product (which means that other firms cannot make the same innovation). Take for example the case of firms with no pre-innovation sales that are competing to introduce a new innovative product.

⁵²² This assumes additionally that the R&D success probability is linear and that the cost of R&D effort x is quadratic ($\mathcal{C}(x) = 5x^2$).

⁵²³ In specific settings where innovation is ‘winner-take-all’ and there are no existing sales at risk from innovation, a merger is neutral on innovation incentives, as firms only benefit from innovation if they are the sole innovator (Sah and Stiglitz, 1987; Federico, Langus, and Valletti, 2017; Gilbert, 2019).

Competition generates a negative innovation externality between rival innovators. This does not operate via the product market (by diverting future sales from rivals), but operates instead at the innovation stage directly (by reducing the probability of winning of rival firms). A merger leads the merging parties to internalise this externality and tends to reduce their innovation efforts. In this case, there are no price effects from the merger, and hence no potential countervailing effects on innovation incentives.

- 15.54 Finally, one common feature of product innovation is its stochastic nature. In particular for early R&D efforts (focused on the discovery of new products, and testing of efficacy), the probability of successful commercialisation may be quite low. In this setting, greater investment in R&D would typically increase the likelihood of success, and generate negative externalities by making it more likely that product innovation is introduced in the future. For later-stage R&D and commercialisation efforts (focused on regulatory processes, marketing, manufacturing, and distribution), the process may be more deterministic, with firms knowing how much they need to invest to successfully introduce a product to the market. In this case, product innovation is more similar to investment in physical infrastructure (see Subsection 15.5.3).

15.5.2 Investment in process innovation

- 15.55 The case of process innovation (i.e. innovation to reduce production costs for existing products) differs from product innovation in a number of aspects.
- 15.56 The main difference is that process innovation by one firm affects competitors only via its effect on prices. There is therefore no direct investment externality from process innovation. A merger affects incentives to invest in process innovation only indirectly, via its impact on prices⁵²⁴ and output.
- 15.57 The indirect impact of price and output effects on incentives to invest in process innovation is ambiguous. On the one hand, a merger, by increasing prices and lowering output, reduces the incentives to invest in unit-cost reducing process innovation.⁵²⁵ In a simultaneous setting (or where firms cannot observe the outcome of process innovation by their rivals prior to setting prices), where firms invest in process innovation and set prices at the same time, this is the only indirect effect on investment incentives, and a merger therefore reduces investment, and thereby harms consumers.⁵²⁶
- 15.58 On the other hand, in a sequential setting with observable R&D outcomes⁵²⁷, there is also an additional (strategic) effect which typically goes in the opposite direction. This effect captures the fact that, absent the merger, investment in process innovation by one firm leads its rivals to cut their prices to protect their sales, which reduces the profitability of the initial investment. This effect is likely to be weaker post-merger.⁵²⁸ The existing academic literature indicates that the combined impact

⁵²⁴ The impact on prices depends in turn on the pass-through of costs to prices, see e.g. Miklos-Thal and Shaffer (2021).

⁵²⁵ This effect is related to the result of Arrow (1962) for the case of non-drastic process innovation.

⁵²⁶ See Motta and Tarantino (2021) and Bourreau, Jullien, and Lefouili (2026).

⁵²⁷ While this is a realistic assumption for the case of product innovation, it may be less relevant in the case of process innovation, as firms typically cannot observe the cost of their rivals and the outcome of their efforts to reduce costs.

⁵²⁸ For a discussion, see Motta and Tarantino (2021) and Moraga-González and Motchenkova (2026). The strategic effect disappears in the merger-to-monopoly case, leading to stronger innovation incentives from a merger (Moraga-González and Motchenkova, 2026).

of these effects results in a reduction in overall investment incentives in some economic settings,⁵²⁹ but an increase in others.⁵³⁰

- 15.59 As for the case of product innovation, the existing academic literature shows that the combined impact of the effects included under Box 1 of our framework is likely to reduce ECW also for the case where firms compete in process innovation.⁵³¹ That is, even in cases where a merger induces more process innovation and lower production costs, the simultaneous increase in future prices associated with the merger is likely to lead to a net reduction in ECW.
- 15.60 The intuition for this result rests on the fact that the impact on investments of the elimination of the strategic effect is stronger when price effects are also strong. Therefore their combined impact on consumer welfare is unlikely to offset the harm from the decrease in investment due to the reduction in output that follows a merger.
- 15.61 Mergers may affect innovation incentives also in settings where firms race to be the first to develop a new technology to reduce costs (i.e. in a patent race). As for the case of a patent race for a product innovation (discussed in the previous Section), a direct investment externality also arises in the case of process innovation because the first firm to innovate prevents its rivals from benefiting from a potential subsequent innovation, thereby imposing a negative externality on them. The case of (sufficiently) non-drastic process innovation has also been studied in the literature, suggesting the presence of a ‘market-power persistence’ effect (whereby an incumbent firm has stronger innovation incentives than a rival firm to protect its existing profits, and prevent a negative price externality in the case of successful entry).⁵³² In this case a merger between the incumbent and the rival firm would reduce investment incentives by reducing or eliminating the threat of competitive entry.

15.5.3 Investment in physical infrastructure

- 15.62 The third category of investment relates to physical infrastructure as opposed to ideas and knowledge.⁵³³ Examples here include investments in a manufacturing plant, IT infrastructure (e.g. data centres or compute capacity), telecommunication networks (e.g. in terms of coverage and capacity), and in transport networks (e.g. investments to enter a new airline route).
- 15.63 Investment in physical infrastructure may be aimed at improving the quality of an existing product or introducing a new product (e.g. entering a new geographic market). In this case, the effects are similar to those that we have discussed for the case of product innovation. The investment externality flows through its effects in the product market in this case, as investment diverts profitable sales from rivals.

⁵²⁹ See the results and simulations in Motta and Tarantino (2021).

⁵³⁰ See the discussion and simulations in Moraga-González and Motchenkova (2026).

⁵³¹ Motta and Tarantino (2021) show that a merger where firms compete in process innovation reduces consumer welfare in a simultaneous setting (where innovation and prices are set at the same time) in a broad class of demand models (see discussion in Volume 2, Chapter 3). Moraga-González and Motchenkova (2026) also report simulations for the sequential duopoly case of process innovation indicating that a merger is likely to harm consumers, even in cases where investment incentives increase (e.g. in a logit demand model).

⁵³² See Gilbert and Newbery (1982), Reinganum (1983), and discussion in Tirole (1988). Vickers (1985) discusses the more general case with multiple incumbents, showing that this may give rise to a free-riding problem in deterring entry.

⁵³³ Similar considerations would apply to other types of deterministic investment or entry (e.g. acquisition of a licence to enter a market).

- 15.64 In some cases, however, investments in physical infrastructure may be more similar to process innovation. This is the case for investments to reduce capacity constraints and to reduce the effective costs of serving consumers.⁵³⁴ In such cases, there may be no direct investment externalities, and the impact of a merger on investment incentives flows only from its effect on prices and output. Conversely, there are also models of investments in capacity where users face congestion costs. In such cases investment in capacity makes a firm more attractive to customers by reducing congestion costs, and thus leads to a direct investment externality on rivals (similar to models of product innovation).⁵³⁵
- 15.65 A difference between innovation and physical investments for the purposes of the assessment of effects in Box 1 relates to the fact that the outcome of physical investment is typically more predictable. That is, firms are generally able to forecast more accurately what the impact will be of a given investment (e.g. the investment to enter a new market will cost X million, and firms know that by incurring this cost they will be able to enter the market—even if they may still face commercial uncertainty about the ultimate cost or profitability of entry). This difference does not have a material impact on the analysis of dynamic merger effects under Box 1. However, there are relevant differences between innovation and physical investments for effects under Box 2, which we discuss in Chapter 16.
- 15.66 The adverse dynamic effects of a merger under Box 1 may manifest themselves in lower levels of investment (including discontinuation of a pre-merger entry plan) or higher prices for future products resulting from the investment (e.g. higher prices for data packages with greater speed and capacity in a telecommunication market).

15.6 Reactions by non-merging firms

- 15.67 The discussion of the internalisation of negative externalities under Box 1 has so far focused on the behaviour of the merging firms, with and without the merger. This is because the merger internalises externalities between these firms, and this mechanism is a primary force behind the overall impact of a merger.
- 15.68 However, in most cases, mergers take place in markets with additional competitors other than the merging firms (i.e. mergers to monopoly are a rare occurrence). It is therefore relevant to also look at the impact of a merger on non-merging firms, as this will be a factor in determining the overall impact on consumers and competition.
- 15.69 The reaction of non-merging firms to a merger partially rests on whether the key parameters of competition (prices and investments) are 'strategic substitutes' or 'strategic complements'.⁵³⁶ For example, in common models of competition, prices are typically strategic complements: if a rival increases prices, this diverts demand to other competitors, inducing them to price higher too.⁵³⁷

⁵³⁴ If firms are capacity constrained, they face an opportunity cost of serving a given customer in addition to the cost of production (if profit margins are positive). Greater capacity reduces this opportunity cost.

⁵³⁵ See López (2025).

⁵³⁶ Actions are strategic substitutes if one firm's increase in a choice variable (such as output or investment) leads rivals to reduce theirs (e.g. as in Cournot quantity competition). They are strategic complements if one firm's increase leads rivals to increase theirs (e.g. as in Bertrand price competition). The distinction matters for merger analysis because it determines whether rival firms amplify or dampen the merged entity's post-merger change in investment.

⁵³⁷ See e.g. Stenbacka and Tombak (2003).

- 15.70 For investments, the assessment is more complex. There are common settings (e.g. markets where firms can offer parallel innovations, and are not racing to invest in a single, winner-take-all, innovation) where investments are typically strategic substitutes, implying that stronger investment by one firm makes the returns for investments lower for rival firms (as this makes it more likely that firms can only catch up with their rivals when they invest, rather than leaping ahead of them).⁵³⁸ Moreover, in the context of a merger, investment by non-merging rivals is affected not only by the change in the investment level of the merging parties, but also by changes in price, which may also induce the rivals to invest more even when merging parties invest less.⁵³⁹
- 15.71 However, there are models of investment competition, where investments are strategic complements. This is the case in some (but not all) patent race settings, where stronger innovation efforts by a participant in the race induce other participants to invest more as well, to increase their chances of winning the contest.⁵⁴⁰ This assumption may apply in practice to some markets with strong ‘winner-take-all’ effects (e.g. due to network effects), where firms compete to be the first to offer a new technology and capture a large share of the market.
- 15.72 In part because of these complexities, the guidance from the economic literature on the impact of mergers in dynamic settings on non-merging parties is not as extensive as the one available in pre-merger duopoly settings (where there is no other competitor in addition to the merging firms).
- 15.73 However, some insights are available and can guide the conceptual framework.
- 15.74 Existing academic work suggests that in a symmetric pre-merger setting with separate investment and pricing stages, and in cases where the merger reduces the investments of the merged entity, ‘outsiders’ to the merger (i.e. non-merging parties) usually react by increasing their investments. Assuming the proposed merger is profitable, the reaction of outsiders is, however, typically not sufficient to avoid a reduction in ECW.⁵⁴¹ Therefore, taking into account the reaction of non-

⁵³⁸ E.g. in Federico, Langus, and Valletti (2018), investments in product innovation are strategic substitutes if the rewards from ‘escaping competition’ are larger than those from ‘catching up’ with rivals. If this is the case, greater investment by rivals makes it more likely that a firm will be ‘catching up’, reducing investment incentives. D’Annunzio et al. (2026a) also consider the case where investments are strategic substitutes.

⁵³⁹ As discussed in D’Annunzio et al. (2026a) in a sequential setting (i.e. with observable R&D efforts) this leads to an upward shift in the best response function of non-merging parties.

⁵⁴⁰ See for example the model of Lee and Wilde (1980) reviewed in Volume 2, Chapter 2. However, in other models of patent races (e.g. Loury, 1979) innovation efforts are strategic substitutes. This is also the case in efficient procurement models (Pietola, Tarantino, and Zenger, 2026).

⁵⁴¹ In several papers, outsiders increase their investment in response to a merger, but overall investment still falls due to the reduction in the investment by insiders. This is the case in the simultaneous setting of Motta and Tarantino (2021) where prices and investment are set at the same time, for a broad class of demand functions (see discussion in Volume 2, Chapter 3). It is also the case in several sequential settings where firms first compete in investment, and then in price, see: the simulations reported for a ‘3-to-2’ merger in Motta and Tarantino (2021), under Constant Elasticity of Substitution (CES) demand and in Salop circular model (in the Salop model, total investment does not vary post-merger, as total output is fixed); the simulations reported for a ‘4-to-3’ merger in Federico, Langus, and Valletti (2018), for the case of linear demand (the working paper version of the article reports similar results for a ‘3-to-2’ merger); and the simulations reported by López (2025) for a linear quality-enhancement model and for a linear capacity sharing model, for both a ‘3-to-2’ and a ‘4-to-3’ merger. On the other hand, D’Annunzio et al. (2026a) show that there are cases in a ‘3-to-2’ merger in a Salop model where both the insiders and the outsider increase investment post-merger, and cases where the insiders invest more and the outsider invests less (for sufficiently low product differentiation). ECW falls in these simulations even in cases where investments increase for all firms (and hence total investment is higher).

merging parties does not invalidate the predictions about the likely sign of the dynamic effects under Box 1 (even though it may have implications for the magnitude of effects—see Section 15.7).

- 15.75 The presence of significant asymmetries pre-merger may affect these predictions. For example, if the merger brings together two weaker firms (i.e. firms with significantly higher costs than incumbents with a larger share of the market), there are some indications in the academic literature that the reaction of the non-merging parties may be sufficiently strong to increase overall investment and prevent a reduction in ECW.⁵⁴²

15.7 Magnitude of potential dynamic effects under Box 1

- 15.76 We conclude our review of dynamic effects included under Box 1 by commenting on factors that can affect their expected magnitude. This is particularly relevant if there is a need to balance countervailing effects under Box 2. We return to this issue when discussing a taxonomy of possible cases and sources of evidence (in Chapter 20).
- 15.77 At a conceptual level, the magnitude of the direct effects under Box 1 depends on the size of the negative externalities internalised by the merger. This is in turn a function of the profitability of sales and the strength of future price and innovation diversion between the merging parties. The higher the expected profit margins and diversion ratios, the higher the likely adverse effects under Box 1.
- 15.78 The overall impact of adverse effects on ECW under Box 1 will tend to be larger from a merger that brings together two out of a limited number of players with investment capabilities and projects aimed at similar consumer needs.⁵⁴³ The existing theoretical and structural empirical academic literature has also found that the adverse dynamic effects of mergers tend to be larger in concentrated markets.⁵⁴⁴
- 15.79 Adverse effects will also be higher if a merger allows an incumbent with market power to protect high existing profit margins from the threat of competition from a potential entrant. A higher degree of market power of the incumbent is therefore associated with stronger effects under Box 1.

⁵⁴² See Vuong and Baranes (2021) for the case of significant cost asymmetries, and a merger between the two high-cost firms. In this case, potential adverse effects under Box 1 may be weaker in the first place, as the merger would bring together relatively smaller competitors.

⁵⁴³ See US Merger Guidelines (p. 39): 'Where firms are two of a small number of companies with specialized employees, development facilities, intellectual property, or research projects in a particular area, competition between them will have a greater impact on their incentives to innovate'.

⁵⁴⁴ See the simulations reported by Motta and Tarantino (2021) on consumer surplus, for different levels of the number of players pre-merger (between 3 and 10), for the Constant Elasticity of Substitution demand model; the simulations reported by Federico, Langus, and Valletti (2018) on total innovation (as captured by the notion of first-order stochastic dominance) for different levels of the number of players pre-merger (between three and eight) and for a linear demand model; the simulations reported by López (2025), comparing the impact of a 3-to-2 merger to the impact of a 4-to-3 merger (for both the quality enhancement and capacity sharing model); results of Igami and Uetake (2020), indicating larger consumer welfare losses for 2-to-1 and 3-to-2 mergers, compared to mergers in markets with more competitors; Guthmann and Rahman (2026), who find increasing welfare harm from mergers as the number of firms pre-merger drops below 4; and Elliott, Hounghonon, Ivaldi, and Scott (2025), who find that consumer welfare drops as the number of firms drops below 8, with particularly sharp falls as the number of firms goes below approximately 4-5. On the other hand, D'Annunzio et al. (2026a) compare a 2-to-1 merger to a 3-to-2 merger in simulations for a Salop model of demand, finding that in some cases consumer harm in a 3-to-2 merger is larger than in a 2-to-1 merger.

- 15.80 Effects will also be larger if innovation is an important parameter of competition (e.g. consumers place a high value on innovation) and if the merging firms are strong innovators.

16 **Box 2: procompetitive dynamic effects of horizontal mergers**

Key points

- Box 2 of our framework captures the potential procompetitive dynamic effects from a horizontal merger, and is split between the internalisation of spillovers (Box 2A) and dynamic synergies (Box 2B).

Box 2A: internalisation of spillovers

- When the fruits of innovation cannot be fully captured by the original innovator and benefit rivals, a merger may internalise positive externalities between competitors and boost innovation incentives. This mechanism may apply to knowledge-based innovation, but not to investment in physical infrastructure (which is excludable).
- However, mergers may also internalise the impact of negative externalities from spillovers—as the merged entity becomes more concerned about spillovers to non-merging firms. This means that the net effect on innovation incentives is ambiguous and depends, at least partly, on whether the effect stemming from *internal* spillovers between merging parties exceeds that from *external* spillovers from merging to non-merging parties.
- The academic literature indicates that a merger can boost innovation and benefit consumers in the presence of spillovers that are concentrated between the merging parties. However, insights and guidance are more limited for oligopoly settings in which spillovers arise not only between merging firms but also from merging firms to their rivals.

Box 2B: dynamic synergies

- Dynamic synergies reflect an improvement in the capabilities of the merged entity, arising from the combination of complementary assets and economies of scale.
- We identify five main sources of such synergies: (i) *technology transfers*, where the non-rivalrous nature of innovation allows one party's innovations to be shared with and exploited by the other, increasing both incentive and diffusion effects; (ii) *reductions in R&D costs or increases in R&D effectiveness*, through elimination of duplicative costs (to the extent these reduce incremental R&D costs), integration of complementary assets, or bridging asymmetries between a strong acquirer and a weaker innovator; (iii) *reduction in investment costs or increases in investment effectiveness*, for example via sharing of infrastructure or other complementary assets; (iv) *production cost efficiencies*, including economies of scale and elimination of double marginalisation, which can raise innovation incentives by increasing post-investment profits; and possibly (v) *easing of financial constraints*, enabling an acquired firm (e.g. a small innovative entrant) to bring innovation to market that it could not otherwise fund.

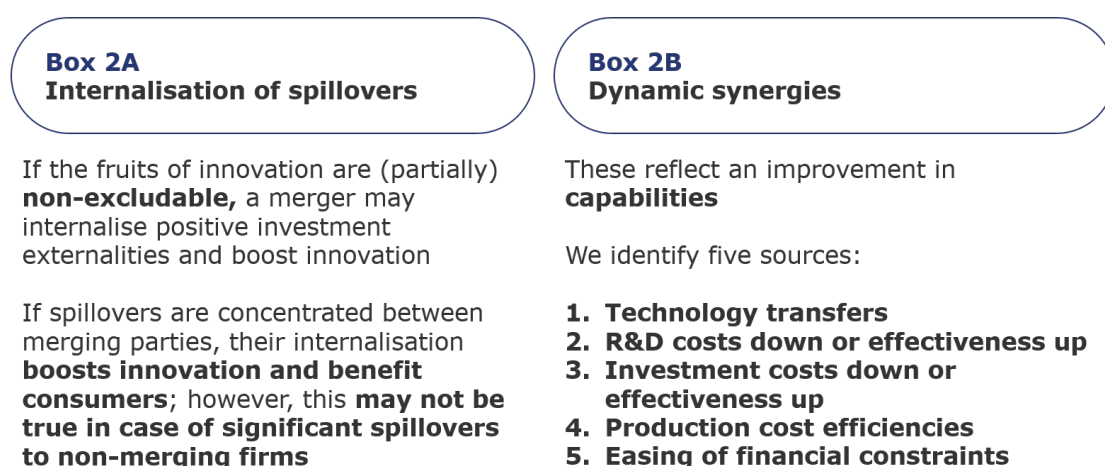
Applying the Three-Limb Test to Box 2 effects

- The well-established three-limb test for efficiencies provides a suitable substantive framework to evaluate dynamic procompetitive effects, under an ECW standard: (i) Benefits to consumers: dynamic synergies should pass through to consumers via enhanced investment or lower prices; (ii) Merger

specificity: benefits must not be achievable through less anticompetitive alternatives, with the economic assessment reflecting the feasibility and likelihood of such alternatives; and (iii) Verifiability: evidence standards should be consistent with those applied to harm under Box 1.

- 16.1 As set out in Chapter 12, we identify two distinct sources of procompetitive dynamic effects under Box 2: the internalisation of spillovers (Box 2A); and dynamic synergies that improve the capabilities of the merging parties (Box 2B). These effects may mitigate or outweigh adverse effects under Box 1, depending on the circumstances, as we discuss in this Chapter.
- 16.2 An overview of the procompetitive dynamic effects from horizontal mergers as considered under Box 2 is provided in Figure 16.1.

Figure 16.1 Box 2: procompetitive dynamic effects of horizontal mergers



Note: This figure sets out the distinction between Box 2A (internalisation of spillovers) and Box 2B (dynamic synergies), as part of the overall procompetitive dynamic effects in horizontal mergers.

Source: Oxera.

16.1 Box 2A: internalisation of spillovers

16.1.1 Overview of the impact of internalisation of spillovers

- 16.3 In some settings, mergers can address externalities in a way that benefits consumers. For example, more competitive behaviour by one firm can increase the profits earned by other firms, giving rise to a positive externality that is internalised by a merger. A standard (static) example is the competitive interaction between producers of complementary goods (which can give rise to a 'Cournot effect'—see further in Chapter 18). In this case, a merger can lead to lower prices.⁵⁴⁵
- 16.4 In the context of dynamic merger effects, a potential source of positive externalities between merging firms is the possibility of *spillovers* from innovation. These can apply to both product and process innovation. As set out in Section 13.1, if innovation is partially non-excludable (e.g. in the presence of weak IP rights), then innovation by one firm may benefit other firms (in the same or different product

⁵⁴⁵ Another common example of a static positive externality is the case of vertically related goods and the elimination of double marginalisation (see Chapter 8).

markets). This generates a positive externality between firms, via knowledge spillovers.⁵⁴⁶

- 16.5 A merger between innovators allows the merging firms to internalise this positive externality, and may lead to a positive effect on innovation incentives. This positive effect reflects the fact that, after a merger, each merging firm would internalise the fact that their innovation efforts increase the profits of the other merging party, thus increasing innovation incentives. This effect can apply also if the innovators are active in separate product markets, but is even stronger in the case of competitors in the same market (as in this case, absent the merger, innovation by one firm would benefit its competitor, reducing the profits that can be gained by the innovator, and hence further lowering the incentives to innovate compared to the merger).
- 16.6 Another distinguishing feature of innovation is that it is a cumulative process. That is, innovation by one firm often builds on innovation ('prior art') by previous inventors. This creates another source of complementarity between rival firms. It can lead to an intertemporal spillover across firms, and a positive externality, between the early inventor and the subsequent ones. While this effect can help the diffusion of knowledge absent the merger,⁵⁴⁷ a merger would internalise this positive externality, potentially leading to greater incentives to invest in the first innovation.⁵⁴⁸
- 16.7 Spillovers may, however, also lead the merging firms to internalise spillover-related *negative externalities*. These externalities result from the fact that the merged entity is more concerned (than each of the merging firms pre-merger) about spillovers to non-merging firms, as it has more profits at stake when non-merging firms benefit from spillovers and become more competitive. To illustrate this mechanism, consider a three-firm setting pre-merger (with Firms A, B, and C), where spillovers from A benefit both B and C. Pre-merger, A does not internalise the negative impact of its spillovers to C on the profits of B. On the one hand, a merger between A and B internalises this negative externality, reducing the innovation incentives of A (as it now has more exposure to competition from C). On the other hand, the merger also internalises the positive externality caused by spillovers from A to B. The net effect of the merger on innovation incentives is in principle ambiguous and depends on the decomposition between internal spillovers (between the merging parties) and external spillovers (from the merging firms to non-merging firms). This in turn depends on the asymmetry of spillovers between firms, with pro-innovation effects of a merger more likely if the impact of internal spillovers is larger than that of external spillovers. In the polar case of merger-to-monopoly, there are no external spillovers, and this mechanism unambiguously enhances the innovation incentives of the merged entity. That said, a merger-to-monopoly typically leads to stronger adverse effects on consumers under Box 1.
- 16.8 The internalisation of spillovers from a merger may overcome the detrimental impact of a merger on innovation incentives and ECW under Box 1, depending on the strength of the spillovers, on the degree of internalisation enabled by the merger, and on the effects through non-merging parties. In the polar case of merger-to-monopoly, the economic literature shows that a merger can benefit

⁵⁴⁶ This includes potential information spillovers, which arise if successful innovation by one firm can help other firms identify which research paths have a higher likelihood of success. For a paper modelling this effect, see Das, Mayskaya, and Nikandrova (2026).

⁵⁴⁷ See the notion of 'symbiotic competition' modelled by Guthmann and Rahman (2026).

⁵⁴⁸ As noted by Régibeau and Rockett (2019), in a sequential context, a merger may also address hold-up concerns by the firm engaging in sequential innovation, and also enhance incentives in the follow-on innovation.

consumers if spillovers are strong, and their internalisation is merger-specific.⁵⁴⁹ This insight can extend to an oligopoly setting where spillovers are concentrated between the merging parties, with sufficiently small external spillovers to non-merging firms. The analysis is, however, more complex in the more general oligopoly case, with material internal and external spillovers.⁵⁵⁰

- 16.9 Dynamic merger effects under Box 2A are related to those under Box 1 as they capture the *direct* change in the *incentives* of the merged entity that is brought about by the change of ownership associated with the merger (without requiring any changes in the capabilities of the merged entity—these are instead captured under Box 2B). On the other hand, there are also similarities between Box 2A and Box 2B, in that they both capture mechanisms which are potentially procompetitive, and can counter the impact of effects under Box 1.

16.1.2 Factors determining the relevance and strength of spillovers

- 16.10 As set out above, the positive effects of the internalisation of spillovers can be expected to be more relevant if the merging parties capture a significant fraction of the spillovers that each of them generates absent the merger (i.e. if spillovers to non-merging firms are limited). For example, if a large fraction of spillovers from innovation by Firm A benefits Firm B (and vice versa) then a merger between the two firms will lead to a positive effect on their innovation incentives under Box 2A, and this effect becomes stronger as that fraction increases. Moreover, if Firms A and B are close competitors, the negative effect of the spillover on the firm generating it will also be stronger, leading to a more significant pro-innovation effect from this channel (at the same time adverse effects under Box 1 are typically stronger in this case). Similarly, intertemporal knowledge spillovers are more likely to be relevant if the two merging firms frequently build on each other's innovations absent the merger (relative to other competitors).
- 16.11 By contrast, if spillovers from innovation are diffuse, then a horizontal merger between two product market competitors may not lead to merging firms internalising a significant fraction of the overall spillovers, and may even reduce innovation incentives as external spillovers to non-merging parties harm the other merging firm, and the merger internalises this effect.
- 16.12 A key determinant of the presence and importance of knowledge spillovers is the degree of excludability of the results from successful innovation, and the effectiveness of IP rights.⁵⁵¹ Strong and effective IP rights (through patents and other defensive mechanisms) make it more likely that investors will be able to appropriate the benefits of their innovation, and less likely that spillovers will be significant.⁵⁵²
- 16.13 The effectiveness of the licensing market is an important determinant of the significance of the internalisation of inter-temporal spillovers. In circumstances where ex ante licensing is feasible and effective, considerations arising from the sequential nature of innovation are less important. For example, this would be the

⁵⁴⁹ D'Aspremont and Jacquemin (1988), Leahy and Neary (1997), and López and Vives (2019).

⁵⁵⁰ Atallah (2016) and Pietola, Tarantino, and Zenger (2026), D'Annunzio, Jullien, Lefouili, and Madio (2026b).

⁵⁵¹ See discussion in US DOJ & FTC (2010), 'Horizontal Merger Guidelines', 9 August. ('The Agencies also consider the ability of the merged firm to appropriate a greater fraction of the benefits resulting from its innovations. Licensing and intellectual property conditions may be important to this enquiry, as they affect the ability of a firm to appropriate the benefits of its innovation', p. 31). Note that this effect is no longer identified in the US Merger Guidelines.

⁵⁵² See discussion of the concept of 'appropriability' in Shapiro (2012).

case if licensing agreements can be negotiated between early innovators and follow-on innovators before the former incur significant sunk R&D investment cost, such that early innovators can obtain at least part of the value of successive innovations, and hold-up of follow-on innovation is not a significant concern.⁵⁵³

- 16.14 Whether positive externalities from spillovers apply depends on the type of innovation or investment concerned. Positive externalities via spillovers may be relevant to investment in R&D given the properties of innovation and knowledge (see discussion under Box 2B in Section 16.2 below). They are unlikely to apply to investment in physical infrastructure, as this is typically an excludable asset (i.e. use charges can readily be levied). That is, the owner of the asset is generally able to exclude other firms from using it (or charge them for doing so), and hence no positive spillovers are generated by firms investing in their infrastructure. While spillovers are typically less salient in infrastructure investment than in R&D investment, other forms of positive spillovers may arise with the former as well, for example, those resulting from third-party access obligations in network industries.⁵⁵⁴
- 16.15 Relatedly, an important factor in the assessment of the internalisation of knowledge spillovers relates to *merger specificity*. That is, are the merging parties able to internalise knowledge spillovers through other, less anticompetitive, means than a merger? For example, can they enter into an R&D cooperation agreement, which allows them to share the benefits of innovation, without coordinating their conduct in the product market (as would be the case in a merger), or sign a licensing agreement?⁵⁵⁵ We briefly discuss the applicability of the three-limb test for efficiencies to dynamic merger effects under Box 2A in Section 16.3 below, including a broader discussion of the relevance of merger specificity.

16.2 Box 2B: dynamic synergies

16.2.1 Overview of dynamic synergies

- 16.16 The second element of procompetitive dynamic effects under Box 2B captures any improvement in the joint capabilities of the merged entity that is enabled by the merger. This is about the *enhanced ability* of the merged entity to be a more effective competitor in the market, thanks to the combination of assets brought about by the transaction.⁵⁵⁶ This includes cost savings enabled by a merger, but also broader improvements in the effectiveness of investments. Contrary to effects under Box 1 and Box 2A, dynamic synergies do not reflect a *direct* change in *incentives* brought about by the change of ownership that follows a merger.⁵⁵⁷
- 16.17 As for effects under Box 2A, dynamic synergies can enhance investment incentives, to the benefit of consumers. They can also increase the effectiveness and diffusion

⁵⁵³ See Régibeau and Rockett (2019), who point out that IPR owners are typically unable to extract the full value of their technology through arm's-length licensing, suggesting that a merger would lead to higher incentives for future innovation through this mechanism. They also suggest that the absence of pre-merger licensing or evidence that arm's-length licensing does not provide high returns to the licensor would imply that this type of merger-specific benefits is more likely.

⁵⁵⁴ See Cambini and Jiang (2009).

⁵⁵⁵ For a discussion of the risk of coordination in the product market from an R&D cooperation see Sovinsky (2022).

⁵⁵⁶ This follows the definition in Farrell and Shapiro (1990).

⁵⁵⁷ To recall, as noted in footnote 495, Box 2A formally captures the fact that two previously separate objective functions are now brought together, while Box 2B reflects the impact of changes in the parameters of the optimisation problem faced by firms (e.g. lower costs, or greater impact from innovation).

of innovation, leading to a *direct* beneficial effect on ECW. These effects can counter or offset any adverse dynamic effects under Box 1, if sufficiently significant. Some of these effects (e.g. reduction in incremental costs of production) may also directly feed into lower prices, offsetting both static and dynamic price effects.

- 16.18 The scope for dynamic synergies is generally broader than the static equivalent, since dynamic merger effects generally materialise over a longer time horizon than static effects. A longer time horizon allows firms to adjust their investment and production processes, and capture greater potential gains from the combination of assets brought about by a merger. This implies that dynamic merger effects relate to a broader set of competitive variables (e.g. investment decisions and not just price). For example, a merger may lead to a reduction in investment costs, which does not affect short-run pricing decisions (and hence has no impact on static effects) but increases investment incentives (and thereby affects dynamic effects).
- 16.19 Several of the dynamic synergies discussed in this Section can be seen as forms of *economies of scale* or *scope* enabled by a merger. However, a generic increase in scale resulting from a merger does not necessarily lead to a dynamic synergy. As we set out in this Section, this depends on the *specific mechanism* (if any) through which greater scale is able to feed into synergies and ultimately benefit consumers.
- 16.20 An overarching issue relating to dynamic synergies is the extent to which a merger is required to achieve the benefits from the synergies, as opposed to contractual solutions. We discuss this point in Section 16.3, in the context of merger-specificity.
- 16.21 Based on the academic literature on dynamic synergies and case practice, we discuss below potential sources for dynamic synergies, distinguishing between five cases:
- technology transfers (or knowledge sharing);
 - reduction in R&D costs or increases in R&D effectiveness (e.g. through asset complementarities);
 - reductions in investment costs or increases in effectiveness of investment, including through infrastructure sharing or asset complementarities;
 - reduction in production costs;
 - reduction in financial or resource constraints.

16.2.2 Technology transfers (or knowledge sharing)

- 16.22 A merger enables an innovator to share the outcome of its innovation with the other merging party without the need for contracting. Given the non-rivalrous nature of innovation, this does not reduce the availability of the innovation to the original innovator, but may allow the other merging firm to exploit the same innovation to enhance its position in the market (by improving the quality of its product or reducing its cost). As in the case of spillovers, technology transfers rest on the specific properties of innovation (notably its typically non-rivalrous nature). They are therefore not directly applicable to investments in physical infrastructure, even though a similar economic effect may apply to that type of investment as well via the sharing of infrastructure (see discussion in Subsection 16.2.4).
- 16.23 Technology transfers differ from spillovers under Box 2A, since in this case the sharing of knowledge is *enabled* by the merger and does not take place absent it. By contrast, spillovers occur also absent the merger (i.e. they are involuntary), and the merger enables their internalisation.

- 16.24 Technology transfers can have two distinct procompetitive effects: an *incentive* effect, and a *diffusion* effect.⁵⁵⁸ The former captures the fact that the original innovator will typically face stronger incentives to invest if it can capture greater benefits post-merger, due to the additional profits captured (appropriated) by the other merging firm. The latter reflects the fact that, even holding innovation efforts constant, innovation is more effective as it applies to multiple products post-merger (and hence reaches a wider group of customers).
- 16.25 In the case of product innovation, both effects are premised on the existence of some form of differentiation between the merging firms, implying that innovation by one firm can be used by the other merging party to improve its products, and meet differentiated customer needs (hence broadening the effective reach of a given innovation). Differentiation between merging parties may be the result of product differentiation, but may also arise from geographic complementarities in distribution assets (implying that innovation by one firm can reach broader distribution through the other merging party). For example, by acquiring a geographically differentiated target, companies may be able to expand the distribution of their products more effectively to other geographies, in turn increasing their customer base from which to generate payoffs from innovation.
- 16.26 For the case of process innovation, the procompetitive effect of technology transfers may also be driven by a volume effect, with the merger enabling a given process innovation to be applied to a larger volume base. This can lead to an incremental benefit compared to what each merging party can achieve on its own.⁵⁵⁹
- 16.27 Both the incentive and the diffusion effects from technology transfers can be expected to benefit consumers, and can therefore mitigate or offset the adverse dynamic effect of mergers under Box 1.
- 16.28 Technology transfers are more likely to be present if IP rights are effective (which means that knowledge spillovers are limited absent the merger) and if the technology licensing market is imperfect (imperfect licensing implies that absent the merger an innovator cannot fully capture the value that its innovation might create for rival firms). With effective licensing, the additional value from technology transfers enabled by the merger is likely to be limited compared to the value that can be captured with external licensing absent the merger.
- 16.29 Knowledge sharing is more likely to arise and have a stronger impact if a given innovation can be shared effectively between the two merging firms. This will depend on the nature of the innovation (e.g. how upstream and general it is, and hence more easily applicable to different products) and on the nature of the products of the two merging firms. If they rely on similar technology (say, production of batteries for EVs) then innovation by one firm will be more applicable to the production process of the other. However, if the firms compete in the product market using different technologies (while addressing similar consumer needs), then the extent of greater technology diffusion after the merger may be more limited.
- 16.30 As reviewed in Chapter 5, the academic literature has examined the impact of voluntary technology transfers. It identifies the qualitative conditions for this effect to be at work,⁵⁶⁰ and it models it in specific environments. This literature shows that

⁵⁵⁸ See discussion in Régibeau and Rockett (2019).

⁵⁵⁹ See results of Denicolò and Polo (2021).

⁵⁶⁰ Notably in Régibeau and Rockett (2019).

a merger can benefit consumers by enabling voluntary technology transfers and hence increasing innovation incentives or promoting diffusion.⁵⁶¹

- 16.31 As noted above, post-merger technology transfers may allow an innovator to appropriate higher returns from its innovation. This is an aspect of appropriability enhanced by a merger, and is related to other channels which can boost appropriability under Box 1 and Box 2A—see Box 16.1.



Box 16.1 Concepts of appropriability

Appropriability reflects the extent to which an innovating firm captures the value its innovation creates. A horizontal merger can increase appropriability through three distinct mechanisms, which correspond to different elements of our conceptual framework. We focus here on the case of product innovation for simplicity.

1. Market power in product markets (Box 1). A merger allows firms to set the prices of their products jointly, internalising negative pricing externalities at work pre-merger.

Relaxing product market competition allows the merging parties to retain more of the surplus that a demand-enhancing product innovation generates, both by allowing the merging firms to increase prices of innovative products, and by allowing them to optimally set the prices of their other substitute products to divert sales to innovative products. This increases appropriability.

We capture this effect in **Box 1**. The literature reviewed in Chapter 5 indicates that, when an increase in market power is the only source of increased appropriability, a merger is unlikely to increase consumer surplus (absent synergies).

2. Internalising spillovers (Box 2A). If innovation is imperfectly excludable, a firm's R&D leads to a positive externality on its rival through imitation and knowledge spillovers. The merger internalises this effect, potentially increasing appropriability. The scope for this effect is more limited where pre-merger appropriability is already high, through strong Intellectual Property Rights limiting imitation from rivals. A merger may also allow firms to internalise inter-temporal spillovers in a sequential setting (with late innovators learning from early innovators).

In an oligopoly setting, this channel does not always lead to greater appropriability as any spillovers to non-merging firms remain in place after the merger and may become more harmful for the merged entity (as it now internalises the harm from spillover to non-merging parties on multiple products).

We capture this effect under **Box 2A** of our framework, and note that this mechanism may increase innovation incentives from a merger, to the benefit of consumers.

⁵⁶¹ The impact of knowledge sharing is studied in Denicolò and Polo (2021), Mukherjee (2022) and Moraga-González and Motchenkova (2026). These papers show that innovation sharing can enhance investments and benefit consumers in models where merging firms are horizontally differentiated or offer a homogeneous good with increasing marginal costs. There are also settings where this effect may not actually increase innovation incentives or benefit consumers (this is the case for the Hotelling model studied in Moraga-González and Motchenkova (2026)).

3. Sharing technology and innovation across products (Box 2B). Even without spillovers, a firm may fail to appropriate an innovation's full value because it cannot license it efficiently to others, due to contractual imperfection and reflecting the fact that technologies may be hard to value or transfer. Pre-merger this may leave a superior technology at least partially under-exploited. A merged entity may be able to apply a technology across its combined product base, raising both the diffusion of existing and future technologies, and the return to investing in future innovations (thus increasing appropriability). We capture this effect under **Box 2B** of our framework, and note that this mechanism benefits consumers, if present.

Source: Oxera.

16.2.3 Reductions in R&D costs or increases in R&D effectiveness

- 16.32 A merger may also enable firms competing in R&D to reduce their R&D costs, or increase the effectiveness of R&D (in terms of the probability of successful introduction of a new product, or the impact of innovation on quality and costs). This can increase innovation incentives and ultimately lead to more innovation, to the benefit of consumers.
- 16.33 A merger may reduce R&D costs or increase R&D effectiveness, via a number of mechanisms, including the following.
- *Efficiency gains in R&D through the elimination of duplicative activities or costs.* For example, in the case of R&D in pharmaceutical markets, post-merger firms may be able to avoid the duplication of some components of their research process (e.g. pre-clinical toxicology, data infrastructure, and regulatory expertise). The merged entity may also be able to centralise some R&D efforts and consolidate clinical-trial management teams. The reduction of duplicative R&D expenditure may lower the costs associated with a given R&D project, and enhance innovation incentives to invest in that project. This will be the case if the duplicative activities relate to avoidable costs of achieving a given R&D output, rather than relating to fixed costs at the level of the firm, that are independent of the level of R&D activities (e.g. central R&D functions and overheads). A reduction in the cost of a new product development or project would typically strengthen the incentives to invest in that product development, and hence increase the probability that more R&D projects are started and new products introduced.
 - *Complementarities in R&D capabilities.* Merging firms may have complementary assets in R&D, including human capital, research facilities, access to inputs, and development expertise. By bringing together these complementary assets, the firms may reduce the cost of innovation, increase the probability of innovation, or increase the impact of innovation.
 - *Complementarities in distribution and marketing assets.* Merging firms may also have complementary assets in distribution and marketing. By bringing together these complementary assets, the firms may increase the diffusion of innovation to customers (i.e. trading parties of the merged entity), and strengthen innovation incentives.
 - *Asymmetries between merging parties.* In some cases, one of the merging parties may have superior R&D capabilities to the other (e.g. a strong R&D player buying a smaller firm with an innovative idea, but with limited R&D assets or

expertise).⁵⁶² The merger may therefore result in lower R&D costs or a higher probability of successful innovation for the weaker player, making it more likely that its innovation will reach the market. Contrary to the other two mechanisms, this effect is one-sided (it applies to the innovation output of the weaker player only). This effect is additional to any reduction in financial constraints from which a smaller player may benefit (see Subsection 16.2.6).

- 16.34 The impact of reductions in incremental R&D costs and increases in R&D effectiveness has been studied in the academic literature. These mechanisms can increase the innovation incentives of the merging parties and, if strong enough, result in an overall increase in consumer welfare.⁵⁶³
- 16.35 When looking at merger-specific evidence on R&D cost efficiencies it is important to distinguish between reductions in R&D expenditure resulting from the elimination of duplicative costs (e.g. activity or project-specific fixed costs), and overall cost reductions due to the elimination of competing R&D activities. While the former can be a procompetitive mechanism as explained above, the latter reflects a *reduction in innovation efforts* by the merged entity, and is therefore the consequence of an anticompetitive effect of the merger to be captured under Box 1. We return to this distinction in Subsection 16.3.1 in the discussion of benefits to consumers.

16.2.4 Reductions in investment costs or an increase in the effectiveness of investment

- 16.36 A merger may enable firms to reduce their investment costs or increase the effectiveness of investment. This is the analogue of the first and second categories of synergy described in the previous sub-section, applied to investment in physical infrastructure rather than investment in R&D. The mechanisms are broadly similar, including notably the impact of asset complementarities.
- 16.37 A specific mechanism through which a merger may reduce the (incremental) cost of investment or increase the effectiveness of a given investment is via infrastructure sharing between the merging firms. The possibility to share infrastructure may provide merging firms with stronger incentives to invest in additional infrastructure, due to economies of density (e.g. in the case of investments to expand geographical coverage). This assumes that this effect is merger-specific and cannot be achieved contractually, as we discuss later in this Section.

⁵⁶² For example, in the case of a pharmaceutical merger, the acquirer may have more expertise than the target in the processes to obtain approval to distribute the drug based on the target's innovation, and in producing and distributing information to physicians on the efficacy of the new drug.

⁵⁶³ See for example: Cunningham, Ederer, and Ma (2021), which allow for the possibility that the incumbent buyer has a higher success probability than the target (which lowers the incidence of 'killer acquisitions'); Federico, Langus, and Valletti (2018), which model both a reduction in incremental R&D costs and an increase in the effectiveness of R&D (both effects can lead to an increase in innovation by the merging parties and, if sufficiently strong, an increase in consumer welfare); D'Annunzio, Jullien, Lefouili, and Madio (2026a), which show that efficiency gains in R&D increase the innovation incentives of the merging firms; Moraga-González and Motchenkova (2026), which model complementarities in R&D investments (which increase innovation incentives by the merging parties compared to a benchmark with no efficiencies); Pietola, Tarantino, and Zenger (2026), which model both reductions in incremental R&D costs (these induce higher innovation by the merging parties and can lead to higher consumer welfare unless there is a merger to monopoly), and efficiencies in product development, resulting in a better product post-merger (this can also increase innovation incentives and benefit consumers, unless there is a merger-to-monopoly); and Padilla, Piccolo, and Reynolds (2024), which consider synergies in fixed cost of entry, in the case of mergers between laggards.

- 16.38 This effect is likely to apply if merging parties can avoid duplicating some elements of infrastructure investment post-merger (and hence reduce the incremental cost of investment), over and above what may be mandated by regulation. Examples may include some of the costs of roll-out in the case of telecommunication markets (e.g. base stations in the case of mobile networks).⁵⁶⁴ Similarly, merging firms may be able to capture network allocation efficiencies, leading to better allocation of network resources, assets, sites, and spectrum. This can improve network capacity (for a given level of investment).⁵⁶⁵
- 16.39 Depending on the underlying technology and demand conditions, merging firms may be able to share infrastructure without reducing their joint capacity or quality of service (relative to the counterfactual), allowing them to serve a larger customer base for a given level of investment. The larger customer base could be the result of the merging firms offering differentiated products or having access to different customer bases to be upgraded to a new product.
- 16.40 Infrastructure sharing may therefore lower the effective unit (i.e. per user) cost of investments (or, equivalently, result in more capacity and greater quality for a given level of investment) and hence lead to higher incentives to invest for the merged entity.⁵⁶⁶ For example, sharing of mobile stations can lower the unit costs of rolling out a new technology (say, 5G) and hence increase incentives to invest in additional coverage or capacity. Similarly, sharing of active telecommunication equipment can lower the unit cost of offering a given level of quality of service (e.g. speed and capacity) and hence again enhance incentives to invest. This effect benefits consumers (potentially offsetting the impact of negative effects under Box 1), by leading to greater deployment of a new technology (hence enhancing quality).
- 16.41 By contrast, infrastructure sharing will not benefit consumers if it merely results in avoiding the duplication of a fixed cost at the level of the firm (e.g. an overhead), but does not affect the incremental cost of investment, or the quality of service offered.⁵⁶⁷
- 16.42 The academic literature on the effects of reductions in incremental investment costs shows that reductions in incremental investment costs brought about by a merger

⁵⁶⁴ These can be seen as *economies of density*, reflecting the fact that a denser network of consumers can be reached post-merger for a given level of investment (see discussion in UPS/TNT Express (2013, EU), in a context with static merger effects). Elliott, Hounghonon, Ivaldi, and Scott (2025) discuss *economies of density* in the mobile telecommunications network, indicating that combining networks reduces average transmission distances, increasing download speed (or reducing cost) for a given level of network investment. If demand fluctuates and is stochastic, combining networks and users smooth congestion when demand fluctuates, resulting in *economies of pooling*. This means that more can be achieved with the same fixed investment—or the same at a lower investment cost—when combining separate networks. The paper finds that economies of scale are driven largely by economies of pooling, rather than economies of density.

⁵⁶⁵ See discussion in Vodafone/Three (2024, UK), discussed in Annex A2.10.

⁵⁶⁶ This implies that infrastructure sharing needs to lower the Long Run Average Incremental Cost (LRAIC) of providing a given level of capacity, and hence increase incremental margins. LRAIC includes both the variable and fixed costs of providing a given level of capacity. This is a broader notion than Average Variable Cost, which captures the change in cost for a discrete change in output (at a given level of capacity), and does not include fixed costs which cannot be avoided in the short-run.

⁵⁶⁷ This is the case for infrastructure investments which do not vary with the level of output or capacity, e.g. overheads or a basic layer of infrastructure which firms need to offer independently of their output (for example, the infrastructure required to offer a minimum level of national coverage in line with regulatory obligations).

can increase the investment of merging parties and ultimately increase consumer welfare.⁵⁶⁸

- 16.43 Potentially beneficial effects from infrastructure sharing have been discussed extensively in case practice, notably in the case of mergers in the telecommunication industry (see case examples and case studies in Annex A2). An important question relates to *merger-specificity*: could merging parties achieve similar efficiency gains through a realistic and less anticompetitive arrangement, compared with the merger? In practice, in mobile telecommunication markets, this entailed an assessment of the feasibility and impact of *network sharing agreements* (NSAs). These are contractual arrangements which can enable competing parties to share part of their infrastructure, while in principle retaining price competition in the product market.⁵⁶⁹ Similarly, for the case of fixed-line networks, the assessment of network sharing may require an assessment of whether firms can engage in co-deployment of passive infrastructure, to avoid inefficient duplication. More generally, the assessment of merger-specificity for this type of efficiency centres on a comparison between the efficiency of a merger and alternative contractual arrangements.⁵⁷⁰

16.2.5 Production cost efficiencies

- 16.44 A merger may lead to direct reductions in production costs by enabling economies of scale in production,⁵⁷¹ or by combining complementary assets (e.g. production costs may fall as a result of the elimination of double marginalisation⁵⁷²).⁵⁷³
- 16.45 Cost efficiencies in production can directly mitigate or offset the effects of a merger on prices of future products (if sufficiently persistent). They can also have a positive indirect effect on innovation incentives.
- 16.46 The positive innovation effect results from the fact that the lower production costs increase post-investment profits (assuming imperfect pass-through of cost reductions into prices), and hence tends to increase the returns from investment. This effect is straightforward in the case of innovation in new products.⁵⁷⁴ It can also apply to innovations to improve existing products, where a reduction in production costs will increase both pre- and post-innovation profits. If innovation is

⁵⁶⁸ See simulations reported by López (2025), both for quality-enhancing investments (which increase the quality of competing products) and a capacity-sharing model (where investments increase capacity and reduce congestion). These show that reductions in incremental investment costs benefit consumers especially in settings with high investment intensity.

⁵⁶⁹ For a concrete example of the analysis of feasibility and effect of NSA as an alternative to a merger, see discussions in cases Telefónica Deutschland/E-Plus (2014, EU) and Hutchison 3G/Italy/Wind/JV (2016, EU). Relevant issues in the assessment include whether Network Sharing Agreements enable firms to capture the same or similar benefits as those associated with the merger, whether they entail contractual frictions undermining their effectiveness, and whether they may be associated with adverse competition effects (e.g. loss of investment or product market competition).

⁵⁷⁰ In some cases, mandated access regulation may also make some of the benefits from network coordination non-merger specific.

⁵⁷¹ This typically needs to go beyond a rationalisation of assets, and result in an improvement in the *joint* production possibilities of the merged entity (see Farrell and Shapiro, 1990).

⁵⁷² For example, in the case of mobile telephony mergers, this can be the result of the elimination of roaming charges (Orange/Jazztel (2015, EU)).

⁵⁷³ We are not referring here to reduction in production costs due to greater investment in process innovation. That is a separate procompetitive mechanism captured under the dynamic synergies discussed above (e.g. reductions in R&D costs, increases in R&D effectiveness) or under positive effects resulting from spillovers (Box 2A).

⁵⁷⁴ D'Annunzio et al. (2026a).

demand-enhancing, the post-innovation effect tends to be larger, resulting in a net increase in innovation incentives.⁵⁷⁵

16.2.6 Reduction in financial or resource constraints

- 16.47 A merger may lead to dynamic synergies by mitigating the financial constraints of one or both of the merged firms.⁵⁷⁶
- 16.48 For example, one of the merging parties may be financially constrained pre-merger for several reasons. It may be a startup with limited existing profits and limited access to external finance (due to standard asymmetric information concerns). The startup may therefore not be able to incur the required R&D costs to introduce a new innovation to the market. Acquisition by a player subject to a less-stringent financial constraint (e.g. a well-resourced incumbent, or an outside investor) may therefore allow the innovation to be successfully introduced to the market.⁵⁷⁷ More generally, acquisition of a financially constrained firm by a firm that is not subject to financial constraints may enable incremental investments by the acquired firm which would not take place otherwise. Similarly, a merger may prevent an inefficient liquidation of otherwise productive assets through a fire sale, or mitigate debt-overhang problems that may discourage firms from investing in otherwise NPV-positive projects since a substantial share of the resulting gains accrues to existing creditors rather than shareholders.
- 16.49 A central question in such a setting is whether the financially constrained target has access to alternative sources of external funding (e.g. through external investors, including via IPOs, or debt relief). The nature of the banking system and maturity of capital markets in the markets where the merger takes place will be relevant to this question. A related question is why a change of ownership would be more effective than access to external finance or an alternative acquisition to resolve a financial constraint. This relates to the more general issue of merger-specificity (we return to this issue below in Subsection 16.3.2, and in the discussion of acquisitions of small innovative entrants in Chapter 20).
- 16.50 Another potential setting in which the presence of financial constraints may lead to dynamic synergies is when both merging firms are financially constrained (relative to the size of available and profitable investment projects). In this case, if the merger leads to fixed cost savings (e.g. by avoiding duplication of fixed costs), it may enable the merged entity to improve profitability and its price of new debt, increasing the ability to raise debt and finance investments. However, in practice this requires the measurement of the effect on fixed-cost savings on the price of debt in the first instance, and then the ability and incentive to use the improved price of debt to fund new investments—as opposed to channelling savings into higher returns to shareholders.
- 16.51 Generally, the strength and depth of financial markets in the markets where the merging firms are active is an important determinant of the likelihood and impact of financial constraints pre-merger, and for the scope of financial synergies through a merger. Stronger and more sophisticated financial markets reduce the likelihood

⁵⁷⁵ See Bourreau, Jullien, and Lefouili (2026).

⁵⁷⁶ For two concrete applications, see the discussion of financial constraints in cases *Hutchison 3G/O2 Ireland* (2014, EU) and *Hutchison 3G Italy/Wind/JV* (2016, EU).

⁵⁷⁷ Motta and Peitz (2021) model a setting of this kind, identifying conditions when a merger would lead to an 'efficient upgrade' by an incumbent, and increase consumer welfare. This requires the startup to be financially constrained, and the incumbent to face sufficient incentives to introduce a new product (despite the cannibalisation of sales of its existing product). The sources of financial constraints are, however, not modelled in this paper.

that financial constraints are significant pre-merger, and that a merger would lead to financial synergies.

16.3 Applicability of three-limb test to effects captured under Box 2

- 16.52 Under the three-limb test of the EC Horizontal Merger Guidelines, for efficiencies to be taken into account they have to *benefit consumers*, be *merger-specific*, and be *verifiable*.⁵⁷⁸
- 16.53 We set out in this Section how this framework can be applied and adapted to the case of effects under Box 2A and 2B. In doing so, we focus on the *substantive* aspects of the economic assessment of possible procompetitive effects, without commenting on procedural questions around the burden of proof for demonstrating that the three criteria are met in a given case, or on legal interpretations of the three-limb test.
- 16.54 Overall, from a substantive perspective, the three-limb test for the assessment of efficiencies set out in existing guidelines provides a valid framework for the evaluation of procompetitive effects under Box 2A and Box 2B, under an ECW standard.
- 16.55 While the discussion in this Section is largely conceptual, in Chapter 20 of the main report we discuss practical examples and case studies in the review of the taxonomy of potential cases with dynamic effects.

16.3.1 Benefits to consumers

- 16.56 The effects discussed above under Box 2A and Box 2B have the potential to benefit consumers, by enhancing investment incentives, increasing the diffusion and impact of innovation, and potentially lowering future prices.
- 16.57 As is the case for static efficiencies, key issues in the determination of benefit to consumers from spillovers and dynamic synergies are:
- whether dynamic benefits (e.g. higher innovation due to a dynamic synergy) are likely to feed through into greater competition to the benefit of consumers;
 - whether savings in investment costs reflect an anticompetitive reduction of output; and
 - the timeliness of dynamic synergies.
- 16.58 We discuss each question in turn.

Pass-through to consumers of dynamic synergies

- 16.59 As discussed above, the range of cost savings which can lead to dynamic synergies is broader than its static equivalent. While for a static analysis only variable unit cost savings are typically passed through to consumers (at least to some extent), in the case of R&D or investment costs, a merger may lead to savings in categories of costs that do not affect pricing decisions in the short-run but which are variable in innovation effort (e.g. a lowering of activity or project-specific fixed costs may lead to a larger number of R&D projects).⁵⁷⁹ Lower R&D costs can benefit consumers by leading to stronger investment incentives, and hence result in higher quality or lower production costs in the future. The impact of this type of cost savings needs

⁵⁷⁸ The CMA Merger Guidelines and the US Merger Guidelines rely on a similar framework.

⁵⁷⁹ These are costs that are fixed in the level of output, but that vary with investment effort (e.g. project-specific R&D costs).

to be evaluated case-by-case, requiring an assessment of the nature of the R&D and investment process, of how costs vary with different levels of investment and with the number of projects, and of the impact of the potential synergies.

- 16.60 The nature of pass-through to consumers of dynamic synergies from a merger will depend on the nature of the synergy (e.g. is it a demand-side or supply-side synergy?) and on market conditions, including the intensity of competition post-merger. Supply-side synergies (e.g. lower future variable production costs due to greater investment in process innovation) are subject to a standard cost pass-through analysis, typically resulting in partial pass-through to consumers (as firms retain some of the benefits from the reduction in variable costs). Demand-side synergies from greater investments (i.e. the introduction of new or improved products) are also subject to a pass-through analysis, as the merged entity is likely to adjust prices accordingly to claw back part of the resulting benefits.⁵⁸⁰ The analysis here is similar to the standard approach to the pass-through of cost savings, with some of the demand-side benefits expected to be captured by the merging firms and some by the consumer.
- 16.61 Two general categories of economic techniques can be used to measure consumer valuation (i.e. willingness to pay) for higher-quality products.⁵⁸¹ First, stated-preference methods elicit consumers' valuations directly through hypothetical choices or questions. Examples include contingent valuation surveys, which ask respondents their willingness to pay for a specified quality improvement; and discrete-choice surveys (or conjoint surveys), which require respondents to choose between products with different price and quality attributes.⁵⁸² Second, revealed-preference methods infer consumer valuations from observed market behaviour using sales data and econometric techniques. These include hedonic pricing models, which estimate the implicit value of quality attributes based on variations in prices, sales, and product characteristics; and demand estimation, which estimates how purchasing or switching responds to price and quality differences.

Distinguishing between dynamic synergies and savings from reductions in the level of investment activities

- 16.62 Whether a dynamic synergy reflects a genuine reduction in investment costs that can benefit consumers through higher future quality or lower future production costs, or is the result of an anticompetitive reduction in investment activities may be central to the assessment of dynamic synergies. Suppose that after a merger, the merged entity plans to discontinue a significant part of overlapping R&D programmes and hence benefit from a large reduction in R&D costs.⁵⁸³ Is this a dynamic synergy that can ultimately benefit consumers (i.e. a procompetitive effect under Box 2B), or instead the result of a reduction of innovation efforts after the merger (i.e. the outcome of an anticompetitive effect under Box 1)?
- 16.63 The relevant distinction here is between a 'shift' in the R&D cost curve and a movement along the R&D cost curve (see Figure 16.2). If the merger results in a shift in the R&D cost curve (implying that R&D costs are lower for any given level

⁵⁸⁰ See discussion in Deutsche Börse/NYSE Euronext (2012, EU), paras 1178–1186 (on lower implementation and maintenance costs) and paras 1234–1242 (on collateral benefits).

⁵⁸¹ Further evidence on customers' willingness to pay may be derived from ordinary-course company assessment or investors' expectations concerning demand for new or improved products.

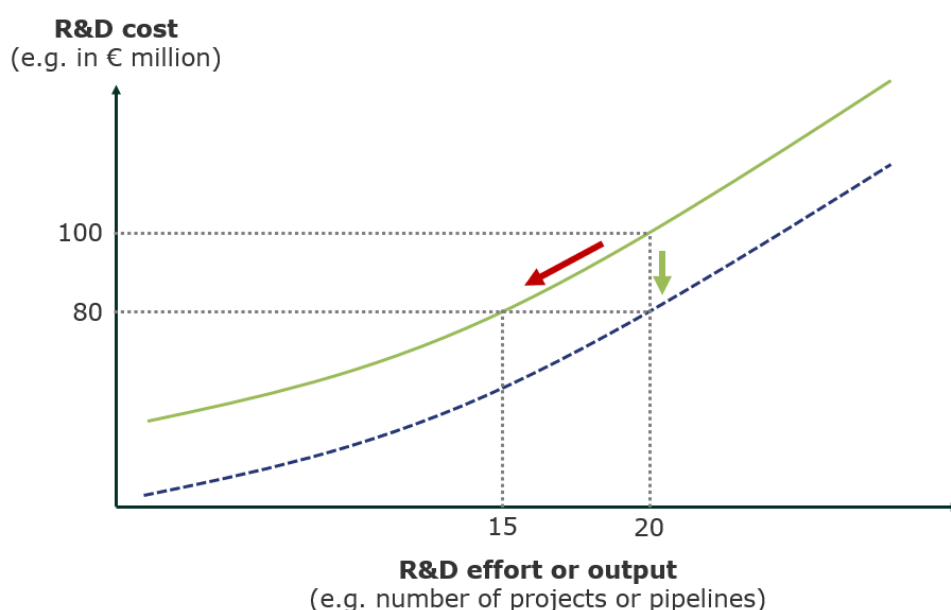
⁵⁸² Contingent valuation surveys may suffer from significant hypothetical bias, with consumers stating different valuations than they actually have. Discrete-choice surveys reduce the decision to a discrete choice, which reduces the risk of hypothetical bias.

⁵⁸³ We use the case of investment in R&D as an illustrative example. The same logic applies to investment in physical infrastructure.

of future expected R&D output, or equivalently that higher expected output can be achieved at the same cost), then the merger results in a dynamic synergy that benefits consumers (and can offset any adverse effects under Box 1).

- 16.64 By contrast, a movement along the pre-merger R&D cost curve of each firm (implying lower R&D efforts and lower R&D costs) would instead be evidence of an anticompetitive reduction in investment due to the merger.
- 16.65 In practice, assessing this question requires an evaluation not only of the nature of the R&D cost savings resulting from a merger, but also of the likely level of expected innovation output projected post-merger.

Figure 16.2 Movement along vs shift in the R&D cost curve



Note: This figure illustrates the distinction between (i) a movement along the R&D cost curve (moving from A to B), as the reduction in R&D cost is linked to an anticompetitive reduction in R&D effort or output, and (ii) a shift in the R&D cost curve (moving from A to C), which is procompetitive as it is likely to result in higher R&D effort or output. The numbers are for illustrative purposes only.

Source: Oxera.

Timeliness

- 16.66 For static efficiencies, the current guidelines from competition authorities (e.g. the EC Horizontal Merger Guidelines and the US Merger Guidelines) indicate that efficiencies should be realised over a short time period to be taken into account for the purposes of balancing against any anticompetitive effect. This is because static harm from a merger is likely to take place in the short run (even though it may persist over time), as prices can typically be changed quickly after the implementation of a merger (at least for new customers, transactions, and contracts). To counteract this harm to consumers, efficiencies should therefore be sufficiently timely.⁵⁸⁴

⁵⁸⁴ See discussion of Orange/MásMóvil (2024, EU) in Annex A2.9.

- 16.67 This logic needs to be adapted to the case of dynamic merger effects, applying a *consistent* approach in the analysis of dynamic harm and dynamic benefits. Depending on the case, it may be that the harm to consumers from a loss of dynamic competition between the merging parties materialises only over time, when new products are introduced or would have been introduced in the counterfactual absent the merger. This is the case even if innovation has already been reduced in the short term. Similarly, the beneficial effects from a merger may take time to be realised (even if the dynamic synergy itself takes place in a timely fashion—e.g. R&D efforts increase soon after the merger).
- 16.68 In this context, to assess correctly the impact of the merger on ECW, the competitive analysis of dynamic effects needs to adopt longer timeframes than static analysis. In principle, if the flows of harm and of benefits have different timelines, the two flows can be discounted at the same rate over time, to ensure a consistent comparison (attributing greater weight to effects that occur earlier in time)—see discussion of balancing in Chapter 17. This may be hard to implement in practice, but the general principle of consistency in the timeframe adopted in the evaluation of anti- and procompetitive effects flows directly from the ECW standard adopted in our conceptual framework.

16.3.2 Merger-specificity

- 16.69 Whether beneficial dynamic effects are merger-specific is a key consideration in the overall evaluation of dynamic merger effects. From an economic perspective, this notion captures the need for a *causal link* between the merger and possible procompetitive effects. As for the case of static efficiencies, beneficial dynamic effects are merger-specific if the same or similar benefits cannot be obtained by the merging parties through realistic and less-anticompetitive alternatives (including notably contractual solutions).
- 16.70 There are several potential alternatives to a merger by which firms may be able to capture similar beneficial dynamic effects, including:
- R&D JVs, to internalise knowledge spillovers (under Box 2A) or enable technology transfers (under Box 2B);
 - licensing of IP, to capture the benefits of technology transfers;
 - cooperation at the wholesale level, to avoid duplication of fixed costs and capture economies of density (e.g. NSAs, and co-deployment partnerships);
 - (in the case of a small innovative entrant) securing finance to develop a given innovative project, via other options such as outside investors not active in the market targeted by the project.
- 16.71 Key considerations in the economic assessment of merger-specificity are:
- whether the alternative to the merger can allow firms to capture a level of beneficial dynamic effects (e.g. cost savings or quality improvements) similar to the level under the merger;
 - the *feasibility* and *likelihood* of the alternative arrangement (e.g. are there contractual imperfections or frictions that can be overcome through a merger, and which make a contractual alternative less likely or less effective?); this may in turn reflect the *incentives* faced by the merging parties to pursue the alternative arrangement absent the merger;⁵⁸⁵

⁵⁸⁵ Typically, the higher the benefits achievable under Box 2, the stronger will be the incentives for the merged firms to capture them, through either a merger or an alternative arrangement.

- whether the alternative arrangements are *common* or *practical* business practices (e.g. based on evidence from other markets or other firms), and not just theoretical possibilities;
 - whether the alternative arrangement is *less anticompetitive* than the merger, and would therefore deliver a superior outcome for consumers (i.e. higher ECW).
- 16.72 From an economic perspective, the assessment of merger specificity can be framed as part of the overall impact of Box 2 effects on ECW. For example, if an alternative and less anticompetitive arrangement is unlikely to be feasible, then beneficial dynamic merger effects under Box 2 are more likely, and would have a stronger positive impact on ECW. In turn, this would make it more likely (all else equal) that the beneficial dynamic effects outweigh the adverse dynamic effects from the merger under Box 1, and hence that $\Delta ECW > 0$. As for the case of assessment of effects under Box 1, adopting an ECW framework for Box 2 effects does not require an estimation of the probability of each of the relevant outcomes (e.g. the outcome where an alternative contractual arrangement is feasible, and one where it is not). It, however, implies that consideration should be given to the likelihood that a given procompetitive effect is merger specific, and to its overall impact on ECW.
- 16.73 Evidence on past attempts to engage in cooperation (including both successful and unsuccessful examples) to capture dynamic benefits in the absence of a merger may be informative on the feasibility and impact of alternatives to the merger. Typically, a less anticompetitive alternative to a merger will be less profitable for merging firms than the merger (as it preserves product market competition), and hence may not be their preferred course of action. Evidence that in the past firms have discarded alternatives to the merger in favour of the option of merging should be considered in this light, and does not necessarily mean that those alternatives to the merger would not be feasible or practical.⁵⁸⁶
- 16.74 Similar considerations apply to the assessment of whether the target firm may be able to be acquired by alternative (less anticompetitive) buyers absent the merger. This situation may arise in a case where the incumbent buys a small innovative entrant, but the entrant could have been sold to an outside investor (or a firm not present in the market where the incumbent is active), absent the merger. The incumbent may face incentives to offer a higher acquisition price than the outside investor if the merger eliminates a potential competitor and protects its existing profits (see illustration in Chapter 20). Evidence that a small innovative entrant has a preference to sell to the incumbent (or that the incumbent won a competitive acquisition process for the asset) does not mean that absent the merger an alternative sale to other investors would not be viable.⁵⁸⁷ An assessment of the presence of realistic and less anticompetitive alternative buyers in the context of the evaluation of beneficial dynamic effects under Box 2 is similar to the current approach for a 'failing firm defence' under the 2004 Horizontal Merger Guidelines.

16.3.3 Verifiability

- 16.75 The evaluation of verifiability of dynamic benefits is again similar to the case of static efficiencies. It will turn on the type of evidence available to substantiate the presence of benefits.

⁵⁸⁶ See discussion in several European Commission decisions on mobile telephony mergers, e.g. Hutchison 3G/Wind Italy (2016, EU).

⁵⁸⁷ For a discussion of this point in the context of counterfactual analysis in EC merger cases, see Amelio et al. (2018).

- 16.76 A key principle here is the *consistency* between the type of evidence available on adverse dynamic effects under Box 1, and the evidence of benefits under Box 2 (controlling for the availability of information on each type of effect). For example, if evidence of harm is largely qualitative (because it is difficult to quantify forward-looking harm on new, future, products) then the evidence for Box 2 may also only be qualitative (for the same reasons). By contrast, for static effects, it may be possible to obtain quantitative evidence on both adverse and beneficial effects.
- 16.77 Future investment plans of the merged entity are typically an important source of evidence for dynamic synergies. The challenge with this evidence is its probative value, as these plans are often drawn up in the context (or anticipation) of the merger control proceedings, and may be influenced by a desire to secure approval by competition authorities. Forward-looking plans will have higher probative value if they are based on documents or evidence produced in the ordinary course of business, if they reflect past synergies obtained in similar contexts, and if they inform the valuation of the business that is being acquired (e.g. that valuation underpins the financial consideration for a target).
- 16.78 For the case of internalisation of knowledge spillovers under Box 2A, evidence that firms have benefited from spillovers from each other in the past (including for the case of sequential innovation), or refrained from investing because of concerns of spillovers to the other merging firm, is likely to be relevant (albeit not necessary) to the assessment of the verifiability of the corresponding dynamic benefits of the merger. Similarly, evidence on the strength of IP rights is an important determinant of the verifiability of benefits under Box 2A.
- 16.79 For the case of potential benefits from technology transfers under Box 2B, the feasibility, nature and prevalence of technology licensing absent the merger is likely to be relevant (including in the case of sequential innovation that may be enabled or facilitated by a merger).

17 Balancing Box 1 and Box 2 effects

Key points

- Balancing between Box 1 (adverse) and Box 2 (beneficial) dynamic effects is required where both types of effect have been established to the requisite standard.
- A consistent framework must be applied across Box 1 and Box 2 in terms of *evidence standards, timeframe, and treatment of uncertainty*. The ECW standard provides a suitable benchmark for symmetric balancing of adverse and beneficial effects.
- Beneficial effects under Box 2 must offset the impact on ECW of effects under Box 1 (resulting from both effects on price of future products and investment effects). This implies that a net positive effect on investment incentives alone is insufficient to conclude that a merger increases ECW, as it may not entirely offset adverse price effects.
- Merger simulation techniques can quantify and balance dynamic harm and benefits, including identifying the critical level of Box 2 benefits needed to offset Box 1 harm. However, they are data-intensive and unlikely to be feasible in many cases. Where claimed benefits fall clearly below the critical level to outweigh Box 1 effects, a full assessment of Box 2 effects is not required. Conversely, where there are limited Box 1 effects, it may be possible to conclude that quantified Box 2 effects outweigh any realistic harm.
- Where quantitative tools are unavailable or unsuitable, a qualitative balancing exercise is necessary, with more in-depth analysis required for cases where the effects on both sides appear material and of comparable orders of magnitude. A 'sliding scale' approach can be used, requiring stronger evidence of Box 2 effects where Box 1 effects are expected to be large, and vice versa. The degree of market power of the acquiring firm is often a key consideration when applying a sliding scale approach to balancing. Relevant qualitative evidence may include information that rivals to the merging firms are expected to benefit from a horizontal merger (indicating that efficiencies are insufficient to offset consumer harm or not merger-specific), while evidence of harm to rivals would support the opposite conclusion.
- Where a merger produces static and dynamic effects in opposite directions, balancing is required between static effects (typically more immediate and certain) and dynamic effects (which may be larger but also more uncertain and occurring over a longer time horizon). Conceptually, an ECW framework accommodates this trade-off through probability-weighting and discounting of merger effects over time. A sliding scale approach is suitable here too, with a higher evidentiary bar for dynamic benefits in the presence of clearly established static harm, and conversely concrete evidence of dynamic harm required to override established static benefits.

- 17.1 In this Chapter we conclude our review of potential dynamic effects from horizontal mergers with a discussion of how to balance potential anticompetitive effects under Box 1 and potential procompetitive effects under Box 2. Balancing is required in a situation where both types of effects have been established to the requisite standard (e.g. effects established under Box 2 are merger-specific, verifiable and beneficial to consumers).

- 17.2 We discuss four conceptual issues in turn (and refer to Chapter 20 for more practical considerations, based on our taxonomy of cases):
- consistency of approach across Box 1 and Box 2;
 - implications of the ECW standard for balancing dynamic harm and benefits;
 - economic tools for balancing, and alternative approaches; and
 - balancing between dynamic and static effects.

17.1 Consistency of approach

- 17.3 The first overarching point to ensure the correct balancing of dynamic harm and benefits is the need to adopt a *consistent approach* in the evaluation of both effects. This applies to the standard of evidence, the timeframe adopted in the analysis, the assessment of the likelihood of each type of effect and of the underlying parameters of competition (e.g. the probabilities of entry by a potential competitor, of successful innovation, or of an efficiency being realised), and how uncertainty is accounted for in the overall assessment. We have already emphasised this point in the review of dynamic benefits in Chapter 16.
- 17.4 An ECW standard provides a consistent framework for balancing, and a benchmark to trade off potentially negative and positive effects. In some cases balancing under an ECW standard can be carried out using quantitative techniques (see discussion in Section 17.3), while in other cases it will need to be a largely qualitative exercise. Even in the latter case, adopting an ECW benchmark supports a consistent balancing of opposite effects, and ensures that a like-for-like comparison is achieved. Under an error-cost framework, the ECW standard implies that the costs of Type I and Type II errors are correctly evaluated and balanced (see discussion of the error cost framework in Box 14.2).
- 17.5 Balancing under an ECW framework is explicitly adopted in the academic literature on dynamic mergers, including in theoretical papers and structural simulations. Some papers explicitly model effects under both Box 1 and Box 2, and trade these two effects against each other by reference to the change in ECW resulting from the merger.⁵⁸⁸
- 17.6 The ECW standard effectively applies a ‘sliding scale’ approach to balancing, with larger expected effects on one side of the ledger (in terms of likelihood of realisation, and magnitude conditional on being realised) requiring larger compensating effects on the other side.

17.2 Implications of an Expected Consumer Welfare standard

- 17.7 An important implication of an ECW standard, and of our definition of dynamic merger effects, is that it includes effects on both investment and prices of new or improved products. This has implications for balancing. In particular, it means that beneficial effects under Box 2 need to offset the *overall* harm to consumers resulting from the combination of price and investment effects under Box 1. As discussed in Chapter 15, merger effects under Box 1 in some cases lead to increased future prices and lower investments, with both channels contributing to a reduction in ECW. In other cases, Box 1 effects imply that both prices and investment increase, so that the investment effect moderates the harm to consumers from higher prices alone. Overall, this indicates that to reach a conclusion that a merger does not result

⁵⁸⁸ See in particular Federico, Langus, and Valletti (2018), and Moraga-González and Motchenkova (2026).

in a decrease in ECW it is not sufficient to establish that the combined impact of Box 1 and Box 2 effects will result in higher investment. Any pro-investment effect under Box 2 needs to be sufficiently large to outweigh the entirety of the reduction in ECW under Box 1, accounting for the *combined* impact of future price and investment effects.

- 17.8 For example, a merger with a potential competitor working on an innovative project may raise a concern that post-merger the project will be discontinued (to avoid cannibalising the incumbent's existing sales). Dynamic synergies in the form of lower development costs for the incumbent will mitigate and potentially remove the risk of project discontinuation. However, even if the project is not discontinued, the merger would reduce ECW by removing future price competition between the incumbent and the potential competitor. To compensate for such harm, dynamic synergies therefore need to increase the probability of innovation sufficiently compared with the counterfactual (see the illustration of this example in Chapter 20, Box 20.4).

17.3 Economic tools for balancing and alternative approaches

- 17.9 Merger simulations are economic techniques to carry out a quantitative assessment of dynamic harm and benefits from a merger. Depending on the case (including data availability), these tools can support a balancing exercise, and provide at least an order of magnitude for harm and benefits, and a measure of the net effect.
- 17.10 Competition authorities (including the European Commission) have relied on merger simulations to balance static effects in several merger precedents. Merger simulation techniques have been used in markets with price competition between differentiated products,⁵⁸⁹ homogeneous goods with capacity constraints,⁵⁹⁰ and in bidding markets.⁵⁹¹ In each of these types of case, the Commission evaluated static efficiency claims made by the parties, and used simulation models to assess the net effects of the merger on prices (which, in a static setting, are also informative of effects on consumer welfare).
- 17.11 Merger simulation models can also be used to assess and balance dynamic merger effects. This requires an assessment of the pro-investment effects of a merger, of the impact of additional investment on consumer welfare (e.g. if investment leads to higher quality, how much do consumers value quality?), and of the net impact on consumers (taking into account the price effects of the transaction). There are examples of the use of merger simulation techniques to assess dynamic effects, though these are rarer and more complex than the static examples.⁵⁹²
- **Case example: Vodafone/Three (2024, UK).**⁵⁹³ In this case the merging parties claimed that the merger would lead to beneficial dynamic effects, in the form of greater network capacity and quality. Two merger simulations were put

⁵⁸⁹ This includes several of the mobile telephony mergers reviewed by the Commission: Hutchison 3G/O2 Ireland (2014), Telefonica/O2 DE (2014), Orange/Jazztel (2015), Hutchison 3G/Wind Italy (2016), H3G/O2 UK (2016), T-Mobile/Tele 2 (2018), and Orange/MásMóvil (2024). In some cases, simpler pricing pressure tools have been used to balance harm and efficiencies.

⁵⁹⁰ See Outokumpu/Inoxum (2013, EU) and INEOS/Solvay/JV (2014, EU).

⁵⁹¹ See General Electric/Alstom (Thermal Power – Renewable Power & Grid Business) (2015, EU).

⁵⁹² See simulations of the consumer benefits from greater investments put forward by the merging parties and evaluated by the Commission in Hutchison 3G/Wind Italy (2016). Another relevant Commission example is General Electric/Alstom (Thermal Power – Renewable Power & Grid Business) (2015), where the Commission's quantitative analysis captured the consumer harm from the risk of product discontinuation (which is similar to a dynamic harm from a loss of innovation).

⁵⁹³ See detailed discussion of this case in Annex A2.10.

forward to quantify these effects and balance them against price effects. One was based on a calibrated capacity model with congestion effects similar to the model of López (2025). This model was used to test different assumptions on the level of margins and synergies and on the functional forms of investment and of demand. The model was used to predict the impact of the merger on ECW under different scenarios. However, the CMA ultimately concluded that while the model was coherent, it provided a restrictive framework to predict the impact of the merger on ECW. To reach its conclusion that the merger was anticompetitive (absent remedies) the CMA relied instead on a static model of price effects, a detailed assessment of whether the merger would lead to improvements in network quality, and a qualitative balancing of harm and benefits.

- 17.12 The challenge with the use of calibrated merger simulations to assess dynamic effects is that they require extensive inputs, and are sensitive to input assumptions and calibration choices. Therefore, depending on the case at hand, they may not be feasible or conclusive. Even where they are used, they will not provide an exact prediction of merger effects, but rather an indication of the potential magnitude of harm and benefits, and of whether a balancing of the effects provides a sufficiently clear indication of the likely net effect.
- 17.13 Quantitative techniques can also be used to provide an indication of the *critical* level of dynamic benefits required to offset the likely harm from the transaction. If actual benefits (or if the benefits claimed by the merging parties) are well below the critical level identified by a merger simulation of harmful effects under Box 1, then this indicates that the merger is likely to reduce ECW.⁵⁹⁴ In some cases, this approach can short-cut an in-depth assessment of the beneficial effects under Box 2, under the three-limb test for efficiencies. If the claimed benefits are above the critical level identified by the merger simulation, then a more in-depth assessment of the potential benefits under the three-limb test would be required.
- 17.14 If merger simulation techniques are not available or appropriate, then a *qualitative* balancing of harm and benefits may be required. Even if qualitative, this would still require an evaluation of the likely orders of magnitude of adverse effects under Box 1, and beneficial effects under Box 2, under a sliding scale approach. In the absence of quantification, the likely magnitude of effects under Box 1 may be assessed on the basis of the market power of the merged entity and the impact of the transaction on market structure. For example, if a merger leads to a strong adverse impact on concentration and brings together two close and significant competitors, then larger efficiency gains are required (see discussion of magnitude of Box 1 effects in Section 15.7).⁵⁹⁵ If effects under Box 2 from that merger appear minor (e.g. the evidence indicates that R&D cost savings or the resulting increase in quality are limited, or that only a small fraction of the benefits are merger-specific), a qualitative balancing may be more straightforward.⁵⁹⁶ The converse holds for a case with limited impact under Box 1 (e.g. a merger of distant competitors in a market that is not concentrated), and demonstrably material positive effects under Box 2 (e.g. through clear asset complementarities).

⁵⁹⁴ This can also be assessed in terms of the critical pass-through rate required for synergies to fully compensate consumers for the harm.

⁵⁹⁵ For the case of static competition in prices, Nocke and Schutz (2025) show that synergies required for a merger not to harm consumers are smaller the more competitive is the market before the merger, and larger for mergers involving larger firms. They also show that absent synergies, the increase in concentration is an appropriate measure of the market power effect of the merger. Similar insights are likely to carry across to mergers with dynamic effects.

⁵⁹⁶ In Deutsche Börse/NYSE Euronext (2012), the Commission carried out a qualitative balancing of harm and efficiencies. This approach was upheld on appeal by the General Court.

- 17.15 For the intermediate cases, where effects under Box 1 and Box 2 appear to be both material and of similar orders of magnitude, then a more in-depth and case-specific assessment of the evidence is likely to be necessary. This may require at least partial quantification of some of the key elements of dynamic harm and dynamic synergies, to assess their potential magnitude. Asymmetries between harms and benefits in terms of their impact over time and potential irreversibility can also play a role in the balancing.
- 17.16 A review of the market context in which the merger takes place may be a useful complementary source of qualitative evidence on a merger's likely net effect on consumers, and hence on the balancing of Box 1 and Box 2 effects. A horizontal merger that is profitable for the merging parties and benefits consumers will tend to reduce the profits of the main competitors to the merging parties (as it will make the merged entity more competitive, and divert profitable sales from their rivals).⁵⁹⁷ Evidence that rivals are expected to benefit from a horizontal merger (e.g. as indicated by the reaction of stock markets and financial analysts to its announcement,⁵⁹⁸ by communications with investors, or by internal documents) could therefore be indicative that efficiencies are insufficient to prevent dynamic harm.⁵⁹⁹ Such evidence may also suggest that the efficiencies are not merger-specific, and that the financial market (and rivals) expect them to be realised also in the absence of the merger. Conversely, if there is evidence suggesting that the key rivals are expected to suffer from a horizontal merger, this may be an indication that efficiencies resulting from the merger outweigh any dynamic harm.

17.4 Balancing static and dynamic effects

- 17.17 As a merger may give rise to both static and dynamic effects, balancing may be required between static harm and dynamic benefits, or between static benefits and dynamic harm (see illustration in Box 17.1).

17.4.1 Static harm and dynamic benefits

- 17.18 The first of these cases (static harm, dynamic benefits) captures a situation where the merger gives rise to a static loss of competition, without sufficient static efficiencies (such that the static balancing points to harm), and to a dynamic benefit (implying that any adverse dynamic effects under Box 1 are outweighed by dynamic benefits under Box 2). For example, a merger may lead to adverse price effects by bringing together two close competitors, at the same time as resulting in dynamic efficiencies in the form of higher capacity or quality due to a reduction in incremental investment costs (e.g. through infrastructure sharing).⁶⁰⁰

⁵⁹⁷ This is the case in standard static models of quantity competition (e.g. Farrell and Shapiro, 1990), or of pricing with differentiated products (Motta, 2004; Nocke and Schutz, 2025; Caradonna, Miller, and Sheu, 2025). However, it may not be the case in more complex settings, e.g. in markets where firms offer multiple qualities and a merger may lead to product repositioning (see Johnson and Rhodes, 2021).

⁵⁹⁸ See Stillman (1983). Financial market indicators may be informative if the expected financial performance of the non-merging party is sufficiently exposed to the market(s) affected by the merger (e.g. a high share of profits of the non-merging party is derived from the market affected by the merger).

⁵⁹⁹ For a discussion of this effect, in the context of a quantitative evaluation of the merger, see the Commission decision in Hutchison 3G/Wind Italy (2016).

⁶⁰⁰ Vodafone/Three (2024, UK) can be seen as an example of this case, even though the CMA decided that an investment remedy would still be required to ensure that the dynamic efficiencies would materialise (see description of this case in Annex A2.10).

- 17.19 In a situation of this type, the static harm will often be more *immediate, certain, and concrete* than the forward-looking dynamic benefit. By contrast, the dynamic benefit may be *larger* in absolute terms, if the merger leads to more innovation or investment output over time, and if consumers value the associated benefits in terms of higher quality and variety.⁶⁰¹
- 17.20 An ECW standard can in principle accommodate this inter-temporal trade-off by placing more weight on more immediate and more certain effects, but also taking into account the magnitude and significance of later effects (by discounting them over time). This can be captured by the comparison of the Net Present Value (NPV) of the flow of ECW under the merger, compared with the counterfactual (see illustration in Box 17.1).⁶⁰² While precise quantification of the NPV of the flow of ECW over time is likely to be difficult in practice, the concept illustrates how to engage in inter-temporal balancing, incorporating both static and dynamic effects, and recognising their relative impact on consumers.



Box 17.1 Balancing static harm and dynamic benefits

This Box illustrates how to conceptually balance static and dynamic effects, under an ECW framework. We consider two hypothetical cases.

In Case 1 we assume that the flow of consumer welfare in a market is constant at €100 million in the counterfactual. The merger leads to harm in the short-run (a reduction of consumer welfare to €90 million) in Years 1–3, but then it is associated with dynamic synergies, which lead to higher quality products, and thereby increase consumer welfare to €120 million from Year 4 to Year 10. We assume that the beneficial impact of dynamic synergies is uncertain, and is realised with 50% probability. Assuming an annual discount rate of 10%, the merger therefore leads to a drop in the NPV of ECW of about €7.2 million (over a ten-year period).

Case 2 is like Case 1 but the probability of the benefits from a dynamic synergy from the merger is higher (60% rather than 50%), and the discount rate is lower (7% rather than 10%), implying that future gains are more valuable. In this case, the merger leads to an increase in the NPV of ECW of about €8.6 million (over a ten-year period), reflecting the fact that the merger is more likely to lead to beneficial dynamic effects, and that consumers place greater value on future outcomes.

Case 1: negative effect on ECW 50% probability of dynamic synergies, 10% discount rate (€ million)

⁶⁰¹ This contrast may also take place within a dynamic assessment, if the effects under Box 1 occur before the effects under Box 2.

⁶⁰² We note that discounting is common practice in the context of cost-benefit analyses (CBA) in public policy. As set out in the EU's CBA guide, the relevant discount rate should be separated from the probability weighting (to avoid double-counting uncertainty) and should be applied symmetrically to harms and benefits. See European Commission (2014), 'Guide to Cost-Benefit Analysis of Investment Projects: Economic appraisal tool for cohesion policy 2014–2020', December. The CBA guide also makes the distinction between the Financial Discount Rate (FDR), which is based on companies' opportunity cost of capital, and the Social Discount Rate (SDR), which is generally lower—see Annex I and II in the CBA Guide. For more detail on the appropriate discount rate in CBA, see Oxera (2002), 'A social time preference rate for use in long-term discounting', Report for the Office of the Deputy Prime Minister, Department for Transport, and Department of the Environment, Food and Rural Affairs, 17 December.

Year	1	2	3	4	5	6	7	8	9	10
Counterfactual	100	100	100	100	100	100	100	100	100	100
Factual (merger)										
with efficiencies	90	90	90	120	120	120	120	120	120	120
without efficiencies	90	90	90	90	90	90	90	90	90	90
expected value	90	90	90	105	105	105	105	105	105	105
<i>Undiscounted ΔECW</i>	-10	-10	-10	5	5	5	5	5	5	5
Discounted ΔECW	-10.00	-9.09	-8.26	3.76	3.42	3.10	2.82	2.57	2.33	2.10
NPV ΔECW	-7.24									

Case 2: positive effect on ECW
60% probability of dynamic synergies, 7% discount rate (€ million)

Year	1	2	3	4	5	6	7	8	9	10
Counterfactual	100	100	100	100	100	100	100	100	100	100
Factual (merger)										
with efficiencies	90	90	90	120	120	120	120	120	120	120
without efficiencies	90	90	90	90	90	90	90	90	90	90
expected value	90	90	90	108	108	108	108	108	108	108
<i>Undiscounted ΔECW</i>	-10	-10	-10	8	8	8	8	8	8	8
Discounted ΔECW	-10.00	-9.35	-8.73	6.53	6.10	5.70	5.33	4.98	4.66	4.30
NPV ΔECW	8.58									

Source: Oxera.

- 17.21 A balancing between static and dynamic effects is likely to be more information-intensive than a more standard static balancing, or even than a balancing of only dynamic effects. It requires information on how consumers affected by the merger value near-term relative to future effects, and potentially also on their appetite for risk (e.g. should consumers be modelled as risk averse?). While these considerations may play a role also within a purely dynamic assessment, they are more prominent in a situation with differences in the direction of static and dynamic effects.
- 17.22 The sliding scale principle can also be applied in a situation with static harm and dynamic benefits. In such a case, balancing of dynamic benefits is likely to be necessary if there is evidence of a clear dynamic benefit from the merger, shown under the three-limb framework for efficiencies. Evidence of significant and relatively timely dynamic benefits would need to be more concrete the larger, more certain, and more immediate is the static harm associated with the merger. More generally, the same considerations developed in Section 17.3 on indicators for the orders of magnitude of effects under Box 1 and Box 2 apply equally to this situation.

17.4.2 Static benefits and dynamic harm

- 17.23 This is the flipside of the previous case, with static benefit and dynamic harm. For example, a merger may bring together two complementary firms, giving rise to consumer benefits, at the same time as entrenching the position of one of the two

merging firms, and deterring future entrants, leading to future consumer harm. As this case captures the risk of foreclosure or entrenchment from acquisition of complementary assets, we return to it in the discussion of non-horizontal mergers in Chapter 18 and the discussion of entrenchment in Chapter 19.

- 17.24 Broadly, the same balancing considerations as in the previous case arise here, under a sliding scale approach. In this case, the static benefit tends to be more immediate, likely, and concrete than the dynamic harm. Concrete evidence on the dynamic harm from the merger would therefore be required to offset the static benefits, and lead to an overall competitive concern.

18 Dynamic effects of non-horizontal mergers

Key points

- A non-horizontal merger brings together firms producing complementary goods, encompassing both vertical and conglomerate mergers.
- Non-horizontal mergers may give rise to anticompetitive dynamic foreclosure effects (Box 3), and thereby reduce rivals' investment or innovation, to the detriment of consumers.
- Key anticompetitive channels under Box 3 include: (i) hold-up of rival investment; (ii) restoring market power where opportunism prevents full rent extraction; (iii) raising rivals' costs via a refusal to deal; (iv) deterring multi-level or multi-product entry; (v) dynamic leveraging to defend profits from pre-existing market power; and (vi) depriving rivals of a valuable input or complement, to prevent them from becoming a stronger competitor.
- Non-horizontal transactions may also lead to procompetitive dynamic effects (Box 4), including: (i) internalisation of positive price and investment externalities (Box 4A); (ii) dynamic synergies (Box 4B); and (iii) resolution of contractual imperfections (Box 4C).
- As non-horizontal mergers tend to generate efficiencies from the integration of complementary activities, potentially harmful and beneficial effects are often concurrent and intertwined, typically requiring an integrated economic assessment.

- 18.1 This Chapter examines the potential dynamic effects of non-horizontal mergers. We define a non-horizontal merger as a transaction that brings together two firms producing complementary goods. This definition includes both *vertical* mergers and *conglomerate* mergers.
- 18.2 In the case of *vertical* mergers, the two merging firms are respectively active in an upstream market and a downstream market within the same supply/value chain, with firms in the upstream market competing to provide an input to firms in the downstream market. In this setting, the upstream good and the downstream good are complementary, since a lower price or higher quality for the upstream input product translates into higher demand for the downstream product. Vertical mergers may lead to *input* or *customer* foreclosure effects.
- 18.3 In the case of *conglomerate* mergers, firms are active in related markets, and are selling complementary goods to their customers. We consider here both demand-side complementarities (e.g. a lower price or higher quality for good A increases the demand for good B) and supply-side links (e.g. joint production of goods A and B resulting in lower costs). We consider here cases where there is a substantial degree of overlap between customers in the two related markets.
- 18.4 We include two categories of potential dynamic non-horizontal effects:
- **Box 3: dynamic foreclosure effects**, which reduce investment, innovation, or entry from rivals to the merged entity, to the detriment of consumers; and
 - **Box 4: procompetitive non-horizontal effects**. Within Box 4, we distinguish between: the benefits from internalising positive externalities between the merging parties (Box 4A); dynamic synergies (Box 4B); and benefits from the resolution of contractual imperfections (Box 4C).

- 18.5 This categorisation provides an analytical separation between foreclosure effects and procompetitive non-horizontal effects. As we discuss further below, these two types of effect are often concurrent and intertwined, often requiring an integrated economic assessment. Our structure is analogous to the one we use for horizontal effects (see Chapters 15–17) even though there are important conceptual differences between horizontal and non-horizontal dynamic merger effects. In particular, horizontal mergers raise more direct concerns about the elimination of competition between the merging firms, while potential anticompetitive effects from non-horizontal mergers are typically indirect, in particular for the case of dynamic foreclosure (see discussion in next Section).
- 18.6 We further discuss potential entrenchment effects from non-horizontal mergers in Chapter 19, as part of our more general discussion of entrenchment.

18.1 **Box 3: Dynamic foreclosure effects**

- 18.7 A non-horizontal merger may lead to anticompetitive dynamic effects through the *foreclosure of rivals* to the merged entity. In our conceptual framework, a merger leads to dynamic foreclosure if it *enables or facilitates conduct aimed at reducing investment (broadly defined to include entry) by existing and potential rivals to the merging parties, to the ultimate detriment of consumers*.⁶⁰³ Foreclosure can reduce the effective competitive constraint exercised by non-merging parties, and hence result in consumer harm. Dynamic foreclosure conduct may be profitable for the merged entity because it reduces investment by its rivals.
- 18.8 Contrary to anticompetitive effects from horizontal mergers (Box 1), dynamic harm in the case of non-horizontal mergers does not *directly* flow from a reduction in competition between the merging parties, but is instead the result of the impact of the merger on competitors to the merged entity. The impact on consumers is therefore less direct than in the case of horizontal mergers, as it is typically mediated through the competitive constraint exercised by third parties, rather than resulting directly from the elimination of rivalry between the merging firms. We review below the possible theories of harm supporting foreclosure concerns, on the basis of insights from the academic literature (see Chapter 8).
- 18.9 Foreclosure effects need to be distinguished analytically from the impact of efficiencies from non-horizontal mergers (Box 4 effects), since efficiency-enhancing mergers may also reduce the incentives faced by non-merging parties to invest or expand, but typically benefit consumers. We restrict our definition of foreclosure to mechanisms that are capable of leading to consumer harm on the basis of conduct aimed at foreclosing rivals to enhance or protect the market power of the merged entity. We identify these mechanisms in light of the insights from the academic literature. We discuss the possibility that efficiency benefits in the short term translate into harm over time in the next Chapter of the main report.

18.1.1 **Dynamic foreclosure conduct**

- 18.10 A merged entity may be able to dynamically foreclose rivals through a range of conduct (depending on the merger and on the applicable anticompetitive mechanism).
- In the case of *vertical mergers*, this may include an outright refusal to supply inputs to downstream rivals (potentially via technical integration, product design,

⁶⁰³ As per the definition that we use in the case of horizontal mergers, we rely on the concept of ‘investment’ to also capture innovation, R&D, and entry.

and lack of interoperability), and softer or more indirect forms of input foreclosure, including quality degradation⁶⁰⁴ or an increase in input prices compared to the counterfactual (a ‘raising rivals’ costs’ strategy). Vertical mergers may also enable customer foreclosure, where the merged entity refuses to deal with rival upstream firms, degrades the support provided to upstream firms, or degrades or restricts access to customers.

- In the case of *conglomerate mergers*, the merged entity may be able to foreclose rivals by bundling or tying the products brought together by the merger (as with the case of a refusal to deal, tying could be implemented via technological integration, product design, and lack or degradation of interoperability). Foreclosure may also take place through mixed bundling, with the merged entity increasing the prices of its standalone products and offering discounts for bundled purchases.⁶⁰⁵

18.11 Foreclosure conduct can only lead to adverse dynamic effects if it reduces rivals’ ability or incentives to invest, ultimately harming consumers.⁶⁰⁶ This may be the case through a number of economic mechanisms including:

- a reduction in the addressable demand by rivals (hindering the ability to cover the costs of investment, especially in the presence of substantial economies of scale or network effects);
- a reduction in the effectiveness or profitability of investment by rivals (through degraded support from the merged entity, higher risk of hold-up from other trading partners, or weaker competition in a vertically related market);
- restricted access to critical R&D inputs or complements, preventing rivals from investing;
- higher prices or reduced availability of critical R&D inputs or complements, lowering the profitability or feasibility of investment by rivals; and
- higher prices for other inputs or complements, lowering margins in the product market, and hence reducing the profitability of demand-enhancing investment by rivals.

18.12 Some of these mechanisms are mediated through higher prices for inputs or complements, while others directly affect the profitability of investments.

18.13 In some cases, foreclosure protects or strengthens a position of pre-existing and significant market power. We capture these cases within our treatment of anticompetitive foreclosure effects, and distinguish them from any additional entrenchment concerns, which we define in the next Chapter of the main report.

18.1.2 Anticompetitive channels based on dynamic foreclosure

18.14 A specific economic mechanism is required to establish that a non-horizontal merger gives rise to or strengthens the *ability* and *incentive* of the merged entity to engage in dynamic foreclosure conduct, resulting in adverse dynamic *effects* on consumers.

18.15 We rely on the existing academic literature to identify several such mechanisms. These mechanisms provide an *integrated* account of how a non-horizontal merger may lead to anticompetitive foreclosure, and do not necessarily decompose the

⁶⁰⁴ For example, through slower roll-out of upgrades or (in the context of digital products) interfering with use of Application Programming Interfaces (APIs).

⁶⁰⁵ Bundled discounts arising from the internalisation of pricing externalities are covered under procompetitive effects from non-horizontal mergers (Box 4A).

⁶⁰⁶ The reduction in investment by rivals due to foreclosure is distinct from a reduction due to the increase in investment by the merged entity (through strategic substitutability).

analysis into separate analytical steps on ability, incentives, and effects. We map the implications of these anticompetitive mechanisms to the ability-incentives-effects framework traditionally used by competition authorities in the next Section of this Chapter (for the part on ability and incentives) and in Section 18.3 (for the part on effects on consumers).

- 18.16 Based on the academic literature we identify several anticompetitive foreclosure channels. These may apply to a specific merger depending on the circumstances. We distinguish between mechanisms which rest on dynamic foreclosure incentives (i.e. the conduct is profitable for the merged entity because it deters investment by rivals), and mechanisms based on static foreclosure incentives which have knock-on effects on dynamic competition (e.g. because they lead to higher prices or lower availability for R&D inputs needed by rivals to the merged entity, and hence deter their future investments).
- 18.17 Foreclosure mechanisms resting on *dynamic foreclosure incentives* include:
- **Hold up of rival investment through commitment to appropriate a high share of returns of investment or through quality degradation.**⁶⁰⁷ A vertical merger may enable the merged entity to commit to appropriate a high share of the returns of investment by a downstream rival (e.g. through an aggressive bargaining strategy), exposing the downstream rival to hold-up by upstream rivals and reducing its investment incentives. Even absent such commitment by the merged entity, the integrated entity may have ex post incentives to degrade the quality of support to the downstream rival when doing so benefits the downstream affiliate.⁶⁰⁸ This exposes the downstream rival to input foreclosure by the unintegrated upstream firms. The same mechanism can apply (upside-down) to the case of customer foreclosure, where upstream firms compete in investment, but require support from downstream firms (e.g. in the form of distribution services). In this case, a vertical merger can lead the integrated entity to degrade the support provided to the unintegrated upstream rival, exposing it to hold-up by the non-integrated downstream firm.
 - **Deterring multi-level or multi-product entry.**⁶⁰⁹ A non-horizontal merger can deter entry if it forces potential competitors to enter at multiple levels of the value chain or in multiple markets. This may be the case if the merged entity can tie its products or refuse to deal with non-integrated competitors, and entry is subject to an uncertain R&D process. Under these conditions, the non-horizontal merger and subsequent foreclosure conduct reduces the probability of entry, as entry would need to occur in multiple products to be successful. The ability to implement bundling or tying conduct on a lasting basis (e.g. through technical or product integration, or by establishing a reputation to engage in the conduct over time) is a requirement for this mechanism to be at work. Under some conditions (e.g. with sufficiently inelastic demand), even without bundling or tying, a non-horizontal merger may deter entry by competitors, harming consumers.

⁶⁰⁷ This mechanism is based on the work of Allain, Chambolle, and Rey (2016). The main setting studies the case of input foreclosure, while customer foreclosure is captured in an extension (see Section 4.3 of the paper). This paper was partially motivated by the European Commission decision in TomTom/Tele Atlas (2008).

⁶⁰⁸ Degradation in the quality of support provided to the upstream rival can also take the form of the abuse of commercially sensitive information, as noted by Allain, Chambolle, and Rey (2016).

⁶⁰⁹ This mechanism is based on Choi and Stefanadis (2001) in a non-merger context, and Etro (2019) with a specific application to conglomerate mergers. The EC Non-Horizontal Merger Guidelines refer to this theory of harm at para. 112.

- **Dynamic leveraging to defend profits from market power in a core market.**⁶¹⁰ A non-horizontal merger may enable or facilitate a dynamic leveraging strategy by the merging firms. By committing to bundle complementary products or to refuse to deal with unintegrated rivals, a merged entity may be able to deter entry into a potentially contestable (secondary) market, to protect its profits in a related (primary) market where it holds a substantial degree of market power. This mechanism assumes that entry in the primary market is possible only in the future, once entry into the secondary market has taken place. This may be the case if entrants need to first establish their position in an adjacent market, to gain market traction and credibility to then threaten the incumbent in its core market. Foreclosure enables the merged entity to protect its market power in the primary market, by depriving potential competitors of the profits required to cover a fixed cost of entry, or by preventing them from capturing network externalities in the secondary market.⁶¹¹
- **Depriving a potential competitor of a valuable input or complement.**⁶¹² An incumbent firm with market power may merge with a provider of a valuable input or component (e.g. an element of an ecosystem), to deprive an existing smaller firm of access to that component. The incumbent firm may place a higher value on the acquisition of the input provider, as doing so protects the incumbent from greater competition and increases its quality advantage over its competitors. Post-merger, the incumbent firm may refuse to license the input to the smaller competitor, to prevent it from catching up with its superior quality level. Over time, this type of non-horizontal merger may strengthen the incumbent's position, and may induce future providers of inputs or complements to direct their innovation towards products that benefit incumbent firms rather than challengers.

18.18 Mechanisms based on *static effects on prices for R&D inputs* and indirect investment effects include:

- **Enhancing market power in the presence of opportunism.**⁶¹³ Absent vertical integration, a firm may not be able to fully extract the profits from its upstream market power if contract terms are confidential, and if it cannot commit to restrict output when dealing with multiple downstream customers. In this setting, vertical integration and foreclosure of unintegrated downstream rivals can enhance upstream market power. Foreclosure may take the form of outright refusal to deal with downstream rivals or higher input prices (if downstream rivals have access to a less efficient upstream alternative). This mechanism can deter investment by rivals if it relates to an important R&D input, and hence makes future investment less profitable or simply not feasible. It may also deter entry into a new market. This mechanism may generally apply to settings where a firm with market power is not able to fully capture the profits of its market power absent integration (e.g. due to the presence of sector

⁶¹⁰ This mechanism is based on the work of Carlton and Waldman (2002), for the case of horizontally related complementary goods, and Fumagalli and Motta (2020), for the case of vertically related goods. Neither of these two papers models a merger, so the application of their results to a merger is not straightforward. However, some of their insights into when foreclosure is profitable and harmful are likely to apply also in the merger context.

⁶¹¹ Foreclosure can also deter upstream entry by making the downstream market less competitive, and hence reducing upstream prices—see Fumagalli and Motta (2020).

⁶¹² This mechanism is based on Bryan and Hovenkamp (2020). A similar mechanism is highlighted by Benkert, Letina, and Liu (2025) in the context of acquihires.

⁶¹³ This mechanism is based on the work of Hart and Tirole (1990), and Rey and Tirole (2007). The EC Non-Horizontal Merger Guidelines refer to this theory of harm at para. 44 (including footnote 5). For an alternative foreclosure mechanism based on higher unit prices to the unintegrated rival, see Reisinger and Tarantino (2015).

regulation, or contractual imperfections). Foreclosure of rivals in an adjacent market may also be a way to transfer the rents from market power to a market where constraints on rent-extraction do not apply (e.g. market power can be transferred to an unregulated downstream market using the input from a regulated upstream market).

- **Raising rivals' costs through a refusal to deal.**⁶¹⁴ A vertical merger can increase the input price faced by unintegrated downstream rivals if the merged entity is able to refuse to deal with unintegrated downstream rivals (e.g. through technical degradation of interoperability or by establishing a reputation to refuse to deal over time). By reducing the available alternatives, the merged entity may increase the market power held by non-integrated upstream firms and in turn raise the cost of downstream rivals. This anticompetitive mechanism can have dynamic implications based on the same reasoning applied to the previous mechanism (i.e. if it relates to an important R&D input).
- 18.19 An incumbent firm may also acquire a potential competitor in an adjacent market that may enter its core market in the future⁶¹⁵ (or may enable entry by a third party⁶¹⁶). While this may superficially appear to be a non-horizontal merger, the adverse dynamic effect in this case would not rest on foreclosure of rivals to the merging parties, and would instead be based on the loss of competition from a potential competitor. It is therefore best seen as a horizontal effect, under Box 1 of our conceptual framework (see Chapter 15).
- 18.20 Box 1 and Box 3 effects may also apply at the same time. For example, a merger may allow an incumbent firm to acquire a target currently active in a complementary market, and which would have been a potential competitor in the incumbent's core market (Box 1), while also facilitating conduct aimed at foreclosing other future potential competitors currently active in that complementary market (Box 3).
- 18.21 As noted, the anticompetitive foreclosure mechanisms discussed above involve specific conduct by the merged entity which has the effect of excluding or harming a rival (either dynamically or through higher R&D input prices). They are not simply based on a competitive response by rivals to procompetitive actions taken by the merging firms (i.e. greater investments, or lower prices), whereby rival firms optimally reduce their level of output or investment. We return to this issue when discussing the potential dynamic procompetitive effects of a merger under Box 4.

18.1.3 The 'ability' and 'incentive' framework

- 18.22 Dynamic foreclosure effects can be analysed under the well-established 'ability' and 'incentive' framework (e.g. as employed by existing guidelines on non-horizontal effects). This framework can be adapted to the case of dynamic effects, by reference to the mechanisms listed above. We return to the issue of overall effects of foreclosure conduct in Section 18.3.
- 18.23 Issues that would typically be analysed to establish the *ability* to dynamically foreclose competitors include the following:
- The *degree of market power* held by the merged entity on the product (i.e. input or complement) that is being used to foreclose rivals. Foreclosure is an issue only if rivals have limited or imperfect alternatives to the merged entity, and cannot easily by-pass it. For the case of customer foreclosure, the larger the

⁶¹⁴ This mechanism is based on Ordover, Saloner, and Salop (1990).

⁶¹⁵ See for example, Motta and Peitz (2021).

⁶¹⁶ For example, if the acquired firm represents a valuable customer acquisition channel, or entry point.

share of demand that is being internalised by the merged entity, the stronger is the ability to foreclose rivals. The degree of market power of the merged entity is also likely to play a significant role in the assessment of the effects of a non-horizontal merger (see discussion in Section 18.3)

- The *competitive significance* of the product (i.e. input or complement) that is being used to foreclose rivals. If the input or complement is necessary or important to being able to offer a competitive downstream product or bundle of products, or to engage in R&D activities, then the merged entity may be able to dynamically foreclose its rivals.
- Whether the merged entity can *implement the required foreclosure conduct on a lasting basis*. Under some of the mechanisms for dynamic foreclosure that we have reviewed above, the integrated entity needs to be able to 'commit' (*ex ante*) not to deal with rivals or to tie its products. If entry by a rival takes place despite this conduct, it may be optimal (*ex post*) to deal with a more efficient rival or to offer an unbundled product, undermining a foreclosure strategy. The merging parties may be able to implement foreclosure conduct on a lasting basis by engaging in technological integration of their complementary products, or by establishing a reputation to engage in the foreclosure conduct, over time or over multiple markets. In the absence of the ability to pre-commit to the conduct, some (but not all) of the economic mechanisms identified above are not applicable, or less likely to result in competitive harm.
- Whether there are sufficiently effective *regulatory or legal constraints* on the foreclosure conduct, including limits from sector-specific regulation (e.g. Digital Markets Act), and ex post competition law (e.g. Article 102). This may deter some of the more blatant forms of foreclosure conduct, or render them less effective.
- Whether competitors to the merged entity can deploy effective *counter-strategies*, e.g. by partnering with other firms, or by engaging in alternative mergers.
- Whether *market conditions are conducive* to an effective foreclosure strategy in light of the specific anticompetitive mechanism, e.g. whether markets are characterised by significant supply or demand-side economies of scale (including network effects) which make it easier to marginalise or foreclose smaller competitors, and to reduce the future competitive constraint that they are able to exercise.

- 18.24 The key issue to analyse under *incentives* is whether foreclosure conduct is *profitable* for the merged entity. This effectively requires setting out and testing an economic mechanism explaining why foreclosure can be expected to be profitable, as in the mechanisms that we have reviewed in Subsection 18.1.2. While the list of mechanisms provided above is not exhaustive, it captures the main insights identified in the economic literature, and as such provides good guidance for when a merger may lead to dynamic foreclosure incentives.
- 18.25 Foreclosure incentives do not necessarily follow from a non-horizontal merger of firms with market power, as the merged entity may find it more profitable *not* to foreclose rivals and instead extract the benefits from its market power by dealing with efficient rivals (this is the essence of the so-called 'Chicago critique').
- 18.26 We discuss the possible *effects* of foreclosure conduct on consumer welfare below, in connection with the balancing of effects under Box 3 and under Box 4.

18.2 **Box 4: Procompetitive non-horizontal effects**

18.27 Non-horizontal mergers may give rise to a number of potential procompetitive dynamic effects.⁶¹⁷ We distinguish between three categories for analytical clarity.

18.2.1 **Box 4A: internalisation of positive price and investment externalities**

- 18.28 A merger between producers of complementary products tends to lead to the internalisation of *positive externalities* between merging parties. This is the counterpart of the *negative externalities* internalised by a horizontal merger (Box 1). These positive externalities stem from the fact that lower prices or higher quality of the product offered by one of the merging parties will typically increase the demand for the product of the other merging party. For example, in a vertical relationship, a lower input price by an upstream producer lowers the cost and boosts the demand faced by downstream producers (for a given downstream mark-up). Similarly, investment by an upstream producer to improve the quality of an input (or lower its cost) generates a positive externality on downstream producers, in the form of higher demand for their product. The upstream firm may not be able to capture these positive externalities through a contract (e.g. due to non-verifiability of its investment efforts).
- 18.29 The presence of these positive externalities means that vertical separation can result in prices that are too high (due to double-marginalisation) and investments that are too low (as firms do not internalise the positive impact of their investment on producers of complementary goods), relative to the joint profit-maximising outcome of the integrated entity. A non-horizontal merger can optimally lower prices and increase investments, by internalising these positive externalities.
- 18.30 This mechanism can lead to beneficial dynamic effects by enhancing the merged entity's incentives to invest. This can be the direct result of the internalisation of investment externalities. It can also be the indirect effect of lower prices for R&D inputs (which enhance the profitability of a given R&D investment), and of lower prices for other inputs (which increase the profitability of demand-enhancing investment, as with the case of production efficiencies).
- 18.31 These effects are likely to be present if markets are not too competitive (implying that firms earn positive margins and that changes in demand lead to variations in profits); if demand is price-elastic (which leads to inefficient double-marginalisation absent a merger); and if firms cannot resort to contractual solutions to internalise positive externalities absent a merger (which relates to the issue of merger-specificity). In the case of pricing externalities, firms may be able to rely on two-part tariffs to mitigate the incidence of double-marginalisation, depending on market conditions (e.g. the presence of uncertain demand may make this difficult). In the case of investment externalities, an RJV may be an effective alternative to a merger.

⁶¹⁷ We abstract here from static efficiencies from non-horizontal mergers (e.g. elimination of double margins, Cournot effects, and benefits from one-stop shopping), unless they also result in dynamic effects. Static efficiencies would, however, be reflected in the overall balancing of potential harm and benefits from a merger. For a broad review of the benefits from non-horizontal mergers see European Commission (2005), 'The efficiency-enhancing effects of non-horizontal mergers', Report by RBB Economics for DG Enterprise and Industry, European Commission.

18.2.2 Box 4B: dynamic synergies

- 18.32 A non-horizontal merger can lead to dynamic synergies similar to those associated with horizontal mergers (see Section 16.2), especially in connection to asset complementarities, economies of scope, and financial synergies.
- 18.33 Complementarities of assets between the merging parties are more likely to be relevant in the case of non-horizontal mergers (compared to horizontal mergers), given their very nature. Depending on the type of complementarities and the nature of the investment process, these complementarities may enhance the effectiveness of innovation or reduce its variable costs.
- 18.34 A non-horizontal merger may also allow the merged entity to diversify its portfolio, resulting in financial synergies (see discussion in Chapter 9).
- 18.35 That said, some of the potential dynamic synergies identified in the case of horizontal mergers (e.g. elimination of duplicative activities, economies of scale, infrastructure, or technology sharing) may be less applicable to the case of non-horizontal mergers, as these tend to bring together different technologies and activities where the scope for these synergies may be more limited.

18.2.3 Box 4C: resolution of contractual imperfections

- 18.36 Finally, non-horizontal mergers may allow firms to resolve contractual imperfections which limit their incentives to invest. A standard example of contractual imperfection is the problem of investment hold-up between firms with market power. If contracts are incomplete (implying that trading parties cannot contract on a given level of investment, e.g. because investment is not verifiable), then each party faces incentives to under-invest, to avoid being held up by the other party. This refers to a situation where, after a (sunk or irreversible) investment by one party has taken place, the other party can extract a high share of the returns from the investment (exploiting its bilateral bargaining power). A canonical example of investment hold-up is the case of relationship-specific investment, which implies that the party undertaking the investment can only trade with one party once the investment has taken place, and hence does not have an outside option. A non-horizontal merger can resolve the investment hold-up problem, leading to greater investment incentives. This type of procompetitive effect is more likely to apply to vertical mergers than conglomerate mergers, since in vertical mergers firms are directly trading with each other.
- 18.37 A non-horizontal merger may also help address other contractual frictions between trading parties (e.g. risk of imitation in situations where the value of technology is non-verifiable; lack of effective coordination between upstream investment and downstream applications, due to asymmetric information or transaction costs; risk of free-riding by competitors to the party undertaking an investment).
- 18.38 For example, vertical integration may be a mechanism to increase appropriability in situations where an upstream innovator cannot credibly disclose the value of its technology to a downstream firm without risking imitation. It may be difficult under such circumstances to negotiate an efficient licence agreement due to non-verifiability of the value of information.
- 18.39 Another example of the benefits of vertical integration is a situation where the commercial success of an upstream input (e.g. an operating system) depends on downstream complementary applications, and vice versa. A contractual solution may fail to fully align incentives for coordinated product development.

18.3 Balancing anti-/procompetitive dynamic effects of non-horizontal mergers

- 18.40 The net dynamic effect on ECW from a non-horizontal merger is the result of the combined impact of anticompetitive effects under Box 3, if any, and procompetitive effects under Box 4.
- 18.41 A balancing of the two categories of effect is often required, if dynamic foreclosure effects under Box 3 are present. This follows from the fact that a non-horizontal merger, by the very nature of the transaction, is more likely to be associated with efficiencies resulting from the combination of complementary assets or products than a horizontal merger.⁶¹⁸ The internalisation of positive externalities in non-horizontal mergers (Box 4A) is the counterpart to the internalisation of negative competitive externalities in horizontal mergers (Box 1), and may give rise to beneficial effects, to be balanced against potential adverse effects under Box 3 (with the key difference being that Box 1 effects are anticompetitive while Box 4 effects are procompetitive).⁶¹⁹ Compared to horizontal mergers, it is therefore in turn more likely that a non-horizontal transaction will be associated with benefits.
- 18.42 Moreover, as noted at the outset in this Chapter, competitive harm under Box 3 is based on a less direct mechanism than harm from the loss of horizontal competition under Box 1, indicating that a balancing of countervailing dynamic effects should typically play a more prominent role before reaching a conclusion that the merger substantially lessens competition. It also suggests that there may often be a need for an *integrated* economic approach to the evaluation of dynamic harm and dynamic benefits, as the effects are often concurrent, intertwined, and driven by underlying changes in the incentives of the merged entity.
- 18.43 The significance of the degree of market power of the merging parties is likely to play an important role in the determination of the dynamic effects of a non-horizontal transaction, and in the balancing of potential harm and benefits. It influences not only the ability and incentives to foreclose rivals (as discussed in Section 18.1), but also the effects of foreclosure on ECW. If the potentially foreclosed rivals represent the only or primary source of competition to the merged entity, then the transaction is more likely to result in competitive harm. This suggests the application of a 'sliding-scale' principle to non-horizontal effects, in a manner similar to that discussed for horizontal effects (Chapter 17). We further discuss the role of the significance of pre-existing market power in the context of entrenchment effects below.
- 18.44 It is also possible that a non-horizontal merger gives rise to an inter-temporal trade-off between short-run benefits (e.g. in terms of static efficiencies from bringing together complementary assets or products) and medium-/long-term harm (in the form of the risk of future foreclosure of rivals). This makes the balancing exercise more complex, as discussed in Section 17.4.

⁶¹⁸ Based on insights from the academic literature, there are some theoretical settings where only Box 3 effects may be present, with no need to balance them against benefits under Box 4. However, these settings capture relatively specific circumstances (e.g. cases where there is no investment hold-up absent a merger, and the merger instead creates hold-up concerns), which would need to be verified on a case-by-case basis.

⁶¹⁹ See discussion in Shapiro (2025).

19 Entrenchment

Key points

- We define entrenchment as a dynamic effect of a merger that increases barriers to entry and expansion faced by current or future rivals of the merged entity, and weakens the competitive constraint that they are able to exercise, to the ultimate detriment of consumers.
- Our definition of entrenchment explicitly excludes anticompetitive mechanisms already captured under Box 1 (horizontal effects) and Box 3 (dynamic foreclosure). The competitive harm in entrenchment cases flows from the fact that the merger deters entry or distorts innovation by non-merging rivals—distinguishing it from effects under Box 1 (which captures the direct loss of competition between the parties) and Box 3 (which requires specific conduct by the merged entity, aimed at foreclosing rivals).
- We identify examples of entrenchment mechanisms based on recent academic literature on dynamic merger effects. These include mechanisms where a merger may: reduce future entry incentives by expanding the customer base of an incumbent (and thus reducing contestability); lower the incentives by weaker rivals to the merged entity to acquire innovative entrants (by increasing the advantage of the market leader), and hence reduce entry incentives; expand ecosystems and allow for cross-market steering of users; allow an incumbent to engage in defensive strategies and deter innovation by rivals; or enable incumbents to deter multi-homing by consumers, and hence discourage rival entry.
- The degree of market power of the acquirer matters for determining whether a merger may lead to entrenchment and ultimately harm consumers. Anticompetitive effects are more likely if the acquirer has significant or substantial market power, since this tends to reinforce the channels identified in the literature. From a policy and error-cost perspective it is therefore appropriate to restrict entrenchment concerns only to acquirers with a high degree of market power (reflecting economies of scale, network effects, learning-by-doing, breadth of ecosystems, or the cumulative effect of past acquisitions).
- We discuss whether a reduction in entry incentives resulting from procompetitive merger effects in the short-run (as captured by effects under Box 2 and Box 4) could also lead to entrenchment and harm to consumers over time. Based on guidance from the literature on static mergers, this is likely to be a rare occurrence, as the countervailing effect on entry by rivals typically would not outweigh the direct benefits from procompetitive effects. We identify conditions where entrenchment stemming from beneficial merger effects in the short-run could nonetheless be envisaged in dynamic settings.
- Where a specific entrenchment mechanism is identified, balancing with the procompetitive effects of the merger may still be required, especially in the case of acquisitions of complementary assets.

- 19.1 Entrenchment is a relatively recent category of anticompetitive dynamic effect discussed both in the academic literature (see Chapter 6) and in policy discussions. As we have reviewed in Chapter 11, the US Merger Guidelines contain a specific guideline on entrenchment of dominance (Guideline 6).

- 19.2 This Chapter discusses how entrenchment concerns fit within our conceptual framework, in light of our characterisation of horizontal and non-horizontal anticompetitive effects (in Chapters 15 and 18 respectively). We first define the concept of entrenchment (Section 19.1), and then discuss examples of potential entrenchment mechanisms based on the recent academic literature, identifying some general insights (Section 19.2). We also discuss whether benefits from a merger in the short-run may lead to entrenchment over time (Section 19.3); and considerations on balancing between harm from entrenchment and merger benefits (Section 19.4).

19.1 Defining the scope of entrenchment effects

- 19.3 A merger may entrench the position of the merged entity. We define entrenchment as a dynamic effect of a merger that increases barriers to entry and expansion faced by rivals of the merged entity, and weakens the competitive constraint they are able to impose in the future, to the ultimate detriment of consumers.
- 19.4 We define entrenchment narrowly to explicitly exclude anticompetitive mechanisms captured under Box 1 and Box 3. Entrenchment can be thought of as a sub-set of the mechanisms through which a merger may lead to adverse dynamic effects, in addition to those captured under Box 1 and Box 3. The competitive harm in entrenchment cases flows from the fact that the merger itself deters entry or distorts innovation by non-merging rivals—distinguishing it from effects under Box 1 (which captures the direct loss of competition between the parties) and Box 3 (which requires specific conduct by the merged entity, aimed at foreclosing rivals). Entrenchment effects may therefore apply to a specific merger concurrently with Box 1 or Box 3 effects.
- 19.5 From a policy perspective, it is appropriate to restrict entrenchment concerns to mergers that enable a firm to obtain or strengthen a position of significant or substantial market power. This is also appropriate under an error-cost approach. We discuss below the guidance from the economic literature and from economic principles on this question. In economic terms this implies that entrenchment effects would only apply to market settings where a firm has or acquires a significant degree of market power. Depending on the case, this may be the result of significant economies of scale, direct and indirect network effects, learning-by-doing effects (including through access to data), access to multiple products (in an ‘ecosystem’ of products), and the impact of past acquisitions cementing an incumbency position.
- 19.6 We conceptually distinguish entrenchment effects from a reduction in sales by rivals (and a corresponding increase in sales by the dominant acquirer) that *directly* follows from procompetitive effects under Box 2 and Box 4 of our framework. These effects benefit consumers (e.g. by lowering costs, boosting innovation incentives of the merging parties, or leading to one-stop-shopping efficiencies), at the same time as bolstering the market position of the dominant firm at the expense of its competitors.⁶²⁰ We however discuss whether there are circumstances where a procompetitive effect of a merger in the short-run could lead to entrenchment over a longer period of time to the detriment of consumers.

⁶²⁰ On this point, see the US Merger Guidelines, p. 18: ‘[...] the Agencies distinguish anticompetitive entrenchment from growth or development as a consequence of increased competitive capabilities or incentives. The Agencies therefore seek to prevent those mergers that would entrench or extend a dominant position through exclusionary conduct, weakening competitive constraints, or otherwise harming the competitive process’.

- 19.7 In cases where a specific entrenchment mechanism can be identified, balancing with procompetitive effects of a merger may be required.

19.2 Insights into entrenchment mechanisms identified in the economic literature

- 19.8 We identify examples of potential entrenchment mechanisms based on insights from the recent academic literature, and set out some of the general principles at work in these mechanisms.⁶²¹ These examples fit our definition of entrenchment, as set out in the previous Section.⁶²²

19.2.1 Examples from the economic literature

- 19.9 *Example 1.* In a setting with customer inertia and switching costs, a horizontal merger with an innovative startup can expand the customer base of an incumbent firm for a period of time, and may render the market less contestable for future entrants.⁶²³ Successive acquisitions can entrench the position of the incumbent and deter entry by weakening innovation-for-buyout effects for future potential competitors. This mechanism rests on the fact that acquisitions can, over time, worsen the ‘outside option’ of potential competitors when negotiating a (future) potential acquisition by the incumbent.⁶²⁴
- 19.10 *Example 2.* A mechanism similar to the previous example may be at work in a setting where firms compete by offering bundles of integrated products (ecosystems), and where a dominant ecosystem instead of a smaller competitor (a laggard) acquires an entrant offering an incremental complementary innovation.⁶²⁵ In this case, acquisitions can increase differences in ecosystem size, resulting in weaker competition between ecosystems to acquire future innovative entrants (e.g. startups) and a lower acquisition price, thereby decreasing the incentives to innovate of future startups as the degree of market power of the incumbent grows. A restrictive policy that does not allow market leaders to buy providers of complementary assets may increase innovation incentives.
- 19.11 *Example 3.* In a setting where platforms can steer users to their services through default settings, and where this effect also applies across multiple products in an ecosystem, an incumbent platform may be able to acquire providers of services in connected markets, growing its ecosystem.⁶²⁶ The ability to steer users provides large platforms with the ability to take over complementary products at a discount, because the firm has the outside option of acquiring its target’s competitors instead, and steering users away from the target. By contrast, smaller competing firms have weaker incentives for takeovers, as they have less to gain from cross-market

⁶²¹ For a discussion of possible entrenchment mechanisms, see US Merger Guidelines (Guideline 2.6.A) and Athey, Gross, Marinescu, and Shanefelter (2024).

⁶²² The list of examples presented here is not exhaustive, and is based on the insights from the existing academic literature. There may be other mechanisms that fit our definition of entrenchment. We note, however, that this is a recent area of research, and that the literature only contains limited examples of anticompetitive entrenchment.

⁶²³ This depends on the price formation process in the relevant market. Access to a larger captive customer base may increase the prices set by the dominant firm, partially offsetting the adverse impact on future entrants from having access to a smaller contestable customer base.

⁶²⁴ See Denicolò and Polo (2026).

⁶²⁵ See Bryan and Hovenkamp (2020) and Azzali, Denicolò, and Polo (2026).

⁶²⁶ This concern applies to the case of market leaders in access-point markets (where users start their online journeys) who are able to steer users in these markets towards their products in other markets.

steering. Over time, mergers by the incumbent platform and dominance of access points may lower incentives for entry and innovation.⁶²⁷

19.12 *Example 4.* A merger may allow an incumbent firm to benefit from a ‘kill zone’ effect, which discourages entry by other competitors. For example, a merger with an innovative startup can allow the incumbent to engage in defensive innovation against other potential competitors, deterring their entry and distorting the direction of their innovation.⁶²⁸

19.13 *Example 5.* The prospect of a merger between an incumbent platform and a small innovative entrant may deprive the entrant of beneficial network effects—for example by discouraging multi-homing behaviour by one side of the market (e.g. app developers), in anticipation of acquisition by the incumbent, and integration into the incumbent’s platform.⁶²⁹ This may in turn reduce the entrant’s incentives to enter the market in the first place (also giving rise to a ‘kill zone’ effect).

19.2.2 Insights from the examples

19.14 A number of common elements emerge from the examples of entrenchment in the economic literature.

19.15 The first relates to the role played by the degree of market power held by the acquirer. In most of the examples provided the acquirer holds a significant degree of market power (or a position of dominance, in legal terminology), that has grown over time (e.g. through past acquisitions or access to multiple products). Dominance may allow the acquirer to weaken the outside option of a potential entrant (by making the market less contestable), or to increase its lead relative to a rival acquirer. Both channels enable a dominant acquirer to offer a lower acquisition price for a target, lowering the innovation incentives of future entrants. Dominance also gives rise to stronger first-mover advantages, making it easier for the incumbent to deter future entry. This provides support for limiting a definition of entrenchment effects only to mergers that create or strengthen a dominant position. It is also in line with our ECW standard, and error cost approach, as dominance makes it more likely that any harm from entrenchment would be significant.

19.16 The second general insight is that the mechanisms listed above do not lead to competitive harm through a direct loss of competition between the acquirer and the target (as under Box 1 of our framework), or through foreclosure conduct by the merged entity (as under Box 3 of our framework). The competitive harm instead flows from the impact of the merger itself on the innovation incentives of current or future non-merging rivals (and, in some cases, on the direction of their innovative efforts). This is consistent with our definition of entrenchment.

19.17 The third element is that all the examples rely on a dynamic setting, where entry by competitors is prevented or discouraged. In the first three examples, entrenchment results from a deterioration of the outside option of future entrants, whilst in the last two they are a consequence of a ‘kill zone’ effect discouraging entry. In some cases, the anticompetitive effects only arise over time, as a result of a series of past mergers. By contrast, in a largely static setting, entrenchment is unlikely to occur.

⁶²⁷ See Heidhues, Köster, and Kőszegi (2024).

⁶²⁸ See Teh, Banerjee, and Wang (2026).

⁶²⁹ See Kamepalli, Rajan, and Zingales (2022).

19.3 Short-term benefits vs long-term harm

- 19.18 The mechanisms described above do not rest on a causal link between a short-term efficiency benefiting consumers, and long-term harm. Whilst the merger may be associated in the short-term with some benefits (e.g. integration of complements in the case of Example 2, or steering to better products in the case of Example 3), the source of harm to rivals (and ultimate harm to consumers) is not premised on the existence of these benefits.
- 19.19 This raises the question of whether a merger that leads to short-term benefits to consumers, and by doing so hurts rivals, may also lead to entrenchment over time, to the ultimate detriment of consumers (with the medium to long-term harm outweighing the short-term benefits).⁶³⁰
- 19.20 The existing guidance from the economic literature and from general economic principles indicates that circumstances where this may happen are likely to be rare.
- 19.21 As we have explained in Chapter 14, in common static horizontal models of price competition, the interests of rivals to the merging firms conflict with those of consumers, and a merger is beneficial if and only if it is harmful to merger 'outsiders'.⁶³¹ This is because a merger efficiency benefits consumers if it leads to lower prices and higher output by the merged entity, implying that rivals earn lower profits. If the efficiency is insufficient to benefit consumers, it means that it does not prevent a price increase by the merged entity. Rivals benefit from higher prices and higher sales in this case. This basic logic is likely to broadly apply also to dynamic models of competition, where firms compete in both investment and price (even though there is less guidance from the literature on this setting). This implies that, at least in the short-run, harm to rivals is unlikely to co-exist with harm to consumers (unless there is a specific foreclosure mechanism, in non-horizontal settings, as discussed in Chapter 18).
- 19.22 In dynamic settings, with entry by firms not yet present in the market, or exit by existing rivals to the merged entity, the analysis is more complex. Considering the case of entry, it may be that in the counterfactual some firms would have entered the market in the future, leading to greater competition.⁶³² If a merger takes place, and if it leads to an efficiency, expected profits for future entrants will be lower, deterring their entry (this is the case if their entry costs are sufficiently high, and expected profits from entry drop below the level of entry costs), and leading to lower future competition. The question in this case is whether the countervailing effect of less future entry can more than offset (over time) the direct effect of the benefit to consumers from a merger.
- 19.23 There is again limited guidance from the economic literature on this question, even though there are some relevant insights from models of merger-induced entry (which can be seen as the converse of merger-induced entry deterrence). For the case of *merger-induced entry*, recent work shows that the countervailing effect of entry following an anticompetitive merger is unlikely to offset the harm from that merger, in isolation. If it were, the merger would not be profitable for the merged entity, and would hence be unlikely to take place. In this case, only the combination of merger-induced entry and a minimum level of efficiencies is able to avoid harm to consumers, and yet allow for both profitable entry and a profitable merger.⁶³³

⁶³⁰ We discuss the question of balancing between benefits and harm in the next Section of this Chapter.

⁶³¹ For example, see discussion in Caradonna, Miller, and Sheu (2025).

⁶³² The case where the merger may induce the exit of existing firms is analytically similar.

⁶³³ See Caradonna, Miller, and Sheu (2025)

Whilst the parallel to *merger-induced entry deterrence* is not perfect, some of the insights from this work are likely to be relevant. Namely, if the merger deters entry and if this were to fully offset the benefit to consumers from an efficiency, then this is likely to at least partially restore entry incentives (e.g. by preserving prices at the pre-merger level). In equilibrium, if some entry is deterred, it is therefore unlikely to fully countervail the efficiency benefits from a merger. Also in this case, the direct effect from the merger (in this case, a benefit to consumers) is likely to dominate the countervailing effect through its knock-on impact on entry conditions.

- 19.24 An additional mechanism is therefore required for short-term benefits to result in long-term harm. One possibility could be that a merger takes place at a time when entry conditions are particularly favourable (e.g. in a period of technological transition). If this is the case, the entry deterrence effect of a beneficial merger could shut down a 'window of opportunity' for would-be entrants, which would be then unable to re-enter the market in the future.⁶³⁴ This would enable a merged entity to deter entry in the short-run, and then raise prices in the future, once re-entry is no longer feasible. A similar logic may apply if the market is subject to strong network effects and tipping dynamics. In this case, if the merged entity gains sufficient scale through an efficiency-enhancing merger, future entry may no longer be feasible or attractive, allowing the incumbent to exercise its market power in the future. This mechanism has a 'predatory' flavour to it, in the sense that the merged entity would increase competition in the short-run, deter entry (or induce exit), and then reduce competition in the future once entry is no longer feasible.⁶³⁵ A challenge for these entrenchment mechanisms is to explain why the effect is merger-specific, and why the merged entity (especially if it holds significant market power) would not be able to take unilateral procompetitive actions aimed at deterring entry in the short-run, in order to increase its market power over time.
- 19.25 Overall, the guidance from economic principles suggests that anticompetitive entrenchment through short-term procompetitive effects is likely to be a rare occurrence. A merger that makes future entry less attractive through its procompetitive effect is likely to benefit consumers (as the entry deterrence effects are positively correlated with the benefits to consumers). The countervailing effect of less favourable entry conditions is therefore unlikely to outweigh the direct benefits from the merger. There are conditions which make this mechanism more likely to occur (e.g. if future entry conditions are expected to be harder than current conditions, or if the market is subject to tipping), but even in these cases the competitive concern may not be merger specific.

19.4 Balancing entrenchment effects with potential procompetitive effects

- 19.26 Entrenchment effects may be associated with concurrent competitive benefits. This is the case both for some of the examples described in Section 19.2, and for entrenchment effects that are premised on the existence of procompetitive effects in the short-run (as discussed in Section 19.3).
- 19.27 For the case of the examples described in Section 19.2, benefits are more likely for the case of mergers involving complementary assets. These include the range of benefits already discussed under Box 4 (e.g. including pricing and investment complementarities and benefits from one-stop-shopping). More generally, for cases

⁶³⁴ This effect is more likely if entrants are financially constrained, and cannot 'invest' in entering the market in the short-run, in order to reap future benefits.

⁶³⁵ Contrary to predation, there would be no 'profit sacrifice' in this case, and no need for future recoupment.

where an incumbent with significant market power acquires a smaller competitor, consumers may also benefit in the short term from stronger network effects, and greater diffusion of a new technology.⁶³⁶

- 19.28 The required balancing exercise is similar to the one that we have described in Chapter 17 (for horizontal mergers) and Section 18.3 (for non-horizontal mergers). The distinguishing feature here is that the balancing is likely to involve an asymmetric comparison between short-term benefits and long-term harm, with the short-term benefits being not only more immediate, but also more likely to materialise. For the balancing exercise to result in net competitive harm, the entrenchment mechanism should therefore be sufficiently concrete. Restricting entrenchment concerns to the case of firms with significant market power is appropriate in this context, as this makes it more likely that harm from entrenchment is going to be significant, and capable of offsetting short-term benefits. The sliding scale principle is again applicable here, with balancing more likely to result in net harm the stronger the market power of the merged entity. For some of the mechanisms described in Section 19.2, a history of past acquisitions can also play a role in the balancing, especially if there is evidence that they have resulted in reduced contestability and a greater competitive lead compared to rivals.
- 19.29 In the case of an entrenchment concern premised on the existence of short-term benefits, a balancing exercise is going to be required by definition. This is de facto equivalent to the articulation of a coherent theory of harm, showing that long-term harm is likely to outweigh short-term benefits. We have discussed the possible elements, and challenges, of such a theory of harm in the previous section. Also in this case, the degree of market power of the merged entity is likely to play a significant role, as the loss of future competition through deterrence of future entrants is more harmful the stronger the position of the incumbent.

⁶³⁶ See Kamepalli, Rajan, and Zingales (2022) and Teh, Banerjee, and Wang (2026).

20 Application of the conceptual framework to canonical merger cases

Key points

- We apply our conceptual framework to five canonical case categories (A–E), where our categories are not mutually exclusive. These cases are distinguished by the nature of any overlap (pipeline versus capabilities), the nature of the target (incumbent versus nascent competitor), presence of an acquirer with significant market power, the type of merger (horizontal versus non-horizontal), and the type of investment (R&D versus physical).
- Category A (*pipeline overlaps*) includes R&D pipeline-to-product overlap (e.g. Novartis/GlaxoSmithKline Oncology Business, Elanco Animal Health/Bayer Animal Health Division) (Category A1); R&D pipeline-to-pipeline overlap (e.g. Illumina/PacBio, Adobe/Figma, Pfizer/Seagen) (Category A2); and overlap in physical investment projects (e.g. ENI/EDP/GDP, Ball/Rexam) (Category A3). Box 1 and Box 2 effects may apply to all three subcategories, and vary depending on the nature of the investment process (R&D versus physical).
- Category B (*capability overlaps*) includes overlaps in innovation capabilities, where the merger combines two firms with significant overlapping R&D capabilities (e.g. Deutsche Börse/NYSE Euronext, Applied Materials/Tokyo Electron, Halliburton/Baker Hughes, Dow/DuPont, Bayer/Monsanto) (Category B1); and overlaps in investment capabilities, where the merger combines two firms with significant investment capabilities (e.g. Ryanair/Aer Lingus I & III, Pure Gym/The Gym, Orange/MásMóvil, Vodafone/Three) (Category B2). Again, Box 1 and Box 2 effects may both apply, with broader scope for spillovers and synergies under Box 2 than for Category A cases.
- Significant capability overlap (Category B) is likely to imply pipeline/product overlap (Category A), but the converse does not hold.
- Category C (*acquisitions of nascent competitors*) includes cases where the target has limited or no sales, the nascent competitor may not be a direct rival, the incumbent acquirer typically has significant market power and profits at risk from entry, and the merger may be one of the entrant's primary exit routes (e.g. Mallinckrodt/Synacthen, CDK Global/Auto/Mate, Facebook/GIPHY, and Facebook/Instagram/WhatsApp).
- In Category C, a sliding-scale approach is relevant for Box 1 effects, where the greater the acquirer's market power, the lower the required probability of successful entry to trigger concerns. Box 2 benefits (e.g. technology transfer, access to the incumbent's financial/R&D assets) must be merger-specific, where the availability of alternative buyers or sources of funding for the target is a critical factor.
- Categories A to C capture concerns both on the loss of innovation competition and the loss of potential competition.
- Category D (*dynamic foreclosure*) captures the effects described in Chapter 18 and includes input, customer, and conglomerate foreclosure. Examples include Facebook/GIPHY and Illumina/GRAIL. Effects under Box 3 and Box 4 may apply here.

- Finally, Category E (*entrenchment*) captures the effects described in Chapter 19. It may include specific mechanisms based on customer inertia or switching costs, adverse innovation-for-buyout effects for future rivals, deterrence of multihoming, and competition between ecosystems. This category may coexist with Categories C and D. A recent example is Booking Holdings/eTraveli Group.

20.1 In this Chapter we apply our conceptual framework to specific categories of cases, illustrating them with case examples and case studies. Table 20.1 provides an overview of our categorisation of mergers with dynamic effects and some relevant definitions. Sections 20.2–20.6 cover each of the five categories of canonical cases, first providing an overview of the category (including potential subcategories) and then discussing its link to the relevant dynamic effects of the conceptual framework, and relevant potential evidence.

20.1 Overview of canonical cases of mergers with dynamic effects

20.1.1 Overview of categorisation

20.2 We identify five types of canonical case, and discuss the most relevant dynamic effects and the types and sources of potential evidence for each. The categories are not mutually exclusive, implying that any given case may include elements across multiple categories. Our five categories of canonical cases are set out in Table 20.1 below. The terms used are defined in Subsection 20.1.2. Some of these categories (e.g. dynamic foreclosure and entrenchment) map directly to the corresponding Chapter of the conceptual framework, and are included for completeness.

Table 20.1 Categories of canonical cases with dynamic merger effects

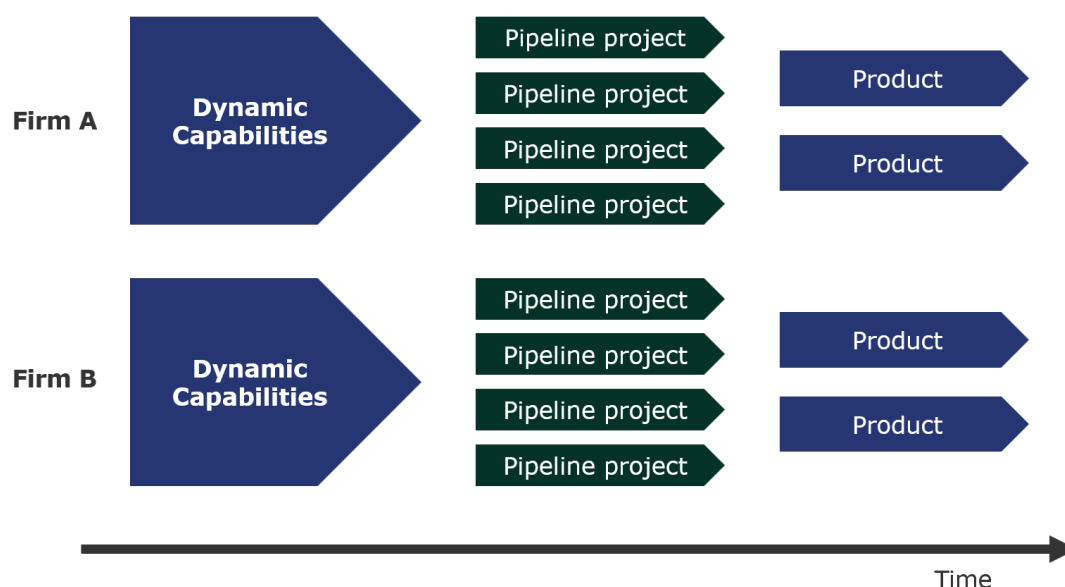
Category	Key features
A: Mergers involving pipeline overlaps	The merger leads to significant overlaps involving R&D pipeline projects (including product-to-pipeline and/or pipeline-to-pipeline overlaps) or physical infrastructure investment projects
B: Mergers involving capability overlaps	The merger leads to significant overlaps in dynamic innovation (R&D) or investment capabilities
C: Acquisitions of nascent competitors	The merger involves the acquisition of a recent entrant, with overlapping pipeline projects or capabilities
D: Mergers raising dynamic foreclosure concerns	The merger enables the merging firms to foreclose rivals (through input, customer, or conglomerate foreclosure), to the potential detriment of consumers
E: Mergers raising concerns about the entrenchment	The merger creates or strengthens a position of significant market power, by reducing market contestability, to the potential detriment of consumers

Source: Oxera.

20.3 The first three categories (A to C) capture potential dynamic horizontal effects. These categories are based on the following stylised structure for the innovation or investment process (value chain), with three vertically connected phases (see Figure 20.1): an ‘upstream’ phase where firms invest in dynamic *capabilities* (see definition in Subsection 20.3.1 below); a subsequent phase where reliance on these capabilities and additional investments results in specific *pipeline* projects; and a final phase where successful pipeline projects lead to the introduction of new or

improved *products* or production technologies. In this stylised representation, over time, access to dynamic capabilities results in pipeline projects and eventually products, with the length of the investment cycle depending on the industry.

Figure 20.1 Stylised investment or innovation value chain



Source: Oxera.

20.4 Our categorisation distinguishes across canonical cases along the following dimensions.

- *The nature of the overlap*: does the merger give rise to significant competitive overlaps, and if so where? This allows us to distinguish between pipeline or product overlaps (Category A); and capability overlaps (Category B).
- *The nature of the target firm*: is it an established (incumbent) firm or a nascent potential competitor? This allows us to distinguish between mergers where both the acquirer and the target are established competitors (Categories A and B); and those related to nascent competitors or startups (Category C).
- *The nature of the acquiring firm*: is it a player with significant market power (e.g. dominant) or not? This distinction matters for the strength and nature of potential adverse effects across all categories, in particular under Category E.
- *Nature of the investment*: does the investment mainly consist of R&D to introduce an innovation (which may be uncertain and is typically non-rivalrous), or is it a physical investment (e.g. in infrastructure, or physical assets) based on an existing technology (which is typically more certain, rivalrous, and excludable)? We distinguish between these cases within Categories A and B. Both types of investment include early-stage efforts and development activities closer to product introduction.
- *The nature of competition between the merging parties*. Are they current or prospective competitors in the same market (Categories A, B, and C), or active in complementary markets (Category D)? Category E (entrenchment) may involve both elements.

20.5 As noted above, the five canonical cases are not mutually exclusive. Indeed, cases raising horizontal effects within Category C (acquisitions of a nascent competitor) and Category E (entrenchment) may also fall within Categories A and B, depending on the identity of the target and the acquirer, and the underlying competitive dynamics in the market. Cases raising non-horizontal effects may also partially

overlap, since Category E (entrenchment) may co-exist with foreclosure concerns under Category D (as explained in Chapter 19). Categories C and D may also overlap, as a nascent competitor may be active in a different, but related, market to that of the acquirer, but over time may grow to become a direct competitive threat to the acquirer.

20.1.2 Definitions

20.6 In our description of canonical cases, we adopt the following definitions.

20.7 **Dynamic capabilities** capture the collection of assets required to be able to invest to address future or new consumer needs, respond to the evolution of markets (e.g. changes in regulation, technology, consumer needs), and introduce new/improved products and technologies over time. The nature of the assets required depends on the industry at hand, and may include R&D assets, data, a user base, infrastructure, branding etc. Dynamic capabilities are not necessarily associated with a given pipeline or product at any given time. They capture firms' ability to innovate, enter new markets, expand geographically, and improve existing processes and products.

20.8 **Pipeline projects** refer to ongoing investment efforts on a project to meet relatively well-identified consumer needs, in a specific relevant market or group of markets. These can include R&D to introduce a new product, or to enhance the quality or lower the costs of an existing product (product and process innovation). These can also include specific investments or initiatives to enter a new product or geographic market. The project may be at an early stage of development (e.g. Phase I in pharmaceuticals), or at a later stage (e.g. Phases II and III in pharmaceuticals). The stage of advancement of the pipeline project affects the size of the investment costs still to be incurred, the degree of uncertainty over the success of the investment, and uncertainty around the scope of the product market(s) targeted with the pipeline. Investment in pipeline projects is broadly defined, and it can include (i) physical investments such as investment in infrastructure (potentially involving large monetary outlays), (ii) investment to introduce new or improved products or new or disruptive business models, and (iii) investment to enter a new product/geographic market in a setting where the costs of entry may include a variety of components (e.g. opportunity costs, management time, working capital, reputational costs, short-term losses to capture network effects).

20.9 **Products** are existing or future products sold in a relevant market.

20.10 **Significant overlaps** (as used in the definition of Categories A and B) refer to the fact that the merging parties are assumed to be two out of a limited number of players with specific dynamic capabilities, or specific pipeline projects or products aimed at the same market.⁶³⁷ Overlaps are significant if the relevant level of the value chain is concentrated, if the acquirer has significant market power, or if the parties are close competitors. We distinguish between two possible types of overlap: overlaps involving at least one pipeline project (Category A), which include both product-to-pipeline and pipeline-to-pipeline overlaps, and overlaps in dynamic capabilities (Category B). The definition of significant overlap is intended as a first screening for the impact of the merger, to facilitate our proposed categorisation—it does not mean that the merger is necessarily associated with significant anticompetitive effects under Box 1.

⁶³⁷ This assumption allows us to focus on cases which may raise competition concerns (under Box 1), and hence may need to be investigated by competition authorities.

- 20.11 A **nascent competitor** is a firm that has recently started its activities, may have only limited or no existing sales, and may not yet have monetised its initial investment in dynamic capabilities or specific pipeline projects. The nascent competitor may not be a current rival of the acquirer, but may grow to become a direct threat over time. Concerns around acquisitions of nascent competitors are often related to potential competition cases, as discussed in Box 20.1.
- 20.12 Finally, for simplicity in our review of each canonical case we describe a merger as the acquisition of a target by an acquirer, assuming that the acquirer is larger than the target (i.e. it has access to more resources and has a stronger incumbency position). This does not affect the substance of the analysis, but facilitates the exposition.



Box 20.1 Relationship between canonical and potential competition cases

The categorisation of canonical cases in the main text includes examples involving potential competition effects in case precedents and existing guidelines (e.g. paras 58–60 of the 2004 Horizontal Merger Guidelines, and Guideline 4 of the US Merger Guidelines).

The typical focus of concerns in potential competition cases is that the merger removes the possibility that entry or expansion by one of the two merging firms would have resulted in greater future competition in the product market (sometimes referred to as ‘actual’ potential competition), or that the merger removes the competitive pressure currently exercised by a potential competitor on an incumbent (sometimes referred to as ‘perceived’ potential competition).

These two effects are captured in our canonical cases. For example, it may be that the potential competitor is pursuing a specific pipeline project to enter a market. A merger with an incumbent would eliminate the possibility of future price competition and may also reduce innovation efforts by the incumbent in response to the entry threat. Both types of effect, if present, are captured by Category A (mergers with pipeline overlaps), or by Category C (if the potential competitor is a nascent competitor).

Similarly, the potential competitor may have acquired or developed dynamic capabilities, enabling it to enter a market in the future, and posing a competitive threat to an incumbent (including by inducing it to innovate to protect its position in the market). Both types of effect, if present, are captured by Category B (mergers with capability overlaps), or by Category C (acquisitions of nascent competitors).

Categories A, B, and C capture broader effects than those sometimes associated with a loss of potential competition, as they also include effects from changes in innovation or investment competition, notably the risk of suppression of the innovation efforts by the potential competitor, and the resulting consumer harm. This broader scope facilitates a consistent treatment of the potential effects of a merger involving a potential competitor.

Source: Oxera.

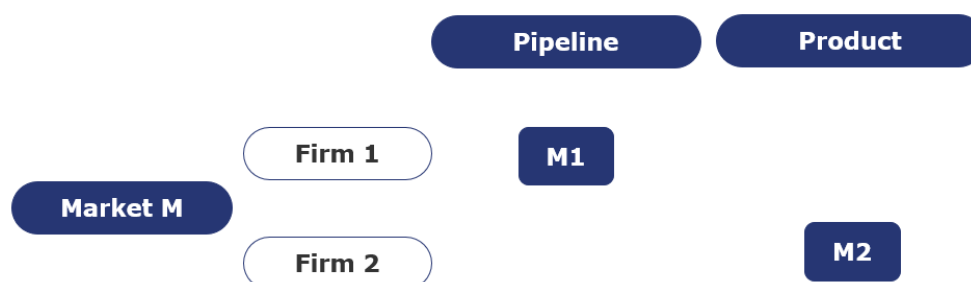
20.2 Category A: mergers with pipeline overlaps

20.2.1 Overview of Category A

- 20.13 Under Category A, the merger leads to a significant overlap in pipeline projects. These may relate to overlaps between existing products and pipeline projects ('pipeline-to-product') and/or overlaps between pipeline projects ('pipeline-to-pipeline').
- 20.14 Category A may also involve overlapping capabilities, but these are not the focus of the analysis for this type of case (e.g. because the market is not significantly concentrated at the capabilities layer). This distinguishes Category A from B.
- 20.15 Category A relates both to pipeline products based on R&D efforts ('innovation') and on pipeline investment projects, not based on R&D (e.g. investments in physical infrastructure, including production facilities). The key distinction here relates to the nature and extent of uncertainty about the likelihood of success of the pipeline project (typically more certain in the case of infrastructure projects), and the public-good features of innovation (typically non-rivalrous and potentially non-excludable in the case of innovation projects).
- 20.16 We include under Category A both 'late-stage' pipeline projects, which are close to market introduction and for which a high share of the investment costs may have already been incurred, and 'early-stage' ones, which may still be relatively far from market introduction and subject to greater uncertainty about the direction and success of the project. While the underlying economic mechanisms are similar in both types of case, the manifestation of dynamic effects may differ.
- 20.17 Category A includes potential competition cases in circumstances where the potential competitor is seeking to enter the market of the other merging firm through a specific and well-identified pipeline project (see Chapter 6). This includes both potential entry via innovation (e.g. R&D on a specific pipeline product) and potential entry via investments in or expenditure on physical assets (e.g. manufacturing facilities, sites, stores). As we set out below, the economic mechanisms typically at work in a potential competition case are similar to those involved in cases with pipeline overlaps (even though the effects captured under Category A are broader than those typically associated with potential competition cases—see Box 20.1).
- 20.18 Finally, cases involving pipeline projects are related to the second and third layers of the four-layer framework for dynamic merger effects introduced by the European Commission in Dow/DuPont (2017) (see Annex A1 for a summary of this approach).
- 20.19 For ease of exposition, we identify three subcategories under Category A.
- **Category A1: mergers with pipeline-to-product overlaps in R&D.** In this case, one of the merging parties is working on an R&D pipeline project that will compete with an existing product of the other merging party. This situation is illustrated in Figure 20.2.
 - **Category A2: mergers with pipeline-to-pipeline overlaps in R&D.** Here, both merging parties are working on parallel and competing R&D pipeline projects (which may also be overlapping with existing products). This situation is illustrated in Figure 20.3.
 - **Category A3: mergers with overlaps in physical investment projects.** In this case, one or both of the merging parties are working on a specific physical investment (e.g. an infrastructure project). The investment project of one of the merging firms overlaps with an existing product or an investment project of the other merging firm. A common application is prospective entry (absent the

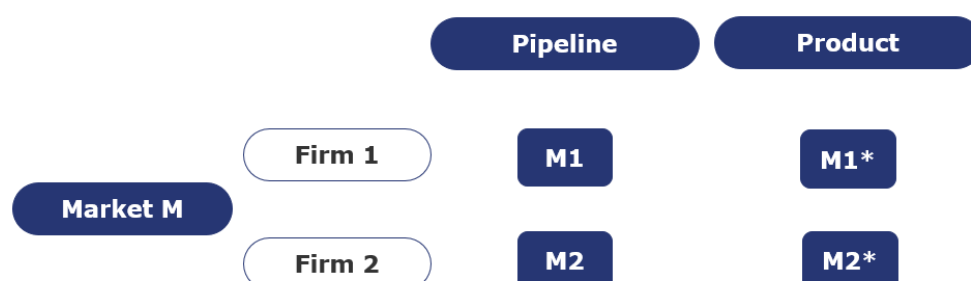
merger) by one of the merging firms in a geographic market where the other firm is present.

Figure 20.2 Category A1: pipeline-to-product overlaps



Note: A merger between Firm 1 and Firm 2 leads to a pipeline-to-product overlap in Market M between Firm 1's pipeline project (M1) and Firm 2's existing product (M2).
Source: Oxera.

Figure 20.3 Category A2: pipeline-to-pipeline overlaps



Note: A merger between Firm 1 and 2 leads to a pipeline-to-pipeline overlap in Market M between pipeline Projects M1 and M2. Here, the merging firms may also sell existing Products M1* and M2* in Market M, and the pipeline projects may be aimed at improving existing quality or lowering costs.
Source: Oxera.

- 20.20 We discuss below (Subsections 20.2.2 and 20.2.3) the principles and evidence for potential dynamic effects under Box 1 and Box 2 for each of these subcategories, together with case studies and examples from past merger control decisions.

20.2.2 Dynamic merger effects under Box 1: principles and evidence

Category A1: R&D pipeline-to-product overlaps

- 20.21 This is the simplest setting to consider dynamic merger effects, where an existing product of one of the merging firms overlaps with a pipeline project of the other merging firm (see Figure 20.2)
- 20.22 The concern under Box 1 is the loss of dynamic competition between the pipeline project and the existing product. This can take the form of a loss of future price competition between the two products (where the pipeline project is successfully introduced), and/or a reduction in R&D efforts in the pipeline product (e.g. a discontinuation of the project, or a lower probability of introduction). A reduction in R&D efforts is more likely for early-stage pipeline projects, where firms have not yet incurred a high share of R&D costs (and hence the incentives to reduce innovation are higher). However, in this case the likelihood of reduced future product market competition is also lower. As discussed in Chapter 15, a reduction in R&D efforts is not a necessary condition for adverse effects under Box 1. That is, the expected harm may entirely materialise in the form of higher future prices of

the existing product and of the future product resulting from the pipeline (see Box 20.2).



Box 20.2 Effects in the case of pipeline-to-product overlaps (*technical material*)

Suppose that there is a market with only one incumbent offering Product A and only one potential competitor with a pipeline project that would allow it to enter the market with a new, differentiated, Product B. We adopt the following notation:

π_m : profits of the incumbent when selling Product A;

π_d : profits of each of the duopolists (if there is entry) when selling Products A and B respectively (assuming symmetry for simplicity);

π_M : profits of the incumbent when selling Products A and B (if there is a merger and if Product B is introduced);

k_e and k_m : fixed cost of project development for Product B for the potential competitor and the merged entity respectively.

We also assume that the potential competitor faces stronger incentives to introduce the product than the incumbent (i.e. $\pi_d > \pi_M - \pi_m$). Cunningham, Ederer, and Ma (2021) and Motta and Peitz (2021) denote this condition as the 'replacement effect' in line with the results of Arrow (1962) and the terminology of Tirole (1988). It applies in several economic settings but not all (see discussion in Volume 2, Chapter 2).

Assuming no synergies ($k_e = k_m$), the merger leads to adverse effects if $k_e < \pi_d$ (i.e. the potential competitor would develop the project absent the merger).

Expected consumer harm can take one of two forms:

a discontinuation of the project if development costs are relatively high ($\pi_M - \pi_m < k_e < \pi_d$); or

a loss of future price competition between Products A and B for lower levels of development costs ($k_e < \pi_M - \pi_m$).

The merger is neutral for higher values of k_e (i.e. when the potential competitor would not develop the project absent the merger).

We return to this example for the case of R&D synergies (i.e. $k_e > k_m$), in Box 20.4, and provide a fuller illustration of the possible effects of the merger.

Source: Oxera. This is a simplified example based on Cunningham, Ederer, and Ma (2021), Motta and Peitz (2021), and Fumagalli, Motta, and Tarantino (2022).

- 20.23 We use the same setting as above to illustrate why an existing player that is threatened by an entrant with a pipeline project may place a higher value on the acquisition of the entrant and can outbid other acquirers (see Box 20.3). This illustrates the standard market power persistence result.



Box 20.3 Incumbent incentives to acquire pipeline competitors

The academic literature on mergers between incumbents and potential competitors shows not only that these mergers are often profitable, but also that an incumbent may face incentives to pay more for the acquisition of the potential competitor

than other bidders who are not active in the same market. This will depend on the degree of market power held by the incumbent, the nature and resources of rival bidders, and on the number of potential entrants. In this illustration we assume for simplicity that there is only one incumbent and one potential competitor (noting that this example does not generalise to a case with multiple competitors or multiple incumbents).

A merger between an incumbent and a potential competitor internalises negative price externalities, preventing a reduction of prices compared with the counterfactual with entry, and hence allowing the incumbent to protect its profits from the threat of entry. This is a manifestation of the 'monopoly persistence' (or 'efficiency') effect,⁶³⁸ which provides an additional inducement for an incumbent to acquire a potential competitor, compared with the incentive faced by a firm whose profits are not at risk from a threat of entry.

We can illustrate this effect by using the example set out in Box 20.2, for the case where the firm with the pipeline project would have introduced Product B absent the merger (i.e. $k_e < \pi_d$), and preserving the assumption of no R&D synergies ($k_e = k_m$). The additional assumption that we need to introduce is that competition dissipates industry profits, implying that the profit of a monopolist offering two products is larger than the profits of two duopolists each offering one product ($\pi_M > 2\pi_d$). This is the case in many common theoretical settings.

Under these conditions, acquisition by the incumbent is profitable. The incumbent can simply replicate the project development strategy of the potential competitor (i.e. introduce Product B), and achieve profits in the product market of $\pi_M - k_e$. The acquisition price needs to compensate the potential competitor for its profits in case of independent entry ($\pi_d - k_e$).⁶³⁹ Net profits from acquisition for the incumbent therefore equal $\pi_M - \pi_d$. This is higher than its profits with no acquisition and entry (π_d) precisely because of the monopoly persistence effect (i.e. $\pi_M > 2\pi_d$).

The incumbent can actually do better than simply replicating the development strategy of the entrant (which further boosts the profitability of the acquisition). As we have seen in Box 20.2, the incumbent may face weaker development incentives than the potential competitor due to the negative investment externality of introducing Product B (as this will divert profits from Product A). If this is the case, there are values of the project development cost k_e for which the incumbent profitably acquires the entrant and discontinues the project development.

Source: Oxera.

- 20.24 Cases with pipeline-to-product overlaps capture the killer acquisition and reverse killer acquisition concerns identified in the academic literature (as reviewed in Chapter 6).
- 20.25 Below, we first set out the types of evidence that may be relevant for the assessment of R&D pipeline-to-product overlaps, and then identify possible sources of evidence. In terms of types of evidence, these include the following.⁶⁴⁰

⁶³⁸ See Tirole (1988).

⁶³⁹ For simplicity, it is assumed that the incumbent has all the bargaining power in the negotiation over the acquisition price with the potential competitor.

⁶⁴⁰ See the findings in Cunningham, Ederer, and Ma (2021) for evidence on some of these effects (e.g. the impact on the number of competitors and of the remaining duration of patent protection).

- Evidence of overlap between the pipeline project and the existing product, i.e. the pipeline project is targeting the same or similar demand and consumer needs as those currently served by the existing product (including cases where the pipeline targets a specific demand subsegment or niche, with a view to use it as a first foothold into the market).
 - Evidence of high concentration (further increased by the merger). For example, there are few other competitors either offering an existing product or working on a competing pipeline project in the same market, or one of the merging parties has significant market power.
 - Evidence of closeness of competition between the pipeline project and the existing product of the merging parties (e.g. introduction of the pipeline project is likely to divert a significant proportion of sales away from the existing product).
 - Evidence of significant profits at risk of cannibalisation by the pipeline project—i.e. profit margins on the existing product are high and are expected to continue to be significant (e.g. due to patent protection).
 - Evidence of specific plans to enter with the pipeline project. This may be evident in the case of structured and relatively public R&D projects (e.g. in pharmaceuticals, through information on clinical trials), but may be less so for confidential and less-visible projects.
 - Evidence on the expected impact of the pipeline product (e.g. is it a major or minor innovation?).
 - Evidence on the incentives and capabilities of the firm with the pipeline product to effectively bring the product to market.
- 20.26 Based on our ECW framework, early-stage pipeline projects may be associated with a significant adverse effect under Box 1 ($\Delta\text{ECW} < 0$) in certain circumstances, even when the probability of success is relatively low (see illustration in Chapter 14).
- 20.27 This may be the case particularly if the acquiring firm has a strong market position, and if successful introduction of the pipeline project would lead to a significant increase in competition to the incumbent or a significant value to consumers.
- 20.28 As listed above, a key consideration when determining the presence of effects under Box 1 is whether the innovator has the capability to successfully introduce the innovation to market. An assessment of overlapping dynamic capabilities is the focus of the analysis under Category B of our canonical cases. What distinguishes Category A from Category B is that a merger in Category A does not necessarily lead to significant overlaps in capabilities, while still resulting in a significant pipeline-to-product overlap. This is possible in some industries given the stochastic nature of R&D, and the long R&D cycles. This means that, at any point in time, the merging parties may happen to be two out of a limited number of players with an existing product and a pipeline project in a given market or area, notwithstanding the fact that the number of firms with capabilities to introduce new products in that market or area may be relatively high.
- **Hypothetical example:** suppose there are five firms with proven capabilities to introduce an innovation to treat a specific type of skin cancer. R&D is costly, stochastic, and subject to long investment cycles (e.g. it takes ten years from discovery to introduce a new treatment). Assume that a merger is proposed at a time when only three out of the five firms have an existing product or pipeline project for the condition. A merger between two of those three firms may raise concerns at the product-pipeline level, but not in terms of the underlying capabilities as the other two firms provide a significant competitive alternative at that level of the innovation value chain. The competition concern from the

merger may apply only over the short to medium term, until rival innovators introduce new pipeline projects.

20.29 Depending on the case, possible sources of evidence include:⁶⁴¹

- public information on the number of existing products and pipeline projects in a given market or area (for industries with structured and relatively public R&D processes);
- information on R&D activities from the merging parties, third parties, or public sources;
- internal documents of merging parties on their pipeline projects (e.g. state of advancement; business plans; expected profitability; expected impact on market outcomes);
- financial documents on current and expected profitability of existing products;
- R&D integration plans for the merged entity (which may reveal information on post-merger R&D efforts on the pipeline project);
- the acquisition price (and related internal documents) may reveal information on the motives for the merger (e.g. is there a defensive motive to protect existing profits from competition?) and on the viability of the pipeline project on a standalone basis;
- evidence that pre-merger the incumbent firm with an existing product perceives the threat of entry by the firm with the pipeline project (as this may be informative on the actual probability of entry).
- evidence that the merging firm with the pipeline project has the potential to bring it to market (i.e. it has the required assets and skills, including R&D resources, managerial skills, financial resources, existing customer base, access to complementary products);

20.30 Particularly tangible evidence for the existence of adverse effects under Box 1 includes evidence that the pipeline project may be suppressed as a result of the merger (e.g. based on internal documents on post-merger plans, or evidence of discontinuations following previous similar acquisitions), and more generally that innovation or R&D output may be reduced. However, this type of evidence may not always be available or reliable, and is not a necessary condition for showing dynamic harm under Box 1, as set out in Chapter 15.

20.31 A number of recent merger cases examined by the Commission and other jurisdictions have raised concerns about pipeline-to-product overlaps, including Thoratec/Heartware (2009, US), Pfizer/Hospira (2015, EU), Novartis/GlaxoSmithKline Oncology Business (2015, EU), General Electric/Alstom (Thermal Power – Renewable Power & Grid Business) (2015, EU), and Elanco Animal Health/Bayer Animal Health Division (2020, EU). Below we discuss Novartis/GlaxoSmithKline Oncology Business and Elanco Animal Health/Bayer Animal Health Division, with a more in-depth description of these cases provided in Annex A2.3 and A2.4 respectively.

- **Case example: Novartis/GlaxoSmithKline Oncology Business (2015, EU).** In this case, the Commission identified concerns in relation to advanced pipelines for Mitogen-activated protein kinase kinase (MEK) and B-Raf Serine/Threonine Kinase (B-Raf) inhibitors,⁶⁴² in particular in advanced melanoma through product-to-pipeline overlaps and in ovarian cancer through

⁶⁴¹ The evidence listed here, while generally informative if available, is not always necessary or sufficient to either prove or disprove the relevant effects considered.

⁶⁴² See Annex A2.3 for further details on these products.

pipeline-to-pipeline overlaps between Phase III programmes. Beyond these, the Commission also identified broader innovation concerns linked to early-stage R&D programmes. It found that overlapping Phase II MEK/B-Raf research programmes competed to deliver future therapies across multiple cancer types. The merger was therefore considered likely to reduce competition in innovation by removing an independent source of R&D and weakening incentives to pursue parallel research programmes. After a Phase I investigation, the merger was approved with structural remedies involving the sale of Novartis' entire MEK/B-Raf programmes (including all necessary IP, data, and support to ensure the continuation of the projects) to the original licensor.

- **Case example: Elanco Animal Health/Bayer Animal Health Division (2020, EU).** The Commission assessed the competitive impact of Elanco's acquisition of Bayer Animal Health in several pharmaceutical product markets for companion animals and production animals using its four-layer competitive assessment framework (see Annex A2.1). In relation to dynamic effects under layers 2 and 3 (advanced and early-stage pipeline products), the Commission identified concerns in Isox-based parasiticides for companion animals as a result of the loss of potential (product and price) competition between the merging parties, as well as the loss of innovation competition resulting from the discontinuation, delay, or redirection of pipeline projects at an early stage of development. High barriers to entry, very limited current or potential alternatives, and likely closeness of competition between the merging parties in a fast-growing market were key considerations. Competition concerns at a broader innovation level (layer 4) did not arise. The merger was approved with remedies following a Phase I investigation.

Category A2: R&D pipeline-to-pipeline overlaps

- 20.32 Effects under Box 1 in the case of pipeline-to-pipeline overlaps are similar to those arising in pipeline-to-product overlap. The main difference stems from the fact that, in the case of pipeline-to-pipeline overlaps, both merging parties are working on competing R&D pipeline projects (i.e. there is innovation competition between the firms). The possibility of a reduction in R&D efforts therefore applies to both merging parties (i.e. under Box 1, both parties may reduce R&D efforts, to internalise negative innovation externalities).
- 20.33 In this context, it is possible that the merging firms are engaged in an innovation race, to be the first to introduce a given new technology. This is particularly relevant in cases where innovation has winner-take-all features (e.g. there are strong network effects and first-mover advantages). In such a case, the threat of innovation by one party may encourage the other party to also innovate, and the merger would reduce or eliminate this innovation rivalry.⁶⁴³ Depending on the case, there may be evidence that these innovation dynamics are at play, and that one or both of the merging firms are increasing their innovation efforts by reacting to the perceived or actual threat of innovation by the rival firm. Conversely, there may also be circumstances where competing innovators react to greater innovation by rivals by reducing their innovation efforts (as more innovation by rivals reduces the reward from innovating, relative to not innovating).⁶⁴⁴ Also in this case, a merger may lead to a reduction in innovation efforts by *both* merging parties, due to the internalisation of negative externalities under Box 1.
- 20.34 As in the case of subcategory A1, a reduction in innovation efforts is not a necessary condition for adverse effects under Box 1. The reduction in future price competition

⁶⁴³ This is a case where innovation efforts are 'strategic complements'.

⁶⁴⁴ This is a case where innovation efforts are 'strategic substitutes'.

between the future products resulting from the pipeline projects may be sufficient to produce significant harmful effects under Box 1.

- 20.35 The other, related, key conceptual difference between subcategories A1 and A2 is that the profitability of existing products (and the risk of innovation by one of the merging parties cannibalising the existing profits of the other merging party) is less relevant for the assessment of the magnitude of potential adverse effects. This is because the negative externalities internalised by the merger also capture the future profits expected to be earned on the future products associated with the pipeline projects. The fact that profits on existing products may be expected to decline (e.g. because of the end of patent protection or other market developments) does not necessarily imply that the internalisation of negative externalities under Box 1 will be less significant.⁶⁴⁵
- 20.36 Depending on the stage of the pipeline projects, it may be more difficult to identify the specific (future) product overlap than in the case of product-to-pipeline overlaps, in particular for early-stage pipelines. This may make it harder to evaluate the concentration in the relevant market and closeness of competition between the merging firms' future products. In these circumstances, the assessment of the loss of dynamic competition may focus mostly on the merging firms' ability or incentives to continue investing in these pipeline projects and on closeness in competition in the underlying innovation capabilities.
- 20.37 The types and sources of evidence are similar between subcategories A1 and A2. The key difference relates to evidence on the nature of innovation rivalry between the merging parties—e.g. do the parties perceive each other as rival innovators in the area targeted by their respective pipeline projects?
- 20.38 A number of recent merger cases examined by the Commission and other jurisdictions have raised concerns about pipeline-to-pipeline overlaps, including: Novartis/GlaxoSmithKline Oncology Business (2015, EU), Illumina/PacBio (2019, UK; 2020 US), Elanco Animal Health/Bayer Animal Health Division (2020, EU), Adobe/Figma (2023; EU, UK), and Pfizer/Seagen (2023, EU).⁶⁴⁶
- **Case example: Illumina/PacBio (2019, UK; 2020, US).** Illumina and PacBio supplied next-generation sequencing (NGS) systems (advanced technologies that enable rapid sequencing of DNA and RNA fragments). Illumina specialised in short-read sequencing systems that process large volumes of short DNA strands simultaneously, while PacBio focused on long-read sequencing technology that analyses longer DNA strands at a slower pace and higher cost. The merger was investigated by the CMA and the FTC which raised similar concerns on the loss of innovation competition between the merging parties as a result of the removal of the competitive threat they posed to each other. Following these parallel investigations, the parties abandoned the transaction in January 2020.
- The CMA's analysis centred on dynamic competition concerns in the rapidly growing and evolving market of NGS systems. While the parties' products were largely complementary, the CMA emphasised that this did not preclude substitutability in some contexts. It found that, given the parties' converging innovation trajectories and R&D devoted to narrowing the gap between their respective technologies, the parties were expected to compete more closely in

⁶⁴⁵ Indeed, it may be that innovation externalities are stronger in relation to future, innovative products relative to current products (since future products are expected to command higher margins).

⁶⁴⁶ Annex A2 includes an in-depth review of all five cases.

the future. Both authorities noted that Illumina and PacBio had engaged in innovation competition for several years, and that, as the parties' sequencing systems increasingly overlapped, this race for innovation would have intensified absent the merger. In addition, given the importance Illumina attached to having a presence in long-read sequencing and its development plans, the CMA considered that Illumina had the ability and a clear incentive to research long-read technologies with a view to launching its own long-read system. The CMA concluded that the merger would therefore eliminate not only current and future competitive pressure between the merging parties in the NGS market, but also innovation competition, including as a result of the threat of entry.

- **Case example: Pfizer/Seagen (2023, EU).** The Commission assessed Pfizer's acquisition of Seagen using the four-layer competitive framework introduced in Dow/DuPont (2017) (see Annex A1). Under the first three layers of the assessment, the Commission found that the merging parties' portfolios were largely complementary, since their products typically had different mechanisms of action, and strong competitive pressure from other firms remained. Because of this lack of direct competition, the Commission did not expect the merger to result in the discontinuation, delay, or reorientation of the parties' ongoing and overlapping lines of research, pipeline projects, or current marketed products, nor to create incentives for price increases. The Commission also found that there was no risk of reduced incentives to innovate or of changes in the direction of R&D in the broader oncology innovation space (under layer 4).
- **Case example: Adobe/Figma (2023; EU, UK).** This proposed transaction, which was investigated in parallel by the DOJ, the Commission, and the CMA, involved the combination of Adobe's dominant desktop-based suite of creative design software with Figma's pioneering web-based collaborative product design tool. The CMA and the Commission identified two theories of harm, both involving dynamic competition concerns. In product design software, where both parties competed, the merger would not only eliminate the existing competition between Figma Design (the market leader) and Adobe, but it would also eliminate Adobe's pipeline project, and broader innovation competition between the parties. Adobe's pipeline project, developed in response to Figma's threat, was cancelled ahead of the merger announcement (a possible instance of a 'reverse killer acquisition'). In the creative design software space, the merger would eliminate Figma as a potential competitor to Adobe's dominant Illustrator and Photoshop. In these markets, the Commission's analysis centred on the elimination of perceived and actual potential competition from Figma, while the CMA focused on the loss of dynamic competition in product development and innovation between the merging parties (finding that Figma had the ability and incentive to expand from the adjacent product design software market into Adobe's core markets).

Category A3: overlaps in physical investment projects

- 20.39 Subcategory A3 relates to overlaps brought about by physical investment projects. This may include overlaps with existing products (as in A1) or pipeline-to-pipeline overlaps (as in A2). The key difference here is that the pipeline projects relate to investments based on a pre-existing technology, rather than R&D to introduce an innovation. This may include investments in infrastructure, production plants, or other physical assets, e.g. network expansion into a new geographic area; a new manufacturing facility in a new market; or entry into a new route (in the case of transport markets).
- 20.40 One key distinction between physical investment and R&D pipeline projects is that the degree of uncertainty about the success of the physical investment may be

lower.⁶⁴⁷ For example, compared with R&D in the pharmaceutical industry, considerations on the highly uncertain efficacy and safety of a new product at an early stage of R&D will not be present. There may nevertheless be commercial uncertainty about the profitability of the investment (e.g. due to uncertainty on market demand and impact of entry).

- 20.41 The fact that the uncertainty of a physical investment project is typically lower than that of an R&D pipeline project does not fundamentally alter the economic mechanisms at work in Box 1. A merger would still lead to the internalisation of negative externalities at work absent the merger (including both price and investment externalities). Adverse effects may result from the loss of future price competition and/or the possibility that the physical investment project is halted or scaled back. The investment project is more likely to be discontinued if there is limited differentiation between the existing/future products of the two merging firms, and hence the merged entity would not benefit significantly from introducing a new product to the market and would face stronger cannibalisation concerns.
- **Hypothetical example:** suppose that Airline A is active on a route (Brussels—Rome), and that Airline B has a plan to enter the same route. Entry on the route would lower prices and improve choice for consumers (e.g. by increasing the frequency of flights). A merger between Airline A and Airline B may lead the merged entity to decide not to enter the route with Airline B. This would harm consumers by leading to higher prices and lower frequency of service relative to the counterfactual. The merged entity may also decide to still enter with Airline B (e.g. because Airline B has a different brand positioning than Airline A, and would allow it to serve a broader set of consumers). In this case, adverse effects would flow mainly from the elimination of price competition between the two airlines. Effects under Box 1 cover both cases.
- 20.42 The scope of evidence for Box 1 effects for physical investment projects may differ from the case for R&D pipeline projects. It may be that the investment process is less structured or standardised than in the case of R&D, at least compared with industries with relatively clear R&D patterns (e.g. pharmaceutical and agrochemicals). The gathering of evidence on an investment project will need to be adapted accordingly, on a case-by-case basis.
- 20.43 Relevant evidence is similar to that for Categories A1 and A2 in terms of concentration effects, closeness of competition, and existing or future profits at risk of cannibalisation by the investment project(s). Evidence that the merging firm working on the investment project is a credible and well-resourced company with the ability and incentives to bring the project to market will also be relevant. This will often be associated with an underlying overlap in dynamic capabilities, under Category B (see discussion below).

⁶⁴⁷ The other key distinction relates to the potential public good nature of innovation relative to investment. This is discussed below when considering the Box 2 procompetitive effects.

- 20.44 A number of past merger cases examined by the Commission have raised concerns about physical investment project overlaps, including ENI/EDP/GDP (2004)⁶⁴⁸ and Ball/Rexam (2016).⁶⁴⁹

20.2.3 Dynamic merger effects under Box 2: principles and evidence

- 20.45 Mergers involving pipeline projects may lead to beneficial dynamic effects under Box 2. We distinguish between R&D pipeline projects (categories A1 and A2) and physical investment projects (Category A3), since the type of potential beneficial effects differs across the two cases (as discussed in Chapter 16).

Box 2 effects under Categories A1 and A2

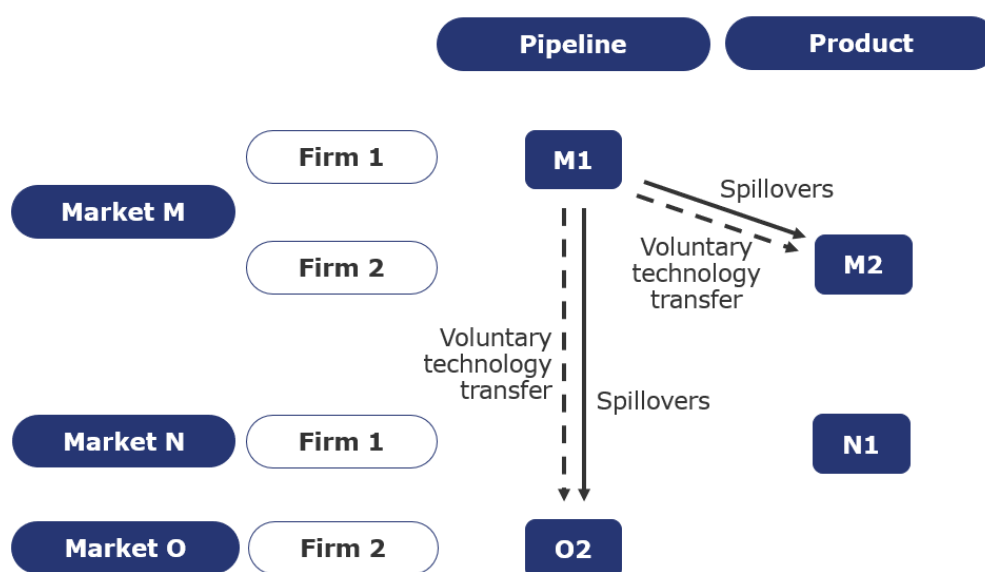
- 20.46 A merger with pipeline overlaps may allow firms to internalise knowledge spillovers (under Box 2A). The most immediate source of them relates to in-market spillovers between R&D on the pipeline project of one of the merging firms, and the existing product or competing pipeline project of the other merging firm (see Figure 20.2).
- 20.47 Taking the example shown in Figure 20.4, it may be that R&D by Firm 1 on pipeline Project M1 allows Firm 2 to improve the quality of its existing Product M2 through knowledge spillovers. This would increase the incentive to invest in R&D on Product M1 after the merger (especially if M1 is a relatively early-stage pipeline, where a significant share of R&D costs is still to be incurred).
- 20.48 There may also be cross-market spillovers, with R&D on pipeline Project M1 benefiting pipeline Project O2.⁶⁵⁰ This too could increase R&D incentives on Product M1 if the merger takes place.
- 20.49 It is also possible that spillovers apply to the broader innovation capabilities of both merging parties (even if they are not aimed at the same product markets) and hence may apply to future R&D efforts across the two merging firms. We consider this case in Category B, where we focus more explicitly on the effects of mergers between firms with significant overlaps in dynamic capabilities.

⁶⁴⁸ This merger concerned a proposed concentration combining major Portuguese gas and electricity assets, which the Commission assessed for vertical and horizontal effects and ultimately prohibited. In this case, the Commission found that GDP was well positioned to enter the electricity generation market in Portugal, due to its incumbency position in gas, and that it had concrete plans to enter. This finding was confirmed on appeal by the Court of First Instance (Judgment of the Court of First Instance, 21 September 2005, T-87/05, para. 188).

⁶⁴⁹ This merger concerned a proposed acquisition of Rexam by Ball that the Commission reviewed over concerns it would reduce competition in Europe's metal beverage can and bottle markets, ultimately clearing it subject to divestment commitments. One concern related in particular to Rexam's plans to enter the Polish market absent the merger.

⁶⁵⁰ The focus here is on spillovers which can lead to Box 2A effects in the overlap Market M.

Figure 20.4 Spillovers and voluntary transfers with pipeline overlaps



Source: Oxera.

- 20.50 Relevant evidence for knowledge spillovers includes information on the strength of IP rights (e.g. in relation to the innovation on M1 in this specific illustration), and documentary evidence that Firm 1's incentives to invest in M1 were constrained by the risk of spillovers to the competing Product M2, or that the merging firms have benefited from spillovers from each other in the past (e.g. M2 and O2 have benefited from spillovers from M1). Additional useful evidence could also include information in documents produced in the ordinary course of business or from industry experts that spillovers are a prevalent feature of the industry, and are taken into account by firms when making investment decisions, and that there are limited options to internalise them absent the merger. This would need to supplement specific information about the extent of spillovers between the merging parties, relative to spillovers to non-merging parties. Evidence on spillovers to non-merging parties would be relevant to determine the overall impact of the internalisation of the impact of spillovers on the innovation incentives of the merged entity (see discussion in Chapter 16).⁶⁵¹
- 20.51 A merger with pipeline overlaps may also give rise to Box 2B effects, across the multiple mechanisms reviewed in Chapter 16. For example, the merger may enable voluntary technology transfers between the R&D pipeline projects of one of the merging parties, and the pipeline projects or existing products of the other merging firm.
- 20.52 Taking again the example in Figure 20.4 successful innovation in pipeline Project M1 could be shared post-merger with existing Product M2 and pipeline Project O2, increasing their quality or reducing their costs. This can have a positive diffusion effect (i.e. increase the quality of Product M2) and could also increase the incentive to innovate in the pipeline Project M1. Relevant evidence for this type of mechanism includes the conditions of licensing markets (e.g. could Firm 1 have licensed its technology to Firm 2 absent the merger?), information on the underlying technology

⁶⁵¹ Assessing empirically the level and importance of involuntary spillovers is a delicate exercise as the larger the extent of such spillovers, the lower the realised R&D investment levels and thereby also the level of realised spillovers.

for the various products (e.g. how feasible is it to transfer technologies from pipeline Project M1 to existing product M2?), and R&D integration plans post-merger.

- 20.53 The merger may also give rise to complementarities in R&D, production, distribution, and marketing. For example, Firm 2 may have stronger R&D capabilities than Firm 1, and hence may be able to increase the likelihood of success of pipeline Project M1. Similarly, it may have stronger distribution assets, increasing the diffusion and profitability of pipeline Project M1 if it were to be introduced to the market. These effects would increase the incentives to invest in M1 after the merger and potentially increase its market impact. Relevant evidence for these types of effect would include R&D integration plans post-merger, and factual evidence on the complementarities of assets across the merging firms.
- 20.54 R&D synergies can offset harm under Box 1, if sufficiently strong. For example, they can reduce or remove the incidence of a 'killer acquisition', and lead instead to a greater probability of innovation and/or a greater value of innovation to consumers. If this effect is sufficiently strong, it can also overcome the loss of price competition on future products from the merger (see illustration in Box 20.4 below).
- 20.55 For all these types of dynamic benefit under Box 2, merger specificity needs to be evaluated. There may be realistic, alternative, and less anticompetitive arrangements allowing the merging firms to capture similar dynamic benefits, e.g. via RJVs, licensing arrangements, and potentially mergers with alternative buyers. We return to this last alternative in the discussion of Category C ('Acquisitions of nascent competitors') as it is a particularly relevant factor in that case.



Box 20.4 R&D efficiencies in the case of pipeline-to-product overlaps (technical material)

R&D synergies may take a variety of forms. For example, a merger may reduce the cost of developing a pipeline project (say, because the acquirer has superior R&D assets compared to the target). In the example set out in Box 20.3, R&D synergies of this type would imply that $k_m < k_e$. This can benefit consumers if the synergies are sufficiently strong.

We illustrate this effect by reference to the figure below, which shows four scenarios depending on the levels of k_e and k_m . In areas I and II the merger is anticompetitive (as illustrated in Box 20.3). In area I, harm comes from discontinuation of the pipeline post-merger; in area II the pipeline is not discontinued but harm comes from loss of (future) price competition. However, in area IV (where $k_e > \pi_d$ and $k_m < \pi_M - \pi_m$), the merger benefits consumers as the potential competitor would not introduce Product B, but the merged entity instead does (by virtue of the incumbent's lower development costs). Note that synergies need to be sufficiently strong for this result to hold (i.e. it is not enough that $k_m < k_e$).⁶⁵²

In the specific example described above, synergies lead to consumer benefits *only* if the merger with no synergies does not lead to harm (i.e. there are no Box 1 effects). However, this is not a general result, and there may be a trade-off between effects under Box 1 and Box 2B.

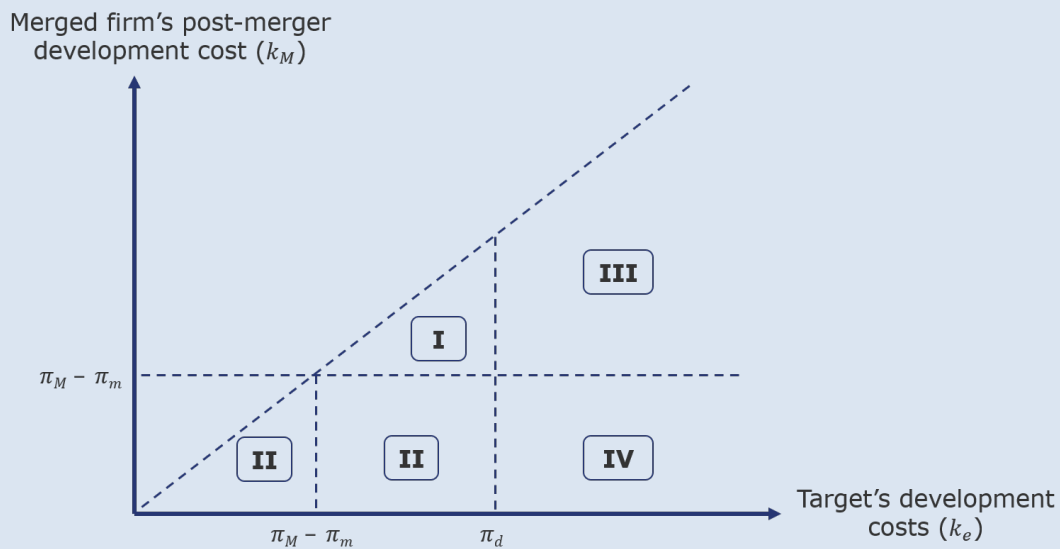
For example, the competition authority may face uncertainty over the level of development costs k_e . Assume that there is a probability p_1 that $k_e < \pi_d$ and hence

⁶⁵² In area III, the merger is neutral for consumers, but it may not take place as the incumbent would not have an incentive to buy the potential competitor.

the potential competitor would introduce Product B.⁶⁵³ In this case, in the counterfactual absent the merger, ECW would equal $p_1 W_d + (1 - p_1) W_m$ (where W_d is the level of consumer welfare if Product B is introduced to the market, and W_m is the level of consumer welfare absent entry, where $W_d > W_m$, since entry lowers prices and leads to greater product variety).

The merger, by leading to R&D synergies, may increase the probability of the introduction of Product B, from p_1 to p_3 (this is the case in this illustrative example if the reduction in development costs from R&D synergies more than compensates for the lower incentives due to cannibalisation). In this case, ECW with the merger equals $p_3 W_M + (1 - p_3) W_m$, where W_M is the level of consumer welfare where Product B is introduced by the merged entity, and where $W_d > W_M > W_m$, reflecting the fact that a monopolist with two products is better for consumers than a single-product monopolist, but worse than a duopolist.

The merger would lead to $\Delta ECW > 0$ if the increase in the probability of product introduction with the merger is sufficiently large, i.e. $\frac{p_3}{p_1} > \frac{W_d - W_m}{W_M - W_m}$.



Area I: Dynamic harm via discontinuation of Product B

Area II: Dynamic harm via loss of price competition between Product A and Product B

Area III: The merger is neutral

Area IV: Dynamic benefits

Note: This illustrates the dynamic effects of a pipeline-to-product overlap depending on target's development costs and the merged firm's post-merger development costs.

Source: Oxera.

⁶⁵³ Alternative formulations to obtain qualitatively similar effects include a setting where there is uncertainty on whether the potential competitor has access to sufficient financial resources to fund the development cost k_e (as in Motta and Peitz, 2021), or where development costs are symmetric between the incumbent and the potential competitor, product introduction is stochastic (i.e. if the development cost is incurred there is a probability p that the product is successfully introduced) and there are synergies in the R&D process (p is lower for the incumbent relative to the entrant). This set-up is studied by Cunningham, Ederer, and Ma (2021).

Box 2 effects under Category A3

- 20.56 The key difference in relation to Box 2 effects between Categories A1 and A2 on the one hand, and Category A3 on the other hand is that Box 2A (spillovers) typically does not apply to physical investments, and that voluntary technology transfers under Box 2B are also often not relevant. This is because investments in physical assets typically do not lead to knowledge improvements that can be shared across firms (i.e. investment is rivalrous and excludable). That said, there are possible spillover mechanisms that also affect physical investments (e.g. spillovers resulting from access obligations in telecommunications).
- 20.57 A number of the benefits under Box 2B can apply to mergers with overlaps in physical investment projects. Considering the illustration in Figure 20.3, the firm owning the existing product or one of the investment projects (e.g. project M2) may have access to complementary assets which make the investment project of the other firm (e.g. project M1) more effective. For example, a firm entering a market with a differentiated product in a new geographic area may have a limited distribution network in that area and may face barriers to building or accessing an effective network. If the acquirer has an existing distribution network, it may face incentives to still offer the new product (despite any cannibalisation effects under Box 1), and the introduction of the new product may have greater beneficial effects on consumers (by reaching more users). Complementarities in other assets (e.g. marketing) would have similar effects.
- 20.58 Production synergies between the competing products (e.g. M1* and M2* in Figure 20.3) would also benefit consumers (by lowering production costs and thus prices), and would enhance the incentives to continue the underlying investment in the investment Project M1 after the merger (by increasing the profitability of the investment).
- 20.59 Finally, in the specific case of physical investment project overlaps, the merged entity may benefit from investment synergies, making it less costly or more effective to bring forward the parallel investment projects with the merger. In particular, the merging parties may be able to share some elements of the required infrastructure (e.g. duplicative fixed costs), lowering the overall cost of the investment projects.⁶⁵⁴ For example, this may apply to investment in the roll-out of more advanced telecommunication infrastructure (e.g. 5G or fibre) in a new geographic area, in order to upgrade existing consumers from a legacy product to a new product.
- 20.60 Evidence of Box 2B effects for mergers with physical investment project overlaps is similar to that discussed above for Categories A1 and A2, and it may include documentary evidence on asset complementarities across the two firms, information from post-integration investment plans, and engineering data on the feasibility and impact of integrating parallel investment plans.
- 20.61 Table 20.2 provides an overview of Box 1 and 2 effects relevant in Category A mergers, together with their principles and evidence.

⁶⁵⁴ This type of synergy may apply to two specific physical investment projects (under Category A3) or to the broader investment capabilities of the merged entity (under Category B). We return to this issue below when discussing cases under Category B.

Table 20.2 Category A: Mergers with pipeline overlaps

	Box 1		Box 2A		Box 2B	
	Principles	Evidence	Principles	Evidence	Principles	Evidence
A1: R&D pipeline-to-product overlap	Loss of future price competition and/or reduction in R&D efforts on pipeline project	Significant concentration or market power in relevant market Closeness of competition High profit margins at risk of cannibalisation Evidence on ability and incentives to enter the market Evidence of suppression of R&D efforts	Internalisation of involuntary spillovers	Weak IP rights Evidence that risk of imitation from other merging party has reduced innovation incentives Evidence of past spillovers across competing R&D programs (including spillovers between the merging firms, and from the merging firms to non-merging firms)	Voluntary technology sharing Complementarities in R&D Complementarities in production, distribution, marketing	Post-merger R&D integration plans Historical evidence on similar R&D synergies
A2: R&D pipeline-to-pipeline overlap	Loss of future price competition and/or reduction in R&D efforts on pipeline projects	<i>As in A1</i> —with main difference that profit margins at risk of cannibalisation include expected profit margins of future products	<i>As in A1</i>	<i>As in A1</i>	<i>As in A1</i>	<i>As in A1</i>
A3: physical investment project overlap	Loss of future price competition and/or reduction in investment efforts on pipeline project	<i>As in A1 and A2</i>	<i>Generally not applicable</i>	<i>Generally not applicable</i>	Infrastructure sharing Complementarities in production, distribution, marketing	<i>As in A1</i>

Source: Oxera.

20.3 Category B: mergers with capability overlaps

20.3.1 Overview of Category B

- 20.62 Category B of our categorisation involves firms with a significant overlap in dynamic capabilities. The concept of dynamic capabilities captures a broad set of assets that enable firms to continuously improve their product offerings to meet the needs of consumers, in future product markets.
- 20.63 We include here both R&D assets that enable firms to discover and introduce innovations to the market (Category B1), and assets that allow firms to invest in new products or enter new markets other than through innovation (Category B2). This class of cases captures the fourth layer in the analytical framework developed by the European Commission in Dow/DuPont (2017) (see Annex A1).
- 20.64 For both subcategories, this group of cases covers situations where the two merging firms are two out of a limited number of firms with significant capabilities to invest or expand in a given industry or area (which may correspond to one or multiple current relevant markets), and where there are significant and durable barriers to entry at the capabilities level (implying that the overlaps between capabilities are expected to persist over time).
- 20.65 The competitive concern here relates to the potential loss of dynamic rivalry between the two merging firms (and resulting loss of future product market competition), over and above any loss of competition in relation to the current product or pipeline overlaps that may be observable at the time of the merger.
- 20.66 In Category B1 (R&D), dynamic capabilities encompass a comprehensive set of assets that enable firms to discover, develop, and commercialise new products and processes. These may include:
- IP and technological access;
 - skilled personnel (scientists, engineers);
 - know-how;
 - physical infrastructure (laboratories, specialised equipment);
 - regulatory expertise and distribution networks;
 - proven track record of successful product introductions and/or of addressing complex customer requirements;
 - access to an existing customer base (to validate and cross-sell new innovations, and upgrade existing products);
 - access to data;
 - access to complementary assets (e.g. key inputs);
 - significant direct or indirect network effects (e.g. from access to data or to a broad user base);
 - access to finance, including from profitability and cashflows from current products;
 - established customer relationships and brand reputation;
 - access to sufficient economies of scale in R&D and production.
- 20.67 These combined assets may position the merging firms as particularly well-equipped innovators within specific technological trajectories or areas. These trajectories are likely to be broader than existing product markets at the capabilities stage.
- 20.68 The definition of dynamic R&D capabilities is related to the definition of R&D markets contained in the US IP Guidelines, the definition of R&D poles in the EC Horizontal

Cooperation Agreement Guidelines, and the notion of innovation spaces in the EC Notice on Market Definition (see Chapter 11).

- 20.69 In Category B2 (investment), dynamic capabilities may include a set of assets that partially overlap with those under Category B1, including:
- skilled personnel (e.g. engineers);
 - physical infrastructure (e.g. manufacturing plants; telecommunication network);
 - distribution networks;
 - proven track record of successful expansions into new products or areas, and/or of addressing complex customer requirements;
 - access to scarce resources (e.g. sites, spectrum, landing slots);
 - vertical integration (where access to competitive inputs is important);
 - access to an existing customer base (to cross-sell new products);
 - significant direct or indirect network effects (e.g. from access to data or to a broad user base);
 - access to finance, including from profitability on current products;
 - established customer relationships and brand reputation;
 - access to sufficient economies of scale in investment and production.
- 20.70 These combined assets may position the merging firms as particularly well-equipped to offer new or improved products and/or expand in new geographic areas, for example by adopting and rolling-out a new technology in the case of the telecommunication market, or entering a new route in the case of transport markets. We illustrate specific examples in Subsection 20.3.2.
- 20.71 Firms with significant overlapping capabilities under Category B1 and B2 are likely to have overlapped also at the product and pipeline level in the past, and are likely to be overlapping at the product or pipeline level at the time of the merger (as in Category A). Indeed, a track record of significant past and current pipeline or product overlaps is an important indicator of a significant overlap at the level of capabilities. Conversely, as discussed above, it is possible that significant overlaps at the pipeline or product level at the time of the merger (as in Category A) are not associated with significant overlaps at the level of capabilities. If this is the case, the key competition concerns raised by the merger relate to the pipeline or product overlaps, but not to the underlying overlaps in capabilities. In this sense, the relationship between Categories A and B is asymmetric: a significant overlap in capabilities under Category B implies a likely overlap under Category A, but the converse does not hold.
- 20.72 A nascent competitor may have unique or significant capabilities that allow it to compete with an incumbent with a more established market presence. We deal with this case specifically under Category C. We can therefore think of the merging firms under Category B as being relatively well-established and not recent entrants.

20.3.2 Dynamic merger effects under Box 1: principles and evidence

Category B1: Overlaps in innovation capabilities

- 20.73 Harmful effects under Box 1 capture the loss of future price competition and/or the reduction in innovation on future products that the merging firms can be expected to introduce absent the merger. In the counterfactual, two firms with overlapping innovation capabilities would compete to divert profitable sales from each other by coming up with new, innovative products in similar areas, and by competing in the corresponding product market. A merger between two out of a few firms with competing innovation capabilities would internalise these negative externalities. The

merger can therefore lead to a reduction in the incentives to initiate new R&D efforts in overlapping R&D areas and/or to a loss of future price competition.

- 20.74 These effects would apply both to the product or pipeline overlaps between the merging parties identifiable at the time of the merger (as in Categories A1 and A2) and to potential future overlaps that are not visible or cannot be identified at the time of the merger. While the latter element may sound abstract, if a merger brings together two out of a limited number of players with significant overlapping capabilities, there can be a credible concern that it would lead to a loss of dynamic competition, over and above the loss of competition relating to existing product or pipeline overlaps. In this case, evidence of past and current product or pipeline overlaps may be an indication of a broader and more structural overlap in the upstream innovation capabilities. The review of the pipeline or product overlaps at the time of the merger provides a snapshot in time of the result of the underlying dynamic rivalry between the merging firms.
- 20.75 On evidence, several proxies can be used to establish the significance of Box 1 effects arising from a merger of two firms with overlapping innovation capabilities.
- 20.76 An important first step is to establish that both parties have significant innovation capabilities, looking at each of the factors listed in para. 20.60, adapted for the industry at hand. Quantitative and qualitative measures can be useful here, such as historical R&D spend, R&D headcount, and patent filings (including text analysis and quality-adjusted measures). Proxies for market concentration (e.g. how many players have similar and equally significant capabilities) can be obtained using similar metrics.
- 20.77 Indirect measures of negative externalities or diversion effects can be obtained by looking at the importance, nature and frequency of past product and pipeline overlaps (e.g. do the parties have a track record of competing with each other in a given R&D trajectory, with few other competitors?). These may be particularly informative if innovation capabilities in the relevant overlap area are durable. Similarly, evidence on overlapping patent portfolios may also be a useful indicator of competing capabilities, by looking at the specific R&D trajectories associated with a given patent family (e.g. by considering information filed for each patent). In markets where tenders are common, evidence from participation and success in tenders with particularly complex requirements may also be informative. There may also be documentary evidence that the merging parties perceive each other as significant innovation competitors and are reacting to each other's innovation efforts.⁶⁵⁵ Evidence that the merging parties are or have been involved in an 'innovation race' with each other, to be the first to bring a given innovation to market, is also a relevant indicator of overlap in the underlying innovation capabilities.
- 20.78 Given the medium- to long-term nature of possible competition concerns arising from overlaps in R&D capabilities, evidence on the presence of significant and durable barriers to entry may be especially relevant to the competition assessment.
- 20.79 Overall, given the intrinsic difficulties of associating capability overlaps with specific existing product markets, the evaluation of the significance of R&D capabilities and of the overlap between the merging parties is likely to place more emphasis on supply-side factors of possible future product markets than on the demand side.

⁶⁵⁵ As for the case of pipeline/product overlaps, in some settings greater innovation by competing firms may induce a firm to innovate more. However, this is not a necessary condition for adverse effects under Box 1.

- 20.80 In some cases, post-merger R&D integration plans may also provide direct evidence of the suppression of combined R&D efforts and expected innovation output post-merger, and hence may provide a concrete indication of a reduction of innovation efforts. As for cases under Category A, this is a sufficient condition for dynamic harm under Box 1, but it is not necessary.
- 20.81 Competition agencies have intervened in a number of high-profile cases involving overlapping innovative capabilities over the past several years, including mergers in the agrochemical sector (Dow/DuPont (2017, EU)⁶⁵⁶ and Bayer/Monsanto (2018; EU, US)⁶⁵⁷), semiconductor manufacturing (Applied Materials/Tokyo Electron (2015, US)⁶⁵⁸), oilfield services (Halliburton/Baker Hughes (2016; EU, US)⁶⁵⁹), gas turbine equipment (General Electric/Alstom (Thermal Power – Renewable Power & Grid Business) (2015, EU)⁶⁶⁰), and stock exchanges (Deutsche Börse/NYSE Euronext (2012, EU)⁶⁶¹). The broad range of sectors shows that concerns about overlapping R&D capabilities are not confined to sectors traditionally associated with R&D competition (e.g. pharmaceutical or agrochemicals).
- **Case example: Applied Materials/Tokyo Electron (2015, US).** The DOJ reviewed the merger between Applied Materials and Tokyo Electron, two leading semiconductor manufacturing equipment suppliers, finding concerns about overlapping innovation capabilities. While the companies competed in specific tools with some pipeline-to-product overlaps, the DOJ's primary concern was that both firms were uniquely positioned—often as the two best or among the three best options—to develop next-generation deposition and etch semiconductor tools for high-volume manufacturing. Their extensive capabilities, assets, experience, and track record made them the preferred development partners for leading-edge semiconductor manufacturers, and the merger would have eliminated competition for next-generation semiconductor equipment development.
 - **Case example: Halliburton/Baker Hughes (2016, US).** The DOJ sued to block the 2016 Halliburton/Baker Hughes merger, which would have combined two of the three largest global oilfield services firms, citing concerns beyond traditional market concentration metrics. While the DOJ identified anticompetitive effects in 23 distinct product markets, a central concern was the loss of dynamic competition and innovation between the merging parties. The DOJ found that Halliburton, Baker Hughes, and their main competitor Schlumberger formed a 'persistent innovation leadership' trio that drove technological advancement through head-to-head competition on 'game-changing' technologies in areas like dissolvable frac plugs and drilling automation. The DOJ rejected the proposed remedies because the divestiture package would not replicate Baker Hughes' full capabilities and dynamic competitive position. The Commission had opened a Phase II investigation of this case, and the parties abandoned the transaction before a decision was adopted.
 - **Case example: Deutsche Börse/NYSE Euronext (2012, EU).** In 2012, the Commission blocked a merger between Deutsche Börse and NYSE Euronext due to concerns it would create a near-monopoly in European exchange-traded financial derivatives. In relation to dynamic effects, the Commission found that the merging firms were close competitors in introducing new products and

⁶⁵⁶ For a description of the R&D concerns, see Coublucq, Kovo, and Valletti (2023).

⁶⁵⁷ See case study in Annex A2.6.

⁶⁵⁸ For a description of this case, see Hill, Rose, and Winston (2015).

⁶⁵⁹ For a description of this case, see Chugh et al. (2016).

⁶⁶⁰ For a description of the R&D concerns, see Coublucq and Federico (2023).

⁶⁶¹ For a discussion of this case, see Kühn, Lorincz, Verouden, and Wilpshaar (2012).

innovations in technology, processes, and market design, in a context with ‘winner-take-all’ dynamics where competition drove innovation. The Commission was concerned that the merger would reduce innovation incentives and eliminate price competition during new-product launches. The General Court upheld the Commission’s decision on appeal, including the finding of a loss of innovation competition between Deutsche Börse and NYSE Euronext (in relation to both new product introduction, and competition in technology, process, and market design).

- 20.82 A common theme from these cases is that they took place in sectors characterised by continuous and costly innovation, high barriers to entry, and the presence of few effective innovators. Where remedies were accepted by the competition authorities, these have typically involved the divestment of significant innovation capabilities (over and above the sale of specific products or pipeline projects), aimed at replacing the loss of an independent innovator brought about by the merger (e.g. this was the case in Dow/DuPont (2017, EU) and Bayer/Monsanto (2018; EU, US)).

Category B2: Overlaps in investment capabilities (other than innovation)

- 20.83 Adverse effects under Box 1 in the case of mergers with significant overlaps in investment capabilities are similar to those in Category B1. The concern is that the merger brings together two firms that are well positioned to compete for future customer needs and/or enter each other’s core markets. Depending on the industry, this competitive dynamic may be driven by the ability to continuously deploy new technologies or the ability to adjust product offerings to meet the evolution of consumer demand (e.g. by entering new markets or segments). Adverse effects include higher future prices and/or lower likelihood of investment or entry.

- **Hypothetical example:** assume that two merging firms have only partially overlapping geographical footprints and are meeting similar consumer needs in each of their areas. For example Firm 1 is active in Markets A, B, and C, and Firm 2 is active in Markets A, D, and E. The two firms overlap in one market (Market A), and are the best placed potential competitors to enter each other’s non-overlapping markets (i.e. Firm 1 is the best-placed potential competitor in Markets D and E, and Firm 2 is the best-placed competitor in Markets B and C). This is driven by the dynamic capabilities of each firm, in terms of access to inputs, brand recognition, and market proximity. There is also evidence that Firm 1 had specific plans to enter Market D. The merger would eliminate dynamic competition between the merging firms, increasing expected future prices in markets where the two firms would have overlapped (over and above the current overlaps), and/or reducing the likelihood of entry in non-overlapping markets.

- 20.84 Evidence of Box 1 effects includes:

- information on the underlying capabilities of the merging firms, e.g. access to scarce inputs, brand recognition, market position, incumbency advantage, access to existing infrastructure, and user base;
- history of past competition between the merging firms, across different markets and products over time (including history of entry in each other’s market);
- concentration of the relevant dynamic capabilities;
- closeness of competition between the two merging firms (e.g. do they have a similar business model and are they addressing similar consumer needs?).

- 20.85 Past merger cases in this category include: Ryanair/Aer Lingus I and III (2007, 2013; EU) and Pure Gym/The Gym (2014, UK).

- **Case example: Ryanair/Aer Lingus I and III (2007, 2013; EU).** The Commission prohibited Ryanair’s first attempt to take over Aer Lingus in 2007

due to concerns about 35 existing route overlaps out of Dublin, and about the loss of Ryanair as a potential competitor on 15 additional routes served by Aer Lingus out of Dublin and Cork. Part of the concern was that both airlines had a strong position in Ireland (including branding) and a significant base at Dublin Airport. When Ryanair attempted to acquire Aer Lingus again in 2013, the Commission again blocked the transaction, finding that the number of existing route overlaps between the merging airlines had increased to 46 routes. This was consistent with the potential competition concerns of the 2007 decision, even though the new overlaps partly differed from the 15 additional routes where potential competition concerns were identified at the time (in particular out of Dublin). Moreover, the new overlaps identified in the 2013 decision also included routes where Aer Lingus entered into Ryanair's routes (and not only the other way around). This evidence on the evolution of product market competition over time was consistent with potential adverse effects from a merger between competitors with overlapping dynamic capabilities.

- **Case example: Pure Gym/The Gym (2014, UK).** In Pure Gym/The Gym, the CMA at Phase I found that the merging parties were the two leading budget gym players and were each significantly larger than the third and fourth-largest players. Their expansion plans were also materially more significant than those of other budget gym players and, in the context of a fast-growing sector, meant that they would open a large number of new gyms in the near future. As the CMA subsequently noted in its Retail Mergers Commentary,⁶⁶² a key concern in this case was the loss of competition between the merging parties at the national or network level to enter and open new sites in new local areas, beyond the loss of potential competition in certain specific local areas (i.e. those where they had specific plans to enter at the time of the merger). The CMA's assessment focused on testing three cumulative conditions: (i) that the merging parties would have been likely to enter into a number of local overlaps absent the merger; (ii) that the merging parties would be substantially more likely to enter into these overlaps than other competitors; and (iii) that such entry would have led to substantially greater competition. Relevant evidence the CMA considered in the assessment of this concern included: the parties' and competitors' expansion plans; their history of successful entry in each other's markets; the ability to obtain similarly prominent locations, and extent of current overlaps, and the closeness of competition (including in site type and location). Based on this evidence, the CMA considered that the merger would result in changed incentives to open new sites, which in turn might lead to a loss of dynamic competition at a national level including site selection and entry alongside a loss of future competition across price, quality, range, or service.⁶⁶³

- 20.86 The mobile telecommunication mergers examined by the Commission over the past 10–15 years (including Orange/MásMóvil (2024)—discussed in Annex A2.9) can also be seen as examples of mergers of firms with significant overlapping investment capabilities. In mobile telecommunication markets, only a limited number of players have access to the resources (spectrum, sites, and existing user base) to be able to effectively deploy new technologies (e.g. 4G, LTE, 5G, etc.). However, in these cases the Commission largely raised static concerns (about higher prices on current products) and assessed the dynamic effects of the mergers mostly under the lens of potential benefits (under Box 2B). This implicitly assumed that the mergers would not lead to a reduction in investment (e.g. slower roll-out of new technologies), possibly due to the potential benefits from sharing infrastructure. However, the Commission also found that most of the benefits could be achieved by the merging parties through less anticompetitive means of cooperation and hence were not

⁶⁶² CMA (2017), 'Retail Mergers Commentary', April, CMA62, paras 1.20–1.22.

⁶⁶³ The deal was abandoned shortly after the decision to refer the merger to Phase II.

merger-specific. In Orange/MásMóvil (2024), the Commission considered in some detail the merging parties' claims around investment synergies—see next subsection and Annex A2.9.

20.3.3 Dynamic merger effects under Box 2: principles and evidence

Category B1: overlaps in innovation capabilities

- 20.87 Beneficial effects under Box 2 for the case of mergers leading to overlaps in innovation capabilities are similar in nature to those reviewed in Section 20.3 for the case of pipeline overlaps (Categories A1 and A2). The key additional dimension is that spillovers, voluntary technology sharing, and R&D complementarities may apply to the underlying innovation capabilities of the merged entity. They may therefore affect future pipeline projects and products, over and beyond the pipeline-product overlaps identified at the time of the merger review. While this effect in principle may also be at work in the case of pipeline-to-product overlaps (in particular if the merger brings together firms with significant overlapping capabilities), this aspect is likely to be more central to the analysis for cases in Category B1, where the focus is squarely on the impact of combining two firms with significant and competing innovation capabilities.
- 20.88 The scope for the spillovers (Box 2A), voluntary technology sharing and R&D complementarities is typically broader in the case of integration of rival innovation capabilities. It may apply over a longer time period, once the R&D activities are fully integrated, and over a broader range of products.
- 20.89 Relevant evidence here includes again information on R&D integration plans, on the nature of R&D in the industry (e.g. are there significant duplicative R&D costs?), on the impact on R&D of past mergers in the industry, on the strength of IP rights, on R&D policy in the industry (e.g. are there existing policy tools to internalise spillovers?), and on licensing conditions. Additional information can be obtained from documentary evidence that in the past merging firms benefited from spillovers from each other (across R&D projects enabled by their capabilities) and have refrained from R&D investment in order not to favour the other merging party. As for categories A1 and A2, evidence on the presence and extent of spillovers to non-merging parties is also relevant.
- 20.90 Firm-specific evidence on complementarities in R&D and other assets (distribution, marketing, footprint) is also relevant. For reasons explained in Section 15.6, evidence on the impact of the merger on non-merging parties may also be informative on the overall effect of the transaction.

Category B2: overlaps in investment capabilities

- 20.91 As for Category A3 (overlaps in physical investment projects), Box 2A effects on spillovers typically do not apply to cases with overlaps in investment capabilities (Category B2) because of excludability. However, beneficial effects under Box 2B may apply, including through infrastructure sharing.
- 20.92 Most of the effects are similar in nature to those discussed for Category A3 for the case of specific physical investment projects. Beyond synergies relating to specific projects, merging firms with overlapping investment capabilities can be expected to integrate their future investment plans, and seek to capture synergies from sharing infrastructure and avoiding duplicative fixed costs. This is similar to the mechanism set out in Section 20.3 in the context of specific pipeline overlaps, but applied more generally to the overall future investment plans of the merging firms. As for the case of innovation capabilities, the scope for synergies is typically greater in this case, as it is likely that joint investment plans can be optimised more effectively over time and across all future projects of the merged entity. That said, alternative

feasible forms of cooperation (e.g. network sharing in the case of mobile telecommunications) may be particularly relevant in the case of a merger of firms with overlapping investment capabilities, as a strategic alternative to a merger.

20.93 Examples of cases where competition authorities have examined Box 2B effects in depth in cases with overlap in investment capabilities include several mergers in the mobile telecommunication sector (see general discussion in Subsection 16.2.4 and case studies on Orange/MásMóvil (2024, EU) and Vodafone/Three (2024, UK) in Annex A2).

- **Case example: Orange/MásMóvil (2024, EU).** In March 2022, Orange and MásMóvil announced their intention to create a JV combining their Spanish fixed and mobile telecommunications operations. The Commission found that the merger would have created Spain's largest operator by number of retail customers, with market share increments of up to 20%, and estimated a level of the Gross Upward Pricing Pressure Index (GUPPI) for the two merging parties in a range of 10–20%, or even higher in some instances, suggesting a significant impact of the transaction on final market prices across different retail markets. The parties claimed variable and fixed cost efficiencies, including from the elimination of wholesale access costs, and network coverage efficiencies through incremental FTTH (fibre-to-the-home) and 5G roll-out over ten years. The Commission applied a four-year timeliness threshold to efficiencies, noting that competitive harm would affect consumers immediately post-merger. As regards cost savings, the Commission accepted only variable cost savings arising from FTTH consolidation, mobile site consolidation, and the elimination of FTTH wholesale access costs, while it considered that fixed cost savings would not affect pricing incentives or incremental network investment. As regards network coverage efficiencies, the Commission rejected them as not verifiable and not merger-specific, also noting that only a small subset of customers would benefit from the increased network quality resulting from the incremental roll-out. The Commission's conclusion was that the efficiencies that were verifiable, merger-specific, and likely to benefit consumers were insufficient to offset the likely price effects of the merger. The merger was eventually approved subject to divestment remedies aimed at creating a new market player.
- **Case example: Vodafone/Three (2024, UK).** The JV between Vodafone and Three involved the third- and fourth-largest MNOs in the UK. The CMA found that the merger would remove two close competitors in a concentrated market, both at the wholesale and the retail levels, leading to material upward impact on prices. A core aspect of the assessment was the analysis of the dynamic efficiencies put forward by the merging parties, resulting from the combination of their networks and spectrum, and of whether they were sufficient to offset the adverse price effects identified. The CMA considered that the merger would lead to 'rivalry-enhancing efficiencies' in the form of improved network quality (better coverage, lower congestion, and faster speeds) and capacity expansion, which would in turn place downward pressure on prices. While some benefits would occur in the short-term (particularly from mutual network access for the merging parties' customers), many of the benefits would take time to materialise. While the CMA found that these efficiencies were merger-specific and would benefit UK customers, it concluded that the merged entity lacked sufficient incentive to fully implement the joint network investment plan required to achieve these efficiencies. Through a qualitative balancing exercise, the CMA also concluded that the efficiencies which were likely to materialise would not be sufficient to offset the price increases expected to result from the merger. The merger was eventually cleared subject to a behavioural remedy package which included legally binding network investment commitments intended to secure the claimed efficiencies and short-term price protections for customers.

20.94 Table 20.3 provides an overview of the Box 1 and 2 effects in Category B mergers.

Table 20.3 Category B: mergers with capability overlaps

Box 1			Box 2A		Box 2B	
	Principles	Evidence	Principles	Evidence	Principles	Evidence
B1: overlap in innovation capabilities	Loss of future price competition and/or reduction in R&D efforts on current and future pipeline projects	Significant concentration in innovation capabilities	Internalisation of involuntary spillovers	Weak IP rights Evidence that risk of imitation from other merging party has reduced innovation incentives Evidence of past spillovers across competing R&D programs (including between merging parties, and to non-merging parties)	Voluntary technology sharing Complementarities in R&D Complementarities in production, distribution, marketing	Post-merger R&D integration plans Historical evidence on similar R&D synergies
		Closeness of competition in capabilities High barriers to entry at capabilities level High expected profit margins on future products Evidence of past or current pipeline or product overlaps Evidence of suppression of R&D outputs post-merger				
B2: overlaps in investment capabilities	Loss of future price competition and/or reduction in investment or entry efforts	Significant concentration in investment capabilities Closeness of competition in capabilities High barriers to entry at capabilities level High profit margins on future products Evidence of reduction of investment or entry post-merger	<i>Generally not applicable</i>	<i>Generally not applicable</i>	Infrastructure sharing Complementarities in production, distribution, marketing	Post-merger investment integration plans Historical evidence on similar synergies

Source: Oxera.

20.4 Category C: acquisition of nascent competitors

- 20.95 Category C of our structure of canonical cases captures specific dimensions of a merger that may apply to cases with pipeline or capability overlaps. It involves an existing incumbent acquiring a nascent competitor who has a competing pipeline project or competing but emerging capabilities. This category may therefore overlap with Category A or Category B, depending on the case at hand. It is useful, however, to discuss it separately, as it raises specific issues on possible adverse effects under Box 1, and on the scope for benefits under Box 2 (see Table 20.4 for a summary of the different effects). We focus here on the acquisition of nascent firms that are potential competitors to the acquirer, that is, mergers that lead to horizontal effects and the internalisation of negative externalities under Box 1.
- 20.96 Cases in Category C relate to firms that are making efforts to enter a market, or have recently entered, with *competing* emerging capabilities or pipeline projects. These nascent competitors may have limited or no sales or profits at the time of the acquisition and may be seeking an 'exit route' for the initial investment (e.g. VC investment). Precisely because the nascent firm has only emerging capabilities, a key question in the assessment of the merger is to establish whether it is actually a potential competitor to the incumbent. The recent policy debate on acquisitions of small innovative entrants (often referred to as 'startups') by large players in digital or pharmaceutical markets typically relates to this type of merger.
- 20.97 The distinguishing features of these cases are:
- a strong asymmetry between the acquirer and the target, in terms of financial resources and assets;
 - merging with an incumbent firm is one of the possible exit routes for the initial investment in the new firm;
 - often, but not always, the acquirer has a strong position in the market targeted by the nascent competitor (e.g. significant market power).

20.4.1 Box 1 effects

- 20.98 Adverse effects under Box 1 are conceptually similar to those identified already for Categories A and B. The merger would eliminate competition between the acquirer and the nascent competitor, leading to higher future prices and/or lower investment. Harm to investment may manifest itself in the discontinuation of current or future pipeline projects that the nascent competitor would have otherwise invested in.
- 20.99 It is also possible that the incumbent may be reacting to the threat of entry by the nascent competitor by itself investing to improve the quality of its products or introduce new products and protect its future profits.⁶⁶⁴ By removing the threat of entry, the merger would also reduce the investment incentive of the incumbent.
- 20.100 As for Box 1 effects under Categories A and B, a reduction in investment incentives is a sufficient but not a necessary condition for a reduction in ECW following the merger. That is, it is also possible that the entirety of the loss of ECW under Box 1 results from a loss of future price competition between the two merging firms, with no change in investment efforts compared to the counterfactual.

⁶⁶⁴ This is the case, for example, if investment by the potential competitor increases the profitability of investment by the incumbent relative to not investing (i.e. investment efforts are 'strategic complements'). It may also be the case if the incumbent is investing to pre-empt entry.

- 20.101 If the acquirer has significant market power, and hence significant profits at risk if the nascent competitor were successfully to enter, the negative price and investment externalities internalised by the merger may be large. This would tend to increase the loss of ECW under Box 1. This effect would also increase the value to the incumbent of acquiring the nascent competitor, in order to defend its existing rents. This market power persistence effect (see illustration in Box 20.3) may allow the incumbent effectively to outcompete rival acquirers in the bidding process to purchase the nascent competitor.
- 20.102 By the same logic, an incumbent with market power may fear that a rival bidder would acquire the entrant, and make it into a more effective competitor (e.g. by relaxing financial constraints, or exploiting other complementarities). This too would increase the expected negative externalities internalised by the merger, leading the incumbent to place a higher value on the acquisition of the nascent competitor relative to the rival bidder.
- 20.103 Especially in situations where the incumbent has a significant market position, it may be that the nascent competitor is not initially entering in head-to-head competition with the incumbent, but may instead deploy alternative and more complex entry strategies. For example, the nascent competitor may be developing a complementary or only partially overlapping product, as this may be an easier way to establish a foothold in the market. Over time, however, the nascent competitor may have the ability and incentives to threaten the position of the incumbent, by exploiting complementarities with its initial product offering (e.g. supply- or demand-side economies), adding features to its offerings, and serving additional customer segments. There may therefore be evidence that a current supplier of *complements* may become (or intends to become) a supplier of *substitutes* over time.⁶⁶⁵
- 20.104 Nascent competitors may also facilitate ecosystem competition, where combinations of products from multiple providers could provide an alternative to the products of an incumbent with market power, and partially constrain it. Nascent competitors may be a particular threat to the incumbent in periods of technology transitions (e.g. a switch from desktop to mobile applications), which provide opportunities for an entrant to differentiate its product based on alignment with a new technology, enabling it to capture network effects that otherwise protect the incumbent from competition.⁶⁶⁶ Therefore, a merger that may appear non-horizontal (e.g. in Category D of our taxonomy of cases) can raise horizontal issues and also be captured in Category C. Indeed, a merger with a potential competitor in a complementary market may at the same time remove a nascent competitor and enable foreclosure of other rival firms.
- 20.105 The acquisition of a nascent competitor may allow an incumbent with market power to remove the competitive threat from a nascent competitor along several competitive dimensions. This may be the case even in the absence of a clear and direct pipeline or product overlap at the time of the merger (under Category A), with the main underlying overlap instead involving the firms' capabilities (under Category B).
- 20.106 All of the mechanisms described above are ultimately 'horizontal' in nature and would generate negative externalities between the merging parties in the

⁶⁶⁵ See, for example, the discussion in Motta and Peitz (2021).

⁶⁶⁶ For a discussion, see Guideline 2.6.A of the US Merger Guidelines.

counterfactual. They therefore fall under Box 1 of our characterisation of potential dynamic effects from a merger.⁶⁶⁷

- 20.107 Especially in the case of a nascent competitor, there may be significant uncertainty at the time of the merger about whether it would grow into a significant competitive constraint on the incumbent. Nonetheless, if the acquirer has significant market power, the impact of successful entry on competition and consumer welfare could be significant. Our ECW framework handles this trade-off using a sliding-scale approach. That is, the greater the market power of the acquirer (and hence the beneficial impact of entry), the lower is the required probability or scale of successful entry by the nascent competitor to raise concerns under Box 1. This approach de facto allows merger control to intervene in cases where an incumbent with market power is acquiring a nascent competitor at an early stage, before it has grown into an effective competitive force.⁶⁶⁸
- 20.108 Evidentiary requirements for effects under Box 1 in the case of acquisitions of nascent competitors are likely to be particularly challenging, in the absence of a clear horizontal overlap and evidence of competitive impact by the nascent competitor. Concrete and cogent evidence (including from internal documents) is therefore particularly important in this category of cases.
- 20.109 Based on the types of effect discussed above, relevant types of evidence include the following.⁶⁶⁹
- Information and evidence on the market power of the acquirer, e.g. as proxied by relevant shares of the market, user base, strength of network effects, and profitability.
 - Information and evidence on the specific capabilities and ongoing projects of the nascent competitor, e.g. how much market traction is the nascent competitor already experiencing, has it come up with a promising innovation or business model, is it already attracting a significant user base, and does it have significant capabilities in a specific trajectory?
 - Information and evidence that the entry strategy of the nascent competitor may enable it to eventually challenge the incumbent in its core market, e.g. what is the mechanism linking current entry in an adjacent (or complementary) market with future entry into the incumbent's core market?
 - Information and evidence on the intent of the acquirer in purchasing the nascent competitor, e.g. is there evidence of a defensive rationale, to forestall a competitive threat either directly by the nascent competitor or indirectly if the nascent competitor is acquired instead by a more established player?
 - Information from the acquisition price paid by the incumbent, e.g. does it suggest a defensive premium to protect existing profits?
 - Evidence on the presence of alternative buyers or exit routes for the nascent competitor, e.g. have other firms made an offer for the target; does it have alternative financing routes, including through an IPO; and what is the evidence from internal financial forecasts of the target, and from past and future external funding rounds? This evidence is particularly relevant for nascent competitors

⁶⁶⁷ We discuss potential non-horizontal effects below, in relation to both entrenchment and foreclosure.

⁶⁶⁸ By analogy, US jurisprudence allows for a similar approach under Section 2 of the Sherman Act (*United States v. Microsoft Corp.*, 253 F.3d 34, 79 (D.C. Cir. 2001) (en banc) (per curiam): 'It would be inimical to the purpose of the Sherman Act to allow monopolists free reign to squash nascent, albeit unproven, competitors at will').

⁶⁶⁹ The evidence listed here, while generally informative if available, is not always necessary or sufficient to either prove or disprove the relevant effects considered.

with limited resources and significant investment needs, to establish whether the nascent competitor would pose a future competitive constraint on the acquiring firm absent the merger. It can also be relevant if the nascent competitor possesses unique complementary assets that an alternative buyer could use to compete against an incumbent with market power.

20.4.2 Box 2 effects

- 20.110 Acquisitions of nascent competitors can give rise to beneficial effects under Box 2 of our framework.
- 20.111 With respect to spillovers (Box 2A), these may be present if the acquirer and the target are able to capture spillovers from each other. In particular, the acquirer may be in a position to benefit from spillovers from the smaller target, which would reduce the target's innovation incentives in the counterfactual. The converse can also hold, if the nascent competitor already has sufficient capabilities to benefit from spillovers from the incumbent firm, and there are few other rivals with similar capabilities.
- 20.112 With respect to dynamic synergies under Box 2B, these will mostly rest on the existence of complementarities between the acquirer and the target, and the potential for the nascent competitor to benefit from the incumbent's stronger position in the market.
- 20.113 For example, in the case of innovation, post-merger successful innovation from the target could be shared with the product(s) of the incumbent, reaching more users. This would have both positive diffusion and incentive effects (as set out in Chapter 16). The feasibility of knowledge sharing will depend on the specific characteristics of the two products and on the innovation project of the nascent competitor (e.g. can it be effectively transferred and applied to the product and user base of the incumbent?).
- 20.114 Similarly, post-merger the nascent competitor could benefit from the likely superior R&D, distribution, marketing, and financial capabilities of the incumbent. This may increase the probability, effectiveness, and reach of any innovation introduced by the nascent competitor.
- 20.115 A key consideration in the assessment of these beneficial effects revolves around *merger-specificity*. For the specific case at hand, a key question is whether the nascent competitor has access to alternative means to capture similar synergies as those that it could obtain via the merger. Here, too, the existence of alternative buyers and less-anticompetitive alternative means to finance the nascent competitor is likely to be important (see discussion in Chapter 16). For example, an alternative industrial buyer, perhaps in an adjacent market to the incumbent's, may have sufficient R&D and other assets to enable the innovation of the nascent competitor to effectively reach the market. In this case, some or all of the potential benefits under Box 2 may not be merger-specific, as discussed in Chapter 16.

20.4.3 Entrenchment effects

- 20.116 Potential entrenchment effects (described in Category E, see Section 20.6) would also need to be considered in the case of acquisition of a nascent competitor by an incumbent with significant market power. These may amplify dynamic concerns under Box 1, and need to be traded off against potential beneficial effects under Box 2.

Table 20.4 Category C: acquisitions of nascent competitors

	Principles	Evidence
Box 1	Loss of future price competition and/or loss of innovation/investment/entry	Significant market power of acquirer Target has realistic prospects of challenging core market of incumbent (ability/incentives)
Box 2A	Spillovers between merging firms, and spillovers to non-merging firms (for the case of innovation)	Past evidence of risk of spillovers constraining incentives IP rights regime
Box 2B	Voluntary technology transfers, especially from nascent competitor to incumbent (for the case of innovation) Assets of incumbent (e.g. R&D, finance, distribution) can make products of target more effective	Technological compatibility between products Merger specificity of potential benefits (including possibility of alternative acquirers or sources of finance)
Entrenchment	Adverse effect on investment by future competitors	<i>See below (Category E)</i>

Source: Oxera.

20.4.4 Case examples

20.117 Past competition or merger cases involving nascent competitors include Facebook/Instagram/WhatsApp (2020, US), Mallinckrodt (2017, US), CDK Global/Auto/Mate (2018, US), and Facebook/GIPHY (2021, UK). An in-depth description of the last case is included in Annex A2.1.

- **Case example: CDK Global/Auto/Mate (2018, US).** The FTC found that CDK Global's acquisition of Auto/Mate was driven by a pre-emptive motive to eliminate a growing competitive threat. The FTC alleged that when Auto/Mate was put up for sale in 2016, CDK Global concluded it could eliminate a strong current competitor that was threatening to become an even more disruptive rival. CDK Global's primary concern was that if Auto/Mate fell into the hands of a well-financed buyer willing to invest additional resources, it would become an even more effective competitor. According to the FTC, this defensive motivation was evidenced by CDK Global offering a price well in excess of its original standalone valuation of Auto/Mate, with CDK Global's post-merger plans revealing intentions to substantially downgrade features and service, raise prices, and prevent larger customers from migrating. The FTC found that the acquisition would eliminate Auto/Mate as an innovative disruptor that had consistently expanded its customer base through aggressive pricing and differentiated products, placing pressure on CDK Global's pricing and margins—particularly as Auto/Mate developed features attractive to larger dealerships, representing an increasing threat to CDK Global's customer base.
- **Case example: Mallinckrodt (2017, US).** The FTC found that Questcor (subsequently acquired by Mallinckrodt) illegally maintained its monopoly over

ACTH drugs⁶⁷⁰ in the US by acquiring exclusive rights to Synacthen, a nascent competitive threat, in 2013.⁶⁷¹ Questcor owned Acthar, the only ACTH product sold in the US, and had demonstrated substantial market power through repeated price increases and significant profitability. Despite substantial uncertainty about regulatory approval in the US, Synacthen (already approved in Canada and Europe) posed a credible competitive threat. The FTC determined that Questcor's late entry into the bidding process and successful outbidding of three rivals was a defensive strategy to pre-emptively eliminate competition and protect its strong market position.

- **Case example: Facebook/Instagram/WhatsApp (2020, US).** Facebook/Instagram and Facebook/WhatsApp were both approved unconditionally by competition agencies in Europe (Facebook/Instagram by the UK OFT in August 2012; and Facebook/WhatsApp by the European Commission in October 2014). The FTC subsequently sued Facebook (now Meta) in December 2020 for illegal monopolisation, alleging that both acquisitions were part of a systematic strategy to eliminate threats to Facebook's personal social networking monopoly, which is protected by high barriers to entry including strong network effects. The FTC's theory focused on the threat from the 'shift to mobile,' where Facebook had 'enormous disadvantages' compared to 'mobile first' entrants like Instagram, which was seen as potentially evolving into a social networking competitor or being used by others (like Apple) as a foothold to challenge Facebook. Internal documents revealed Facebook's defensive motivation, stating that the purpose of the acquisition would be 'neutralizing a potential competitor' while acknowledging that Instagram could 'easily add pieces of their service that copy what we're doing now,' posing a 'real issue'. Similarly, in the case of WhatsApp, Facebook was concerned about the competitive threat from mobile messengers, because of a risk of a move into core social networking product areas. Facebook was also concerned that another company could buy WhatsApp and develop it into a social network service. In November 2025, the US District Court for the District of Columbia ruled in favour of Meta Platforms in the case brought by the FTC, largely because the FTC had not shown that Facebook continued to hold market power. The FTC appealed the judgment in January 2026.
- **Case example: Facebook/GIPHY (2021, UK).** In May 2020, Facebook (now Meta) acquired GIPHY, the leading provider of searchable GIF libraries to social media platforms. GIPHY had developed an innovative advertising service allowing brands to promote GIFs within users' search results. As set out in its 2021 Phase II final report on the transaction, the CMA was concerned by the removal of GIPHY as an innovative potential competitor to Facebook in display advertising, valuing its innovation efforts despite modest current revenues and uncertain commercial success. The CMA identified innovation concerns in display advertising, including the risk of a reduction in Facebook's incentives to innovate due to the elimination of innovation competition between the parties, and the elimination of innovation by GIPHY (noting that Facebook discontinued GIPHY's innovative advertising service upon completion of the merger). In addition, the CMA identified dynamic feedback-loop effects where, by losing a source of advertising revenue through partnerships with GIPHY, rival social media platforms would have reduced their investment to attract users in social media, thus lessening their competitive pressure on Facebook. The CMA was also concerned that Facebook could degrade social media rivals' access to GIPHY's

⁶⁷⁰ ACTH refers to adrenocorticotrophic hormone, a hormone that stimulates the adrenal glands to make cortisol.

⁶⁷¹ While this case was brought under Section 2 of the Sherman Act, it effectively describes the acquisition of a nascent competitor which could have been the subject of merger control.

library through input foreclosure, reducing their attractiveness to social media users and their competitiveness in social media. The CMA identified dynamic effects on innovation in social media as a result of foreclosure, with both social media rivals and Facebook facing reduced incentives to innovate (see discussion in next Section). The CMA also pointed to the feedback loop effect of rivals' weakened competitive position in social media into display advertising, which would have exacerbated the horizontal effects in display advertising.

20.5 Category D: dynamic foreclosure

20.118 Cases in Category D capture the non-horizontal effects articulated in Chapter 18, including both anticompetitive effects under Box 3 and procompetitive effects under Box 4.

20.119 The analysis and relevant evidence on *ability* and *incentives* to engage in dynamic foreclosure is captured in the specific chapter on dynamic foreclosure in our conceptual framework (Chapter 18), and hence we do not repeat here. We present instead two recent case examples.

20.120 Recent examples of dynamic foreclosure concerns are the Illumina/GRAIL (2022, EU; 2023, US) case (see also detailed description in Annex A2.8) and the second theory of harm in Facebook/GIPHY (2021, UK) (see also detailed description in Annex A2.1). Other recent conglomerate merger cases raised dynamic effects as part of broader foreclosure concerns (e.g. Microsoft/LinkedIn (2016, EU), Qualcomm/NXP Semiconductors (2018, EU), Arianespace/ASL (2016, EU), and Meta/Kustomer (2022, EU)).

- **Case example: Illumina/GRAIL (2023, US).**⁶⁷² The transaction involved the vertical integration of Illumina, the leading supplier of Next Generation Sequencing systems for genetic and genomic analysis, with GRAIL, a customer of Illumina using NGS systems to develop multi-cancer early detection (MCED) tests. The market for MCED tests was an emerging market at the time of the merger, and was projected to grow substantially in the coming years. The concern raised by the FTC (as well as the European Commission in a parallel review) was that the merger would have enabled and incentivised Illumina to foreclose GRAIL's potential rival providers of MCED tests from access to its NGS technology, an essential input they needed to develop and eventually market their own tests. The FTC's assessment of the merger centred on harm to innovation competition among firms that, with the exception of GRAIL, were not yet present in the market but were engaged in R&D in relation to MCED tests for subsequent commercial sale. By foreclosing rival suppliers of MCED tests, Illumina would have stifled innovation competition and increased barriers to entry in the MCED market, creating an unlevel playing field and ensuring that GRAIL would ultimately win the innovation race. The FTC emphasised that the harm to R&D competition was current and immediate. Reduced innovation, lower quality, or lower availability of competing MCED tests would harm patients, depriving them of the benefit of multiple competitors pursuing diverse scientific approaches in an effort to design the best-performing product. The merging parties' efficiency claims (including dynamic efficiencies in R&D and commercialisation) were dismissed.

⁶⁷² The European Commission investigated the transaction in parallel but its prohibition decision (2022) was eventually annulled by the European Court of Justice in 2024, which ruled that the Commission lacked jurisdiction. The prohibition decision was not published, and we therefore only summarise the FTC investigation here.

- **Case example: Facebook/GIPHY (2021, UK).** In its investigation of Facebook’s (now Meta) acquisition of GIPHY, alongside the concerns related to the removal of an innovative nascent competitor to Facebook in display advertising, the CMA assessed vertical effects in social media, through input foreclosure of rival social media platforms. It articulated its foreclosure theory of harm with significant focus on innovation, in relation to both the foreclosure mechanisms and the dynamic effects of the merger. The concern was that Facebook could leverage GIPHY’s strong position in GIF libraries to foreclose rival social media platforms, including by limiting their access to GIPHY’s future innovations, reprioritising innovation to its rivals’ disadvantage, or imposing onerous data-sharing requirements. The vertical effects in social media included dynamic effects on innovation. First, rivals would have more limited opportunities to innovate or improve platform features that use GIFs as an input. Second, by gaining access to rivals’ data via GIPHY, Facebook would be able to spot emerging competitive threats or trends in social media earlier and narrow its innovation efforts only to target the areas of greater threat.

20.6 Category E: entrenchment

- 20.121 Category E cases cover entrenchment effects described in Chapter 19. As discussed, entrenchment effects may apply concurrently with potential horizontal and non-horizontal effects applicable to the previous categories (A to D).
- 20.122 Chapter 19 discusses the examples of potential entrenchment effects from the academic literature, and the market conditions that are more likely to give rise to these effects. As for the case of foreclosure, we do not repeat that discussion here, as it maps directly to our categorisation of mergers with dynamic effects.
- 20.123 A prominent recent example of Category E is the ‘ecosystem’ theory of harm set out in Booking Holdings/eTraveli Group.
- **Case example: Booking Holdings/eTraveli Group (2023, EU).** This case related to the acquisition by Booking (the largest platform in Europe for online hotel bookings) of eTraveli (a provider of online flight bookings). The Commission found that online flight bookings were an important customer-acquisition channel for online hotel bookings. It alleged that the merger would have allowed Booking to increase its market position in online hotel bookings, through the addition of online flight bookings to its integrated service and cross-selling of accommodation to flight customers (reinforced by the presence of customer inertia and the strength of Booking’s brand). The Commission concluded that by increasing traffic and sales on Booking’s platform, the transaction would have reinforced network effects and would have increased barriers to entry and expansion in the market for online hotel bookings, strengthening the dominant position of Booking to the detriment of hotels and possibly end-customers. The Commission did not accept the efficiency claims made by the merging party (e.g. the elimination of double-marginalisation on online flights).

PART D SECTORAL ANALYSIS

21 Introduction and overview of the sectoral analysis

- 21.1 Part D of the main report presents a sectoral evaluation of three strategic sectors. Its purpose is to provide a sector-specific assessment of the current challenges and bottlenecks to investment and innovation in the EU, and of how different policy tools including merger control can help address such challenges.⁶⁷³
- 21.2 Each sectoral analysis Chapter is structured around the following three sections:
- a high-level overview of the sector, including investment and innovation trends in Europe, and benchmarking these against international comparators;
 - a discussion of the main policy tools to promote innovation and investment, as considered in existing reports and other relevant publications;
 - a third section covering merger control, including both a conceptual discussion of its role in influencing investment and innovation in the sector, and a practical overview of the indicators and analyses relevant for merger control in each sector. This third section builds on the results and insights covered in Part B and Part C of the Study, and makes reference to the case studies in Annex A2 where relevant.
- 21.3 The sectoral analysis covers the following three sectors: biopharmaceuticals, quantum computing (QC), and clean technologies. These sectors have been chosen because of their strategic importance, their emerging or rapidly evolving nature, the relevance of investment and innovation in these sectors, and (potential) M&A activity.

Biopharmaceuticals

- 21.4 Biopharmaceuticals are medicinal products derived from biological sources (such as living cells or organisms) using biotechnology, including vaccines, monoclonal antibodies,⁶⁷⁴ and gene therapies.⁶⁷⁵ Biopharmaceuticals have a growing prominence within the pharmaceutical industry—itself a key sector of the global economy.
- 21.5 Biopharmaceuticals have emerged as a leading driver of both innovation and commercial growth, spurred by the unique therapeutic potential of biologics and sustained R&D investment. This dynamism has fuelled considerable M&A activity, particularly involving small innovative companies which play a central role in driving sectoral innovation. Within this framework, merger control has the potential to promote innovation-enhancing acquisitions (e.g. supporting development and commercialisation), while deterring or preventing acquisitions that, for example, suppress future potential competitors.

Quantum computing

⁶⁷³ This Part follows from the tender specifications—European Commission (2025), '[Study on dynamic merger effects](#)', accessed 21 July 2026.

⁶⁷⁴ Monoclonal antibodies are lab-made proteins designed to mimic the body's natural immune system and seek out and bind to very specific targets (such as abnormal proteins, cancer cells, or viruses) to neutralise them, block their growth, or flag them for destruction by the immune system. They are termed 'monoclonal' because they are identical genetic clones derived from a single parent cell.

⁶⁷⁵ Gene therapies are treatments that modify, replace, add, or silence genes within a patient's cells to treat, prevent, or potentially cure disease by addressing its underlying genetic cause.

- 21.6 Quantum computing exploits quantum mechanical phenomena such as superposition and entanglement⁶⁷⁶ to process information much faster than classical computers for certain complex problems. Although the technology remains at an early stage of development, its disruptive potential is considerable and could affect a wide range of sectors. While the sector depends on significant R&D investment, it stands to potentially generate substantial value across the entire economy.
- 21.7 As the market gradually matures, mergers are beginning to materialise within the industry, and it can reasonably be expected that consolidation trends will occur over time. While various policy tools may be used to support the development of this nascent industry, merger control may have an important prospective role to play in stimulating or protecting innovation trends and the overall competitiveness of the industry. Particular attention is warranted with respect to the innovation potential of small, innovative players and access to key inputs by existing players and potential entrants across markets, thus reducing risks of entrenchment and foreclosure.

Clean technologies

- 21.8 Clean technologies include a broad range of products and processes aimed at supporting decarbonisation, improving resource efficiency, and enabling the transition to a more sustainable economy. This sector is of growing strategic importance given its role in meeting climate and energy objectives, its rapid pace of development, and its increasing economic significance across a wide range of industries.
- 21.9 Clean technologies are characterised by strong innovation dynamics, significant investment needs, and growing global competitive pressures. There is increasing policy attention to the need to accelerate the employment of green technologies, and to develop renewable energy and manufacturing.⁶⁷⁷ Against this background, merger control may play an important role by allowing consolidation that helps firms capture economies of scale, preventing acquisitions that eliminate emerging innovators, scrutinising vertical mergers that could restrict access to critical inputs, and taking account of sustainability-related factors in the competitive assessment, where relevant.

⁶⁷⁶ Superposition is the ability of a qubit (the basic unit of information in a quantum computer) to represent multiple possible values at the same time rather than only 0 or 1 (as in a standard bit); entanglement is a quantum phenomenon in which two or more qubits become correlated such that the state of one is intrinsically linked to the state of the others, regardless of the distance between them.

⁶⁷⁷ Ribera, T. (2025), '[Speech by Executive Vice-President Teresa Ribera at the 2025 CRA Brussels Conference](#)', accessed 21 July 2026.

22 Biopharmaceuticals

Key points

- Biopharmaceuticals are the EU's fastest-growing pharmaceutical subsector, with the European market valued at about €160 billion in 2025 (around 36% of the total pharmaceutical sector) and growing significantly faster than conventional small-molecule drugs. However, the EU faces widening productivity and R&D investment gaps compared to the US and China.
- Innovation activity has shifted decisively from traditional pharmaceutical incumbents towards small emerging biopharmaceutical companies, whose share of trial launches rose from 29% to 59% between 2011 and 2021, making acquisitions of small innovative companies (such as startups) by large incumbents a market dynamic with potential implications for merger policy.
- Key policy tools to boost the competitiveness of the EU biopharmaceutical sector—improved financing, streamlined regulation, and stronger innovation clusters—are well established. Merger control remains a tool which is not prominent in the mainstream policy debate.
- Merger control can nevertheless have an important impact on biopharmaceutical innovation: acquisitions of complementary small innovative companies can accelerate pipeline development, while killer acquisitions and large-to-large consolidations risk suppressing R&D diversity, inventor productivity, and nascent competition.
- Competition authorities have reviewed several biopharmaceutical mergers, focusing on quantitative and qualitative evidence around pipeline overlaps and closeness of product and innovation competition.

22.1 Background and market developments

22.1.1 Industry background

- 22.1 Within pharmaceuticals,⁶⁷⁸ biopharmaceuticals represent one of the most innovation-intensive and rapidly expanding subsectors. Unlike conventional small-molecule drugs, which are chemically synthesised, biopharmaceuticals are large, complex molecules derived from living biological sources such as cells, proteins, and nucleic acids. What makes biopharmaceutical drugs particularly significant is their ability to target disease mechanisms with a precision that traditional pharmaceuticals cannot match, enabling treatments for conditions that were previously untreatable or poorly managed. This therapeutic potential is the primary driver behind the rapid increase in R&D investment: global biopharmaceutical sales are currently growing approximately three times faster than sales of small-molecule

⁶⁷⁸ The pharmaceutical industry is a major sector of the global economy. In the EU specifically, Gross Value Added (GVA) reached €311 billion in 2022 through direct, indirect, and induced contributions—an increase of 46% compared to 2016. Focusing on direct GVA alone, this rose from €107.6 billion in 2016 to €162.7 billion in 2022 (+51%). The pharmaceutical sector also supports 2.3 million jobs across the EU economy, the majority of which are highly skilled and well remunerated (Source: PwC (2024), 'Economic footprint of the pharmaceutical industry in Europe', November, pp 3–13). Indeed, 15% of the sector's direct workforce is engaged in R&D (Source: Draghi (2024), Part B, p. 192).

drugs,⁶⁷⁹ and spending on biologic drugs has nearly doubled since 2015 while spending on small-molecule drugs has fallen in both nominal and real terms.⁶⁸⁰

- 22.2 In Europe, the biopharmaceutical market was valued at €159.7 billion in 2025 and is projected to reach €166.4 billion in 2026⁶⁸¹—representing approximately 36% of the broader European pharmaceutical market, which stood at €437.9 billion⁶⁸² in 2025.⁶⁸³ The European biopharmaceutical segment is forecast to grow at a CAGR⁶⁸⁴ of 6.8% through to 2033,⁶⁸⁵ outpacing the broader pharmaceutical sector (CAGR of 5.9%).⁶⁸⁶

22.1.2 R&D intensity and innovation in biopharmaceuticals

- 22.3 R&D is fundamental to the biopharmaceutical industry, which is considerably more R&D intensive than comparable sectors. In 2019, OECD countries' pharmaceutical industry (including biopharmaceutical companies) invested over 30% of its Gross Value Added (GVA) in R&D—significantly more than other innovation-driven industries such as electronics and optical (23.5%), and aerospace manufacturing (14.7%).⁶⁸⁷
- 22.4 The pharmaceutical sector has undergone a structural shift in innovation leadership. Whereas large incumbents dominated R&D investment and innovation pipelines historically, their share of trial launches more than halved from 59% in 2011 to 28% by 2021.⁶⁸⁸ Conversely, emerging biopharmaceutical companies increased their share from 29% in 2011 to 59% in 2021.⁶⁸⁹ This significant shift underscores the critical and expanding role that emerging biopharmaceutical companies play as drivers of innovation in the sector, with implications for industry consolidation patterns and merger dynamics. Given the increasing importance of small innovative companies in the biopharmaceutical innovation landscape, the strategic rationale for large incumbents to access external innovation sources through acquisitions has intensified.
- 22.5 Table 22.1 below provides an overview of the main players active in the biopharmaceuticals space in 2026.

Table 22.1 Overview of the main players in biopharmaceuticals in 2026

Rank	Name	Country	Status
1	Roche	Switzerland	Incumbent

⁶⁷⁹ Drug Discovery & Development (2026), '[Moving beyond the binary: how the convergence of small and large molecules is reshaping pharmaceutical R&D](#)', accessed 21 July 2026.

⁶⁸⁰ The Foundation for Research on Equal Opportunity (2024), '[No Contest: Small Pharma Innovates Better than Big Pharma](#)', accessed 21 July 2026.

⁶⁸¹ Fortune Business Insights (2026), '[Biopharmaceutical Market](#)', accessed 21 July 2026.

⁶⁸² Converted from \$ to € using the ECB average exchange rate for 2025.

⁶⁸³ Healthcare Web Wire (2025), '[Europe Pharmaceutical Market Report Growth 2025-2034: CAGR, Leaders & Strategic Shifts](#)', accessed 21 July 2026.

⁶⁸⁴ Compound Annual Growth Rate.

⁶⁸⁵ Grand View Horizon (2026), '[Europe Biopharmaceutical Market Size & Outlook, 2026-2033](#)', accessed 21 July 2026.

⁶⁸⁶ Grand View Horizon (2026) (full reference in footnote 685). Note that the pharmaceutical industry CAGR includes biopharmaceuticals, so the difference would be larger by comparing biopharmaceuticals and pharmaceuticals excluding biopharmaceuticals.

⁶⁸⁷ OECD (2023), '[Health at a Glance 2023](#)', accessed 21 July 2026.

⁶⁸⁸ Draghi (2024), Part B, p. 192.

⁶⁸⁹ Draghi (2024), Part B, p. 192.

Rank	Name	Country	Status
2	Novo Nordisk	Denmark	Incumbent
3	AstraZeneca	UK	Incumbent
4	Regeneron	US	Incumbent
5	Vertex Pharmaceuticals	US	Incumbent
6	Amgen	US	Incumbent
7	Gilead Sciences	US	Incumbent
8	Recursion Pharmaceuticals	US	Challenger
9	Insilico Medicine	Hong Kong	Challenger
10	Isomorphic Labs	UK	Challenger
11	Generate:Biomedicines	US	Challenger
12	XtalPi	China	Challenger

Note: the table differs from the original one in the source as companies falling outside biopharmaceuticals have been excluded (e.g. diagnostics, lab technologies). The company status represents an Oxera elaboration on the original source: incumbents are defined as established commercial-stage pharma companies with revenues typically above €1 billion, whereas challengers are defined as less mature, specialised companies with significantly lower revenues (typically not exceeding €100 million).

Source: Prezent AI (2026), '[Top 30 Biotech Companies Shaping the Industry in 2026](#)', accessed 21 July 2026.

22.1.3 Productivity, sectoral importance, and challenges

- 22.6 Productivity in the pharmaceutical sector (including biopharmaceuticals) is relatively high: in Europe, GVA per hour worked is approximately three times that of the European economy overall,⁶⁹⁰ a pattern consistent across EU member states. Moreover, productivity growth in the manufacture of pharmaceuticals outpaces that of other sectors.⁶⁹¹
- 22.7 Against this backdrop, the pharmaceutical sector is an industry capable of generating substantial value added and growth in Europe. It already represents around 12% of the EU's exports, and with the global market for medicines (including biopharmaceutical products) projected to expand in the coming years, its prominence within the EU industrial landscape is set to increase even further.⁶⁹² In 2020, the biopharmaceutical sector accounted for 5% of value added to the EU economy from all manufacturing, and over 20% for Belgium and Denmark.⁶⁹³
- 22.8 It is also a sector of geostrategic importance, as evidenced by the Covid-19 pandemic, which showed the importance of an agile and scalable biopharmaceutical industry to produce vaccines.
- 22.9 However, the overall EU pharmaceutical industry faces challenges, as productivity gaps with major global competitors are significant. In 2022, GVA per worker in the US biopharmaceutical industry exceeded €381,779 (compared with approximately

⁶⁹⁰ Europe here includes EU27 plus the UK, Norway, and Switzerland.

⁶⁹¹ PwC (2024), p. 15 (full reference in footnote 678). Note that while figures for biopharmaceuticals specifically are not available, it is reasonable to assume that they do not differ materially from overall pharmaceuticals.

⁶⁹² Precedence Research (2025), '[Pharmaceutical Market Size, Share and Trends 2025 to 2034](#)', accessed 21 July 2026.

⁶⁹³ Draghi (2024), Part B, p. 187.

€69,300 across all US industries),⁶⁹⁴ while it stood at around €197,200 in the EU27 biopharmaceutical industry (compared with an average of around €60,000 across all sectors).⁶⁹⁵ These figures illustrate that, while US productivity exceeds EU27 productivity by around 15% across industries as a whole, the gap widens considerably in biopharmaceuticals, where US productivity is almost twice that of the EU27.⁶⁹⁶ Besides, data on pharmaceutical R&D investments in the US and Europe (EU plus Norway, Switzerland, and the UK) shows that while R&D investments were comparable in value between the US and Europe in 2014, they have since then diverged substantially: the US increased R&D investments from approximately €30 billion in 2010 to €71.5 billion in 2022, while Europe only increased its investment levels from €27.8 billion to €46.2 billion over the same period.⁶⁹⁷ These diverging trajectories are also reflected in global biopharmaceutical market shares, where North America accounted for more than 50% of the market in 2025, almost twice as large as Europe's share (26%).⁶⁹⁸ While many of the current incumbents in the biopharmaceutical industry are European (see Table 22.1), Europe is lagging behind the US in terms of productivity and R&D investments, indicating that R&D investments and productivity are mainly driven by smaller companies (as also shown in Subsection 22.1.2). This has potential policy implications for merger control, which are discussed in Section 22.3.

- 22.10 China has also emerged as a major global competitor. Starting from a very low base, it has achieved significant growth with an average increase in annual pharmaceutical R&D expenditure of 20.7% between 2010 and 2022. From 2015 to 2019, expenditure rose from €4.3 billion to €12.6 billion, a higher growth rate than that of any OECD country.⁶⁹⁹ China's investments are also efficient: while its pharmaceutical R&D spending represented approximately 25% of European investment levels in 2022, China achieved a number of New Molecular Entities (NMEs) comparable to Europe's in 2022, and exceeded Europe's NME output in 2023.⁷⁰⁰

22.2 Policy tools

- 22.11 Several policy papers and reports (including the Draghi Report—see Box 22.1) have identified key tools which can help unlock competitiveness and growth for the EU biopharmaceutical industry. Broadly, these include: improving the financing environment, simplifying regulatory processes, and fostering human capital and development of innovation clusters.

⁶⁹⁴ TEconomy Partners and PhRMA (2024), 'The Economic Impact of the U.S. Biopharmaceutical Industry: 2022 National and State Estimates', May, p. 9. Converted from \$ to € using the ECB average exchange rate for 2022.

⁶⁹⁵ PwC (2024), p. 15 (full reference in footnote 678).

⁶⁹⁶ We note that these are nominal GDP levels, which do not correct for differences in price level or potential market power between the EU and US.

⁶⁹⁷ PwC (2024), p. 17 (full reference in footnote 678). According to the pharmaceutical industry, 50% of European R&D investment would be at risk of being diverted to the US, see EFPIA (2025), '[Pharma CEOs alert President von der Leyen to risk of exodus to the U.S.](#)', accessed 21 July 2026.

⁶⁹⁸ Fortune Business Insights (2026), '[Biopharmaceuticals Market Size, Share & Forecast, 2026–2034](#)', accessed 21 July 2026.

⁶⁹⁹ OECD (2023) (full reference in footnote 687).

⁷⁰⁰ PwC (2024), p. 17 (full reference in footnote 678). Note that NMEs are new drugs with a novel active ingredient which are marketed for the first time: their discovery represents a key indicator of R&D productivity.



Box 22.1 Insights from the Draghi Report on biopharmaceuticals

The Draghi Report calls for a significant increase in public R&D investment in pharmaceuticals, including support for world-class innovation hubs in life sciences focused on advanced therapy medicinal products (ATMPs). It also recommends mobilising private R&D investment and bolstering the supporting environment, for instance through competitive public-private partnership funding calls targeting startups and research teams active in disruptive innovation.

The report also recommends streamlining the set-up and management of multi-country clinical trials across the EU, including the introduction of binding EU-level coordination mechanisms between national ethics committees and model templates for trial administration. It further calls for expediting market access through coordinated action by medicines agencies, health technology assessment authorities, and public payers—with the aim of aligning guidance on clinical evidence requirements, pricing, and reimbursement across Member States.

The report highlights the need for a skilled workforce equipped with expertise spanning both pharmaceuticals and emerging technologies such as AI—including data scientists, bioinformatics experts, and AI specialists—as a critical enabler of sectoral competitiveness. It also recommends supporting a number of world-class innovation hubs in life sciences, particularly for ATMPs, to concentrate talent, R&D capacity, and public investment in strategically significant clusters across the EU.

Source: Draghi (2024), Part B, Section 1, Chapter 9; and Section 2 Chapter 1.

- 22.12 The public and private R&D funding bases are both smaller and more fragmented in the EU than in the US. Europe's share of global pharmaceutical R&D investment has declined from 37% in 2010 to 32% in 2020,⁷⁰¹ while US biotechnology companies attract approximately five times the venture capital of their European counterparts. While this gap has been partly attributed to structural differences between US and EU pension systems,⁷⁰² several avenues exist to address the shortfall, notably through expanding the EU venture capital ecosystem, increasing banking sector financing capacity, and adopting a more coherent approach to public funding.
- 22.13 On regulation, bringing innovative products to market in the EU remains lengthier and more complex than in other jurisdictions. The review speed of the European Medicines Agency (EMA) remains slightly slower than that of the Food and Drug Administration (FDA),⁷⁰³ and post-approval procedures for national pricing and reimbursement add further delays. Streamlining multi-country clinical trial coordination, harmonising approval and reimbursement timelines across member states, and strengthening the scientific dialogue between regulators and industry—an area where the FDA currently outperforms the EMA—are key tools to support European competitiveness. Stable incentive frameworks and robust intellectual property protection are equally important as they improve the stability of the operating environment of firms.⁷⁰⁴
- 22.14 On skills and clusters, several reports recommend increasing the focus of EU investment towards centres of excellence and innovation, which should be identified

⁷⁰¹ EFPIA (2025), '[A comparative analysis of biopharmaceutical strategies in 10 countries](#)', November, accessed 21 July 2026.

⁷⁰² Amsterdam Data Collective and Danish Bio (2025), 'Bridging the Gap – Transforming EU Biotech Policy', June, p. 25.

⁷⁰³ Han and Kesselheim (2025).

⁷⁰⁴ Draghi (2024), Part B, p. 199.

and consolidated with EU public support, with the goal of developing a limited number of world-class innovation hubs. EU innovation hubs remain fragmented along national lines, preventing the emergence of world-class ecosystems. EFPIA's comparative analysis of biopharmaceutical strategies across ten leading countries highlights talent pools as a pivotal pillar for the industry, with skill shortages identified as a material constraint on growth. Concentrating EU investment into a limited number of centres of excellence—facilitating cross-industry, government, and academic collaboration—has been recommended as the most effective means of achieving critical mass.⁷⁰⁵

22.3 Merger control

- 22.15 While merger control does not appear as a primary driver for unlocking the competitiveness of the EU biopharmaceutical industry in the policy debate, it still plays a relevant role. Merger control can impact innovation in this sector by discouraging or preventing mergers and acquisitions that are harmful to innovation and encouraging those that are beneficial. Most of the literature on pharmaceutical mergers has focused on their innovation effects. Merger control also matters for future product market competition, in line with our conceptual framework. Additionally, other parameters of competition may be at work too (e.g. supply chain resilience).
- 22.16 This Section discusses three specific mechanisms through which merger control can support innovation and competitiveness in the sector.

22.3.1 Accounting for the role of small innovative entrants

- 22.17 Small innovative companies (referred to as 'startups' by some in the literature) are a key driver of biopharmaceutical innovation. As detailed in Chapter 7, the academic literature identifies two potential effects of merger control on small innovative entrants.
- 22.18 Several studies document 'killer acquisitions' in the overall pharmaceutical sector—whereby large incumbents acquire nascent competitors primarily to discontinue their pipeline projects and neutralise future product market competition. Research finds that mergers involving overlapping drug development projects carry a 23% lower probability of continued development compared with non-overlapping ones, that at least 5–7% of acquisitions qualify as killer acquisitions,⁷⁰⁶ and that a significant share of transactions involving substitutable projects show no clinical justification for post-deal discontinuation.⁷⁰⁷ It also shows that small firms tend to experience sharp R&D declines in the first post-merger year.⁷⁰⁸
- 22.19 However, the acquisition of small innovative entrants can enhance innovation when the target of the acquisition lacks the financial or operational capacity to independently commercialise its pipeline projects. In such cases, larger firms can provide the regulatory expertise, capital, and operational scale needed to bring projects to market—and early evidence suggests that in 79% of acquisitions involving small targets, the expected rate of discovery of New Molecular Entities

⁷⁰⁵ EFPIA (2025) (full reference in footnote 701).

⁷⁰⁶ Cunningham, Ederer, and Ma (2021).

⁷⁰⁷ Lear (2024).

⁷⁰⁸ Danzon, Epstein, and Nicholson (2007).

increased,⁷⁰⁹ with acquired projects in non-oncology markets showing higher late-stage progression.⁷¹⁰

22.3.2 Promoting synergies and complementarities but maintaining parallel research paths

- 22.20 Academic research focusing on the pharmaceutical industry identifies patterns in the way M&As affect innovation. Firms can merge to access external innovation and combine complementary capabilities—a strategy particularly valuable for companies facing declining pipelines, with evidence of portfolio improvements and greater post-merger innovation, for example where prior partnerships existed⁷¹¹ and where technological synergies were present.⁷¹²
- 22.21 However, mergers between incumbents may reduce the diversity of parallel research pathways that foster biopharmaceutical innovation,⁷¹³ leading to a decline in patent filings.⁷¹⁴

22.3.3 Taking human capital outcomes into consideration in merger analysis

- 22.22 As noted in Section 22.2, talent pools and workers' skills are crucial to the biopharmaceutical industry. While promoting hubs and centres of excellence represents the primary avenue for providing the workforce the sector requires, merger policy may also have a role to play. Inventors are the underlying drivers of innovation, and their optimal allocation across biopharmaceutical firms may be affected by mergers. Indeed, research suggests that inventors' productivity may improve or decline with acquisitions, depending on the R&D focus of the merging parties.
- 22.23 When mergers diminish inventors' social status and centrality,⁷¹⁵ or where inventors' expertise is more specific to the target company (and hence less adaptable),⁷¹⁶ inventors are more likely to leave the firm, become inactive,⁷¹⁷ or experience substantial productivity decline following a merger.⁷¹⁸ A study by Ornaghi and Cassi drawing on European Patent Office (EPO) patent data identifies both a higher rate of inventors exiting the innovation market altogether and a higher rate of inventors departing from their current company to join other research laboratories. Both of these movements are also associated with a reduction in patenting activity for each inventor who exits the market or departs the firm. Furthermore, even inventors who remain within the firm are not insulated from these effects, and likewise experience a decline in productivity.⁷¹⁹
- 22.24 Just as acquisitions of small firms involving overlapping pipeline projects appear riskier than acquisitions involving complementary ones in terms of innovation outcomes, acquisitions of firms with similar technological focus are more likely to result in degraded productivity outcomes for inventors. By contrast, inventors with

⁷⁰⁹ Munos (2009).

⁷¹⁰ Wabiszewski (2025).

⁷¹¹ Higgins and Rodriguez (2006).

⁷¹² Malek, Seldeslachts, and Veugelers (2025).

⁷¹³ Danzon et al. (2007), Munos (2009), Comanor and Scherer (2013).

⁷¹⁴ Haucap, Rasch, and Stiebale (2019).

⁷¹⁵ Paruchuri, Nerkar, and Hambrick (2006).

⁷¹⁶ Verginer and Riccaboni (2025).

⁷¹⁷ Munos (2009), Ornaghi and Cassi (2026).

⁷¹⁸ Verginer and Riccaboni (2025), Ornaghi and Cassi (2026).

⁷¹⁹ Ornaghi and Cassi (2026).

skills complementary to the acquiring firm's R&D focus tend to maintain or even enhance their productivity post-acquisition.⁷²⁰

22.3.4 Practical assessment of mergers in the biopharmaceutical sector

- 22.25 Mergers in the biopharmaceutical sector lend themselves to the taxonomy of cases set out in Part C, Chapter 20. It is generally possible to characterise the overlap between merging parties in terms of specific pipeline projects (Category A) and in underlying innovation capabilities (Category B) in pharmaceutical markets. Within Category A, it is also feasible to distinguish between pipeline-to-product overlaps (Category A1) and pipeline-to-pipeline overlaps (Category A2). This can help structure the analysis of concentration and closeness. Depending on the circumstances, the merger may involve a nascent competitor (Category C). Finally, as we review below, cases in the broader biopharmaceutical space (including upstream technologies) can also give rise to foreclosure concerns (Category D).
- 22.26 It is also typically possible to construct a variety of quantitative indicators to capture the competitive significance of merging parties and their competitors, in terms of the underlying innovation capabilities. Building on the academic literature and industry practice, these can include indicators on inputs (R&D investment, R&D facilities, and human capital),⁷²¹ outputs, research intensity,⁷²² or R&D productivity.⁷²³ Output indicators include patent applications,⁷²⁴ citation-weighted patents,⁷²⁵ the number of NMEs newly discovered (where a greater number indicates higher levels of innovation), and Biologics License Applications (BLAs, which are formal requests for authorisation to place a biological product on the market in interstate commerce in the US). It is typically also feasible to document pipeline progression across development phases,⁷²⁶ project termination rates,⁷²⁷ and clinical trial counts, to provide further insight into the nature of R&D in the sector, and on the innovative trajectories of the merging parties and their competitors.⁷²⁸
- 22.27 Competition authorities have reviewed several mergers in the broader biopharmaceutical space in recent years, relying on some of the evidence listed above.⁷²⁹ Table 22.2 provides a high-level summary of the practical assessment of the biopharmaceutical and biotechnology cases discussed in Annex A2.

⁷²⁰ Verginer and Riccaboni (2025).

⁷²¹ Ornaghi (2009), Haucap, Rasch, and Stiebale (2019).

⁷²² Basant and Jaiswal (2022).

⁷²³ Ornaghi (2009).

⁷²⁴ Haucap, Rasch, and Stiebale (2019).

⁷²⁵ Haucap, Rasch, and Stiebale (2019). While patent activity can be a useful indicator of innovation, empirical findings from Beneish, Harvey, Tseng, and Vorst (2022) suggest that it can underestimate total innovation activity by the merging firms, as it fails to capture the impact of unpatented innovation.

⁷²⁶ Grabowski and Kyle (2008), Schutz (2023).

⁷²⁷ Informa Pharma Consulting (2020), Cunningham, Ederer, and Ma (2021).

⁷²⁸ Richman, Mitchell, Vidal, and Schulman (2017), Malek, Seldeslachts, and Veugelers (2025).

⁷²⁹ While the focus of this Section is the biopharmaceutical sector (i.e. therapeutics), case studies drawn from biotechnology—particularly those centred on diagnostics—are also included here given the extensive overlaps between the two fields. Biotechnology firms active in diagnostics and biopharmaceutical ones share a common technological foundation, relying on many of the same core capabilities: genomics, proteomics, assay development, biomarker identification, and precision biology. Beyond shared technology, these sectors are closely integrated within the same value chain, as diagnostics tools are increasingly essential to biopharmaceutical drug development, enabling patient stratification, companion diagnostic co-development, and clinical trial design. The relationship

Table 22.2 Indicators used in previous biopharmaceutical cases

Merging parties	Categorisation from Part C	Qualitative and quantitative indicators
Pfizer/Seagen (2023, EU)	A2 (R&D pipeline-to-pipeline overlaps)	R&D spending; number of players active; number of products under development; industry growth outlook; research stages of competitors, indication by indication; internal documents about pipeline strategy; information about molecules, mechanisms of action, and mechanisms of disease; competitors' and Key Opinion Leaders' (KOL) testimonies.
Illumina/GRAIL (2022, EU; 2023, US)	D (vertical merger with dynamic foreclosure)	Market shares; internal documents on competitive intelligence and strategy, including on competitors' research paths; internal documents on profit expectations and on business strategy; reference to the company's past conduct as a behavioural indicator; reference to competitors' testimony on innovation competition; ability and incentive for Illumina to foreclose GRAIL's rivals.
Illumina/PacBio (2019, UK; 2020, US)	A2 (R&D pipeline-to-pipeline overlaps)	Market shares; HHI; internal strategy documents, investor presentations, SEC filings, sales plans, technical assessments; respondents' own characterisation of each other; acquisition price; assessment of the innovation trajectory by looking at cost per genome, throughput, read accuracy, read length; patent portfolio; pipeline products; third-party submissions, including by sequencing service providers and customers of sequencing instruments.

Source: Oxera.

is also vertical in nature: a diagnostic insight frequently precedes and directly informs a therapeutic intervention, making diagnostics firms natural upstream partners to biopharmaceutical ones.

23 Quantum computing

Key points

- While quantum computing technologies are at a development stage, they are expected to bring about major breakthroughs across a wide range of industries.
- The QC market is approaching a commercial tipping point, with 2026 forecast to be a pivotal year, and market value projected to reach €21–29 billion by 2035. However, technical hurdles remain significant and earlier timelines have proved over-optimistic.
- The EU is relatively well-positioned in QC research, ranking second to China in overall research output and second to the US in 'leading' research output, while hosting over 100,000 quantum-ready experts. Yet it faces critical challenges in translating scientific excellence into commercial scale, with EU innovative entrants being younger, smaller, and raising roughly one third of the private investment secured by their US counterparts.
- The principal policy tools for EU QC competitiveness partly overlap with those identified in the biopharmaceutical sector: they involve consolidating public funding while reducing fragmentation and attracting private investment, as well as expanding the QC talent pipeline. While the EU has taken steps in these directions, implementation remains limited, and EU QC firms are already turning to US public markets for financing.
- The QC industry is attracting increasing scrutiny from competition authorities. While largely untested so far, merger control may become increasingly relevant to support the growth and development of QC as more consolidation is expected, with horizontal pipeline-to-pipeline and capability overlaps already visible in recent deals, and potential competition concerns as large incumbents from other sectors acquire QC firms, potentially enabling them to foreclose rivals' access to transformative technology.

23.1 Background and market developments

23.1.1 Industry background

- 23.1 Quantum computing is part of a broader family of quantum technologies (QT), which also include quantum sensing and quantum communication. QC draws on the principles of quantum physics to carry out tasks that ordinary computers either cannot do at all or only inefficiently.⁷³⁰ A conventional computer stores and handles data using 'bits'—units that are always in one of two states, either 0 (off) or 1 (on), whereas quantum computers use 'qubits', whose behaviour differs vastly from that of bits. This is due to two remarkable properties of quantum physics: superposition and entanglement. Superposition means that unlike a classical bit, a qubit does not have to be either 0 or 1; it can exist in both states simultaneously. As for entangled qubits, their states are linked such that the state of one instantly determines the

⁷³⁰ UK Department for Science, Innovation and Technology (2023), 'National Quantum Strategy', March, p. 7; European Commission (2025), 'Quantum Europe Strategy: Quantum Europe in a Changing World', COM(2025) 363 final, July, p. 1.

state of the other. By combining these two properties, quantum computers can explore many possible solutions far more quickly than classical computers.⁷³¹

- 23.2 QC remains in its early stages. The current generation of hardware, known as Noisy Intermediate-Scale Quantum (NISQ) devices, is still unable to solve industry-relevant problems due to intrinsic noise levels.⁷³² Indeed, the very properties of qubits—superposition and entanglement—also make them extremely sensitive to their environment. Any vibration, slight change in temperature, or other disturbance can cause a qubit to lose its quantum state, thereby interrupting the computation in progress. The central challenge is, therefore, to stabilise qubits throughout the computation process. Early 2020s predictions that the field would have matured beyond this stage by 2030 are now recognised as over-optimistic. Significant hardware challenges persist and competition from classical computing has been fiercer than expected (as it has continued to improve), slowing down the development of QC.⁷³³
- 23.3 Nevertheless, commercial momentum is increasing. 2026 is forecast to be a pivotal year, with first movers evolving from pilots to enterprise applications. Such applications include portfolio optimisation and cryptography in the financial service sector, the acceleration of drug testing and experimentation in the pharmaceutical sector, as well as solutions to complex logistics challenges in the transport sector, such as routing constraints and scheduling conflicts.⁷³⁴ QC's transformative potential indeed spans optimisation problems⁷³⁵ across a large array of sectors—from encryption, finance, logistics, and automotive to AI, chemistry, pharmaceuticals, energy, and manufacturing.⁷³⁶ Over 300 organisations, including Airbus and JPMorgan Chase, already collaborate with QT firms.⁷³⁷ Security and defence applications are also expected to be significant.⁷³⁸ McKinsey's latest estimates project the QC market at €21–29 billion by 2035,⁷³⁹ with the broader QT sector expected to create 600,000 new jobs by 2040.⁷⁴⁰ The downstream impact on other industries is expected to be greater still: McKinsey projects that the value at stake from QC, through the value created via the technological advancements in the post-quantum era, reaches 1.1–2.3 trillion euros.⁷⁴¹ In a study from 2021, it was anticipated that around 80% of the value created by QC would accrue to end users such as biopharmaceutical and financial services companies, the remainder flowing to QC industry players themselves.⁷⁴²

⁷³¹ Gamble (2019).

⁷³² European Commission (2025), 'Future Directions for Quantum Technology in Europe: An analysis of policy questions', October, p. 13.

⁷³³ BCG (2024), '[The Long-Term Forecast for Quantum Computing Still Looks Bright](#)', accessed 21 July 2026.

⁷³⁴ McKinsey (2026), '[McKinsey Quantum Technology Monitor 2026: A commercial tipping point](#)', 28 April, accessed 21 July, p. 16.

⁷³⁵ Gamble (2019).

⁷³⁶ BCG (2022), '[Can Europe Catch Up with the US \(and China\) in Quantum Computing?](#)', August, p. 5, accessed 21 July. The report focuses specifically on the EU's capabilities in the quantum computing sector relative to those of China and the US, identifying key gaps and the means to address them.

⁷³⁷ McKinsey (2026) (full reference in footnote 734).

⁷³⁸ European Commission (2025), p. 12 (full reference in footnote 732).

⁷³⁹ McKinsey (2026), p. 8 (full reference in footnote 734). Converted from \$ to € using the ECB average exchange rate for 2026 (averaged from January to April).

⁷⁴⁰ Venegas-Gomez (2020).

⁷⁴¹ McKinsey (2026), p. 11 (full reference in footnote 734). Converted from \$ to € using the ECB average exchange rate for 2026 (averaged from January to April).

⁷⁴² BCG (2021), '[What Happens When "If" Turns to "When" in Quantum Computing?](#)', accessed 21 July 2026.

- 23.4 The competitive landscape features both small QC-specialist innovative companies (including startups) and large diversified technology incumbents. In the US, leading players include IBM, Google, AWS, and Rigetti (which, contrary to the other players listed here, is a QC-focused company).⁷⁴³ The US government is increasingly getting involved in the industry, announcing in May 2026 its intent to take equity stakes in nine QC companies.⁷⁴⁴ In China, diversified players such as Huawei Cloud and Tencent remain active following Alibaba's and Baidu's withdrawal from the sector in 2024.⁷⁴⁵ Smaller firms specialising in QT are also present, such as Origin Quantum, Qasky, or CIQTEK. Key EU players are generally more specialised. They include IQM, Pasqal, Quandela, C12 Quantum Electronics, and Multiverse, operating principally in France, Germany, the Netherlands, Finland, and Spain.⁷⁴⁶ Table 23.1 presents the 24 main players in the quantum computing sector as observed in March 2026.

Table 23.1 Overview of the main players in QC in 2026

Rank	Name	Country	Technology	Status
1	IBM Quantum	US	Superconducting	Listed
2	Google Quantum AI	US	Superconducting	Listed
3	IonQ	US	Trapped-ion	Listed
4	Quantinuum	UK	Trapped-ion	Currently going public
5	PsiQuantum	US	Photonic	Not listed
6	Rigetti Computing	US	Superconducting	Listed
7	D-Wave	Canada	Quantum annealing and gate-model	Listed
8	Xanadu	Canada	Photonic	Not listed
9	IQM ⁷⁴⁷	Finland	Superconducting	Not listed
10	QuEra Computing	US	Neutral-atom	Not listed
11	Pasqal ⁷⁴⁸	France	Neutral-atom	Not listed
12	Atom Computing	US	Neutral-atom	Not listed
13	Infleqtion	US	Neutral-atom	Listed
14	Intel Quantum	US	Silicon spin	Listed
15	Alice & Bob	France	Cat qubit	Not listed
16	Oxford Quantum Circuits	UK	Superconducting	Not listed
17	Quantum Circuits Inc	US	Superconducting	Listed (acquired by D-Wave)
18	SandboxAQ	US	AI and quantum security	Not listed

⁷⁴³ McKinsey (2026) (full reference in footnote 737).

⁷⁴⁴ Miller, J. and Peel, M. (2026), 'U.S. quantum investment includes groups tied to Trump administration', Financial Times, 21 May.

⁷⁴⁵ IEEE (2024), '[Alibaba and Baidu Cash Out on Quantum Computing Stakes](#)', accessed 21 July 2026.

⁷⁴⁶ Digital Europe (2024), 'The EU's critical tech gap', June, pp 62–64.

⁷⁴⁷ IQM recently announced a Special Purpose Acquisition Company (SPAC) merger to list on Nasdaq. See IQM (2026), '[IQM, a Global Leader for Quantum Computing, to Become the First Listed European Quantum Company, Through Merger with Real Asset Acquisition Corp](#)', accessed 21 July 2026.

⁷⁴⁸ Pasqal recently announced a SPAC merger to list on Nasdaq. See Quantum Insider (2026), '[Pasqal Latest Quantum Company to Announce Intentions to Go Public, Valued at \\$2 Billion](#)', accessed 21 July 2026.

Rank	Name	Country	Technology	Status
19	Photonic Inc	Canada	Silicon spin and photonic	Not listed
20	Classiq	Israel	Quantum software	Not listed
21	Multiverse Computing	Spain	Quantum software and optimisation	Not listed
22	Q-CTRL	Australia	Quantum control and infrastructure	Not listed
23	Nord Quantique	Canada	Superconducting	Not listed
24	Silicon Quantum Computing	Australia	Silicon spin	Not listed

Note: This ranking was published on 24 March 2026.

Source: Quantum Navigator (2026), '[Top 25 Quantum Computing Companies 2026](#)', accessed 21 July 2026. Although Quantum Navigator notes that it lists the top 25 companies, the list only includes 24 companies.

- 23.5 So far, the main Big Tech players⁷⁴⁹ have not been particularly active in acquisitions of QC startups (or small innovative companies more generally), with Google's acquisition of Atlantic Quantum being one exception⁷⁵⁰. Rather than outright acquisitions, incumbents have favoured ecosystem-building: IBM selects business-to-business small innovative companies that slot into its existing enterprise client network,⁷⁵¹ while Intel has taken seed positions in promising ventures.⁷⁵² That said, M&A activity is growing. Between 2016 and 2024, approximately 5% of QT companies globally underwent an acquisition, sometimes involving cross-border transactions.⁷⁵³ This points to a maturing market: 72% of QC customers are now privately owned companies, a marked shift from when public-sector research labs dominated.⁷⁵⁴ The shift is observable in the EU as well: in early 2026, IQM (Finland) and Pasqal (France) each announced Special Purpose Acquisition Company mergers to list on Nasdaq.⁷⁵⁵ As both firms told investors, besides raising capital, these operations sought to gain the financial leverage needed to acquire other QC firms ahead of expected industry consolidation.⁷⁵⁶

23.1.2 R&D intensity and innovation in QC

- 23.6 The EU is relatively well-positioned in the global race for leadership in QC.⁷⁵⁷ It ranks second only to China in overall research output, and second to the US in 'leading' research output,⁷⁵⁸ with EU scientists contributing more frequently than any other country to quantum-relevant publications.⁷⁵⁹ The EU also hosts the

⁷⁴⁹ Includes Alphabet (Google), Apple, Meta (Facebook), Amazon, and Microsoft (GAFAM).

⁷⁵⁰ Google (2025), '[We're scaling quantum computing even faster with Atlantic Quantum](#)', accessed 21 July 2026.

⁷⁵¹ Quantum Insider (2025), '[IBM Backs Startups That Fit Its AI-and-Quantum Playbook](#)', accessed 21 July 2026.

⁷⁵² Intel Capital (2026), '[Our Investment in Q-Factor: Building a Million-Qubit Quantum Computer](#)', accessed 21 July 2026.

⁷⁵³ OECD and EPO (2025), 'Mapping the global quantum ecosystem', December, paras 87–88.

⁷⁵⁴ McKinsey (2026) (full reference in footnote 737).

⁷⁵⁵ Quantum Insider (2026) (full reference in footnote 748); IQM (2026) (full reference in footnote 747).

⁷⁵⁶ ComputerWeekly (2026), '[European Union deep tech plan too late for quantum champions IQM and Pasqal](#)', accessed 21 July 2026.

⁷⁵⁷ European Commission (2025), p. 5 (full reference in footnote 730).

⁷⁵⁸ Digital Europe (2024), p. 62 (full reference in footnote 746).

⁷⁵⁹ McKinsey (2024), 'Quantum Technology Monitor', April, p. 40.

highest absolute number of quantum-ready experts worldwide (over 100,000, or 231 per million inhabitants) alongside strong academic and research infrastructure, multiple Nobel Prize winners,⁷⁶⁰ and a rapidly expanding offering of quantum-focused research programmes and master's degrees.⁷⁶¹

- 23.7 From an industry standpoint, the EU ranks third globally in value added in products building on the QC technology,⁷⁶² accounts for 32% of quantum companies worldwide (with an industry largely based on small innovative companies)⁷⁶³ and hosts 39 quantum computers, matching the US, while uniquely exploring all five major QC technologies in parallel.⁷⁶⁴ Public investment is also substantial. Of the €49 billion committed worldwide to date (including upcoming initiatives), the US accounts for approximately 13.5% of the total, while the EU represents more than 20%.⁷⁶⁵ Between 2014 and 2024, the EU has funded approximately 1,000 quantum projects totalling around €2 billion, of which some 28% was directed specifically at QC.⁷⁶⁶ The EU's strengths are summarised by the Quantum Europe Strategy as combining world-class research, scientific excellence, a vibrant base of small innovative companies, and a strong public investment structure.⁷⁶⁷ However, the large-scale investments envisioned by China could significantly reshape the global competitive landscape for advanced research and commercialisation.⁷⁶⁸

23.1.3 Productivity, sectoral importance, and challenges

- 23.8 Notwithstanding its strengths, the EU faces three main challenges. First, public funding efforts remain insufficiently coordinated.⁷⁶⁹ At the national level, European countries have not achieved the degree of stakeholder alignment seen in the US or China. Moreover, within the EU, Member States tend to work in 'silos'.⁷⁷⁰ As noted in the EU Innovation Policy report, shifting a greater share of funding to the EU level would broaden the pool of eligible projects and reviewers, strengthen the integrity of the evaluation process, and reduce the reputational risk associated with the inevitable failures that ambitious R&D entails.⁷⁷¹
- 23.9 Second, the EU's strong research base translates into only modest patenting activity. China leads globally in QT patents, followed by the US, Japan, and the EU. In QC more specifically, the EU also ranks fourth at around 6% of total patents. Notably, 17% of QT patents worldwide remain held by universities and research institutions, reflecting the sector's early stage, while 35% are owned by companies active in other industries—such as finance—positioning themselves to leverage QC in their core operations.⁷⁷²

⁷⁶⁰ Draghi (2024), Part B, p. 80.

⁷⁶¹ McKinsey (2024), p. 41 (full reference in footnote 759).

⁷⁶² Digital Europe (2024), p. 62 (full reference in footnote 746).

⁷⁶³ European Commission (2025), p. 6 (full reference in footnote 732).

⁷⁶⁴ Computing platforms are based on superconducting circuits, trapped ions, neutral atoms, photonics, diamonds, or spin qubits. While the EU explores all of them, the US does not have any machine based on the spin qubits technology. See European Commission (2025), pp 5–6 (full reference in footnote 730).

⁷⁶⁵ European Commission (2025), p. 55 (full reference in footnote 732).

⁷⁶⁶ European Commission (2025), pp 44–46 (full reference in footnote 732).

⁷⁶⁷ European Commission (2025), p. 2 (full reference in footnote 730).

⁷⁶⁸ Quantum Insider (2025), '[China Launches \\$138 Billion Government-Backed Venture Fund, Includes Quantum Startups](#)', accessed 21 July 2026.

⁷⁶⁹ European Commission (2025), p. 2 (full reference in footnote 730).

⁷⁷⁰ BCG (2022), p. 1 (full reference in footnote 736).

⁷⁷¹ European Policy Analysis Group (2024), 'EU Innovation Policy – How to escape the middle technology trap', April, p. 7.

⁷⁷² European Commission (2025), pp 49–50 (full reference in footnote 732).

- 23.10 Third, and most critically, EU small innovative companies struggle to scale up. As the market matures, most investments are now targeting later-stage companies.⁷⁷³ Yet, EU companies are younger (60% founded post-2018, compared with 29% of US firms and 50% of Chinese companies) and smaller (large and medium firms represent only 13% of EU QT companies, against 29% in the US and 82% in China).⁷⁷⁴ Overall, they attract far less private investment: US QC firms raise almost three times more than their EU counterparts.⁷⁷⁵ In fact, the US and Canada—and, to a lesser degree, the United Kingdom and Israel—are the only countries where private investment in quantum technologies is broadly on a par with public funding. Elsewhere, in China, EU member states, Japan, and South Korea, private investment amounts to less than a tenth of public funding. According to Digital Europe (a trade association representing digitally transforming industries in Europe), this disparity reflects both a greater appetite for risk among American investors and the fact that the world's largest corporations active in quantum computing—including Intel, Google, and IBM—are headquartered in the US.⁷⁷⁶ Cross-border investment flows compound the problem: while QC attracts a 44% share of cross-border deals, US-based small innovative companies generally attract most of this funding.⁷⁷⁷ The decisions of IQM and Pasqal to pursue US listings in early 2026 illustrate the concern that the EU is failing to fund its own 'deep-tech' champions through to commercialisation.⁷⁷⁸
- 23.11 Commercial applications of quantum computing remain nascent, making it difficult to compare EU productivity in this space with that of its counterparts. For the time being, patenting activity, investment levels, and relevant skills serve as more reliable metrics for gauging the prospects of this sector across different geographies. The Draghi Report explicitly notes that the EU relies on third countries for over 80% of its digital products, services, infrastructures, and intellectual property.⁷⁷⁹ Given that 70% of the new value created in the world economy over the next ten years will be digitally enabled, developing leadership in next-generation materials would generate significant value. Advanced technologies are likely to unlock productivity in the years ahead, and quantum computing stands out as a particularly strong candidate to improve the EU's position, as it is expected to raise significant cross-sectoral opportunities and the EU is relatively well positioned.

23.2 Policy tools

- 23.12 Various industry reports identify three principal policy tools to strengthen the EU's position in QC: consolidating public funding and attracting private investment towards small innovative companies, as well as expanding the quantum talent pipeline. The EU has already taken steps in these directions—most notably through the €1 billion Quantum Technologies Flagship launched in 2018 and the 2025 Quantum Europe Strategy (see Box 23.1 below), with a Quantum Act planned for 2026—though implementation remains at an early stage.



Box 23.1 The five pillars of the Quantum Europe Strategy

⁷⁷³ European Commission (2025), p. 52 (full reference in footnote 732).

⁷⁷⁴ European Commission (2025), p. 48 (full reference in footnote 732).

⁷⁷⁵ Digital Europe (2024), p. 63 (full reference in footnote 746).

⁷⁷⁶ Digital Europe (2024), p. 60 (full reference in footnote 746).

⁷⁷⁷ European Commission (2025), p. 53 (full reference in footnote 732).

⁷⁷⁸ ComputerWeekly (2026) (full reference in footnote 756).

⁷⁷⁹ Draghi (2024), Part B, pp 67–68.

The 2025 Quantum Europe Strategy sets out five pillars that aim to strengthen Europe's leadership, resilience, and industrial capacity across the quantum ecosystem:

- Consolidating excellence across Europe to lead in quantum science and its industrial transformation;
- Developing sustainable, scalable, coordinated infrastructure hubs to support production, design, and application development;
- Securing supply chains and the industrialisation of quantum technologies through investments in startups and scaleups;
- Integrating secure, sovereign quantum capabilities into Europe's space, security, and defence strategies;
- Building a diverse, world-class workforce through coordinated agile education and training systems and programmes and promoting talent mobility across the EU.

Source: European Commission (2025), '[Quantum Europe Strategy: Quantum Europe in a Changing World](#)', accessed 21 July 2026.

- 23.13 Several studies point to the need for the EU to consolidate its public funding by both sustaining investment levels and reducing fragmentation across the bloc. The OECD identifies at least two grounds for such an approach: the significant positive externalities that QC generates for downstream industries, and the chronic under-investment by private players owing to the long development timelines and high degree of technological risk.⁷⁸⁰ The BCG notes that public support of this kind should operate at the EU scale on both the demand and supply sides. On the demand side, public institutions could become early customers of quantum small innovative companies—following the US model—while economic incentives should be created to encourage incumbents to adopt quantum applications and reduce the risk associated with such early adoption, for instance through a quality label for QC firms akin to that developed for high-risk AI applications.⁷⁸¹ On the supply side, the EU should build upon the foundations laid by its Quantum Strategy, which aims to coordinate Member States' R&D agendas, pool resources around shared themes, and build critical mass. The BCG recommends that the governance framework underpinning such coordination should involve stakeholders from across the sector and be more inclusive than the Quantum Flagship, whose coordination team the BCG criticises for lacking representation from national science ministries and research agencies.⁷⁸²
- 23.14 Second, as noted by the OECD and other industry reports, the EU should ensure that it attracts private investment. As QC approaches commercialisation, private investment becomes essential to bridge the gap between research and industrialisation.⁷⁸³ The EU Quantum Strategy seeks to address the risk of EU startups failing to attract the private investments needed for their development

⁷⁸⁰ OECD and EPO (2025), p. 88 (full reference in footnote 753).

⁷⁸¹ BCG (2022), p. 21 (full reference in footnote 736).

⁷⁸² European Commission (2025), p. 4 (full reference in footnote 730); BCG (2022), p. 20 (full reference in footnote 736).

⁷⁸³ European Quantum Industry Consortium (2024), 'strategic Industry Roadmap', January, p. 174, OECD and EPO (2025), p. 55 (full reference in footnote 753).

through the support of the European Innovation Council Fund, the European Investment Bank, the European Tech Champions Initiative, as well as through first-loss guarantees and co-investment schemes via InvestEU, while also tackling fragmentation in financial services and removing barriers to cross-border venture capital.⁷⁸⁴ However, the US listings of IQM and Pasqal in early 2026 suggest that these policies have not yet borne fruit. It therefore appears necessary to further attract international capital to the EU; to that end, the BCG suggests streamlining screening processes for foreign investment or preauthorising foreign funds to invest in startups.⁷⁸⁵ Alongside financing, the European Quantum Industry Consortium underlines the importance of strengthening IP protection and standardisation: spinoffs are more attractive to investors when they own their IP, and active participation in standardisation processes confers a competitive advantage.⁷⁸⁶ To achieve this, the Consortium recommends that adequate financial support and effective communication of the benefits of standardisation be provided to EU firms, with the European Patent Office taking the lead in harmonising patent protection and actively encouraging SMEs to file patents.⁷⁸⁷

- 23.15 Third, experts highlight that the EU should expand its talent pipeline. Although the EU already has over 100,000 quantum-ready experts,⁷⁸⁸ demand is set to outpace supply significantly, with 600,000 new jobs expected in the QT sector by 2040.⁷⁸⁹ The Quantum Europe Strategy envisages a centralised training portal, apprenticeship programmes, and initiatives to foster innovation.⁷⁹⁰ The BCG underlines that the talent pipeline should also be shaped by business needs, arguing that academic expertise must be translated into engineering and commercial skills. Given the global competition for talent, the BCG further recommends that the EU offer incentives to attract skilled workers from abroad, including fast-track visas, direct monetary incentives, and tax exemptions.⁷⁹¹
- 23.16 Merger control could also play a role in fostering innovation in the QC sector. As noted in one policy report, 'European quantum companies [must] remain attractive targets for European and foreign investors alike, while maintaining Europe's strategic capability in the field. The balanced approach involves VC investments as well as future mergers and acquisitions'.⁷⁹² The reference to 'future' M&As is to be understood in a context where, as described in Section 23.1, significant consolidation may materialise in the near future. In order to sustain growth and innovative developments, mergers in this sector are likely to require careful scrutiny, particularly in the context of potential acquisitions of small, QC-focused firms by large tech incumbents.⁷⁹³ Two recent investigations by competition authorities are noteworthy. In March 2026, the Italian Competition Authority launched a market investigation into the QC sector;⁷⁹⁴ a month later, the FTC started to examine the proposed merger between IonQ and SkyWater

⁷⁸⁴ European Commission (2025), pp 13–14 (full reference in footnote 730).

⁷⁸⁵ BCG (2022), p. 21 (full reference in footnote 736).

⁷⁸⁶ European Quantum Industry Consortium (2024), pp 164 and 173 (full reference in footnote 783).

⁷⁸⁷ European Quantum Industry Consortium (2024), pp 159 and 173 (full reference in footnote 783).

⁷⁸⁸ Draghi (2024), Part B, p. 80.

⁷⁸⁹ Venegas-Gomez (2020).

⁷⁹⁰ European Commission (2025), pp 17–18 (full reference in footnote 730).

⁷⁹¹ BCG (2022), p. 24 (full reference in footnote 736).

⁷⁹² European Quantum Industry Consortium (2024), p. 176 (full reference in footnote 783).

⁷⁹³ Klepper (2002) shows that nascent, innovation-intensive industries often undergo a 'shakeout', as early experimentation gives way to consolidation around a smaller number of leading firms. A similar dynamic may emerge in QC as the sector matures, making future acquisitions and broader consolidation particularly important to scrutinise.

⁷⁹⁴ AGCM (2026), '[The Italian Competition Authority launches market investigation and call for inputs into Quantum Computing](#)', accessed 21 July 2026.

Technology.⁷⁹⁵ Such heightened scrutiny indicates the growing relevance of competition policy in this field.

23.3 Merger control

23.17 To the best of our knowledge, only one merger has to date been subject to review by a competition authority worldwide (the FTC, though the case remains at a preliminary stage),⁷⁹⁶ and no academic literature specifically addresses merger effects in this sector. However, drawing on academic work in patent-intensive industries,⁷⁹⁷ the maturation of QC into commercially viable products and services is likely to drive a consolidation process, as small firms and incumbents may have increasing incentives to engage in M&As to access innovation opportunities and leverage efficiencies of scale respectively. In this context, we consider recent merger activity in the sector and apply the conceptual framework developed in Part C. Given QC's high innovation intensity and the multiplicity of parallel research paths being currently pursued by a mix of large tech players and specialised small innovative players, a horizontal merger framework to assess dynamic effects in the presence of pipeline and capability overlaps is relevant (see Subsection 23.3.1). The wide range of potential QC applications may also give rise to risks of foreclosure effects, whereby incumbents in a range of industries might acquire QC firms to take control of a strategic technology and foreclose existing and future rivals' access to it (see Subsection 23.3.2). These potential theories of harm are dynamic, affecting both future innovation and future product market competition, with implications for ECW under our framework. Merger control may also need to account for other parameters of competition in this area, such as supply chain resilience.⁷⁹⁸

23.3.1 Horizontal considerations: pipelines, products, and capabilities

23.18 According to industry experts, mergers in the QC sector have so far been primarily driven by talent acquisition (acquihring) and strategic shifts in technological focus (commonly referred to as 'technology pivots').⁷⁹⁹ Given the early stage of this technology, this suggests that such mergers might have involved pipeline-to-pipeline and capability overlaps (respectively in Categories A and B of our taxonomy—see Part C). The three deals examined below illustrate this.

23.19 In Google's acquisition of Atlantic Quantum (October 2025), the two parties were pursuing divergent superconducting qubit technologies: while Google, IBM, Rigetti, and others had relied on transmons, Atlantic Quantum had been developing fluxonium qubits. On the one hand, a potential competition concern could have been the pipeline-to-pipeline overlap between the merging parties and the consequent risk that one research path might have been discontinued. On the other hand, the merger could also create the opportunity to focus complementary research efforts on the more promising research path, thus accelerating its development, or to combine the two paths, noting that researchers have explored hybrid approaches integrating fluxonium qubits with transmon tunable couplers. D-Wave's acquisition

⁷⁹⁵ CPI (2026), '[FTC Targets \\$15.5B IonQ–SkyWater Chip Tie-Up With New Request](#)', accessed 21 July 2026.

⁷⁹⁶ CPI (2026) (full reference in footnote 795).

⁷⁹⁷ Lee (2020).

⁷⁹⁸ Resilience is often discussed in the context of advanced technologies, with an emphasis on the need for the EU to strengthen its strategic autonomy (see Draghi (2024), Part B, p. 67).

⁷⁹⁹ Quantum Global Intelligence (2026), '[Three Quantum Computing Hardware Acquisitions – What's Behind Them?](#)', accessed 21 July 2026. Note that other mergers have taken place since then, for example with Viewbix completing the acquisition of Quantum X Labs in March 2026. See Quantum Global Intelligence (2026), '[Viewbix Completes Acquisition of Quantum X Labs, Pivots to Quantum Technology](#)', accessed 21 July 2026.

of Quantum Circuits Inc. (January 2026) may follow a similar logic: D-Wave had been pursuing fluxonium-based superconducting qubits, but its post-merger roadmap indicates a pivot towards the target's dual-rail qubit technology, suggesting that the acquirer's original research path may not be continued. The IonQ/Oxford Ionics merger (September 2025), for its part, enabled IonQ to transition from laser-based to microwave-based qubit control, improving fidelity and overall technology quality.⁸⁰⁰

- 23.20 As set out in Chapter 15, in horizontal mergers involving pipeline or capability overlaps, the key dynamic risks stem from the loss of future product market competition and a reduction in overall incentives to carry out R&D efforts and innovate, ultimately depriving consumers of enhanced products and services. These risks must be weighed against potential dynamic benefits, arising from voluntary technology sharing or dynamic synergies, as the examples above suggest.
- 23.21 In practice, measures of the innovative activity of the merging firms can be informative. On the input side, VC funding may serve as a useful proxy for R&D expenditure in the startup context (as R&D expenditure is typically financed this way). To assess firms' proximity in terms of innovation capabilities, authorities could also examine the degree of similarity between the skill sets of the engineers they employ. On the output side, relevant measures include patent counts, citation-weighted patent quality, the radicalness of patent portfolios (defined as the share of technological sub-classes that a patent cites but is not classified in) and technological proximity between the parties, which captures whether firms are patenting in similar fields.⁸⁰¹
- 23.22 Relevant analytical tools include the concept of 'Technology Business Zones' as developed in the academic literature.⁸⁰² This metric captures business affinity between technological fields: where firms patenting in one field systematically also patent in another, that co-occurrence signals a business relationship between the two, even if they are technologically distinct. A sufficiently strong pattern of this kind suggests that the two fields may belong to the same Technology Business Zone, irrespective of whether the precise nature of the underlying relationship, in terms of products, supplier links, or other commercial functions, can be determined. Alongside technological proximity, which captures whether firms patent in similar fields, such tools can help competition authorities assess the competitive dynamics between the merging parties, and the competitive importance of their rivals.
- 23.23 In the transactions described above, industry experts have highlighted the 'acquihiere' dimension as a significant underlying driver: given the very limited talent pool in the sector, acquiring a company that already possesses the relevant skills and know-how represents an efficient means of rapidly expanding one's business.⁸⁰³ This approach confirms the importance of looking into the merging firms' capabilities to assess how closely the parties compete and whether their innovation capabilities are materially overlapping.
- 23.24 Finally, while the phenomenon of 'killer acquisitions' has not yet been a material concern in QC—given that the early stage of development and limited commercial applications reduce incumbents' incentive to suppress research to protect existing sales—it may become more relevant in the medium term as the market matures. Similar competition concerns may arise in the context of partnerships which, while

⁸⁰⁰ Quantum Global Intelligence (2026) (full reference in footnote 799).

⁸⁰¹ Berger, Calligaris, Greppi, and Kirpichev (2025).

⁸⁰² Cheng, Jin, Leccese, Lee, and Wagman (2023).

⁸⁰³ Quantum Global Intelligence (2026) (full reference in footnote 799).

they may fall outside the scope of merger control, could produce similar competitive effects.

- 23.25 Horizontal mergers in the quantum sector may also generate efficiency gains under Box 2A and Box 2B, including the internalisation of spillovers and dynamic synergies respectively. As regards the former, mergers in the quantum industry may generate intertemporal spillovers, as firms build on previous innovations by competitors. Mergers can also efficiently combine complementary capabilities, resulting in enhanced post-merger investments as well as broader improvements in the effectiveness of investment. This may occur through efficient asset combination in a sector where hardware is costly, but also through knowledge sharing and technology transfers between the merging parties.

23.3.2 Balancing foreclosure and entrenchment risks with non-horizontal efficiencies

- 23.26 As industry experts note, QC involves significant capital barriers to development, which may potentially result in only a few well-funded entities owning essential QT infrastructures.⁸⁰⁴ The race on the hardware side has not yet been won, and it will be important to preserve contestability when one or a few players emerge, to preserve access to QC as an input to downstream applications.
- 23.27 Moreover, QC has the potential to disrupt a wide range of strategically important sectors, including finance, logistics, and defence. Many businesses are already investing in QC with a view to being well-positioned to deploy the technology when it matures and outperforms classical computers. Various partnerships already exist in this space: France's Crédit Agricole, for instance, concluded a partnership with the French innovative company Pasqal in 2019, and announced recently that it would accelerate the use of its QC technology in financial applications.⁸⁰⁵ Other notable examples include Airbus and JPMorgan, as noted in Section 23.1.
- 23.28 Major players across industries may seek to acquire QC-specialised firms in order to secure access to the technology, and merger control may need to examine potential foreclosure concerns. As detailed in Chapter 18, non-horizontal mergers could prevent the new technology from being integrated across markets and could deprive existing and future rivals of an important complementary asset, thus hindering future entry or expansion and reinforcing the market position of incumbents.
- 23.29 Given that non-horizontal mergers may also produce procompetitive effects, the assessment is likely to require a context-specific balancing exercise, in line with the conceptual framework developed in Part C of the main report. For example, transactions may generate asset complementarities across downstream applications (such as in the logistics or banking sector), or help firms secure access to important inputs (e.g. quantum-computing infrastructure and specialist technical talent).
- 23.30 Finally, competition authorities may need to investigate the competition risks that partnerships may raise—as it has been seen in other industries, most notably AI.⁸⁰⁶

⁸⁰⁴ Perrier and Aboy (2026).

⁸⁰⁵ Maddyne (2026), '[Quantique : Crédit Agricole CIB et Pasqal passent un nouveau cap dans leur collaboration](#)', accessed 21 July 2026.

⁸⁰⁶ In a recent Policy Brief, the European Commission has noted the potential effects of the partnerships concluded between AI foundation model developers and existing digital players. While partnerships may allow smaller and more innovative players to enter and grow in the market, they may also lead to detrimental foreclosure effects; European Commission (2024), 'Competition in Generative AI and Virtual Worlds', Competition Policy Brief, 2024-03, September. The CMA expressed

Examples of such partnerships already exist in the QC sector, including with large tech incumbents, although the reach of these arrangements remains limited.⁸⁰⁷ As the technology matures and the hardware race develops, major players may seek to control access to important inputs through partnerships, potentially disadvantaging rivals and raising barriers to entry, while at the same time accelerating the scaling-up and commercialisation of smaller firms.

similar concerns regarding the ‘interconnected web’ of over 90 partnerships and strategic investments involving GAFAM and Nvidia, noting that such arrangements—connecting key players across the value chain—could exacerbate existing positions of market power. Again, while leveraging increased resources, expertise, and innovation capabilities can play a procompetitive role in the technology ecosystem, these deals also risk undermining rival firms’ ability to compete, with potential entrenchment effects; see CMA (2024), [‘CMA outlines growing concerns in markets for AI Foundation Models’](#), accessed 21 July 2026.

⁸⁰⁷ Microsoft, for example, has entered into a partnership with Atom Computing, amongst others, to integrate Atom Computing’s neutral atom quantum hardware into Microsoft’s Azure Quantum platform. Such collaboration is seen as a step towards achieving reliable QC hardware for commercial use. See Quantum Insider (2024), [‘Microsoft and Atom Computing Partner to Build a Quantum Supercomputer’](#), accessed 21 July 2026.

24 Clean technologies

Key points

- The clean technologies sector is one of the world's fastest-growing industrial sectors, with the global market expanding rapidly over the past decade and annual growth rates of 20% on average.
- R&D and learning-curve effects are a defining competitive dynamic of the sector, with continuous improvements in manufacturing efficiency and material use driving sharp cost reductions across major technologies.
- The European clean technologies sector is confronted with structural challenges: a widening gap with Chinese manufacturers, whose cost advantages reflect decades of integrated policy support and manufacturing scale, and, for some technologies such as solar energy, a deep supply-chain dependency on China across the main clean-technology inputs.
- Key policy tools in clean technologies include trade measures, which can support domestic producers, demand-side instruments, which stimulate market uptake and help domestic manufacturers achieve scale, and improved financing, to lower the cost of capital and support investment in clean technologies production.
- Merger control is an increasingly relevant tool in the clean technologies sector. It can enable firms to scale up and reduce costs through consolidation, while also guarding against killer acquisitions, where incumbents acquire disruptive innovators to neutralise future competitive pressure. It can also scrutinise vertical mergers to prevent foreclosure of access to critical green inputs and infrastructure, while allowing for efficient integration. Finally, it can incorporate sustainability as a parameter of competition, where relevant.

24.1 Background and market developments

24.1.1 Industry background

- 24.1 The clean-technology sector encompasses the development, application, and commercialisation of products, services, and processes broadly aimed at reducing environmental impact, optimising the use of natural resources, producing or storing renewable energy, and capturing or absorbing emissions.
- 24.2 This includes solutions that promote energy efficiency (e.g. smart grids) and support renewable energy generation (e.g. solar, wind, hydro, geothermal, and battery storage).
- 24.3 Leading players in the clean technologies sector range from large corporations such as GE Vernova, Iberdrola, BYD, and Tesla, which are active in electric mobility, energy storage, and renewable energy solutions, Siemens Gamesa and Vestas, which are active in both onshore and offshore wind turbines, up to specialised solar manufacturers including Solarwatt, Canadian Solar, and Jinko Solar, battery innovators such as CATL, and a diverse set of companies developing niche technologies—for example Stegra in low-carbon steel production, Hydrostor in long-duration energy storage, and QuantumScape in next-generation solid-state batteries.

- 24.4 The global market for clean technologies is growing. Over the last decade, it has been expanding at an average rate of 20% annually, reaching close to €1.06 trillion in 2025.⁸⁰⁸ Under conservative assumptions, the market size for clean technologies is expected to equal that of the 2025 global crude oil markets by 2035.⁸⁰⁹ In 2023, clean-tech manufacturing was responsible for approximately 4% of global GDP growth and nearly 10% of investment growth.⁸¹⁰ In the EU, there are also increasing policy calls to accelerate the employment of green technologies, innovate and develop renewable energy and manufacturing.⁸¹¹ However, as pointed out in the Draghi Report (see Box 24.1), significant productivity and cost gaps remain between European clean-tech producers and leading international competitors, in particular Chinese firms. These gaps reflect differences in manufacturing scale, supply-chain integration, energy costs, and accumulated industrial capabilities (see Subsection 24.1.3).

24.1.2 R&D intensity and innovation in clean technologies

- 24.5 Innovation is a major determinant of competitiveness in clean technologies markets. Firms compete not only through scale, which provides unit cost-advantages, but also through continuous process innovations, such as improved manufacturing efficiency and material use, as well as product innovation (e.g. improved product performance or new compounds or materials that result in new or improved products).
- 24.6 Public and corporate R&D spending is steadily increasing. Between 2015 and 2023, it grew by around 6% per year in real terms.⁸¹² In absolute terms, EU-level public funding for clean technologies R&D amounted to €36 billion over the 2021–'27 period,⁸¹³ with the EU accounting for approximately 22% of all international patent families in clean and sustainable technologies between 2017 and 2021, and Germany, France, and the UK as the leading contributors within Europe.⁸¹⁴
- 24.7 Competitiveness in clean technologies is also shaped by strong learning-curve effects and associated cost reductions.⁸¹⁵ These have been driven by mass manufacturing and process improvements which have resulted in major cost reductions and increased productivity in technologies such as solar PV, wind turbines, and batteries over recent years. For example, solar PV module prices have fallen by more than 85% since 2010⁸¹⁶ (and by over 60% between 2019 and 2024 alone) driven by rising manufacturing efficiencies, economies of scale, and intense competition.⁸¹⁷ Battery pack costs (a major input for electric vehicles) have similarly declined sharply, falling by approximately 80% since 2015.⁸¹⁸ Lower costs and prices are a key condition for wider deployment of clean technologies: as they become more affordable, they are deployed at greater scale across more markets.

⁸⁰⁸ Converted from \$ to € using the ECB average exchange rate for 2025.

⁸⁰⁹ IEA (2026), 'Energy Technology Perspectives 2026', March, p. 15.

⁸¹⁰ IEA (2024), 'Advancing Clean Technology Manufacturing', May, p. 7.

⁸¹¹ Speech by Executive Vice-President Teresa Ribera (2025) (full reference in footnote 677).

⁸¹² IEA (2026), p. 205 (full reference in footnote 809).

⁸¹³ Draghi (2024), Part B, p. 123.

⁸¹⁴ European Investment Bank (2024), 'Financing and commercialisation of cleantech innovation', April, p. 23.

⁸¹⁵ IEA (2026), p. 40 (full reference in footnote 809).

⁸¹⁶ IEA (2026), p. 75 (full reference in footnote 809).

⁸¹⁷ IEA (2026), p. 331 (full reference in footnote 809).

⁸¹⁸ IEA (2026), p. 201 (full reference in footnote 809).

This is, for example, reflected in record renewable capacity additions in 2024, with global additions reaching 582 GW, an increase of 19.8% compared to 2023.⁸¹⁹

24.1.3 Productivity, sectoral importance, and challenges

- 24.8 The European clean technologies industry is of significant importance because it is central to achieving the EU's decarbonisation objectives by enabling the development and deployment of low-carbon technologies, being set to contribute to 65% of emission reductions by 2030.⁸²⁰
- 24.9 Given the scale and importance of these decarbonisation objectives, two main challenges emerge for the European clean technologies industry: a widening competitiveness gap with China, and strong concentration and dependency in critical supply chains, which both may increase technology deployment costs and hence lower adoption and deployment of clean technologies in the EU.
- 24.10 On competitiveness, China's competitive edge and lower costs reflect decades of accumulated advantages, including in innovation, large-scale production, manufacturing efficiency, a skilled workforce, integrated supply chains, and access to cheap resources and labour—all reinforced by consistent policy and financial support.⁸²¹ The current cost advantage of Chinese firms is particularly pronounced due to increases in European energy costs caused by geopolitical developments.⁸²²
- 24.11 On supply chain concentration, China accounts for 60–85% of production capacity in key clean technologies supply chains, and over 95% for some production steps.⁸²³ Therefore, the clean technologies sector is increasingly seen through a geostrategic lens, with supply chain resilience becoming a priority concern for policymakers in European economies.⁸²⁴
- 24.12 Additional challenges faced by European clean technologies manufacturers include high operating and capital costs, mainly driven by high energy and labour costs,⁸²⁵ and a persistent skills gap: job vacancy rates in clean technologies manufacturing doubled between 2019 and 2023, and 25% of EU companies reported labour shortages in 2023.⁸²⁶ In addition, the sector has recently been facing globally declining investment dynamics, with global investments falling by 50% between 2023 and 2025. This dynamic is mainly caused by existing overcapacities in production and policy uncertainties, such as the rollback of the 2022 US Inflation Reduction Act, which included substantial cuts to clean technologies subsidies.⁸²⁷

24.2 Policy tools

- 24.13 Several policy papers and reports (including the Draghi Report—see Box 24.1) have identified tools that can help unlock competitiveness and enable growth for the EU clean technologies industry and reduce the gap with Chinese rivals. While different

⁸¹⁹ IRENA (2025), 'Renewable Power Generating Costs in 2024', July, p. 15.

⁸²⁰ Draghi (2024), Part B, p. 116.

⁸²¹ IEA (2026), p. 38 (full reference in footnote 809).

⁸²² IEA (2026), p. 69 (full reference in footnote 809).

⁸²³ For example, China processes over 70% of critical raw materials (such as lithium, cobalt, graphite, and rare earths), and accounts for 95% of global production capacities for wafers as a core input for manufacturing solar cells. See IEA (2026), p. 38 (full reference in footnote 809).

⁸²⁴ IEA (2022), 'securing Clean Energy Technology Supply Chains', August.

⁸²⁵ Draghi (2024), Part B, p. 122.

⁸²⁶ Draghi (2024), Part B, p. 128.

⁸²⁷ Bruegel (2026), '[Understanding the global clean tech manufacturing slowdown](#)', accessed 21 July 2026.

clean technologies are quite heterogeneous in their development and hence their corresponding needs, three policy measures have been identified as key (though to different extents) across all clean technologies sub-sectors—trade policy measures, demand-side policies, and financial instruments.



Box 24.1 Insights from the Draghi Report on clean technologies

The Draghi Report identifies clean technologies as central to Europe's future competitiveness but warns that the EU risks remaining dependent on foreign production despite strong domestic demand and innovation capabilities. It highlights growing global competition—particularly from China and the US—and calls for securing strategic EU manufacturing capacity in sectors such as batteries, solar PV, wind, heat pumps, electrolyzers, and carbon capture utilisation and storage systems (CCUS).

The report recommends accelerating the implementation of the Net-Zero Industry Act and strengthening demand-side support through public procurement, Contracts for Difference auctions, and expanded EU financing instruments to scale clean technologies manufacturing and mobilise private investment. It also calls for simpler and faster access to EU funding instruments and for stronger coordination between Member States to avoid fragmented industrial support schemes across the EU.

It also emphasises the importance of innovation and skills, calling for clean technologies to become a strategic priority in future EU R&D programmes, alongside targeted support for breakthrough innovation, coordinated industrial projects, and measures to address shortages of specialised talent across the clean technologies sector. In addition, the report stresses the importance of securing resilient supply chains for critical raw materials and strengthening Europe's capacity to retain industrial know-how, intellectual property, and high-skilled workers within strategic clean technologies industries.

Source: Draghi (2024), Part B, Section 1, Chapter 5.

- 24.14 On trade policy measures, the EU and other major economies have increasingly turned to defensive trade policies to protect domestic clean technologies industries from low-cost foreign competition (particularly from China). For example, the US, the EU, and Canada have deployed anti-dumping and countervailing duties targeting Chinese imports of solar PV panels, wind turbines, and electric vehicles, aiming to 'level the playing field' for domestic producers, with the number of trade-restrictive measures in place worldwide rising almost sixfold in less than a decade, from just over 600 in 2015 to nearly 3,300 in 2024.⁸²⁸ The economic rationale is that domestic industries, faced with foreign competitors benefiting from state subsidies and lower production costs through large-scale manufacturing, cannot currently compete on price. Temporary trade barriers may prevent market displacement, preserve strategic manufacturing capacity, and allow domestic producers sufficient time to achieve cost competitiveness through investment and scale.
- 24.15 On demand-side policies, clean technologies growth continues to rely heavily on measures that stimulate market demand. This is particularly important in clean technologies markets, which are characterised by high upfront capital costs, long investment horizons, and strong learning effects. In such markets, demand-side policies can increase deployment, expand market size, and help firms achieve the

⁸²⁸ IEA (2026), p. 28 (full reference in footnote 809).

scale needed to lower costs and reach competitive production levels. Relevant instruments include purchase incentives for electric vehicles, renewable energy targets, energy efficiency standards, carbon pricing, and public procurement programmes.⁸²⁹ In the EU, many of these elements are covered as part of broader industrial policy initiatives such as the Net-Zero Industry Act and the Clean Industrial Deal, which aim to further strengthen demand by improving market conditions and supporting the expansion of domestic clean technologies ecosystems.⁸³⁰

- 24.16 Clean technologies deployment and manufacturing are capital-intensive activities in which the financing environment is a critical determinant of competitiveness. Especially in light of recently declining investments in the clean technologies sector (as discussed in Section 24.1), financial instruments such as tax credits, concessional loans, and investment guarantees can help attract capital to European plants to reach competitive scale.⁸³¹ This holds in particular for innovative firms at the scale-up stage where financing gaps and risk aversion by private investors can limit access to capital for commercially promising but technologically uncertain projects. For these firms, public policy measures—besides their role in promoting positive externalities associated with clean technologies—help close financing gaps by supplying direct capital, but also by crowding in private finance and providing a quality signal to the market.⁸³²
- 24.17 Merger control is also becoming an increasingly relevant tool in the clean technologies sector.⁸³³ The key challenge is to allow transactions that support decarbonisation, deployment, and circularity, while preventing mergers that weaken incentives to innovate in clean technologies or reduce access to key green inputs and infrastructure. Merger control also serves to protect future price competition, which in turn may affect the competitiveness of the value chain. The following section discusses three mechanisms through which merger control can support innovation and competition in the clean technologies sector.

24.3 Merger control

24.3.1 Scaling up through consolidation while preventing ‘green’ killer acquisitions

- 24.18 The large gap between European clean technologies firms and Chinese rivals can be largely attributed to significant cost advantages of Chinese manufacturers. These advantages stem from a combination of lower labour and energy costs, extensive state subsidies, but also China’s manufacturing scale which allows Chinese firms to produce at significantly lower unit costs.
- 24.19 Merger control can support clean technologies firms by enabling them to benefit from economies of scale and reduce costs. In turn, this may allow further investment in innovation, and accelerate technological development and the uptake of emerging clean technologies. This argument may in particular apply to European

⁸²⁹ IEA (2026), p. 367 (full reference in footnote 809).

⁸³⁰ European Commission (2025), ‘The Clean Industrial Deal: A joint roadmap for competitiveness and decarbonisation’, COM(2025) 85 final, February.

⁸³¹ IEA (2026), p. 367 (full reference in footnote 809).

⁸³² Draghi (2024), Part B, p. 4.

⁸³³ Academic research on M&As in clean technologies shows that mergers and acquisitions can redirect firms’ innovation trajectories: when companies acquire firms with clean-technology capabilities, especially across technologically distinct domains, acquirers tend to shift their innovation activity towards the target’s knowledge base and increase subsequent clean patenting (Newham, Jaggi, and Posth, 2026).

clean technologies innovators at the startup stage, which are facing significant financing gaps in comparison to US firms, and for which M&As may be an opportunity to close this gap.⁸³⁴

- 24.20 At the same time, merger control should guard against so-called ‘green’ killer acquisitions, where incumbents acquire disruptive clean technologies innovators in order to slow down or neutralise competitive pressure. This may be relevant in the clean technologies industry in which many innovators are small firms (with 95% of all enterprises in clean technologies-related manufacturing being classified as SMEs)⁸³⁵ and where some transactions may fall below standard notification thresholds.⁸³⁶ The risk is that, where an incumbent acquires an innovative clean technologies target with the purpose of discontinuing its research and pre-empting future competition, the technology in question is not developed or deployed, or may be significantly delayed.⁸³⁷

24.3.2 Accounting for potential harm and efficiencies in vertical mergers

- 24.21 Supply chains in the clean technologies sector are characterised by a high degree of interdependence and concentration across upstream and downstream stages. Typically, upstream firms supply raw materials, components, or infrastructure to downstream firms that manufacture or distribute final products and services. As a result, vertical mergers may be of particular significance in the sector. They can raise twofold competition concerns, in line with the discussion of foreclosure concerns in Part C of the main report (Chapter 18).
- 24.22 First, vertical integration may enable the input foreclosure of rival firms’ access to critical raw materials or components required for the development and manufacture of clean technologies and products.⁸³⁸ Where the merged entity has both the ability and incentive to restrict supply, downstream rivals may face higher costs, reduced competitiveness, and weaker incentives or ability to invest in innovation and scale-up. Second, such mergers may also give rise to customer foreclosure, resulting in upstream competitors facing reduced sales opportunities and diminished incentives to innovate or expand, which may slow down the development of new technologies.
- 24.23 For example, in Siemens/Gamesa (2017), the European Commission examined whether the merged entity (which manufactured both gearboxes and generators as inputs for wind turbines) would have the ability and incentive to foreclose rival wind turbine manufacturers from access to those components post-transaction. Similarly, in General Electric/LM Wind Power (2017)—a vertical merger between the wind turbine manufacturer GE and the blade supplier LM—the Commission assessed in detail whether the merged entity would have the ability and incentive to restrict blade supply to rival turbine manufacturers, examining the profitability of an input

⁸³⁴ European Investment Bank (2024) (full reference in footnote 814).

⁸³⁵ OECD (2025), ‘The evolution of cleantech manufacturing Implications for FDI and SME linkages across Europe’, October, p. 6.

⁸³⁶ European Commission (2023), ‘EU Green Mergers & Acquisitions Deals—How Merger Control Contributes to a Sustainable Future’, Competition Merger Brief, 2023-01, February, p. 6.

⁸³⁷ The European Commission has explicitly identified vigilance against ‘green’ killer acquisitions as a priority in a recent competition merger brief. See European Commission (2023) (full reference in footnote 836).

⁸³⁸ This concern is of particular relevance given the high concentration of critical raw material inputs in a small number of upstream producers and processors, predominantly in China.

foreclosure strategy and the extent to which downstream demand could be diverted to the merged entity.⁸³⁹

- 24.24 On the other hand, vertical mergers may also promote the implementation of more innovative, efficient, and profitable manufacturing processes across the supply chain. Where vertical integration secures reliable access to critical inputs, it can reduce transaction costs, lower supply chain uncertainty, and enable longer-term investment in process improvements and R&D. In the clean technologies sector specifically, vertical integration can enhance the circular use of raw or recycled materials, enabling companies to adopt more resource-efficient and low-carbon practices across broader segments of the value chain.⁸⁴⁰
- 24.25 For example, in Aurubis/Metallo (2020)—a case concerning access to copper scrap in the EEA—the parties claimed efficiencies from better valorisation of scrap through complementary technologies and from additional metal recovery and environmental benefits. The case was ultimately cleared unconditionally, with the Commission recognising that some of the first set of alleged efficiencies might partly materialise and be passed on to customers.⁸⁴¹

24.3.3 Sustainability as a relevant parameter of competition

- 24.26 Merger control can support the competitiveness of the European clean technologies sector and of the overall value chain by taking into account sustainability as a non-price factor (including features such as energy efficiency, emission intensity, and lifecycle emissions, material or resource efficiency, durability, and recyclability), when there is evidence that these are an important dimension of future competitive significance of firms (including their cost competitiveness), or that customers in the relevant markets consider these factors important.⁸⁴²
- 24.27 This assessment typically has a direct interplay with the relevant regulatory regime. For example, if emissions are priced under a cap-and-trade mechanism (e.g. the European Emissions Trading System), and the price of emission is set to rise in the future (as emission targets become more stringent), then technologies which enable stronger carbon abatement can be expected to become more important over time, and acquire greater competitive significance. Similarly, if regulatory standards around environmental objectives (e.g. air emissions, circular economy, energy efficiency) are expected to become stricter over time, technologies better able to meet these standards can be expected to play a more important role over time. A forward-looking approach to merger control can recognise these competitive dynamics, and seek to protect competition and innovation in valuable clean technologies.

⁸³⁹ In both Siemens/Gamesa (2017) and General Electric/LM Wind Power (2017), the Commission combined qualitative indicators (such as the extent of multi-sourcing or switching possibilities, and the availability of alternative suppliers) with quantitative indicators (such as market shares, shares of affected sales, and capacity data). It also examined whether any foreclosure strategy would be economically viable, including by considering foregone revenues and the likely downstream price impact of restricting access to the input.

⁸⁴⁰ European Commission (2025), p. 36 (full reference in footnote 830).

⁸⁴¹ European Commission (2023), p. 4 (full reference in footnote 836).

⁸⁴² OECD (2021), 'Environmental considerations in competition enforcement', November, p. 6. Recent Commission practice recognises that sustainability can constitute a relevant parameter of competition in clean technologies markets, to be incorporated when defining relevant markets or assessing closeness of competition—see discussion in European Commission (2023), pp 3–4 (full reference in footnote 836). Where consumers directly value sustainability, this readily falls within the scope of an ECW framework.

- 24.28 For a practical example of sustainability as a parameter of competition, in *Norsk Hydro/Alumetal* (2023, EU), the Commission assessed whether demand for low-carbon advanced aluminium alloys (driven by the automotive sector's decarbonisation targets) was sufficiently distinct to warrant separate market treatment, noting that the significant majority of market participants confirmed a significant and growing demand for low-carbon products. In the competitive assessment, sustainability considerations played a role as a non-price parameter of competition: specifically, the Commission investigated whether Alumetal was a potentially important competitive force in the supply of low-carbon advanced alloys made from recycled scrap, a route that was cheaper and less carbon-intensive than Hydro's primary production. The Commission also noted that particularly innovative firms or recent entrants may qualify as important competitive forces and may exert significant competitive pressure on incumbent firms in the market.⁸⁴³

⁸⁴³ In *Norsk Hydro/Alumetal* (2023), the Commission combined qualitative indicators (such as evidence of growing demand for low-carbon advanced alloys, internal documents and market feedback on Alumetal's scrap-based know-how, cost-efficiency, and potential importance as an innovative competitive force, and the closeness of competition in specific alloy segments) with quantitative indicators (such as market shares, HHI/delta HHI, market reconstruction data, and price comparisons between Hydro and Alumetal).

ANNEXES

A1 European Commission's four-layer framework

- A1.1 Following the two high-profile agrochemical mergers in 2017 and 2018,⁸⁴⁴ the European Commission has in several cases followed a four-layer analytical framework to assess dynamic horizontal concerns around innovation and R&D.⁸⁴⁵ This framework has since been applied and further developed in subsequent merger assessments, with a focus on pharmaceutical mergers.⁸⁴⁶
- A1.2 The four-layer framework aims to assess how mergers affect competition at different stages of the innovation process, taking into account expected longer-term effects on product market competition and innovation. The four layers are not applied cumulatively: a merger may raise competition concerns even if harm is identified at only one or two layers. Additionally, although the framework consists of four discrete layers, in practice there may exist a continuum between the different layers.⁸⁴⁷
- A1.3 In this Chapter, the focus of each layer is discussed based on our understanding of the Commission's decisional practice. Figure A1.1 at the end provides a general overview of each layer.

A1.1 First layer: overlaps between existing products

- A1.4 The first layer simply focuses on the effects of the merger on **overlapping products** of the parties. This layer primarily relates to static competition concerns, on price, output, or quality, based on product market competition taking place at the time of the merger. This layer does not capture any future product market competition or innovation competition beyond static effects.

A1.2 Second layer: overlaps involving advanced pipeline products

- A1.5 The second layer examines competition involving products that are not yet commercialised, but are at an advanced stage of development. This includes both product-to-pipeline and pipeline-to-pipeline overlaps. A **product-to-pipeline**

⁸⁴⁴ Dow/DuPont (2017) and Bayer/Monsanto (2018).

⁸⁴⁵ Although the framework was set out in Dow/DuPont (2017), paras 272–281 and Bayer/Monsanto (2018), para. 49, some of the underlying logic was already developed in earlier cases, e.g. Novartis/GlaxoSmithKline Oncology Business (2015).

⁸⁴⁶ The four-layer framework has since been used in the assessment of the following mergers: Bristol-Myers Squibb/Celgene (2019), para. 22; AbbVie/Allergan (2020), para. 19; AstraZeneca/Alexion Pharmaceuticals (2021), para. 12; Elanco Animal Health/Bayer Animal Health Division (2020), para. 136; and Pfizer/Seagen (2023), para. 20. Additionally, several cases since 2017 have relied to different degrees on the principles behind the four layers, even if they do not cite the framework explicitly. This includes for example Takeda/Shire (2018) (involving layer 2 competitive assessment), CSL/Vifor Pharma (2022) (involving layer 2 and layer 3 competitive assessments) and Novozymes/Chr. Hansen Holding (2023) (involving layer 2 and layer 4 competitive assessments). Further examples include case Siemens/Gamesa (2017), BASF/Bayer divestment business (2018), concerning the remedy divestment of assets to BASF in Bayer/Monsanto (2018), Nidec/Whirlpool (Embraco Business) (2019), and Siemens/Alstom (2019). Finally, the recently updated market definition notice by the Commission includes a discussion on market definition in the case of significant R&D, which aligns with the Commission's four-layer framework by referring to both market- and pipeline-to-pipeline products (layers 2 and 3) and to the concept of innovation spaces (layer 4). See EC Market Definition Notice, Section 4.3.

⁸⁴⁷ For example, the EC Market Definition Notice discusses how R&D activities may lie on a continuum between clearly identifiable pipeline products and more exploratory innovation efforts. Where research has become targeted but remains capable of leading to multiple alternative outcomes, the appropriate approach to market definition will depend on how far the R&D process has progressed along that continuum. See EC Market Definition Notice, Section 4.3.

overlap occurs when one merging party already has a product on the market and the other party has a pipeline product in advanced development, with both products targeting the same market. A **pipeline-to-pipeline overlap** occurs when both companies are developing competing products or services that are in advanced development but not yet on the market.

- A1.6 To determine whether projects are in advanced stages of development, the European Commission has relied on the R&D process in the industry at hand. For example in the case of the pharmaceutical industry, different phases are typically identified: Phase I tests drug safety and dosage in a small group of people; Phase II explores therapeutic efficacy in a small, selected patient group; Phase III confirms safety and effectiveness in a large patient population (hundreds to thousands); and Phase IV monitors adverse reactions post-approval.⁸⁴⁸ Advanced stages of development typically include late-stage R&D or clinical trials, usually in **Phase III or IV of clinical trials**.⁸⁴⁹
- A1.7 Table A1.1 provides a breakdown of each phase and the expected time to market and likelihood of success across pharmaceutical development phases. Advanced stages of development can therefore be interpreted as an expected time to market of at most three years and with a more-likely-than-not success rate (i.e. at least 50%).

Table A1.1 Pharmaceutical development phases

Phases	Expected time to market and likelihood of success
Phase I	Ca. 8–10 years before product is marketed, ca. 10% chance of success
Phase II	Ca. 4–5 years before product is marketed, ca. 30% chance of success
Phase III	Ca. 3 years before product is marketed, ca. 50% chance of success
Phase IV	Post-marketing phase conducted after regulatory approval

Note: This table shows the expected time to market and likelihood of success for different development phases.

Source: Glaxo Wellcome/SmithKline Beecham (2000), para. 70, see also discussion in Dow/DuPont (2017), para. 289.

- A1.8 At the R&D stages considered under the second layer, development is typically sufficiently advanced to identify the future product market in which the pipeline products are expected to compete. The Commission has assessed potential competition by considering, first, whether the merger may adversely affect future product market competition, including prices, assuming overlapping pipeline products were brought to market; and, second, whether the transaction increases the likelihood that one or more pipeline projects would be discontinued, thereby reducing future product choice.^{850,851}

⁸⁴⁸ See European Commission (2024), 'Assessing Innovation Competition in Pharma Merger', Competition Policy Brief, 2024-01, April.

⁸⁴⁹ Phase I tests drug safety and dosage in a small group of people. Phase II explores therapeutic efficacy in a small, selected patient group. Phase III confirms safety and effectiveness in a large patient population (hundreds to thousands). Phase IV monitors adverse reactions post-approval. See European Commission (2024) (full reference in footnote 848).

⁸⁵⁰ European Commission (2024), 'Non-price Competition: EU Merger Control Framework and Case Practice', Competition Policy Brief, 2024-01, April.

⁸⁵¹ Layer 2 concerns were found in, for example, Novartis/GlaxoSmithKline Oncology Business (2015) and Elanco Animal Health/Bayer Animal Health Division (2020).

A1.3 Third layer: early-stage innovation efforts

- A1.9 The third layer focuses on R&D projects at an early stage of development. In the case of the pharmaceutical market, this typically includes **Phase I or Phase II pipelines**—i.e. expected time to market between around 4–10 years and a likelihood of success between around 10–30%.
- A1.10 This layer is typically more complex than the preceding layers, as the relevant future product market may not be clearly identified. Early-stage R&D efforts often do not translate into clearly defined products, and it may also be unclear whether such innovation will fall into an already existing specific market, or give rise to a new one.⁸⁵² In this context, the European Commission has assessed whether the merger is likely to lead to the ‘discontinuation, delay, or redirection’ of important R&D projects as a result of combining the parties’ research activities. This can cause harm to consumers in the affected markets over the longer term, by reducing future product choice and weakening competitive pressure on rivals. Such harm may reinforce the competitive harm arising from the loss of rivalry between the parties and can materialise repeatedly over time as potential future interactions are eliminated.⁸⁵³ This harm could be sustained over time if merging parties have several overlapping early-stage research projects which could be discontinued post-merger. This is because reduced innovation effort today means fewer sources of competitive pressure in future markets (i.e. less competition between innovative products in the future) causing the competitive harm to persist over time.

A1.4 Fourth layer: structural reduction in overall innovative effort

- A1.11 Under the fourth layer, the European Commission further broadens the scope of its analysis by introducing the concept of **innovation spaces**.⁸⁵⁴ By innovation spaces, the Commission has referred to areas of research and development where innovation activities do not yet target a specific existing or clearly identifiable future product market.⁸⁵⁵ At this stage, R&D takes place at an early phase of the innovation process, before the contours of a product market have crystallised. In such circumstances, while firms may still compete in innovation, that competition is better understood as taking place within broader innovation spaces rather than within defined product markets.
- A1.12 Rather than focusing on specific overlaps between existing products or individual pipeline projects, the Commission has considered under layer four whether the transaction alters the overall incentives and capacity to innovate in the broader market. Even where no direct product-level or pipeline-level overlaps can be identified, and even where the merger does not lead to the immediate discontinuation, delay, or redirection of specific R&D projects, the Commission may consider whether a merger reduces the number of independent sources of innovation, weakens competitive pressure to invest in research and development, or diminishes incentives to pursue long-term or high-risk innovation. This layer therefore captures harm arising from a general reduction in the intensity or diversity of innovation efforts in the market.

⁸⁵² For a similar discussion in the context of market definition, see EC Market Definition Notice, Section 4.3.

⁸⁵³ Dow/DuPont (2017), para. 283.

⁸⁵⁴ European Commission (2024), p. 20 (full reference in footnote 848).

⁸⁵⁵ European Commission (2024), p. 6 (full reference in footnote 850).

Figure A1.1 European Commission's four-layer framework for dynamic effects

Layer 1: existing products

- Current product market overlaps
- Static effects on actual competition
- Innovation competition in existing products

Layer 2: advanced pipeline products

- Late-stage R&D
- Market-pipeline and pipeline-pipeline overlaps
- Foreseeable future product competition and discontinuation

Layer 3: early-stage innovation

- Discontinuation, delay or redirection of early-stage R&D
- Weakened innovation incentives and loss of future product market competition
- Focus on future product competition

Layer 4: innovation spaces

- Structural reduction in overall innovation
- Focus on market-wide innovation competition
- Fewer independent R&D efforts and lower diversity of innovation

Note: In the case of pharmaceutical mergers, late-stage R&D in layer 2 typically refers to Phase III–IV pipelines (i.e. later in the development pipeline), while early-stage R&D in layer 3 typically refers to Phase I–II pipelines (i.e. earlier in the development pipeline).

Source: Oxera, based on European Commission decisional practice.

A2 Case studies

A2.1 This annex sets out the case studies considered, illustrating the key aspects related to dynamic effects based on the relevant competition authorities' published decisions and commentary on such cases. The cases discussed include the following:

- Case 1: Facebook/GIPHY (2021, UK)
- Case 2: Adobe/Figma (2023; EU, UK)
- Case 3: Novartis/GlaxoSmithKline Oncology Business (2015, EU)
- Case 4: Elanco Animal Health/Bayer Animal Health Division (2020, EU)
- Case 5: Pfizer/Seagen (2023, EU)
- Case 6: Bayer/Monsanto (2018, US)
- Case 7: Illumina/PacBio (2019, UK; 2020, US)
- Case 8: Illumina/GRAIL (2023, US)
- Case 9: Orange/MásMóvil (2024, EU)
- Case 10: Vodafone/Three (2024, UK)

A2.1 Case 1: Facebook/GIPHY (2021, UK)

- Authority: CMA
- Relevant product market(s): UK display advertising services; global social media services
- Outcome: full divestiture of GIPHY⁸⁵⁶

A2.2 In May 2020, Facebook acquired GIPHY for around \$400 million. Facebook operates the world's largest social media platforms with approximately 98% of revenue derived from display advertising. At the time of the merger GIPHY was the leading provider of searchable GIF libraries to social media platforms, where GIFs are an engagement feature used by social media users in messaging and Stories. Approximately half of GIPHY's user traffic was generated through Facebook's own platforms, primarily Instagram. Since 2017 GIPHY had begun monetising its products, offering an innovative 'Paid Alignment' advertising service which allowed brands to promote their 'branded' GIFs within users' search results on social media platforms. Upon completing the merger, Facebook terminated all GIPHY's Paid Alignment activities (in fact GIPHY's revenue-generating operations were not part of the transaction).⁸⁵⁷

A2.3 After a nearly two-year-long process, the CMA identified—in its first Phase II decision published in November 2021—innovation concerns in display advertising, including the risk of a reduction in Facebook's incentives to innovate due to the elimination of innovation competition between the parties, and the elimination of innovation by GIPHY. The CMA also found concerns of input foreclosure of Facebook's rivals in social media. The parties were ordered to unwind the transaction and in May 2023 Facebook sold GIPHY to Shutterstock, an online stock image provider.

⁸⁵⁶ The first Phase II decision was published by the CMA in November 2021 and the second, following an appeal to the Competition Appeal Tribunal, was published in October 2022. References throughout this Section are to the first CMA decision—see Facebook/GIPHY (2021, UK). For commentary on the case see for example Bon et al. (2023); Walker, M. (2023), '[The UK Facebook/GIPHY case: Taking dynamic competition seriously](#)', accessed 21 July 2026; and Spiegel (2026).

⁸⁵⁷ Facebook/GIPHY (2021, UK), paras 2.10–2.14.

A2.1.1 Loss of dynamic competition in display advertising

- A2.4 The CMA's more novel theory of harm, which reflected the updated framework set out in the revised 2021 Merger Assessment Guidelines, centred on the loss of dynamic competition in innovation/product development in display advertising through the removal of GIPHY as a potential competitor to Facebook.⁸⁵⁸
- A2.5 While GIPHY had only recently started monetising its products through advertising and its revenues were still modest, the CMA focused on GIPHY's innovation capabilities, trajectory and potential to expand its advertising business on its own or in collaboration with other social media platforms, building on an already large user base. The assessment examined the impact of the merger in the context of Facebook's significant market power in display advertising and the high barriers to entry.⁸⁵⁹
- A2.6 The CMA acknowledged the uncertainty regarding the commercial success at scale for GIPHY's Paid Alignment model, but considered that GIPHY's innovation efforts had significant competitive value regardless of the ultimate product success. In particular, GIPHY's advertising product focused on an area of potential revenue growth for display advertising over which Facebook and other social media platforms had shown an interest (i.e. monetisation of messaging and Stories) and it would have been a close competitor to Facebook's display advertising at least for some advertising budget. As such, GIPHY's innovation activities in display advertising, and the competitive threat that it would increasingly pose, would have spurred innovation by Facebook and other social media platforms, increasing the likelihood of new advertising products emerging.⁸⁶⁰
- A2.7 The CMA assessment of this theory of harm focused on dynamic direct effects on innovation (business-stealing effects), whereby the elimination of GIPHY as an innovative potential competitor in display advertising would result in reduced incentives to innovate for Facebook (as well as other social media platforms).⁸⁶¹ However, the CMA also noted dynamic feedback loop effects in social media as a result of network effects in a two-sided platform, as discussed in Subsection A2.1.3.

A2.1.2 Vertical foreclosure in social media

- A2.8 The CMA's second theory of harm related to vertical effects in social media, through input foreclosure of rival social media platforms.⁸⁶² The CMA considered that Facebook could leverage GIPHY's strong position in GIF libraries to foreclose rival platforms through limiting their access to GIPHY's GIF libraries and future innovations, degrading service quality, reprioritising innovation to its rivals' disadvantage, or imposing onerous data-sharing requirements. This would reduce rivals' attractiveness to social media users, thereby reducing their competitive pressure on Facebook in social media.
- A2.9 The vertical effects in social media also included dynamic effects on innovation through two possible mechanisms. First, rivals would have more limited opportunities to innovate or improve platform features that use GIFs as an input. Second, by gaining access to rivals' data via GIPHY, Facebook may spot emerging

⁸⁵⁸ Facebook/GIPHY (2021, UK), Chapter 7.

⁸⁵⁹ Facebook/GIPHY (2021, UK), paras 7.45–7.48.

⁸⁶⁰ Facebook/GIPHY (2021, UK), paras 5.22–5.24, and 7.14–7.21.

⁸⁶¹ Facebook/GIPHY (2021, UK), para. 7.168.

⁸⁶² Facebook/GIPHY (2021, UK), Chapter 8.

competitive threats or trends in social media earlier and narrow its innovation efforts to target the areas of greater threat.⁸⁶³

A2.1.3 Multi-sided platform effects

- A2.10 The CMA assessment recognised that in multi-sided platform markets characterised by network effects, dynamic effects amplify through feedback loops.
- A2.11 On one hand, GIPHY's Paid Alignment model would have generated advertising revenues for social media platforms partnering with GIPHY, notably Facebook's rivals, enabling greater investment to attract users and intensifying competition in social media markets. On the contrary, with GIPHY under its control, Facebook would be able to reinforce its strong position in this space.⁸⁶⁴
- A2.12 On the other hand, input foreclosure in social media, by weakening social media rivals, including their ability to develop their platforms in the future, would have allowed Facebook to strengthen its position across social media and display advertising, thus exacerbating the effects on competition in display advertising arising from the loss of dynamic competition.⁸⁶⁵

A2.1.4 Efficiencies

- A2.13 The parties claimed post-merger dynamic efficiencies. In particular, they argued that, following the acquisition of GIPHY, Facebook could enhance user experience by significantly investing in additional GIPHY services and by pursuing further integration of GIPHY's library into Facebook's services, thereby allowing Facebook to offer more innovative products to users. This would in turn put competitive pressure on rivals to innovate.
- A2.14 The CMA rejected these claims as insufficiently substantiated and not merger-specific. The CMA also noted that, whether or not any such improvements by Facebook would create pressure on rivals to innovate and improve their own products, the competitive response by rivals would be undermined relative to the counterfactual because those rivals would be foreclosed from GIPHY, an important input to such innovation. In other words, Facebook's rivals in social media would no longer have access to the same opportunities as Facebook with regard to developing their own GIF-related innovations, which would lessen their ability to compete.⁸⁶⁶

A2.2 Case 2: Adobe/Figma (2023; EU, UK)

- Authority: European Commission (M.11033), CMA
 - Relevant product market(s): product design software, vector editing software, raster editing software
 - Outcome: withdrawal
- A2.15 In September 2022, Adobe announced its intention to acquire Figma for approximately \$20 billion. Adobe is the leading provider of desktop-based creative design software, with long-standing market leading products in vector editing (Illustrator) and raster editing (Photoshop), and is also present in product design software (Adobe XD). Figma, which was founded in 2012 and was a pioneer in

⁸⁶³ Facebook/GIPHY (2021, UK), paras 8.110 and 8.161–8.163.

⁸⁶⁴ Facebook/GIPHY (2021, UK), para. 44.

⁸⁶⁵ Facebook/GIPHY (2021, UK), paras 8.167–8.169.

⁸⁶⁶ Facebook/GIPHY (2021, UK), paras 8.12, 8.162, and 9.110–9.111.

developing collaborative web-based software technology, is the market leader in product design software (Figma Design).⁸⁶⁷

- A2.16 The transaction attracted parallel scrutiny from the US FTC, the European Commission, and the UK CMA. Following the Commission's Statement of Objections and the CMA's Provisional Findings in November 2023, the transaction was abandoned in December 2023. Figma successfully completed an IPO on the New York Stock Exchange in July 2025. The discussion which follows draws upon the CMA's Provisional Findings, the only published decision in this case, and the Commission's published commentary on the case, outlining commonalities and differences between the approaches adopted by the two authorities.^{868,869}
- A2.17 Both the Commission and the CMA considered two theories of harm. The first theory of harm was substantially equivalent and centred on the loss of competition between the merging firms on their current offerings and on product development/innovation in product design software, a market where the merging firms were the two larger players at the time of the merger.⁸⁷⁰ The second theory of harm related to vector and raster editing software, two of Adobe's core markets where Figma was not present yet. The two authorities articulated the second theory differently. The Commission's concern centred on the elimination of Figma as a potential competitor and the resulting strengthening of Adobe's dominance in these markets, while the CMA's concern centred on the loss of dynamic competition between the merging parties in product development and innovation in vector and raster editing software.⁸⁷¹

A2.2.1 Loss of competition in product design software

- A2.18 The horizontal theory of harm in product design software focused on the loss of competition between the merging parties' existing products, Adobe XD and Figma Design, as well as on the loss of competition in product development and innovation.
- A2.19 The first limb of the theory of harm focused on closeness of competition between the market leader, Figma Design, and its weaker second, Adobe XD, in the context of Figma's significant market power and the very limited alternatives available. The second limb related instead to the dynamic aspects, focusing in particular on Adobe's product development trajectory in product design in the counterfactual.
- A2.20 Recognising certain weaknesses of Adobe XD relative to Figma Design, Adobe had initiated Project Spice, aimed at developing a web-based collaborative product design tool as a competitive response to Figma and as the next iteration of Adobe

⁸⁶⁷ Product design software tools are used mainly to design websites, mobile applications, and other digital products. The European Commission refers to the relevant market as the supply of interactive product design tools (as the digital products involve some degree of user interaction), while the CMA as the supply of all-in-one product design (as the merging parties' products cover all main stages of product design). For simplicity, we will refer to product design software. With creative design software, users can create or edit digital assets, such as photos (so-called 'raster images'), graphic illustrations (so-called 'vector images'), and videos.

⁸⁶⁸ Adobe/Figma (2023, UK).

⁸⁶⁹ The European Commission published a press release of its Statement of Objections over Adobe's proposed acquisition of Figma and a Merger brief outlining its views on the case, see European Commission (2023), 'Commission sends Adobe Statement of Objections over proposed acquisition of Figma', Press Release, 17 November; and European Commission (2024), 'Adobe/Figma: Much Ado(he) About Nothing?', Competition Policy Brief, 2024-09, September. The case is also discussed in Atzeni et al. (2024).

⁸⁷⁰ European Commission (2023) (full reference in footnote 869); Adobe/Figma (2023, UK).

⁸⁷¹ European Commission (2023) (full reference in footnote 869); Adobe/Figma (2023, UK), para. 9.1.

XD. Investments in Adobe XD were significantly reduced in February 2022 (Adobe XD was put in 'maintenance mode') and Project Spice was scaled down and then cancelled a few days before the merger announcement. Both competition authorities investigated whether these decisions were motivated by the merger and considered that absent the merger Adobe would have continued to compete in product design software, including through its innovation efforts, whether by way of Adobe XD, Project Spice, or any potential successor product, and would have remained a close competitor to Figma.⁸⁷²

- A2.21 It was noted that Adobe would have a strong incentive to continue investing in the market for product design tools and to address existing competition, in order to protect its market position against the threat posed by competing products (such as Figma Design), as well as in order to increase the sales of its Creative Cloud product bundle.⁸⁷³ The CMA pointed to Adobe's efforts in product design over the years, including through its continued provision of Adobe XD and its development of Project Spice, as clear contemporaneous evidence that Adobe had the ability and incentive to continue its efforts in product design.⁸⁷⁴
- A2.22 In the context above and consistently with some evidence, Adobe's acquisition of Figma could be seen as an alternative to further developing its own product design software and, as such, a 'reverse killer acquisition', leading to less innovation and less (current and future) competition in the product design market.⁸⁷⁵

A2.2.2 Elimination of potential competition (European Commission) and loss of dynamic competition (CMA) in vector and raster editing software

- A2.23 In relation to the second theory of harm, while the European Commission's and the CMA's concerns both related broadly to the elimination of Figma as a potential competitor to Adobe in vector and raster editing software, the approaches of the two authorities differed in the way they articulated the framework and impact of the merger.
- A2.24 The Commission's concern centred on the elimination of Figma as a nascent competitive threat to Adobe in its core vector and raster editing software markets, with Figma potentially being particularly well-placed to use its position in the adjacent product design market, at the boundaries of Adobe's ecosystem, to innovate and enter Adobe's core markets. It was noted that one of the main competitive risks that incumbents with a wide ecosystem of products face is the emergence of nascent competitors active not at the core but rather at the boundaries of their ecosystem. By offering complementary services a new entrant could build up scale and unlock its own network effects at the fringes of the incumbent's ecosystem that it could later leverage to enter the core market(s).⁸⁷⁶ The Commission investigated this concern using the 'potential competition' framework set out in its horizontal merger guidelines, whereby Adobe would have strengthened its significant position in vector and raster editing tools by eliminating Figma as a potential competitor. It appears that the Commission was of the view that Figma already exerted a significant constraining influence on Adobe and that it was significantly likely to enter these markets and grow into an effective competitive

⁸⁷² European Commission (2023) (full reference in footnote 869), and European Commission (2024), pp 2–3 (full reference in footnote 869). Adobe/Figma (2023, UK), para. 8.374.

⁸⁷³ European Commission (2024), p. 3 (full reference in footnote 869).

⁸⁷⁴ European Commission (2023) (full reference in footnote 869), and European Commission (2024), pp 2–3 (full reference in footnote 869). Adobe/Figma (2023, UK), para. 8.374.

⁸⁷⁵ European Commission (2024), p. 3 (full reference in footnote 869). Adobe/Figma (2023, UK), para. 3.27.

⁸⁷⁶ European Commission (2024), pp 3–4 (full reference in footnote 869).

force, suggesting that the loss of both perceived and actual potential competition was a concern.⁸⁷⁷

- A2.25 The CMA's assessment focused on the loss of dynamic competition in product development and innovation between the merging parties in vector and raster editing software, in the context of Adobe's significant and enduring market power in these markets. The assessment recognised that Figma's current vector and raster editing functionalities were limited and that only limited investment had been made to develop them. However, it was noted that Figma had already taken concrete steps (including some limited investment) to develop its product offering (incrementally and through third-party extensions) and it had the ability and incentive to develop these functionalities over time from its presence at the boundaries of Adobe's core markets in the adjacent product design market.⁸⁷⁸ Alongside its organisational and technical capabilities, Figma's incentives were assessed considering customer overlap, workflow adjacency, size of the opportunity, and benefits of multi-market strategy, which overall pointed to a 'good strategic fit' with Figma's business.
- A2.26 The CMA acknowledged the uncertainties around Figma's successful development of vector and raster editing, but it deemed that for Figma to be perceived as a credible threat by Adobe, thus spurring an innovation response, it was not necessary to establish that its successful entry within a particular timeframe would be more likely than not.⁸⁷⁹ The CMA also recognised that there was some uncertainty around the exact timescale of Figma's development in vector and raster editing software, with these potentially not being the top development priority in the near term, but considered them to remain a near to medium-term priority.⁸⁸⁰
- A2.27 The CMA's decision noted evidence that Adobe perceived Figma as a threat to its core markets for vector and raster editing, and its flagship products Illustrator and Photoshop, as well as evidence of Adobe's competitive response to Figma through product development which sought to defend Adobe's wider Creative Cloud suite (e.g. the development of web versions of Illustrator and Photoshop, and the prioritisation of certain features in the desktop versions).⁸⁸¹

A2.2.3 Multi-market effects

- A2.28 The CMA considered the effects of network externalities, product complementarity, customer adjacency and multi-market presence as reinforcing the horizontal theories of harm (by increasing the merging parties' ability and incentive to enter or expand into each other's markets).⁸⁸²

A2.2.4 Dynamic efficiencies

- A2.29 Based on the CMA's provisional decision, it appears that the parties did not raise any efficiency argument. However, the parties claimed that the rationale for the merger centred on enabling Figma's web-based collaboration tools to reach Adobe's significantly larger customer base (through bundling and cross-selling), using Figma's web-based collaboration technology to innovate new products for

⁸⁷⁷ European Commission (2023) (full reference in footnote 869).

⁸⁷⁸ Adobe/Figma (2023, UK), paras 9.107, 9.288, 9.362–9.370.

⁸⁷⁹ Adobe/Figma (2023, UK), paras 9.64–9.68.

⁸⁸⁰ Adobe/Figma (2023, UK), paras 9.286–9.287.

⁸⁸¹ Adobe/Figma (2023, UK), paras 9.360–9.361.

⁸⁸² Adobe/Figma (2023, UK), paras 9.117–9.123.

customers, and improving Adobe's emerging social content creation application (Adobe Express).⁸⁸³

A2.3 Case 3: Novartis/GlaxoSmithKline Oncology Business (2015, EU)

- Authority: European Commission (M.7275)
- Relevant product market: targeted oncology therapies, in particular MEK and B-Raf inhibitors
- Outcome: Phase I clearance with remedies

A2.30 In November 2014, Novartis announced that it would acquire GlaxoSmithKline's oncology business, which included both marketed drugs and pipelines for the treatment of advanced cancers. Novartis was already a major global player in oncology, while GlaxoSmithKline had a limited presence in oncology markets (notwithstanding that it had successfully launched several high-profile cancer drugs).

A2.31 The transaction was reviewed by the European Commission, which raised horizontal effects concerns as a result of the parties' product-to-pipeline and pipeline-to-pipeline overlaps in MEK and B-Raf inhibitors in two specific oncology markets as well as, more broadly, their overlap in early R&D programmes on MEK and B-Raf inhibitors potentially relevant across several oncology markets.⁸⁸⁴ Following a Phase I investigation, the Commission approved the merger subject to a structural remedy.

A2.3.1 Advanced pipeline overlaps

A2.32 Consistent with its approach to market definition in the oncology space, the Commission identified two key affected markets, namely targeted therapies for MEK and B-Raf protein inhibitors used for the treatment of advanced melanoma and ovarian cancer.

A2.33 In advanced melanoma targeted therapies, the Commission identified product-to-pipeline overlaps, as Novartis' late-stage MEK and B-Raf inhibitors (MEK162 and LGX818) would have competed with GlaxoSmithKline's already-approved Tafenlar and Mekinist (as monotherapies and combination therapies).⁸⁸⁵ Roche was the only competitor present, so the merger would have led to a reduction from three to two independent MEK/B-Raf competitors.⁸⁸⁶

A2.34 In ovarian cancer, specifically low-grade serous carcinoma, the merging firms were running Phase III trials for MEK inhibitors (Novartis' MEK162 and GlaxoSmithKline's Mekinist), resulting in a pipeline-to-pipeline overlap, with AstraZeneca the only other competitor.

A2.35 The Commission's theory of harm was that the transaction would likely lead Novartis to disinvest or delay its own MEK pipelines because the acquired GlaxoSmithKline assets were more advanced and would cannibalise the returns from continuing Novartis' programs. The concern was that a merger-induced shift in incentives would substantially weaken future market competition, reducing the likelihood of

⁸⁸³ Adobe/Figma (2023, UK), para. 3.25.

⁸⁸⁴ European Commission (2024), p. 12 (full reference in footnote 848).

⁸⁸⁵ In the later formalised four-layer competitive framework, this would be identified as a layer 2 concern.

⁸⁸⁶ Novartis/GlaxoSmithKline Oncology Business (2015), para. 57.

new treatment options being developed and ultimately resulting in harm to consumers.

A2.3.2 Early R&D program overlaps

- A2.36 Beyond the specific product markets, the Commission also raised a concern in relation to the wider innovation markets for developing MEK and B-Raf protein inhibitors.⁸⁸⁷ The merger would combine two of the only three global research programmes working on MEK and B-Raf inhibitors (GlaxoSmithKline, Novartis, and Roche), which were being developed to treat several cancer types.⁸⁸⁸ In its assessment, while recognising that clinical outcomes remained uncertain at that stage, the Commission considered that the merging parties' Phase II research programmes were aimed at developing substitutable therapies and therefore competed with each other.^{889,890}
- A2.37 Before the merger, Novartis and GlaxoSmithKline were both independently developing MEK and B-Raf drugs and competed to push their projects forward. After the transaction, the merged entity would no longer face the same incentives to pursue both research lines in parallel. With only three firms active in this field, the loss of an independent research effort would significantly weaken innovation in an already concentrated market. The Commission concluded that the transaction would reduce the number of future products likely to reach the market and limit the variety of therapies available to patients in terms of safety, tolerability, and suitability for different medical needs, while also reducing future price competition.⁸⁹¹
- A2.38 Given the more advanced stage of development of GlaxoSmithKline's Mekinist and Tafinlar for the treatment of advanced melanoma, the Commission considered it likely that the merged entity would prioritise the development of GlaxoSmithKline's research programmes on these inhibitors also in other types of cancer, while abandoning or delaying Novartis' competing research programmes. The Commission pointed out that this would also likely negatively affect areas where GlaxoSmithKline's and Novartis' research programmes did not currently compete, such as in uveal melanoma where only Novartis was active with a Phase III MEK pipeline programme. Following the prioritisation of GlaxoSmithKline's research programmes over Novartis' in overlap areas, the expected return on the sales of Novartis' product only for this type of cancer might not justify the continuation of this programme.⁸⁹²

A2.3.3 Remedies

- A2.39 To address these concerns, the Commission cleared the merger subject to a structural remedy. Novartis was required to sell its entire MEK/B-Raf programme, including MEK162 and LGX818, to Array BioPharma, the original licensor. The transfer included all necessary intellectual property, data, and support to ensure the projects could continue, allowing Array to complete the ongoing Phase III clinical studies.

⁸⁸⁷ Federico, Scott Morton, and Shapiro (2020).

⁸⁸⁸ Including colorectal cancer, non-small-cell lung cancer, advanced melanoma brain metastases, and uveal melanomas.

⁸⁸⁹ Novartis/GlaxoSmithKline Oncology Business (2015), para. 108.

⁸⁹⁰ This would be identified as a layer 3 concern in the four-layer competitive framework formalised by the European Commission later in its practice.

⁸⁹¹ Novartis/GlaxoSmithKline Oncology Business (2015), paras 104–105.

⁸⁹² Novartis/GlaxoSmithKline Oncology Business (2015), paras 106 and 111.

- A2.40 Following the divestiture, Array BioPharma launched MEKTOVI and BRAFTOVI in 2018, which were approved in the US and the EU for advanced melanoma, and later for non-small-cell lung cancer, competing with GlaxoSmithKline's Tafinlar and Mekinist.

A2.4 Case 4: Elanco Animal Health/Bayer Animal Health Division (2020, EU)

- Authority: European Commission (M.9554)
 - Relevant product market: pharmaceutical products for companion animals and production animals
 - Outcome: Phase I clearance with remedies
- A2.41 In August 2019, Elanco Animal Health announced its proposed acquisition of the Bayer Animal Health (BAH) division, with the transaction valued at US\$7.6 billion. Elanco Animal Health is a US-based animal health company and BAH comprised Bayer's animal health business. At the time of the merger both Elanco Animal Health and BAH developed, manufactured, and marketed veterinary products for pets and livestock worldwide, with the transaction leading to the creation of the second-largest animal health company globally (after Zoetis).
- A2.42 The European Commission, alongside other competition authorities, reviewed the transaction and in June 2020 approved it, subject to divestment remedies.⁸⁹³ The main overlaps between the merging parties arose in animal pharmaceuticals for both companion animals (CA) and production animals (PA) across multiple veterinary therapeutic areas.⁸⁹⁴ The Commission assessed the merger using the four-layer competitive assessment framework,⁸⁹⁵ and identified horizontal competition concerns in relation to current market overlaps (layer 1)⁸⁹⁶ as well as advanced and early-stage pipeline overlaps (layers 2 and 3). In what follows, the discussion will focus on the assessment of the dynamic effects of the merger.

A2.4.1 Pipeline overlaps

- A2.43 The Commission identified dynamic concerns in relation to Isox-based ecto/endecto parasiticides for CA. Isox products are a new chemical class of fast-acting oral or topical parasiticides used to treat fleas and ticks in CA; they are sold as prescription-only at a higher price than more traditional ecto/endecto treatments and highly valued for their wide spectrum of indications, long duration of efficacy, and ease of administration. Isox constitutes a rapidly growing, innovation-driven chemical class and it is supplied by very few firms.

⁸⁹³ The deal was also approved subject to remedies in the US and Australia, among other countries.

⁸⁹⁴ As regards product market definition, based on precedents, the Commission divides animal health products into three categories, for companion animals (CA) and production animals (PA) respectively: biologicals (e.g. vaccines); pharmaceuticals; and feed supplements. Within CA/PA pharmaceuticals, the product market is segmented into parasiticides, otitis products, antimicrobials, endocrine therapies, anti-inflammatories, and analgesics. The product markets may be further segmented based on a number of factors (such as species, active ingredient/chemical class, mode of administration, duration of efficacy).

⁸⁹⁵ This is discussed in detail in Annex A1.

⁸⁹⁶ The Commission identified horizontal effects concerns (layer 1) in several marketed CA and PA segments, including CA endoparasiticides, with combined shares above 50% in several Member States, CA long-acting otitis products, where the merger would create a monopoly (Osumia/Neptra), and PA anticoccidials for sheep and cattle, characterised by high combined shares, limited generic presence, and high barriers to entry.

A2.44 In the Isox markets, Elanco Animal Health was already active with Credelio (launched in 2017), and both merging parties had a number of Isox pipelines, giving rise to both product-to-pipeline overlaps and pipeline-to-pipeline overlaps. The Commission raised concerns around the loss of potential (product and price) competition between the merging parties' current/future Isox-based products in relation to advanced pipeline overlaps (layer 2) as well as the loss of innovation competition resulting from the discontinuation, delay, or redirection of early-stage pipeline projects overlapping with the other party's existing/pipeline products (layer 3). The Commission found that the merging parties' Isox-based products would likely be close competitors in a market with a very limited number of current/potential competitors, absence of generic competition for several years and very high barriers to entry. In this context, the significant risk of cross-cannibalisation would limit the merged entity's incentives to develop and market in parallel BAH's and Elanco Animal Health's Isox products, leading to a reduction in innovation and potentially less choice and higher prices.⁸⁹⁷

A2.4.2 Broad innovation concerns

A2.45 Outside the Isox markets, the Commission did not find broader innovation concerns related to a reduction in innovation capabilities in specific innovation spaces (over and beyond the competitive situation for specific marketed and pipeline products). While barriers to entry into innovation in the animal health space are high (particularly for parasitocides), Elanco Animal Health and BAH were smaller than the top three global players in R&D (i.e. Zoetis, Merck, and BI) and medium-sized companies had the capability of bringing new products to market across all relevant product areas. Finally, the Commission noted evidence suggesting that the merger may have a neutral or even positive effect on innovation, with the combination of the parties' R&D activities/capabilities potentially creating synergies and accelerating product development.⁸⁹⁸

A2.4.3 Remedies

A2.46 The Commission approved the transaction subject to divestments which removed all overlaps between Elanco Animal Health and BAH in the markets where competition concerns were identified. These included Elanco Animal Health's products and/or pipelines in relation to otitis for pets and antioctocidals for ruminants at global level, and BAH's products and/or pipelines in parasitocides for pets in the EEA/UK, including all the necessary assets such as applicable licenses, contracts, and brands, as well as relevant studies and data.

A2.5 Case 5: Pfizer/Seagen (2023, EU)

- Authority: European Commission (M.11177)
- Relevant product market: oncology targeted therapies, in particular those used for the treatment of bladder, breast, cervical, colorectal, leukaemia, lymphoma, and lung cancers
- Outcome: Phase I clearance

A2.47 In March 2023, Pfizer announced its proposed acquisition of Seagen. Pfizer is a pharmaceutical company active worldwide in the research, development,

⁸⁹⁷ Elanco Animal Health/Bayer Animal Health Division (2020), paras 135–152. The Commission did not reach a conclusion on the exact scope of the product market (i.e. whether it was limited to Isox-based products only or it included all ecto/endecto products), as it found competition concerns on both a narrower and a wider basis.

⁸⁹⁸ Elanco Animal Health/Bayer Animal Health Division (2020), paras 315–320.

manufacturing, and marketing of innovative medicines. Seagen is a biotechnology company that develops and commercialises antibody-drug conjugate (ADC) therapies to treat cancer. The transaction concerned oncology markets characterised by intensive research activity and high barriers to entry. In October 2023, following a Phase I investigation, the European Commission unconditionally cleared the merger, finding that it would not significantly impede effective competition.

A2.5.1 Existing product, advanced pipeline, and early R&D program overlaps

- A2.48 The Commission applied the four-layer competitive assessment framework⁸⁹⁹ used in other cases to examine the transaction's effects on competition.⁹⁰⁰ The relevant product markets identified were ADC therapies for the treatment of bladder, breast, cervical, colorectal, leukaemia, lymphoma, and lung cancer, as these were the areas in which the parties were active and where potential overlaps could arise.
- A2.49 Within these relevant markets, the investigation showed that Pfizer and Seagen rarely competed directly. Under the first three layers of the Commission's assessment, their portfolios were found to be largely complementary, since their products typically had different mechanisms of action and were therefore more likely to be used in combination rather than as substitutes. Furthermore, in all markets examined, strong competitive pressure from other firms remained. Because of this lack of direct competition, the Commission did not expect the merger to result in the discontinuation, delay, or reorientation of the parties' ongoing and overlapping lines of research, pipeline projects, or current marketed products, nor to create incentives for price increases (i.e. no concerns under layer 1, 2, or 3). As a result, the transaction was not expected to reduce options for patients or weaken competitive pressure in any of the cancer markets identified.

A2.5.2 Innovation spaces

- A2.50 The Commission also examined, under the fourth layer of its assessment of innovation effects, whether the merger could reduce innovation and incentives to innovate in oncology, but found no evidence supporting such a risk. The Commission concluded that the oncology sector was characterised by exceptionally intense and diversified research activity, with a large number of firms pursuing new treatments across almost every cancer type.⁹⁰¹ This high level of fragmentation generated strong competitive pressure, and meant that the transaction was not likely to influence the overall pace or direction of research and development in the sector.⁹⁰²
- A2.51 The evidence in this case suggested that the transaction would have a neutral or even slightly positive effect on innovation.⁹⁰³ In particular, market experts noted that the merger was likely to create R&D synergies, by combining Pfizer's global research, development, and commercial capabilities with Seagen's specialised antibody-drug conjugate technology, enabling more efficient scaling, coordination, and progression of Seagen's pipeline projects.⁹⁰⁴

⁸⁹⁹ This is discussed in detail in Annex A1.

⁹⁰⁰ Pfizer/Seagen (2023), para. 20.

⁹⁰¹ European Commission (2024), p. 20 (full reference in footnote 848).

⁹⁰² Pfizer/Seagen (2023), para. 183.

⁹⁰³ Pfizer/Seagen (2023), para. 185.

⁹⁰⁴ Mergersight (2023), ['Pfizer's \\$43bn Acquisition of Seagen'](#), accessed 21 July 2026.

A2.6 Case 6: Bayer/Monsanto (2018, US)

- Authority: DOJ
- Relevant product market: 17 distinct agricultural markets
- Outcome: clearance with remedies

A2.52 In September 2016, Bayer announced its proposed acquisition of Monsanto for \$66 billion. Bayer and Monsanto were, alongside DowDuPont and Syngenta, major global players in agricultural products. In May 2018, the US DOJ filed a complaint alleging that the merger would lessen current and future competition in 17 distinct agricultural product markets, while simultaneously filing a proposed settlement that, if approved by the court, would resolve the DOJ's concerns. The DOJ's complaint identified both horizontal and vertical theories of harm involving static effects and dynamic effects on innovation.⁹⁰⁵ Following parallel investigations by the DOJ and the European Commission⁹⁰⁶ the transaction was approved subject to a large package of divestments.

A2.6.1 Horizontal and vertical effects

A2.53 The DOJ found that in each relevant market the merger would substantially lessen competition either through the loss of head-to-head competition between the merging parties in highly concentrated markets or, in the case of certain seed treatments, through input foreclosure. The anticompetitive effects identified included higher prices, lower quality, and reduced choice as well as a significant negative impact on innovation, as discussed in Subsection A2.6.2.⁹⁰⁷

A2.6.2 Innovation effects

A2.54 The DOJ identified two potential dynamic concerns: threats to future competition and harm to innovation competition.

A2.55 First, the DOJ identified several markets where future competition between the two parties would be reduced post-merger. Both Bayer and Monsanto had several research pipelines in the markets for herbicide-tolerant cotton traits and GM soybeans, that could lead to new or increased competition absent the merger.⁹⁰⁸ Bayer was developing new foundational herbicides and related herbicide-tolerant traits for soybeans and corn that would rival Monsanto's Roundup Ready-based systems, while Monsanto was pursuing improvements to its Roundup formulations.⁹⁰⁹

A2.56 Second, the DOJ identified instances where the merger would have a significant negative impact on innovation. In integrated weed-management systems—those combining seeds, traits, and crop protection products, supported by digital-farming technologies and other services—Bayer and Monsanto were two out of four

⁹⁰⁵ The DOJ classified the agricultural product markets identified into three broad categories: genetically modified (GM) seeds and traits, crop protection (including foundational herbicides and seed treatments), and vegetable seeds. Bayer/Monsanto (2018, US), para. 14.

⁹⁰⁶ The Commission also looked at the merger between Bayer and Monsanto. The concerns were similar to those found by the US DOJ, and included innovation concerns in traits, herbicides, and weed management systems. The Commission found that the merging parties were close competitors in these innovation areas, and the merger would have eliminated Bayer as a key challenger to Monsanto's dominant position in traits and in weed management. See Bayer/Monsanto (2018, US). For a discussion, see also: Coublucq, Kovo, and Valletti (2023).

⁹⁰⁷ Bayer/Monsanto (2018, US), para. 59.

⁹⁰⁸ Bayer/Monsanto (2018, US), paras 25 and 31.

⁹⁰⁹ Bayer/Monsanto (2018, US), paras 35–37 and 60–61.

companies leading R&D efforts in the industry and capable of developing innovative systems in competition with each other.⁹¹⁰ The DOJ complaint also identified similar innovation concerns in certain vegetable seed markets where competition relied significantly on the parties' innovative efforts to introduce new varieties through traditional plant breeding.

- A2.57 By combining the competing R&D programs for weed-management systems and vegetable seeds, the transaction would reduce incentives to innovate. The DOJ Competitive Impact Statement explained that without the merger each merging firm had strong incentives to innovate because of the prospect of capturing market share from their rival. By contrast, with the merger innovation efforts would be reduced due to the internalisation of cannibalisation effects (i.e. the gains from innovative products would cannibalise the merged entity's own products rather than capture rival market share). The DOJ complaint concluded that this loss of innovation rivalry would likely diminish future product development and reduce available options for farmers in the future, especially in the field of integrated solutions where only DowDuPont and Syngenta would be capable of exerting real competitive pressure.⁹¹¹

A2.6.3 Remedies

- A2.58 To preserve competition, Bayer was required to divest to BASF SE the businesses that competed directly with Monsanto and those that could enable the merged entity to raise rivals' input prices. Bayer also transferred intellectual property and research capabilities to enable BASF to replace it as a leading innovator, along with additional assets necessary to provide BASF with the scale and scope to compete effectively.

A2.7 Case 7: Illumina/PacBio (2019, UK; 2020, US)

- Authority: CMA, FTC
 - Relevant product market: next-generation sequencing systems
 - Outcome: withdrawal
- A2.59 In November 2018, Illumina announced its proposed acquisition of PacBio for \$1.2 billion. Illumina and PacBio both provided next-generation sequencing (NGS) systems, which are advanced technologies that enable rapid and simultaneous sequencing of millions of DNA and RNA fragments. In particular, Illumina specialised in short read NGS systems, processing a large number of short DNA strands simultaneously, whereas PacBio specialised in long read systems, processing longer strands of DNA at a slower pace and higher cost.⁹¹² Short read and long read NGS systems are highly differentiated products along several dimensions, including read accuracy, read length, and throughput/cost (although the relative importance of each factor depends on the customer, research question, or project).
- A2.60 The transaction was investigated by the CMA and the FTC. Both agencies raised horizontal effects concerns as a result of the loss of competition between the

⁹¹⁰ Bayer/Monsanto (2018, US), p. 10: 'Absent the merger, Bayer and Monsanto would have each incentives to pursue these competing pipeline projects because any new innovation developed would help win market share from the other. In contrast, the merged firm will have different incentives due to heightened concerns that new innovation would simply cannibalize sales'.

⁹¹¹ Bayer/Monsanto (2018, US), para. 60.

⁹¹² Short read sequencers, also referred to as second generation sequencing, started to emerge in 2005, while long read sequencers, also referred to as third generation sequencing, have been developed since 2010. Consistently with the CMA's provisional findings, 'NGS systems' is used here to encompass both sequencing systems.

merging parties, with a clear focus on competition over innovation. The two authorities articulated the theories of harm slightly differently, reflecting the different frameworks in their respective merger guidelines.⁹¹³ The CMA assessed the risk of horizontal effects as a result of the reduction of current and future competition in areas where Illumina and PacBio overlapped or were likely to overlap in the future; and/or as a result of the elimination of potential competition from Illumina in the long read segment. The FTC's theories of harm centred on the removal of PacBio as a competitive threat to Illumina and the elimination of current and future competition between Illumina and PacBio, while also noting that the acquisition was presumed likely to enhance market power given the high levels of concentration in the NGS market. Following the parallel investigations by the two competition authorities, the parties abandoned the transaction in January 2020.

- A2.61 In what follows, we discuss the CMA's findings, with a focus on the dynamic aspects of the assessment, given the FTC only published a short public version of their complaint (which is consistent with the CMA's findings).

A2.7.1 Loss of current and future competition in NGS systems

- A2.62 The competitive assessment of the impact of the merger was carried out in the context of a dynamic, rapidly growing and evolving market for NGS systems, where innovation (in terms of both improved features and new products) was a key parameter of competition.⁹¹⁴ The market was highly concentrated, with Illumina holding a substantial degree of market power (approximately 80% global market share), and characterised by high barriers to entry.⁹¹⁵
- A2.63 With high differentiation between sequencing technologies, the CMA recognised that for a significant portion of the current market, the merging parties' NGS systems were seen by customers as complements rather than substitutes. However, the CMA found that the parties were competing in some situations currently and that they were likely to compete more closely in the future given their converging innovation trajectories, with R&D devoted to improving each party's technology to narrow the gap between them and address a wider range of uses, applications, and/or projects.⁹¹⁶
- A2.64 The CMA noted that PacBio was regarded by Illumina as a disruptive force in the market and a competitive threat, and that recent improvements to the efficiency of PacBio's technology (including the launch of Sequel II) resulted in some partial switching by customers from Illumina to PacBio, placing competitive pressure on Illumina to improve its own technology in response. While acknowledging some uncertainty around the timing and extent of convergence, the CMA considered that, as the parties' sequencing systems increasingly overlapped, this race for innovation would have intensified absent the merger.⁹¹⁷
- A2.65 Regarding the dynamic effects of the merger, in addition to the reduction in future competition between the merging parties in the market for NGS systems, the CMA emphasised that the merger, by eliminating the threat of PacBio on Illumina (and vice versa), which was a factor currently driving their innovation efforts, would have led to a reduction in R&D/innovation. In addition, the CMA noted that the merger would likely result in a shift in the direction of the parties' R&D. In particular, PacBio

⁹¹³ Both authorities assessed the transaction under their respective 2010 merger guidelines (i.e. prior to the revised merger guidelines coming into force).

⁹¹⁴ Illumina/PacBio (2019, UK), paras 8.39 and 8.63.

⁹¹⁵ Illumina/PacBio (2019, UK), paras 8.270–8.272 and 9.64.

⁹¹⁶ Illumina/PacBio (2019, UK), para. 8.297.

⁹¹⁷ Illumina/PacBio (2019, UK), paras 8.287–8.299 and 8.332.

would be incentivised to focus its R&D towards uses where its systems would be complementary to those of Illumina instead of uses where it would compete with them.⁹¹⁸

A2.7.2 Loss of potential competition from Illumina in long read

- A2.66 The second limb of the horizontal effects concerns focused on the long read segment of the NGS market and potential harm to customers with a strong preference for long read technologies for their specific projects.
- A2.67 The CMA noted that Illumina had recognised the benefit of long read technology and its growth potential, with the rationale for the merger being driven by its desire to supply long read systems. The CMA's assessment focused on Illumina's ability and incentive to enter the long read segment by developing its own technology. The CMA found that Illumina was well placed, with respect to other firms, given its resources, the extent of its customer relationships, well-established distribution networks, and history of commercialisation to develop and launch such a system. While recognising the uncertainty around when and if Illumina would have launched a commercial long read system absent the merger, given the importance Illumina attached to having a presence in long read sequencing and its development plans, the CMA considered that Illumina had a clear incentive to research long read technologies with a view to launching its own long read system.⁹¹⁹
- A2.68 As to the dynamic effects resulting from the loss of potential competition from Illumina, the CMA identified effects in terms of reduction in the potential future number of options for customers and projects that require a long read technology as well as in terms of a reduction in the incentives to compete and innovate of other competitors as a result of the removal of the threat of entry by Illumina.⁹²⁰

A2.7.3 Efficiencies

- A2.69 The merging parties presented several efficiency claims and the CMA carried out an extensive assessment using the framework set out in its merger guidelines. Efficiency claims included (among other things) the acceleration of innovation and the potential for improving PacBio's commercial operations through a wider distribution of PacBio's products using Illumina's global infrastructure. While the CMA found that the merged entity would likely have the ability to improve on PacBio's commercial operations, and to speed up the development of PacBio's technology through higher levels of investment and existing know-how, it questioned its incentive to do so, the pass-through to customers, and the merger-specificity of the potential efficiencies, ultimately dismissing them.⁹²¹

A2.8 Case 8: Illumina/GRAIL (2023, US)

- Authority: FTC
 - Relevant product markets: next-generation sequencing systems; multi-cancer early detection tests
 - Outcome: full divestiture of GRAIL
- A2.70 In September 2020, Illumina announced its proposed acquisition of GRAIL for \$8 billion. Illumina was an established supplier of next-generation sequencing (NGS)

⁹¹⁸ Illumina/PacBio (2019, UK), paras 8.333–8.334.

⁹¹⁹ Illumina/PacBio (2019, UK), paras 8.300–8.307.

⁹²⁰ Illumina/PacBio (2019, UK), paras 8.307 and 8.334.

⁹²¹ Illumina/PacBio (2019, UK), paras 9.93 and 9.145.

systems for genetic and genomic analysis,⁹²² while GRAIL was a customer of Illumina using NGS systems to develop multi-cancer early detection (MCED) tests, a new and emerging market at the time of the merger.

A2.71 The transaction was investigated by both the US FTC and the European Commission.⁹²³ In March 2023, the FTC issued its final opinion concluding that the merger would significantly lessen competition in the market for MCED tests, as a result of vertical foreclosure by Illumina of GRAIL’s MCED test competitors. The FTC’s assessment of the merger centred on harm to innovation competition among firms not yet present in the market but engaged in research and development of MCED tests for subsequent commercial sale. The FTC rejected Illumina’s proposed behavioural remedies, concluding that only structural remedies—specifically the full divestiture of GRAIL—could adequately address the dynamic competitive concerns.⁹²⁴ Illumina completed the divestiture of GRAIL in April 2024 and GRAIL became a publicly traded company in June 2024.

A2.8.1 Vertical foreclosure of R&D competitors

A2.72 The transaction concerned two vertically related markets: the supply of NGS systems (upstream input) and the development and commercialisation of MCED tests (downstream application),⁹²⁵ as shown in Figure A2.1.

A2.73 MCED tests are blood-based screening tests that can detect multiple cancers in asymptomatic individuals at an early stage, relying on NGS technology as a critical input. The FTC found that, at the time of the merger, GRAIL was the only supplier with an MCED test commercialised in the market (Galleri) and was in an innovation race with other cancer screening companies to develop and improve these tests. While the FTC noted that rival tests were at different stages of development and tests could ultimately differ in their features and performance, it pointed out that MCED test developers were on the same ‘racetrack’, with intense competition in R&D between them.⁹²⁶

A2.74 The core competition concern raised by the FTC (as well as the Commission) was vertical foreclosure by Illumina of GRAIL’s MCED test rivals with a particular focus on the dynamic effects of the merger on innovation and future developments in the market for MCED tests. The FTC found that Illumina had both the ability and the incentive to foreclose downstream rivals, noting that Illumina was the only supplier of short read NGS technology with the characteristics that MCED test developers needed (in terms of accuracy, throughput, and cost) and that it would have earned substantially more profit on the sale of GRAIL tests (currently the only commercialised product) than by supporting rival test developers. The increased

⁹²² For an explanation on NGS systems, see the Illumina/PacBio case study.

⁹²³ Illumina/GRAIL (2023, US). The European Commission’s prohibition decision (2022) was eventually annulled by the Court of Justice (2024), which ruled that the Commission lacked jurisdiction. The Commission’s decision was not published, but a high-level summary of the case can be found in the press release accompanying the prohibition (European Commission (2022), ‘Mergers: Commission prohibits acquisition of GRAIL by Illumina’, Press Release, 6 September). The focus of this case study will therefore be on the FTC’s assessment, noting that the two authorities investigated the same theory of harm. For commentary on the case and its implications, see Alderman and Blair (2023), and Hole, LeGower, Lipsitz, and Nevo (2024).

⁹²⁴ Illumina/GRAIL (2023, US), pp 66–73 and pp 97–98.

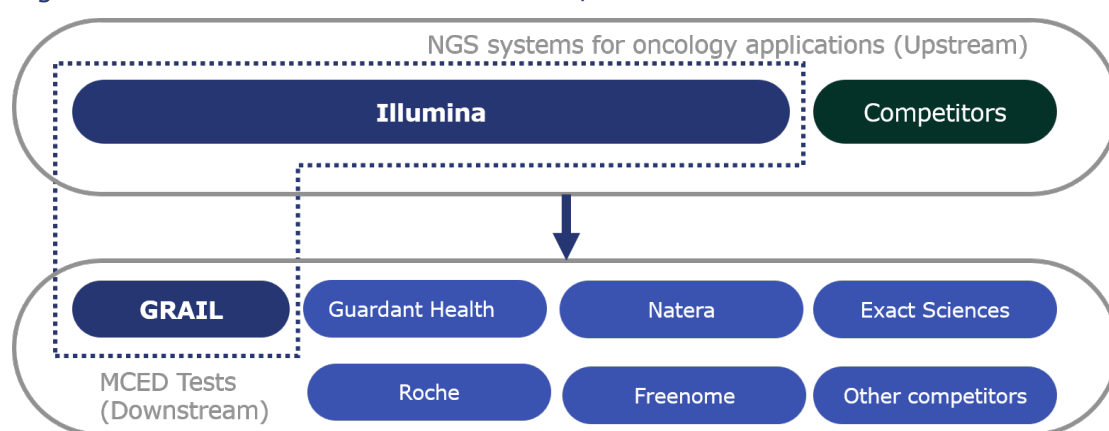
⁹²⁵ Illumina/GRAIL (2023, US), pp 30–32.

⁹²⁶ Illumina/GRAIL (2023, US), pp 30–32.

incentive to foreclose brought about by the merger was further amplified by the projected growth and size of the MCED test market.⁹²⁷

- A2.75 In terms of dynamic effects, by foreclosing MCED test rivals (through raising their costs, degrading their access to the NGS technology and services, withholding/delaying/limiting their access to new or improved technology, or reducing cooperation), Illumina would have stifled current innovation competition and increased barriers to entry, creating an unlevel playing field and ensuring that GRAIL would ultimately win the innovation race.⁹²⁸ The FTC emphasised that the harm to R&D competition was current and immediate, although the effects on commercialised sales of MCED tests (in terms of higher prices, reduced choice, and lower quality) might not be felt immediately. Reduced innovation, lower quality, or lower availability of competing MCED tests would harm US patients, depriving them of the benefit of multiple competitors pursuing diverse scientific approaches in an effort to design the best performing product.⁹²⁹

Figure A2.1 Market structure in Illumina/GRAIL



Note: Illumina's competitors include for example Thermo Fisher Scientific and Danaher. GRAIL's competitors were still developing in order to enter the market. Source: Oxera.

A2.8.2 Dynamic efficiencies

- A2.76 The merging parties put forward a number of efficiency claims. Dynamic efficiencies included: efficiencies in R&D, through the combination of complementary expertise and an acceleration/expansion of GRAIL's R&D as a result of Illumina's greater financial resources; and market access acceleration for GRAIL's Galleri test (e.g. by speeding up the regulatory approval process) resulting from using Illumina's management capabilities. All efficiency claims were dismissed by the FTC as not sufficiently substantiated and not merger-specific and the FTC noted that, even if they were, the merging parties had not demonstrated that they would have been passed on to consumers (noting the current absence of competitive pressure on GRAIL in the downstream market).⁹³⁰

⁹²⁷ Illumina/GRAIL (2023, US), pp 36 and 47–57.

⁹²⁸ Illumina/GRAIL (2023, US), pp 43–47 and pp 58–59. The FTC notes (p. 47): 'In sum, Illumina's acquisition of GRAIL increases the risks to GRAIL's competitors of doing business with Illumina, the only viable alternative for NGS for MCED developers, thereby increasing barriers to entry at the downstream level of the market'.

⁹²⁹ Illumina/GRAIL (2023, US), pp 56 and 59.

⁹³⁰ Illumina/GRAIL (2023, US), pp 74–79 and 86.

A2.9 Case 9: Orange/MásMóvil (2024, EU)

- Authority: European Commission (M.10896)
- Relevant product market: retail and wholesale telecommunication markets
- Outcome: Phase II clearance with remedies

A2.77 In March 2022, Orange and MásMóvil announced their intention to create a joint venture combining their mobile and fixed telecommunication businesses in Spain. At the time, Orange supplied mobile and fixed telecommunication services to residential and business customers as well as wholesale customers, while MásMóvil was a hybrid mobile network operator (MNO) providing fixed and mobile services to residential customers. As a hybrid MNO, MásMóvil relied partly on its own mobile network and partly on a national roaming agreement with Orange. The transaction raised concerns in four markets: retail mobile services, fixed internet access services, retail multiple-play bundles, and fixed-mobile convergent (FMC) bundles.⁹³¹ The transaction would have created the largest operator in terms of both mobile and broadband retail customers in Spain with a market share increment of up to 20% across all relevant retail markets. The European Commission estimated a level of the Gross Upward Pricing Pressure Index for the two merging parties in a range of 10–20%, or even higher in some instances, suggesting a significant impact of the transaction on final market prices.⁹³² Concerns around the foreclosure of mobile virtual network operators were instead dismissed. In February 2024, the Commission approved the transaction subject to divestment remedies.

A2.9.1 Efficiency assessment framework

A2.78 A key dynamic aspect of the case was the assessment of efficiencies claimed by the merger parties. In earlier telecommunication mergers, the Commission had accepted that mergers could generate some efficiencies, notably through the elimination of wholesale access costs for at least one of the parties, where such savings were verifiable, merger-specific, and likely to benefit consumers.⁹³³ In this instance, the parties submitted a large number of cost synergy and integration cost items over a 10-year period as well as a plan to improve fibre-to-the-home and 5G network coverage in Spain.

A2.79 The Commission concluded that only efficiencies realised within a four-year time horizon could be considered timely, in line with previous decisions in telecommunication cases.⁹³⁴ However, as it had previously stated, the Commission reiterated that any claimed efficiencies that would be realised in the future must be appropriately discounted when assessing whether such claimed efficiencies would be sufficient to outweigh the competitive harm.⁹³⁵

⁹³¹ ‘Multiple-play bundles’ refer to bundles including at least two of the following products: mobile telecommunication services, fixed telephony services, fixed internet access, and Pay-TV services. ‘FMC bundles’ are a particular type of ‘multiple-play bundles’ including at least one mobile and one fixed component.

⁹³² Orange/MásMóvil/JV (2024), Annex A 2.3.1, para. 31.

⁹³³ See for example Orange/Jazztel (2015, EU), and T-Mobile NL/Tele2 NL (2018, EU).

⁹³⁴ See for example Hutchison 3G UK/Telefónica Ireland (2014, EU), Hutchison 3G Italy/Wind/JV (2016, EU), para. 1391, and T-Mobile NL/Tele2 NL (2018, EU), para. 892.

⁹³⁵ Hutchison 3G UK/Telefónica Ireland (2014, EU).

A2.9.2 Cost efficiencies

- A2.80 The Commission assessed cost savings separating those which excluded Elimination of Double Marginalisation (EDM) and those related to EDM.
- A2.81 First, cost savings excluding EDM were divided into four categories: fixed network, mobile network, transmission network, and non-network synergies.⁹³⁶ The Commission considered that only variable cost savings related to FTTH consolidation and to the consolidation of mobile sites met the cumulative test for efficiencies. Fixed cost savings failed the test as they were found not to affect the parties' pricing incentives or their incentives to invest in incremental network roll-out.
- A2.82 Second, regarding EDM-related cost savings, the Commission considered that only variable cost savings from the elimination of FTTH wholesale access costs met the cumulative test. Variable cost savings related to the transmission network agreements were found not to benefit consumers as they did not vary with the number of subscribers and, as such, were unlikely to be passed on.
- A2.83 Overall, the Commission's conclusion was that the limited proportion of the claimed cost efficiencies that were verifiable, merger-specific, and likely to benefit consumers fell short of offsetting the likely substantial price effects of the transaction.⁹³⁷

A2.9.3 Network coverage efficiencies

- A2.84 On network coverage efficiencies, the Commission found that the efficiency claims for both the incremental FTTH roll-out and the incremental 5G roll-out were not verifiable as the JV's roll-out plans were not binding. Additionally, the Commission considered that the transaction would not increase the merged entity's ability and incentive to invest in incremental FTTH and 5G network roll-out. As regards ability, it considered that the merger parties were financially sound undertakings with access to capital (including via financial lease agreements).⁹³⁸ As regards incentives, it argued that migrating customers to the merged entity's own networks brought no extra benefit compared to the pre-merger scenario; in fact the merger parties would lose the benefit from attracting subscribers from each other, which could actually reduce the incentive to invest in areas where both parties competed pre-merger.⁹³⁹
- A2.85 The Commission also pointed out that the claimed network roll-out efficiencies did not meet the merger-specificity condition as the merger parties could rely on less-anticompetitive alternatives to achieve them, such as FTTH roll-out by third parties, network co-deployment (with each other or third parties), financial lease agreements, RAN-sharing, and several different types of network sharing agreements (e.g. VAS, NRA, RaaS). As to the parties' arguments that the roll-out plans would not be as attractive for either party standalone or through other forms of cooperation, the Commission argued that less anticompetitive alternatives can be '*realistic and attainable*' even if not as profitable for the merging parties as the full merger.⁹⁴⁰
- A2.86 The Commission acknowledged that in areas where a network of equivalent quality already existed, incremental roll-out could in principle exert downward pressure on

⁹³⁶ Non-network synergies were related to (i) sales and marketing, (ii) customer care, and (iii) general expenses, personnel expenses, and IT.

⁹³⁷ Orange/MásMóvil/JV (2024), para. 1739(a).

⁹³⁸ Orange/MásMóvil/JV (2024), para. 1687(a).

⁹³⁹ Orange/MásMóvil/JV (2024), para. 1687(c).

⁹⁴⁰ Orange/MásMóvil/JV (2024), para. 1690(b).

prices. However, it considered that this would not occur in practice because the parties and most rivals set retail prices nationally rather than offering lower prices in areas covered by their own infrastructure compared to the prices offered in areas where they relied on wholesale access. Furthermore, while customers in areas lacking high-quality network coverage could potentially benefit from new infrastructure deployment, the Commission found that due to Spain's existing network coverage, only a limited proportion of consumers would actually gain access to improved services through the incremental roll-out.⁹⁴¹

A2.87 While the claimed efficiencies resulting from the incremental roll-out were not accepted, the Commission noted that, even if they were, they would only materialise in the medium term and would benefit only a small subset of retail customers. As such, they would be unlikely to outweigh the substantial likely price effects that would be felt immediately by retail consumers across Spain.⁹⁴²

A2.9.4 Remedies

A2.88 The Commission approved the merger with a remedy package consisting of divestments of spectrum rights in favour of Digi, the largest MVNO in Spain, with the goal of creating a new MNO.^{943, 944} The remedies also included a commitment to make a binding offer to enter into a capacity-based National Roaming Agreement (NRA) with the new MNO. This provided Digi with the right to exercise the NRA Option with an opt-in mechanism until a certain date.

A2.10 Case 10: Vodafone/Three (2024, UK)

- Authority: CMA
- Relevant product market: retail and wholesale mobile services
- Outcome: Phase II clearance with remedies

A2.89 In June 2023, Vodafone and Three (owned by CK Hutchison) announced their intention to create a joint venture combining their mobile businesses in the UK. They were the third and fourth UK mobile network operators (MNOs) respectively, after BTEE and Virgin Media O2. In December 2024, the CMA concluded that the merger would remove two close competitors in a concentrated market, both at the wholesale and the retail levels.⁹⁴⁵ In relation to wholesale mobile services to mobile virtual network operators, the merger would reduce the number of MNOs from four to three, in a market with high switching costs and incumbency advantages. In relation to retail mobile services, the merging parties were placing competitive pressure on each other and other mobile operators, with Three being the cheapest MNO and having improved its network quality.⁹⁴⁶ Using merger simulation tools, the CMA found a material upward impact on retail prices, with the merged entity predicted to increase Three's tariffs by 5.5% on average and Vodafone's tariffs by

⁹⁴¹ The decision also notes that consumers' valuation of quality can be assessed in terms of willingness to pay and that internal documents prepared by the Parties in the ordinary course of business consider consumers' willingness to pay for 5G network services. See *Orange/MásMóvil/JV* (2024), para. 1728.

⁹⁴² *Orange/MásMóvil/JV* (2024), para. 1694(d).

⁹⁴³ At the moment of the transaction, Digi relied on Telefónica's mobile network to provide mobile services and was building its FTTH network to provide fixed services.

⁹⁴⁴ *Orange/MásMóvil/JV* (2024), para. 96.

⁹⁴⁵ *Vodafone/Three* (2024, UK). For a commentary on the case, see Akhoundova et al. (2025).

⁹⁴⁶ *Vodafone/Three* (2024, UK), paras 8.276–8.279 and 9.286.

2.6%. BTEE and VMO2 would also increase their prices by 0.6% and 0.4% respectively on average.⁹⁴⁷

- A2.90 A core aspect of the assessment was a detailed analysis of whether the dynamic efficiencies put forward by the merging parties, resulting from the combination of their networks and spectrum, were sufficient to offset the adverse price effects identified. Following the CMA's findings, the merger was approved subject to a behavioural remedy package which was designed to secure the dynamic efficiencies claimed by the parties. In what follows, we will focus on the efficiency and remedy assessments.

A2.10.1 Efficiency assessment

- A2.91 The central aspect of the dynamic efficiencies claimed by the merging parties was the improvement in network quality resulting from the integration of their networks (and related investments) which would accelerate the deployment of 5G SA across the UK. They claimed that without the merger they would remain 'sub-scale' compared to the other two MNOs, and therefore would be unable to invest sufficiently.⁹⁴⁸
- A2.92 Using its framework to assess efficiencies, the CMA considered that the merger would lead to rivalry-enhancing efficiencies. The merged firm would have more sites and spectrum (per site) than each mobile operator on a standalone basis. This would improve network quality in terms of better coverage, less congestion, and faster speeds. The merged firm would also have incentives over time to fill its increased capacity by making more attractive offers to customers and could incrementally expand capacity at a lower cost. The CMA also considered that the merger would lead to network quality improvements for VMO2 through the network sharing agreement with Vodafone and the acquisition of some spectrum from Vodafone upon approval of the merger.⁹⁴⁹
- A2.93 While the CMA found that these dynamic efficiencies were merger-specific, would benefit UK customers, and could be achieved within a sufficiently timely short-to-medium term, it had concerns around whether the merging parties would have the incentive to deliver the efficiencies in full in a timely manner.⁹⁵⁰ The CMA found that the merging parties' financial modelling (under alternative reasonable assumptions) was consistent with a smaller scale implementation of their network integration plan. The CMA also assessed the two merger simulation models submitted by the merging parties to support their claimed efficiencies. While it noted that the models helped shape its thinking on the trade-off between the effect of quality increases and price changes on consumer welfare and helped to refine aspects of its own

⁹⁴⁷ The CMA noted that these estimates were likely to be underestimates as the merger simulation did not account for the impact on the retail market from the loss of competition in the wholesale market. In particular, faced with less competitive wholesale terms, MVNOs would be less able to compete in the retail market, particularly in the low-cost segment where they tended to operate. Vodafone/Three (2024, UK), paras 8.315 and 8.335–8.336.

⁹⁴⁸ Vodafone/Three (2024, UK), paras 14.10–14.13.

⁹⁴⁹ Vodafone/Three (2024, UK), paras 14.213–14.218. It is worth noting that some of the network changes expected as a result of the merger—those that would enable Three UK customers to access the Vodafone UK network and vice versa, and to deploy additional spectrum through sharing of the merging parties' combined holdings—would result in short term quality improvements (in particular in coverage and reduction of congestion) over a significant part of the UK.

⁹⁵⁰ Vodafone/Three (2024, UK), para. 14.243.

modelling, the CMA had a number of reservations about the models' predictive value and robustness to alternative assumptions.⁹⁵¹

- A2.94 The CMA therefore concluded that the merged entity would have the ability and incentive to make some of the network improvement investments, but not the full scale they claimed, noting that the merged entity could/would re-assess the scale, prioritisation, and pace of network investments in light of future market circumstances. In particular, it noted that the merging parties may have incentives to retain fewer sites than claimed, particularly in low and mid traffic areas, given the considerable cost savings this would generate. Finally, the CMA carried out a qualitative balancing exercise of the likely harm (retail and wholesale price increases in the shorter term) and benefits (network quality and capacity improvements in the longer term) and concluded that the efficiencies which were likely to materialise would not be sufficient to offset the price increases expected to result from the merger.⁹⁵²

A2.10.2 Remedies

- A2.95 The CMA accepted a proposed behavioural remedy package (the 'Network Commitment Package') with the twofold objective of addressing the competition concerns in the retail and wholesale markets and securing the full-scale network quality improvements claimed by the merging parties.⁹⁵³ The remedy primarily involved legally-binding network investment commitments by the merged entity to undertake the proposed network investment plan over a period of eight years. The merged entity's and VMO2's greater capacity would lead to downward pressure on prices as well as lower unit costs of expanding capacity in the longer term. It would also lead to significant network quality improvements across a range of key parameters (including coverage, congestion, and effective speed). This would in turn induce a competitive response from BTEE. The CMA, however, recognised that it would take time for these competitive benefits to materialise and, as such, it imposed time-limited price protections for customers at both the retail and wholesale levels.⁹⁵⁴

⁹⁵¹ For a commentary on the assessment of the merging parties' quantitative modelling, see Akhoundova et al. (2025).

⁹⁵² Vodafone/Three (2024, UK), paras 14.225–14.227, 14.232, 14.236–14.237, 14.245, and 14.252.

⁹⁵³ Vodafone/Three (2024, UK), para. 16. The remedy was described as an enabling measure that works 'with the grain of competition' by fundamentally changing the merged entity's incentives (see also Akhoundova et al., 2025).

⁹⁵⁴ Vodafone/Three (2024, UK), paras 16.208–16.216.

A3 European Commission cases with dynamic merger effects

- A3.1 This annex provides an overview of all European Commission merger cases in which dynamic effects were an important element of the Commission's assessment. It provides the basis for the summary statistics provided in Section 2.3.
- A3.2 To identify Commission decisions relevant to the analysis of dynamic effects in merger control, a two-stage process was employed.
- A3.3 In the first stage, we inputted a list of prompts and keywords related to dynamic effects into an LLM, which performed text-based analysis on all final decisions published by the Commission, i.e. those falling under Art. 6 and Art. 8. The keywords and prompts covered various aspects of innovation and dynamic competition. The LLM was instructed to flag excerpts as relevant only where they appeared within the Commission's own assessment—excluding, for example, claims made by the merging parties or third parties. For a decision to be classified as relevant, dynamic effects did not need to constitute a key determinant of the outcome, but rather needed to have been considered by the Commission in their assessment to a sufficient extent.
- A3.4 In the second stage, the LLM-identified decisions were subject to manual review to validate and refine the classification. For cases that had been withdrawn, the relevant press release on the withdrawal was additionally reviewed.
- A3.5 The sample covers merger cases up to and including 2025, dated by date of final outcome.

Table A3.1 List of European Commission dynamic merger cases

Number	Merging Parties	Outcome	Year	Sector
M.3354	Sanofi-Synthélabo/Aventis	Phase I remedies	2004	Pharmaceuticals
M.3440	ENI/EDP/GDP	Phase II prohibition	2004	Other (Energy)
M.3445	Microsoft/Time Warner/ContentGuard	Phase II withdrawal	2005	Tech
M.3687	Johnson & Johnson/Guidant	Phase II remedies	2005	Medical devices
M.3796	Omya/J.M. Huber PCC	Phase II remedies	2006	Other (Industrial minerals)
M.3916	T-Mobile Austria/tele.ring	Phase II remedies	2006	Telecom
M.3923	AMI/Eurotecnica	Phase II withdrawal	2005	Other (Chemicals)
M.3998	Axalto/Gemplus	Phase I clearance	2006	Tech
M.4150	Abbott/Guidant	Phase I clearance	2006	Medical devices
M.4153	Toshiba/Westinghouse	Phase I remedies	2006	Other (Industrial machinery)
M.4187	Metso/Aker Kvaerner	Phase II remedies	2006	Other (Industrial machinery)
M.4439	Ryanair/Aer Lingus	Phase II prohibition	2007	Aviation
M.4731	Google/DoubleClick	Phase II clearance	2008	Tech
M.4747	IBM/Telelogic	Phase II clearance	2008	Tech
M.4854	TomTom/Tele Atlas	Phase II clearance	2008	Tech
M.4942	Nokia/Navteq	Phase II clearance	2008	Tech
M.5253	Sanofi-Aventis/Zentiva	Phase I remedies	2009	Pharmaceuticals
M.5440	Lufthansa/Austrian Airlines	Phase II remedies	2009	Aviation
M.5476	Pfizer/Wyeth	Phase I clearance	2009	Pharmaceuticals
M.5529	Oracle/Sun Microsystems	Phase II clearance	2010	Tech
M.5675	Syngenta/Monsanto's Sunflower Seed Business	Phase II remedies	2010	Agrochemicals
M.5830	Olympic/Aegean Airlines	Phase II prohibition	2011	Aviation
M.5865	Teva/Ratiopharm	Phase I remedies	2010	Pharmaceuticals
M.5984	Intel/McAfee	Phase I clearance	2011	Tech
M.6166	Deutsche Börse/NYSE Euronext	Phase II prohibition	2012	Other (Financial infrastructure)
M.6203	Western Digital Ireland/Viviti Technologies	Phase II remedies	2011	Tech
M.6214	Seagate/HDD Business of Samsung	Phase II clearance	2011	Tech
M.6258	Teva/Cephalon	Phase I remedies	2011	Pharmaceuticals
M.6266	Johnson & Johnson/Synthes	Phase II remedies	2012	Pharmaceuticals

Number	Merging Parties	Outcome	Year	Sector
M.6314	Telefónica UK/Vodafone UK/Everything Everywhere/JV	Phase I clearance	2012	Tech
M.6410	UTC/Goodrich	Phase II remedies	2012	Other (Aerospace)
M.6497	Hutchison 3G Austria/Orange Austria	Phase II remedies	2012	Telecom
M.6564	ARM/Giesecke & Devrient/Gemalto/JV	Phase I clearance	2012	Tech
M.6663	Ryanair/Aer Lingus III	Phase II prohibition	2013	Aviation
M.6765	Precision Castparts/Titanium Metals	Phase I clearance	2012	Other (Titanium products)
M.6796	Aegean/Olympic II	Phase II clearance	2013	Aviation
M.6944	Thermo Fisher Scientific/Life Technologies	Phase I clearance	2013	Other (Laboratory equipment)
M.6992	Hutchison 3G UK/Telefónica Ireland	Phase II remedies	2014	Telecom
M.7000	Liberty Global/Ziggo	Phase II remedies	2014	Telecom
M.7018	Telefónica Deutschland/E-Plus	Phase II remedies	2014	Telecom
M.7217	Facebook/WhatsApp	Phase I clearance	2014	Tech
M.7265	Zimmer/Biomet	Phase II remedies	2015	Medical devices
M.7275	Novartis/GlaxoSmithKline Oncology Business	Phase I clearance	2015	Pharmaceuticals
M.7276	GlaxoSmithKline/Novartis Vaccines Business (Excl. Influenza)/Novartis Consumer Health Business	Phase I clearance	2015	Pharmaceuticals
M.7278	General Electric/Alstom (Thermal Power – Renewable Power & Grid Business)	Phase II remedies	2015	Other (Power equipment)
M.7326	Medtronic/Covidien	Phase I clearance	2014	Medical devices
M.7419	TeliaSonera/Telenor/JV	Phase II withdrawal	2015	Telecom
M.7421	Orange/Jazztel	Phase II remedies	2015	Telecom
M.7477	Halliburton/Baker Hughes	Phase II withdrawal	2016	Other (Oilfield services)
M.7559	Pfizer/Hospira	Phase I clearance	2015	Pharmaceuticals
M.7567	Ball/Rexam	Phase II remedies	2016	Other (Packaging)
M.7612	Hutchison 3G UK/Telefónica UK	Phase II prohibition	2016	Telecom
M.7637	Liberty Global/BASE Belgium	Phase II remedies	2016	Telecom
M.7724	ASL/Arianespace	Phase I clearance	2016	Other (Space launch)
M.7746	Teva/Allergan Generics	Phase I clearance	2016	Pharmaceuticals
M.7758	Hutchison 3G Italy/Wind/JV	Phase II remedies	2016	Telecom
M.7822	Dentsply/Sirona	Phase I remedies	2016	Medical devices
M.7917	Boehringer Ingelheim/Sanofi Animal Health Business	Phase I clearance	2016	Pharmaceuticals

Number	Merging Parties	Outcome	Year	Sector
M.7932	Dow/DuPont	Phase II remedies	2017	Agrochemicals
M.7962	ChemChina/Syngenta	Phase II remedies	2017	Agrochemicals
M.7975	Mylan/Meda	Phase I remedies	2016	Pharmaceuticals
M.7995	Deutsche Börse/London Stock Exchange	Phase II prohibition	2017	Other (Financial infrastructure)
M.8060	Abbott Laboratories/St Jude Medical	Phase I clearance	2016	Medical devices
M.8084	Bayer/Monsanto	Phase II remedies	2018	Agrochemicals
M.8124	Microsoft/LinkedIn	Phase I clearance	2016	Tech
M.8222	Knorr-Bremse/Haldex	Phase II withdrawal	2017	Other (Vehicle braking systems)
M.8306	Qualcomm/NXP Semiconductors	Phase II remedies	2018	Tech
M.8314	Broadcom/Brocade	Phase I clearance	2017	Tech
M.8401	J&J/Actelion	Phase I clearance	2017	Pharmaceuticals
M.8523	BD/Bard	Phase I clearance	2017	Medical devices
M.8677	Siemens/Alstom	Phase II prohibition	2019	Other (Rail equipment)
M.8713	Tata Steel/ThyssenKrupp/JV	Phase II prohibition	2019	Other (Steel)
M.8792	T-Mobile NL/Tele2 NL	Phase II clearance	2018	Telecom
M.8851	BASF/Bayer Divestment Business	Phase I clearance	2018	Agrochemicals
M.8864	Vodafone/Certain Liberty Global Assets	Phase II remedies	2019	Telecom
M.8947	Nidec/Whirlpool (Embraco Business)	Phase I clearance	2019	Other (Compressors)
M.8955	Takeda/Shire	Phase I clearance	2018	Pharmaceuticals
M.9076	Novelis/Aleris	Phase II remedies	2019	Other (Metals)
M.9097	Boeing/Embraer	Phase II withdrawal	2020	Other (Aerospace)
M.9162	Fincantieri/Chantiers de l'Atlantique	Phase II withdrawal	2021	Other (Shipbuilding)
M.9294	BMS/Celgene	Phase I clearance	2019	Pharmaceuticals
M.9331	Danaher/GE Healthcare Life Sciences Biopharma	Phase I remedies	2019	Pharmaceuticals
M.9343	Hyundai Heavy Industries Holdings/Daewoo Shipbuilding & Marine Engineering	Phase II prohibition	2022	Other (Shipbuilding)
M.9409	Aurubis/Metallo Group Holding	Phase II clearance	2020	Other (Metals)
M.9424	NVIDIA/Mellanox	Phase I clearance	2019	Tech
M.9461	AbbVie/Allergan	Phase I clearance	2020	Pharmaceuticals
M.9547	Johnson & Johnson/Tachosil	Phase II withdrawal	2020	Pharmaceuticals
M.9554	Elanco Animal Health/Bayer Animal Health Division	Phase I clearance	2020	Pharmaceuticals

Number	Merging Parties	Outcome	Year	Sector
M.9564	LSEG/Refinitiv Business	Phase II remedies	2021	Other (Financial data)
M.9660	Google/Fitbit	Phase II remedies	2020	Tech
M.9827	International Flavors & Fragrances/Nutrition & Biosciences	Phase I clearance	2020	Other (Food ingredients)
M.9945	Siemens Healthineers/Varian Medical Systems	Phase I clearance	2021	Medical devices
M.9987	NVIDIA/ARM	Phase II withdrawal	2022	Tech
M.10078	Cargotec/Konecranes	Phase II remedies	2022	Other (Cargo equipment)
M.10188	Illumina/GRAIL	Phase II prohibition	2022	Medical devices
M.10262	Meta (formerly Facebook)/Kustomer	Phase II remedies	2022	Tech
M.10560	Sika/MBCC Group	Phase I remedies	2023	Other (Chemicals)
M.10615	Booking Holdings/eTraveli Group	Phase II prohibition	2023	Tech
M.10646	Microsoft/Activision Blizzard	Phase II remedies	2023	Tech
M.10783	EQT Future/AM Fresh/SNFL/IFG	Phase I clearance	2023	Other (Plant breeding)
M.10806	Broadcom/VMware	Phase II remedies	2023	Tech
M.10807	Viasat/Inmarsat	Phase II clearance	2023	Telecom
M.10896	Orange/MásMóvil/JV	Phase II remedies	2024	Telecom
M.10920	Amazon/iRobot	Phase II withdrawal	2024	Tech
M.11033	Adobe/Figma	Phase II withdrawal	2023	Tech
M.11043	Novozymes/Chr. Hansen Holding	Phase I clearance	2023	Other (Industrial biotechnology)
M.11071	Deutsche Lufthansa/MEF/ITA	Phase II remedies	2024	Aviation
M.11177	Pfizer/Seagen	Phase I clearance	2023	Pharmaceuticals
M.11481	Synopsys/Ansys	Phase I remedies	2025	Tech
M.11486	Novo Holdings/Novo Nordisk/Catalent	Phase I clearance	2024	Pharmaceuticals
M.11578	Boeing/Spirit	Phase I remedies	2025	Aviation
M.11663	Nokia/Infinera	Phase I clearance	2025	Tech
M.11936	Naspers/Just Eat Takeaway	Phase I remedies	2025	Other (Food delivery)

Note: Year refers to the year of the Art. 6(1)(b) decision in case of Phase I cases or the year of the Art. 8 decision or withdrawal announcement in case of Phase II cases. Phase II includes cases withdrawn in Phase II or where the European Commission's jurisdiction was later annulled (which only applies to M.10188 Illumina/GRAIL).

Source: Oxera.

A4 Cited cases

- A4.1 The list below provides an overview of all the merger control decisions by the European Commission, UK CMA, and US DOJ and FTC that are cited in the Study—including both Volume 1 and Volume 2.
- A4.2 This does not include all merger control decisions identified in our overview of European Commission merger cases with dynamic effects, which are instead listed separately in Annex A3.

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Aurubis/Metallo, COMP/M.9409 (2020), OJEU C/2020/224.

Ball/Rexam (2016), COMP/M.7567.

BASF/Bayer Divestment Business (2018), COMP/M.8851.

Bayer/Monsanto (2018), COMP/M.8084.

Bristol-Myers Squibb/Celgene (2019), COMP/M.9294.

Booking Holdings/eTraveli Group (2023), COMP/M.10615.

CSL/Vifor Pharma (2022), COMP/M.10629.

Deutsche Börse/NYSE Euronext (2012), COMP/M.6166.

Dow/DuPont (2017), COMP/M.7932.

ENI/EDP/GDP (2004), COMP/M.3440.

Elanco Animal Health/Bayer Animal Health Division (2020), COMP/M.9554.

Facebook/WhatsApp (2014), COMP/M.7217.

General Electric/LM Wind Power (2017), COMP/M.8283.

General Electric/Alstom (Thermal Power – Renewable Power & Grid Business) (2015), COMP/M.7278.

Glaxo Wellcome/SmithKline Beecham (2000), COMP/M.1846.

Google/Fitbit (2020), COMP/M.9660.

Hutchison 3G Austria/Orange Austria (2012), COMP/M.6497.

Hutchison 3G Italy/Wind/JV (2016), COMP/M.7758.

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Meta (formerly Facebook)/Kustomer (2022), COMP/M.10262.

Microsoft/Activision Blizzard (2023), COMP/M.10646.

Microsoft/LinkedIn (2016), COMP/M.8124.

Nidec/Whirlpool (2019), COMP/M.8947.

Norsk Hydro/Alumetal (2023), COMP/M.10658.

Novartis/GlaxoSmithKline Oncology Business (2015), COMP/M.7275.

Novozymes/Chr. Hansen Holding (2023), COMP/M.11043.

Orange/Jazztel (2015), COMP/M.7421.

Orange/MásMóvil (2024), COMP/M.10896.

Outokumpu/Inoxum (2013), COMP/M.6471.

Pfizer/Hospira (2015), COMP/M.7559.

Pfizer/Seagen (2023), COMP/M.11177.

Qualcomm/NXP Semiconductors (2018), COMP/M.8306.

Ryanair/Aer Lingus (2007), COMP/M.4439.

Ryanair/Aer Lingus III (2013), COMP/M.6663.

Siemens/Alstom (2019), COMP/M.8677.

Siemens/Gamesa (2017), COMP/M.8134.

Takeda/Shire (2018), COMP/M.8955.

Telefónica Deutschland/E-Plus (2014), COMP/M.7018.

TeliaSonera/Telenor Denmark (2015), COMP/M.7419 (withdrawn).

T-Mobile NL/Tele2 NL (2018), COMP/M.8792.

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Halliburton/Baker Hughes (2016), *United States v. Halliburton Company and Baker Hughes Incorporated*, Complaint, Case No. 1:16-cv-00233, 2016.

A5 Cited guidelines and recommended practices

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European Commission (2026), 'Draft Communication: Guidelines on the assessment of mergers under Council Regulation (EC) No 139/2004 on the control of concentrations between undertakings', 30 April.

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Glossary of abbreviations

The table below provides an overview of abbreviations used in the Study—including both Volume 1 and Volume 2.

Term	Definition
2SLS	Two-stage Least Squares
ACTH	Adrenocorticotrophic hormone
ADC	Antibody-drug conjugate
AI	Artificial intelligence
API	Application Programming Interface
ATMP	Advanced Therapy Medicinal Product
ATP	Antitakeover provisions
BCC	Mutual Cooperative Banks
BINs	Business/industry niches
BLA	Biologics License Application
B-RAF	B-Raf Serine/Threonine Kinase
BTB	Breakthrough Therapy Designations
CA	Companion animals
CAGR	Compound annual growth rate
CAPEX	Capital expenditure
CAR	Cumulative abnormal returns
CBA	Cost-benefit analysis
CCUS	Carbon Capture Utilisation and Storage
CEM	Coarsened Exact Matching
CEPR	Centre for Economic Policy Research
CES	Constant Elasticity of Substitution
CF	Counterfactual Scenario
CMA	Competition and Markets Authority
CPC	Cooperative Patent Classification
CPU	Central Processing Unit
CRSP	Center for Research in Security Prices
DID	Difference-in-Differences
DOJ	Department of Justice (US)
EBITDA	Earnings before interest, taxes, depreciation and amortisation
EC	European Commission
ECW	Expected Consumer Welfare
EDM	Elimination of Double Marginalisation
EEA	European Economic Area
EFPIA	European Federation of Pharmaceutical Industries and Associations
EMA	European Medicines Agency
EPO	European Patent Office
EU	European Union
EU27	27 EU Member States
EUMR	European Union Merger Regulation (Regulation 139/2004)

Term	Definition
FDA	Food and Drug Administration (US)
FDI	Foreign direct investment
FDR	Financial Discount Rate
F&E	Forschung und Entwicklung
FMC	Fixed-mobile Convergent
FTC	Federal Trade Commission (US)
FTTH	Fibre-to-the-home
GAAP	Generally Accepted Accounting Principles
GAFAM	Alphabet (Google), Apple, Meta (Facebook), Amazon, and Microsoft
GDP	Gross domestic product
GEE	Generalised estimating equations
GIF	Graphics Interchange Format
GM	Genetically Modified
GMM	Generalized Method of Moments
GUPPI	Gross Upward Pricing Pressure Index
GVA	Gross Value Added
HDD	Hard Disk Drives
HHI	Herfindahl-Hirschman Index
HTV	Hard to Value
ICM	Internal capital markets
ICN	International Competition Network
IIA	Independence of Irrelevant Alternatives
IL-23	Interleukin-23
IO	Industrial Organisation
IP	Intellectual property
IPO	Initial public offering
IPRs	Intellectual property rights
IRI	Implied Return-on-equity Improvement
IRR	Internal rate of return
IV	Instrumental Variable
JV	Joint Ventures
KOL	Key Opinion Leader
LASSO	Least absolute shrinkage and selection operator
LATE	Local average treatment effect
LCD	Liquid crystal display
LLM	Large language model
LRAIC	Long Run Average Incremental Cost
M&A	Mergers & Acquisitions
MCED	Multi-cancer early detection
MEK	Mitogen-activated protein kinase kinase
MNO	Mobile network operator
MoA	Mechanisms of action
MOPS	US Census Management and Organizational Practice Survey

Term	Definition
MVNO	Mobile virtual network operator
NACE	Nomenclature of Economic Activities
NAICS	North American Industry Classification System
NBER	National Bureau of Economic Research
NGS	Next-generation sequencing
NISQ	Noisy Intermediate-Scale Quantum (devices)
NME	New Molecular Entity
NPV	Net Present Value
NRA	National Roaming Agreement
NSA	Network Sharing Agreement
OECD	Organisation for Economic Co-operation and Development
OFT	Office of Fair Trading
OLS	Ordinary Least Squares
PA	Production animals
PATSTAT	Patent Statistical Database
PPA	Purchase Price Allocation
PRT	Property Rights Theory
QC	Quantum computing
QT	Quantum technologies
R&D	Research and development
RJV	R&D Joint Ventures
ROE	Return on Equity
ROI	Return on Investment
SDR	Social Discount Rate
SIC	Standard Industrial Classification
SIEC	Significant Impediment to Effective Competition
SLC	Substantial Lessening of Competition
SMEs	Small and medium-sized enterprises
SPAC	Special Purpose Acquisition Company
SPRU	Science Policy Research Unit
SUTVA	Stable Unit Treatment Assumption
TCE	Transaction Cost Economics
TFP	Total Factor Productivity
UK	United Kingdom
US	United States
USPTO	United States Patent and Trademark Office
VC	Venture Capital
WTO	World Trade Organization

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Abstract

This Study develops an economic framework for assessing dynamic effects in mergers, focusing on how mergers affect future product market competition, including through their impact on investment, innovation, and entry. After an introduction and context (Part A), the Study sets out a detailed synopsis of the relevant academic literature on dynamic merger effects (Part B), integrating theoretical and empirical insights from industrial economics, financial economics, macroeconomics, and the business literature. The framework (Part C) defines dynamic merger effects and puts forward a substantive standard based on the impact of mergers on Expected Consumer Welfare. It distinguishes horizontal dynamic harms, horizontal dynamic benefits, non-horizontal dynamic foreclosure, non-horizontal dynamic benefits, and entrenchment effects. The Study applies the framework to canonical merger cases, including mergers involving: (i) pipeline overlaps, (ii) capability overlaps, (iii) nascent competitors, (iv) dynamic foreclosure, and (v) entrenchment. A separate sectoral analysis of biopharmaceuticals, quantum computing, and clean technologies (Part D) illustrates the role of merger control in the broader context of these three dynamic sectors.

Résumé

La présente étude élabore un cadre économique d'analyse pour l'appréciation des effets dynamiques des concentrations, en mettant l'accent sur la manière dont celles-ci influent sur la concurrence future sur les marchés de produits, notamment par leur impact sur l'investissement, l'innovation et l'entrée. Après une introduction et une mise en contexte (partie A), l'étude présente une synthèse détaillée de la littérature académique pertinente sur les effets dynamiques des concentrations (partie B), intégrant les enseignements théoriques et empiriques provenant de l'économie industrielle, de l'économie financière, de la macroéconomie et des sciences de gestion. Le cadre d'analyse (partie C) définit les effets dynamiques des concentrations et propose un critère substantiel fondé sur leur impact sur le bien-être attendu des consommateurs. Il distingue les préjudices dynamiques horizontaux, les bénéfices dynamiques horizontaux, le verrouillage dynamique non horizontal, les bénéfices dynamiques non horizontaux et les effets de renforcement de position. L'étude applique ce cadre d'analyse à des cas canoniques de concentrations impliquant : (i) des chevauchements de projets ou produits en développement, (ii) des chevauchements de capacités en termes de ressources et compétences, (iii) des concurrents émergents, (iv) un verrouillage dynamique et (v) un renforcement de position. Une analyse sectorielle distincte portant sur les secteurs de la biopharmaceutique, de l'informatique quantique et des technologies propres (partie D) illustre le rôle du contrôle des concentrations dans le contexte plus large de ces trois secteurs dynamiques.

Kurzfassung

Diese Studie entwickelt einen ökonomischen Rahmen zur Beurteilung dynamischer Effekte von Zusammenschlüssen. Im Mittelpunkt steht die Frage, wie Zusammenschlüsse den künftigen Produktmarktwettbewerb beeinflussen, insbesondere durch ihre Auswirkungen auf Investitionen, Innovationen und Markteintritt. Nach einer Einführung und einer Einordnung in den Kontext (Teil A) bietet die Studie einen detaillierten Überblick über die einschlägige wissenschaftliche Literatur zu dynamischen Zusammenschlusswirkungen (Teil B) und verbindet theoretische und empirische Erkenntnisse aus Industrieökonomik, Finanzökonomik, Makroökonomik und Betriebswirtschaftslehre. Der Rahmen (Teil C) definiert dynamische Zusammenschlusswirkungen und entwickelt einen materiellen Prüfungsmaßstab, der auf den Auswirkungen auf die erwartete Konsumentenwohlfahrt beruht. Er unterscheidet horizontale dynamische Schäden, horizontale dynamische Vorteile, nicht-horizontale dynamische Abschottung, nicht-horizontale dynamische Vorteile und Verfestigungswirkungen. Die Studie wendet den Rahmen auf typische Zusammenschlussfälle an, darunter Zusammenschlüsse mit (i) Pipeline-Überschneidungen, (ii) Überschneidungen bei Fähigkeiten, (iii) aufstrebenden Wettbewerbern, (iv) dynamischer Abschottung und (v) Verfestigung. Eine gesonderte Sektoranalyse zu Biopharmazie, Quantencomputing und sauberen Technologien (Teil D) veranschaulicht die Rolle der Fusionskontrolle im breiteren Kontext dieser drei dynamischen Sektoren.



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